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AIRCRAFT HANDLING QUALITIES DATA

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Prepared by

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16. Abstract Available information on weight and inertia, aerodynamic derivatives, control characteristics, and stability augmentation systems is documented for 10 representative contemporary airplanes. Data sources are given for each airplane. Flight envelopes are presented and dimensional derivatives, transfer functions for control inputs, and several selected handling qualities parameters have been computed and are tabulated for 10 different flight conditions including the power approach configuration. The airplanes documented are the NT-33A, F-104A, F-4C, X-15, HL-10, Jetstar, CV-880M, B-747, C-5A, and XB-70A.			
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SECTION I

INTRODUCTION

The purpose of this document is to provide handling qualities investigators with readily usable data on several representative contemporary aircraft. Included are those data required to obtain transfer functions relating the aircraft's response to control inputs. An analytical description of the aircraft's stability augmentor is also given.

For those aircraft for which complete information was available, the following summarizes the contents and presentation:

1. Flight conditions for which computations are made including:
 - a. Configurations (e.g., fuel load, flaps, gear, etc.)
 - b. Mach/altitude combinations
2. General arrangement
3. Control system description
4. Stability augmentation description
5. Tabulations and/or plots of non-dimensional stability derivatives for trimmed flight
6. Dimensional, mass, and flight condition parameters
7. Dimensional stability derivatives
8. Transfer functions for control inputs
9. Selected handling qualities parameters
10. Data sources

A page number cross index is presented in Table I-1.

The intention has been to make this report completely self-consistent insofar as symbols, nomenclature, definitions, etc. The system used is described in three appendices. Appendix A covers axis systems, symbols and notation, and definitions of nondimensional and dimensional stability derivatives. Appendix B gives the axis system transformations for the derivatives. Appendix C includes the aircraft equations of motion and transfer functions used herein.

TABLE I-1

PAGE NUMBER CROSS INDEX

	NT-33A	F-104A	F-4C	X-15	HL-10	Jetstar	CV-880M	B-747	C-5A	XB-70A
BACKGROUND	6	33	62	109	136	167	194	211	244	274
FLIGHT CONDITIONS	7	34	63	110	139	168	195	212	245	275
GENERAL ARRANGEMENT	8	35	64	111	140	169	196	213	246	276
CONTROL SYSTEM	9	36	65	112	141	170	197	214	247	277
STABILITY AUGMENTATION SYSTEM	-	-	69	113	142	-	-	215	-	278
TRIMMED NON-DIMENSIONAL DERIVATIVES	10	37	70	114	143	171	198	216	248	279
DIMENSIONAL, MASS AND FLIGHT CONDITION PARAMETERS	22	49	82	125	152	183	200	229	261	292
LONGITUDINAL DIMENSIONAL DERIVATIVES	23	50	83	126	153	184	201	230	262	293
LONGITUDINAL TRANSFER FUNCTION FACTORS										
• SAS off										
- Bobweight loop open										
• Pitch axis control	24	51	84	127	154	185	202	231	263	294
• Thrust	25	52	85	-	-	186	203	232	264	295
- Bobweight loop closed										
• Pitch axis control	-	53	86	-	-	-	-	-	265	296
• Thrust	-	54	88	-	-	-	-	-	266	297
• SAS on										
- Bobweight loop open										
• Pitch axis control	-	-	90	127	155	-	-	-	-	299
• Thrust	-	-	92	-	-	-	-	-	-	301
- Bobweight loop closed										
• Pitch axis control	-	-	94	-	-	-	-	-	-	303
• Thrust	-	-	96	-	-	-	-	-	-	305
LONGITUDINAL HANDLING QUALITIES FACTORS	26	55	98	129	156	187	204	233	267	307
LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES	27	56	99	130	157	188	205	234	268	308
LATERAL-DIRECTIONAL TRANSFER FUNCTION FACTORS										
• SAS off										
- Roll axis control	28	57	100	131	158	189	206	235	269	309
- Yaw axis control	29	58	101	132	159	190	207	236	270	310
• SAS on										
- Roll axis control	-	-	102	133	160	-	-	237	-	311
- Yaw axis control	-	-	104	134	162	-	-	239	-	313
LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS	30	59	106	135	164	191	208	241	271	315
DATA SOURCES	31	60	107	136	165	192	209	242	272	316

The aircraft considered in this report span a wide range of sizes, speeds, and uses. In each case, transfer functions and handling qualities parameters were computed for flight conditions which were selected to cover the flight regimes of interest. A nominal configuration (generally cruise) was picked for all up and away flight conditions. For this nominal configuration, plots of trimmed non-dimensional aerodynamic force and moment coefficients are presented. Also, in most cases, a power approach case is presented along with a tabulation of aerodynamic coefficients. The coefficients are based on rigid wind tunnel data, estimated flexible data, or flight test results, depending upon availability. This is indicated by the words "rigid," "flexible," and "flight" on each aero data plot. Also, the axis system is indicated by "stability" for a body-fixed stability axis system or "body" for a body-fixed system aligned with the F.R.L. (Further clarification of axis systems used is given in Appendix A.) Descriptions of control systems and stability augmentation systems are given along with transfer functions. Where a longitudinal control system has a significant effect on the equations of motion (as with a bobweight) the stick-free transfer functions and handling qualities are given.

Transfer functions are always given for body axis motion quantities. Handling qualities parameters are also given in the body axis. All acceleration transfer functions (a_z' and a_y') are for the pilot's position. Thrust transfer functions do not include any engine response characteristics.

A substantial portion of this report is in the form of computer printout. The mnemonics used in this printout are defined in Appendix A.

The handling qualities parameters given in this report represent only a small fraction of those developed over the years. The majority presented here are used in past and present versions of MIL-F-8785. Although only SAS-off values are shown, the definitions given in Appendix A are general and could be used in conjunction with the SAS-on transfer functions to yield SAS-on handling qualities parameters.

While complete coverage of each aircraft including only the "latest" and "best" data would be desirable, the major criterion used was that the data be accessible to the author. This is why only isolated flight conditions are given for some aircraft, and also why, as those people more intimately familiar

with each particular aircraft will recognize, the data presented may represent an early estimate in the design process and perhaps the "nominal configuration" is one which never left the drawing board. The data have been reviewed and, although not all those presented indicate unquestionable trends, those data known to be based on only early "guesstimates" or showing unreasonable trends have been deleted. In some cases data were estimated by the author. As to how well the data can be expected to match the flying aircraft, it is assumed that those for whom this document is intended know well the difficulties of obtaining derivatives from flight test data. Every attempt has been made to insure reliable translation, interpretation, and transcription of the data from their source documents.

The manufacturers of the aircraft described herein can not be held accountable for the information presented, nor would they be bound to concur in any conclusions with respect to their aircraft which might be derived from its use.

SECTION II

NT-33A

NT-33A BACKGROUND

"The NT-33A variable stability airplane (Serial No. 51-4120) is an extensively modified T-33 jet trainer. The elevator, aileron and rudder controls in the front cockpit are disconnected from their respective control surfaces and have been connected to separate servomechanisms that make up an 'artificial feel' system. In addition, the elevator, aileron and rudder control surfaces have been connected to individual servos which can be driven by a number of different inputs. These servos receive their electrical inputs from the artificial feel system (pilot's commands, position or force), attitude and rate gyros, accelerometers, dynamic pressure, α vane and β probe. This arrangement, through a response-feedback system, allows the normal T-33 derivatives to be augmented to the extent that the handling qualities of many existing airplanes, future airplanes or hypothetical research configurations, can be simulated. The original T-33 nose section has been replaced with the larger nose of an F-94 to provide the volume required for the electronic components of the response-feedback system and the recording equipment."*

Transfer functions are given for only the primary surfaces and engine thrust although the NT-33A also has other control surfaces and a range of control crossfeed and feedback combinations.

Aerodynamic data, for the most part, was taken from AFFDL-TR-70-71. However, longitudinal data for the high lift configuration was obtained from LAL 127 and Mach number derivatives from NACA-RM-7116.

Nominal Configuration

230 gal Tip Tanks
 60% Internal Fuel
 $W = 13700 \text{ lb}$
 c.g. at $0.263 \bar{c}$, W.L. 100.2

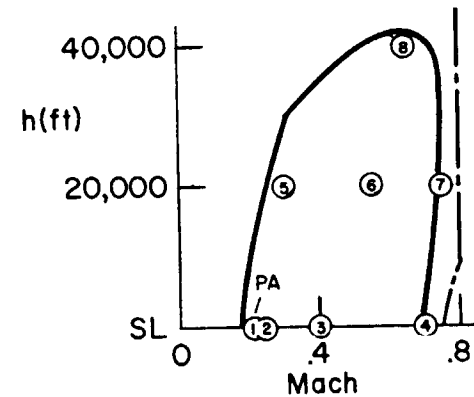
$I_x = 23800 \text{ slug-ft}^2$ $I_y = 21100 \text{ slug-ft}^2$ $I_z = 43800 \text{ slug-ft}^2$ $I_{xz} = 480 \text{ slug-ft}^2$	}	Body Axis
---	---	-----------

Power Approach Configuration

230 gal Tip Tanks
 25% Internal Fuel
 Full Flaps
 Gear Down
 $1.4 V_S$
 $W = 11800 \text{ lb}$
 c.g. at $0.260 \bar{c}$, W.L. 100

$I_x = 12700 \text{ slug-ft}^2$ $I_y = 20700 \text{ slug-ft}^2$ $I_z = 32000 \text{ slug-ft}^2$ $I_{xz} = 480 \text{ slug-ft}^2$	}	Body Axis
---	---	-----------

Flight Envelope



———— Level Flight Envelope (Nominal Configuration)
 - - - - - Speed Restrictions
 (n) Transfer Function Case n

Figure II-1. NT-33A Flight Conditions

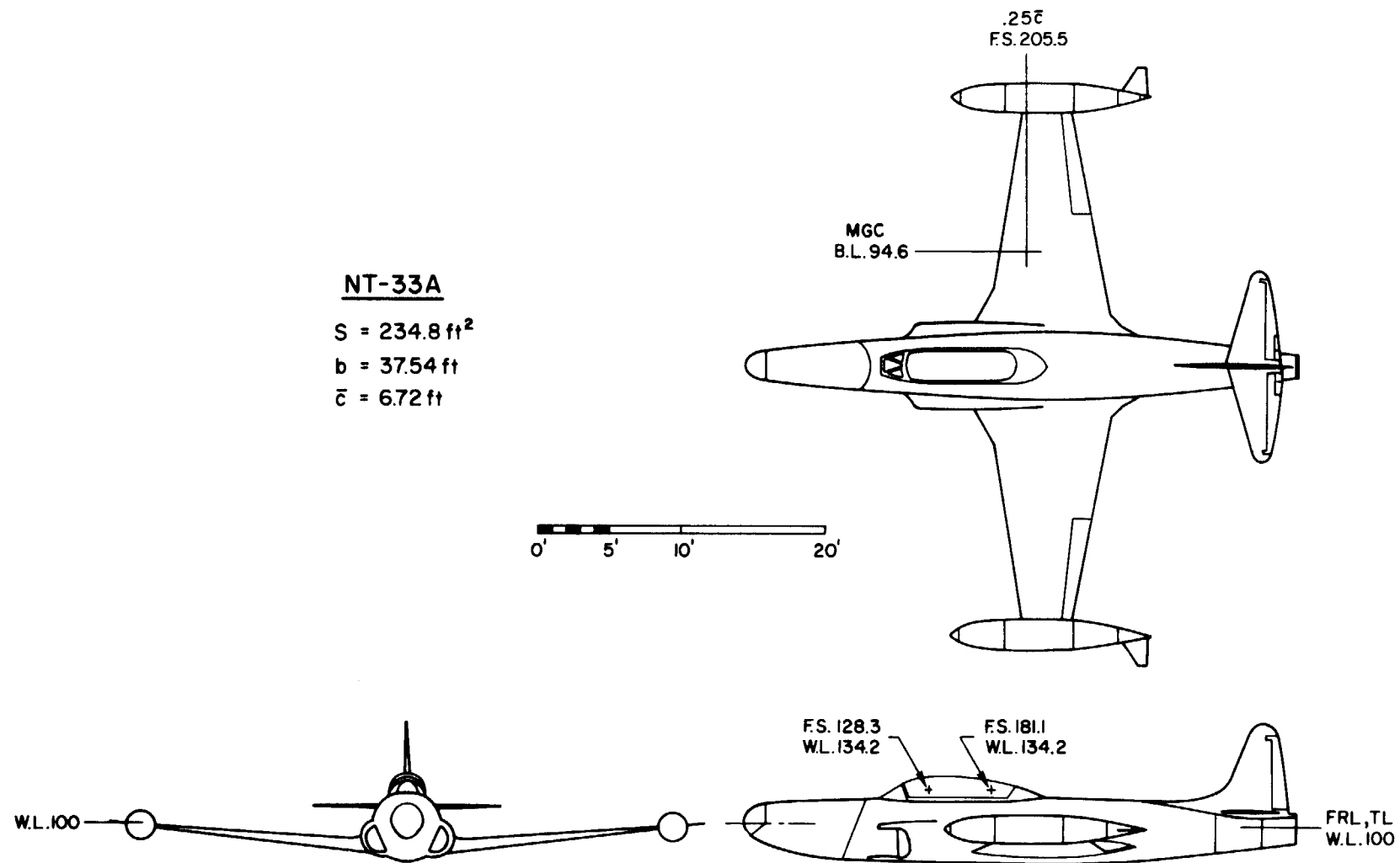
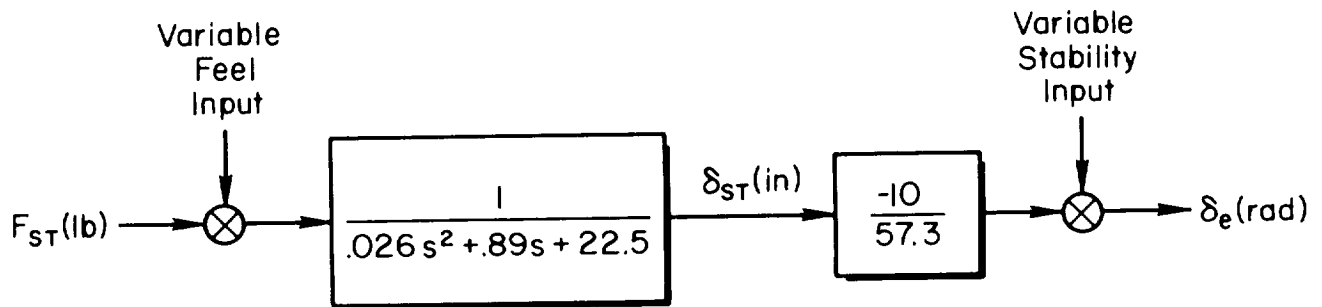


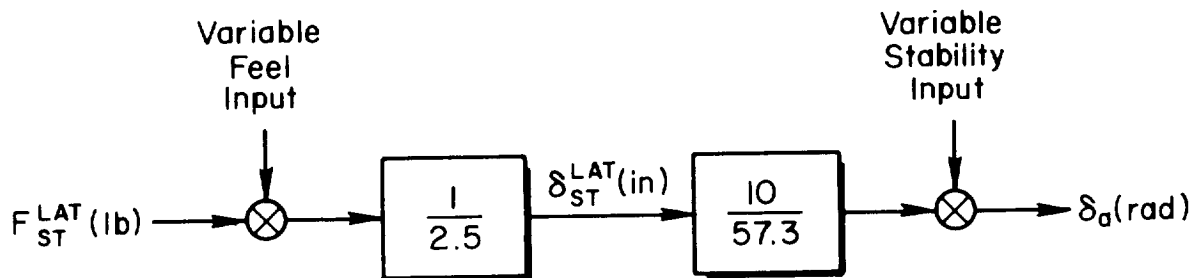
Figure II-2. NT-33 A General Arrangement

NT-33A

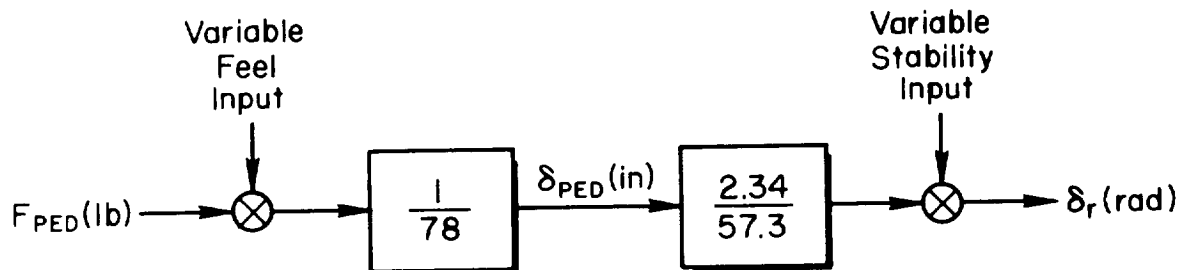
PITCH AXIS



ROLL AXIS



YAW AXIS



Feel system parameter values shown correspond to the "Front Seat Engage" mode (normal NT-33)

Figure II-3. NT-33A Control System

TABLE II-1

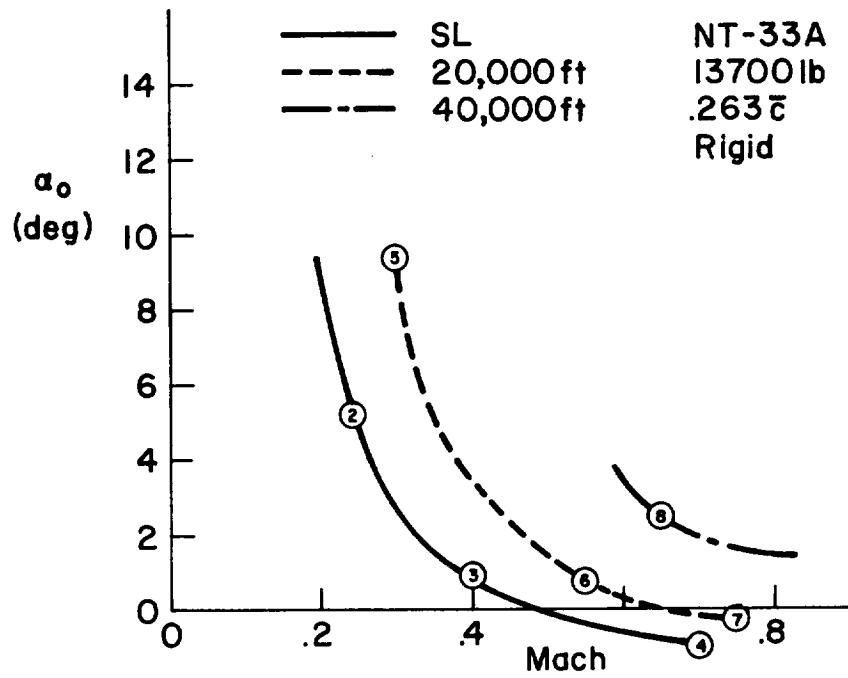
NT-33A

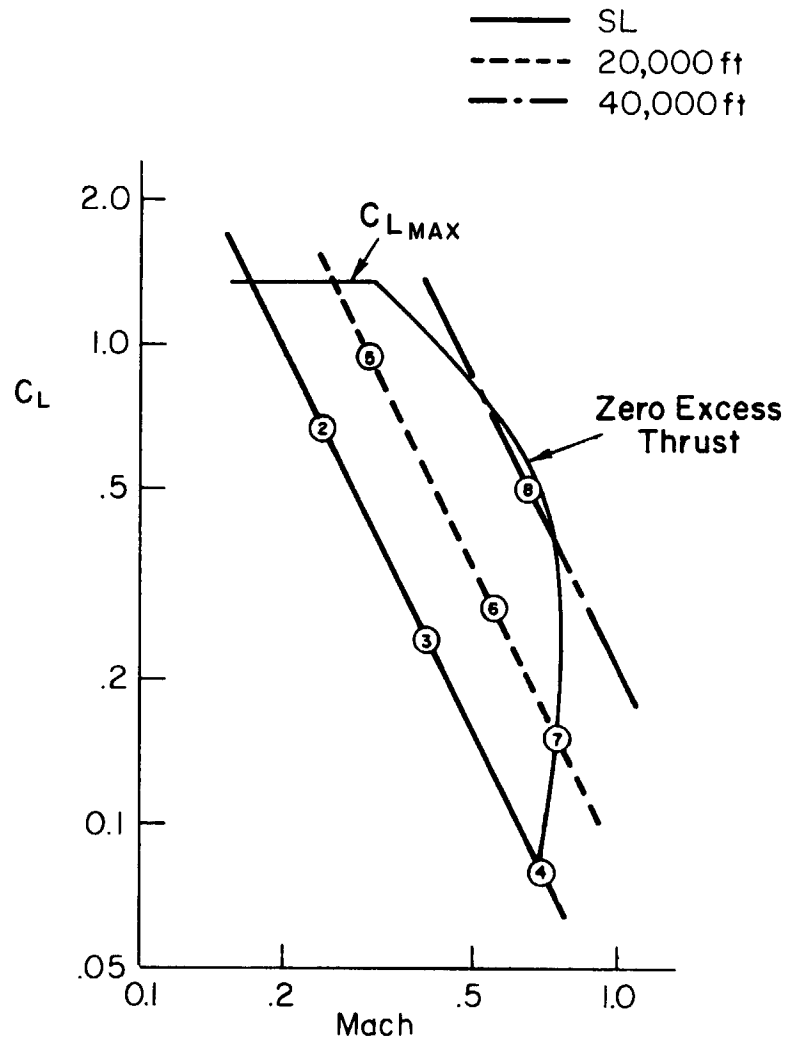
Power Approach Non-Dimensional Stability Derivatives

h = sea level

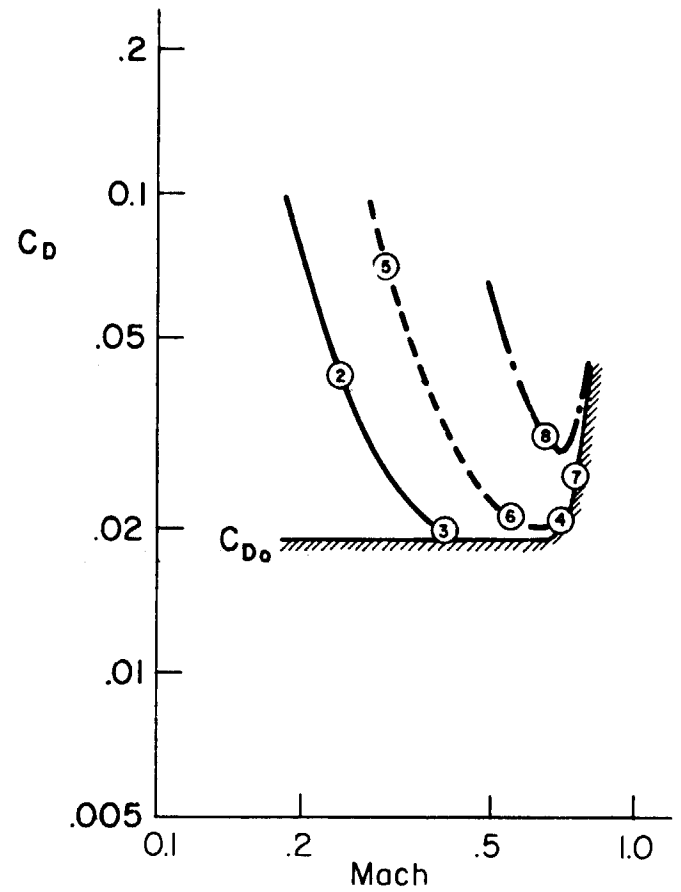
 $V_{T_0} = 228 \text{ ft/sec} = 135 \text{ kt}$ $\alpha_0 = 2.2^\circ$

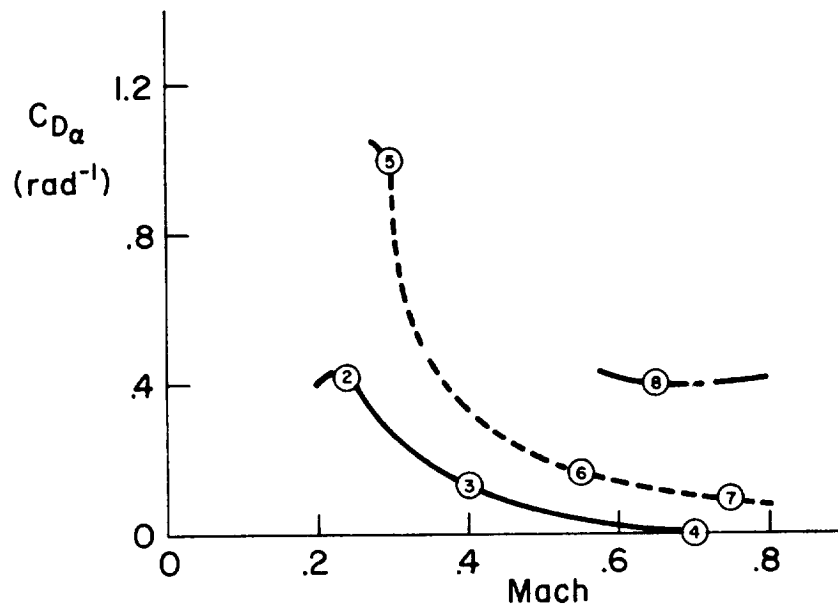
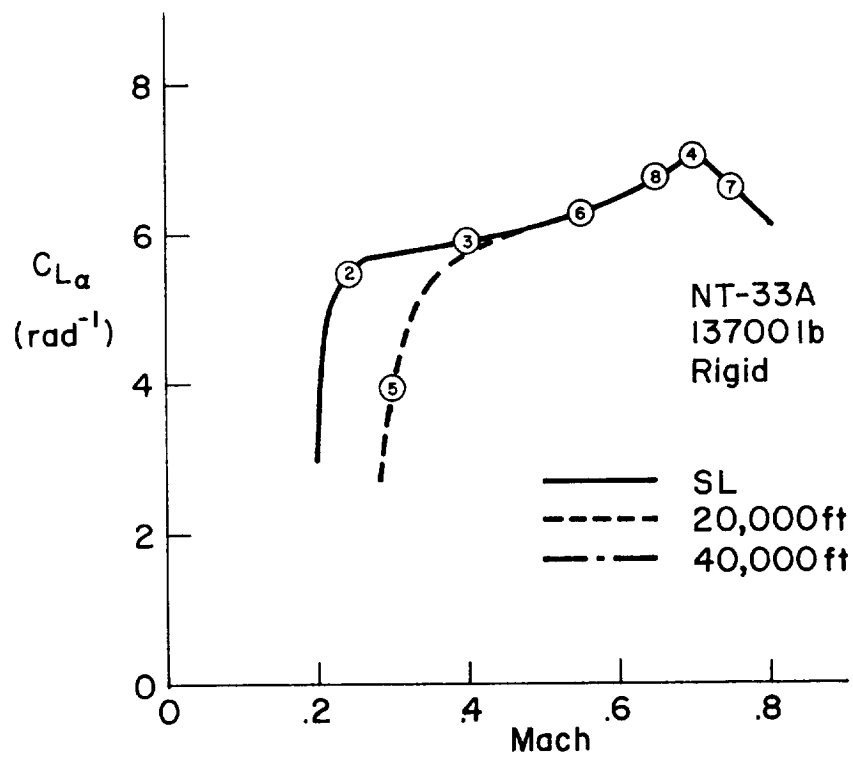
Longitudinal	Lateral-Directional (Stability Axis)
$C_L = .813$	$C_{Y\beta} = -.72/\text{rad}$
$C_D = .135$	$C_{n\beta} = .049/\text{rad}$
$C_{L\alpha} = 5.22/\text{rad}$	$C_{l\beta} = -.127/\text{rad}$
$C_{D\alpha} = .54/\text{rad}$	$C_{lp} = -.57/\text{rad}$
$C_{m\alpha} = -.401/\text{rad}$	$C_{np} = -.045/\text{rad}$
$C_{mq} = -10/\text{rad}$	$C_{lr} = .20/\text{rad}$
$C_{m\dot{\alpha}} = -5/\text{rad}$	$C_{nr} = -.16/\text{rad}$
$C_{L\delta_e} = .34/\text{rad}$	$C_{n\delta_a} = -.009/\text{rad}$
$C_{m\delta_e} = -.89/\text{rad}$	$C_{l\delta_a} = .14/\text{rad}$
	$C_{y\delta_r} = .17/\text{rad}$
	$C_{n\delta_r} = -.073/\text{rad}$
	$C_{l\delta_r} = -.002/\text{rad}$

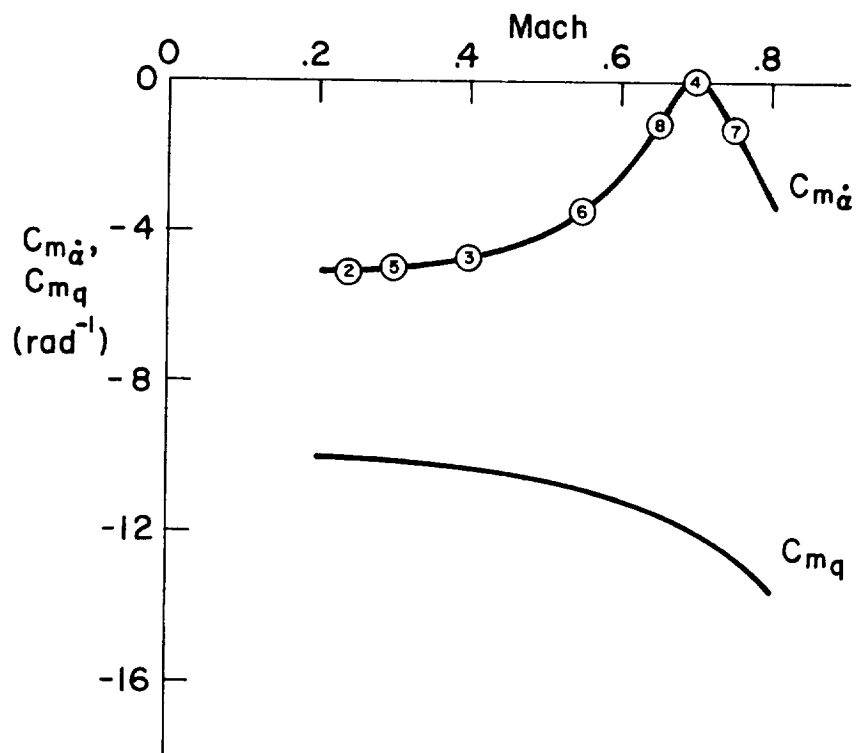
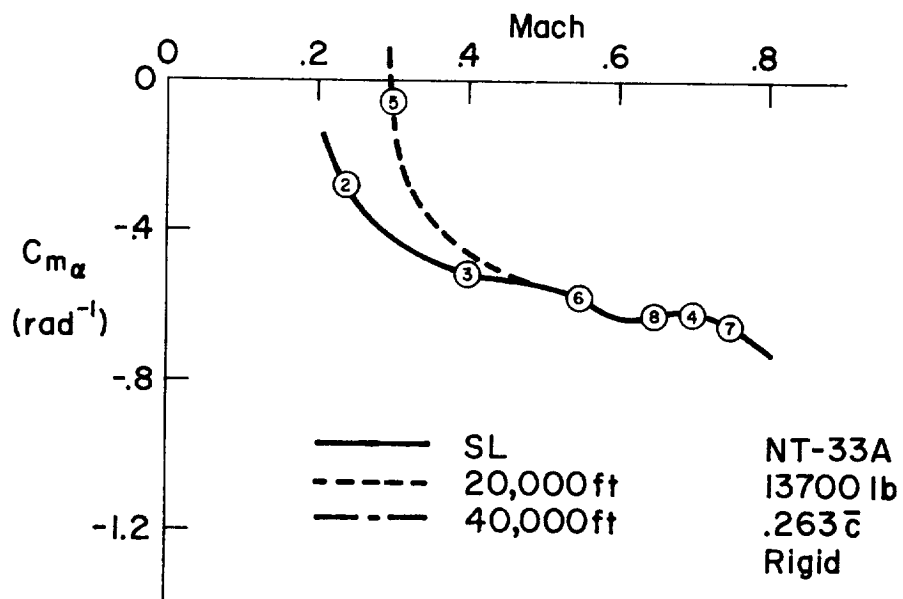


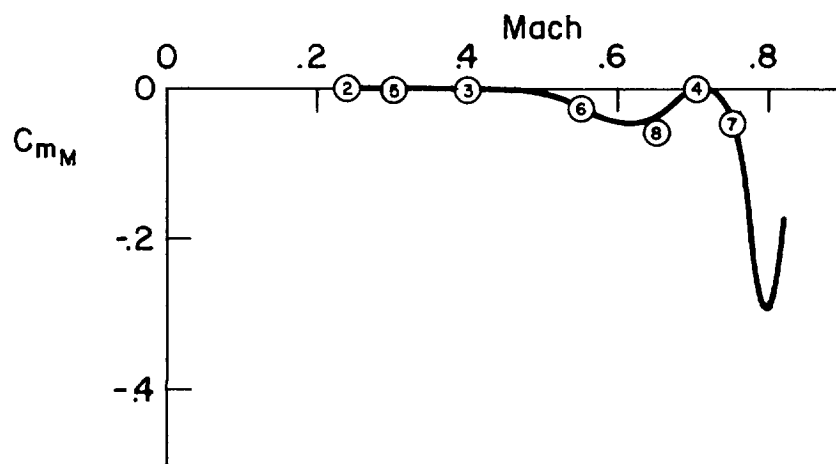
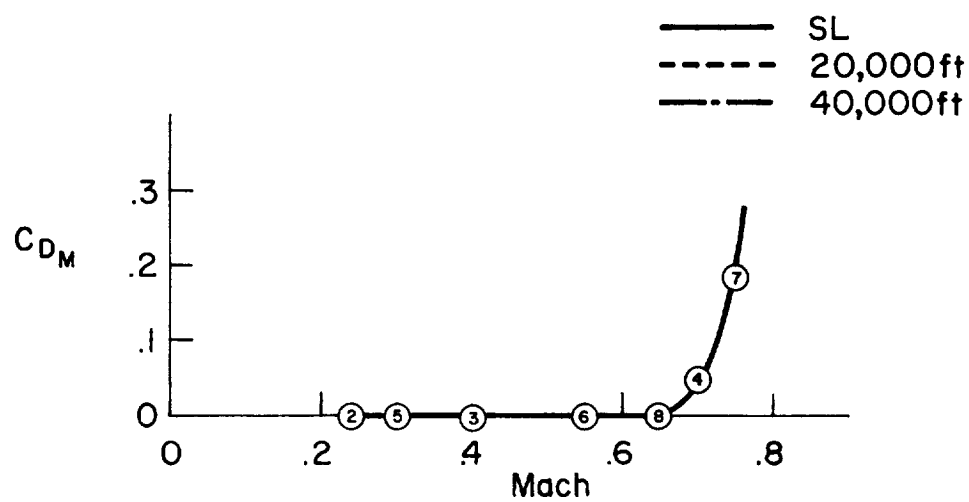
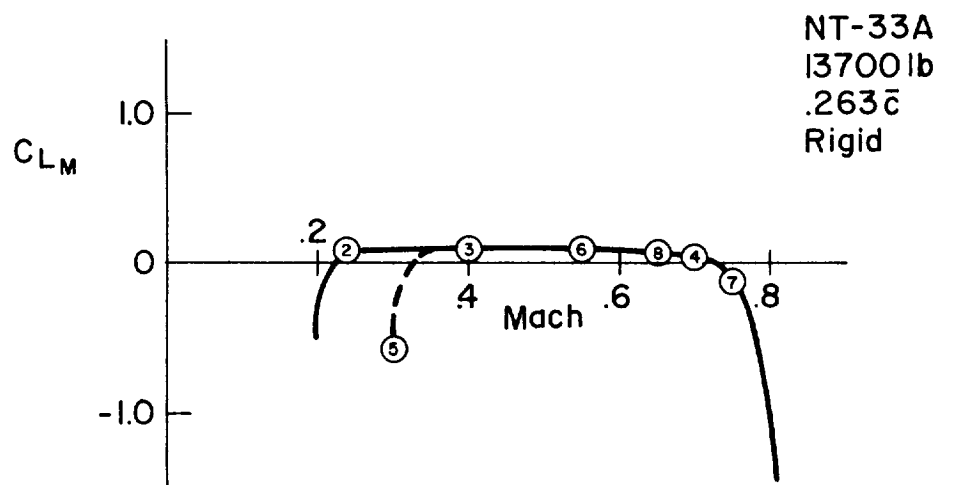


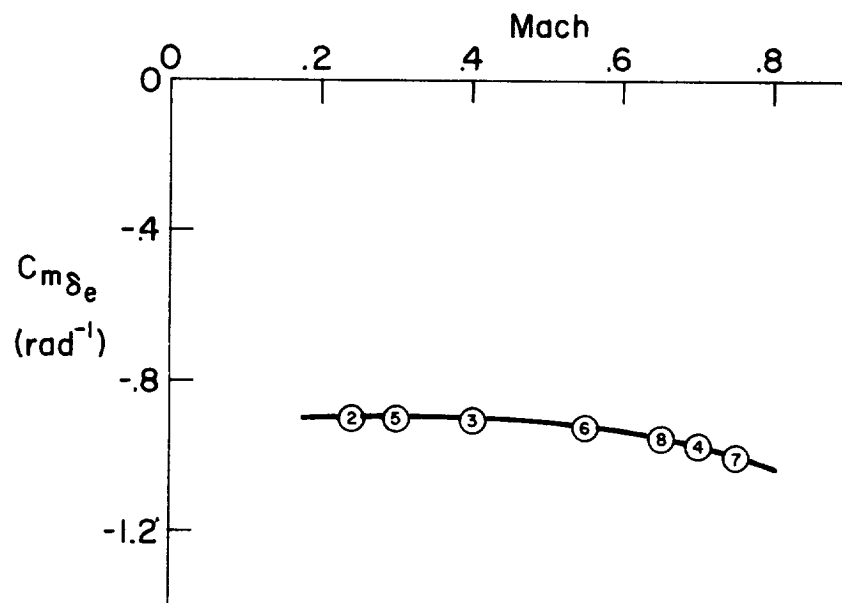
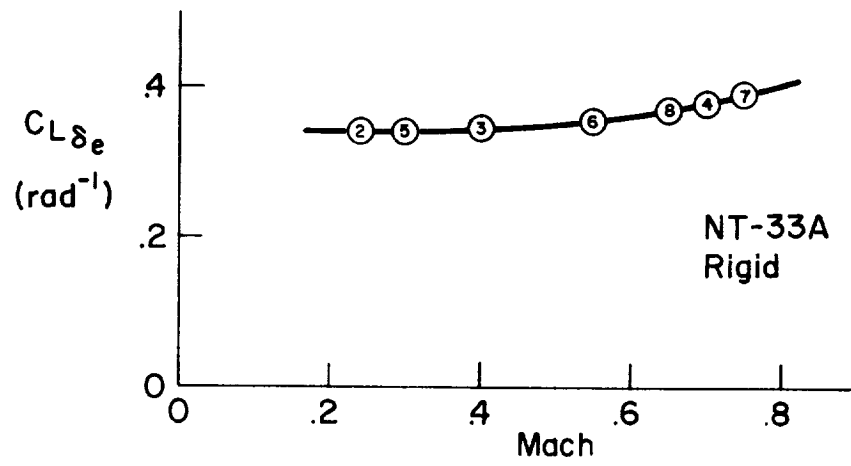
NT-33A
13700 lb

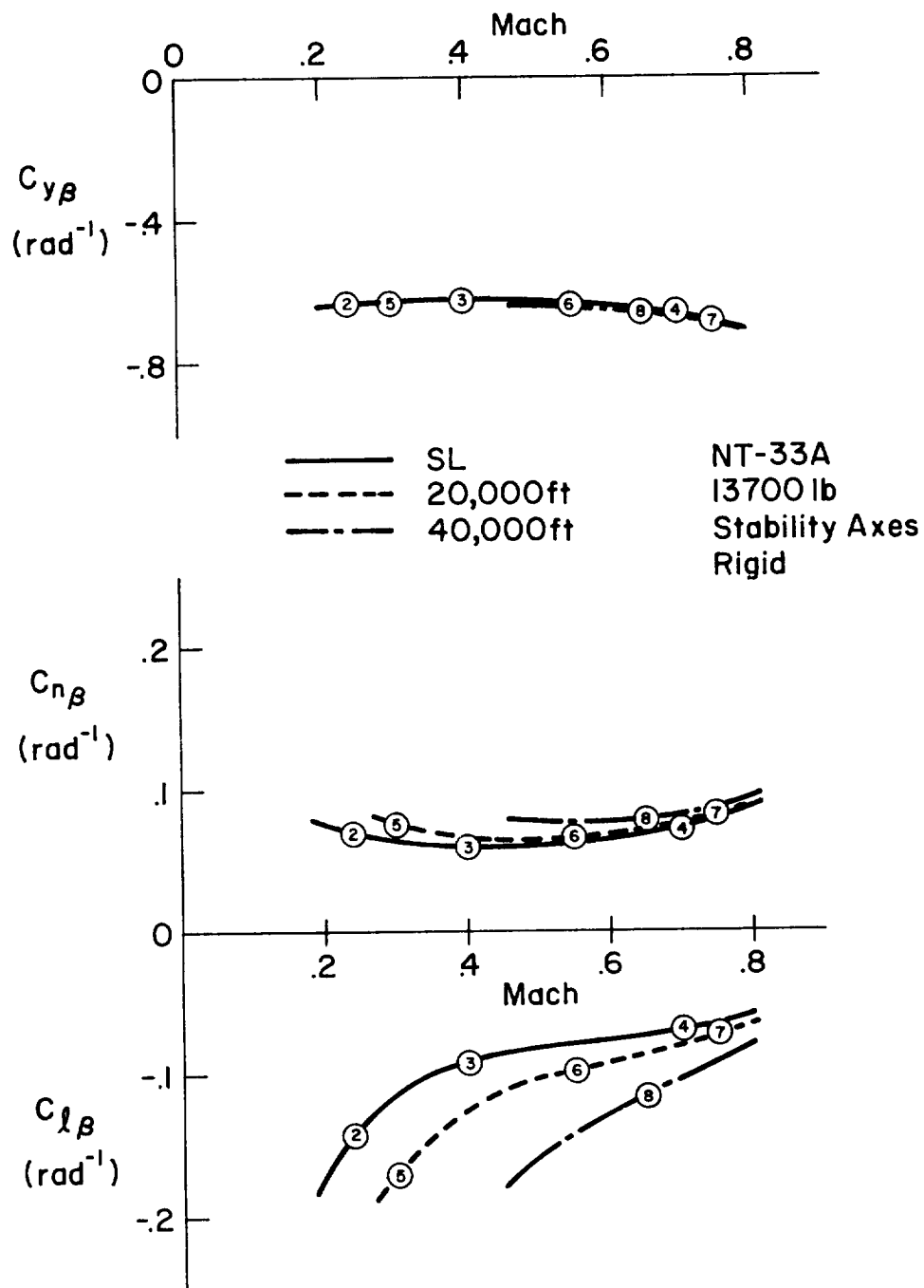


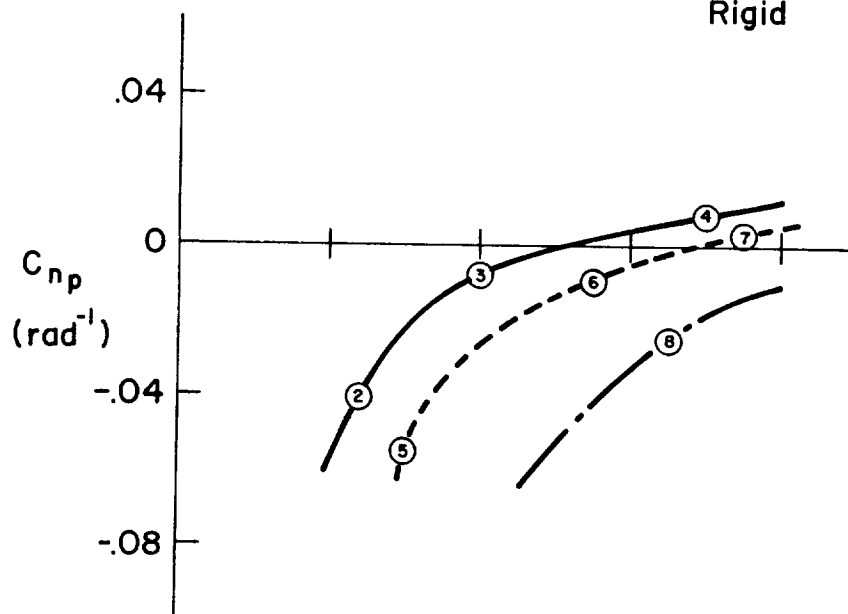
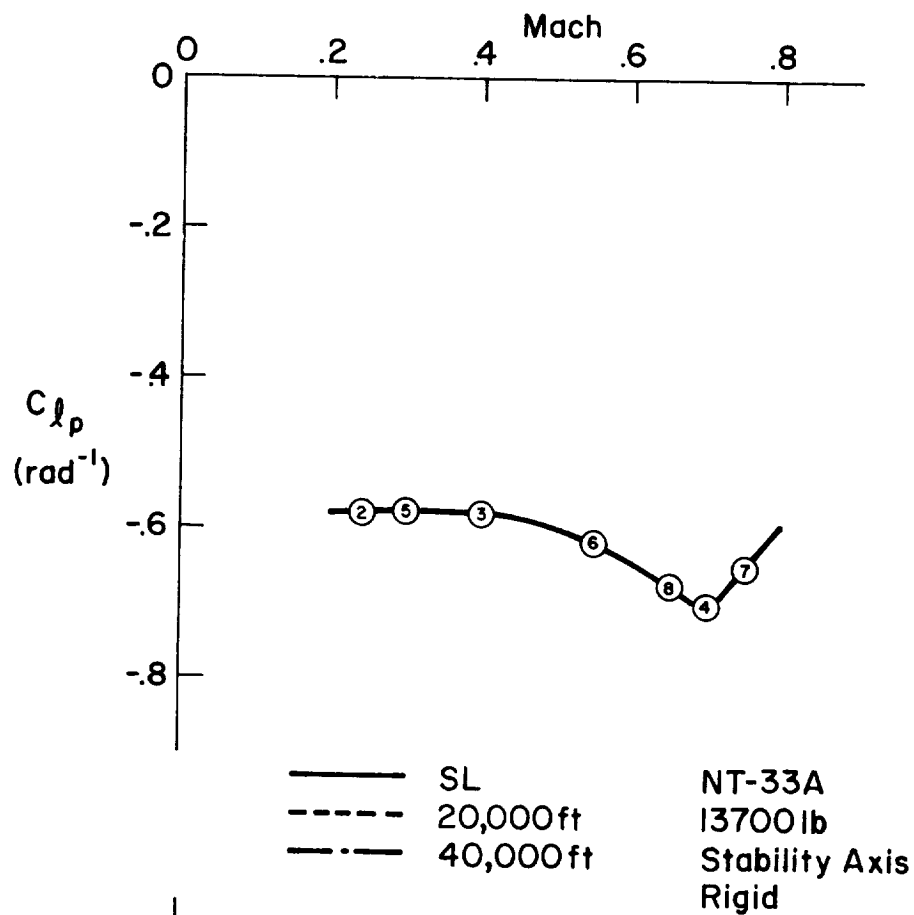


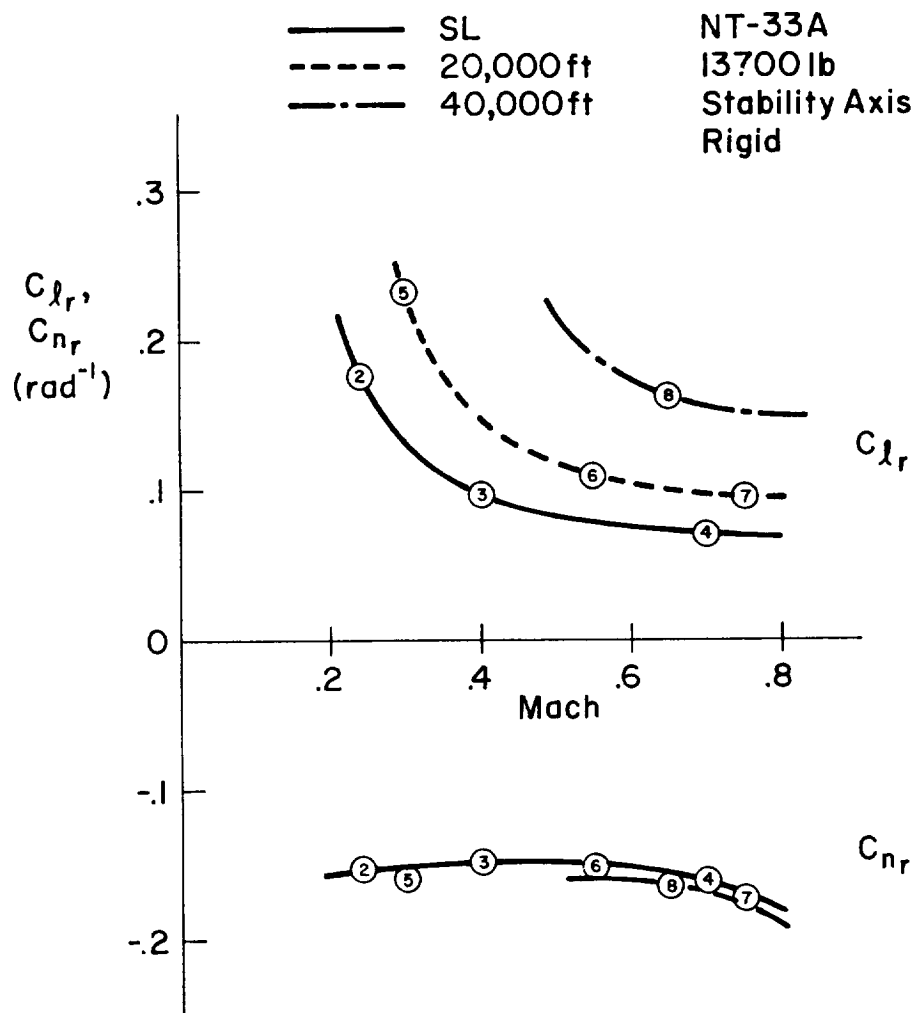


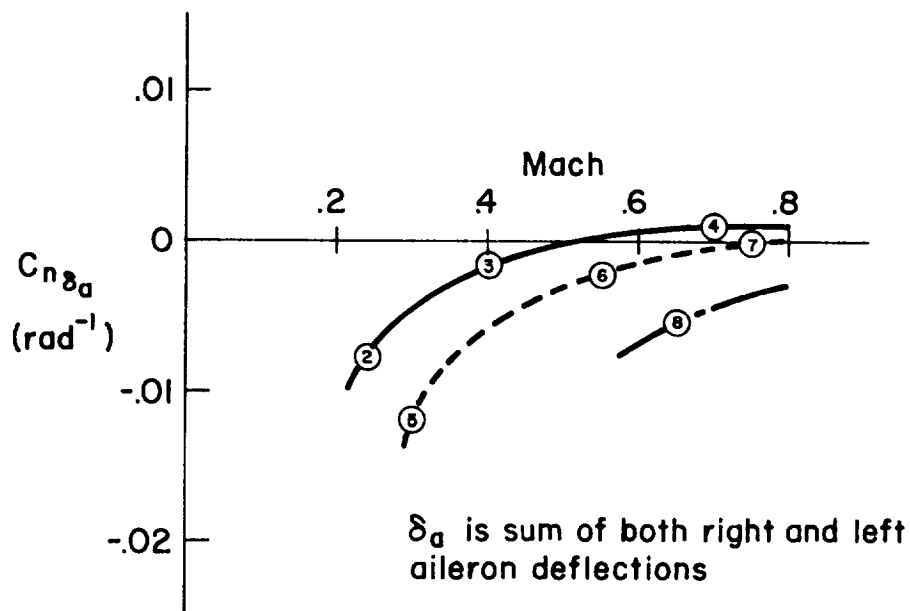
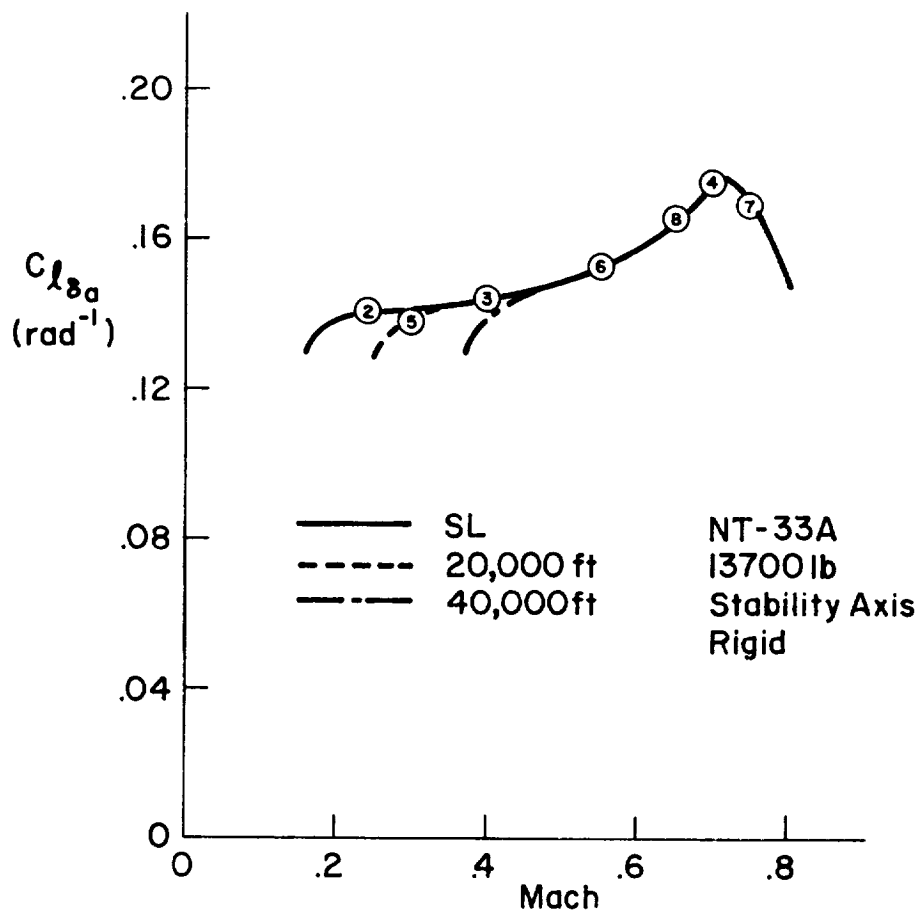


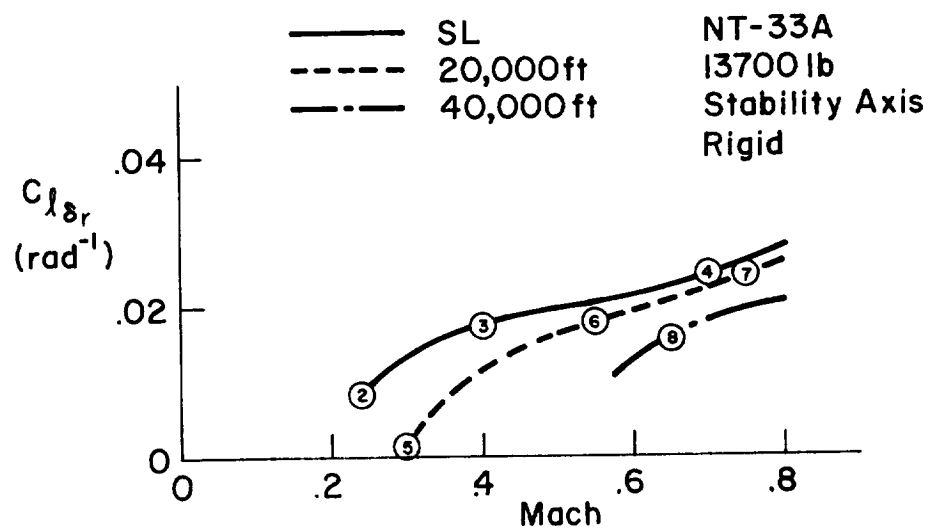
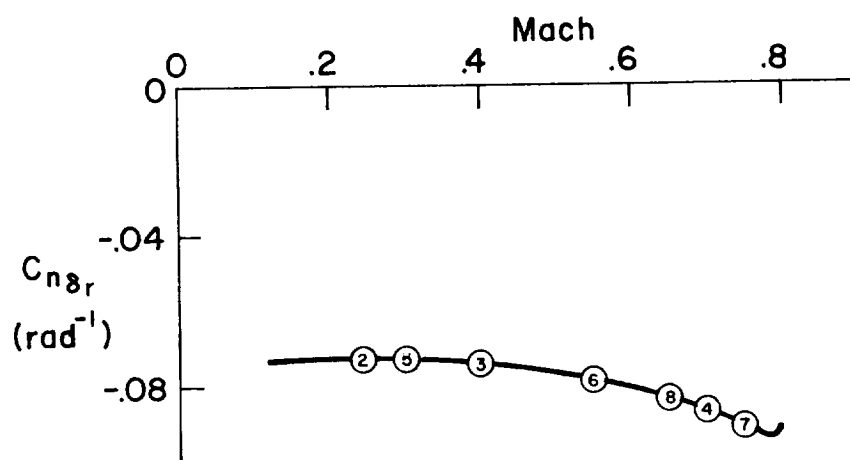
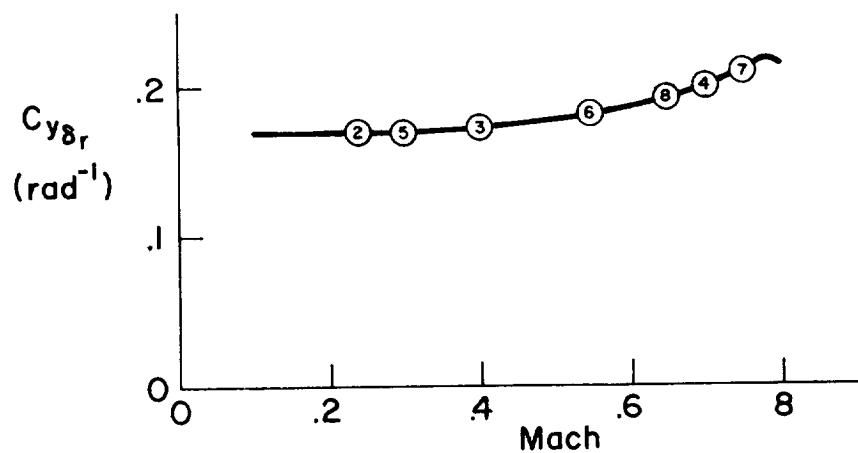












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TABLE II-3

NT-33A LONGITUDINAL DIMENSIONAL DERIVATIVES

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
XU *	-.0391	-.00484	-.0104	-.0415	.00477	-.00735	-.0511	-.00355
ZU *	-.248	-.153	-.128	-.162	-.114	-.107	-.0703	-.0766
MU *	.000318	.000603	.000283	-.000760	.000114	-.000183	-.00151	-.000183
XW	.0815	.131	.0562	-.0211	.0657	.0391	.00986	.0391
ZW	-.936	-.991	-1.73	-3.55	-.451	-1.25	-1.80	-.696
MW	-.00827	-.00669	-.0206	-.0431	-.000728	-.0157	-.0239	-.00861
ZWD	0.	0.	0.	0.	0.	0.	0.	0.
ZQ	0.	0.	0.	0.	0.	0.	0.	0.
MWD	-.00152	-.00149	-.00140	0.	-.000785	-.000541	-.000207	-.887E-4
MQ	-.694	-.806	-1.37	-2.80	-.500	-.981	-1.56	-.535
XDE	.516	1.47	.620	-2.65	1.88	.500	-.432	.996
ZDE	-13.4	-16.2	-44.4	-152.	-11.3	-40.9	-82.4	-23.8
MDE	-4.19	-5.83	-16.0	-52.7	-4.13	-14.2	-28.7	-8.28
XDTH	.00273	.00235	.00235	.00235	.00235	.00235	.00235	.00235
ZDTH	0.	0.	0.	0.	0.	0.	0.	0.
MDTH	0.	.948E-6	.948E-6	.948E-6	.948E-6	.948E-6	.948E-6	.948E-6

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TABLE II-4

NT-33A ELEVATOR TRANSFER FUNCTION FACTORS

Bare Airframe

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
DENOMINATOR								
Z(DET)1	.0948	.0199	.0546	.351	-.00782	.0522	(-.0217)	.0315
w(DET)1	.172	.141	.0933	.0561	.0977	.0678	(.0717)	.0543
Z(DET)2	.622	.680	.548	.484	.887	.398	.380	.268
w(DET)2	1.59	1.62	3.41	6.61	.674	3.19	4.63	2.40
NUMERATORS								
N(U /DE)								
A(U)	.516	1.47	.620	-2.65	1.88	.500	-.432	.996
1/T(U)1	68.0	96.5	177.	2.84	112.	222.	2.15	228.
Z(U)1	.673	.369	.484	(-3.13)	.631	.560	(-6.16)	.545
w(U)1	1.87	1.14	2.80	(249.)	.537	2.23	(280.)	.889
N(W /DE)								
A(W)	-13.4	-16.2	-44.4	-152.	-11.3	-40.9	-82.4	-23.8
1/T(W)1	71.7	97.8	162.	273.	112.	199.	272.	220.
Z(W)1	.115	.0290	.0584	.245	-.0137	.0519	.488	.0373
w(W)1	.186	.135	.0955	.0805	.109	.0774	.0522	.0623
N(TH/DE)								
A(TH)	-4.17	-5.81	-15.9	-52.7	-4.12	-14.2	-28.6	-8.28
1/T(TH)1	.0627	.0258	.0147	.0406	.0123	.0108	.0515	.00794
1/T(TH)2	.890	.955	1.68	3.47	.433	1.20	1.73	.667
N(HD /DE)								
A(HD)	13.4	16.2	44.4	152.	11.5	40.9	82.4	23.8
1/T(HD)1	.0174	-.00440	.00796	.0354	-.0326	.00499	.0501	-.000124
1/T(HD)2	-7.48	-9.06	-15.4	-29.3	-6.54	-14.8	-20.8	-11.8
1/T(HD)3	8.55	10.3	17.4	32.1	7.33	16.1	22.5	12.4
N(AZP/DE)								
A(AZP)	13.7	21.7	59.3	192.	15.5	51.5	105.	30.3
1/T(AZP)1	-.0116	.0145	-.00172	.000660	.00549	-.00134	.000224	.00414
1/T(AZP)2	.0288	-.0191	.00967	.0387	-.0385	.00633	.0499	-.00428
Z(AZP)1	.0507	.0482	.0510	.0734	.0209	.0416	.0454	.0343
w(AZP)1	7.92	8.32	14.2	27.3	5.92	13.7	19.2	10.7

+ + + + + + + + +

TABLE II-5

NT-33A THRUST TRANSFER FUNCTION FACTORS

Bare Airframe

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
DENOMINATOR								
Z(DET)1	.0948	.0199	.0546	.351	-.00782	.0522	(-.0217)	.0315
w(DET)1	.172	.141	.0933	.0561	.0977	.0678	(.0717)	.0543
Z(DET)2	.622	.680	.548	.484	.887	.398	.380	.268
w(DET)2	1.59	1.62	3.41	6.61	.674	3.19	4.63	2.40
NUMERATORS								
N(L/DTH)								
A(L)	.00273	.00235	.00235	.00235	.00235	.00235	.00235	.00235
1/T(U)1	-.00403	-.0124	-.00284	-.000570	-.0214	-.00229	-.000903	-.00366
Z(L)1	.621	.680	.548	.484	.883	.398	.381	.266
w(L)1	1.59	1.62	3.41	6.61	.676	3.19	4.63	2.40
N(W/DTH)								
A(W)	-.000639	-.180E-4	.000143	.000360	.000116	.000297	.000570	.000421
1/T(W)1	.00398	.00529	-.0335	-.00167	-.0415	-.00713	-.000521	-.00815
1/T(W)2	.421	-5.17	-.746	-6.74	-.425	-1.64	-5.23	-.854
N(TH/DTH)								
A(TH)	.159E-6	.127E-5	.954E-6	.948E-6	.125E-5	.963E-6	.945E-6	.957E-6
Z(TH)1	(5.44)	.729	.521	.266	.819	.252	(.899)	.117
w(TH)1	(7.38)	1.74	2.73	3.26	.506	1.89	(-2.77)	1.15
N(HD/DTH)								
A(HD)	.000105	.000213	.369E-4	-.369E-4	.000384	.328E-4	-.123E-4	.000102
1/T(HC)1	7.38	1.96	6.51	-10.9	.242	3.23	-1.24	.849
Z(HD)1	.379	.612	.727	.795	.817	.816	(7.45)	.365
w(HD)1	1.37	1.57	3.70	4.42	1.01	4.29	(-16.1)	3.00
N(AZP/DTH)								
A(AZP)	-.104E-5	-.827E-5	-.649E-5	-.619E-5	-.818E-5	-.629E-5	-.617E-5	-.625E-5
1/T(AZP)1	-.00542	-.0109	-.00112	.000646	-.0173	-.000790	.000217	-.00224
1/T(AZP)2	663.	43.8	44.6	2.91	.351	36.8	-1.04	26.0
Z(AZF)1	.389	.632	.685	(8.44)	(1.06)	.730	.768	.701
w(AZF)1	1.46	1.69	3.38	(51.8)	(32.2)	2.90	16.9	2.20

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TABLE II-6

NT-33A LONGITUDINAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
STICK FIXED								
D(G)/D(U) (DEG/KT)	-.0526	.0131	-.0240	-.118	.0977	-.0150	-.151	.000330
NZA (G/RAD)	6.37	8.05	23.0	83.3	4.26	21.2	41.6	13.1
DE/G (DEG/G)	5.39	3.14	1.79	.565	1.46	1.92	1.02	3.05
CAP (RAD/SEC/SEC/G)	.392	.219	.497	.519	.105	.475	.512	.441
PHUGOID(2) (SEC) (TUCK(2))	--	--	--	--	908.	--	(32.0)	--
1/C(1/10)	2.17	2.53	1.79	1.51	5.25	1.19	1.12	.758
+	+	+	+	+	+	+	+	+

TABLE II-7
NT-33A LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES
(BODY AXIS SYSTEM)

NT-33A LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8 *
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
YV	-.125	-.111	-.181	-.338	-.0696	-.128	-.185	-.0674
YB	-28.4	-30.1	-81.0	-264.	-21.6	-72.7	-144.	-42.4
LB'	-5.49	-4.72	-8.02	-18.0	-4.06	-7.42	-9.89	-5.08
NB'	.667	.940	2.71	10.6	.540	2.60	6.24	1.68
LP'	-2.03	-1.32	-2.15	-4.51	-.820	-1.56	-2.23	-.877
NP'	-.116	-.112	-.0512	.0118	-.103	-.0383	-.0141	-.0428
LR'	.641	.305	.320	.495	.214	.256	.328	.179
NR'	-.207	-.173	-.291	-.561	-.104	-.204	-.318	-.110
Y*CA	0.	0.	0.	0.	0.	0.	0.	0.
L'CA	6.01	4.53	12.6	47.0	3.14	11.7	24.0	7.13
N'CA	.0286	.134	.165	.260	.164	.121	.195	.118
Y*CR	.0295	.0301	.0503	.102	.0185	.0363	.0571	.0195
L'CR	-.0125	.443	1.57	5.89	.287	1.39	3.20	.808
N'DR	-1.24	-1.25	-3.50	-12.6	-.883	-3.21	-6.99	-1.92

TABLE II-8
NT-33A AILERON TRANSFER FUNCTION FACTORS

Bare Airframe
(BODY AXIS SYSTEM)

28

F/C #	1	2	3	4	5	6	7	8
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
DENOMINATOR								
1/T(DET)1	.0318	.0185	.0143	.00465	.0129	.00932	.00333	.00483
1/T(DET)2	2.20	1.47	2.24	4.57	.966	1.66	2.29	.979
Z(DET)1	.0609	.0435	.102	.127	.00638	.0647	.0868	.0251
w(DET)1	1.13	1.26	1.75	3.28	1.16	1.70	2.52	1.41
NUMERATORS								
N(B /DA)								
A(B)	.202	.278	.0333	-.999	.351	.0419	-.320	.193
1/T(B)1	.116	.103	.214	-.946	.0616	.144	.330	.0692
1/T(B)2	7.48	3.30	37.8	1.15	1.56	22.6	-3.01	3.05
N(P /DA)								
A(P)	6.01	4.53	12.6	47.0	3.14	11.7	24.0	7.13
1/T(P)1	-.00522	-.0106	-.00111	.000636	-.0169	-.000781	.000215	-.00222
Z(P)1	.200	.145	.141	.136	.116	.102	.0999	.0687
w(P)1	.849	1.05	1.65	3.30	.868	1.64	2.53	1.33
N(R /DA)								
A(R)	.0286	.134	.165	.260	.164	.121	.195	.118
1/T(R)1	.885	.786	1.75	10.4	.485	1.60	3.86	.828
Z(R)1	(-1.06)	-.673	-.559	-.621	-.450	-.597	-.553	-.482
w(R)1	(-22.0)	2.35	2.98	2.77	1.74	3.02	2.89	2.56
N(PHI/DA)								
A(PHI)	6.01	4.55	12.6	47.0	3.17	11.7	24.0	7.14
Z(PHI)1	.195	.136	.141	.136	.0995	.102	.0999	.0673
w(PHI)1	.848	1.05	1.65	3.30	.874	1.64	2.53	1.33
N(AYP/DA)								
A(AYP)	17.3	13.7	37.0	135.	9.99	34.0	69.4	21.0
1/T(AYP)1	.122	.110	.204	-.356	.0666	.141	.236	.0730
1/T(AYP)2	-1.24	-1.07	-.806	.481	-.587	-.660	-.395	-.604
Z(AYP)1	.437	.407	.269	.121	.460	.226	.126	.236
w(AYP)1	1.38	1.33	1.89	3.53	1.05	1.77	2.66	1.37

+ + + + + + + + +

TABLE II-9

Bare Airframe

F/C #	1	2	3	4	5	6	7	8
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K
M	.204	.242	.400	.700	.300	.550	.750	.650
DENOMINATOR								
1/T (DET) 1	.0318	.0185	.0143	.00469	.0129	.00932	.00333	.00483
1/T (DET) 2	2.20	1.47	2.24	4.57	.966	1.66	2.29	.979
Z (DET) 1	.0609	.0435	.103	.127	.00638	.0647	.0868	.0251
W (DET) 1	1.13	1.26	1.75	3.28	1.16	1.70	2.52	1.41
NUMERATORS								
N(B /DR)								
A(B)	.0295	.0301	.0503	.102	.0185	.0363	.0571	.0195
1/T (B) 1	-.0494	-.0312	-.00728	-.00146	-.0377	-.00664	-.00313	-.00955
1/T (B) 2	2.05	1.36	2.19	4.57	.836	1.60	2.26	.902
1/T (B) 3	42.3	42.9	70.2	122.	49.8	89.2	123.	100.
N(P /DR)								
A(P)	-.0125	.443	1.57	5.89	.287	1.39	3.20	.808
1/T (P) 1	-.00533	-.0107	-.00112	.000641	-.0170	-.000785	.000215	-.00223
1/T (P) 2	8.06	3.12	3.67	5.07	3.10	3.60	3.74	3.05
1/T (P) 3	69.0	-4.00	-4.17	-5.54	-3.83	-4.05	-4.13	-3.42
N(R /DR)								
A(R)	-1.24	-1.25	-3.50	-12.6	-.883	-3.21	-6.99	-1.92
1/T (R) 1	2.12	1.35	2.23	4.58	.730	1.66	2.31	.947
Z (R) 1	.0199	.0724	.0912	.259	.123	.0170	.0822	-.00220
W (R) 1	.605	.620	.469	.343	.737	.463	.355	.486
N(PHI/DR)								
A(PHI)	-.0602	.329	1.51	6.09	.140	1.35	3.23	.724
1/T (PHI) 1	(-.822)	3.35	3.70	5.06	3.90	3.63	3.74	3.16
1/T (PHI) 2	(10.8)	-5.06	-4.30	-5.36	-6.38	-4.15	-4.10	-3.68
N(AYP/DR)								
A(AYP)	-1.40	1.22	4.08	14.8	.799	3.68	7.80	2.03
1/T (AYP) 1	-.0583	-.0519	-.0140	-.00362	-.0602	-.0120	-.00564	-.0154
1/T (AYP) 2	1.36	.680	1.78	4.37	.471	1.25	2.00	.643
1/T (AYP) 3	(-.201)	5.29	7.29	11.4	5.13	7.24	9.38	5.98
1/T (AYP) 4	(5.68)	-6.80	-9.12	-15.2	-6.23	-8.58	-11.2	-6.90

TABLE II-10

NT-33A LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(BODY AXIS SYSTEM)

+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	
M	.204	.242	.400	.700	.300	.550	.750	.650	
DR PERIOD (SEC)	5.57	4.97	3.61	1.93	5.43	3.71	2.50	4.45	
1/C(1/2)	.553	.395	.941	1.16	.0578	.588	.790	.228	
SPIRAL (2) (SEC)	--	--	--	--	--	--	--	--	
P(1)	2.34	2.41	5.18	10.4	2.11	6.29	10.5	5.69	
P(2)	.418	1.22	4.79	10.3	.659	6.04	10.3	5.56	
P(3)	2.00	2.41	5.16	10.4	2.46	6.61	10.4	6.71	
P(2)/P(1)	.179	.505	.924	.983	.313	.961	.981	.977	
P(OSC)/P(AV)	.677	.329	.0384	.00752	.552	.0328	.00677	.0542	
W(PHI)/W(D)	.751	.829	.966	1.01	.755	.970	1.00	.942	
DEL-B-MAX	1.01	.701	.326	.104	.781	.322	.141	.459	
PHI TO BETA, PHASE	-297.	-313.	-313.	48.7	-322.	-320.	38.2	-328.	
PHI TO BETA	2.14	2.07	1.73	1.06	2.44	1.95	1.22	2.16	
PHI TO VE	.539	.438	.223	.0778	.616	.269	.124	.395	
+	+	+	+	+	+	+	+	+	+

NT-33A DATA SOURCES

Hall, G. Warren, and Ronald W. Huber, System Description and Performance Data for the USAF/CAL Variable Stability T-33 Airplane, Air Force Flight Dynamics Laboratory Rept. No. AFFDL TR-70-71, Aug. 1970

Tests of a 1/5 Scale Wind Tunnel Model of the TP-80C Trainer, Lockheed Aerodynamics Laboratory Rept. No. LAL 127, Jan. 23, 1948

Cleary, Joseph W., and Lyle J. Gray, High Speed Wind-Tunnel Tests of a Model Pursuit Airplane and Correlation with Flight-Test Results, NACA-RM-7116, Jan. 21, 1948

Statler, Irving C., et al, The Development and Evaluation of the CAL/Air Force Dynamic Wind Tunnel Testing System; Part 1 — Description and Dynamic Tests of an F-80 Model, AFFDL-TR-66-153, Feb. 1967

Flight Manual, USAF Series T-33A Aircraft, T. O. 1T-33A-1.

SECTION III

F-104A

F-104A BACKGROUND

The F-104A is a single place, lightweight, supersonic air superiority fighter powered by a single turbojet engine with afterburner. The wing has a full span leading edge flap. Trailing edge flaps have a blowing-type boundary layer control system. Control is provided by conventional ailerons and rudder and an all-movable stabilizer. Pitch, roll, and yaw dampers are incorporated, however their effect is not shown here. Pitch and roll controls are fully irreversible while the yaw control is a cable-actuated rudder without boost. A bobweight is used in the longitudinal feel system. Its position is assumed to be at the pilot's location.

The primary source of data was LR 10794. Drag information was obtained from LR-12873.

The nominal configuration used here is the combat loading for the F-104A based on actual weight and balance data. The PA configuration is a typical loading at flight manual approach speeds.

F-104A

Nominal Configuration

Clean, 750 Rounds Ammunition

50% Internal Fuel

W = 16300 lb

c.g. at .070 $\bar{c}$

$I_x = 3549 \text{ slug-ft}^2$
 $I_y = 58611 \text{ slug-ft}^2$
 $I_z = 59669 \text{ slug-ft}^2$
 $\epsilon = 2.76^\circ$

Principal Axis

34

Power Approach Configuration

Clean

20% Internal Fuel

Full Flaps (45°), BLC

Gear Down

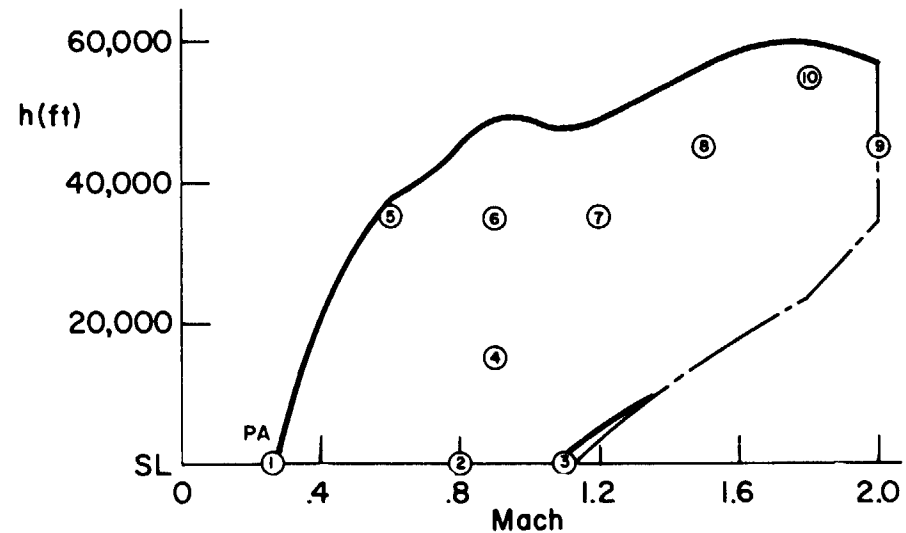
$1.4 V_S$

W = 14126 lb

c.g. at .164 $\bar{c}$

$I_x = 3450 \text{ slug-ft}^2$
 $I_y = 55800 \text{ slug-ft}^2$
 $I_z = 56800 \text{ slug-ft}^2$
 $\epsilon = 2.86^\circ$

Principal Axis



———— Level Flight Envelope (Nominal Configuration)

----- Speed Restrictions

⑩ Transfer Function Case η

Figure III-1. F-104A Flight Conditions

F-104A

$S = 196.1 \text{ ft}^2$

$b = 21.94 \text{ ft}$

$\bar{c} = 9.55 \text{ ft}$

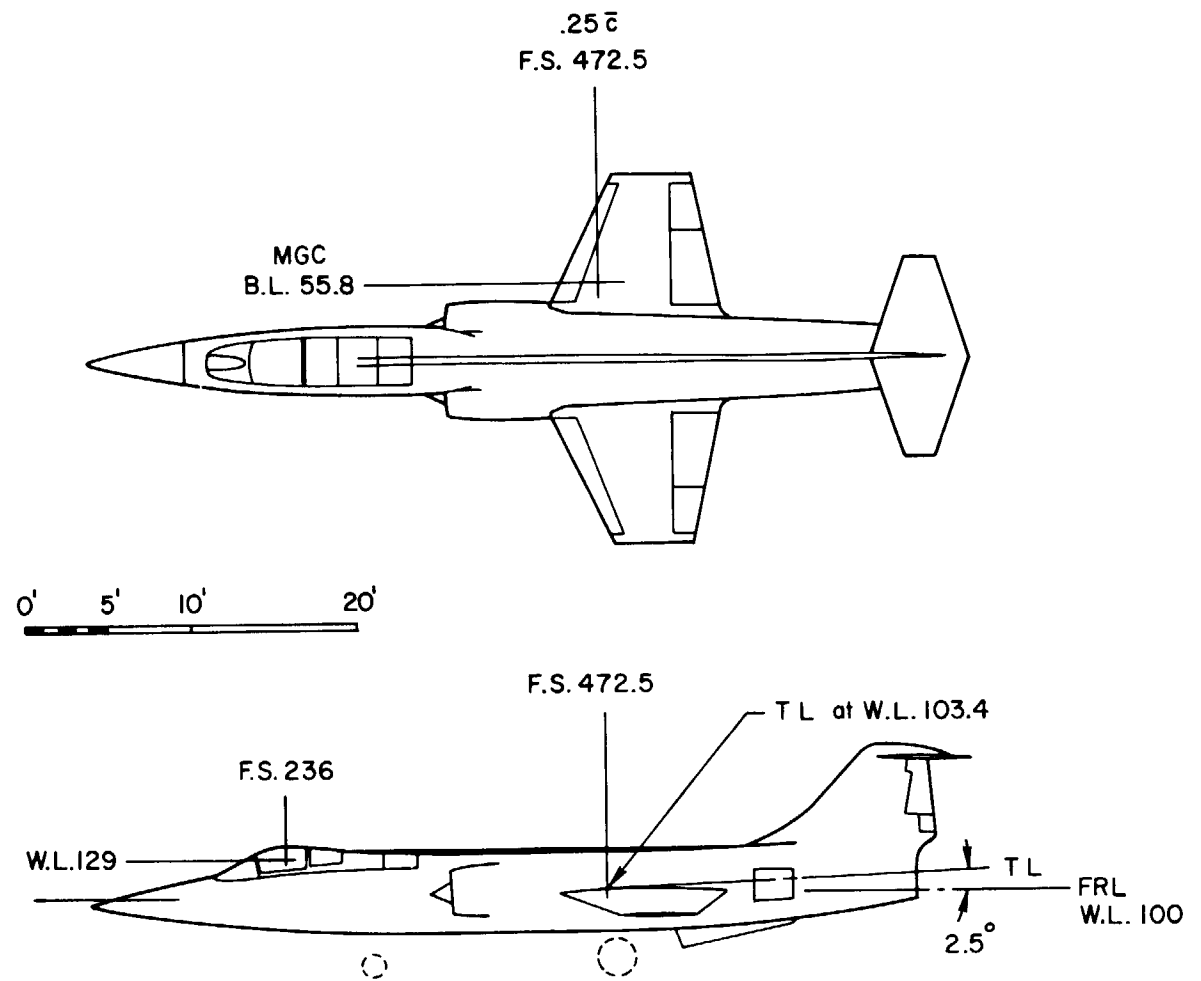
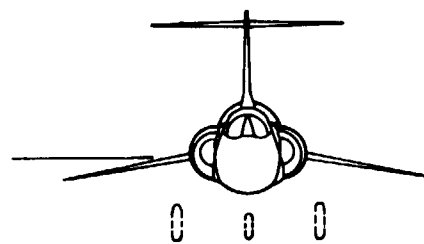
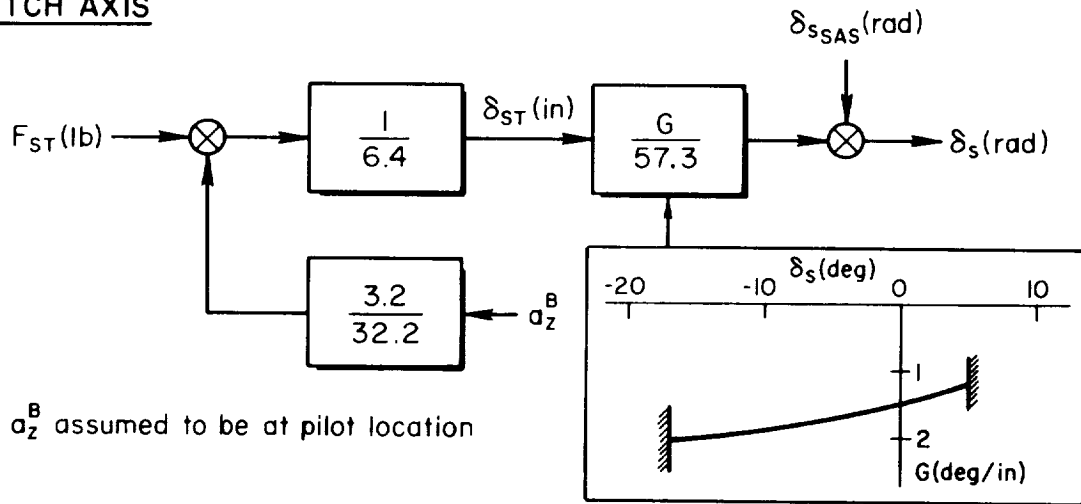


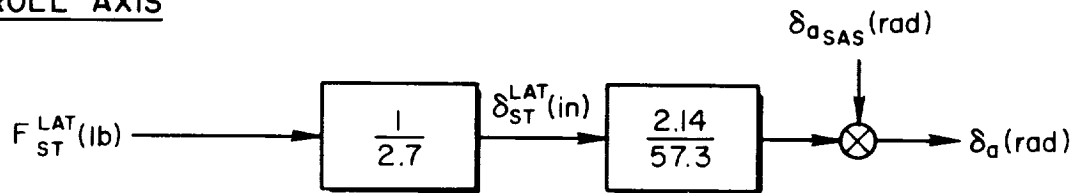
Figure III-2. F-104A General Arrangement

F-104A

PITCH AXIS



ROLL AXIS



YAW AXIS

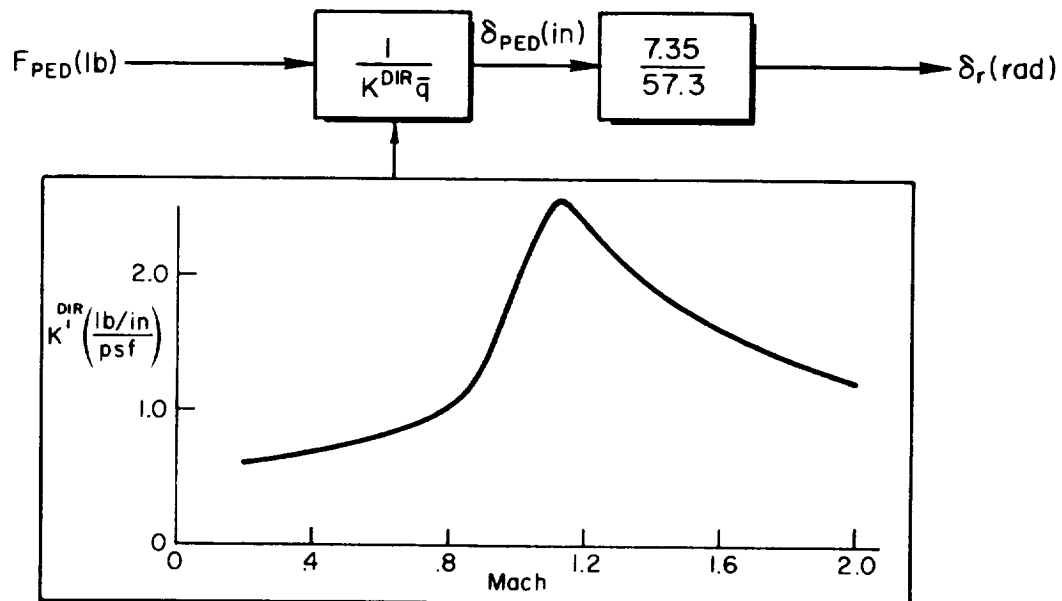


Figure III-3. F-104A Control System

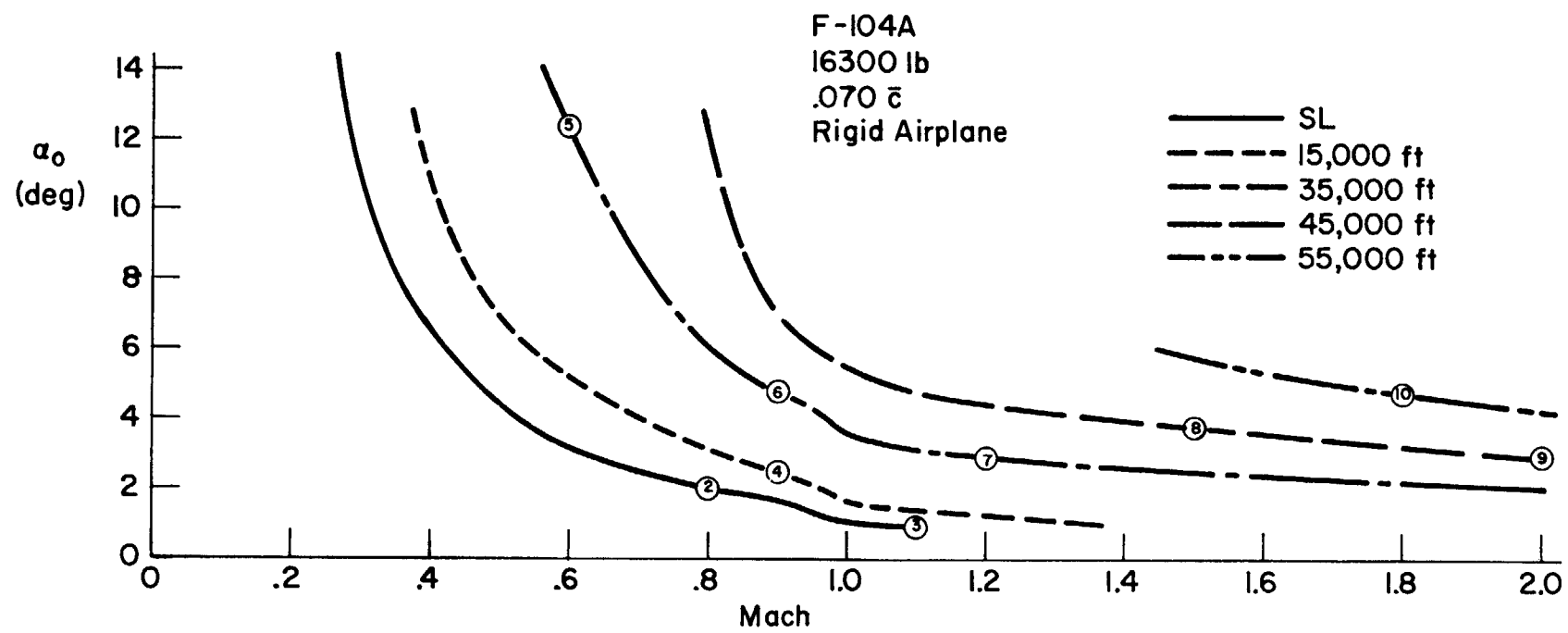
TABLE III-1

F-104A

Power Approach Non-Dimensional Stability Derivatives

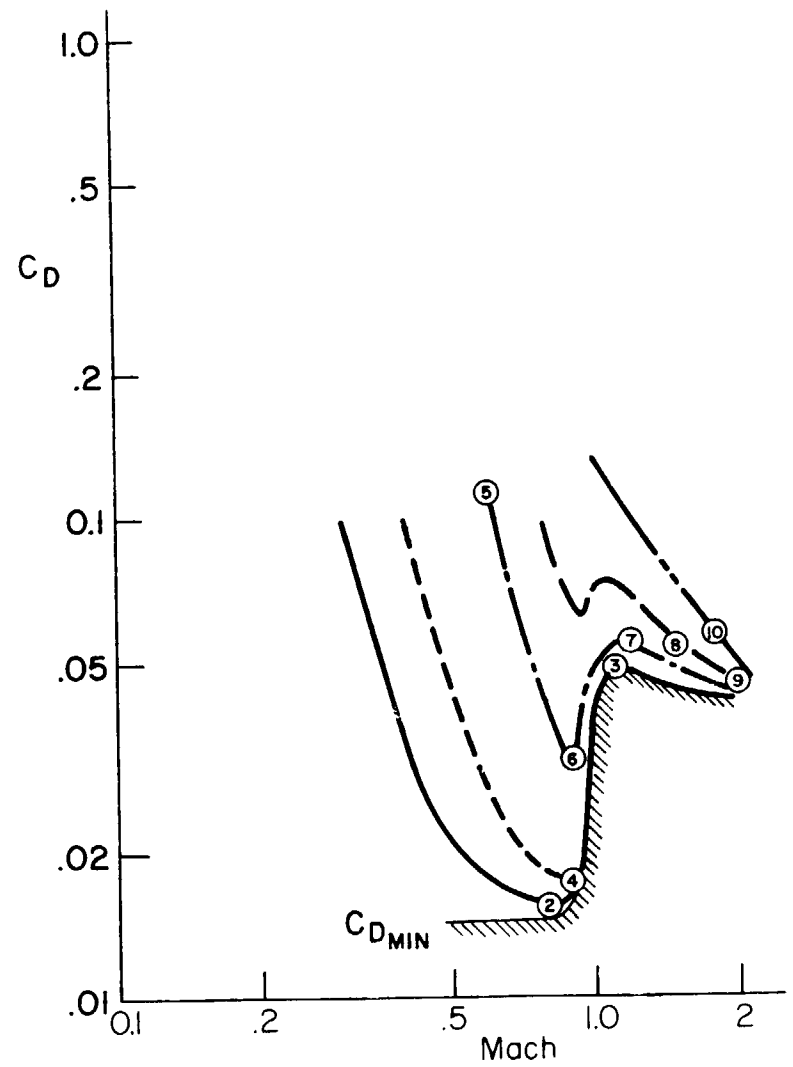
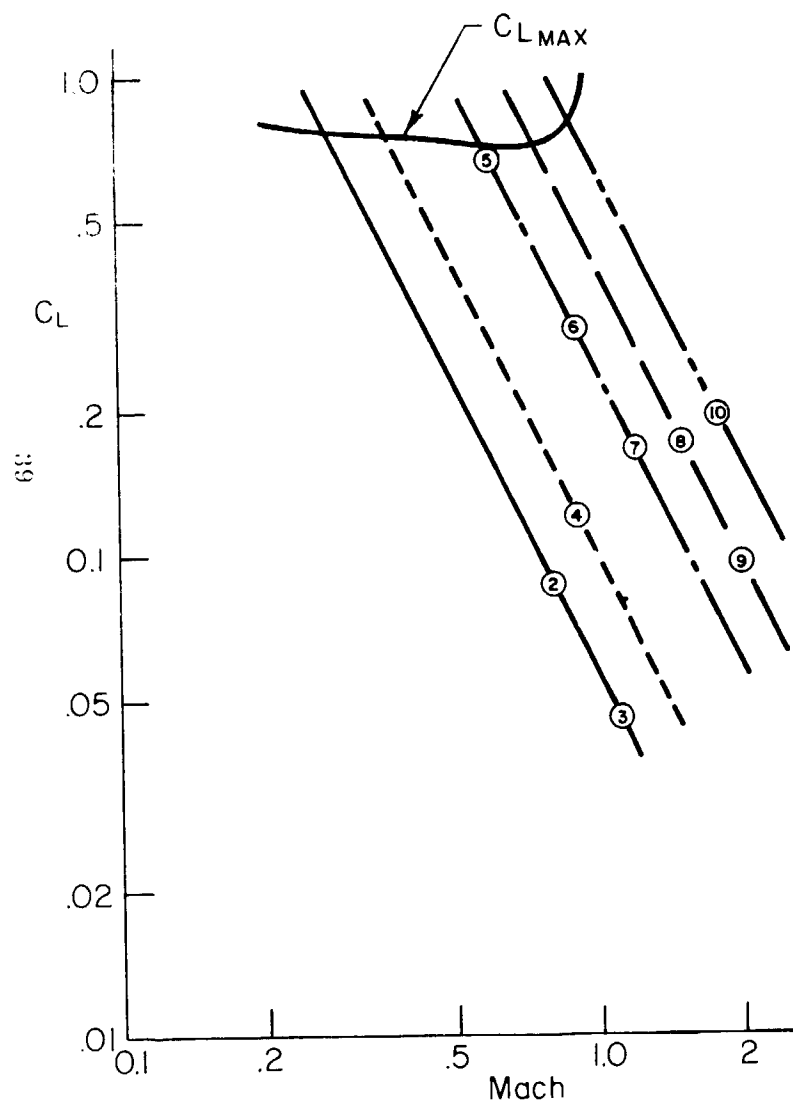
$$\begin{aligned}
 h &= \text{sea level} \\
 V_{T_0} &= 287 \text{ ft/sec} = 170 \text{ kt} \\
 \alpha_0 &= 2.3^\circ \\
 \delta_s &= -7.1^\circ
 \end{aligned}$$

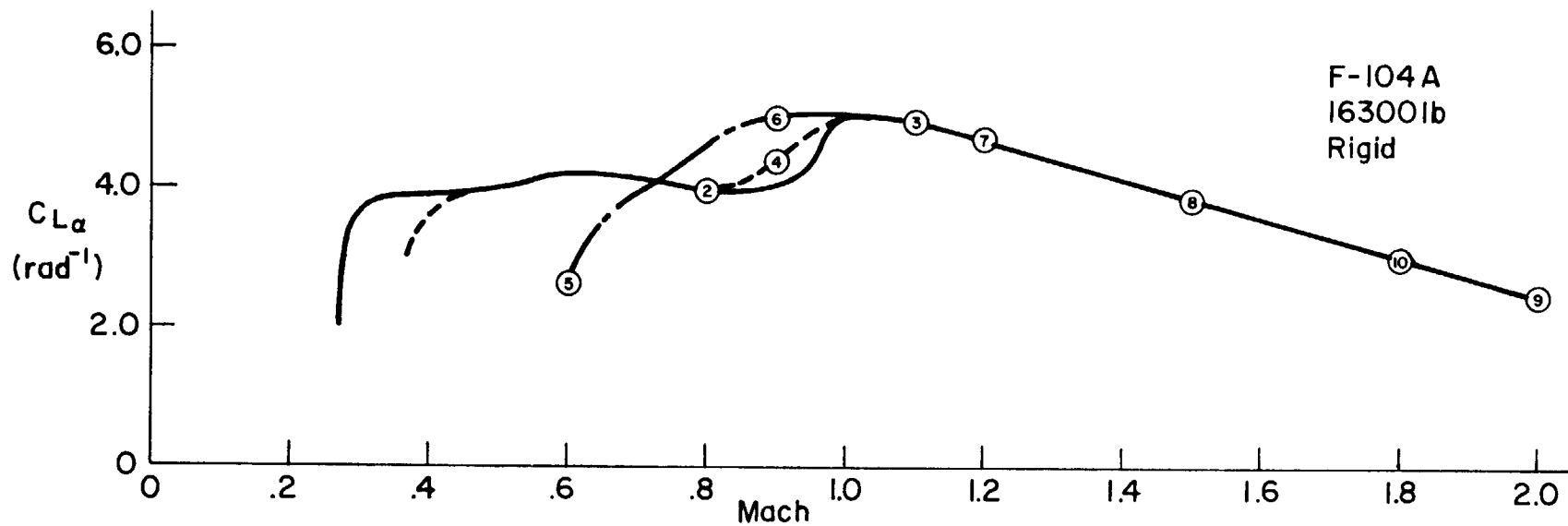
Longitudinal	Lateral-Directional (Stability Axis)
$C_L = .735$	$C_{Y\beta} = -1.17/\text{rad}$
$C_D = .263$	$C_{n\beta} = .50/\text{rad}$
$C_{L\alpha} = 3.44/\text{rad}$	$C_{\ell\beta} = -.175/\text{rad}$
$C_{D\alpha} = .45/\text{rad}$	$C_{\ell p} = -.285/\text{rad}$
$C_{m\alpha} = -.64/\text{rad}$	$C_{n p} = -.14/\text{rad}$
$C_{m\dot{\alpha}} = -1.6/\text{rad}$	$C_{\ell r} = .265/\text{rad}$
$C_{mq} = -5.8/\text{rad}$	$C_{n r} = -.75/\text{rad}$
$C_{L\delta_s} = .68/\text{rad}$	$C_{n\delta_a} = .0042/\text{rad}$
$C_{m\delta_s} = -1.46/\text{rad}$	$C_{\ell\delta_a} = .039/\text{rad}$
	$C_{Y\delta_r} = .208/\text{rad}$
	$C_{\ell\delta_r} = .045/\text{rad}$
	$C_{n\delta_r} = -.16/\text{rad}$
	$C_{Y\delta_d} = .0325/\text{rad}$
	$C_{n\delta_d} = -.025/\text{rad}$
	$C_{\ell\delta_d} = -.0044/\text{rad}$



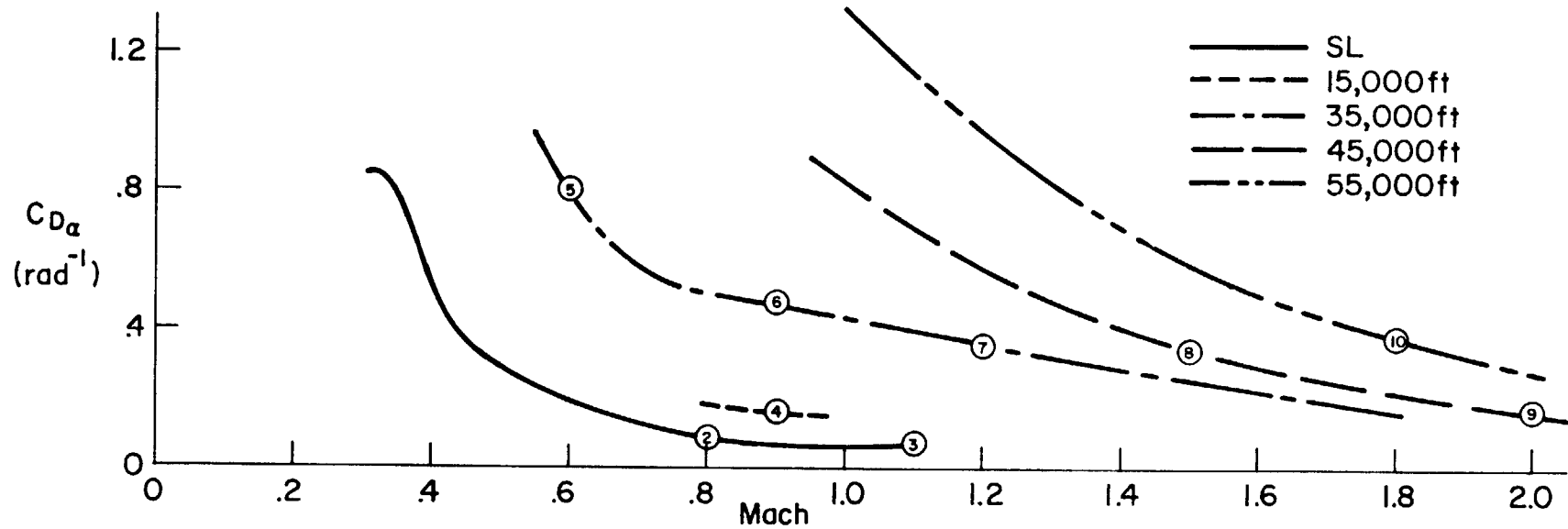
— SL
 --- 15,000 ft
 -.- 35,000 ft
 — 45,000 ft
 -.- 55,000 ft

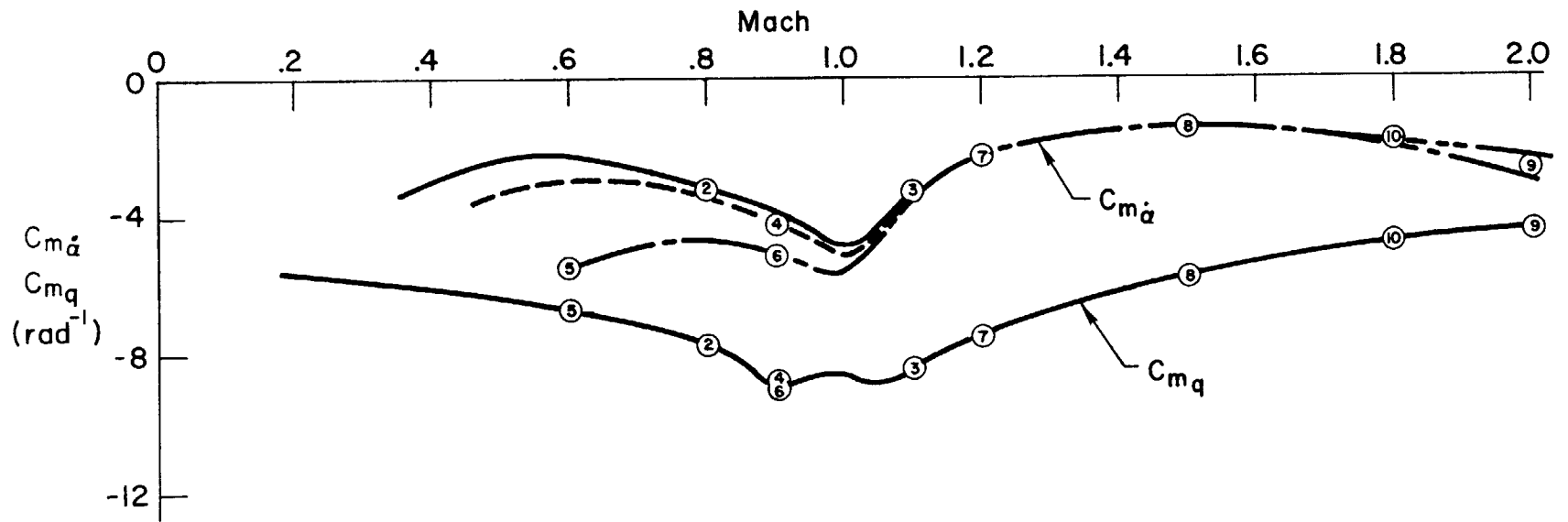
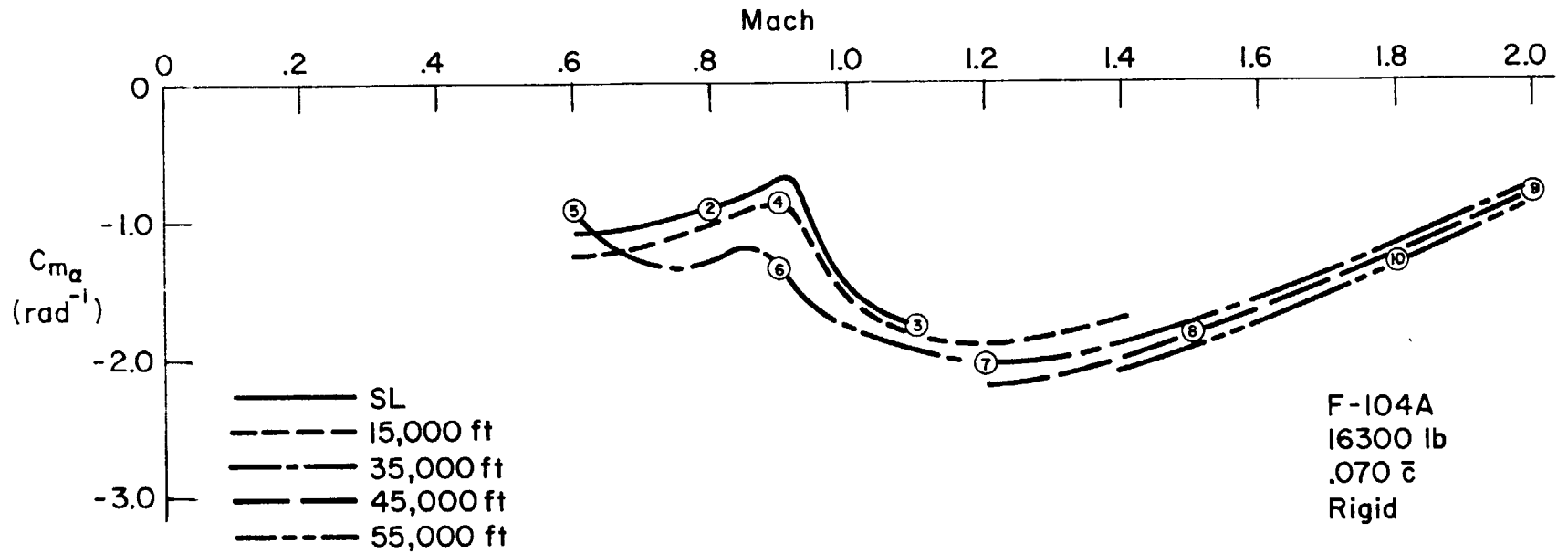
F-104A
 16300 lb

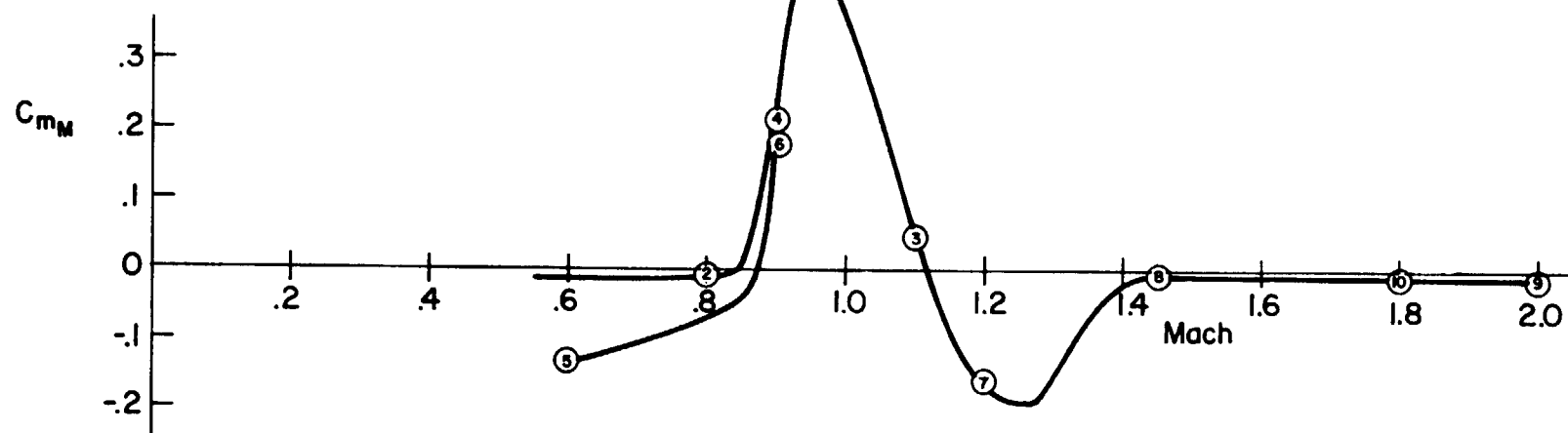
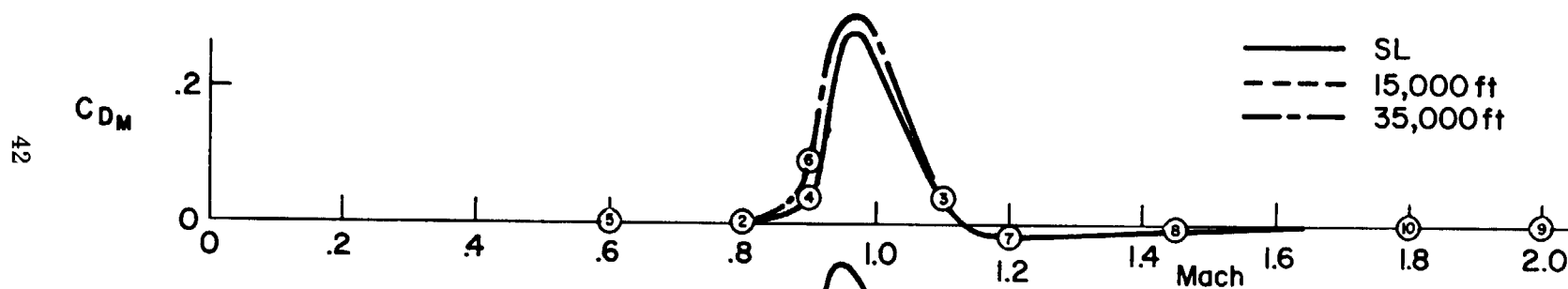
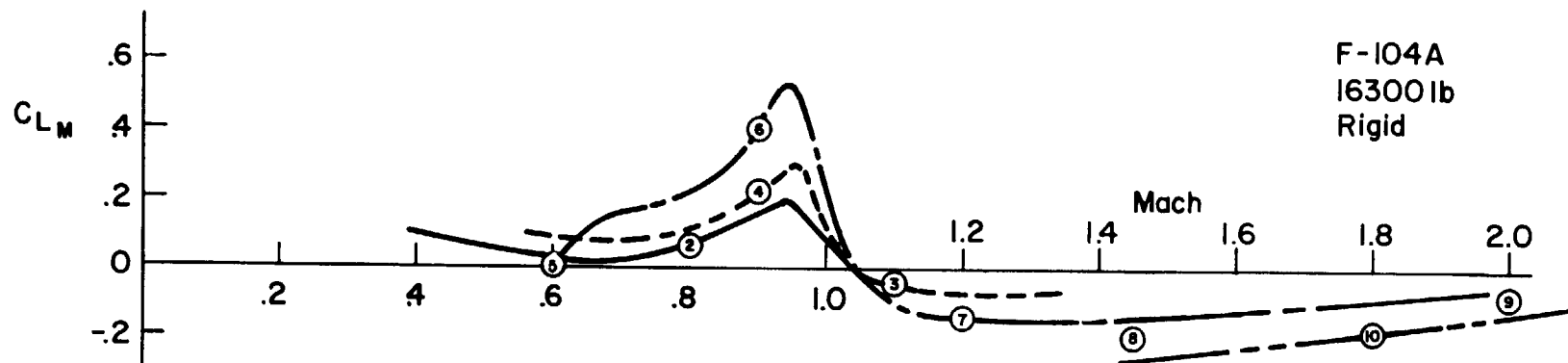


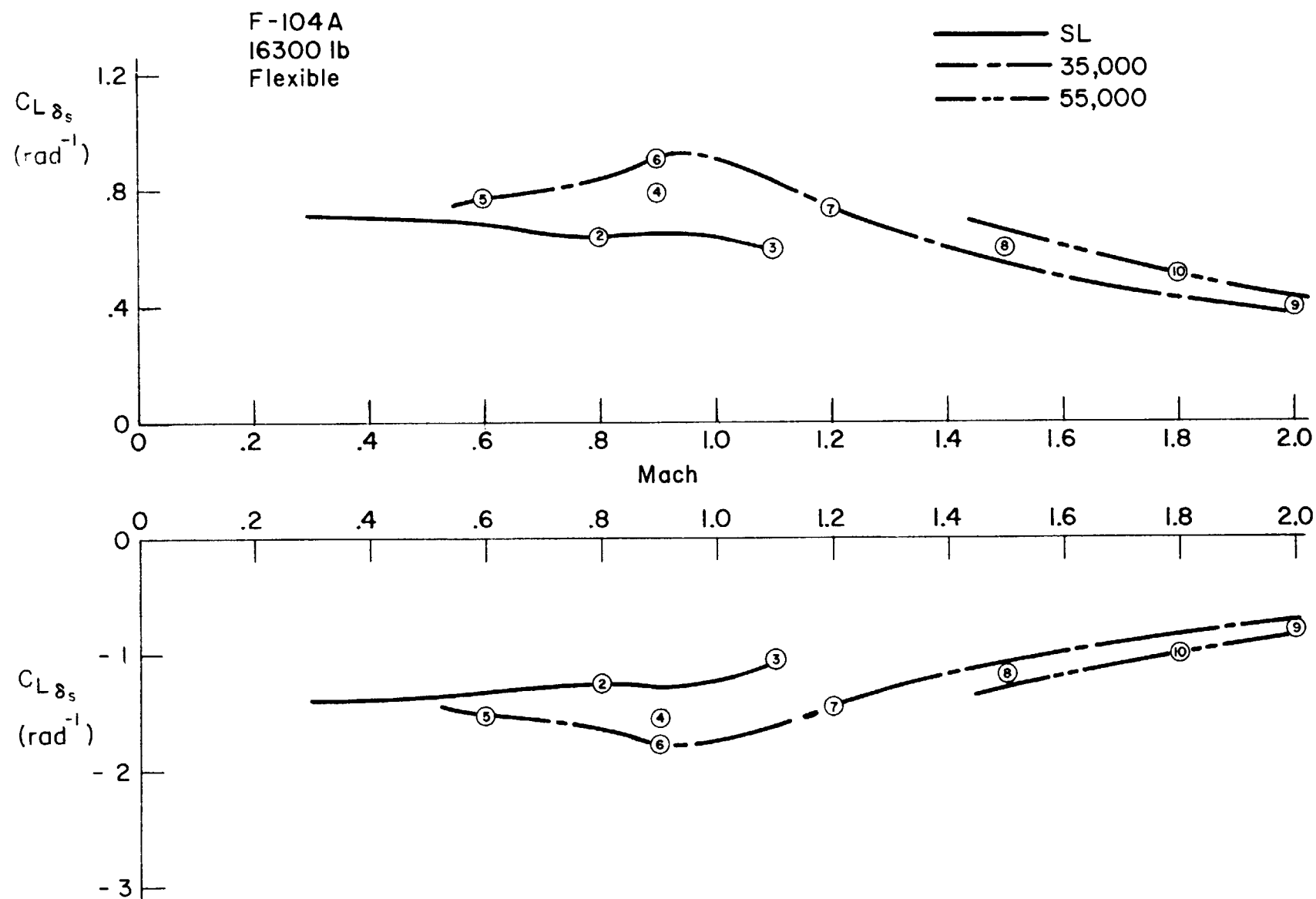


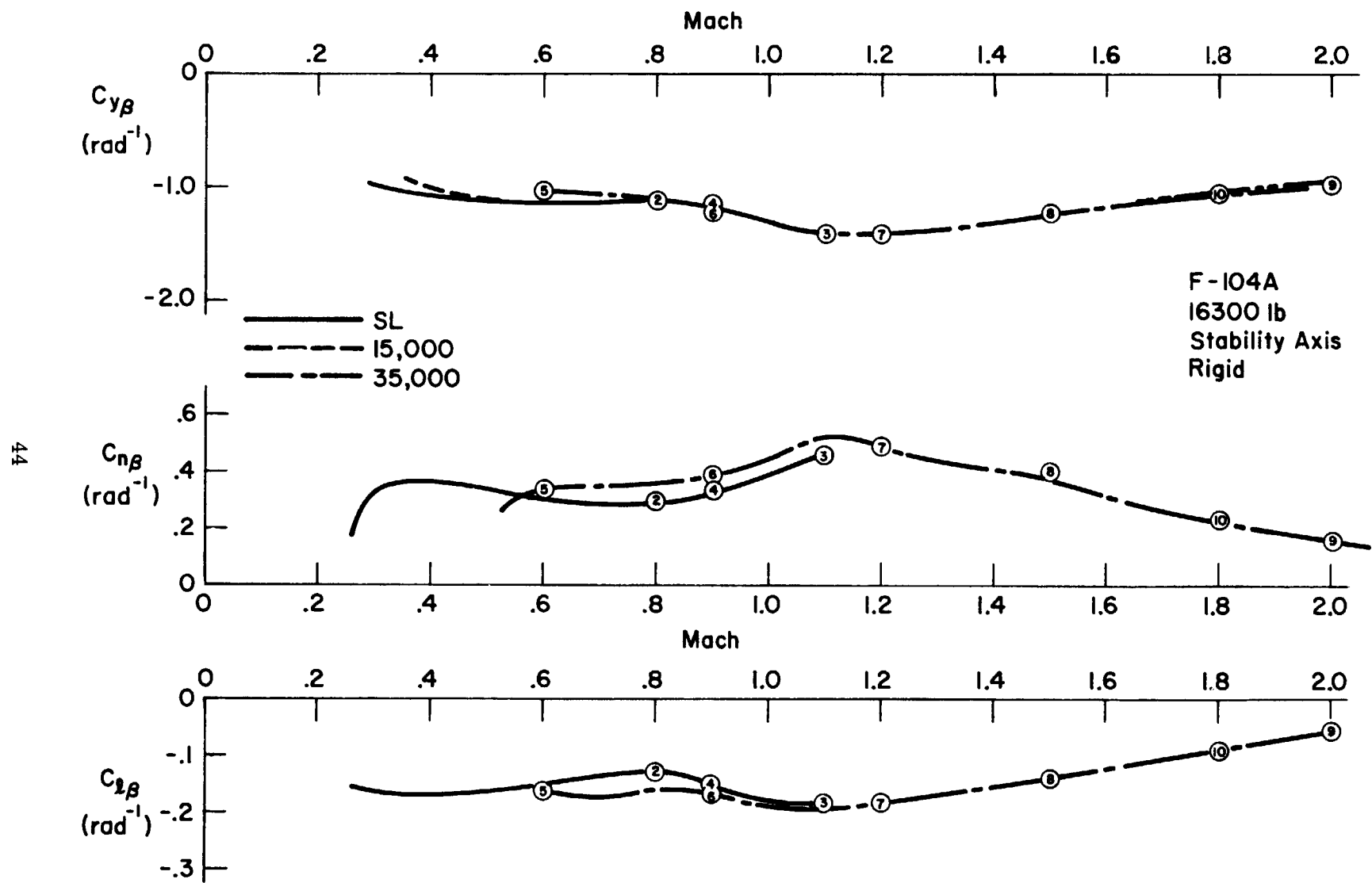
40

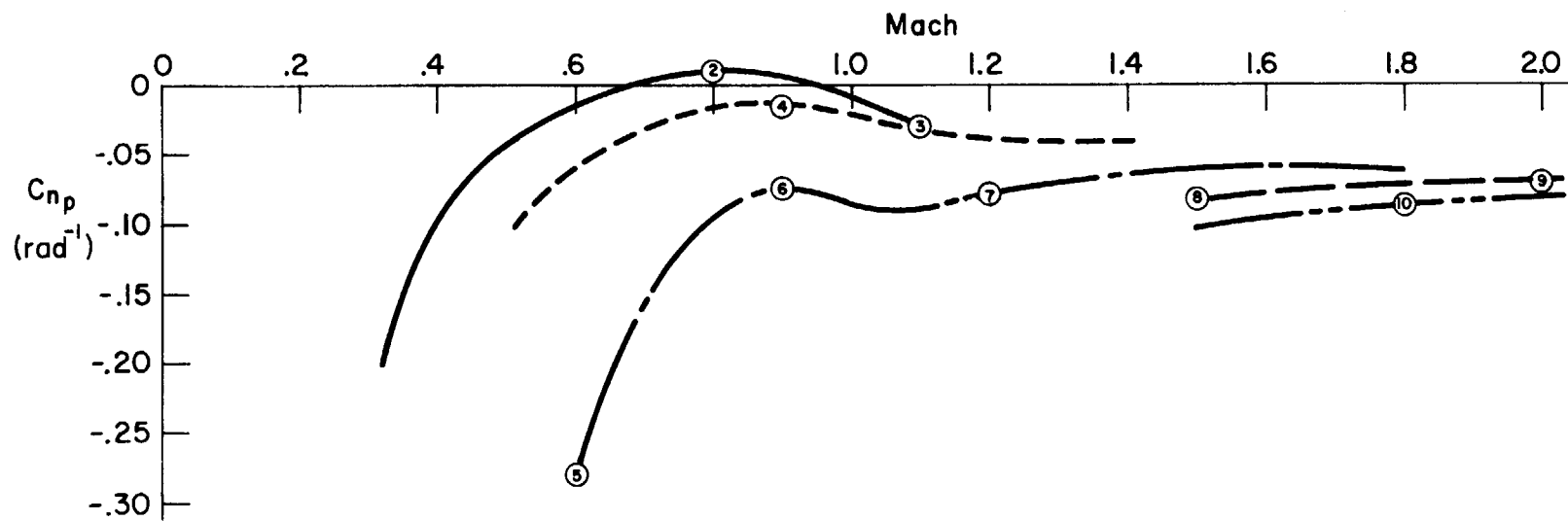
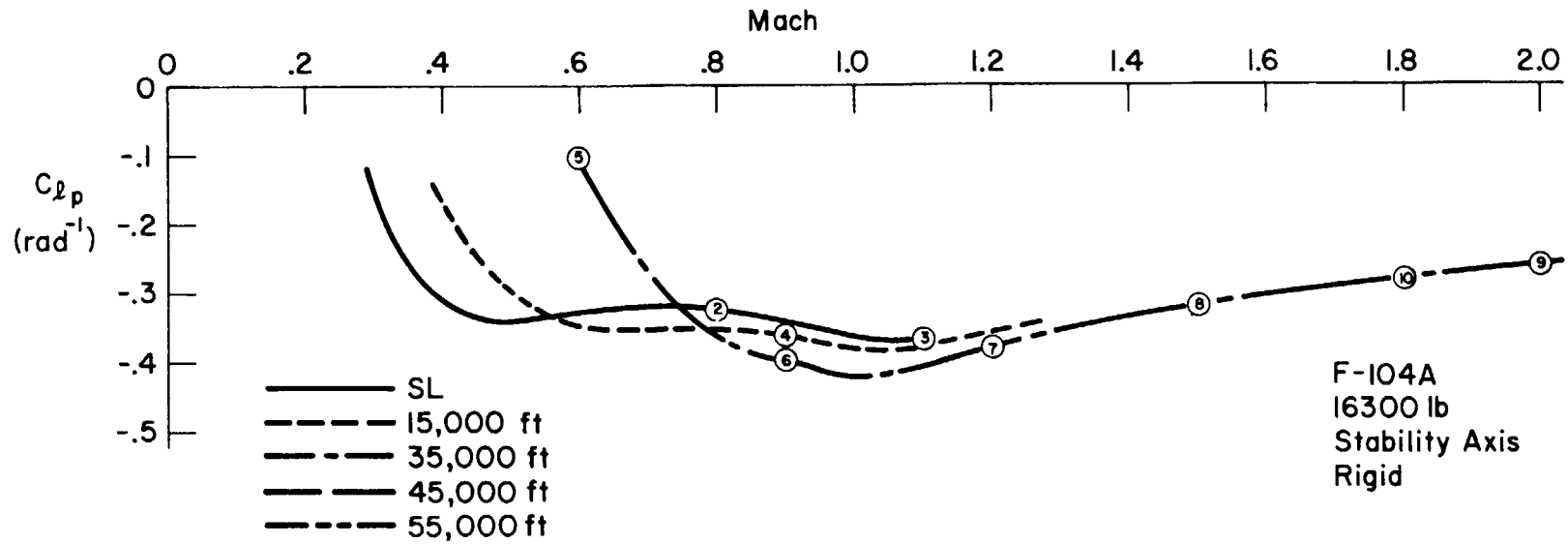


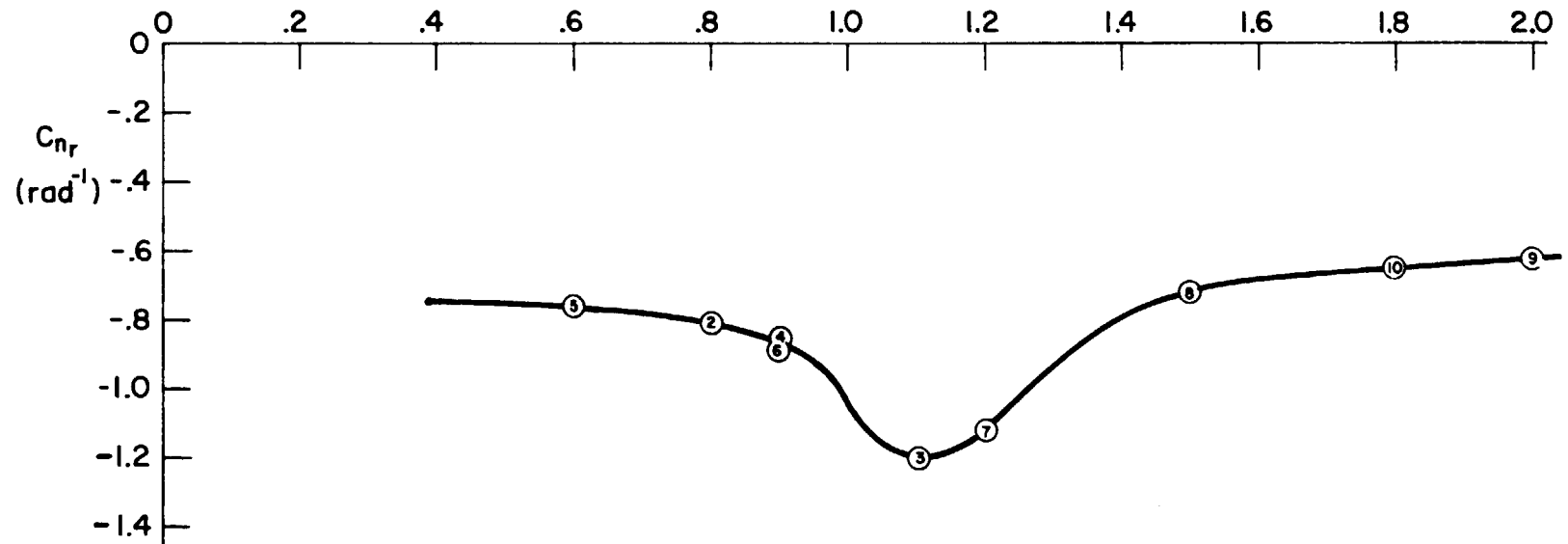
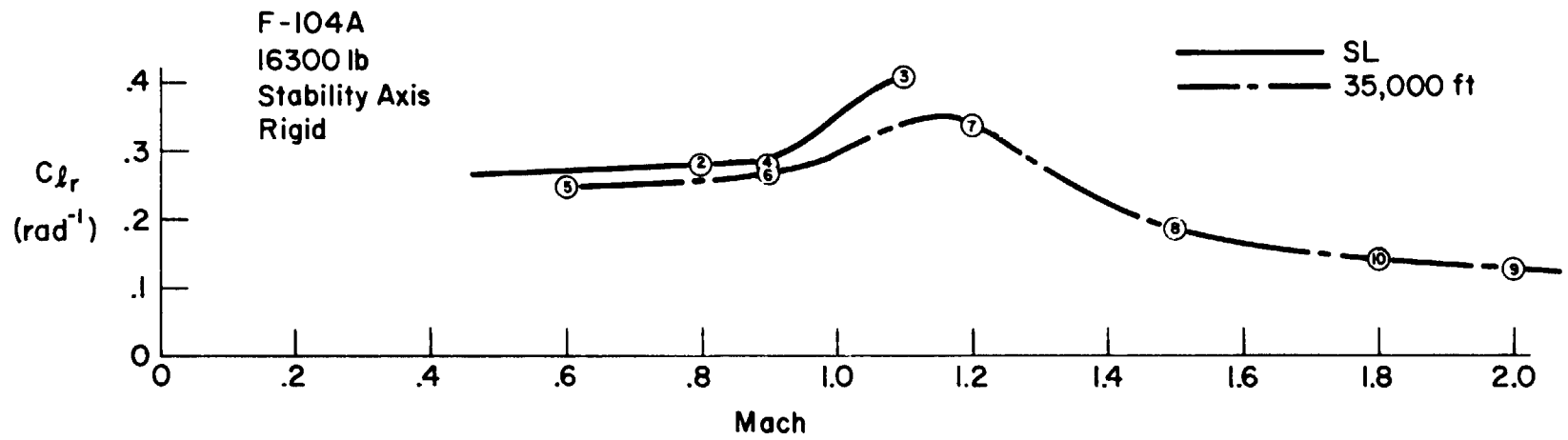


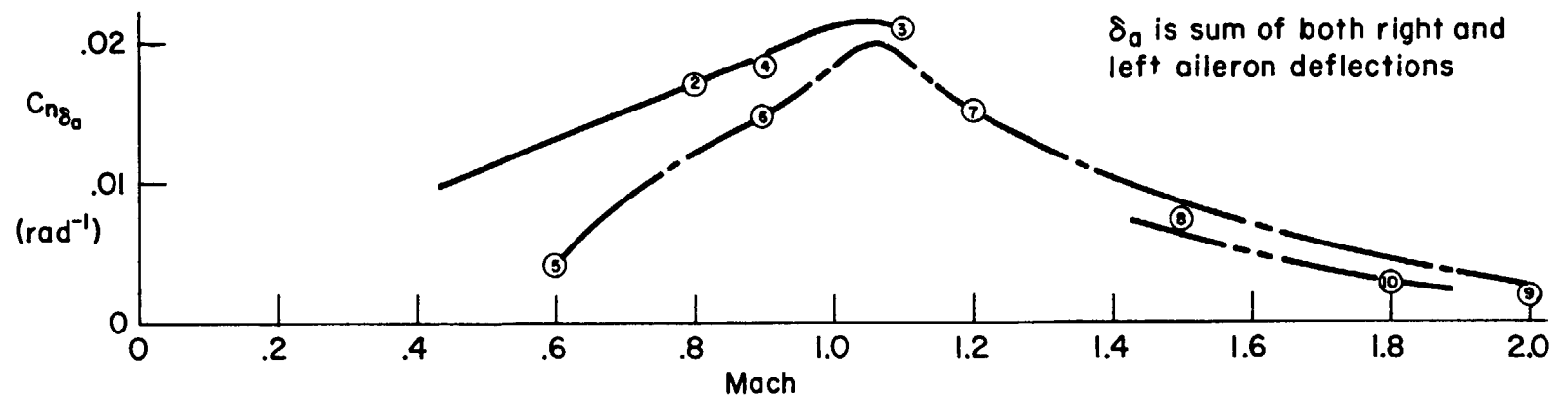
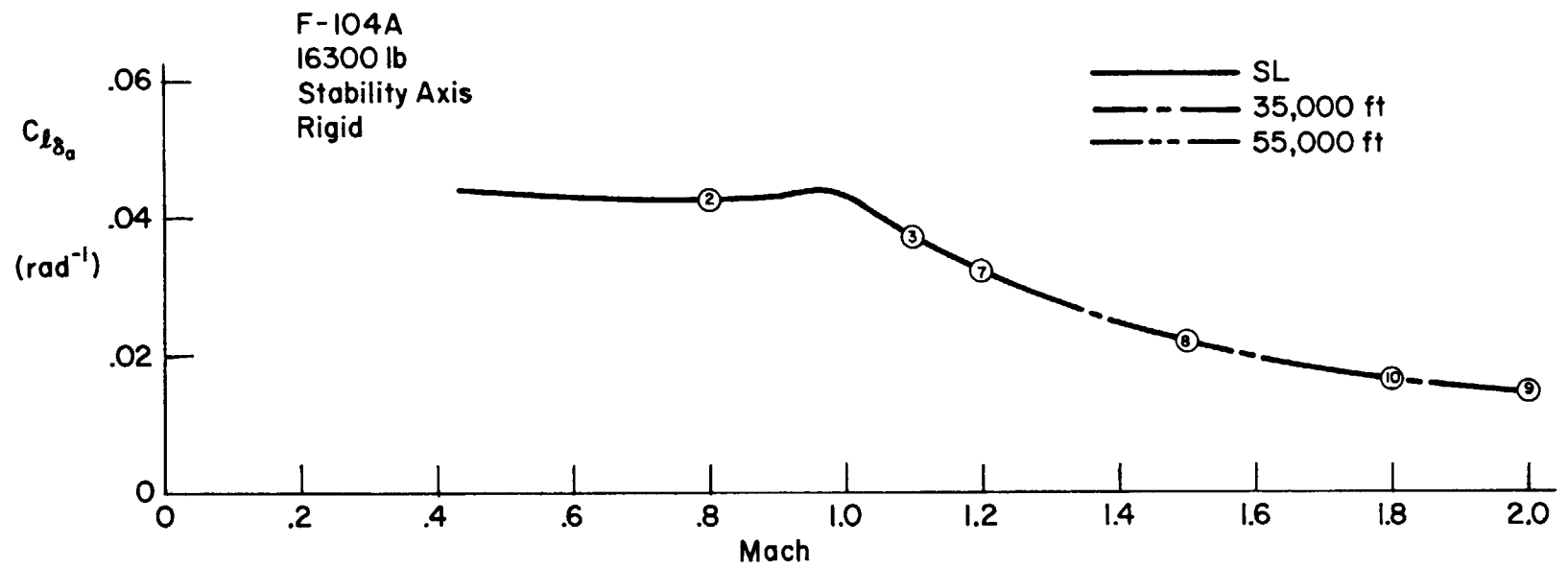












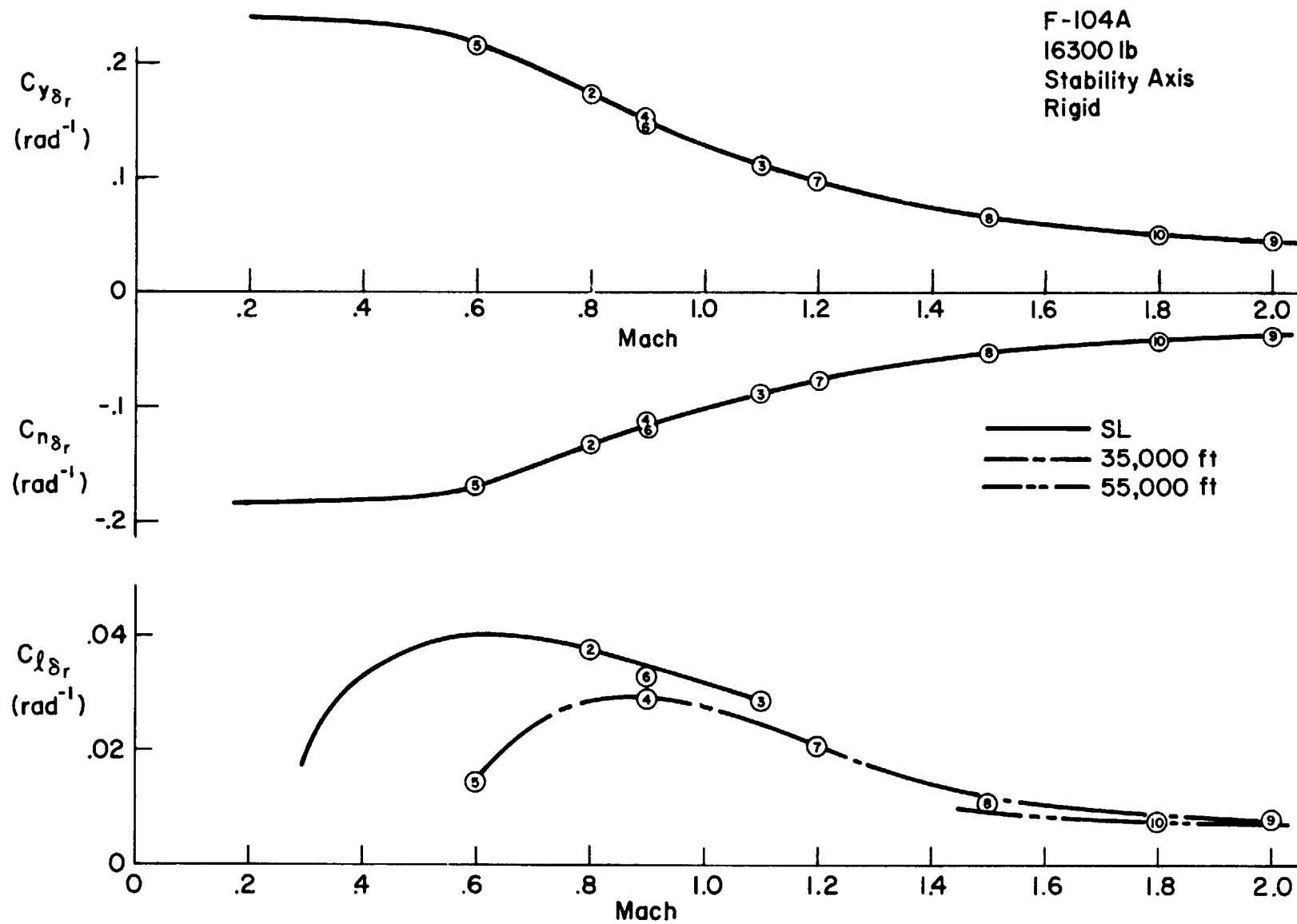


TABLE III-2

F-104A DIMENSIONAL, MASS AND FLIGHT CONDITION PARAMETERS

s = 196.1 sq ft, b = 21.94 ft, $\bar{c}$ = 9.55 ft

F/C #	1	2	3	4	5	6	7	8	9	10
H(FT)	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M(-)	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
VTC(FPS)	287.	893.	1228.	952.	584.	876.	1167.	1452.	1936.	1742.
VTC(KTAS)	170.	529.	728.	564.	346.	519.	692.	860.	1147.	1032.
VTC(KCAS)	170.	529.	728.	465.	199.	311.	432.	445.	591.	433.
W(LBS)	14126.	16300.	16300.	16300.	16300.	16300.	16300.	16300.	16300.	16300.
C.G.(MGC)	.164	.0700	.0700	.0700	.0700	.0700	.0700	.0700	.0700	.0700
IX (SLLG-FT SQ)	3582.	3679.	3679.	3679.	3679.	3679.	3679.	3679.	3679.	3679.
IY (SLLG-FT SQ)	55802.	58613.	58613.	58613.	58613.	58613.	58613.	58613.	58613.	58613.
IZ (SLLG-FT SQ)	56669.	59541.	59541.	59541.	59541.	59541.	59541.	59541.	59541.	59541.
IXZ(SLLG-FT SQ)	2658.	2699.	2699.	2699.	2699.	2699.	2699.	2699.	2699.	2699.
EPSILON(DEG)	-2.86	-2.76	-2.76	-2.76	-2.76	-2.76	-2.76	-2.76	-2.76	-2.76
Q(PSF)	97.8	948.	1792.	677.	126.	283.	503.	489.	869.	436.
QC(PSF)	99.5	1109.	2397.	826.	138.	345.	703.	749.	1440.	706.
ALPHA(DEG)	2.30	2.00	1.00	4.80	12.4	2.50	3.00	3.80	3.00	4.80
GAMMA(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LXP(FT)	19.0	18.1	18.1	18.1	18.1	18.1	18.1	18.1	18.1	18.1
LZP(FT)	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40
ITH(DEG)	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50
XI(DEG)	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50	-2.50
LTH(FT)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
+	+	+	+	+	+	+	+	+	+	+

TABLE III-3

F-104A LONGITUDINAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
XU *	-.0737	-.0117	-.0793	-.0167	-.00221	-.0129	-.0131	-.0125	-.0159	-.0111
ZU *	-.204	-.0332	.0270	-.0199	-.0626	-.0932	.0139	.0277	.0171	.0175
YU *	.000294	.000794	.00399	.00610	.000806	.00228	-.000998	.00124	.000457	.000798
XW	.0631	.0556	.0343	.0898	.0384	.00555	.0100	.0108	.0110	.00699
ZW	-.570	-1.65	-2.32	-1.22	-.242	-.635	-.794	-.508	-.442	-.296
YW	-.00732	-.0305	-.0816	-.0200	-.00617	-.0139	-.0283	-.0199	-.0115	-.0104
ZWD	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ZQ	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MYD	-.000304	-.000580	-.000580	-.000478	-.000297	-.000287	-.000129	-.494E-4	-.883E-4	-.393E-4
MQ	-.317	-1.25	-1.87	-.956	-.220	-.434	-.493	-.293	-.301	-.183
XDS	1.19	8.07	7.27	17.6	8.05	4.35	7.55	7.52	7.04	7.21
ZDS	-29.7	-231.	-416.	-209.	-36.6	-99.6	-144.	-113.	-134.	-85.8
MDS	-4.79	-27.9	-63.0	-33.6	-6.03	-16.3	-23.3	-18.4	-22.2	-13.9
XDTH	.00228	.00197	.00197	.00197	.00197	.00197	.00197	.00197	.00197	.00197
ZDTH	.994E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4
MDTH	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TABLE III-4

F-104A STABILIZER TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DENOMINATOR										
Z(DET)1	.238	.122	.767	.121	.0844	.143	(-.0299)	.716	(-.000333)	.603
W(DET)1	.152	.0504	.0523	.111	.0709	.0839	(.0389)	.00834	(.0156)	.00895
Z(DET)2	.324	.315	.263	.288	.163	.185	.125	.0810	.0967	.0643
W(DET)2	1.51	5.41	10.3	4.54	1.91	3.53	5.78	5.39	4.73	4.26
NUMERATORS										
N(U /DS)										
A(U)	1.19	8.07	7.27	17.6	8.05	4.35	7.55	7.52	7.04	7.21
1/T(U)1	43.8	147.	186.	153.	93.9	143.	189.	236.	320.	282.
Z(U)1	.740	.438	.632	.412	.690	.989	.989	.946	.797	.957
W(U)1	1.25	1.23	1.85	.665	.230	.681	.571	.359	.344	.226
N(W /DS)										
A(W)	-29.7	-231.	-416.	-209.	-36.6	-99.6	-144.	-113.	-134.	-85.8
1/T(W)1	46.6	148.	-.000664	153.	94.1	143.	-.0190	-.0174	-.0103	-.0123
1/T(W)2	(.256)	(.158)	.0791	(.178)	(.0315)	(.123)	.0300	.0266	.0245	.0204
1/T(W)3	(.150)	(.0367)	188.	(.0437)	(.0608)	(.0625)	189.	236.	320.	282.
N(TH/DS)										
A(TH)	-4.79	-37.7	-62.8	-33.5	-6.02	-16.3	-23.3	-18.4	-22.2	-13.9
1/T(TH)1	.104	.0128	.0789	.0178	.0117	.0127	.0134	.0118	.0155	.0106
1/T(TH)2	.496	1.47	2.29	1.09	.195	.550	.620	.386	.373	.233
N(HC /DS)										
A(HC)	29.7	231.	416.	210.	37.5	99.7	144.	114.	135.	86.1
1/T(HC)1	.0504	.0106	.0784	.0132	-.0198	.00399	.0129	.0116	.0153	.0101
1/T(HC)2	-4.65	-13.8	-19.4	-12.2	-3.99	-8.48	-10.5	-9.36	-10.7	-7.96
1/T(HC)3	5.12	15.5	21.9	13.6	4.41	9.18	11.1	9.72	11.2	8.22
N(AZP/DS)										
A(AZP)	61.2	452.	720.	396.	72.3	195.	278.	220.	267.	166.
1/T(AZP)1	-.00775	-.00135	-.000458	-.00311	.00551	-.00297	-.00136	-.00135	-.000839	-.00143
1/T(AZP)2	.0575	.0120	.0789	.0162	-.0262	.00690	.0142	.0128	.0161	.0114
Z(AZP)1	.0887	.0631	.0678	.0498	.0210	.0386	.0390	.0291	.0209	.0192
W(AZP)1	3.41	10.5	15.7	9.35	3.01	6.32	7.79	6.86	7.74	5.83

TABLE III-5

F-104A THRUST TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DENOMINATOR										
Z(DET)1	.239	.123	.767	.122	.0853	.143	(-.0299)	.716	(-.000333)	.604
W(DET)1	.152	.0504	.0523	.111	.0708	.0839	(.0389)	.00834	(.0156)	.00895
Z(DET)2	.324	.315	.263	.289	.165	.185	.125	.0810	.0968	.0643
W(DET)2	1.51	5.41	10.3	4.54	1.91	3.53	5.78	5.39	4.73	4.26
NUMERATORS										
N(U /DTH)										
A(U)	.00228	.00197	.00197	.00197	.00197	.00197	.00197	.00197	.00197	.00197
1/T(U)1	.000361	.000293	.000652	-.00128	-.00947	.119E-8	-.000238	-.000501	-.000144	-.000740
Z(U)1	.323	.316	.263	.293	.170	.187	.124	.0811	.0968	.0645
W(U)1	1.51	5.42	10.3	4.50	1.90	3.52	5.79	5.39	4.73	4.26
N(W /DTH)										
A(W)	.994E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4	.861E-4
1/T(W)1	.00157	.000334	.000679	-.00136	-.00993	.471E-9	-.000242	-.000501	-.000144	-.000740
Z(W)1	(-.118)	.117	.134	.0605	-.0529	-.107	(-4.67)	.0816	.101	.0645
W(W)1	(-4.09)	3.91	10.7	11.5	3.21	6.70	(5.68)	6.43	4.52	5.65
N(TH /DTH)										
A(TH)	-.242E-8	-.000E-8	-.300E-7	.380E-7	.103E-6	.355E-14	.223E-8	.222E-8	.152E-8	.313E-8
1/T(TH)1	24.0	-4.54	13.6	1.49	1.14	1.60	.538	.184	.0481	.144
1/T(TH)2	-64.1	100.	-42.5	273.	9.78	.949E+9	-1972.	330.	-55.5	218.
N(HD /DTH)										
A(HD)	-.795E-5	-.172E-4	-.517E-4	.792E-4	.000339	.146E-10	.172E-4	.448E-4	.172E-4	.792E-4
1/T(HD)1	-64.1	-10.0	-3.51	5.83	.543	.163E+8	-2.94	.151	-.0205	.119
Z(HD)1	.139	.149	.248	-.124	.135	.0769	.453	.0424	.116	.0356
W(HD)1	1.45	4.85	10.8	5.65	1.90	4.46	7.39	5.38	4.75	4.26
N(AZP/DTH)										
A(AZP)	.994E-4	.863E-4	.866E-4	.854E-4	.842E-4	.861E-4	.861E-4	.861E-4	.861E-4	.860E-4
1/T(AZP)1	-.00451	-.00126	-.000458	-.00284	-.0122	-.00161	-.00145	-.00150	-.000767	-.00157
1/T(AZP)2	-4.79	-1.63	-2.25	-5.05	-1.96	-3.65	.982	-.0763	.00466	-.107
Z(AZP)1	.194	.265	.255	.328	.168	.195	.0805	.0868	.0947	.0738
W(AZP)1	1.50	5.38	10.4	5.83	1.99	3.87	5.71	5.40	4.72	4.27

TABLE III-6

F-104A STICK FORCE TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Closed

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DENOMINATOR										
Z(DET)1	.249	.151	.990	.127	.0717	.134	(-.0266)	.776	(-.000450)	.644
W(DET)1	.142	.0387	.0401	.0851	.0683	.0749	(.0363)	.00765	(.0157)	.00832
Z(DET)2	.303	.239	.199	.218	.156	.164	.113	.0754	.0838	.0602
W(DET)2	1.59	6.31	11.3	5.35	1.95	3.76	5.91	5.45	5.02	4.33
NUMERATORS										
N(U /FST)										
A(U)	-.00565	-.0265	-.0213	-.0591	-.0317	-.0161	-.0268	-.0275	-.0252	-.0270
1/T(U)1	43.8	147.	186.	153.	93.9	143.	189.	236.	320.	282.
Z(U)1	.740	.438	.632	.412	.690	.989	.989	.946	.797	.957
W(U)1	1.25	1.23	1.85	.665	.230	.681	.571	.359	.344	.226
N(W /FST)										
A(W)	.141	.760	1.22	.704	.144	.369	.512	.414	.480	.322
1/T(W)1	46.6	148.	-.000664	153.	94.1	143.	-.0190	-.0174	-.0103	-.0123
1/T(W)2	(.256)	(.158)	.0791	(.178)	(.0315)	(.123)	.0300	.0266	.0245	.0204
1/T(W)3	(.150)	(.0367)	188.	(.0437)	(.0608)	(.0625)	189.	236.	320.	282.
N(THE/FST)										
A(THE)	.0227	.124	.184	.113	.0237	.0602	.0829	.0674	.0794	.0523
1/T(THE)1	.104	.0128	.0789	.0178	.0117	.0127	.0134	.0118	.0155	.0106
1/T(THE)2	.496	1.47	2.29	1.09	.195	.550	.620	.386	.373	.233
N(HD /FST)										
A(HD)	-.141	-.761	-1.22	-.707	-.148	-.369	-.513	-.415	-.481	-.323
1/T(HD)1	.0504	.0106	.0784	.0132	-.0198	.00399	.0129	.0116	.0153	.0101
1/T(HD)2	-4.69	-13.8	-19.4	-12.2	-3.90	-8.48	-10.5	-9.36	-10.7	-7.96
1/T(HD)3	5.12	15.5	21.9	13.6	4.41	9.18	11.1	9.72	11.2	8.22
N(AZP/FST)										
A(AZP)	-.290	-1.49	-2.11	-1.34	-.285	-.720	-.988	-.805	-.956	-.625
1/T(AZP)1	-.00775	-.00135	-.000458	-.00311	.00551	-.00297	-.00136	-.00135	-.000839	-.00143
1/T(AZP)2	.0575	.0120	.0789	.0162	-.0262	.00690	.0142	.0128	.0161	.0114
Z(AZP)1	.0887	.0631	.0678	.0498	.0210	.0386	.0390	.0291	.0209	.0192
W(AZP)1	3.41	10.5	15.7	9.35	3.01	6.32	7.79	6.86	7.74	5.83

TABLE III-7
F-104A THRUST TRANSFER FUNCTION FACTORS
SAS Off — Bobweight Loop Closed
(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DENOMINATORS										
Z(DEL)1	.249	.151	.990	.127	.0725	.134	(-.0266)	.777	(-.000450)	.644
w(DEL)1	.142	.0387	.0401	.0851	.0683	.0749	(.0363)	.00765	(.0157)	.00832
Z(DEL)2	.303	.239	.199	.218	.158	.164	.113	.0755	.0824	.0602
w(DEL)2	1.59	6.31	11.3	5.35	1.95	3.76	5.91	5.45	5.02	4.33
NUMERATORS										
U(DEL/DTH)										
A(U)	.00228	.00197	.00197	.00197	.00197	.00197	.00197	.00197	.00197	.00197
1/T(U)1	-.000293	-.000339	.000203	-.00193	-.00967	-.000328	-.000443	-.000624	-.000308	-.000831
Z(U)1	.302	.238	.198	.216	.158	.164	.111	.0730	.0804	.0566
w(U)1	1.59	6.31	11.3	5.33	1.94	3.76	5.92	5.45	5.02	4.33
W(DEL/DTH)										
A(W)	.000101	.926E-4	.966E-4	.921E-4	.873E-4	.893E-4	.905E-4	.896E-4	.902E-4	.889E-4
1/T(W)1	-.00555	.00585	.000650	-.00115	-.00935	.000740	.519E-4	-.000608	-.000310	-.000809
Z(W)1	(.605)	(.244)	(4.20)	.566	.164	.336	(-1.53)	.814	(1.60)	.846
w(W)1	(-4.19)	(10.5)	(17.8)	5.43	2.78	5.18	(11.6)	6.32	(13.8)	5.57
N(THC/DTH)										
A(THC)	.221E-6	.105E-5	.156E-5	.593E-6	.301E-6	.515E-6	.711E-6	.578E-6	.681E-6	.450E-6
1/T(THC)1	(-.531)	(-.429)	(.0874)	2.83	(.669)	(.565)	.532	.142	-.0204	.128
1/T(THC)2	(4.03)	(1.86)	(2.82)	4.53	(1.92)	(3.08)	-5.44	1.46	.256	1.56
N(HD/DTH)										
A(HD)	-.934E-5	-.237E-4	-.622E-4	.732E-4	.000338	-.316E-5	.128E-4	.412E-4	.131E-4	.765E-4
1/T(HD)1	-55.4	-11.4	-4.52	3.78	.495	-79.8	-1.87	.106	-.00763	.0940
Z(HD)1	.159	.680	.363	.0712	.142	.215	.315	.0372	.0709	.0331
w(HD)1	1.41	3.48	7.37	6.62	1.95	4.12	10.0	6.23	8.41	4.67
N(AZF/DTH)										
A(AZF)	.965E-4	.735E-4	.684E-4	.741E-4	.919E-4	.799E-4	.776E-4	.792E-4	.779E-4	.807E-4
1/T(AZF)1	-.00451	-.00126	-.000458	-.00284	-.0122	-.00161	-.00145	-.00150	-.000767	-.00157
1/T(AZF)2	-4.79	-1.63	-2.24	-4.99	-1.95	-3.64	.982	-.0763	.00466	-.107
Z(AZF)1	.193	.261	.253	.325	.168	.194	.0775	.0870	.0943	.0741
w(AZF)1	1.49	5.23	9.98	5.72	1.98	3.83	5.61	5.32	4.64	4.23

TABLE III-8

F-104A LONGITUDINAL HANDLING QUALITIES PARAMETERS

SAS Off

(BODY AXIS SYSTEM)

55	F/C #	1	2	3	4	5	6	7	8	9	10
	H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
	N	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
STICK FIXED											
D(G)/D(U) (DEC/KT)		-.152	-.0319	-.236	-.0397	.0594	-.0120	-.0388	-.0343	-.0460	-.0305
NZA (G/RAD)		4.64	40.3	86.3	32.0	3.62	14.9	22.4	17.4	22.3	12.5
DE/G (DEG/G)		5.83	1.09	1.10	1.07	9.36	2.92	3.66	5.17	2.57	5.92
CAP (RAD/SEC/SEC/G)		.487	.719	1.21	.623	.983	.829	1.40	1.66	.996	1.44
PHUGOID(2) (SEC) (TUCK(2))		--	--	--	--	--	--	(23.2)	--	(2080.)	--
1/C(1/10)		.933	.907	.744	.823	.455	.515	.343	.222	.265	.176
STICK FREE											
FST/KT (LB/KT)		-.223	-.0171	-.0254	-.0875	-.189	-.126	.0345	-.00351	.000317	-.00563
FST/G (LB/G)		23.9	7.86	7.90	7.76	43.1	15.7	18.8	25.3	14.2	28.5

TABLE III-9

F-104A LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/ C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80	
YV	-.178	-.452	-.791	-.328	-.0868	-.149	-.234	-.160	-.170	-.102	
YB	-51.1	-404.	-971.	-312.	-50.7	-130.	-273.	-233.	-330.	-177.	
LB'	-20.9	-146.	-363.	-134.	-32.3	-58.1	-115.	-87.8	-64.3	-52.2	
NB'	2.68	13.6	42.7	9.91	1.06	4.98	11.9	9.79	6.92	4.62	
LP'	-1.38	-4.64	-7.12	-3.63	-.374	-1.77	-2.27	-1.46	-1.59	-.962	
NP'	-.0993	-.188	-.341	-.150	-.0406	-.0943	-.117	-.0604	-.0901	-.0544	
LR'	1.16	3.67	7.17	2.66	1.02	1.08	1.88	.822	.689	.469	
NR'	-.157	-.498	-1.06	-.350	-.0809	-.169	-.292	-.152	-.188	-.106	
Y*CA	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	
L' CA	4.76	49.6	81.5	34.7	6.35	14.8	19.4	12.9	15.8	8.38	
N' CA	.266	3.51	6.50	2.64	.407	1.01	1.49	.902	.890	.517	
Y*CR	.0317	.0719	.0621	.0413	.0179	.0188	.0159	.00847	.00782	.00485	
L' CR	5.35	41.5	57.6	27.6	6.66	11.2	13.1	7.17	8.68	5.04	
N' CR	-.923	-7.07	-8.72	-4.49	-1.18	-1.91	-2.09	-1.52	-1.78	-1.01	
	+	+	+	+	+	+	+	+	+	+	+

TABLE III-10

F-104A AILERON TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DENOMINATOR										
1/T (DET) 1	-.000594	.00711	.00404	.00849	.0172	.00849	.00570	.00368	.00588	.00602
1/T (DET) 2	1.86	4.82	7.86	3.08	.446	2.04	2.41	1.50	1.72	.941
Z (DET) 1	-.0345	.0849	.0732	.136	.0138	.00590	.0453	.0339	.0331	.0373
W (DET) 1	2.10	4.51	7.53	4.50	2.84	2.85	4.29	3.97	3.25	3.00
NUMERATORS										
N(B /DA)										
A(B)	-.0749	-1.78	-5.08	.275	.966	-.369	-.468	-.0432	-.0631	.187
1/T (B) 1	.170	-.308	-.229	-.447	.0864	.317	-.596	.295	.127	.111
1/T (B) 2	-9.28	2.48	3.00	-5.36	.586	-1.13	.843	-4.74	-7.43	1.01
N(P /DA)										
A(P)	4.76	45.6	81.5	34.7	6.35	14.8	19.4	12.9	15.8	8.38
1/T (P) 1	-.00446	-.00124	-.000450	-.00282	-.0121	-.00160	-.00144	-.00147	-.000868	-.00155
Z (P) 1	.103	.123	.142	.0983	.0699	.0656	.0737	.0466	.0612	.0426
W (P) 1	1.97	4.93	8.54	4.49	1.76	3.00	4.55	3.99	3.25	2.80
N(R /DA)										
A(R)	.266	3.51	6.50	2.64	.407	1.01	1.49	.902	.890	.517
1/T (R) 1	1.48	1.08	1.61	.405	.249	.804	.528	.334	.316	.220
Z (R) 1	-.372	.202	.265	.169	-.0646	-.0533	.0591	.0170	-.0252	-.00604
W (R) 1	2.28	3.35	3.83	4.69	3.25	2.44	3.75	3.89	3.14	3.27
N(PHI/DA)										
A(PHI)	4.77	45.7	81.7	34.9	6.44	14.8	19.5	13.0	15.8	8.43
Z (PHI) 1	.101	.123	.142	.0987	.0639	.0655	.0737	.0464	.0610	.0423
W (PHI) 1	1.97	4.92	8.53	4.49	1.78	3.00	4.55	3.99	3.25	2.80
N(AYP/DA)										
A(AYP)	16.5	183.	313.	131.	22.6	53.8	73.6	47.3	53.9	29.5
1/T (AYP) 1	-.278	-.176	-.169	-.154	.111	-.146	-.144	-.114	.116	.0988
1/T (AYP) 2	.343	.721	.961	.705	-.290	.250	.301	.164	-.289	-.167
Z (AYP) 1	.0370	.112	.128	.104	.0760	.0574	.0695	.0444	.0758	.0520
W (AYP) 1	1.96	4.87	8.16	4.49	1.85	2.96	4.48	3.98	3.29	2.76

TABLE III-11

F-104A RUDDER TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DENOMINATOR										
1/T (DET) 1	-.000594	.00711	.00404	.00849	.0172	.00849	.00570	.00368	.00588	.00602
1/T (DET) 2	1.86	4.82	7.86	3.08	.446	2.04	2.41	1.50	1.72	.941
Z (DET) 1	-.0345	.0849	.0732	.136	.0138	.00590	.0453	.0339	.0331	.0373
W (DET) 1	2.10	4.51	7.53	4.50	2.84	2.85	4.29	3.97	3.25	3.00
NUMERATORS										
N(B /DR)										
A(B)	.0317	.0719	.0621	.0413	.0179	.0188	.0159	.00847	.00782	.00485
1/T (B) 1	-.0139	-.00574	-.00100	-.00640	-.0439	-.00267	-.00171	-.00256	.000969	-.000579
1/T (B) 2	2.16	4.94	8.64	3.11	.391	2.02	2.40	1.48	1.69	.938
1/T (B) 3	35.3	119.	156.	165.	144.	128.	175.	235.	285.	294.
N(P /DR)										
A(P)	5.35	41.5	57.6	27.6	6.66	11.2	13.1	7.17	8.68	5.04
1/T (P) 1	-.00447	-.00125	-.000454	-.00283	-.0121	-.00160	-.00144	-.00147	-.000872	-.00155
1/T (P) 2	-.960	-3.32	-3.33	-3.42	2.09	-2.19	-2.49	-2.95	-2.42	-2.37
1/T (P) 3	.976	3.40	3.70	3.47	-2.18	2.23	2.58	2.98	2.58	2.44
N(R /DR)										
A(R)	-.923	-7.07	-8.72	-4.49	-1.18	-1.91	-2.09	-1.52	-1.78	-1.01
1/T (R) 1	2.01	5.41	9.26	.498	.254	1.95	2.27	.397	.477	.236
Z (R) 1	.0299	.493	.627	.966	.0889	.320	.635	.508	.820	.358
W (R) 1	.548	.662	.478	2.22	2.36	.736	.699	1.52	1.03	1.51
N(PHI/DR)										
A(PHI)	5.32	41.2	57.4	27.2	6.40	11.2	13.0	7.07	8.58	4.95
1/T (PHI) 1	.972	-3.36	-3.35	3.47	2.16	-2.21	-2.52	-2.99	-2.45	-2.42
1/T (PHI) 2	-.974	3.39	3.69	-3.49	-2.30	2.23	2.58	3.00	2.58	2.46
N(AYP/DR)										
A(AYP)	4.40	35.8	56.7	24.3	5.13	8.79	12.2	2.04	3.83	2.30
1/T (AYP) 1	-.0277	-.0129	-.00460	-.0144	-.0582	-.0100	-.00748	-.00671	-.000431	-.00323
1/T (AYP) 2	-6.66	-18.6	-22.9	2.25	.209	1.46	2.00	.897	1.40	.622
1/T (AYP) 3	(.611)	(.787)	(.612)	4.85	5.75	3.03	3.74	8.86	8.88	8.20
1/T (AYP) 4	(1.43)	(4.07)	(6.05)	-16.5	-8.08	-10.3	-11.4	-25.9	-20.6	-15.0

TABLE III-12

F-104A LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.257	.800	1.10	.900	.600	.900	1.20	1.50	2.00	1.80
DR PERIOD (SEC)	3.00	1.40	.836	1.41	2.22	2.21	1.47	1.59	1.93	2.10
1/C(1/2)	--	.773	.665	1.24	.125	.0535	.411	.308	.300	.338
SPIRAL (2) (SEC)	1167.	--	--	--	--	--	--	--	--	--
P(1)	2.47	13.2	15.0	11.1	2.60	8.04	8.72	8.51	9.14	7.28
P(2)	1.66	11.2	11.7	10.8	1.37	7.12	8.42	8.48	8.79	7.10
P(3)	2.86	12.8	14.5	11.0	4.89	8.44	9.31	8.61	9.06	7.66
P(2)/P(1)	.671	.847	.783	.973	.525	.885	.966	.997	.963	.975
P(OSC)/P(AV)	.232	.0751	.114	.0111	.466	.0732	.0340	.00450	.0170	.0256
W(PHI)/W(D)	.940	1.09	1.13	.999	.629	1.05	1.06	1.01	1.00	.934
DEL-B-MAX	.170	.0908	.0873	.0302	.261	.0954	.0456	.0129	.0383	.0427
PHI TO BETA, PHASE	-318.	44.3	44.9	390.	-353.	33.6	26.4	18.2	-336.	14.8
PHI TO BETA	3.94	5.31	4.92	5.59	3.98	6.12	5.64	5.28	5.54	5.56
PHI TO VE	.787	.341	.230	.424	.701	.719	.497	.472	.371	.526
	+	+	+	+	+	+	+	+	+	+

F-104A DATA SOURCES

Stability and Control and Handling Qualities, F-104A, Lockheed Rept.
No. LR 10794, 12 Dec. 1955

Andrews, William H., and Herman A. Rediess, Flight-Determined Stability and Control Derivatives of a Supersonic Airplane with a Low Aspect-Ratio Unswept Wing and a Tee-Tail, NASA Memo 2-2-59H, Apr. 1959

Performance, F-104D, Lockheed Rept. No. LR-12873, 1 May 1958

Flight Manual, F-104A and F-104B USAF Series Aircraft, T. O. 1F-104A-1,
15 Dec. 1961

Technical Manual, Flight Controls, USAF Series F-104A and F-104C Aircraft, T. O. 1F-104A-2-8, 15 Mar. 1960

SECTION IV

F-4C

F-4C BACKGROUND

The F-4C is an Air Force tactical fighter whose primary mission is all-weather air-to-air missile combat. Lateral control is achieved by ailerons in combination with spoilers on a swept wing. A swept stabilator provides longitudinal stability and control. Directional stability and control is accomplished through a conventional fin-rudder combination. Landing speed is reduced by full span leading edge flaps and inboard plain trailing edge flaps in conjunction with blowing-type boundary layer control (BLC). Boundary layer control is automatically induced when full flap deflection occurs.

Features distinguishing the USAF F-4C from its Navy counterpart, the F-4B, are:

- Lack of drooped ailerons with flaps down resulting in higher landing speeds.
- Dual flight controls resulting in slightly increased control system inertia.
- Wing bumps to house larger main gear wheels resulting in a slight drag increase.

Data included here was obtained primarily from MAC Report No. 9842. Special emphasis is placed on the longitudinal control system because of its relative complexity when compared to other aircraft. Figure IV-4 has been added to help illustrate this system. Also, care has been taken to retain some of the control system nomenclature used by the manufacturer, e.g., q_B and P_{BF} (see Fig. IV-5).

The Stability Augmentation block diagrams are shown in Fig. IV-7. The roll SAS described is not included in lateral directional SAS on transfer functions since it is faded out with the lateral control stick out of neutral position.

F-4C

NOMINAL CONFIGURATION

4 AIM-7 missiles
60% internal fuel
 $W = 38924 \text{ lb}$
c.g. at $0.289 \bar{c}$, W.L. 27.65

$$\left. \begin{array}{l} I_x = 25001 \text{ slug-ft}^2 \\ I_y = 122186 \text{ slug-ft}^2 \\ I_z = 139759 \text{ slug-ft}^2 \\ I_{xz} = 2177 \text{ slug-ft}^2 \end{array} \right\} \text{body axis}$$

POWER APPROACH CONFIGURATION

2 AIM-7 missiles aft
20% internal fuel
Full flaps, BLC
Gear down
19 units angle of attack
 $W = 33196 \text{ lb}$
c.g. at $0.291 \bar{c}$, W.L. 25.2

$$\left. \begin{array}{l} I_x = 23668 \text{ slug-ft}^2 \\ I_y = 117500 \text{ slug-ft}^2 \\ I_z = 133723 \text{ slug-ft}^2 \\ I_{xz} = 1575 \text{ slug-ft}^2 \end{array} \right\} \text{body axis}$$

FLIGHT ENVELOPE

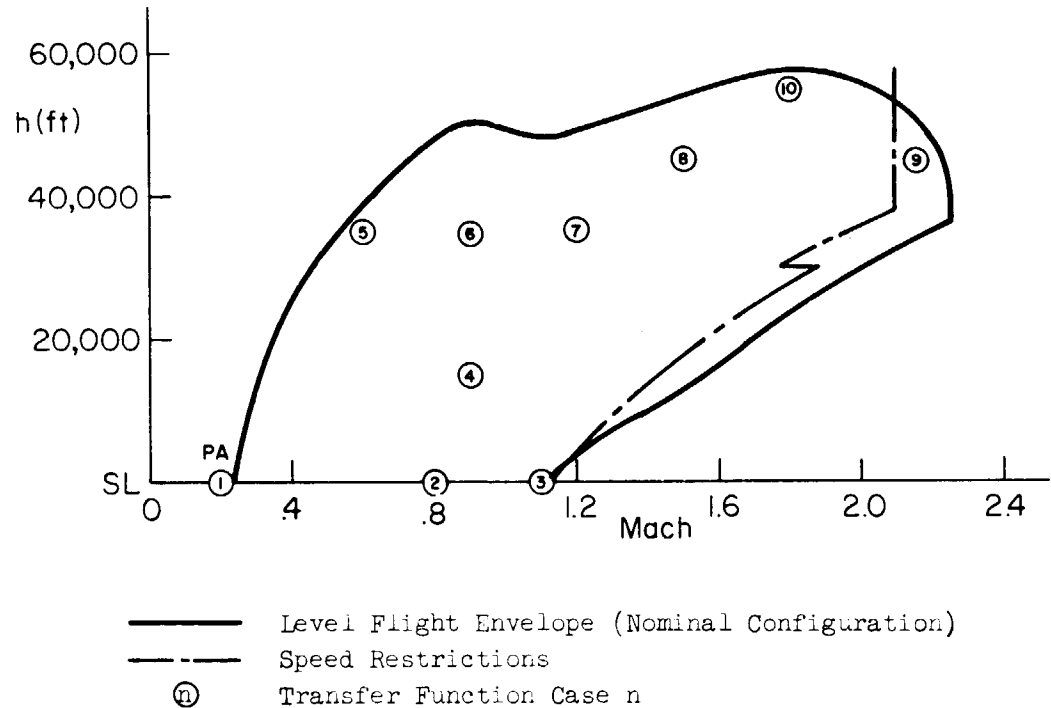


Figure IV-1. Flight Conditions

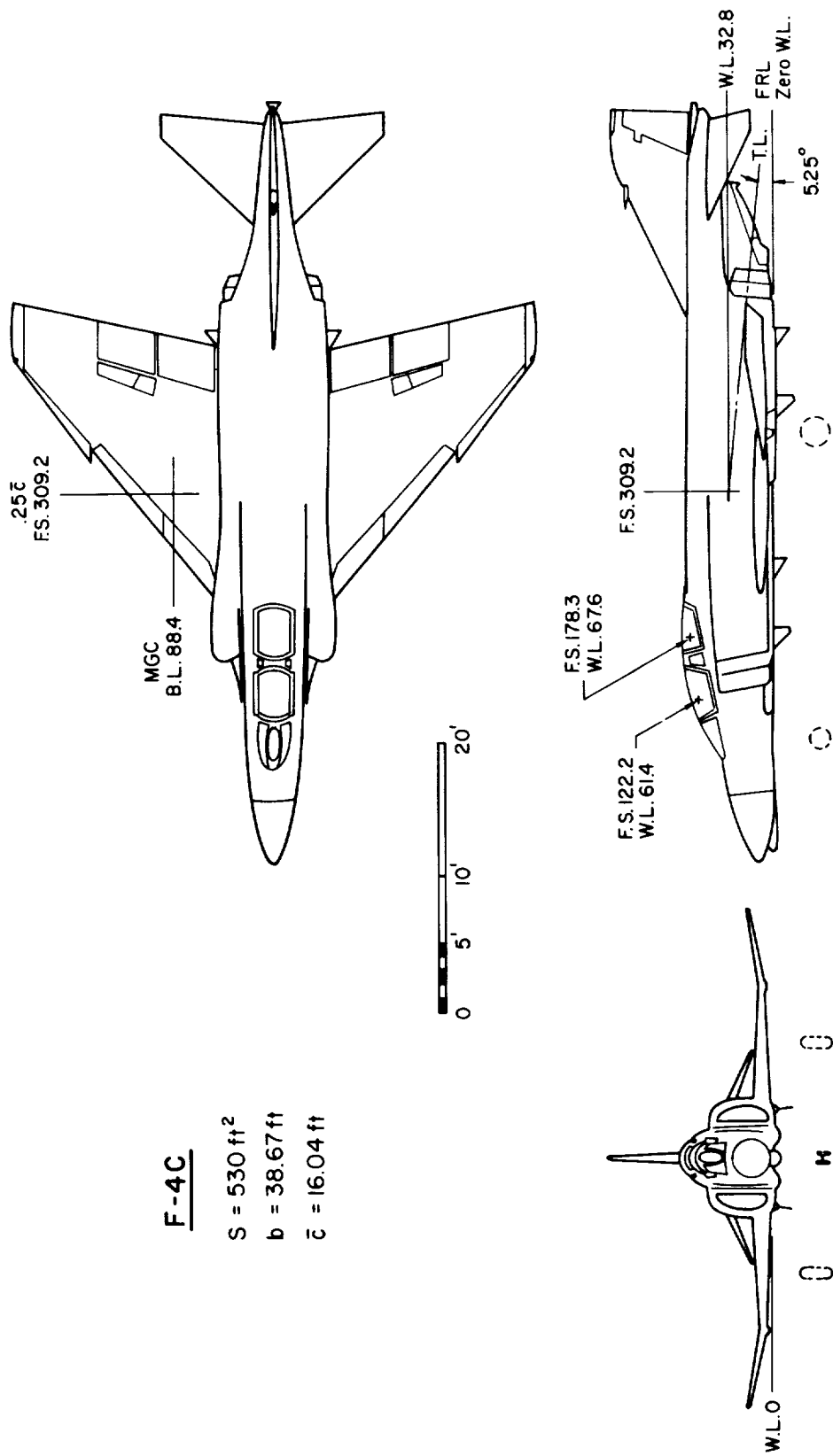
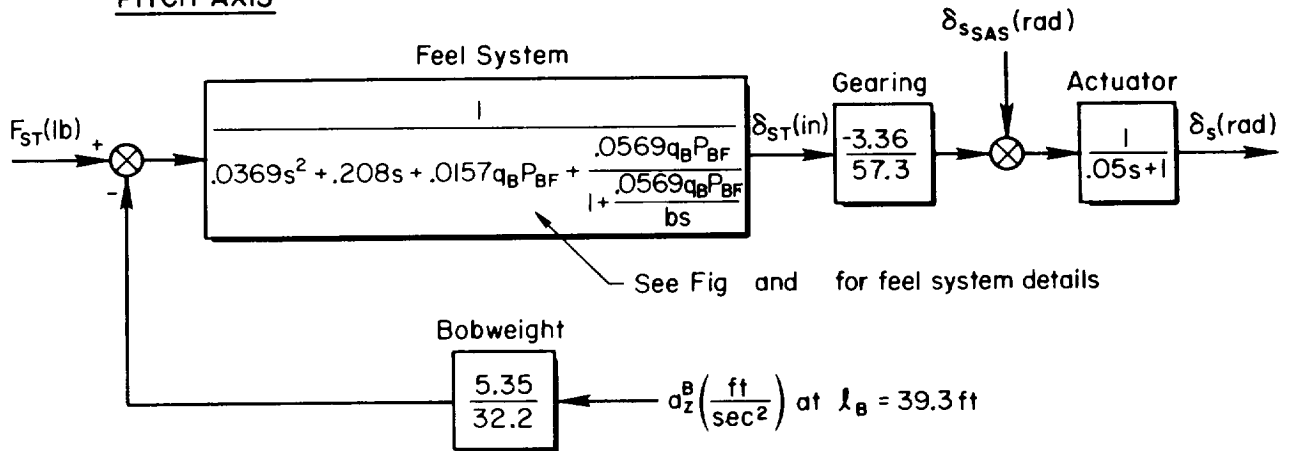


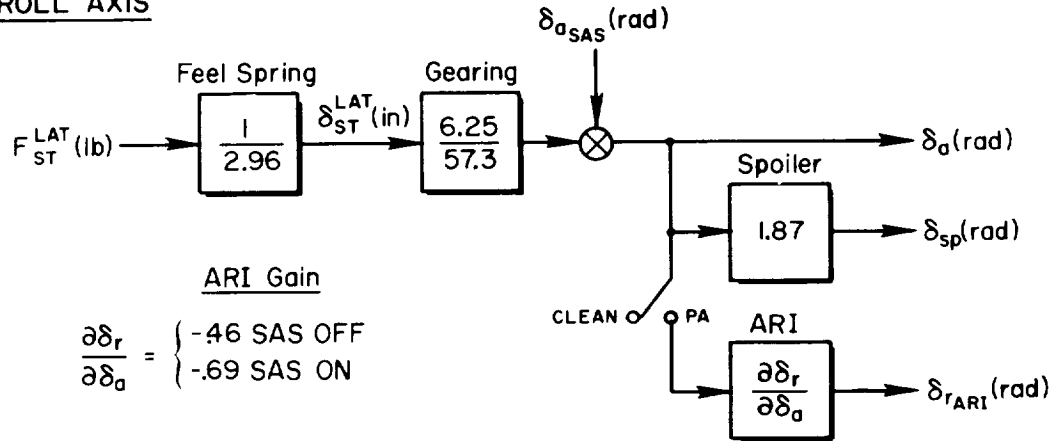
Figure IV-2. F-4C General Arrangement

F-4C

PITCH AXIS



ROLL AXIS



YAW AXIS

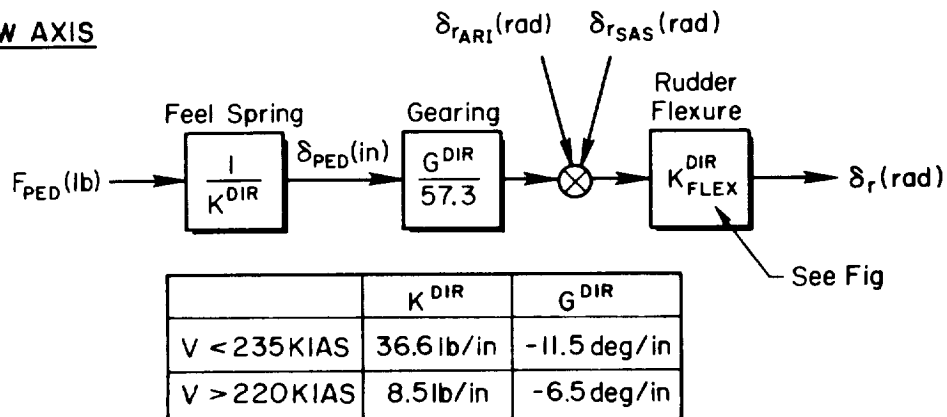


Figure IV-3. F-4C Control System

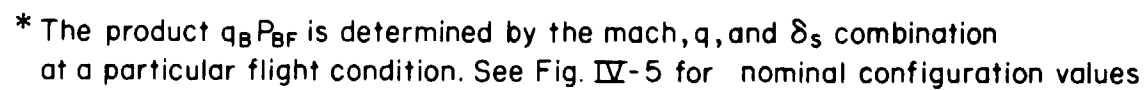


Figure IV-4. F-4C Feel System Schematic

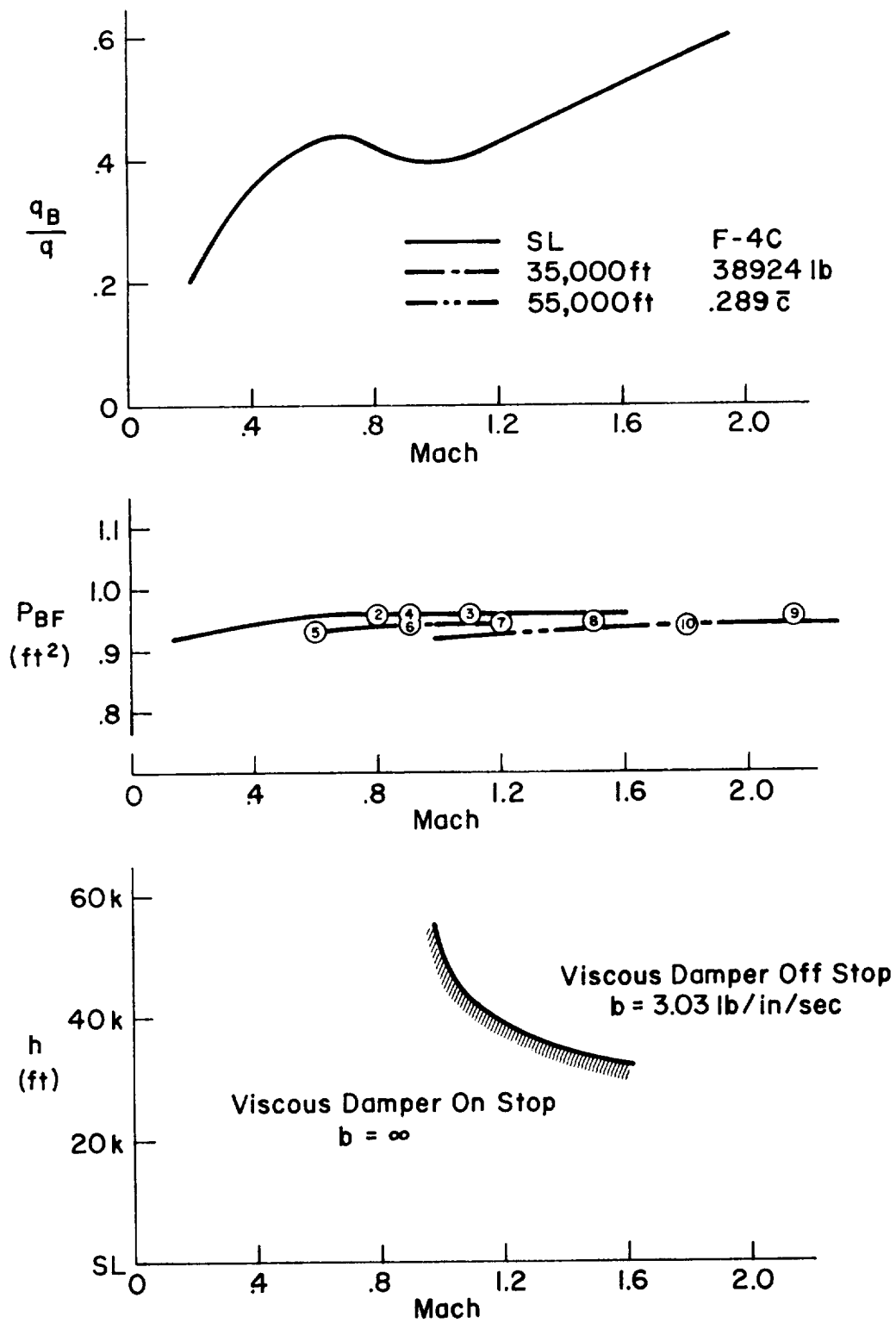


Figure IV-5. F-4C Feel System Parameters

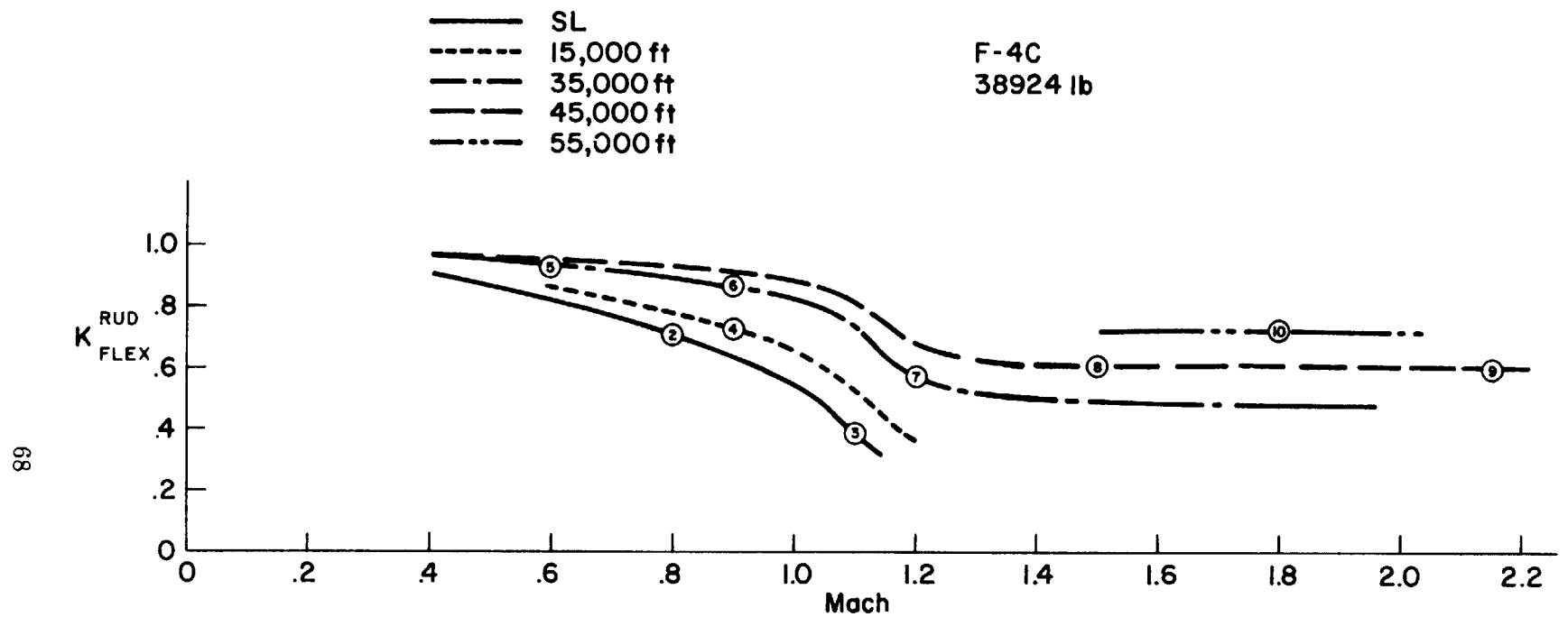
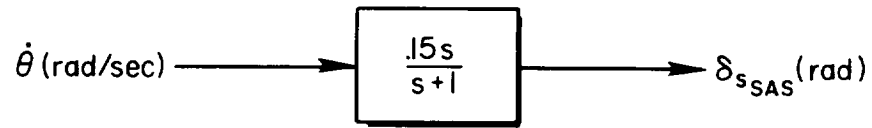


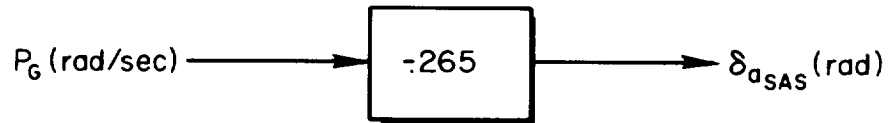
Figure IV-6. F-4C Rudder Flexure Coefficient

F-4C

PITCH SAS



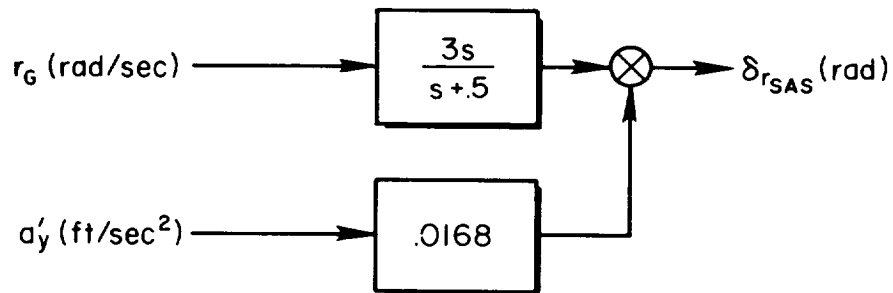
ROLL SAS



$P_G = P$ (Roll rate gyro assumed aligned with FRL)

Note: Roll SAS faded out with lateral control out of neutral

YAW SAS



$$r_G = r \cos(-1.5^\circ) + p \sin(-1.5^\circ)$$

$$a'_y = a_y + 9.9 \dot{r} - .39 \dot{p}$$

Yaw rate gyro inclined 1.5° below FRL and lateral accelerometer at F.S. 198.0 and W.L. 23.0

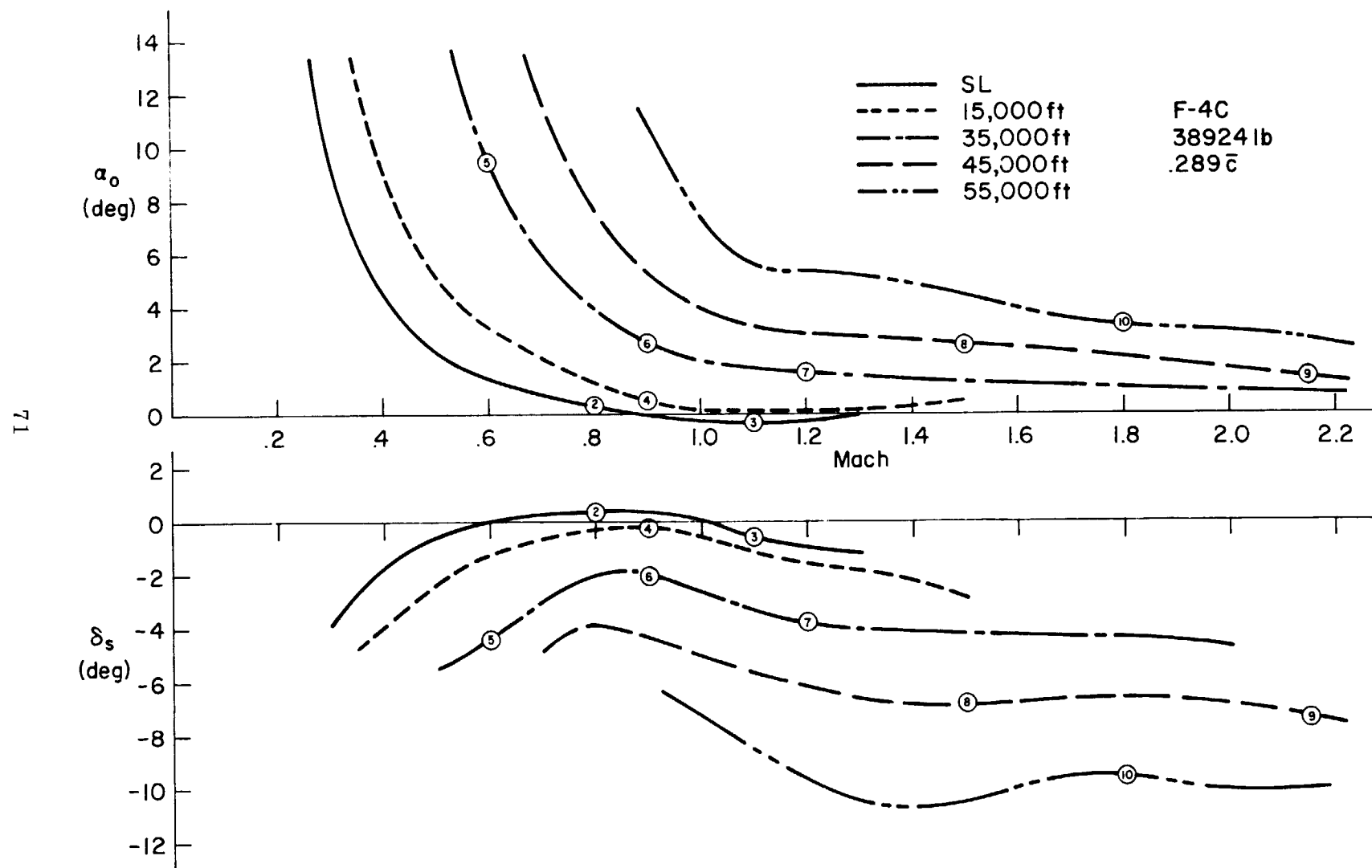
Figure IV-7. F-4C Stability Augmentation

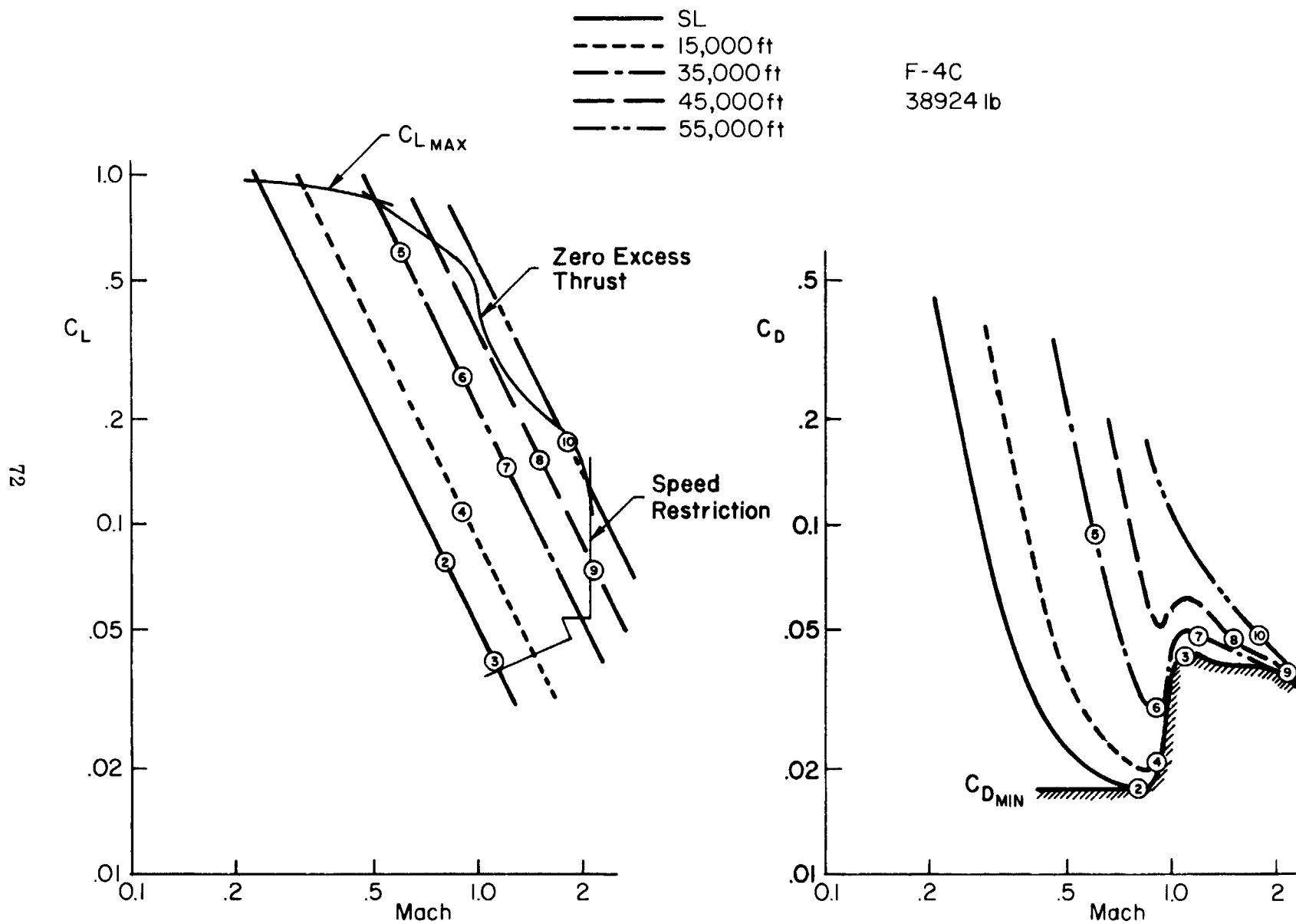
TABLE IV-1

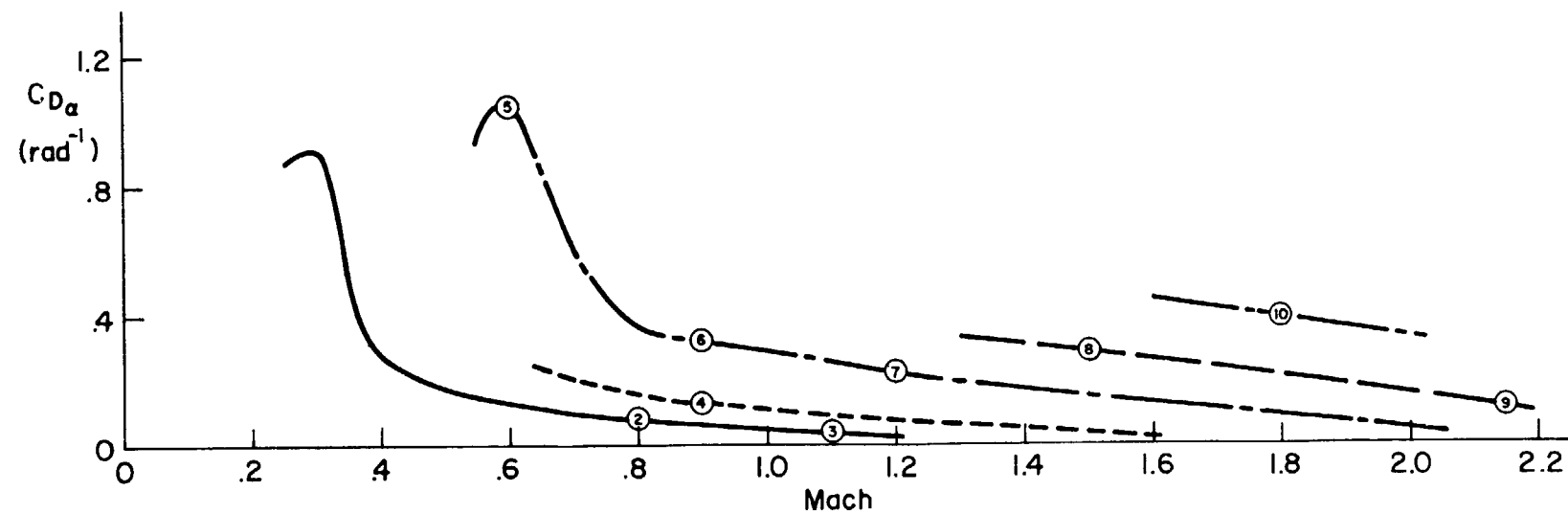
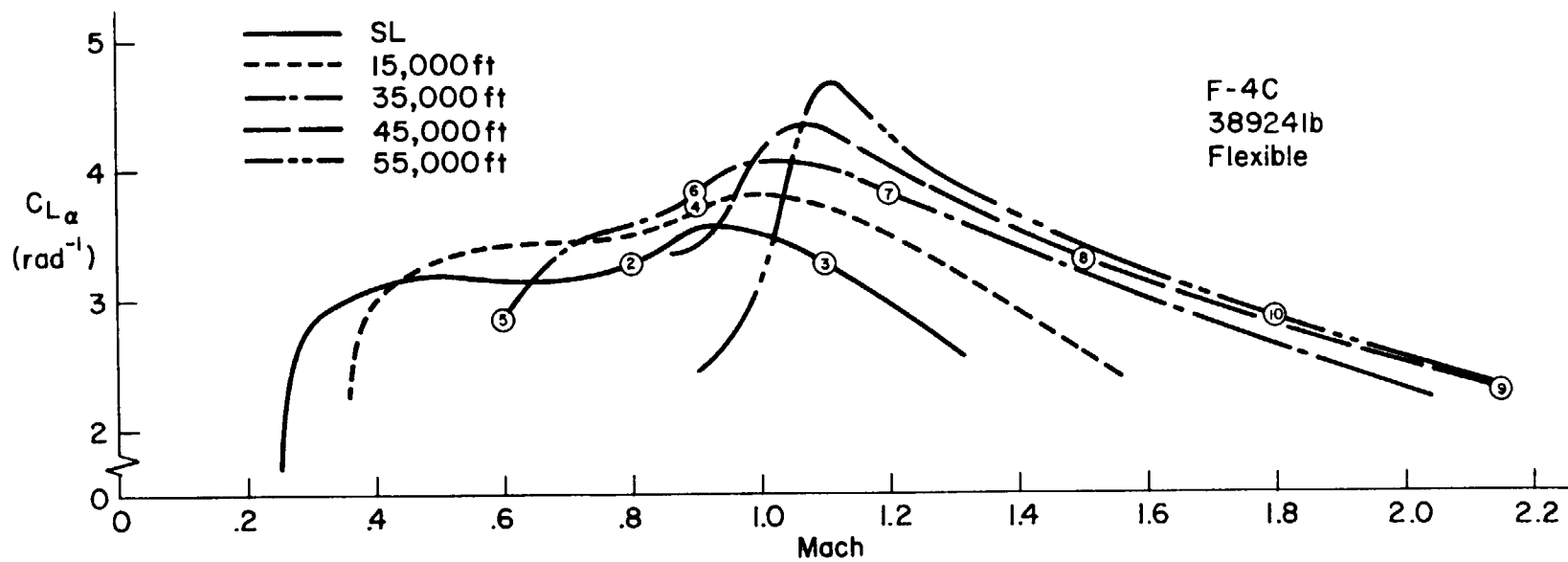
F-4C**Power Approach Non-Dimensional Stability Derivatives**

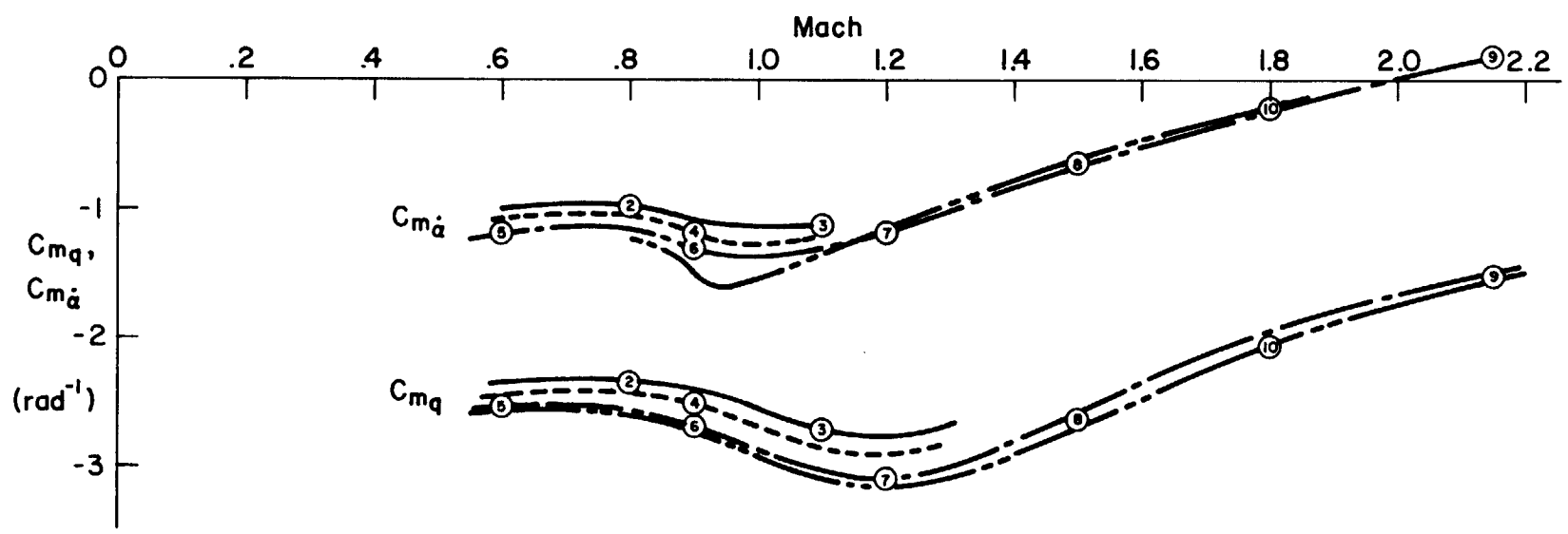
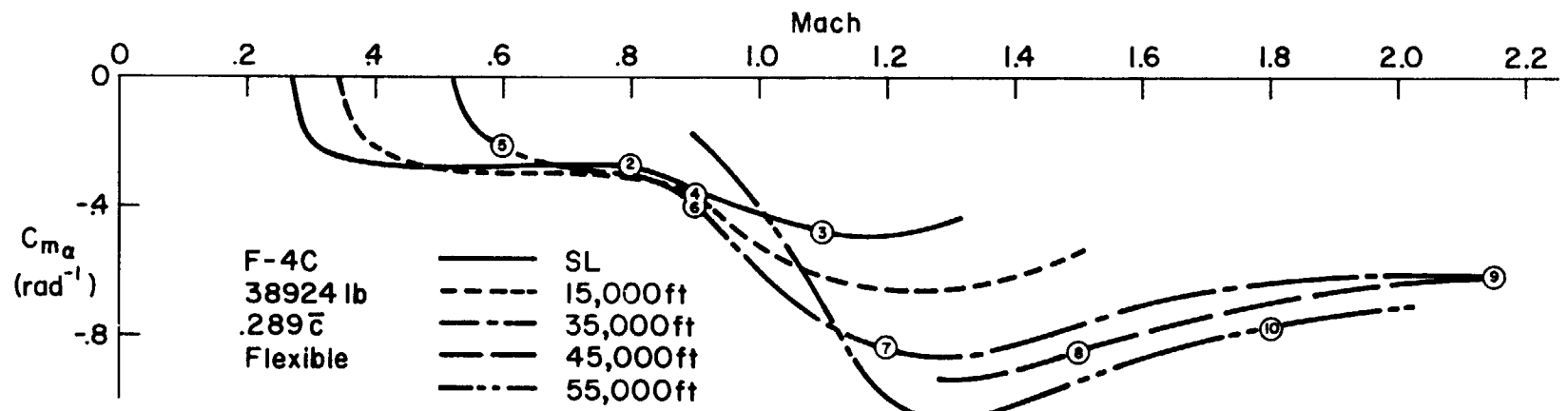
$$\begin{aligned}
 h &= \text{sea level} \\
 V_{T0} &= 230 \text{ ft/sec} = 136 \text{ kt} \\
 \alpha_0 &= 11.7^\circ \\
 \delta_S &= -9.1^\circ
 \end{aligned}$$

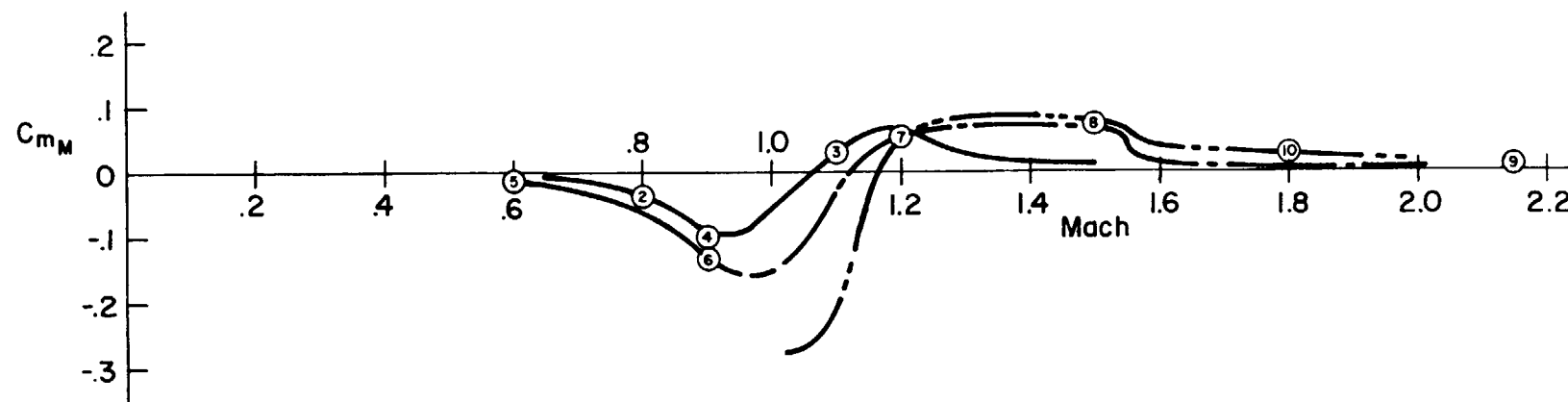
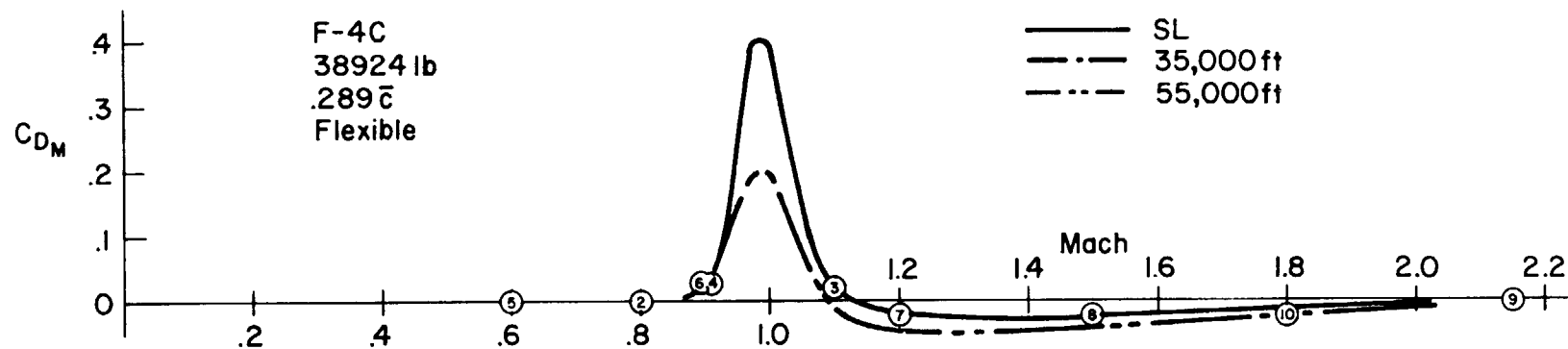
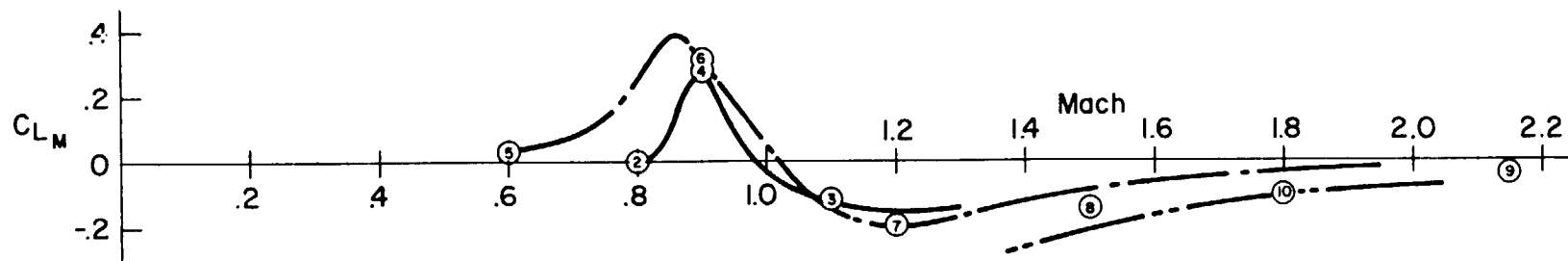
Longitudinal	Lateral-Directional (Stability Axis)	
$C_L = .915$	$C_{Y\beta} = -.655/\text{rad}$	
$C_D = .242$	$C_{n\beta} = .199/\text{rad}$	
$C_{L\alpha} = 2.8/\text{rad}$	$C_{l\beta} = -.156/\text{rad}$	
$C_{D\alpha} = .555/\text{rad}$	$C_{lp} = -.272/\text{rad}$	
$C_{m\alpha} = -.098/\text{rad}$	$C_{np} = -.013/\text{rad}$	
$C_{m\dot{\alpha}} = -.95/\text{rad}$	$C_{lr} = .205/\text{rad}$	
$C_{mq} = -2.0/\text{rad}$	$C_{nr} = -.320/\text{rad}$	
$C_{L\delta_S} = .24/\text{rad}$	$C_{Y\delta_a} = -.0355/\text{rad}$	} Spoiler Effects Included
$C_{m\delta_S} = -.322/\text{rad}$	$C_{n\delta_a} = -.0041/\text{rad}$	
$C_{D\delta_S} = -.14/\text{rad}$	$C_{l\delta_a} = .057/\text{rad}$	
	$C_{Y\delta_r} = .124/\text{rad}$	
	$C_{n\delta_r} = -.072/\text{rad}$	
	$C_{l\delta_r} = -.0009/\text{rad}$	

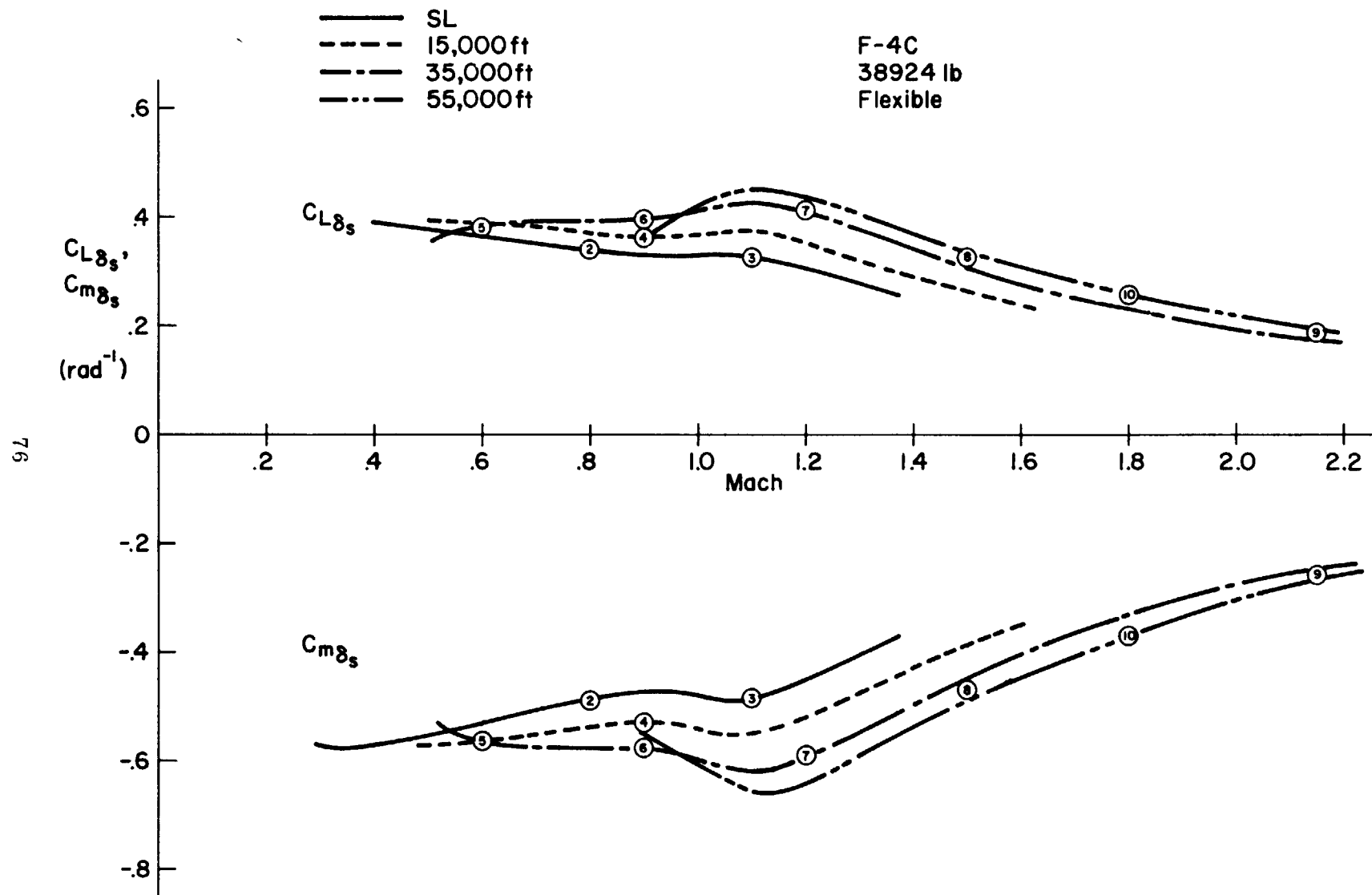


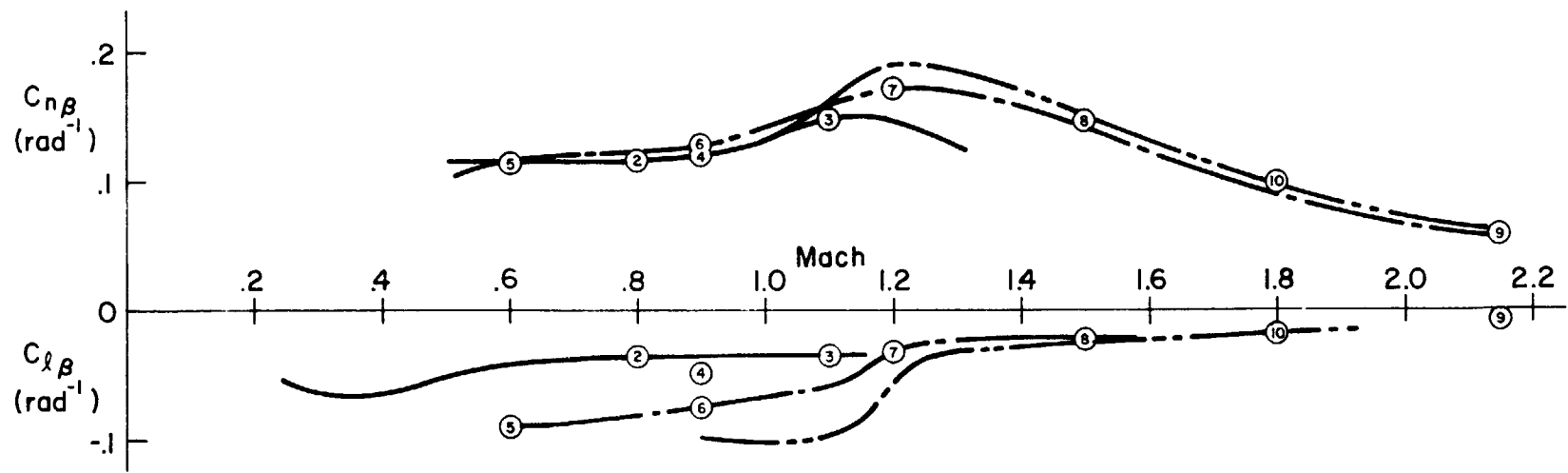
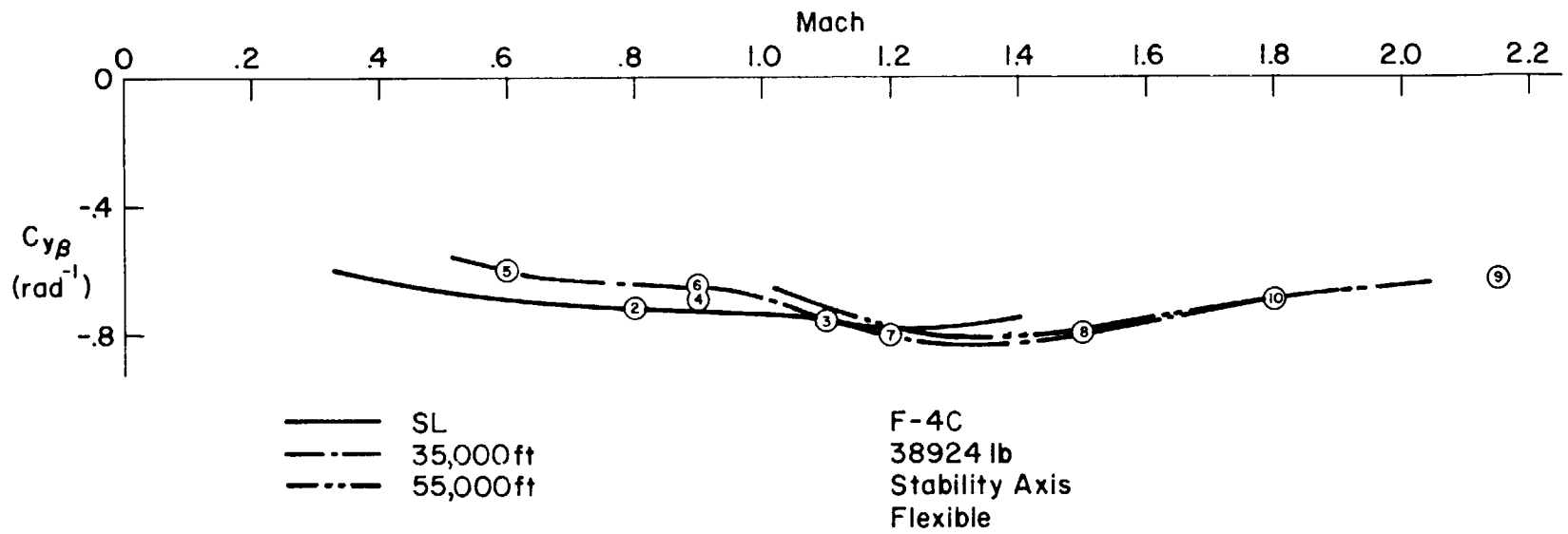


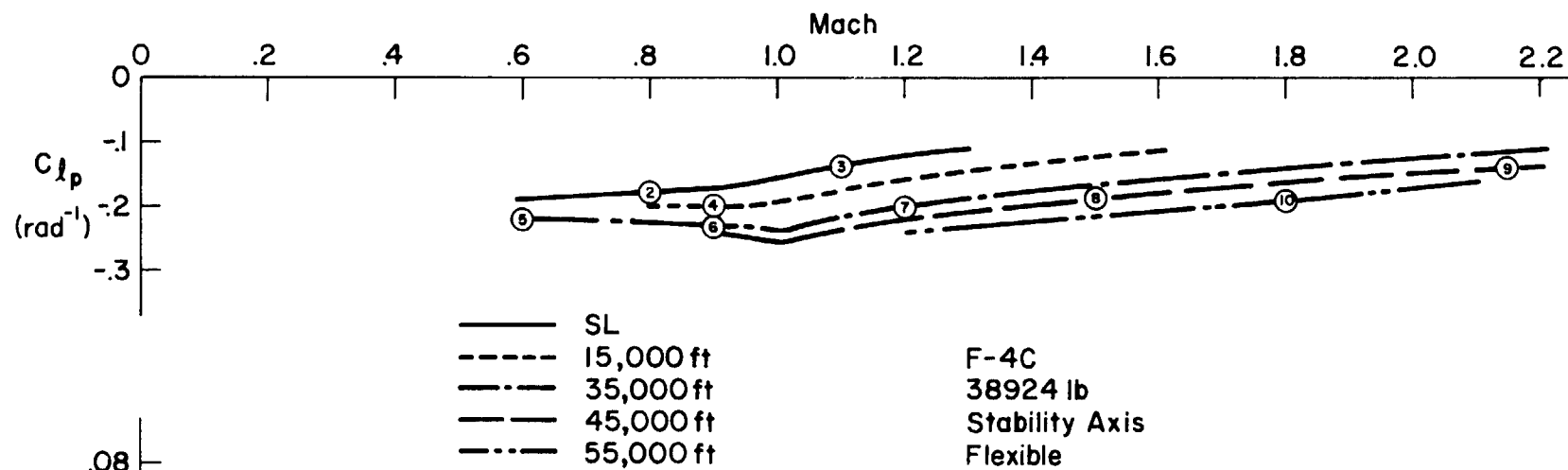




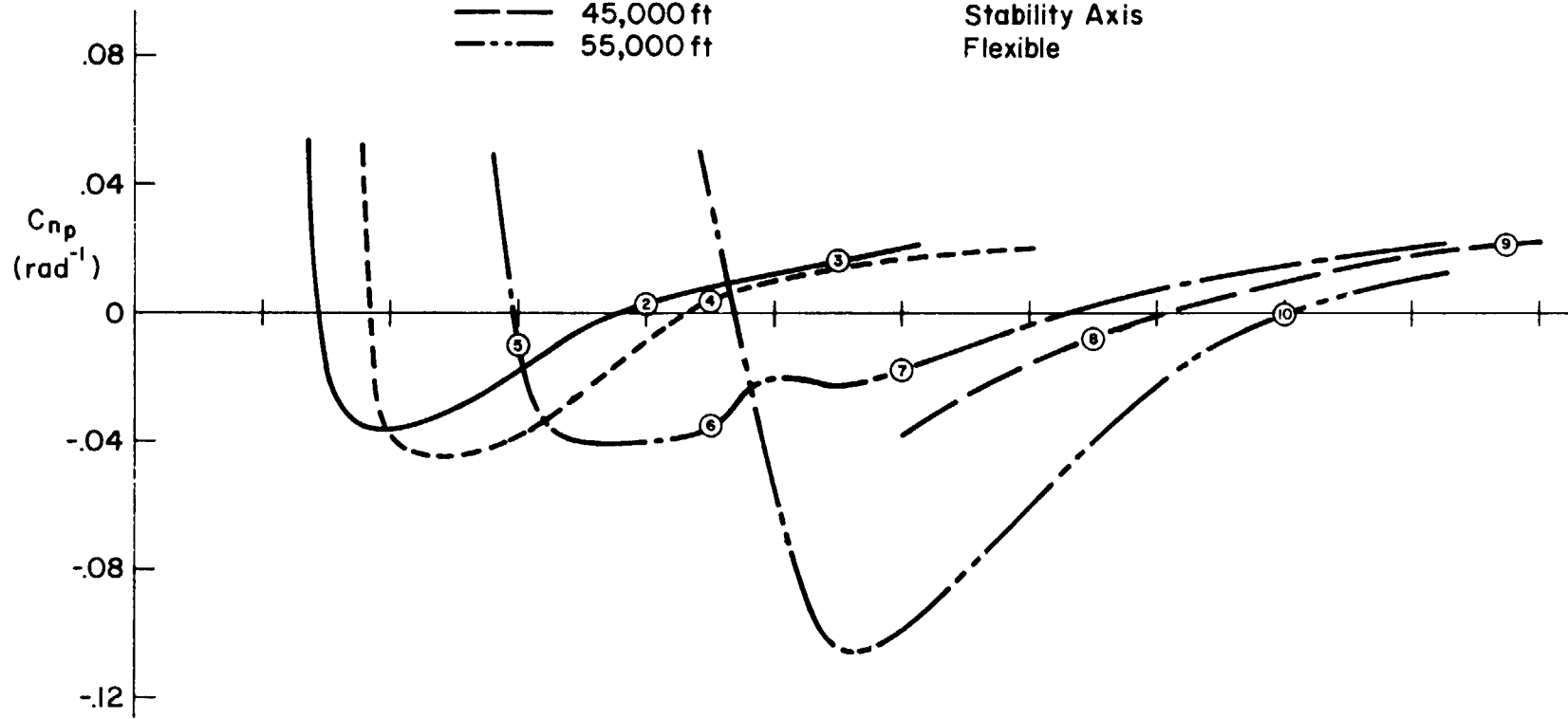


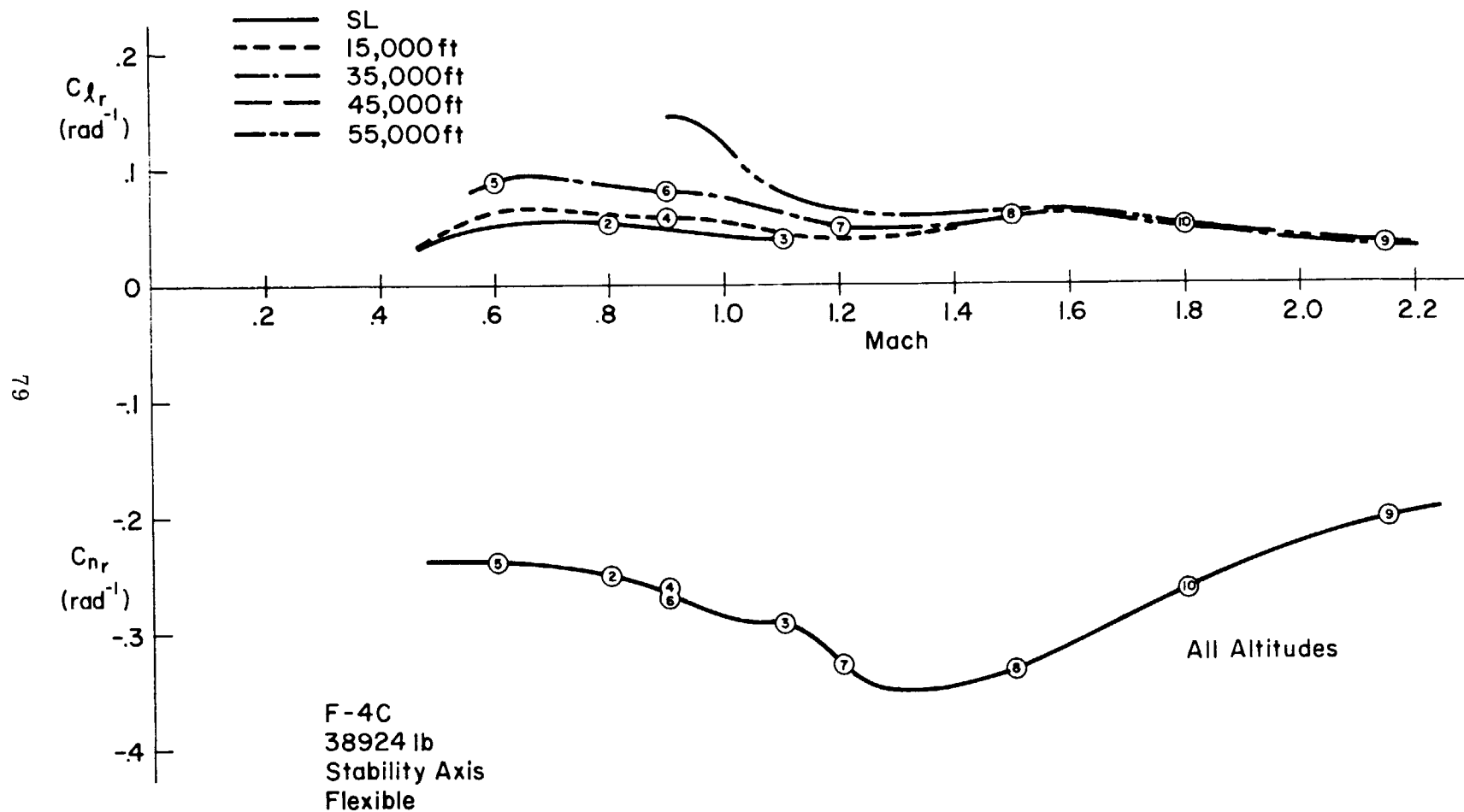


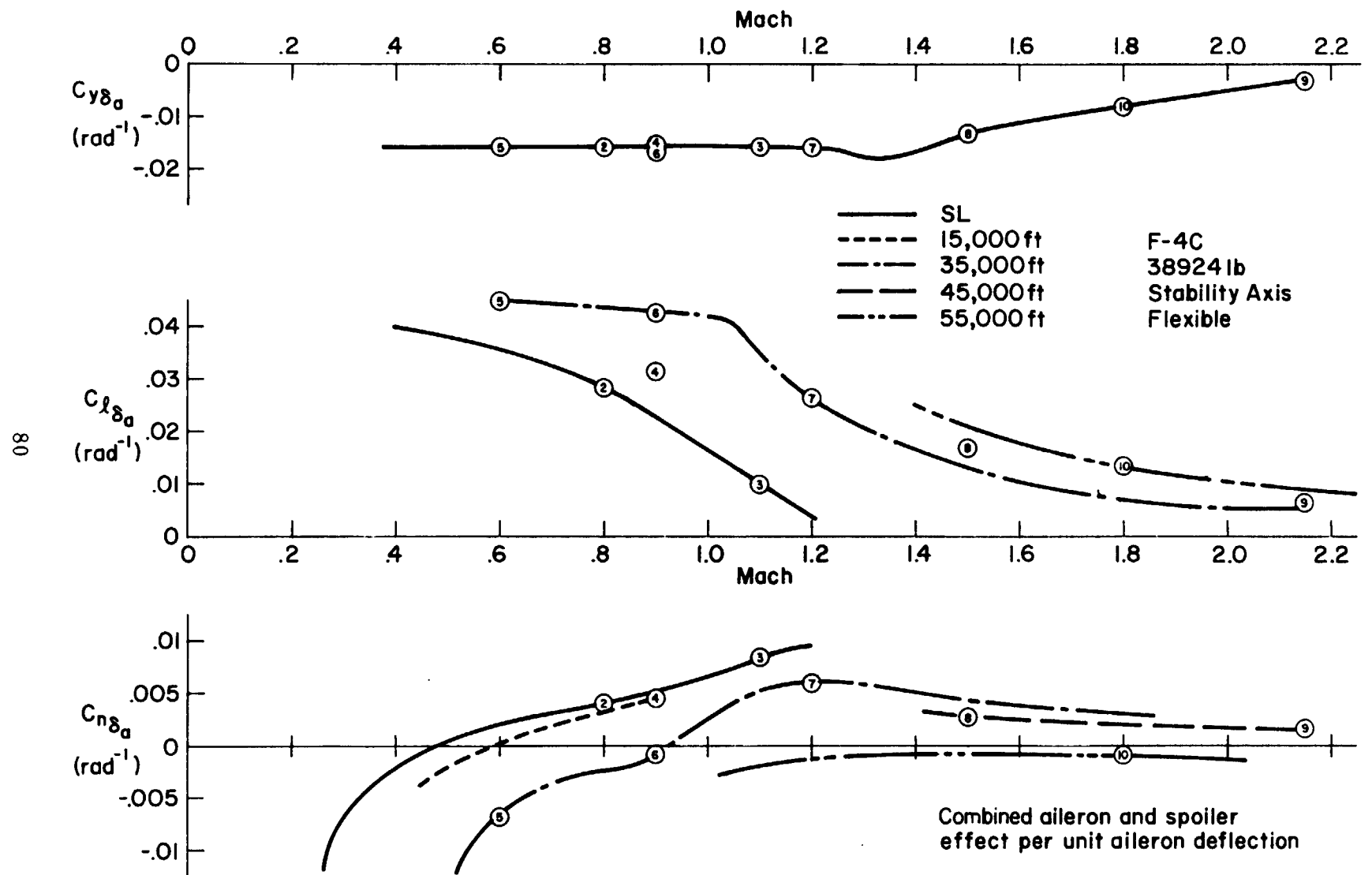




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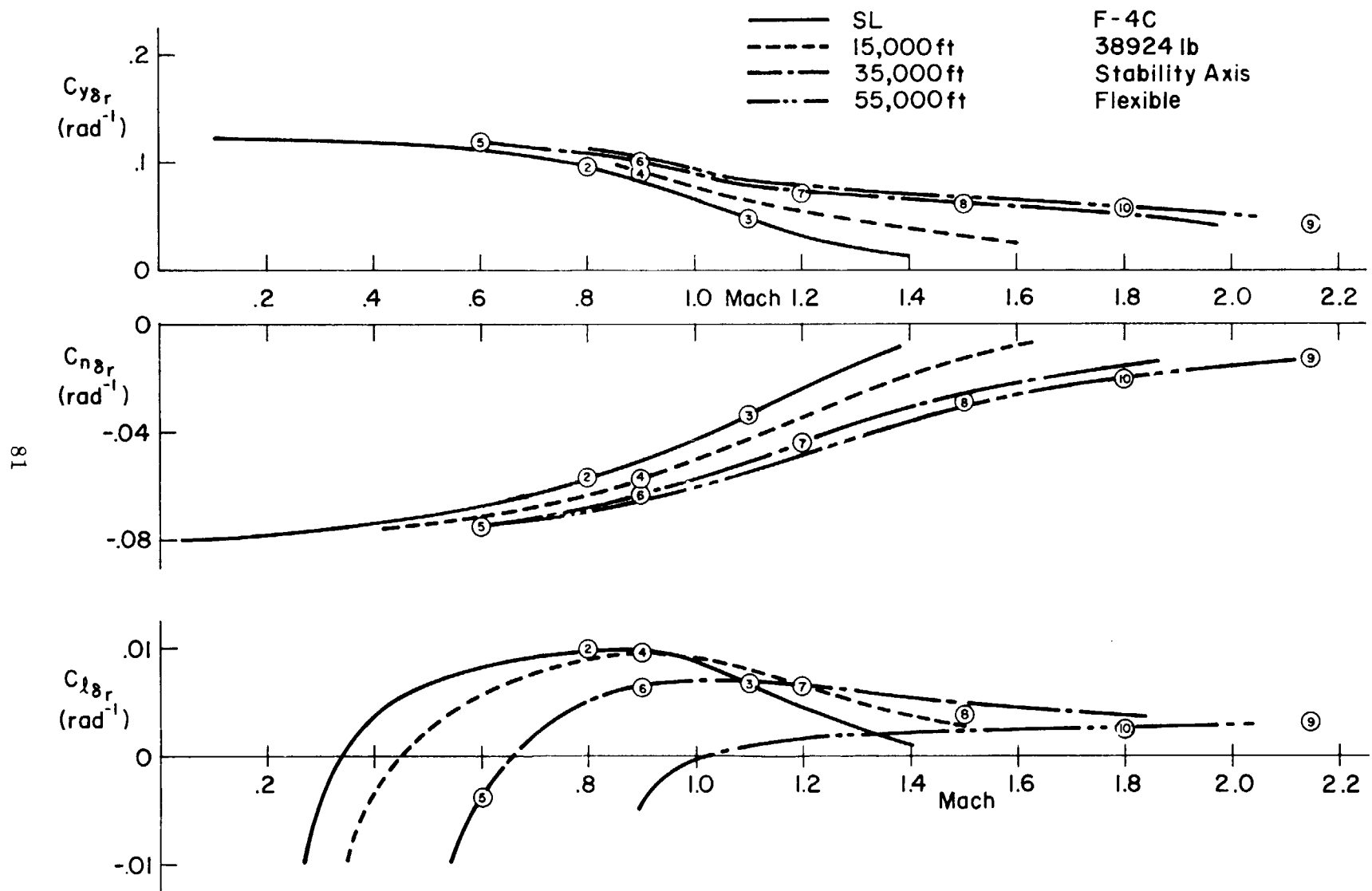


TABLE IV-2

F-4C DIMENSIONAL, MASS, AND FLIGHT CONDITION PARAMETERS

s = 530 sq ft, b = 38.67 ft, $\bar{c}$ = 16.04 ft

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	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H(FT)	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M(-)	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
VTO(FPS)	230.	893.	1228.	952.	584.	876.	1167.	1452.	2081.	1742.
VTO(KTAS)	136.	529.	728.	564.	346.	519.	692.	860.	1233.	1032.
VTO(KCAS)	136.	529.	728.	465.	199.	311.	432.	445.	632.	433.
W(LBS)	33197.	38925.	38925.	38925.	38925.	38925.	38925.	38925.	38925.	38925.
C.G.(PGC)	.291	.289	.289	.299	.289	.289	.289	.289	.289	.289
IX (SLUG-FT SQ)	23669.	25002.	25002.	25002.	25002.	25002.	25002.	25002.	25002.	25002.
IY (SLUG-FT SQ)	117506.	122193.	122193.	122193.	122193.	122193.	122193.	122193.	122193.	122193.
IZ (SLUG-FT SQ)	133730.	139767.	139767.	139767.	139767.	139767.	139767.	139767.	139767.	139767.
IXZ(SLUG-FT SQ)	1575.	2177.	2177.	2177.	2177.	2177.	2177.	2177.	2177.	2177.
EPSILON(DEG)	-1.820	-1.09	-1.09	-1.09	-1.09	-1.09	-1.09	-1.09	-1.09	-1.09
Q(PSF)	62.6	948.	1792.	677.	126.	283.	503.	489.	1034.	436.
QC(PSF)	63.3	1109.	2397.	825.	138.	345.	703.	749.	1687.	706.
ALPHA(DEG)	11.7	.300	-.300	.500	9.40	2.60	1.60	2.60	1.40	3.30
GAMMA(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LXP(FT)	16.3	16.2	16.2	16.2	16.2	16.2	16.2	16.2	16.2	16.2
LZP(FT)	-3.02	-2.81	-2.81	-2.31	-2.81	-2.81	-2.81	-2.81	-2.81	-2.81
ITH(DEG)	5.25	5.25	5.25	5.25	5.25	5.25	5.25	5.25	5.25	5.25
XI(DEG)	5.25	5.25	5.25	5.25	5.25	5.25	5.25	5.25	5.25	5.25
LTH(FT)	-.570	-.370	-.370	-.370	-.370	-.370	-.370	-.370	-.370	-.370
	+	+	+	+	+	+	+	+	+	+

TABLE IV-3

F-4C LONGITUDINAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
XU *	-.0417	-.0159	-.0677	-.0203	.000719	-.00796	-.0135	-.00679	-.0158	-.00528
ZU *	-.177	-.0645	.0226	-.134	-.0689	-.0876	.0105	.0110	-.000992	.000474
MU *	.000743	-.00161	.00329	-.00425	.000511	-.00239	.00292	.00341	.00128	.00175
XW	.130	.00706	-.0107	.00371	.00458	.0158	.00576	.00146	.00387	-.00501
ZW	-.452	-1.54	-2.11	-1.16	-.296	-.547	-.727	-.494	-.694	-.319
MW	-.00182	-.0199	-.0488	-.0175	-.00326	-.00911	-.0248	-.0198	-.0205	-.0133
ZWD	-.00305	-.00271	-.00326	-.00210	-.00104	-.00116	-.00106	-.000358	.977E-4	-.604E-4
ZQ	-2.48	-8.20	-8.72	-6.00	-1.84	-2.89	-4.09	-2.24	-1.27	-1.14
MWD	-.000642	-.000663	-.000729	-.000480	-.000244	-.000267	-.000247	-.840E-4	.259E-4	-.150E-4
MC	-.317	-1.36	-2.20	-.993	-.307	-.487	-.745	-.488	-.404	-.286
XDS	5.98	.739	-1.32	.952	3.42	2.25	2.52	3.21	2.04	2.86
ZDS	-6.65	-141.	-251.	-107.	-20.7	-49.6	-90.4	-70.6	-83.6	-49.6
MDS	-1.46	-32.3	-61.1	-25.0	-4.90	-11.4	-20.7	-16.0	-16.1	-11.2
XDTH	.000965	.000823	.000823	.000823	.000823	.000823	.000923	.000823	.000823	.000823
ZDTH	-.887E-4	-.756E-4	-.756E-4	-.756E-4	-.756E-4	-.756E-4	-.756E-4	-.756E-4	-.756E-4	-.756E-4
MDTH	-.485E-5	-.303E-5	-.303E-5	-.303E-5	-.303E-5	-.303E-5	-.303E-5	-.303E-5	-.303E-5	-.303E-5

TABLE IV-4

F-4C STABILIZER TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
DENOMINATOR										
Z(DET)1	.102	(-.0278)	.639	(-.0612)	.0928	(.0446)	.191	.156	.384	.175
W(DET)1	.191	(.0516)	.0542	(.0741)	.0774	(-.0456)	.0450	.0402	.0220	.0274
Z(DET)2	.607	.393	.324	.308	.259	.224	.162	.102	.0645	.0650
W(DET)2	.757	4.44	7.99	4.24	1.41	2.85	5.43	5.39	6.54	4.84
NUMERATORS										
N(L /DS)										
A(U)	5.97	.737	-1.31	.930	3.42	2.25	2.52	3.20	2.04	2.86
1/T(U)1	11.4	1.94	1.49	1.25	136.	201.	266.	.310	400.	.143
1/T(U)2	(.452)	5.35	-6.31	3.44	(.980)	(.787)	(.965)	.641	(.978)	.584
1/T(U)3	(.561)	197.	304.	218.	(.307)	(.643)	(.783)	328.	(.496)	394.
N(W /DS)										
A(W)	-6.62	-141.	-250.	-107.	-20.6	-49.5	-90.3	-70.6	-83.6	-49.6
1/T(W)1	49.3	204.	-.00320	222.	137.	202.	267.	328.	400.	394.
Z(W)1	.151	.176	(.0711)	.169	.0121	.0964	.852	.290	.731	.184
W(W)1	.156	.0456	(.299.)	.0627	.0627	.0532	.00729	.00911	.0106	.0114
N(TH/DS)										
A(TH)	-1.45	-32.2	-60.9	-24.9	-4.90	-11.4	-20.6	-16.0	-16.1	-11.2
1/T(TH)1	.104	.0162	.0678	.0208	-.000498	.0106	.0131	.00608	.0157	.00460
1/T(TH)2	.379	1.46	1.90	1.08	.282	.505	.618	.407	.388	.260
N(HD /DS)										
A(HD)	7.70	141.	250.	107.	20.9	49.6	90.3	70.6	83.6	49.7
1/T(HD)1	.00726	.0146	.0680	.0169	-.0245	.00335	.0123	.00489	.0151	.00307
1/T(HD)2	-4.21	17.0	-23.8	15.3	5.96	9.99	12.7	11.5	-12.4	-10.1
1/T(HD)3	4.27	-17.5	23.9	-15.7	-6.05	-10.2	-12.9	-11.5	12.5	10.1
N(AZP/DS)										
A(AZF)	17.0	382.	737.	298.	58.7	135.	244.	188.	177.	132.
1/T(AZP)1	-.0514	-.000207	.000137	-.000356	-.000194	-.00287	-.000776	-.00104	-.000385	-.00117
1/T(AZP)2	.0543	.0148	.0679	.0172	-.0243	.00618	.0131	.00590	.0154	.00419
Z(AZP)1	.121	.104	.0917	.0876	.0620	.0625	.0586	.0400	.0294	.0280
W(AZP)1	2.80	10.5	13.9	9.30	3.61	6.09	7.81	7.07	8.56	6.19

TABLE IV-5

F-4C THRUST TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
DENOMINATOR										
Z(DET)1	.102	(-.0378)	.639	(-.0612)	.0928	(.0446)	.191	.156	.384	.175
W(DET)1	.191	(.0516)	.0542	(.0741)	.0774	(-.0456)	.0450	.0402	.0220	.0274
Z(DET)2	.607	.393	.324	.308	.259	.224	.162	.102	.0645	.0650
W(DET)2	.757	4.44	7.99	4.24	1.41	2.85	5.43	5.39	6.54	4.84
NUMERATORS										
N(U /DTH)										
A(U)	.000965	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823
1/T(U)1	.109	.00607	.00176	.00439	.00376	.00295	-.000327	-.000995	-.000421	-.00113
Z(U)1	.694	.393	.322	.309	.371	.244	.173	.125	.0788	.104
W(U)1	.732	4.45	8.00	4.26	1.46	2.88	5.43	5.38	6.54	4.84
N(W /DTH)										
A(W)	-.878 E-4	-.754 E-4	-.754 E-4	-.755 E-4	-.754 E-4	-.755 E-4	-.755 E-4	-.756 E-4	-.756 E-4	-.756 E-4
1/T(W)1	14.3	.00167	-.00175	.000503	24.0	-.000110	-.00300	-.00281	-.00193	-.00279
1/T(W)2	(-.123)	.458	-.788	1.18	(-.807)	.658	-.756	-.897	-.327	-.441
1/T(W)3	(.161)	37.0	51.7	39.1	(.0746)	35.6	47.9	59.2	84.0	70.3
N(TH/DTH)										
A(TH)	-.467 E-5	-.298 E-5	-.298 E-5	-.299 E-5	-.298 E-5	-.300 E-5	-.300 E-5	-.302 E-5	-.303 E-5	-.303 E-5
1/T(TH)1	-.192	.283	-1.21	.517	-.289	.178	.450	.321	.282	.215
1/T(TH)2	.480	1.23	1.28	1.38	.358	.800	-1.13	-1.24	-.630	-.700
N(HD /DTH)										
A(HD)	.000282	.797 E-4	.711 E-4	.826 E-4	.000209	.000113	.985 E-4	.000113	.957 E-4	.000123
1/T(HD)1	1.39	.341	-1.86	1.05	1.45	.681	2.53	1.43	1.08	.655
Z(HD)1	-.552	(-4.91)	(-4.80)	(-4.11)	-.367	(-1.86)	-.293	-.122	-.0978	-.0393
W(HD)1	.516	(6.88)	(8.90)	(5.67)	.793	(2.62)	2.68	3.31	3.29	3.15
N(AZP/DTH)										
A(AZP)	-.120 E-4	-.272 E-4	-.272 E-4	-.270 E-4	-.272 E-4	-.269 E-4	-.268 E-4	-.267 E-4	-.265 E-4	-.266 E-4
1/T(AZP)1	-.0214	-.000139	.000127	-.000256	-.000872	-.00168	-.000768	-.00100	-.000377	-.00106
1/T(AZP)2	16.5	.332	-2.06	.996	-.805	.451	6.65	5.07	3.61	3.30
Z(AZP)1	-.984	(-8.60)	(-6.63)	(-7.34)	(-1.59)	(-4.60)	-.536	-.316	-.294	-.275
W(AZP)1	.815	(11.8)	(15.3)	(10.2)	(5.35)	(6.67)	3.16	3.60	3.43	3.01

TABLE IV-6

F-4C STICK FORCE TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Closed

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
DE NCMINATOR										
1/T(DET)1	20.8	-.0271	26.6	-.0445	22.0	-.0335	24.5	1.49	3.10	1.35
1/T(DET)2		.0412		.0585		.0343		23.2	21.1	22.2
Z(DET)1	.146	(25.3)	.760	(25.0)	-.0455	(23.5)	.205	.159	.431	.169
W(DET)1	.0881		.0454		.0540		.0389	.0291	.0192	.0218
Z(DET)2	.271	.277	.263	.215	.166	.155	.148	.138	.102	.101
W(DET)2	1.15	4.91	7.90	4.60	1.67	3.09	5.12	5.07	6.36	4.63
Z(DET)3	.427	.0192	.000438	.0253	.167	.0751	.0302	.109	.174	.137
W(DET)3	6.01	25.1	38.9	24.6	11.4	16.5	22.1	22.4	34.5	22.5
NUMERATORS										
N(U /FST)										
A(U)	-190.	-23.4	41.7	-29.6	-109.	-71.5	-80.2	-102.	-64.9	-90.9
1/T(U)1	11.4	1.95	1.49	1.25	136.	201.	266.	.310	11.1	.143
1/T(U)2	(.452)	5.35	-6.31	3.44	(.980)	(.787)	(.965)	.641	400.	.584
1/T(U)3	(.561)	197.	304.	218.	(.307)	(.643)	(.783)	4.22	(.278)	4.42
1/T(U)4								328.	(.406)	394.
N(W /FST)										
A(W)	210.	4476.	7961.	3388.	656.	1573.	2869.	2242.	2656.	1575.
1/T(W)1	49.3	204.	-.00320	222.	137.	202.	267.	4.22	11.1	4.42
1/T(W)2			.0711					328.	400.	394.
Z(W)1	.151	.176	(299.)	.165	.0121	.0964	.852	.290	.731	.184
W(W)1	.156	.0456		.0627	.0627	.0532	.00729	.00911	.0106	.0114
N(TH /FST)										
A(TH)	46.2	1024.	1936.	792.	156.	363.	656.	508.	511.	357.
1/T(TH)1	.104	.0162	.0678	.0208	-.000498	.0106	.0131	.00608	.0157	.00460
1/T(TH)2	.379	1.46	1.90	1.08	.282	.505	.618	.407	.388	.260
1/T(TH)3								4.22	11.1	4.42
N(HD /FST)										
A(HD)	-245.	-4476.	-7961.	-2388.	-665.	-1575.	-2870.	-2245.	-2657.	-1579.
1/T(HD)1	.00726	.0146	.0680	.0165	-.0245	.00335	.0123	.00489	.0151	.00307
1/T(HD)2	-.21	17.0	-23.8	15.3	5.96	9.99	12.7	4.22	11.1	4.42
1/T(HD)3	4.27	-17.5	23.9	-15.7	-6.05	-10.2	-12.9	11.5	-12.4	-10.1
1/T(HD)4								-11.5	12.5	10.1

TABLE IV-6 (Concluded)

N(AZP/FST)										
A(AZP)	-540.	-12129.	-23430.	-9456.	-1867.	-4306.	-7765.	-5989.	-5624.	-4209.
1/T(AZP)1	-.0514	-.000207	.000137	-.000356	-.000194	-.00287	-.000776	-.00104	-.000385	-.00117
1/T(AZP)2	.0543	.0143	.0679	.0172	-.0243	.00618	.0131	.00590	.0154	.00419
1/T(AZP)3								4.22	11.1	4.42
Z(AZP)1	.121	.104	.0917	.0876	.0620	.0625	.0586	.0400	.0294	.0280
h(AZP)1	2.80	10.5	13.9	9.30	3.61	6.09	7.81	7.07	8.56	6.19
	+	+	+	+	+	+	+	+	+	+

TABLE IV-7

F-4C THRUST TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Closed

(BODY AXIS SYSTEM)

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	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.50
DE NCMINATOR										
1/T(DET)1	20.8	-.0271	26.6	-.0445	22.0	-.0335	24.5	1.49	3.10	1.35
1/T(DET)2		.0412		.0585		.0343		23.2	21.1	22.2
Z(DET)1	.146	(25.3)	.760	(25.0)	-.0455	(23.5)	.205	.159	.431	.169
W(DET)1	.0881		.0454		.0540		.0389	.0291	.0192	.0218
Z(DET)2	.271	.277	.263	.219	.166	.155	.148	.138	.102	.101
W(DET)2	1.15	4.91	7.90	4.60	1.67	3.09	5.12	5.07	6.36	4.63
Z(DET)3	.427	.0192	.000438	.0253	.167	.0751	.0302	.109	.174	.137
W(DET)3	6.01	25.1	38.9	24.6	11.4	16.5	22.1	22.4	34.5	22.5
NUMERATORS										
N(U /DTH)										
A(U)	.000965	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823
1/T(U)1	.00561	.00339	.00128	.00241	-.00271	.000946	-.000438	-.000299	-.000410	-.00110
1/T(U)2	20.8	25.3	26.6	25.0	22.0	23.5	24.5	1.49	3.10	1.36
1/T(U)3								23.2	21.1	22.2
Z(U)1	.318	.277	.262	.220	.247	.172	.158	.159	.117	.138
W(U)1	1.19	4.92	7.91	4.62	1.71	3.11	5.12	5.05	6.36	4.62
Z(U)2	.432	.0192	.000420	.0254	.169	.0755	.0305	.109	.174	.137
W(U)2	6.02	25.1	38.9	24.6	11.4	16.5	22.1	22.5	34.5	22.5
N(W /DTH)										
A(W)	-.878E-4	-.754E-4	-.754E-4	-.755E-4	-.754E-4	-.755E-4	-.755E-4	-.756E-4	-.756E-4	-.756E-4
1/T(W)1	.0248	.00116	-.00173	.000299	18.4	-.000597	-.00304	-.00293	-.00215	-.00306
1/T(W)2	.396	.518	-.651	1.26	26.3	.744	-.587	-.537	-.222	-.277
1/T(W)3	12.7	20.7	21.9	20.9	(.983)	20.4	21.4	1.21	2.92	1.21
1/T(W)4	21.3	39.3	53.4	40.9	(.0514)	36.8	48.8	21.3	20.5	21.0
1/T(W)5								59.6	84.2	70.5
Z(W)1	.520	.0458	.0242	.0557	.214	.121	.0722	.149	.188	.167
W(W)1	5.25	29.0	38.7	24.3	10.9	16.0	21.5	22.2	34.6	22.4
N(THE /DTH)										
A(THE)	-.467E-5	-.298E-5	-.298E-5	-.299E-5	-.298E-5	-.300E-5	-.300E-5	-.302E-5	-.303E-5	-.303E-5
1/T(THE)1	-.0923	.250	-1.10	.391	-.185	.127	.420	.273	.266	.199
1/T(THE)2	.901	1.11	1.13	1.43	.444	.865	-.949	-.887	-.516	-.516
1/T(THE)3	20.3	23.2	23.8	23.1	21.2	22.2	22.8	1.28	2.95	1.24
1/T(THE)4								21.9	20.6	21.3
Z(THE)1	.478	.0457	.0246	.0556	.197	.110	.0640	.146	.189	.165
W(THE)1	5.04	28.4	38.0	23.8	10.7	15.7	21.1	21.9	34.5	22.2

TABLE IV-7 (Concluded)

N(HD /DTH)										
A(HD)	.000282	.797E-4	.711E-4	.826E-4	.000209	.000113	.985E-4	.000113	.957E-4	.000123
1/T(HD)1	1.12	.349	-1.84	1.10	.734	-1.33	1.94	.527	.785	.363
1/T(HD)2	20.8	-4.32	-4.26	-3.53	22.0	23.4	24.4	2.15	3.39	1.57
1/T(HD)3		5.45	7.06	4.38				23.1	21.1	22.2
Z(HD)1	-.0630	(25.1)	(26.5)	(24.8)	-.0241	.941	-.245	-.0579	-.0494	.00928
W(HD)1	.484				.906	1.30	2.53	3.11	3.13	3.00
Z(HD)2	.435	.0241	.00513	.0308	.169	.0800	.0351	.114	.177	.141
W(HD)2	5.91	29.0	38.7	24.4	11.3	16.3	21.9	22.4	34.5	22.5
N(AZP/DTH)										
A(AZF)	-.120E-4	-.272E-4	-.272E-4	-.270E-4	-.272E-4	-.269E-4	-.268E-4	-.267E-4	-.265E-4	-.266E-4
1/T(AZP)1	-.0214	-.000189	.000137	-.000256	-.00872	-.00168	-.000758	-.00100	-.000377	-.00106
1/T(AZP)2	7.89	.331	-2.13	.991	3.78	.448	4.63	.867	1.94	.895
1/T(AZP)3	24.9	-7.43	-5.54	-6.19	23.1	-3.74	27.0	4.83	5.12	3.53
1/T(AZP)4		8.15	9.66	7.01		4.62		25.1	21.9	23.5
Z(AZP)1	-.847	(28.4)	(31.7)	(27.7)	-.969	(25.2)	-.532	-.281	-.258	-.235
W(AZF)1	.709				1.02		2.81	3.06	3.01	2.60
Z(AZF)2	.622	-.00375	-.0192	.000280	.151	.0481	.00588	.0768	.157	.108
W(AZP)2	7.20	29.9	39.9	25.4	12.1	17.2	23.2	23.3	34.6	23.0
	+	+	+	+	+	+	+	+	+	+

TABLE IV-8

F-4C STABILIZER TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Open

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
DENOMINATOR										
1/T(DET)1	1.24	-.0375	.858	-.0607	1.44	.0447	1.05	.883	1.04	.924
1/T(DET)2	19.8	.0520	6.40	.0746	19.2	-.0454	16.1	1.06	2.34	1.06
1/T(DET)3		.851		.974		1.18		17.2	17.4	18.1
Z(DET)1	.0868	(4.44)	.632	(14.2)	.0890	(17.9)	.189	.155	.384	.175
W(DET)1	.189		.0542		.0775		.0450	.0402	.0220	.0274
Z(DET)2	.672	.940	.620	.824	.464	.568	.477	.339	.250	.247
W(DET)2	.690	10.2	15.3	5.10	1.20	2.77	5.90	5.64	6.88	4.93
Z(DET)3	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
W(DET)3	4.83	27.4	37.1	22.6	9.82	14.5	20.0	21.4	34.5	21.9
NUMERATORS										
N(U /DS)										
A(U)	5.97	.737	-1.31	.930	3.42	2.25	2.52	3.20	2.04	2.86
1/T(U)1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.310	1.00	.143
1/T(U)2	11.4	1.94	1.49	1.25	20.0	20.0	20.0	.641	2.34	.584
1/T(U)3	20.0	5.35	-6.31	3.44	136.	201.	266.	.883	20.0	.924
1/T(U)4	(.452)	20.0	20.0	20.0	(.980)	(.787)	(.965)	1.00	400.	1.00
1/T(U)5	(.561)	197.	304.	218.	(.307)	(.643)	(.783)	20.0	(.978)	20.0
1/T(U)6								328.	(.496)	394.
Z(U)1	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
W(U)1	4.83	27.4	37.1	22.6	9.82	14.5	20.0	21.4	34.5	21.9
N(W /DS)										
A(W)	-6.62	-141.	-250.	-107.	-20.6	-49.5	-90.3	-70.6	-83.6	-49.6
1/T(W)1	1.00	1.00	-.00320	1.00	1.00	1.00	1.00	.883	1.00	.924
1/T(W)2	20.0	20.0	.0711	20.0	20.0	20.0	20.0	1.00	2.34	1.00
1/T(W)3	49.3	204.	1.00	222.	137.	202.	267.	20.0	20.0	20.0
1/T(W)4			20.0					328.	400.	394.
Z(W)1	.151	.176	(299.)	.165	.0121	.0964	.852	.290	.731	.184
W(W)1	.156	.0456		.0627	.0627	.0532	.00729	.00911	.0106	.0114
Z(W)2	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
W(W)2	4.83	27.4	37.1	22.6	9.82	14.5	20.0	21.4	34.5	21.9

TABLE IV-8 (Concluded)

N(THE/DS)										
A(THE)	-1.45	-32.2	-60.9	-24.5	-4.90	-11.4	-20.6	-16.0	-16.1	-11.2
1/T(THE)1	.104	.0162	.0678	.0208	-.000498	.0106	.0131	.00608	.0157	.00460
1/T(THE)2	.379	1.00	1.00	1.00	.282	.505	.618	.407	.388	.260
1/T(THE)3	1.00	1.46	1.90	1.08	1.00	1.00	1.00	.883	1.00	.924
1/T(THE)4	20.0	20.0	20.0	20.0	20.0	20.0	20.0	1.00	2.34	1.00
1/T(THE)5								20.0	20.0	20.0
Z(THE)1	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
w(THE)1	4.83	27.4	37.1	22.6	9.82	14.5	20.0	21.4	34.5	21.9
N(HD /DS)										
A(HD)	7.70	141.	250.	107.	20.9	49.6	90.3	70.6	83.6	49.7
1/T(HD)1	.00726	.0146	.0680	.0165	-.0245	.00335	.0123	.00489	.0151	.00307
1/T(HD)2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.883	1.00	.924
1/T(HD)3	-4.21	17.0	20.0	15.3	5.96	9.99	12.7	1.00	2.34	1.00
1/T(HD)4	4.27	-17.5	-23.8	-15.7	-6.05	-10.2	-12.9	11.5	-12.4	-10.1
1/T(HD)5	20.0	20.0	23.9	20.0	20.0	20.0	20.0	-11.5	12.5	10.1
1/T(HD)6								20.0	20.0	20.0
Z(HD)1	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
w(HD)1	4.83	27.4	37.1	22.6	9.82	14.5	20.0	21.4	34.5	21.9
N(AZP/DS)										
A(AZF)	17.0	382.	737.	298.	58.7	135.	244.	188.	177.	132.
1/T(AZP)1	-.0514	-.000207	.000137	-.000356	-.000194	-.00287	-.000775	-.00104	-.000385	-.00117
1/T(AZP)2	.0543	.0148	.0675	.0172	-.0243	.00618	.0131	.00590	.0154	.00419
1/T(AZP)3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.883	1.00	.924
1/T(AZP)4	20.0	20.0	20.0	20.0	20.0	20.0	20.0	1.00	2.34	1.00
1/T(AZP)5								20.0	20.0	20.0
Z(AZF)1	.121	.104	.0917	.0876	.0620	.0625	.0585	.0400	.0294	.0280
w(AZF)1	2.80	10.5	13.9	9.30	3.61	6.09	7.81	7.07	8.56	6.19
Z(AZF)2	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
w(AZF)2	4.83	27.4	37.1	22.6	9.82	14.5	20.0	21.4	34.5	21.9
	+	+	+	+	+	+	+	+	+	+

TABLE IV-9

F-4C THRUST TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Open

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.9CC	.600	.900	1.20	1.50	2.15	1.80	
DENOMINATOR											
1/T(DET)1	1.24	-.0375	.858	-.0607	1.44	.0447	1.05	.883	1.04	.924	
1/T(DET)2	19.8	.0520	6.40	.0746	19.2	-.0454	16.1	1.06	2.34	1.06	
1/T(DET)3		.851		.974		1.18		17.2	17.4	18.1	
Z(DET)1	.0868	(4.44)	.632	(14.2)	.0890	(17.9)	.189	.155	.384	.175	
W(DET)1	.189		.0542		.0775		.0450	.0402	.0220	.0274	
Z(DET)2	.672	.940	.620	.824	.464	.568	.477	.339	.250	.247	
W(DET)2	.690	10.2	15.3	5.1C	1.20	2.77	5.90	5.64	6.88	4.93	
Z(DET)3	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209	
W(DET)3	4.83	27.4	37.1	22.6	5.82	14.5	20.0	21.4	34.5	21.9	
NUMERATORS											
N(U /DTH)											
A(U)	.000965	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823	
1/T(U)1	.112	.00608	.00177	.00435	.00376	.00295	-.000327	-.000995	-.000421	-.00113	
1/T(U)2	1.31	.851	.858	.974	1.55	1.18	1.05	.883	1.04	.924	
1/T(U)3	19.8	4.45	6.43	14.2	19.2	17.9	16.0	1.06	2.34	1.06	
1/T(U)4								17.2	17.3	18.1	
Z(U)1	.738	.939	.619	.825	.573	.589	.490	.365	.265	.289	
W(U)1	.637	10.2	15.2	5.11	1.20	2.80	5.91	5.64	6.89	4.93	
Z(U)2	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209	
W(U)2	4.83	27.4	37.1	22.6	5.82	14.5	20.0	21.4	34.5	21.9	
N(W /DTH)											
A(W)	-.878E-4	-.754E-4	-.754E-4	-.755E-4	-.754E-4	-.755E-4	-.755E-4	-.756E-4	-.756E-4	-.756E-4	
1/T(W)1	1.02	.00167	-.00175	.000503	.998	-.000110	-.00300	-.00281	-.00193	-.00279	
1/T(W)2	14.8	.429	-.746	.847	(-.798)	.630	-.740	-.883	-.324	-.458	
1/T(W)3	19.4	.959	.918	1.28	(.0738)	1.00	.970	.883	.984	.924	
1/T(W)4	(-.120)	(.103)	26.0	24.3	(.287)	22.2	22.0	.983	2.34	.987	
1/T(W)5	(.159)	(.27.4)	45.7	34.9	(.5.82)	33.5	45.9	21.0	20.6	20.6	
1/T(W)6								58.2	83.4	69.8	
Z(W)1	.584	.994	.0760	.125	.992	.194	.141	.210	.208	.209	
W(W)1	4.83	28.7	37.1	22.6	22.2	14.5	20.0	21.4	34.5	21.9	

TABLE IV-9 (Concluded)

N(THE /DTH)										
A(THE)	-.467E-5	-.298E-5	-.298E-5	-.299E-5	-.298E-5	-.300E-5	-.300E-5	-.302E-5	-.303E-5	-.303E-5
1/T(THE)1	-.192	.283	1.00	.517	-.289	.178	.450	.321	.282	.215
1/T(THE)2	.480	1.00	-1.21	1.00	.358	.800	1.00	.883	-.630	-.700
1/T(THE)3	1.00	1.23	1.28	1.36	1.00	1.00	-1.13	1.00	1.00	.924
1/T(THE)4	20.0	20.0	20.0	20.0	20.0	20.0	20.0	-1.24	2.34	1.00
1/T(THE)5								20.0	20.0	20.0
Z(THE)1	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
W(THE)1	4.63	27.4	37.1	22.6	5.82	14.5	20.0	21.4	34.5	21.9
N(HD /DTH)										
A(HD)	.000282	.797E-4	.711E-4	.826E-4	.000209	.000113	.985E-4	.000113	.957E-4	.000123
1/T(HD)1	.998	.331	.938	.775	.840	.614	.897	.851	.785	.598
1/T(HD)2	2.06	.972	(-.942)	1.26	2.25	.907	5.34	.883	2.14	.924
1/T(HD)3	19.8	-3.83	(2.51)	-3.27	19.2	-1.52	16.4	2.73	2.34	1.44
1/T(HD)4				11.2		4.30		17.4	17.6	18.3
Z(HD)1	-.523	.946	.825	(13.7)	-.311	(18.1)	-.148	.109	.160	.188
W(HD)1	.497	13.7	16.4		.710		2.15	2.78	2.82	2.88
Z(HD)2	.584	.103	.0760	.125	.287	.194	.141	.210	.208	.209
W(HD)2	4.83	27.4	37.1	22.6	5.82	14.5	20.0	21.4	34.5	21.9
N(AZP /DTH)										
A(AZP)	-.120E-4	-.272E-4	-.272E-4	-.270E-4	-.272E-4	-.269E-4	-.268E-4	-.267E-4	-.265E-4	-.266E-4
1/T(AZP)1	-.0214	-.000189	.000137	-.000256	-.000872	-.00168	-.000758	-.00100	-.000377	-.00106
1/T(AZP)2	1.04	.324	.940	.766	.994	.426	.928	.883	.906	.902
1/T(AZP)3		.968		1.24	7.36	1.00		.933	2.34	.924
Z(AZP)1	-.922	(-5.98)	-.880	(-5.22)	(17.5)	(-3.55)	-.392	-.177	-.115	-.133
W(AZP)1	.780		2.76				2.44	2.75	2.47	2.30
Z(AZP)2	.584	.770	.651	.808	-.945	.951	.814	.873	.897	.996
W(AZP)2	4.83	19.0	24.2	17.4	1.03	13.5	15.5	13.6	12.4	11.2
Z(AZP)3	.975	.103	.0760	.125	.287	.194	.141	.210	.208	.209
W(AZP)3	18.6	27.4	37.1	22.6	5.82	14.5	20.0	21.4	34.5	21.9
	+	+	+	+	+	+	+	+	+	+

TABLE IV-10

F-4C STICK FORCE TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Closed

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80	
DENOMINATOR											
1/T(DET)1	1.05	-.0270	.902	-.0443	1.14	-.0335	1.04	.990	1.02	.991	
1/T(DET)2	20.6	.0413	17.9	.0550	21.4	.0344	22.2	1.68	3.34	1.50	
1/T(DET)3		.924		.988		1.07		21.1	18.5	20.7	
Z(DET)1	.143	(21.2)	.755	(22.1)	-.0469	(22.2)	.204	.159	.431	.168	
W(DET)1	.0880		.0454		.0540		.0389	.0291	.0192	.0218	
Z(DET)2	.313	.627	.657	.504	.294	.342	.357	.337	.289	.263	
W(DET)2	1.12	5.53	10.1	4.87	1.56	3.03	5.22	4.98	6.46	4.56	
Z(DET)3	.431	.0181	-.00197	.0256	.170	.0781	.0316	.106	.173	.135	
W(DET)3	6.06	29.4	39.2	24.9	11.5	16.7	22.4	22.6	34.5	22.6	
NUMERATORS											
N(U /FST)											
A(U)	-190.	-23.4	41.7	-29.6	-109.	-71.5	-80.2	-102.	-64.9	-90.9	
1/T(U)1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.310	1.00	.143	
1/T(U)2	11.4	1.94	1.49	1.25	136.	201.	266.	.641	11.1	.584	
1/T(U)3	(.452)	5.35	-6.31	3.44	(.980)	(.787)	(.965)	1.00	400.	1.00	
1/T(U)4	(.561)	197.	304.	218.	(.307)	(.643)	(.783)	4.22	(.978)	4.42	
1/T(U)5								328.	(.496)	394.	
N(W /FST)											
A(W)	210.	4476.	7961.	3388.	656.	1573.	2869.	2242.	2656.	1576.	
1/T(W)1	1.00	1.00	-.00320	1.00	1.00	1.00	1.00	1.00	1.00	1.00	
1/T(W)2	49.3	204.	.0711	222.	137.	202.	267.	4.22	11.1	4.42	
1/T(W)3			1.00					328.	400.	394.	
Z(W)1	.151	.176	(299.)	.169	.0121	.0964	.852	.290	.731	.184	
W(W)1	.156	.0456		.0627	.0627	.0532	.00729	.00911	.0106	.0114	

TABLE IV-10 (Concluded)

N(THE /FST)										
A(THE)	46.2	1024.	1936.	752.	156.	363.	656.	508.	511.	357.
1/T(THE)1	.104	.0162	.0678	.0208	-.000498	.0106	.0131	.00608	.0157	.00460
1/T(THE)2	.379	1.00	1.00	1.00	.282	.505	.618	.407	.388	.260
1/T(THE)3	1.00	1.46	1.90	1.06	1.00	1.00	1.00	1.00	1.00	1.00
1/T(THE)4								4.22	11.1	4.42
N(HD /FST)										
A(HD)	-245.	-4476.	-7961.	-3388.	-665.	-1575.	-2870.	-2245.	-2657.	-1579.
1/T(HC)1	.00726	.0146	.0680	.0165	-.0245	.00335	.0123	.00489	.0151	.00307
1/T(HC)2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
1/T(HC)3	.21	17.0	-23.8	15.3	5.96	9.99	12.7	4.22	11.1	4.42
1/T(HC)4	4.27	-17.5	23.9	-15.7	-6.05	-10.2	-12.9	11.5	-12.4	-10.1
1/T(HC)5								-11.5	12.5	10.1
N(AZP/FST)										
A(AZP)	-540.	-12129.	-23430.	-9456.	-1867.	-4306.	-7765.	-5989.	-5624.	-4209.
1/T(AZP)1	-.0514	-.000207	.000137	-.000356	-.000194	-.00287	-.000776	-.000104	-.000385	-.00117
1/T(AZP)2	.0543	.0148	.0679	.0172	-.0243	.00618	.0131	.00590	.0154	.00419
1/T(AZP)3	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
1/T(AZP)4								4.22	11.1	4.42
Z(AZP)1	.121	.104	.0917	.0816	.0620	.0625	.0585	.0400	.0294	.0280
w(AZP)1	2.80	10.5	13.9	9.30	3.61	6.09	7.81	7.07	8.56	6.19
+ + + + + + + + + +										

TABLE IV-11

F-4C THRUST TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Closed

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80	
DENOMINATOR											
1/T(DET)1	1.05	-.0270	.902	-.0443	1.14	-.0335	1.04	.990	1.02	.991	
1/T(DET)2	20.6	.0413	17.9	.0590	21.4	.0344	22.2	1.68	3.34	1.50	
1/T(DET)3		.924		.988		1.07		21.1	18.5	20.7	
Z(DET)1	.143	(21.2)	.755	(22.1)	-.0469	(22.2)	.204	.159	.431	.168	
W(DET)1	.0880		.0454		.0540		.0389	.0291	.0192	.0218	
Z(DET)2	.313	.627	.657	.504	.294	.342	.357	.337	.289	.263	
W(DET)2	1.12	5.53	10.1	4.87	1.56	3.03	5.22	4.98	6.46	4.56	
Z(DET)3	.431	.0181	-.00157	.0256	.170	.0781	.0315	.106	.173	.135	
W(DET)3	6.06	29.4	39.2	24.9	11.5	16.7	22.4	22.6	34.5	22.6	
NUMERATORS											
N(U /DTH)											
A(U)	.000965	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823	.000823	
1/T(U)1	.00561	.00340	.00128	.00241	-.00271	.000946	-.000438	-.000999	-.000410	-.00110	
1/T(U)2	1.05	.924	.902	.988	1.15	1.07	1.04	.990	1.02	.991	
1/T(U)3	20.6	21.2	17.9	22.1	21.4	22.2	22.2	1.69	3.35	1.51	
1/T(U)4								21.1	18.5	20.7	
Z(U)1	.359	.627	.655	.505	.375	.359	.367	.360	.305	.303	
W(U)1	1.15	5.54	10.1	4.89	1.60	3.06	5.22	4.97	6.46	4.55	
Z(U)2	.436	.0181	-.00200	.0256	.172	.0786	.0319	.107	.173	.135	
W(U)2	6.07	29.4	39.2	24.9	11.5	16.7	22.4	22.6	34.5	22.6	
N(W /DTH)											
A(W)	-.878E-4	-.754E-4	-.754E-4	-.755E-4	-.754E-4	-.755E-4	-.755E-4	-.756E-4	-.756E-4	-.756E-4	
1/T(W)1	.0248	.00116	-.00173	.000299	.998	-.000597	-.00304	-.00293	-.00215	-.00306	
1/T(W)2	.381	.493	-.625	.892	20.2	.719	-.578	-.532	-.221	-.276	
1/T(W)3	1.03	.962	.927	1.32	24.4	1.00	.974	1.01	.988	1.01	
1/T(W)4	13.1	26.2	27.2	24.3	(.974)	22.2	23.2	1.17	2.90	1.19	
1/T(W)5	20.9	33.6	48.0	37.4	(.0509)	35.0	47.0	22.2	21.1	21.5	
1/T(W)6								58.6	83.6	70.0	
Z(W)1	.522	.0483	.0261	.0630	.217	.124	.0751	.151	.188	.167	
W(W)1	5.25	29.0	38.8	24.3	10.9	16.0	21.5	22.2	34.7	22.4	

TABLE IV-11 (Concluded)

[illegible]

TABLE IV-13

F-4C LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES
(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80	
YV	-.0918	-.335	-.466	-.215	-.0566	-.0921	-.151	-.118	-.133	-.0768	
YB	-21.1	-299.	-597.	-205.	-33.1	-80.6	-176.	-171.	-277.	-134.	
LB'	-10.4	-26.3	-47.0	-27.4	-10.7	-18.3	-14.1	-11.7	-9.67	-8.66	
NB'	1.44	15.6	38.2	11.5	1.66	4.97	12.3	9.90	8.37	6.18	
LP'	-1.43	-3.04	-3.11	-2.27	-.799	-1.24	-1.38	-1.00	-1.08	-.757	
NP'	-.0260	-.0372	.0184	-.0240	-.0179	-.0504	-.0378	-.0170	.0153	-.00013	
LN'	.929	.817	.802	.638	.300	.395	.318	.328	.217	.198	
NR'	-.215	-.739	-1.20	-.530	-.134	-.238	-.397	-.309	-.273	-.181	
Y*DA	-.0130	-.00744	-.0102	-.00499	-.00151	-.00227	-.00302	-.00199	-.00169	-.000329	
L'DA	2.74	22.2	15.0	17.5	4.70	0.00	10.9	6.78	5.35	4.67	
N'DA	.416	.923	2.45	.747	.0887	.195	.667	.376	.357	.0567	
Y*DR	.0174	.0442	.0307	.0261	.0113	.0142	.0132	.00988	.00867	.00614	
L'DR	.699	7.32	9.26	5.07	.768	1.95	2.99	1.95	2.57	1.21	
N'DR	-.670	-7.80	-8.80	-5.58	-1.36	-2.61	-3.19	-2.03	-1.86	-1.31	
	+	+	+	+	+	+	+	+	+	+	+

TABLE IV-14
F-4C AILERON TRANSFER FUNCTION FACTORS
SAS Off
(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80	
DENOMINATOR											
1/T(DET)1	.0147	.00469	.00568	.00348	.0173	.00969	.00187	-.000179	.000558	.000226	
1/T(DET)2	1.15	3.10	3.13	2.33	.650	1.33	1.40	.996	1.06	.748	
Z(DET)1	.156	.125	.134	.0972	.0881	.0491	.0727	.0670	.0731	.0535	
W(DET)1	1.82	4.01	6.21	3.45	1.83	2.43	3.57	3.23	2.93	2.58	
NUMERATORS											
N(B /DA)											
A(B)	-.0130	-.00744	-.0102	-.00455	-.00151	-.00227	-.00302	-.00199	-.00162	-.000329	
1/T(B)1	-10.4	-.425	-.0664	-.407	.121	.127	-.487	-.760	-.0660	.155	
1/T(B)2	(.297)	1.77	3.08	1.35	.433	2.70	.704	.930	1.60	.479	
1/T(B)3	(1.05)	111.	249.	121.	-450.	-115.	121.	35.1	134.	-644.	
N(P /DA)											
A(P)	2.74	22.2	15.0	17.5	4.70	0.00	10.9	6.78	5.35	4.67	
1/T(P)1	-.0285	-.000186	.000125	-.000252	-.00908	-.00166	-.000766	-.00100	-.000376	-.00106	
Z(P)1	.152	.136	.135	.109	.0767	.0742	.0788	.0691	.0706	.0522	
W(P)1	1.74	4.11	6.82	3.57	1.36	2.31	3.63	3.25	3.00	2.51	
N(R /DA)											
A(R)	.416	.923	2.45	.747	.0887	.195	.667	.376	.357	.0567	
1/T(R)1	.746	3.08	4.09	2.39	.331	.733	.964	.494	.711	.320	
Z(R)1	.145	-.169	-.200	-.146	-.0560	-.275	-.0224	.0457	.203	-.0241	
W(R)1	1.91	2.16	1.34	2.05	4.03	3.69	2.48	2.92	1.71	5.46	
N(PHI /DA)											
A(PHI)	2.82	22.2	15.0	17.5	4.71	10.0	10.9	6.80	5.35	4.67	
Z(PHI)1	.150	.136	.135	.109	.0722	.0735	.0788	.0691	.0709	.0518	
W(PHI)1	1.74	4.11	6.83	3.57	1.38	2.31	3.63	3.25	3.00	2.51	
N(AYP/DA)											
A(AYP)	12.0	70.7	69.3	56.6	13.8	29.3	38.0	22.3	17.3	13.5	
1/T(AYP)1	-.234	-.311	-.0885	-.243	.125	.119	-.199	.184	-.0738	.104	
1/T(AYP)2	.373	.496	1.27	.393	-.400	-.390	.230	-.195	.485	-.257	
Z(AYF)1	.149	.114	.0935	.0950	.149	.0765	.0710	.0665	.0230	.0766	
W(AYF)1	1.77	4.14	6.40	3.57	1.32	2.34	3.58	3.26	3.28	2.42	
	+	+	+	+	+	+	+	+	+	+	+

100

SAS Off
(BODY AXIS SYSTEM)

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TABLE IV-16
F-4C AILERON TRANSFER FUNCTION FACTORS
SAS On
(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80	
DENOMINATOR											
1/T(DET)1	.00233	-.000122	.00466	.00114	.00289	.00176	.00129	-.000595	.00114	-.000318	
1/T(DET)2	.905	.852	.534	2.72	.453	1.38	.645	.617	.530	.624	
1/T(DET)3	2.03	1.58	3.15	8.01	3.62	5.31	1.38	.092	1.11	.803	
Z(DET)1	.393	(3.20)	.657	.763	.332	.508	.752	.608	.572	.592	
W(DET)1	1.52	(9.95)	6.68	1.05	1.36	1.01	3.38	2.15	3.24	2.48	
NUMERATORS											
N(B /DA)											
A(B)	-.0160	-.00965	-.00937	-.00605	-.00181	-.00294	-.00315	-.00216	-.00187	-.000461	
1/T(B)1	.0939	.0124	.0828	.0150	.0136	.0128	.0299	.0354	.0543	.0223	
1/T(B)2	.887	-1.04	-.154	-1.14	.398	1.62	-1.68	1.10	-.309	.591	
1/T(B)3	-6.92	7.03	4.91	5.78	4.15	10.3	2.24	-4.27	2.13	2.79	
1/T(B)4	8.52	128.	246.	137.	-354.	-48.9	129.	55.1	143.	-393.	
N(P /DA)											
A(P)	2.64	21.8	15.2	17.3	4.68	9.90	10.9	6.75	5.29	4.64	
1/T(P)1	-.0285	-.000187	.000136	-.000294	-.000908	-.00166	-.000757	-.00100	-.000777	-.00106	
1/T(P)2	1.09	.592	.513	.710	2.61	5.33	.602	.594	.578	.636	
Z(P)1	.867	(2.16)	.675	(1.45)	.543	.764	.766	.626	.590	.598	
W(P)1	1.29	(10.2)	7.33	(8.16)	.549	.772	3.48	3.19	3.35	2.42	
N(R /DA)											
A(R)	.547	1.31	2.21	.958	.125	.320	.699	.411	.395	.0850	
1/T(R)1	.471	.481	.495	.468	.302	.417	.456	.400	.423	.300	
1/T(R)2	.904	7.04	5.26	5.87	.719	3.08	1.84	.755	1.36	.623	
Z(R)1	.226	-.0496	-.220	-.0272	.467	.417	.271	.305	.399	.414	
W(R)1	1.69	1.52	1.38	1.38	2.21	1.68	1.93	2.69	1.53	4.47	

TABLE IV-16 (Concluded)

N(PHI/DA)										
A(PHI)	2.76	21.9	15.2	17.4	4.70	9.92	10.9	6.77	5.30	4.65
1/T(PHI)1	1.07	.592	.513	.710	3.59	5.33	.602	.594	.578	.635
Z(PHI)1	.841	(2.16)	.675	(1.49)	.539	.763	.766	.625	.590	.596
W(PHI)1	1.30	(10.2)	7.34	(8.15)	.556	.772	3.48	3.16	3.34	2.42
N(AYP/DA)										
A(AYP)	13.2	74.1	67.1	58.5	14.1	30.4	38.3	22.5	17.4	13.6
1/T(AYP)1	.149	.0293	.111	.0294	.0208	.0212	.0486	.0580	.0790	.0371
1/T(AYP)2	-.370	-.579	-.174	-.515	.508	-.675	-.396	-.372	-.174	-.476
1/T(AYP)3	.925	9.30	2.56	7.52	-.880	5.02	1.17	.759	1.14	.615
Z(AYP)1	.610	.641	.475	.632	(1.31)	.770	.673	.573	.470	.641
W(AYP)1	1.64	2.32	6.00	1.94	(3.16)	1.18	3.33	3.28	3.47	2.63
	+	+	+	+	+	+	+	+	+	+

TABLE IV-17
F-4C RUDDER TRANSFER FUNCTION FACTORS
SAS On
(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80
DEACMINATOR										
1/T(DET)1	.00233	-.000122	.00468	.00114	.00289	.00176	.00129	-.000595	.00116	-.000318
1/T(DET)2	.905	.852	.534	2.72	.453	1.38	.645	.617	.539	.624
1/T(DET)3	2.03	1.58	3.15	8.01	3.62	5.31	1.38	.992	1.11	.803
Z(DET)1	.393	(3.20)	.657	.763	.332	.508	.752	.608	.572	.562
W(DET)1	1.52	(9.95)	6.68	1.05	1.36	1.01	3.38	3.15	3.24	2.48
NUMERATORS										
N(B /DR)										
A(B)	.0166	.0298	.0229	.0204	.0102	.0117	.0113	.00926	.00854	.00594
1/T(B)1	-.0911	-.00161	.00396	-.00256	-.0240	-.00775	.000301	-.00165	.00190	-.00179
1/T(B)2	.500	.500	.500	.500	.500	.500	.500	.500	.500	.500
1/T(B)3	1.26	3.09	3.13	2.30	.750	1.26	1.40	.996	1.05	.752
1/T(B)4	46.4	178.	286.	201.	130.	191.	248.	215.	222.	225.
N(P /DR)										
A(P)	.669	4.93	6.91	3.69	.690	1.62	2.57	1.83	2.53	1.17
1/T(P)1	-.0287	-.000198	.000140	-.000294	-.00911	-.00167	-.000771	-.00100	-.000376	-.000106
1/T(P)2	.500	.500	.500	.500	.500	.500	.500	.500	.500	.500
1/T(P)3	2.53	-3.52	-2.16	4.27	2.91	4.26	-1.58	-1.51	(.0758)	1.79
1/T(P)4	-3.34	3.57	2.92	-4.38	-4.40	-4.59	1.73	1.54	(1.45)	-1.79

TABLE IX-17 (Concluded)

[illegible]

TABLE IX-18

F-4C LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	15 K	35 K	35 K	35 K	45 K	45 K	55 K	
M	.206	.800	1.10	.900	.600	.900	1.20	1.50	2.15	1.80	
DR PERIOD (SEC)	3.49	1.58	1.02	1.83	3.45	2.59	1.76	1.95	2.15	2.44	
1/C(1/2)	1.43	1.15	1.22	.885	.802	.446	.661	.609	.665	.485	
SPIRAL (2) (SEC)	--	--	--	--	--	--	--	3868.	--	--	
P(1)	1.90	7.39	5.51	7.79	2.52	6.80	7.93	6.83	5.22	5.79	
P(2)	--	7.33	5.42	7.59	2.12	6.28	7.90	6.83	5.21	5.75	
P(3)	--	7.52	5.81	7.98	4.00	6.63	7.98	6.85	5.28	5.85	
P(2)/P(1)	--	.992	.984	.938	.844	.924	.996	.000	.997	.993	
P(OSC)/P(AV)	--	.00847	.0217	.0125	.211	.0335	.00355	.000659	.00415	.00588	
w(PHI)/w(D)	.955	1.03	1.10	1.03	.753	.951	1.02	1.01	1.02	.972	
DEL-B-MAX	.0738	.0664	.106	.0757	.338	.145	.0521	.0157	.0559	.0562	
PHI TC BETA, PHASE	29.1	32.4	18.7	28.4	16.4	-335.	18.2	15.5	16.9	-346.	
PHI TC BETA	2.63	1.39	1.16	2.00	2.98	2.79	1.03	1.05	.948	1.21	
PHI TC VE	.657	.0891	.0539	.152	.526	.327	.0910	.0940	.0591	.115	
	+	+	+	+	+	+	+	+	+	+	+

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SECTION V

X-15

X-15 BACKGROUND

The X-15 is a single-place, rocket-powered airplane designed for flight at hypersonic speeds and extreme altitudes. The airplane is carried aloft under the right wing of a B-52 and is launched at an altitude of about 45,000 ft and a Mach number of about 0.80. After launch, the X-15 performs a powered flight mission, followed by a deceleration glide prior to vectoring for a landing. With this operational technique, the airplane is capable of attaining a Mach number of 6 and can be flown to and recovered from an altitude in excess of 300,000 feet.

Flights to high altitudes have been made with all three of the X-15 airplanes in two configurations: the basic and the ventral off. The basic configuration is considered here.

Aerodynamic control is provided through conventional aerodynamic surfaces, with vertical surfaces used for yaw control and the horizontal tail for both pitch and roll control. All of the aerodynamic control surfaces are actuated by irreversible hydraulic systems. Control force is provided by bungee for pilot feel. A conventional center stick is used for pitch and roll control, and rudder pedals are used for yaw control; however, a side-located stick is provided for control of pitch and roll in high-acceleration environments at the option of the pilot. Most of the X-15 missions have been made with the side stick, although the pilots used the center stick on their first flights. Only the center stick control is shown here.

The augmentation system shown in this report consists of angular rate feedback loops about all three axes. In addition to the normal $p \rightarrow \delta_a$ roll SAS loop, there is an $r \rightarrow \delta_a$ feedback known as the YAR loop. The gains for each SAS loop are manually set by the pilot. The SAS-on transfer functions given for this airplane assume maximum gain settings for each loop. This may not have been realistic for actual flights.

The flight conditions considered for this airplane are all for straight and level trimmed flight. This is definitely unrealistic for this airplane; however, the intent here is to show general speed and altitude variation effects.

X-15

Nominal Configuration

Zero Fuel
 Lower Ventral On
 Speed Brakes Retracted
 $W = 15560 \text{ lb}$
 c.g. at $.22 \bar{c}$
 $I_x = 3650 \text{ slug-ft}^2$
 $I_y = 80000 \text{ slug-ft}^2$
 $I_z = 82000 \text{ slug-ft}^2$
 $I_{xz} = -90 \text{ slug-ft}^2$

Flight Envelope

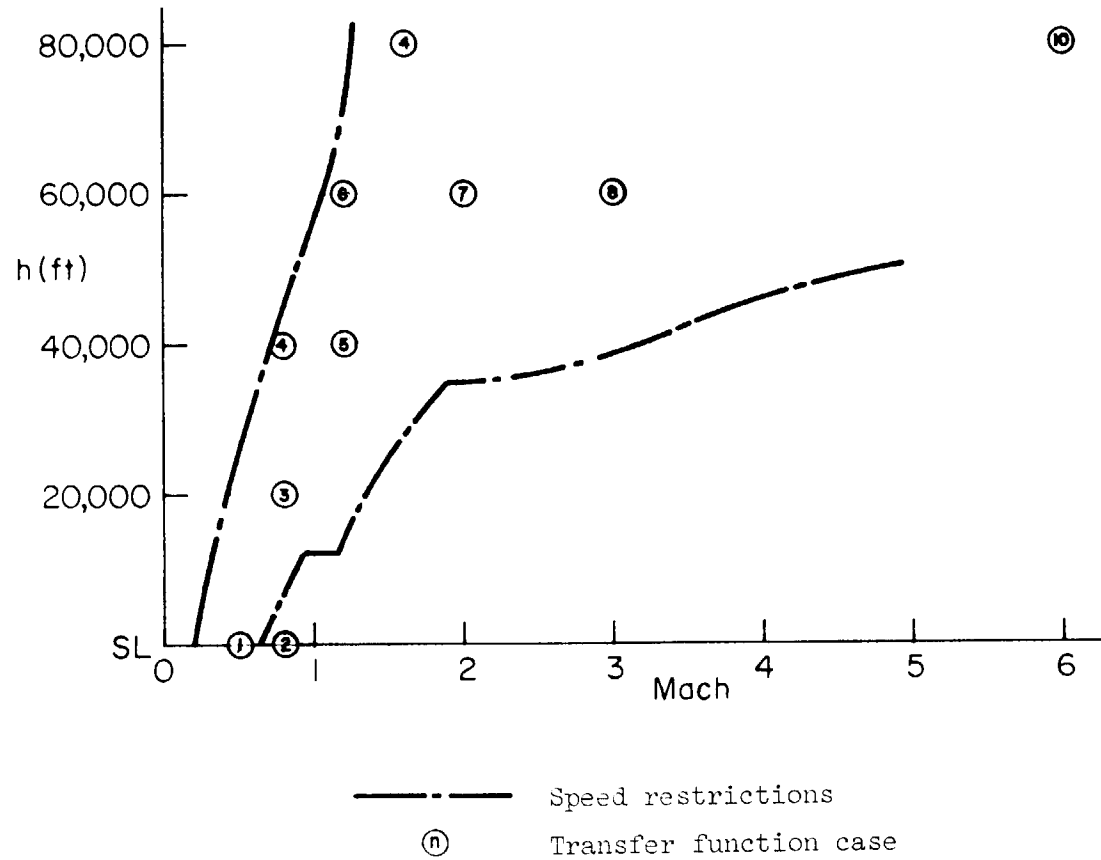


Figure V-1. X-15 Flight Conditions

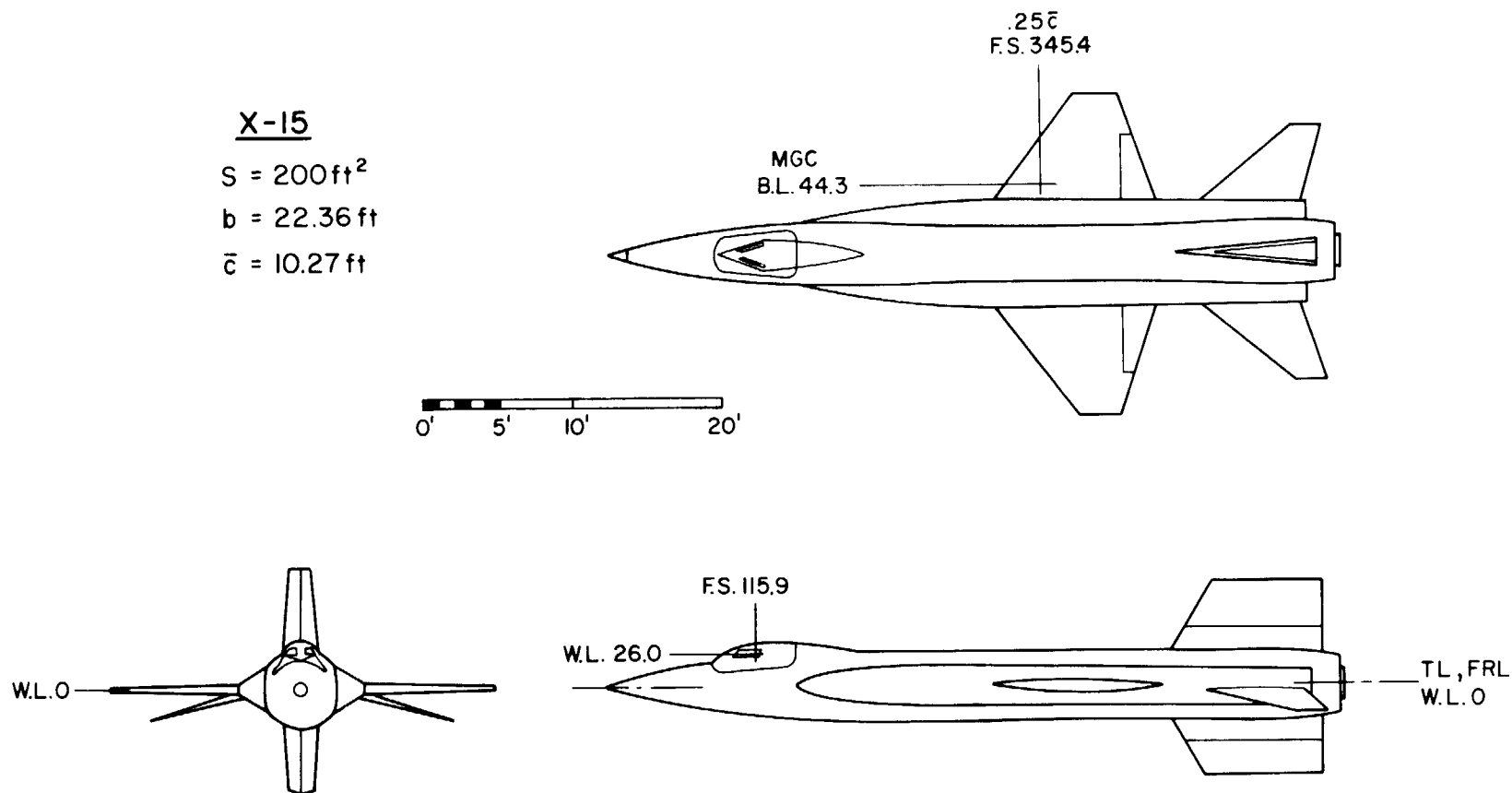
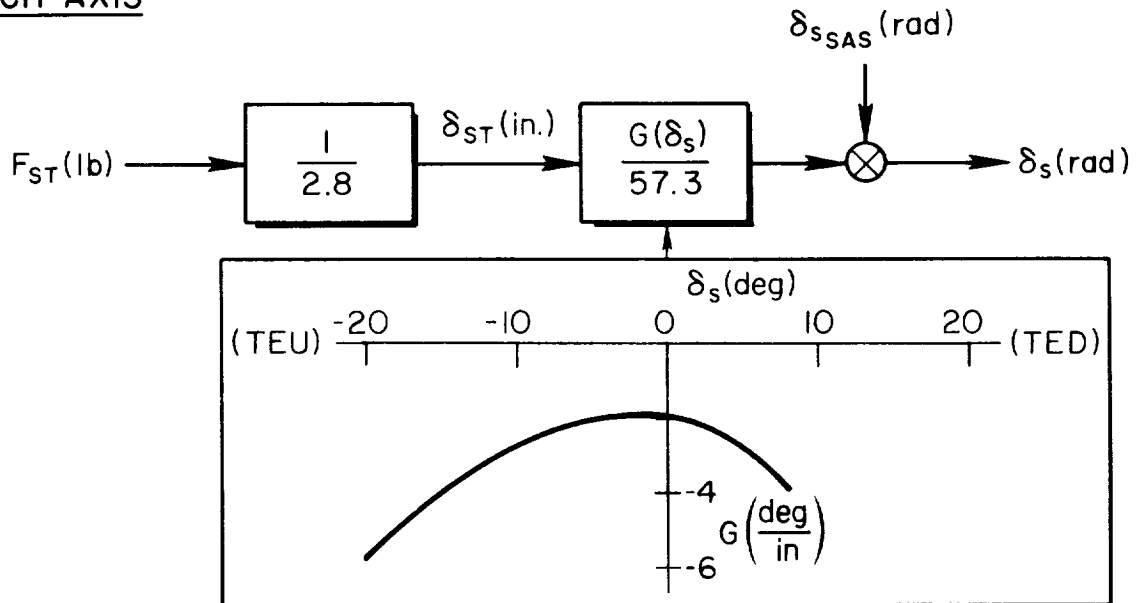


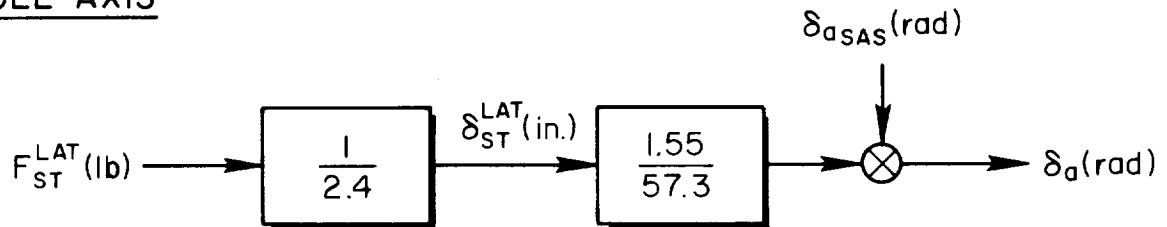
Figure V-2. X-15 General Arrangement

X-15

PITCH AXIS



ROLL AXIS



YAW AXIS

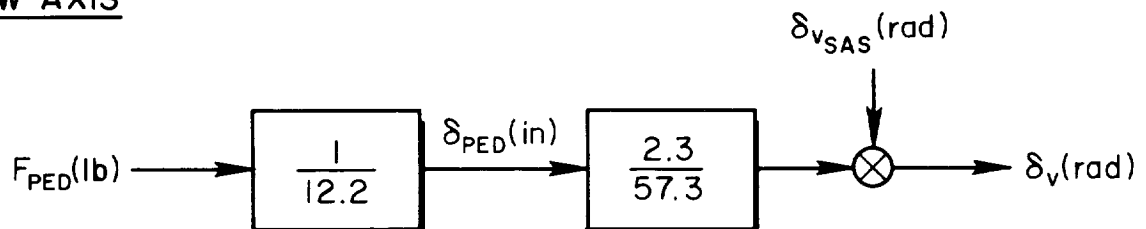
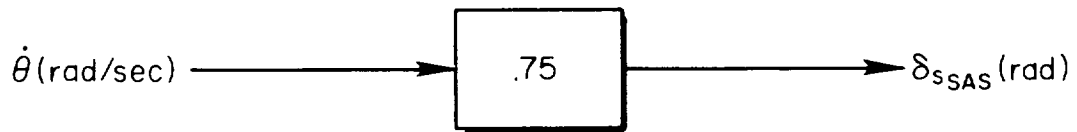


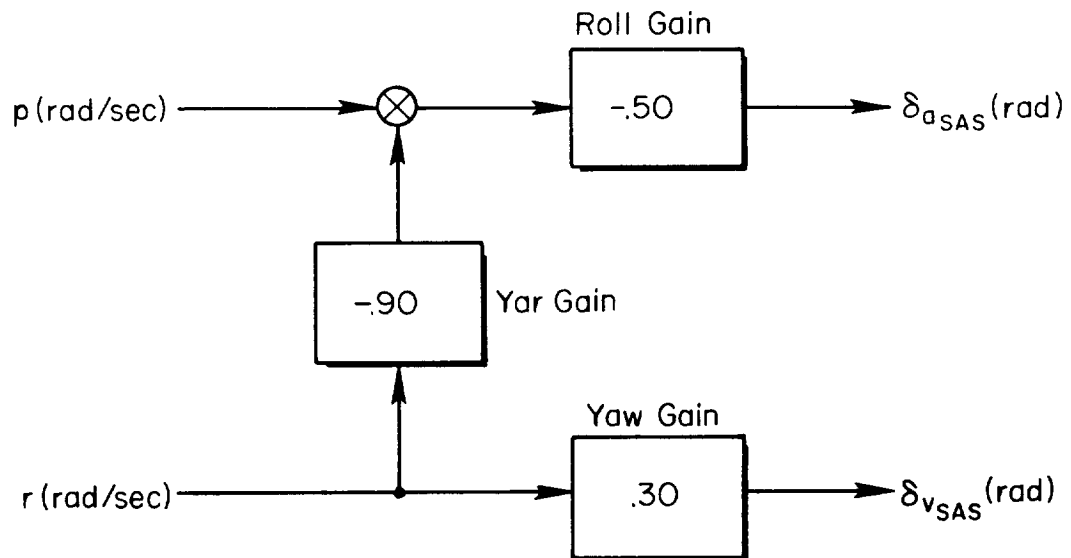
Figure V-3. X-15 Control System

X-15

PITCH SAS



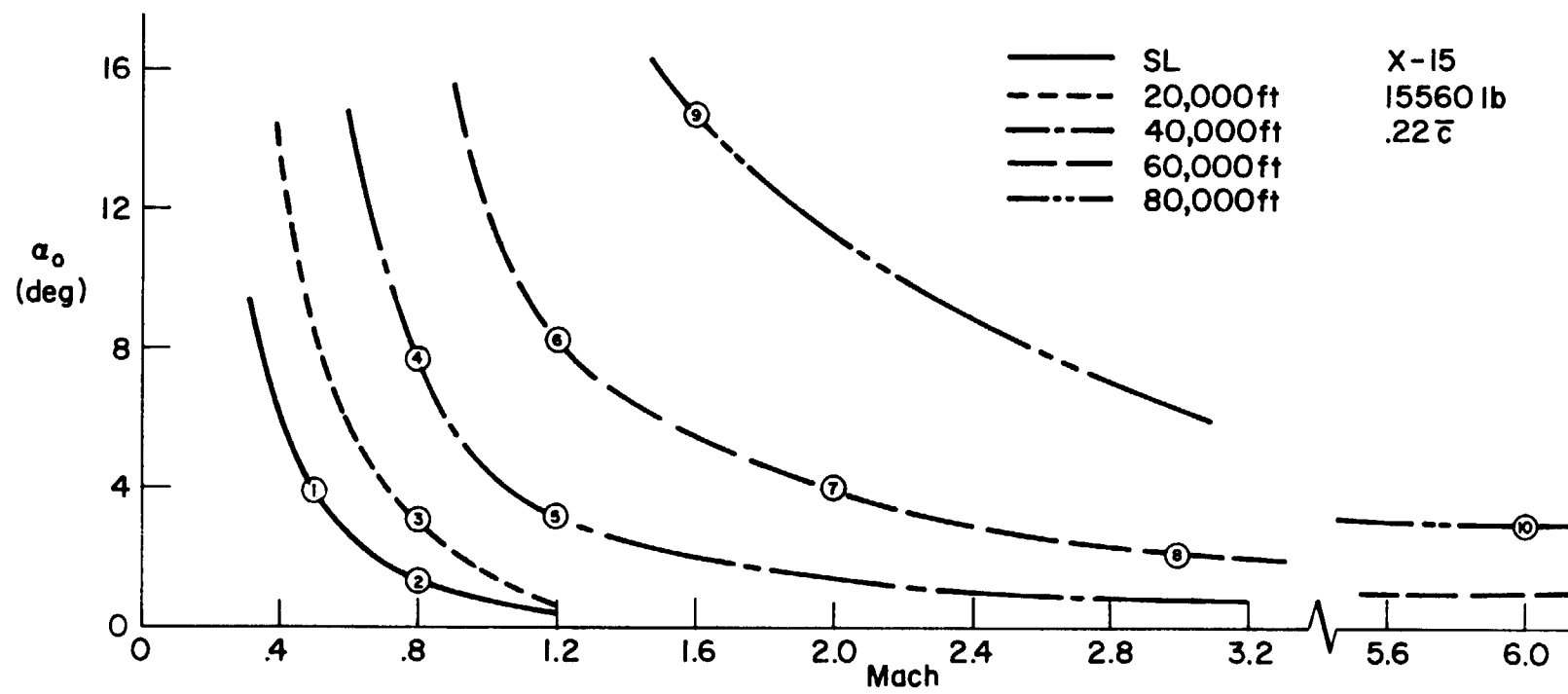
ROLL - YAW - YAR SAS

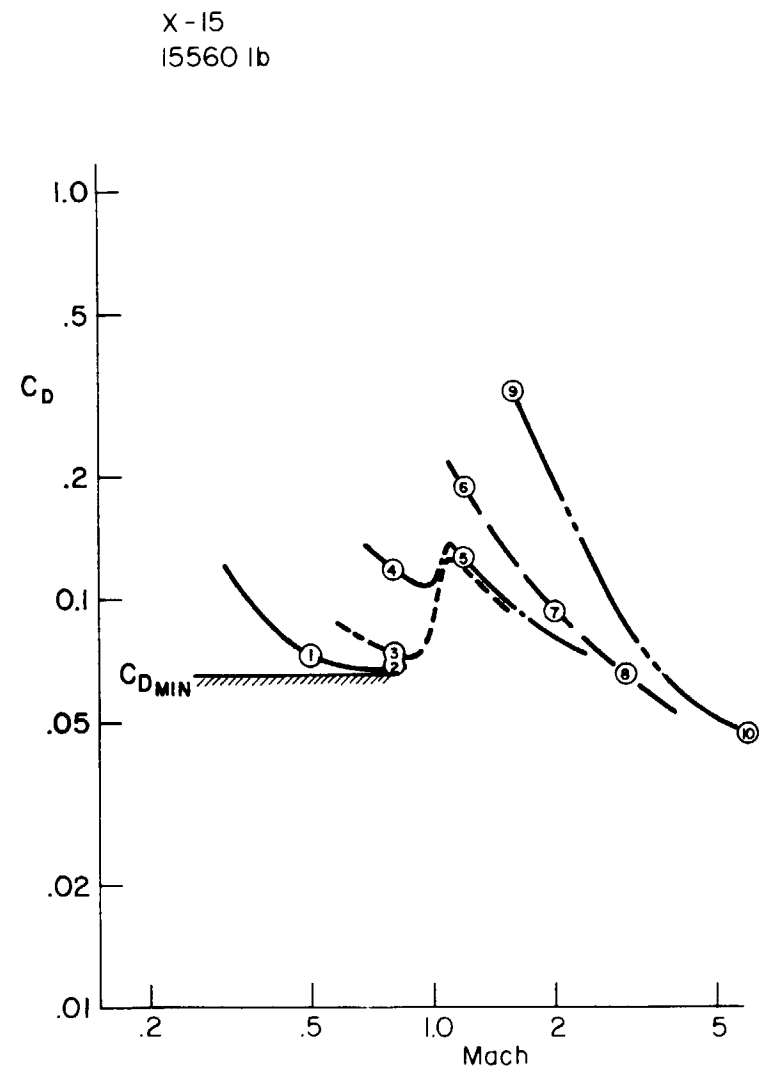
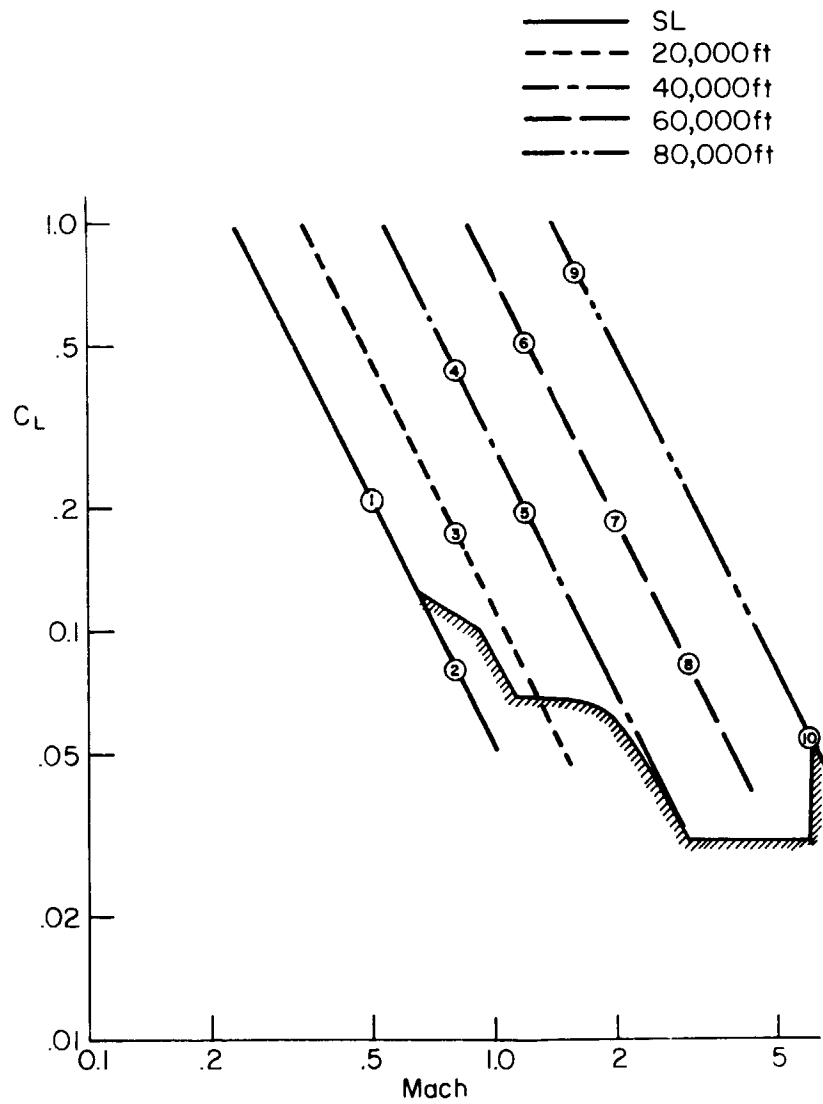


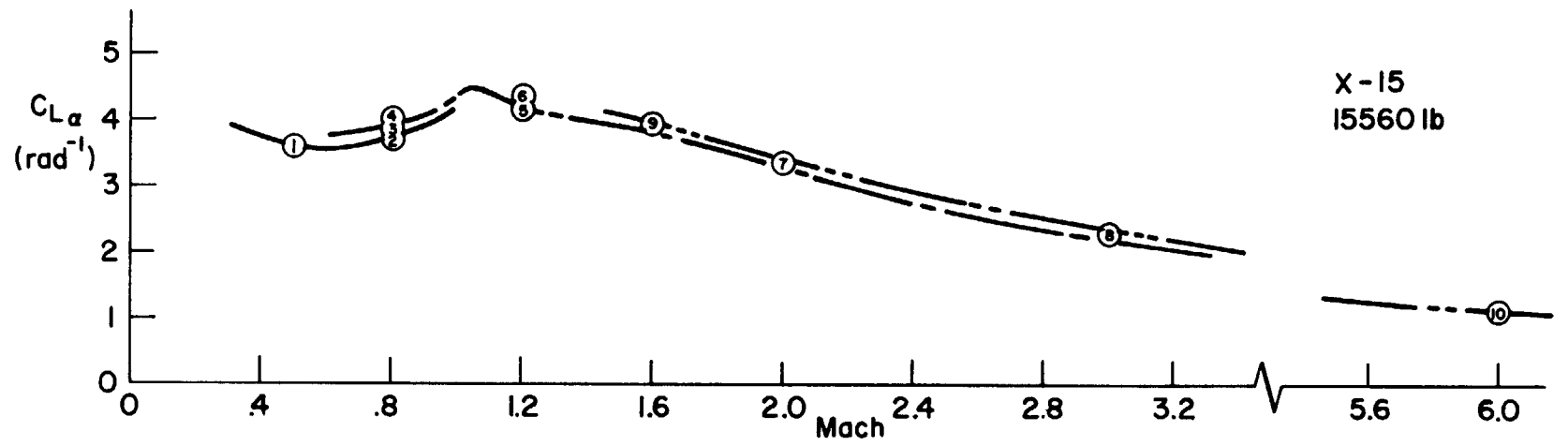
Note:

Gains variable in 10% increments of the maximum values which are shown above. (e.g. roll gains selectable are .05, .10, .15, .20, .25, .30, .35, .40, .45, and .50)

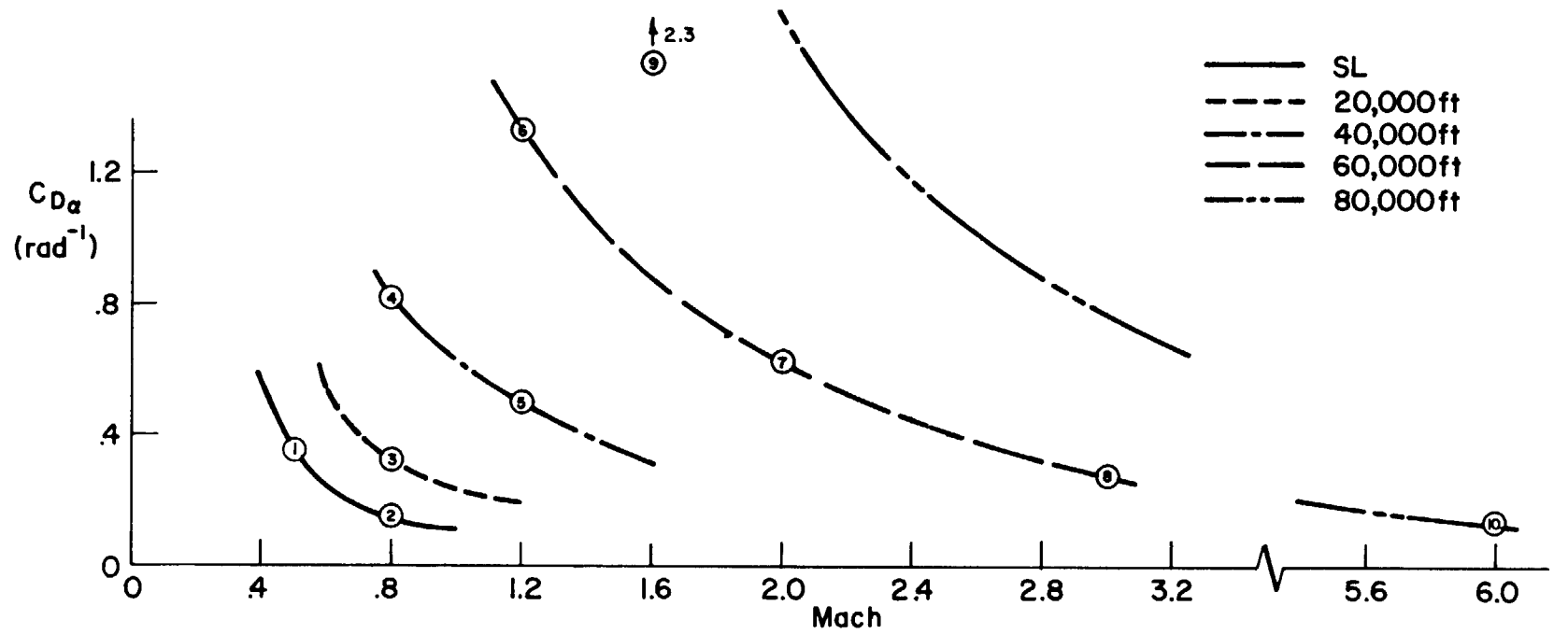
Figure V-4. X-15 Stability Augmentation

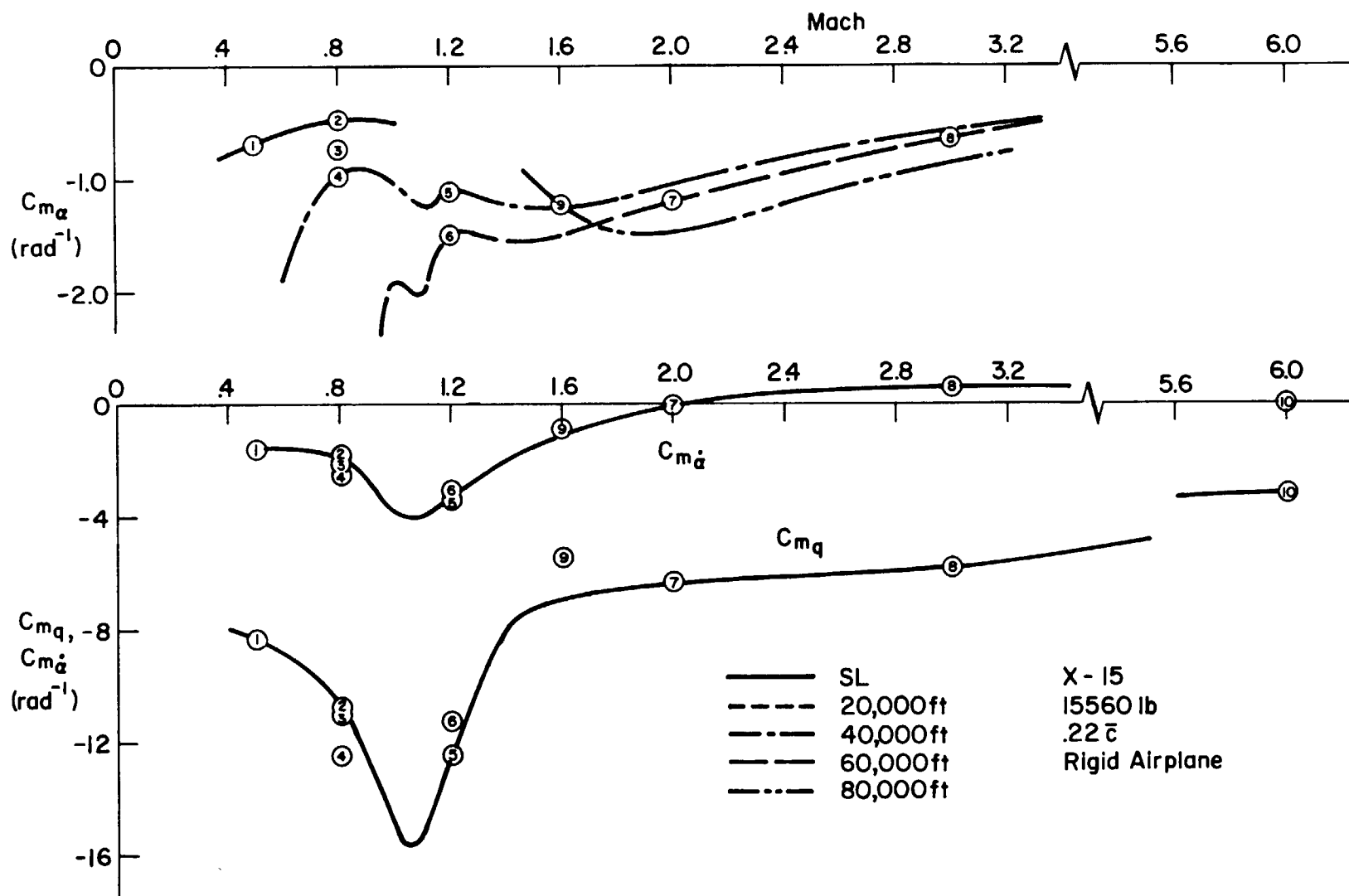


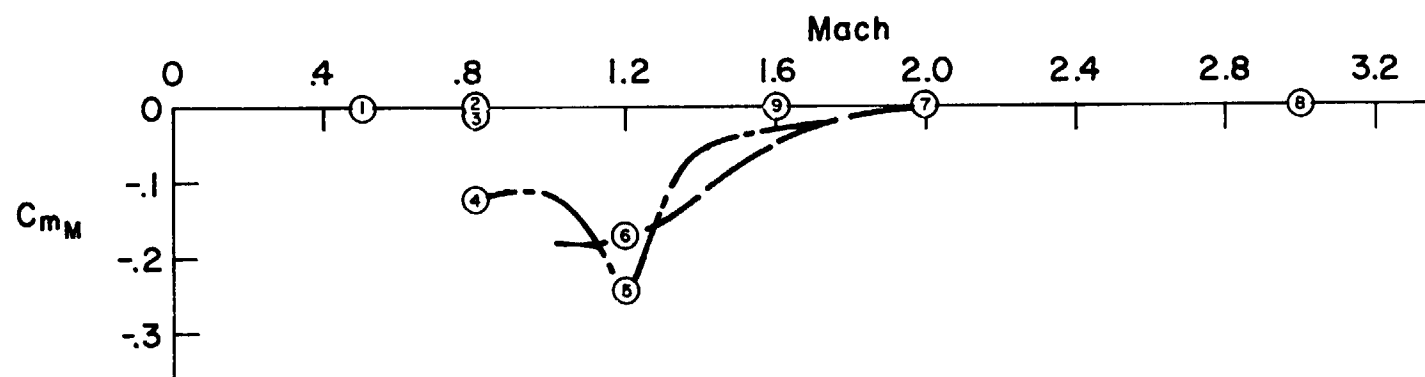
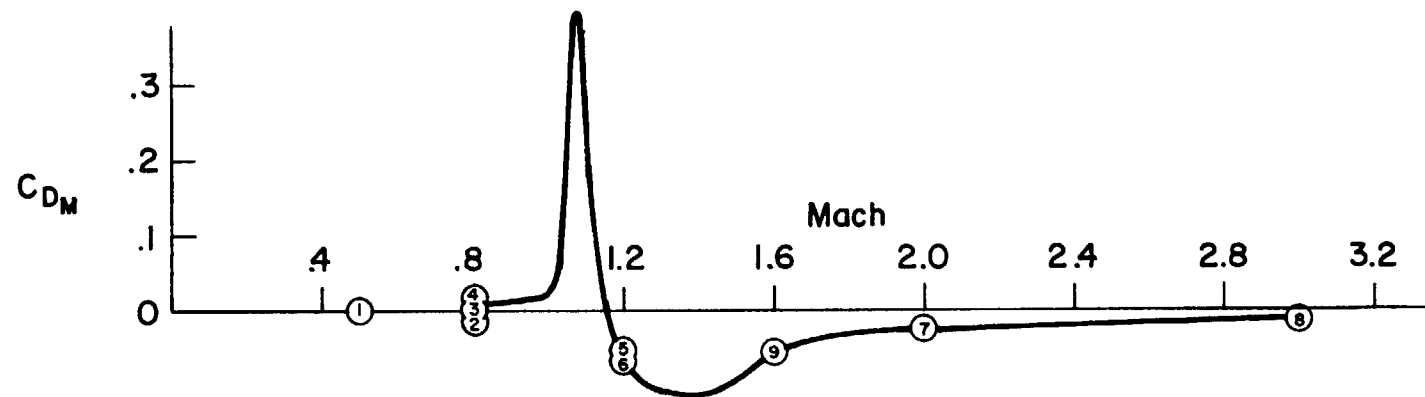
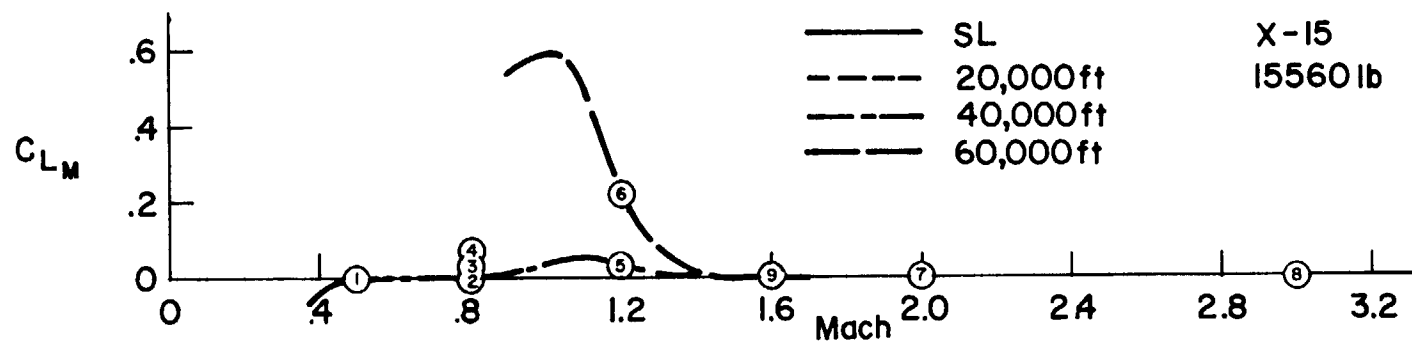


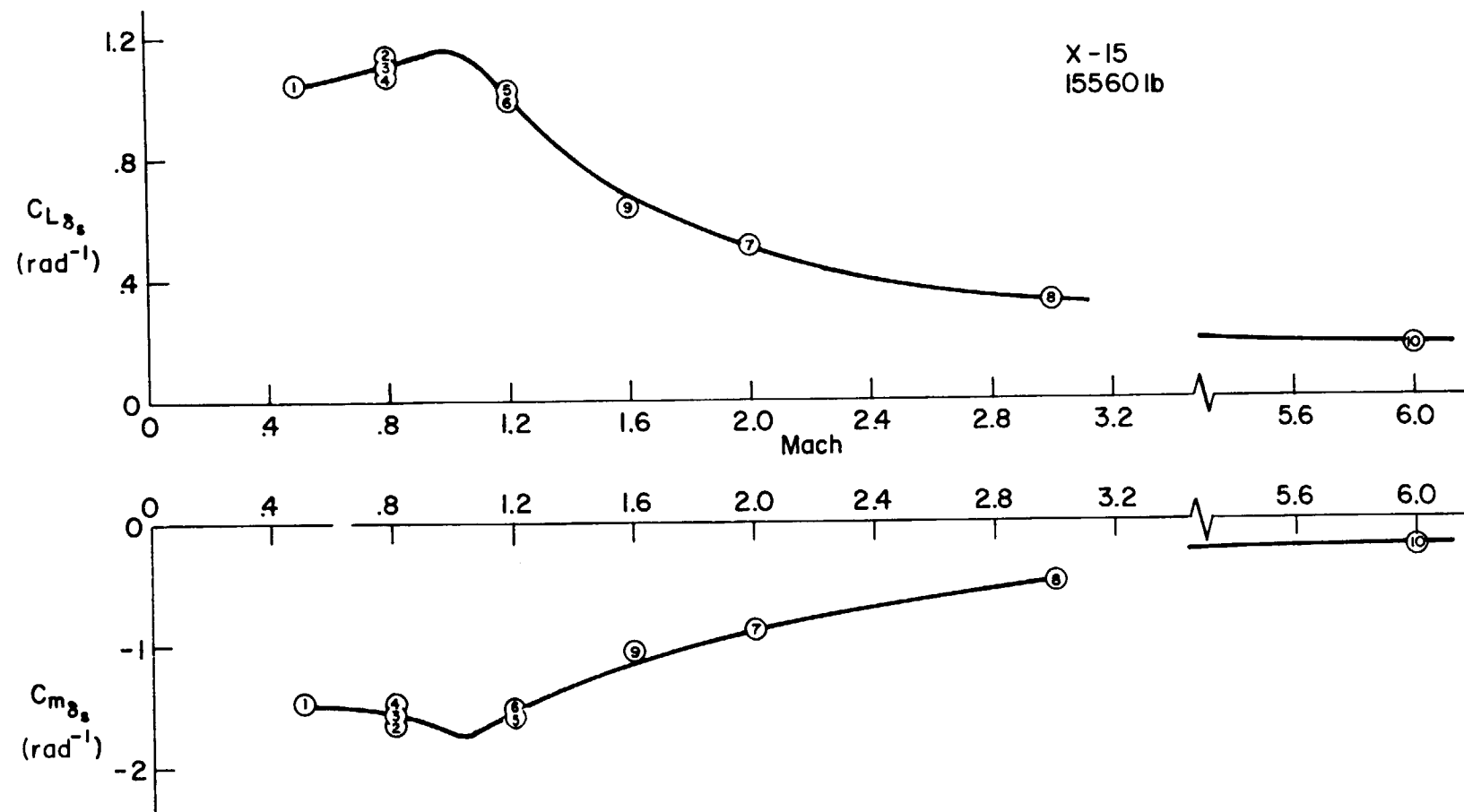


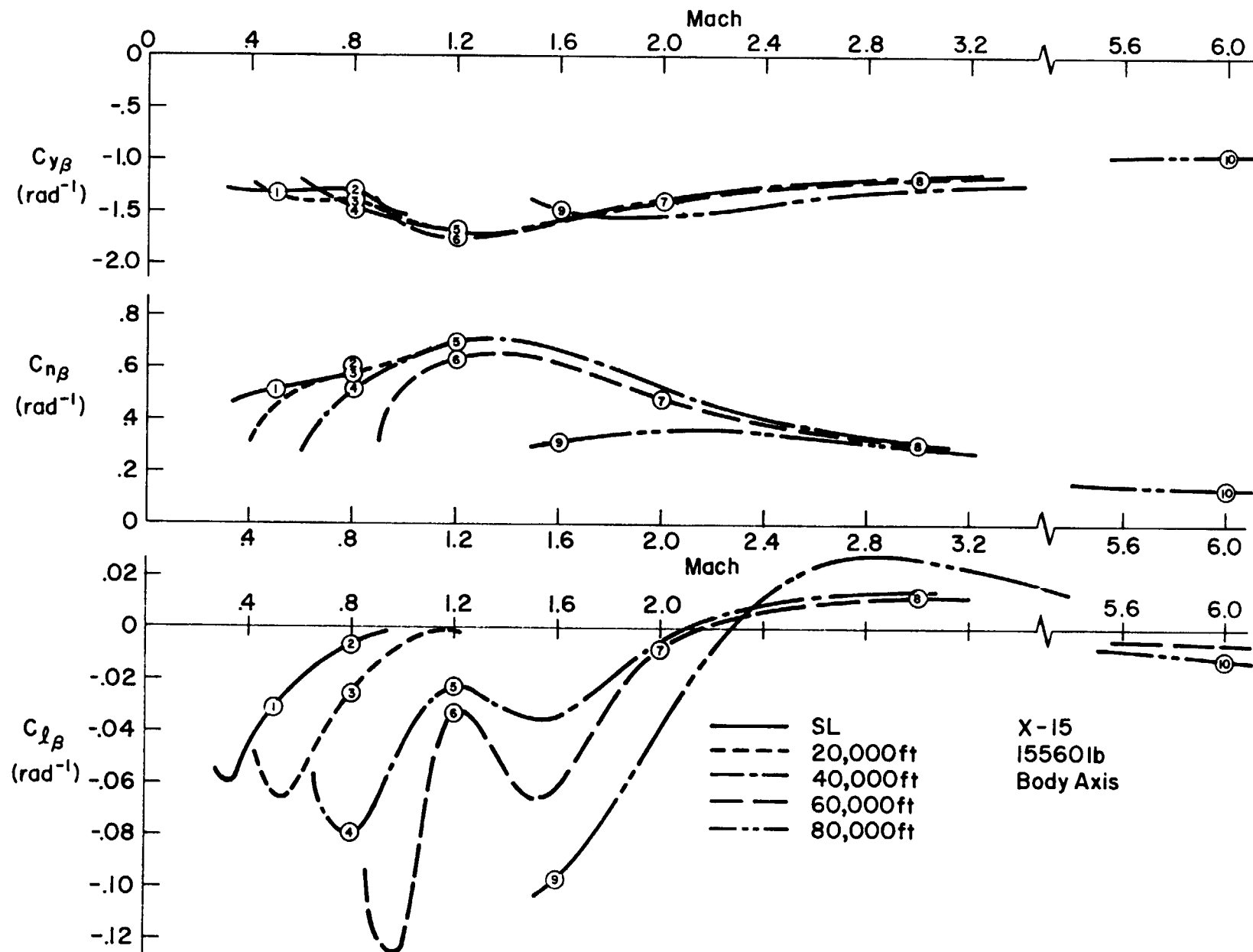
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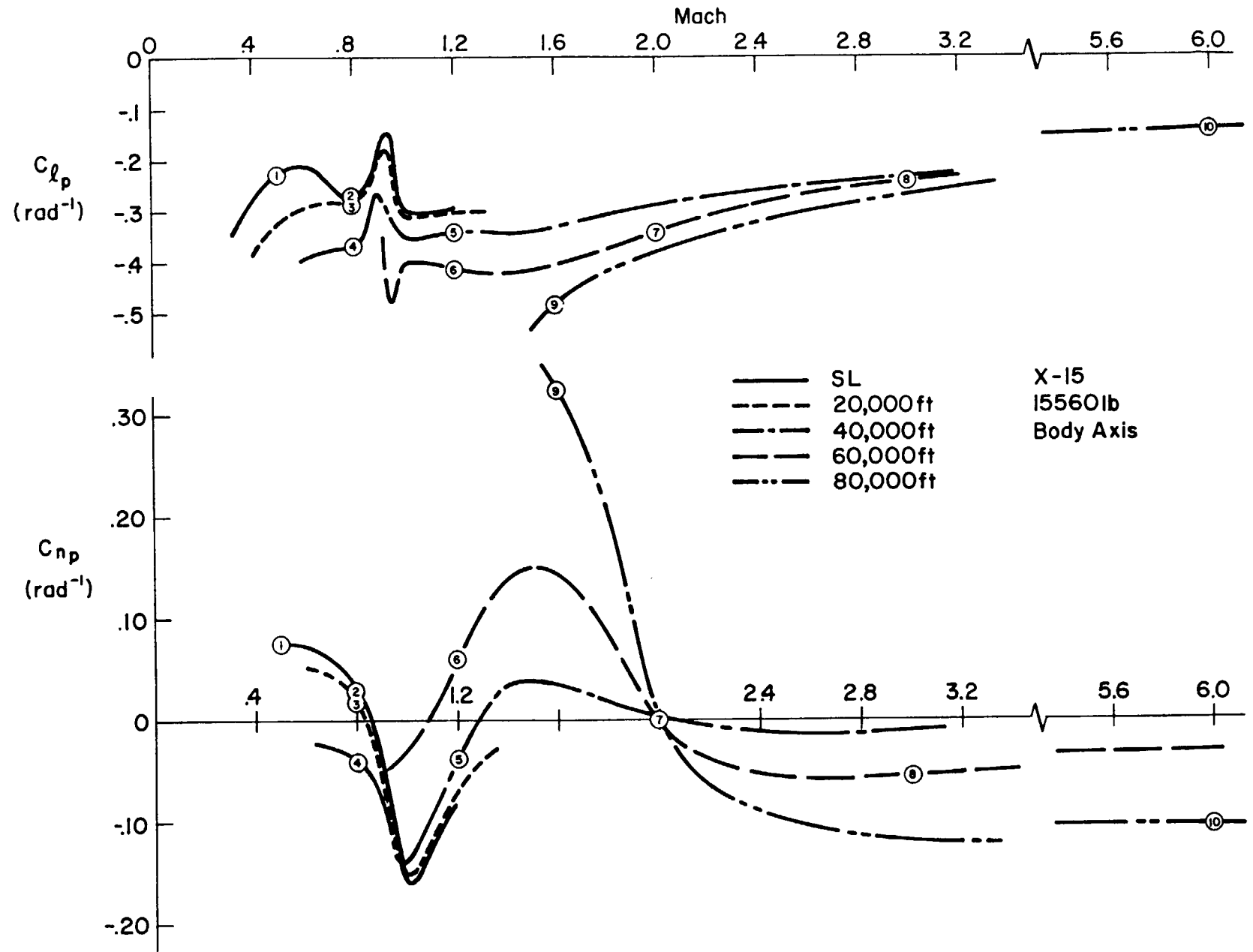


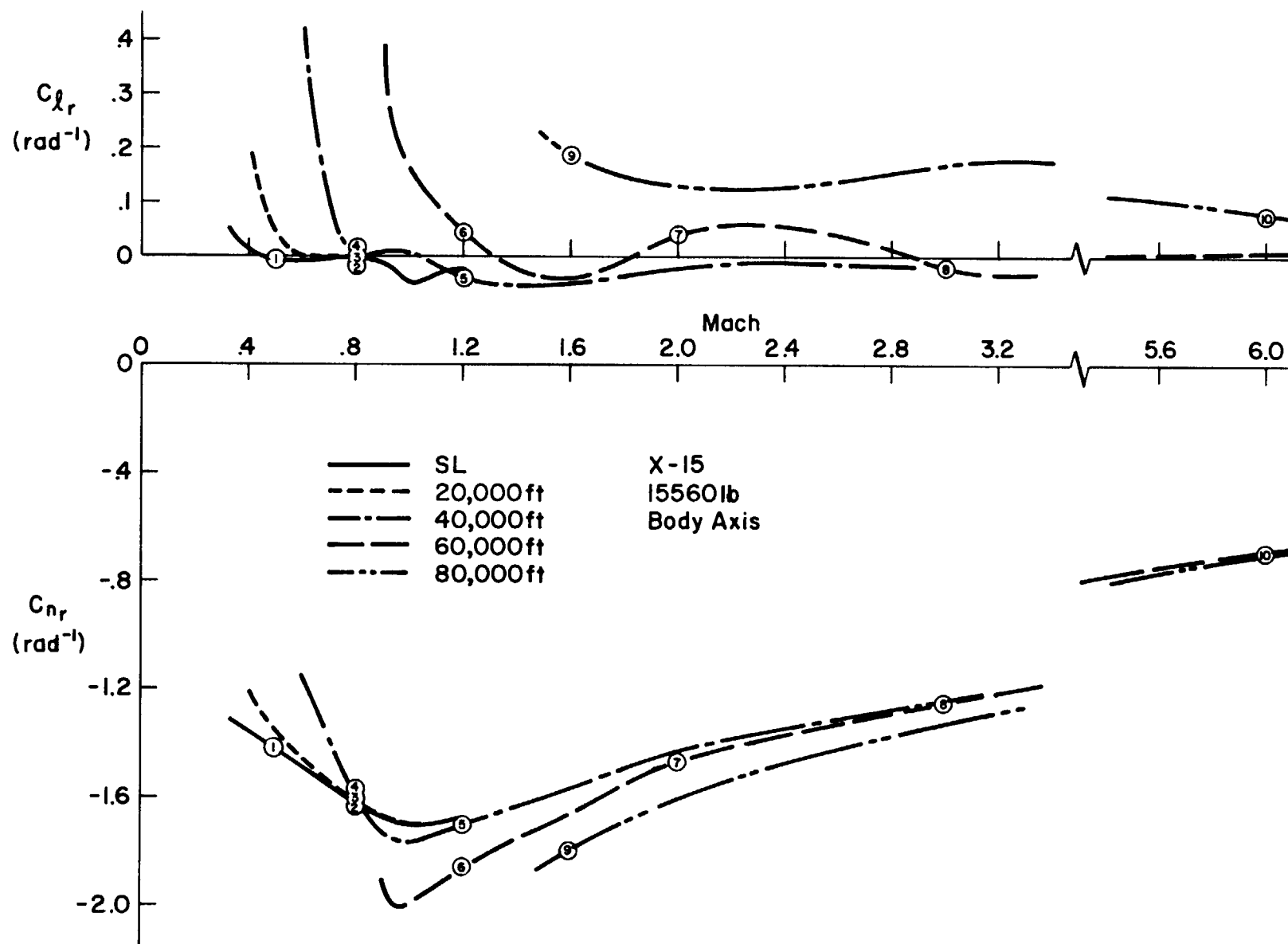




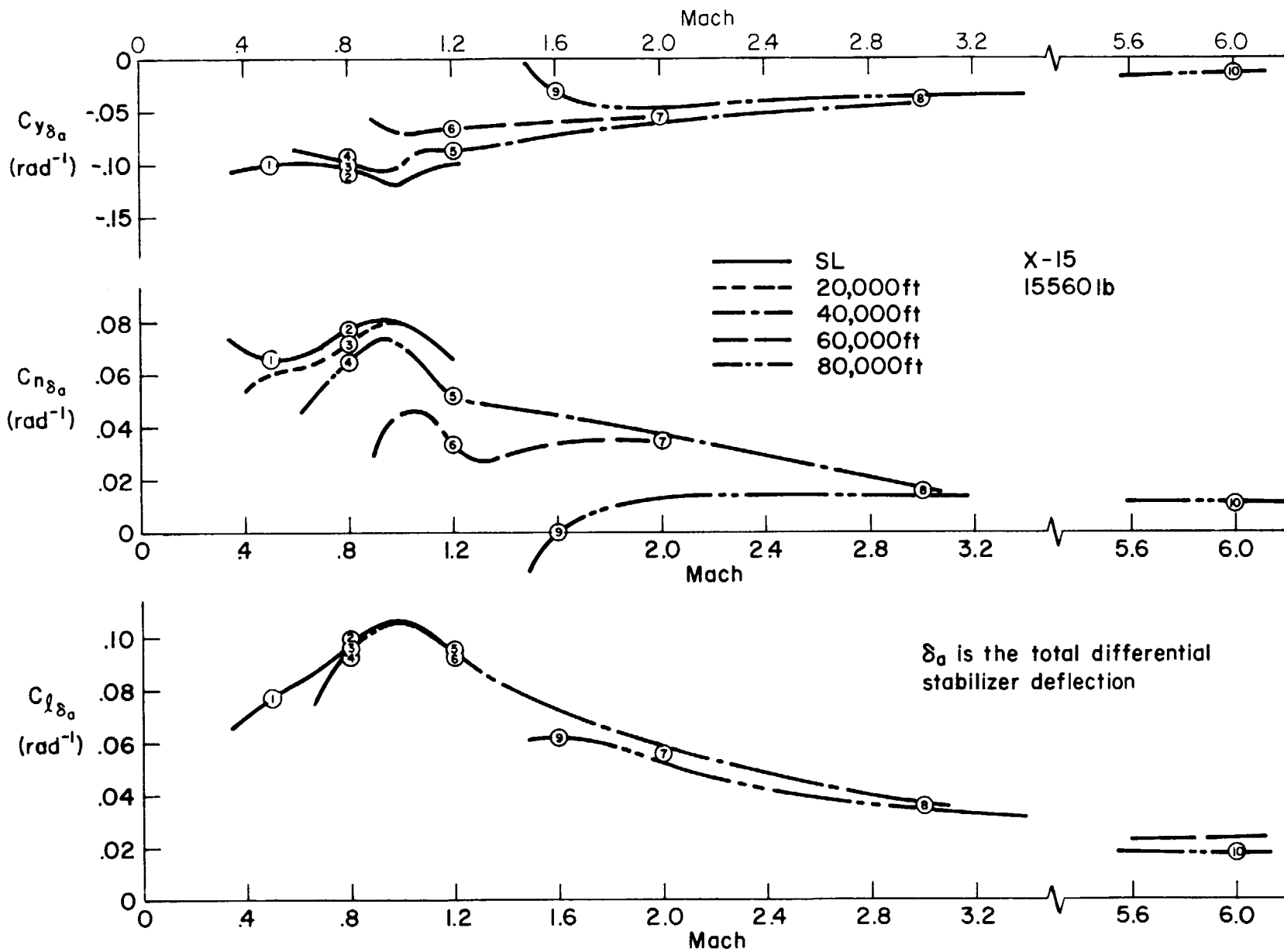








123



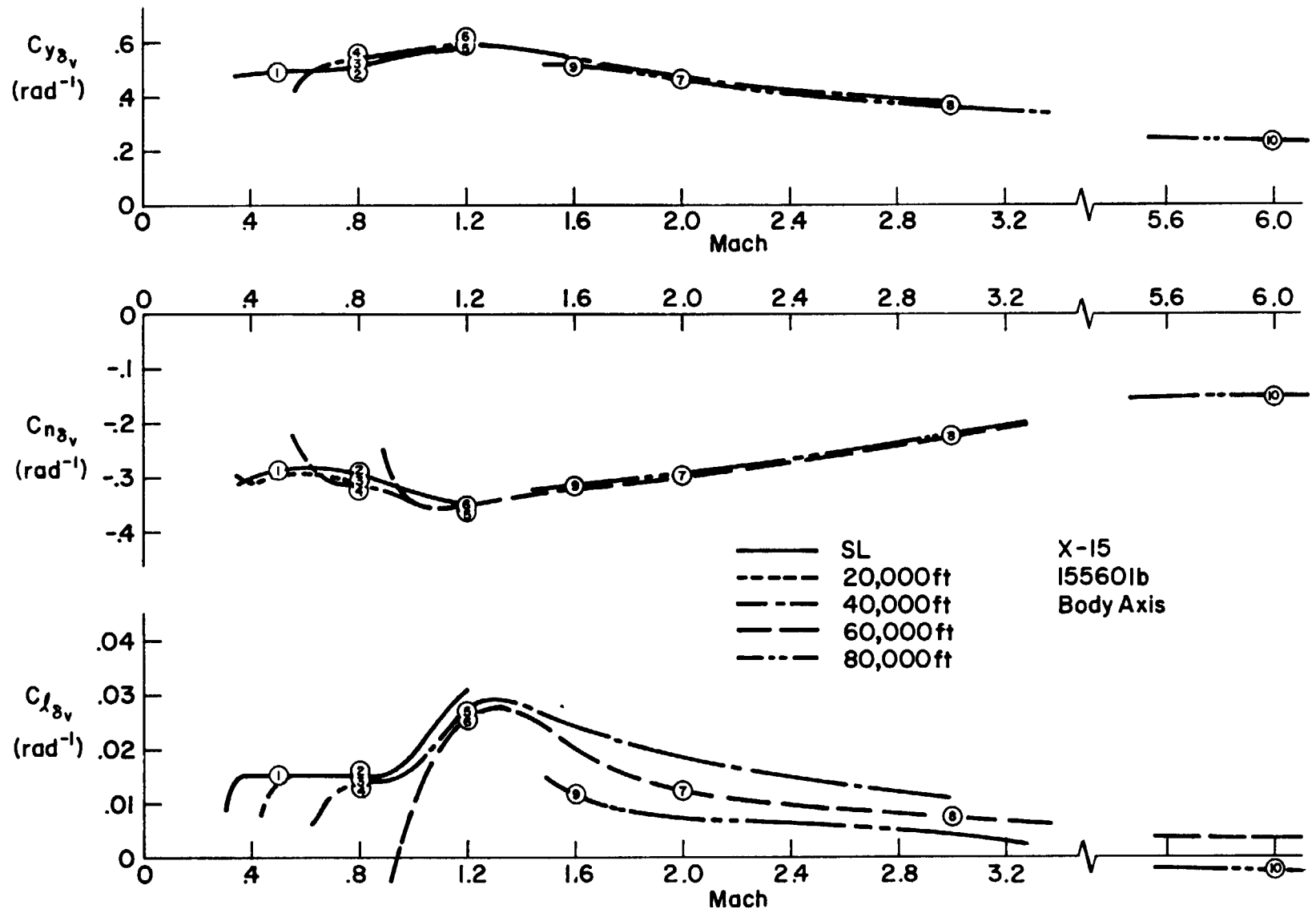


TABLE V-1

X-15 DIMENSIONAL, MASS, AND FLIGHT CONDITION PARAMETERS

$$S = 200 \text{ sq ft}, b = 22.36 \text{ ft}, \bar{c} = 10.27 \text{ ft}$$
[illegible]

TABLE V-2

X-15 LONGITUDINAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
XU *	-.0339	-.0601	-.0292	-.0134	-.0216	-.00516	-.00871	-.0101	-.00111	-.00916
ZU *	-.0471	-.0253	-.0335	-.0323	-.0281	-.0348	-.0117	-.0106	-.0113	-.00551
RU *	.000808	.000278	.000279	.000188	-.00199	.495E-4	.000471	.000210	.000529	.430E-4
XW	.0269	.00105	.0111	.0149	-.00810	-.00893	-.0190	-.0148	-.0127	-.00215
ZW	-1.01	-1.66	-.845	-.398	-.602	-.261	-.311	-.323	-.132	-.121
MW	-.0116	-.0123	-.00945	-.00559	-.00979	-.00511	-.00673	-.00548	-.00202	-.000820
ZWD	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ZU	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MWD	-.000250	-.000282	-.000150	-.923E-4	-.000124	-.472E-4	0.	.894E-5	-.607E-5	0.
MZ	-.735	-1.53	-.755	-.376	-.559	-.194	-.182	-.251	-.0482	-.107
XDS	11.2	9.78	10.4	10.8	9.27	9.21	6.24	4.85	7.11	5.64
ZDS	-160.	-431.	-198.	-79.6	-166.	-63.1	-89.2	-126.	-27.1	-108.
MDS	-13.8	-37.7	-17.4	-7.03	-15.5	-5.96	-9.80	-12.2	-2.85	-8.79
	+	+	+	+	+	+	+	+	+	+

TABLE V-3

X-15 STABILIZER TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	60 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
DENOMINATOR										
Z(DET)1	.247	.716	.338	.173	(-.0469)	.108	.264	.366	.262	.626
W(DET)1	.0773	.0424	.0452	.0419	(.0552)	.0317	.0232	.0158	.0294	.00788
Z(DET)2	.351	.467	.296	.200	.194	.104	.0675	.0683	.0489	.0519
W(DET)2	2.68	3.67	2.91	2.11	3.40	2.43	3.62	4.00	1.81	2.20
NUMERATORS										
N(U /DS)										
A(U)	11.2	9.78	10.4	10.8	9.27	9.21	6.24	4.85	7.11	5.64
1/T(U)1	47.8	1.22	72.6	67.8	.247	.0926	.0795	.0840	.0386	.0548
1/T(U)2	(.849)	1.98	(.928)	(.926)	.996	.420	.741	.913	.221	.212
1/T(U)3	(.853)	78.1	(.739)	(.321)	109.	109.	212.	282.	159.	479.
N(W /DS)										
A(W)	-160.	-431.	-198.	-79.6	-166.	-63.1	-89.2	-126.	-27.1	-108.
1/T(W)1	48.6	79.7	73.3	68.1	109.	109.	212.	282.	159.	479.
Z(W)1	.299	.967	.394	.166	.876	.100	.261	.431	-.0363	.930
W(W)1	.0555	.0310	.0367	.0367	.0124	.0309	.0160	.0117	.0184	.00552
N(TH/DS)										
A(TH)	-13.7	-37.6	-17.3	-7.02	-15.5	-5.96	-9.80	-12.3	-2.85	-8.79
1/T(TH)1	.0344	.0600	.0293	.0138	.0226	.00218	.00688	.00919	-.00270	.00898
1/T(TH)2	.881	1.52	.738	.334	.498	.210	.251	.267	.116	.111
N(HD /DS)										
A(HD)	161.	431.	198.	80.3	166.	63.8	89.4	126.	28.0	108.
1/T(HD)1	.0270	.0586	.0256	.00439	.0209	-.00661	.00482	.00833	-.0121	.00849
1/T(HD)2	-6.03	-10.0	-6.87	-4.46	-6.98	-4.47	-7.13	-8.52	-3.90	-7.21
1/T(HD)3	6.92	11.8	7.75	4.93	7.69	4.74	7.32	8.75	3.99	7.32
N(AZP/DS)										
A(AZP)	98.0	276.	128.	52.4	125.	48.9	95.0	104.	26.6	57.7
1/T(AZP)1	-.00446	-.000827	-.00217	-.00711	-.00157	.00116	-.00134	-.000451	-.00155	-.000297
1/T(AZP)2	.0312	.0593	.0276	.0110	.0223	-.00789	.00605	.00873	-.0100	.00872
Z(AZP)1	.0540	.0411	.0286	.0141	.0135	.0148	.0224	.0166	.0210	.00597
W(AZP)1	8.28	13.6	9.10	5.83	8.44	5.37	7.04	9.53	4.21	9.94

TABLE V-4

X-15 STABILIZER TRANSFER FUNCTION FACTORS

SAS On

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
H	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
M										
DENOMINATOR										
1/T(DET)1	1.53	.00933	1.35	1.29	-.0350	(.0957)	3.24	2.58	(.242)	.946
1/T(DET)2	10.7	.0517	13.4	4.82	.0491	(.0295)	4.58	7.16	(.0285)	5.88
Z(DET)1	.338	(1.89)	.474	.188	(1.53)	.951	.262	.378	.622	.660
W(DET)1	.0514	(29.7)	.0309	.0355	(11.4)	2.62	.0218	.0147	1.87	.00734
NUMERATORS										
N(U /DS)										
A(U)	11.2	9.78	10.4	10.8	5.27	9.21	6.24	4.85	7.11	5.64
1/T(U)1	47.8	1.22	72.6	67.8	.247	.0926	.0796	.0840	.0386	.0548
1/T(U)2	(.849)	1.99	(.928)	(.926)	.996	.420	.741	.913	.221	.212
1/T(U)3	(.853)	78.1	(.739)	(.321)	109.	109.	212.	282.	159.	479.
N(W /DS)										
A(W)	-160.	-431.	-198.	-79.6	-166.	-63.1	-89.2	-126.	-27.1	-108.
1/T(W)1	48.6	79.7	73.3	68.1	109.	109.	212.	282.	159.	479.
Z(W)1	.299	.967	.394	.166	.876	.100	.261	.431	-.0363	.830
W(W)1	.0555	.0310	.0367	.0367	.0124	.0309	.0160	.0117	.0184	.00552
N(TH/DS)										
A(TH)	-13.7	-37.6	-17.3	-7.02	-15.5	-5.96	-9.80	-12.3	-2.85	-8.79
1/T(TH)1	.0344	.0600	.0293	.0138	.0226	.00218	.00688	.00919	-.00270	.00699
1/T(TH)2	.881	1.52	.738	.334	.498	.210	.251	.267	.116	.111
N(HD /DS)										
A(HD)	161.	431.	198.	80.3	166.	63.8	89.4	126.	28.0	108.
1/T(HD)1	.0270	.0586	.0256	.00439	.0209	-.00661	.00482	.00833	-.0121	.00849
1/T(HD)2	-6.03	-10.0	-6.87	-4.46	-6.98	-4.47	-7.13	-8.52	-3.90	-7.21
1/T(HD)3	6.92	11.8	7.75	4.93	7.69	4.74	7.32	8.75	3.99	7.32
N(AZP/DS)										
A(AZP)	98.0	276.	128.	52.4	125.	48.9	95.0	104.	26.6	57.7
1/T(AZP)1	-.00446	-.000827	-.00217	-.00711	-.00157	.00116	-.00134	-.000451	-.00155	-.000297
1/T(AZP)2	.0312	.0593	.0276	.0110	.0223	-.00789	.00605	.00873	-.0100	.00872
Z(AZP)1	.0540	.0411	.0286	.0141	.0135	.0148	.0224	.0166	.0210	.00597
W(AZP)1	8.28	13.6	9.10	5.83	8.44	5.32	7.04	9.53	4.21	9.94

TABLE V-5

X-15 LONGITUDINAL HANDLING QUALITIES PARAMETERS

SAS Off

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
STICK FIXED										
D(G)/D(U) (DEG/KT)	-.0812	-.176	-.0769	-.0132	-.0629	.0198	-.0145	-.0250	.0362	-.0255
NZA (G/RAD)	15.0	41.2	18.8	7.92	17.8	7.37	15.0	24.0	5.32	20.2
DE/G (DEG/G)	1.96	.487	1.47	4.54	2.41	7.71	5.10	3.11	11.5	1.55
CAP (RAD/SEC/SEC/G)	.471	.320	.445	.556	.652	.801	.872	.666	.574	.238
PHUGGIO(2) (SEC) (TUCK(2))	--	--	--	--	(14.8)	--	--	--	--	--
1/C(1/10)	1.02	1.44	.846	.557	.539	.287	.185	.187	.133	.141
+	+	+	+	+	+	+	+	+	+	+

TABLE V-6

X-15 LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
YV	-.357	-.571	-.304	-.137	-.241	-.0951	-.127	-.163	-.0414	-.0997
YB	-197.	-510.	-252.	-106.	-279.	-110.	-246.	-474.	-64.8	-565.
LB'	-12.4	-1.96	-11.7	-16.3	-8.76	-5.33	-2.36	16.6	-12.3	-20.1
NB'	10.4	31.0	13.7	4.89	15.1	5.21	11.1	15.7	1.76	11.2
LP'	-2.54	-3.93	-2.09	-1.16	-1.60	-.738	-1.02	-1.08	-.448	-.507
NP'	.0129	-.00384	-.00862	-.0139	-.0198	-.000503	-.00735	-.0188	.00998	-.0109
LK'	-.184	-.170	-.0830	-.0353	-.245	.0570	.103	-.131	.164	.261
NR'	-.576	-1.05	-.513	-.219	-.356	-.149	-.196	-.251	-.0727	-.106
Y*DA	-.0274	-.0461	-.0217	-.00895	-.0120	-.00353	-.00498	-.00543	-.000840	-.00157
L'DA	35.2	113.	52.2	21.1	46.5	17.8	28.7	42.3	8.05	33.0
N'DA	1.59	4.85	2.09	.778	1.46	.403	.993	1.08	.0579	1.13
Y*DV	.137	.224	.113	.0509	.0821	.0326	.0426	.0503	.0143	.0241
L'DV	5.87	15.0	6.60	2.55	11.9	4.21	5.38	6.88	1.20	-6.54
N'DV	-5.81	-14.9	-7.09	-2.97	-7.50	-2.88	-6.90	-11.7	-1.81	-12.2
	+	+	+	+	+	+	+	+	+	+

TABLE V-7

X-15 ALLERON TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
DE Nominator										
1/T(DET)1	.0149	.00132	.00734	.0176	.00608	-.000447	-.00215	-.00185	.00863	-.000987
1/T(DET)2	2.46	3.93	2.06	.991	1.59	.679	1.01	1.09	.207	.503
Z(DET)1	.148	.144	.110	.0957	.0754	.0623	.0503	.0524	.0792	.0302
w(DET)1	3.36	5.63	3.80	2.64	3.96	2.43	3.35	3.89	2.10	3.50
NUMERATORS										
N(B /DA)										
A(B)	-.0274	-.0461	-.0217	-.00896	-.0120	-.00353	-.00498	-.00543	-.000840	-.00157
1/T(B)1	-27.3	-.347	-26.3	-229.	-93.2	-615.	-202.	-98.7	-2366.	.0479
1/T(B)2	(-.560)	5.43	(-.306)	(.705)	(.551)	(.734)	(.121)	(.984)	(.665)	.727
1/T(B)3	(1.25)	49.2	(1.35)	(.306)	(.634)	(.185)	(.308)	(.466)	(.0700)	-380.
N(P /DA)										
A(P)	35.2	113.	52.2	21.1	46.5	17.8	28.7	42.3	8.05	33.0
1/T(P)1	-.00396	-.000803	-.00201	-.00555	-.00154	-.00403	-.00116	-.000425	-.00530	-.000287
Z(P)1	.140	.143	.109	.0783	.0754	.0544	.0490	.0523	.0456	.0314
w(P)1	3.34	5.63	3.78	2.34	3.93	2.30	3.34	3.92	1.34	3.45
N(R /DA)										
A(R)	1.59	4.85	2.09	.778	1.46	.403	.993	1.08	.0579	1.13
1/T(R)1	.895	1.95	.770	.310	.501	.190	.239	.289	.0786	.105
Z(R)1	.267	.279	.150	.0615	.0559	.0493	.0677	.0152	.110	-.0111
w(R)1	3.96	3.67	4.22	4.45	5.20	5.83	4.74	4.78	8.07	4.27
N(PHI /DA)										
A(PHI)	35.3	114.	52.3	21.2	46.6	17.9	28.8	42.3	8.07	33.1
Z(PHI)1	.141	.144	.109	.0772	.0753	.0534	.0490	.0522	.0434	.0312
w(PHI)1	3.34	5.63	3.78	2.36	3.93	2.32	3.35	3.92	1.38	3.45
N(AYP/DA)										
A(AYP)	91.9	300.	136.	54.1	116.	42.7	72.2	97.6	17.5	84.7
1/T(AYP)1	-.387	-.344	-.235	-.196	-.219	-.136	-.157	-.107	1.31	.0383
1/T(AYP)2	.606	.696	.408	-.376	-.325	-.296	-.192	-.360	-1.41	-.396
Z(AYP)1	.154	.138	.114	.118	.0823	.0997	.0591	.0680	.920	.0777
w(AYP)1	3.33	5.52	3.79	2.31	3.95	2.19	3.27	3.80	.150	3.08

TABLE V-8

X-15 VERTICAL STABILIZER TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
DEALMINATOR										
1/T(DET)1	.0149	.00132	.00734	.0176	.00608	-.000447	-.00216	-.00185	.00863	-.000987
1/T(DET)2	2.46	3.93	2.06	.991	1.59	.679	1.01	1.09	.207	.503
Z(DET)1	.148	.144	.110	.0957	.0754	.0623	.0503	.0524	.0792	.0302
W(DET)1	3.36	5.63	3.80	2.64	3.96	2.43	3.35	3.89	2.19	3.50
NUMERATORS										
N(B /DV)										
A(B)	.137	.224	.113	.0509	.0821	.0326	.0425	.0503	.0143	.0241
1/T(B)1	.0128	.0101	.00807	.00215	.0117	.00181	-.000362	.00237	-.0110	-.00393
1/T(B)2	2.45	3.90	2.05	1.10	1.56	.656	.989	1.08	.368	.495
1/T(B)3	45.9	65.1	66.2	64.7	59.8	106.	171.	237.	144.	492.
N(P /DV)										
A(P)	5.87	15.0	6.60	2.55	11.9	4.21	5.38	6.88	1.20	-6.54
1/T(P)1	-.00417	-.000802	-.00193	-.00561	-.00154	-.00402	-.00115	-.000425	-.00540	-.000287
Z(P)1	(-.988)	.162	.319	(-3.70)	.111	.0673	.0304	.0572	(3.90)	.0550
W(P)1	(1.82)	5.45	1.11	(3.78)	3.11	1.24	2.84	6.62	(-4.17)	6.98
N(R /DV)										
A(R)	-5.81	-14.9	-7.09	-2.97	-7.50	-2.88	-6.90	-11.7	-1.81	-12.2
1/T(R)1	2.49	-.510	-.112	.365	.352	.157	.190	.255	.0790	.106
Z(R)1	.356	(.513)	(.179)	.367	(-.612)	(-.393)	(-.417)	(-.691)	.116	.201
W(R)1	.208	(4.04)	(2.12)	1.17	(1.96)	(1.01)	(1.31)	(1.63)	1.68	1.16
N(PHI /DV)										
A(PHI)	5.46	14.7	6.23	2.15	11.5	3.79	4.90	5.44	.724	-7.18
Z(PHI)1	(-1.10)	.155	.271	(4.04)	.103	.0361	.0135	.0530	(4.99)	.0563
W(PHI)1	(1.79)	5.51	1.14	(-4.19)	3.17	1.32	2.98	6.85	(-5.77)	6.67
N(AYP /DV)										
A(AYF)	-19.8	-46.9	-24.9	-10.7	-19.4	-7.07	-35.5	-58.0	-9.07	-103.
1/T(AYP)1	.00785	.0492	.0100	-.0278	.0238	.00545	.00159	.00518	-.0208	-.00486
1/T(AYP)2	2.60	6.05	2.27	.620	2.69	.787	1.17	1.42	.215	.418
Z(AYF)1	-.0739	-.498	-.0937	.0724	-.0924	.0224	-.00935	-.0362	.0430	.00863
W(AYF)1	4.28	4.47	4.37	4.61	4.70	4.16	4.44	6.67	4.25	7.78

+ + + + + + + + + +

TABLE V-9

X-15 AILERON TRANSFER FUNCTION FACTORS

SAS On

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
DENOMINATOR										
1/T(DET)1	-.0464	-.0329	-.0343	-.0315	-.0277	-.0312	-.0173	-.0131	-.00528	-.00250
1/T(DET)2	19.3	58.2	27.1	11.1	24.0	9.30	14.8	21.6	3.58	16.4
Z(DET)1	.419	.556	.408	.322	.390	.298	.385	.518	.507	.564
w(DET)1	3.37	5.75	3.81	2.33	3.93	2.21	3.32	2.94	1.51	3.52
NUMERATORS										
N(B /DA)										
A(B)	-.0274	-.0461	-.0217	-.00896	-.0120	-.00353	-.00498	-.00543	-.000840	-.00157
1/T(B)1	-33.3	5.02	1.14	.362	.557	.210	.273	.293	.0867	.102
1/T(B)2	(.854)	-3.23	6.23	1.33	5.26	1.12	3.94	10.4	.580	10.7
1/T(B)3	(2.31)	51.9	-35.6	-230.	-99.1	-617.	-207.	-108.	-2266.	-392.
N(P /DA)										
A(P)	35.2	113.	52.2	21.1	46.5	17.8	28.7	42.3	8.05	33.0
1/T(P)1	-.00388	-.000790	-.00198	-.00554	-.00152	-.00400	-.00115	-.000416	-.00533	-.000281
Z(P)1	.410	.553	.397	.273	.374	.248	.365	.500	.248	.547
w(P)1	3.37	5.67	3.81	2.35	2.95	2.31	3.36	2.96	1.35	3.49
N(R /DA)										
A(R)	1.59	4.85	2.09	.778	1.46	.403	.993	1.08	.0579	1.13
1/T(R)1	.895	1.95	.770	.310	.501	.190	.239	.289	.0786	.105
Z(R)1	.267	.279	.150	.0615	.0559	.0493	.0677	.0152	.110	-.0111
w(R)1	3.96	3.67	4.22	4.45	5.20	5.83	4.74	4.78	8.07	4.27
N(PHI /DA)										
A(PHI)	35.3	114.	52.3	21.2	46.6	17.9	28.8	42.3	8.07	33.1
Z(PHI)1	.409	.553	.397	.270	.373	.245	.364	.500	.230	.546
w(PHI)1	3.37	5.67	3.81	2.36	3.96	2.33	3.36	3.96	1.39	3.49
N(AYP/UA)										
A(AYP)	91.9	300.	136.	54.1	116.	42.7	72.2	97.6	17.5	84.7
1/T(AYP)1	-.635	-.533	.488	.239	.312	.164	.208	.228	.0826	.0967
1/T(AYP)2	.668	.820	-.578	-.670	-.656	-.672	-.744	-1.21	-1.65	-1.95
Z(AYP)1	.412	.522	.389	.334	.374	.342	.390	.523	.827	.597
w(AYP)1	3.46	5.58	3.90	2.46	4.10	2.38	3.52	4.35	1.39	4.38

TABLE V-10

X-15 VERTICAL STABILIZER TRANSFER FUNCTION FACTORS

SAS On

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
DENOMINATOR										
1/T(DET)1	-.0464	-.0328	-.0343	-.0315	-.0277	-.0312	-.0173	-.0131	-.00528	-.00250
1/T(DET)2	19.3	58.2	27.1	11.1	24.0	9.30	14.8	21.6	3.58	16.4
Z(DET)1	.419	.556	.408	.322	.390	.298	.385	.518	.507	.564
W(DET)1	3.37	5.75	3.81	2.33	3.93	2.21	3.32	3.94	1.51	3.52
NUMERATORS										
N(B /DV)										
A(B)	.137	.224	.113	.0509	.0821	.0326	.0426	.0503	.0143	.0241
1/T(B)1	-.0492	-.0306	-.0348	-.0428	-.0250	-.0300	-.0160	-.0101	-.0286	-.00544
1/T(B)2	17.5	56.9	25.6	9.37	22.1	7.11	14.0	21.2	2.99	16.4
1/T(B)3	49.0	73.6	69.1	67.2	103.	109.	172.	239.	146.	492.
N(P /DV)										
A(P)	5.87	15.0	6.60	2.55	11.9	4.21	5.38	6.88	1.20	-6.54
1/T(P)1	-.00206	-.00100	.00252	-.00536	-.00173	-.00474	-.00133	-.000458	-.00534	-.000275
1/T(P)2	.230	-.469	.0371	1.17	-.691	-.250	-.427	-1.33	2.09	1.95
1/T(P)3	-15.8	-50.6	-25.5	-12.5	-12.4	-5.26	-16.4	-30.6	-7.86	26.1
N(R /DV)										
A(R)	-5.81	-14.9	-7.09	-2.97	-7.50	-2.88	-6.90	-11.7	-1.21	-12.2
1/T(R)1	20.9	-.0885	-.0147	11.9	-.107	-.0770	-.0670	-.0898	.0863	16.6
Z(R)1	.811	(.189)	(.0987)	.448	(.151)	(.0810)	(.0983)	(.141)	(.087)	.857
W(R)1	.0719	(.63.1)	(.29.2)	.204	(.26.1)	(.9.98)	(.15.8)	(.22.6)	(.3.74)	.0927
N(PHI/DV)										
A(PHI)	5.46	14.7	6.23	2.15	11.5	3.79	4.90	6.44	.724	-7.18
1/T(PHI)1	.219	-.455	.0423	1.06	-.639	-.211	-.391	-1.26	2.01	1.85
1/T(PHI)2	-14.5	-53.2	-28.7	-16.8	-13.9	-7.03	-19.7	-34.5	-14.5	25.2
N(AYP/DV)										
A(AYP)	-19.8	-46.9	-24.9	-10.7	-19.4	-7.07	-35.5	-58.0	-9.07	-103.
1/T(AYP)1	-.0508	-.0289	-.0351	-.0465	-.0243	-.0298	-.0159	-.00994	-.0298	-.00547
1/T(AYP)2	44.9	149.	62.4	24.2	80.4	30.0	26.7	36.1	4.86	17.7
Z(AYP)1	-.0560	-.206	-.0680	.119	-.0320	.0670	.0242	.0106	.492	.0515
W(AYP)1	2.82	3.61	3.01	2.19	3.26	2.24	3.51	5.79	2.50	6.96
	+	+	+	+	+	+	+	+	+	+

TABLE V-11

X-15 LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	20 K	40 K	40 K	60 K	60 K	60 K	80 K	80 K
M	.500	.800	.800	.800	1.20	1.20	2.00	3.00	1.60	6.00
DR PERIOD (SEC)	1.89	1.13	1.67	2.39	1.59	2.59	1.88	1.62	2.88	1.79
1/C(1/2)	1.36	1.32	1.01	.871	.685	.566	.456	.476	.720	.274
SPIRAL (2) (SEC)	--	--	--	--	--	1550.	321.	375.	--	702.
P(1)	13.6	28.8	24.5	15.8	28.2	23.0	--	39.5	3.99	--
P(2)	--	--	--	14.7	--	22.5	--	--	2.96	--
P(3)	--	--	--	15.6	--	23.0	--	--	8.59	--
P(2)/P(1)	--	--	--	.929	--	.982	--	--	.742	--
P(OSC)/P(AV)	--	--	--	.0326	--	.0101	--	--	.360	--
W(PHI)/W(D)	.993	1.00	.997	.894	.994	.954	.998	1.01	.631	.985
DEL-B-MAX	.0324	.132	.0384	.398	.100	.543	.153	.0631	.685	.104
PHI TO BETA, PHASE	22.3	-3.41	17.5	14.2	9.05	13.9	30.0	191.	3.58	7.48
PHI TO BETA	.888	.0391	.699	2.14	.484	.755	.144	1.11	2.46	1.58
PHI TO VE	.0911	.00251	.0662	.318	.0480	.121	.0139	.0709	.472	.0809
	+	+	+	+	+	+	+	+	+	+

X-15 DATA SOURCES

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SECTION VI

HL-10

HL-10 BACKGROUND

The HL-10 is one of a number of lifting body research vehicles. The airplane is typically launched from a B-52 at 0.8 Mach and 45,000 feet. In numerous glide and powered flights the HL-10 has been flown in excess of 1.8 Mach and 90,000 feet.

Following problems involving the loss of roll-control effectiveness, the leading edge of the tip fins was modified. This became known as the Mod II configuration. The information contained here is for the Mod II HL-10.

Pitch and roll control is obtained by elevons and yaw control by a conventional rudder. A subsonic or a transonic configuration is selected using combinations of speed brakes, elevon flaps, and tip fin flaps. These combinations are specified in Fig. VI-1.

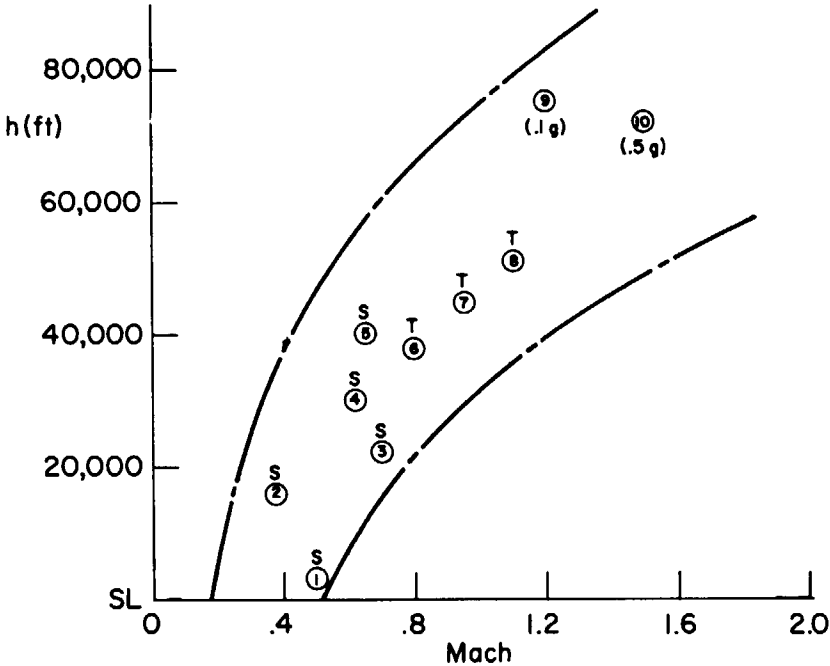
The stability augmentation system consists of angular rate feedback loops about all three axes.

The flight conditions shown correspond to actual flight test points.

Nominal Configuration

Zero fuel (burnout)
Gear up
Transonic or subsonic configuration
depending upon flight condition
W = 6466 lb
c.g. at .517 $\bar{c}$, W.L. 94.4
 $I_x = 1353 \text{ slug-ft}^2$
 $I_y = 6413 \text{ slug-ft}^2$
 $I_z = 7407 \text{ slug-ft}^2$
 $I_{xz} = 399 \text{ slug-ft}^2$ } Body Axis

Flight Envelope



Nominal Envelope Extremes

Transfer Function Case n
(S $\approx$ Subsonic, T $\approx$ Transonic)

Note:

Configuration	Speed Brakes	Elevon Flaps	Tip-Fin Flaps
Subsonic	Zero	Zero	Zero
Transonic	8°	30°	30.5°/32.5°

Figure VI-1. HL-10 Flight Conditions

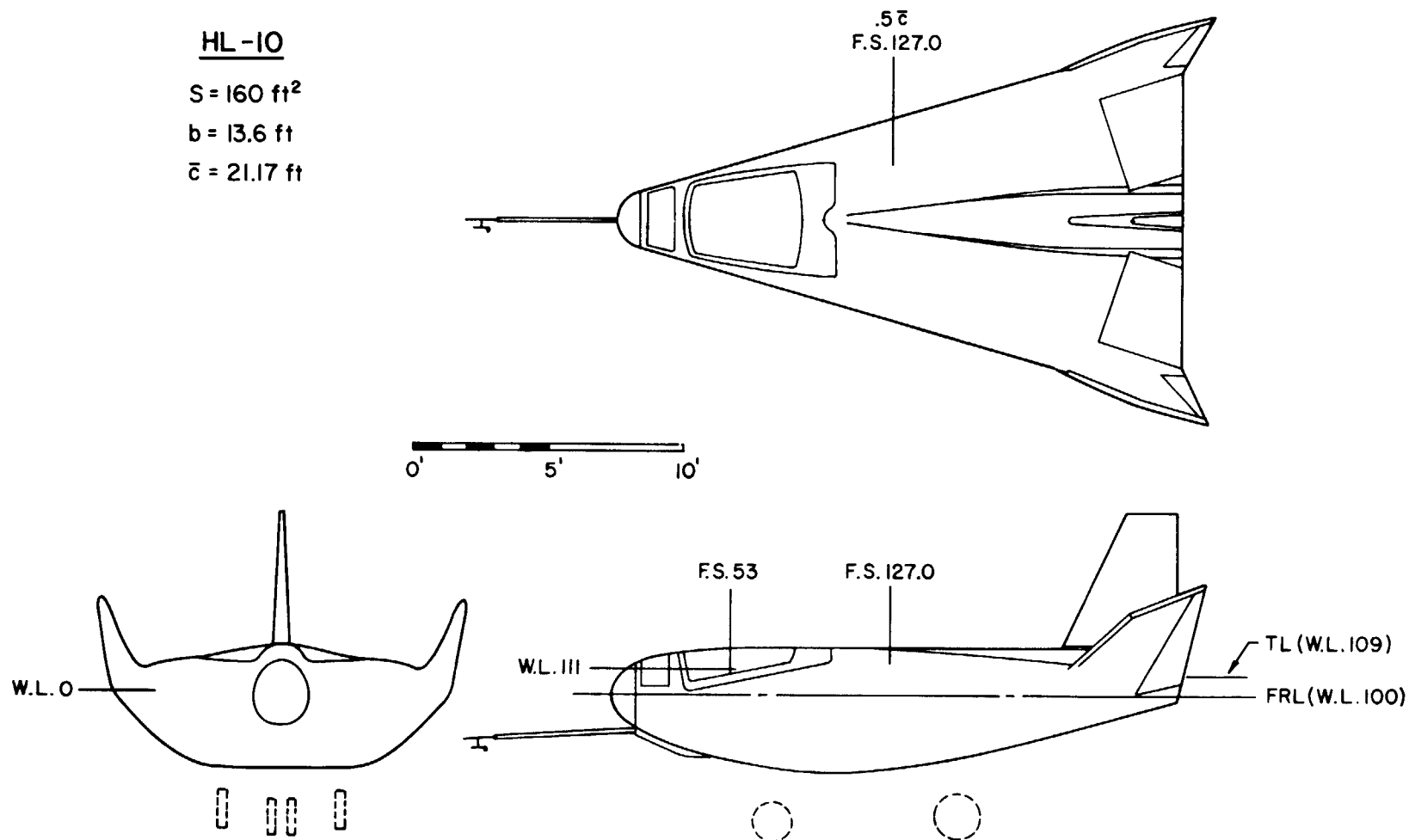
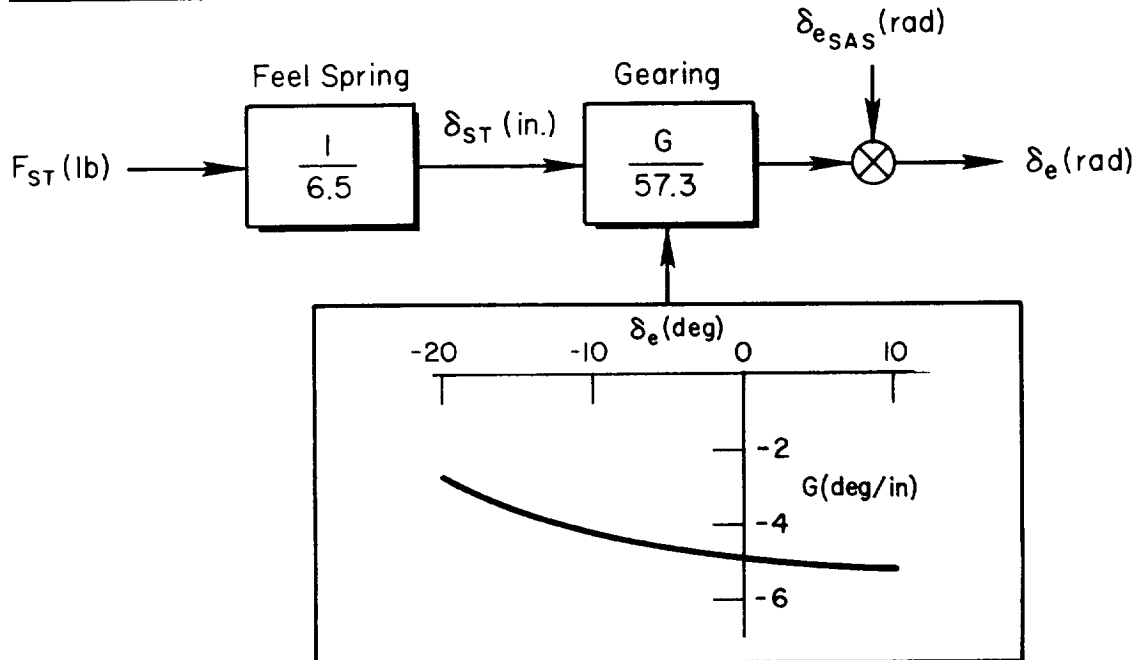


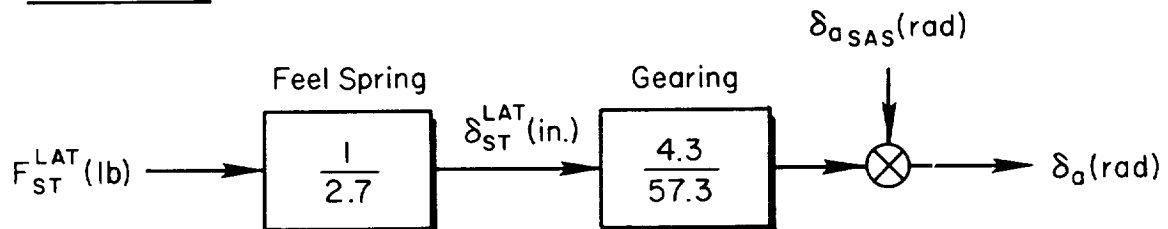
Figure VI-2. HL-10 General Arrangement

HL-10

PITCH AXIS



ROLL AXIS



YAW AXIS

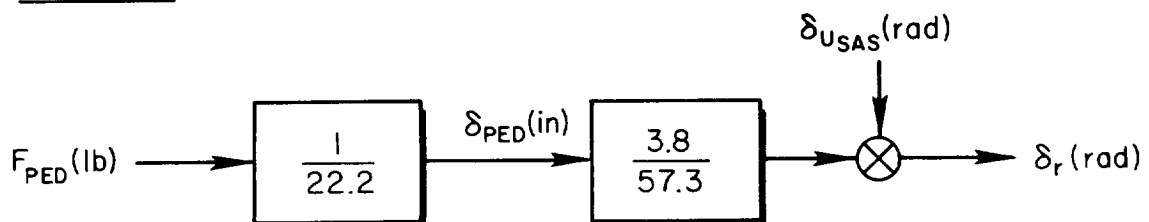
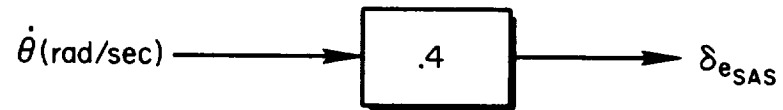


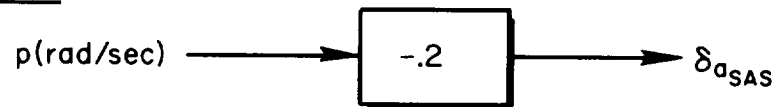
Figure VI-3. HL-10 Control System

HL - 10

PITCH SAS



ROLL SAS



YAW SAS

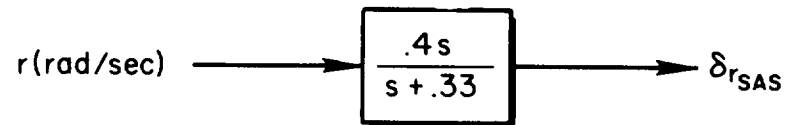
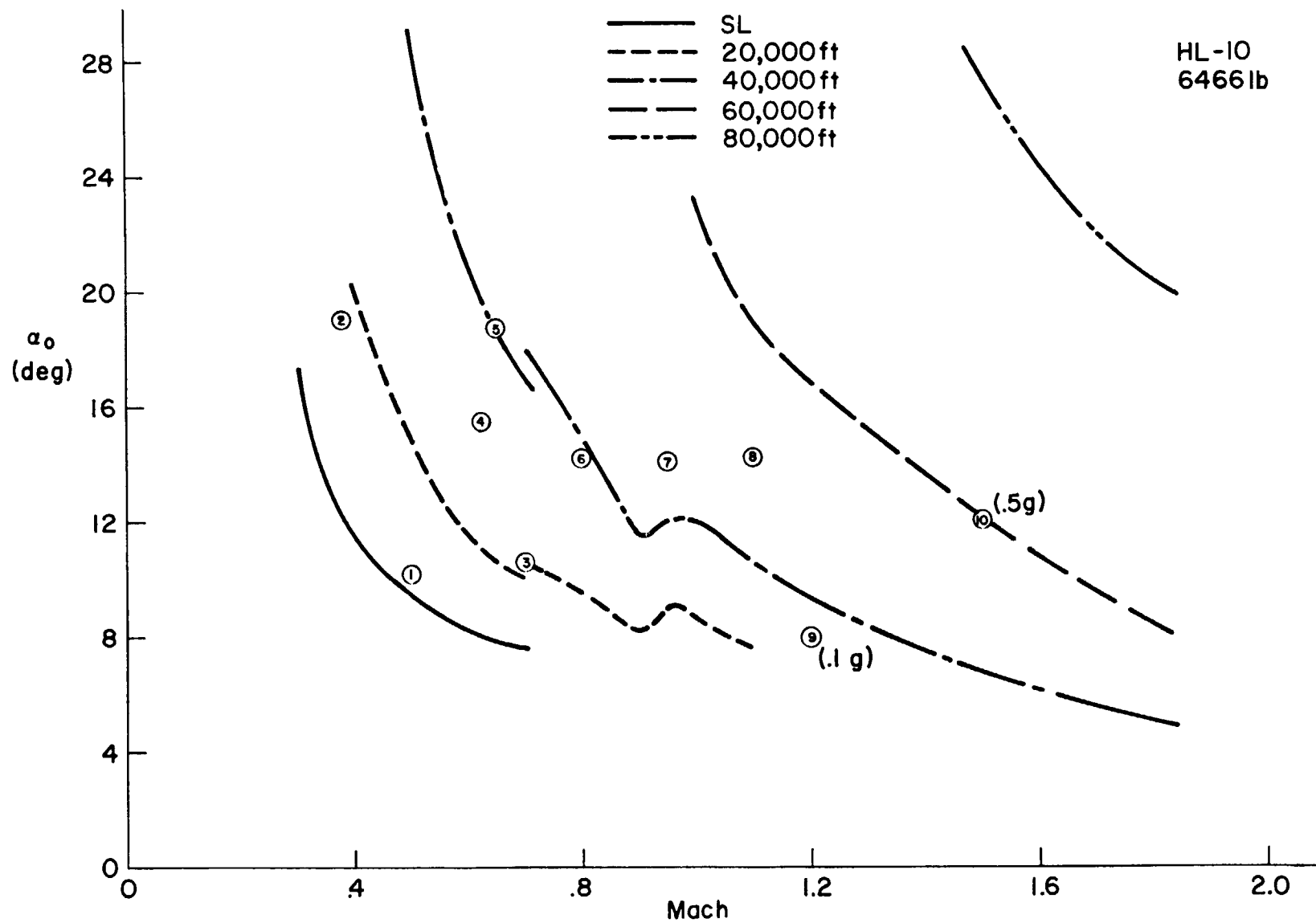
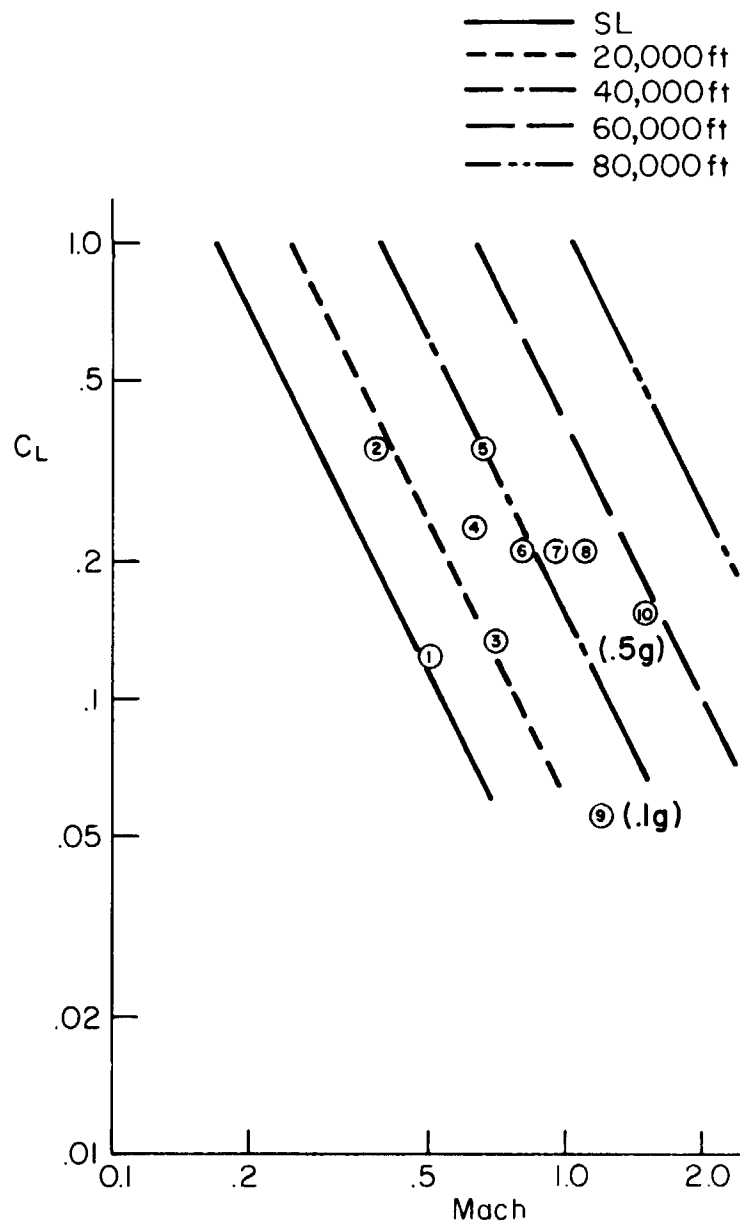
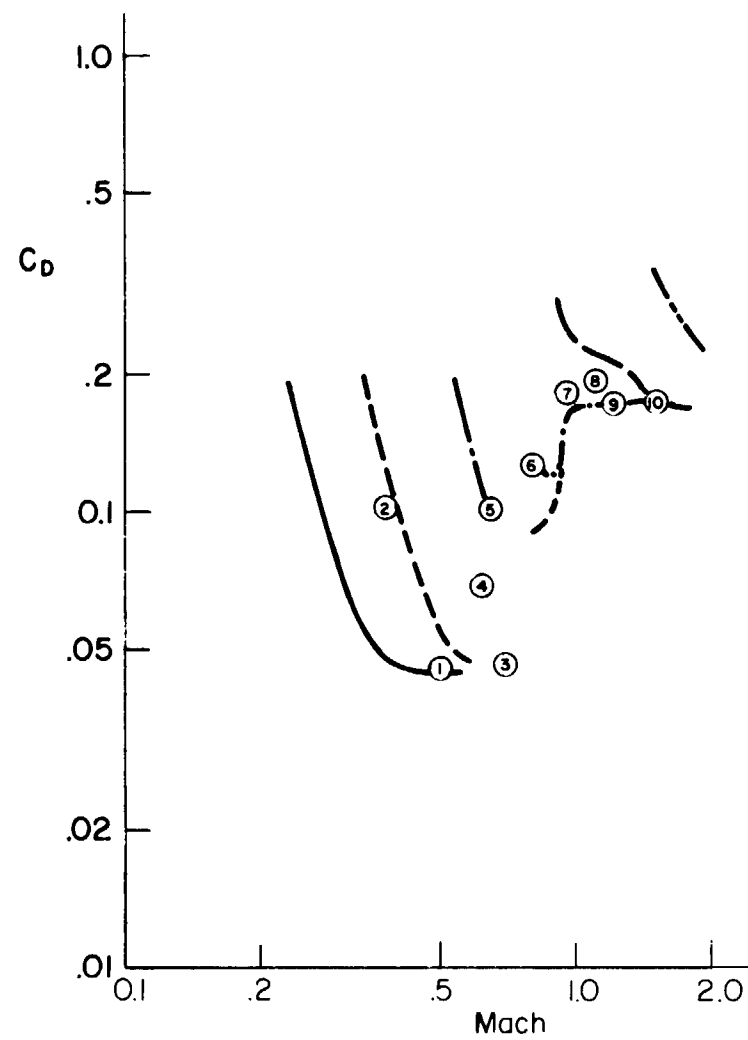


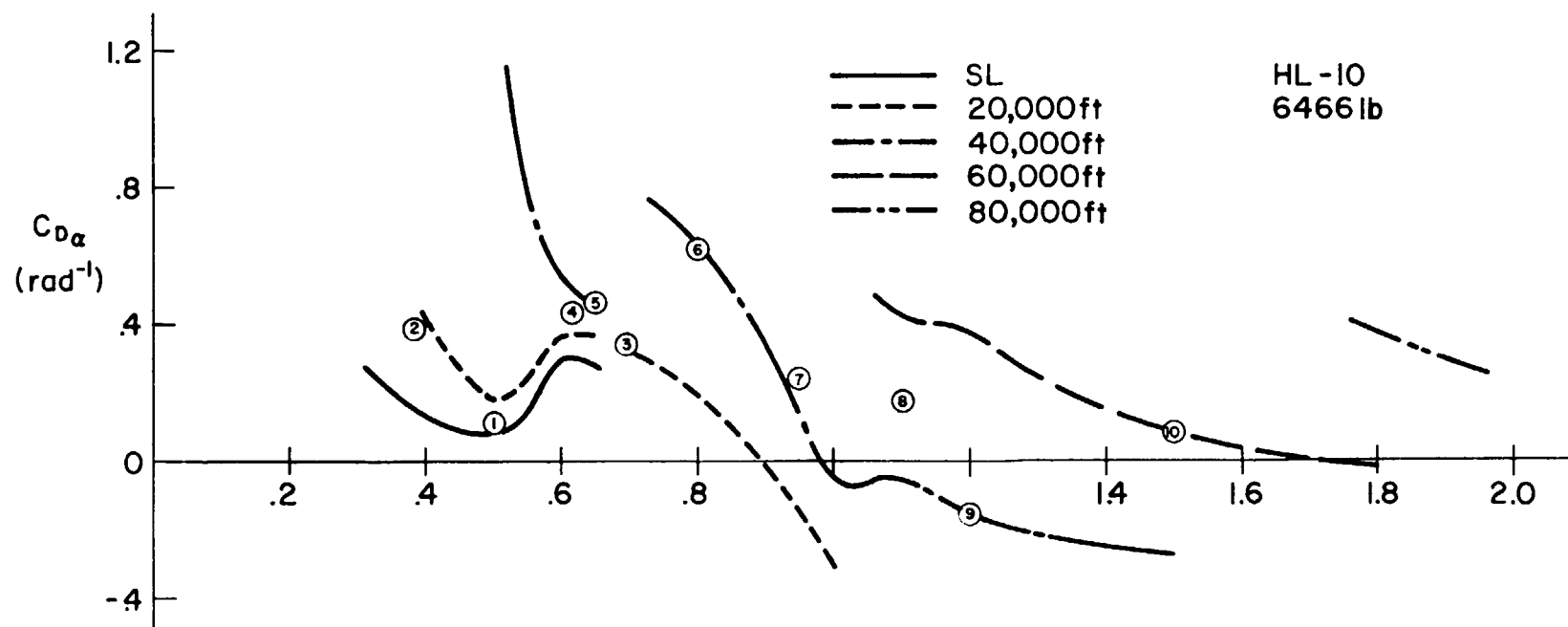
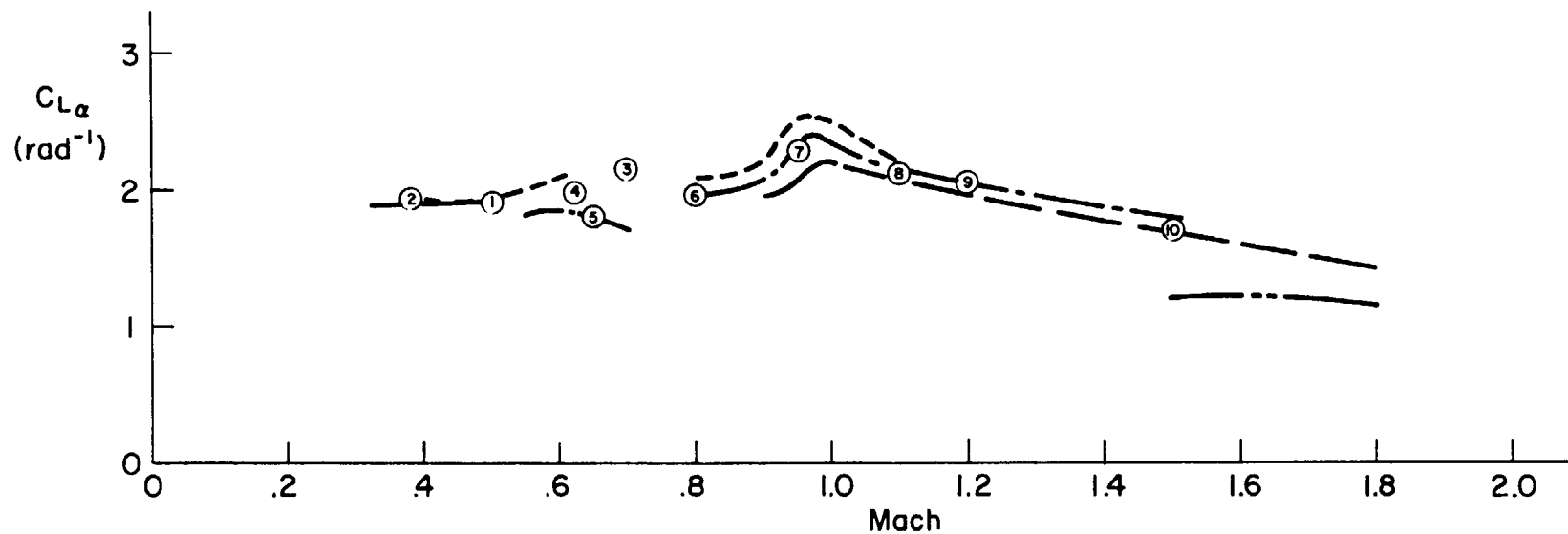
Figure VI-4. HL-10 Stability Augmentation

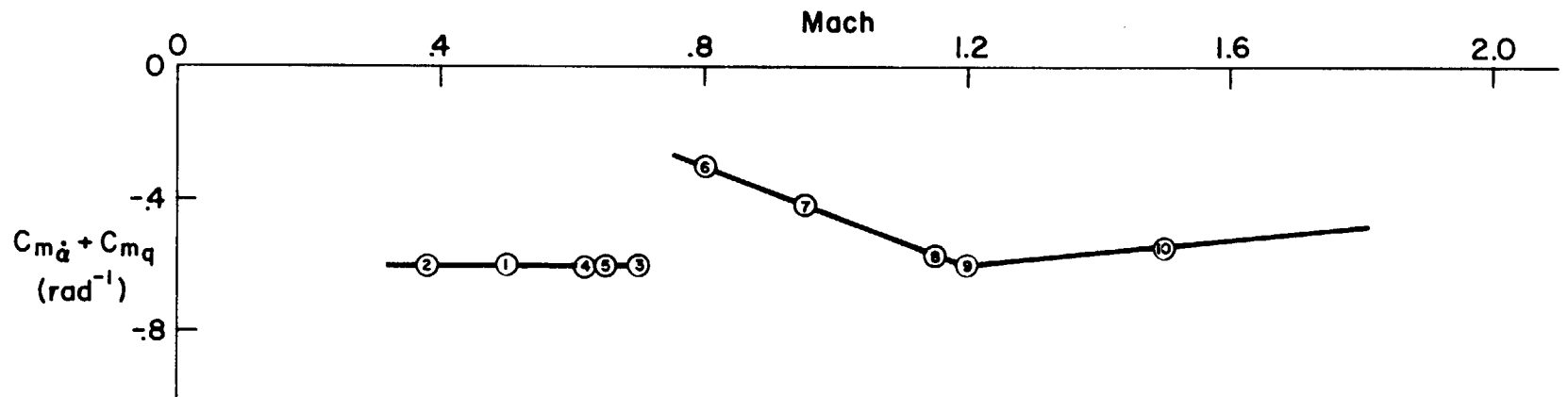
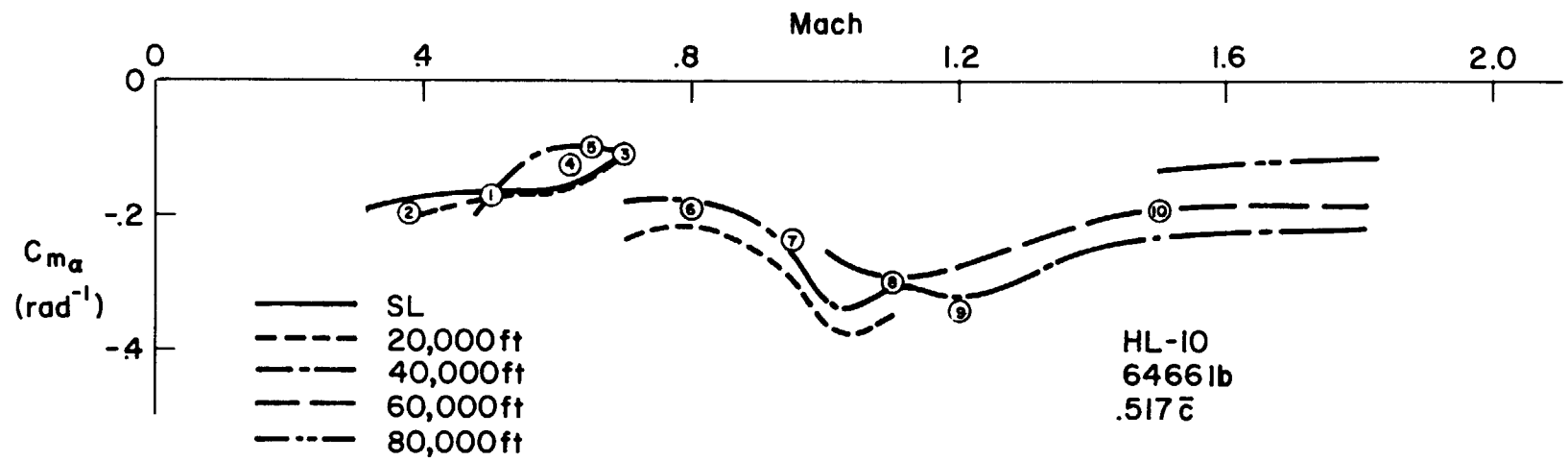


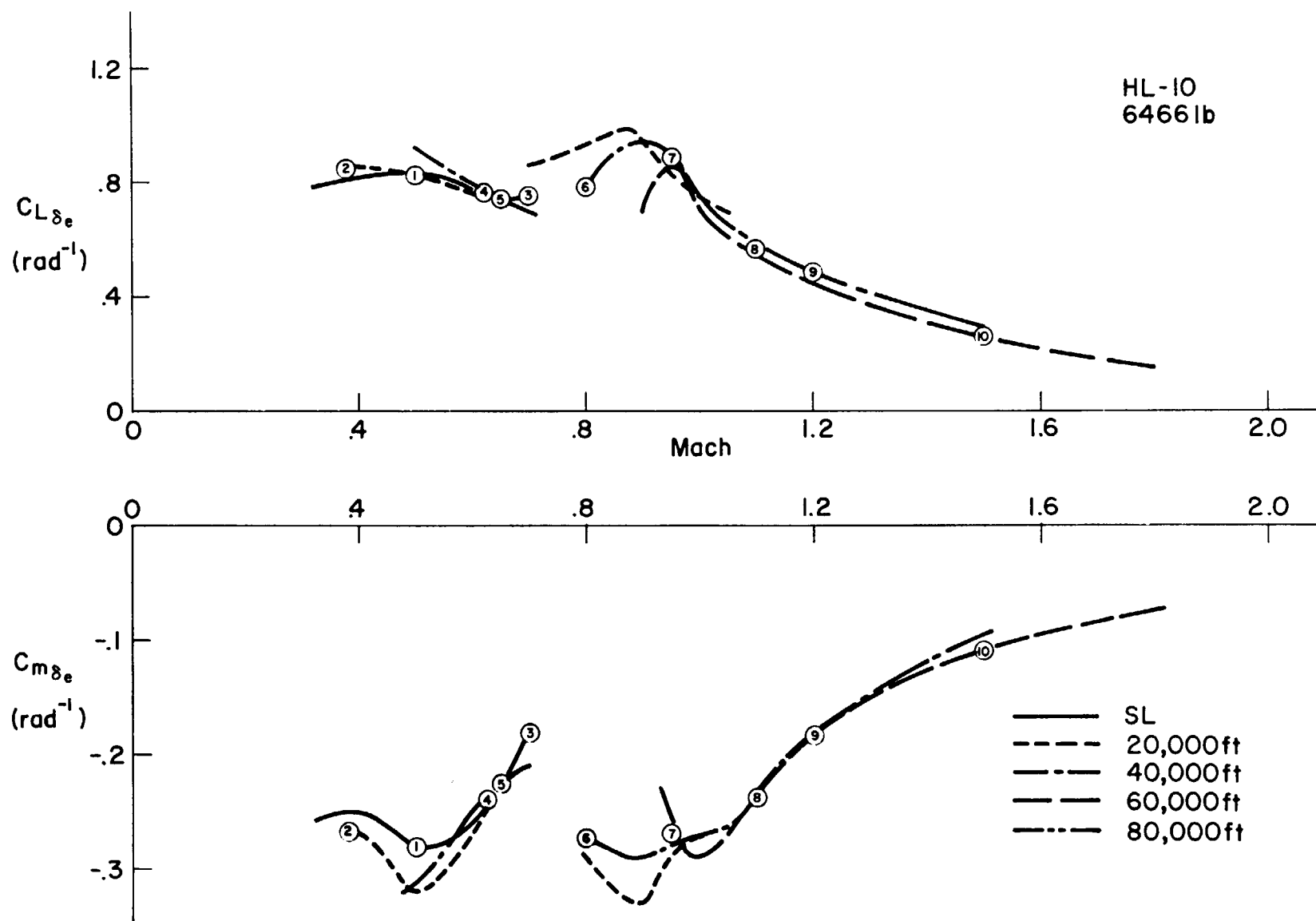


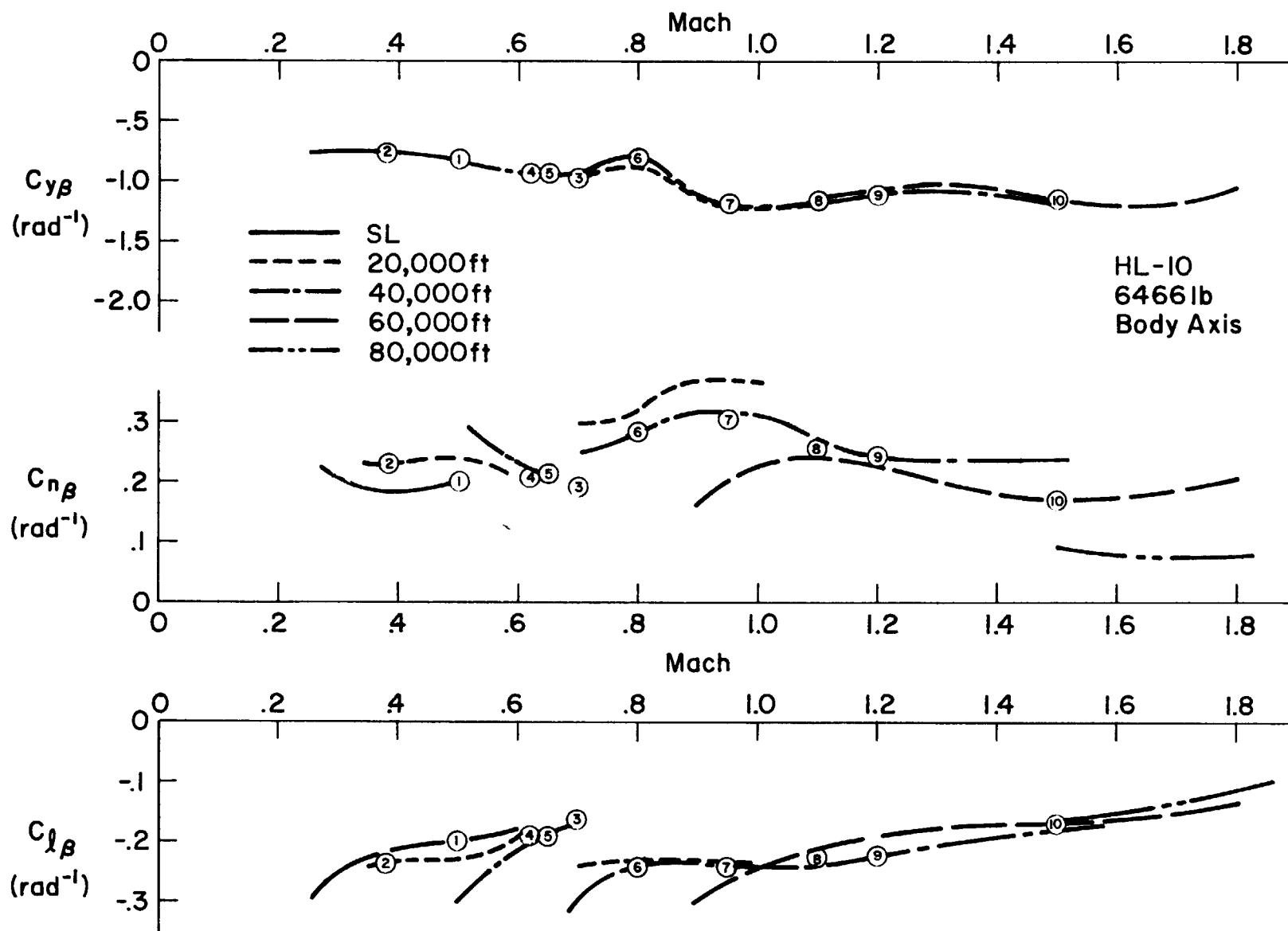
HL-10
6466 lb

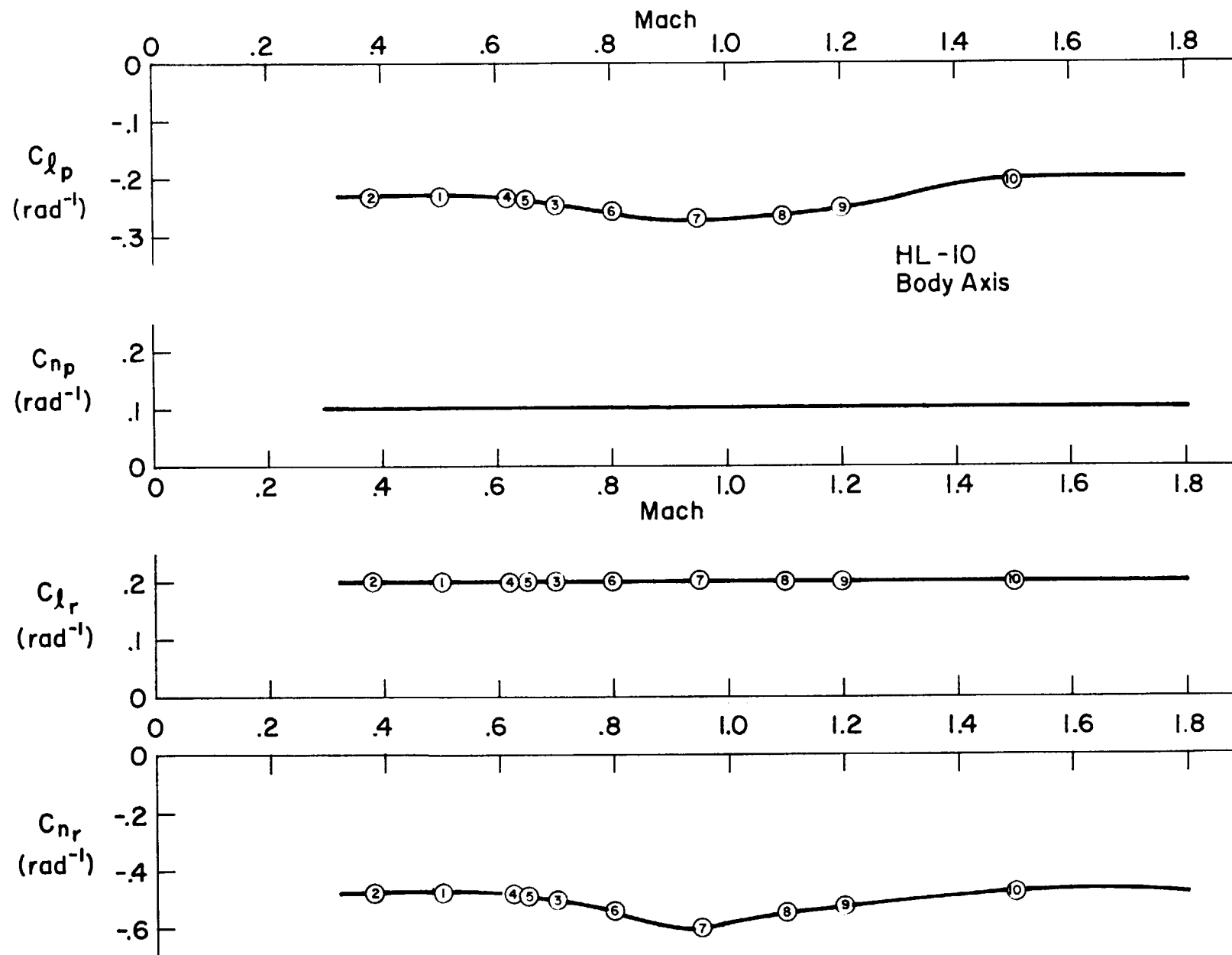


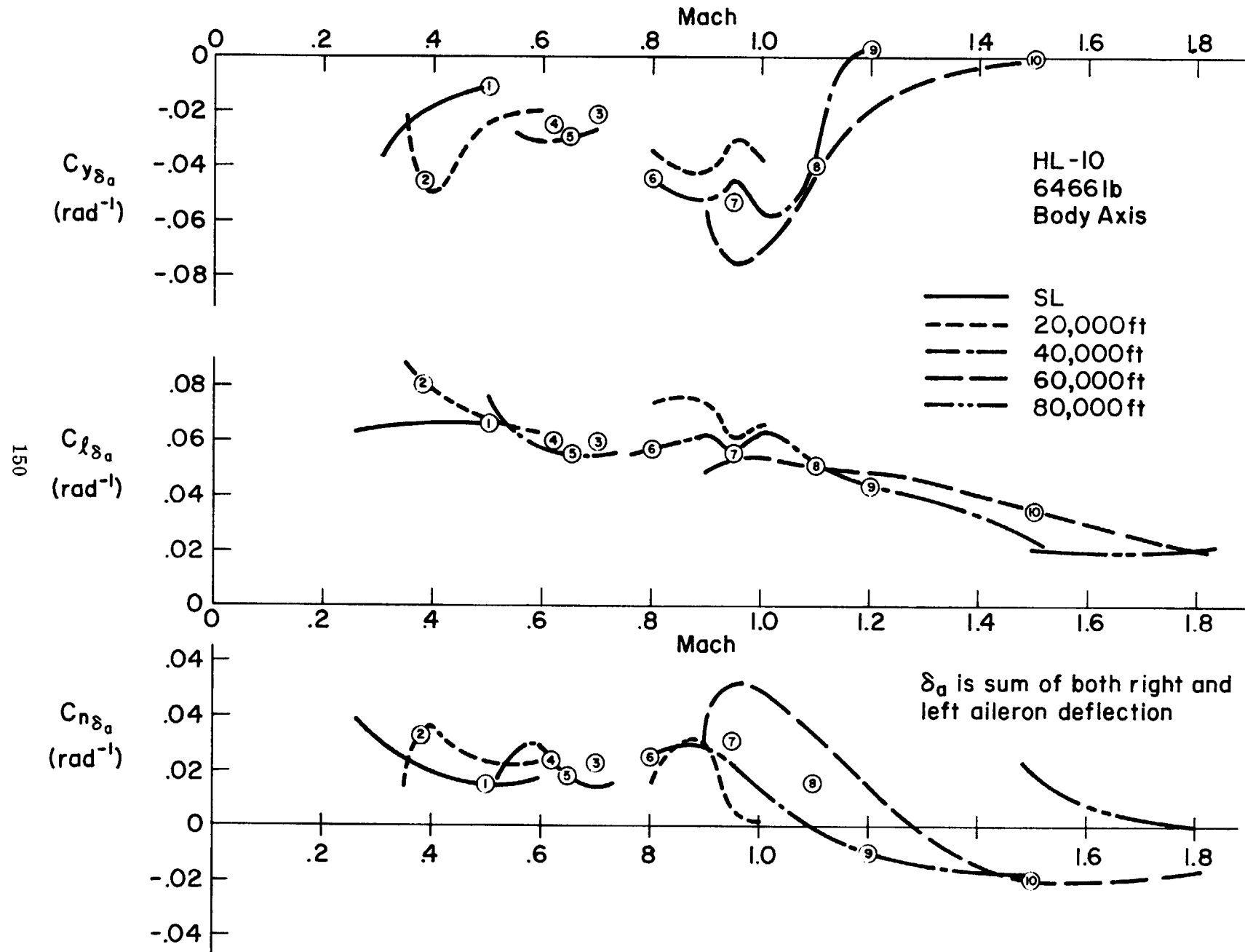












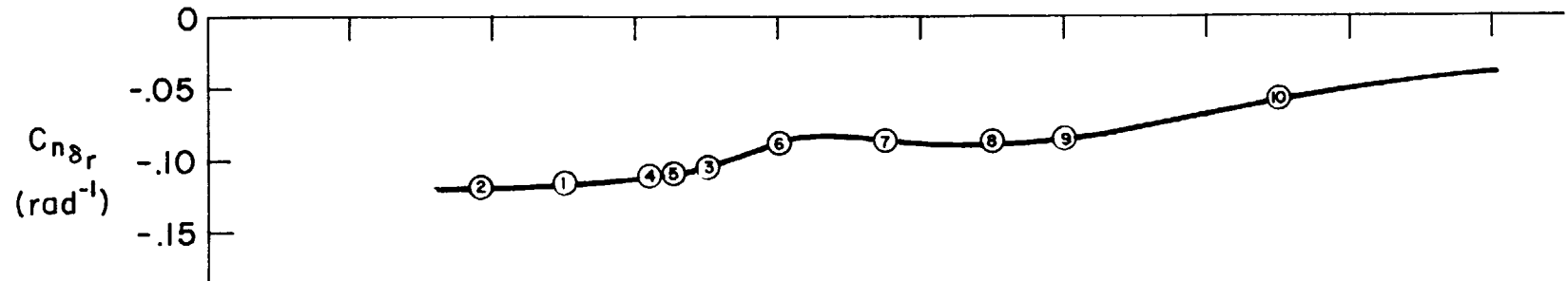
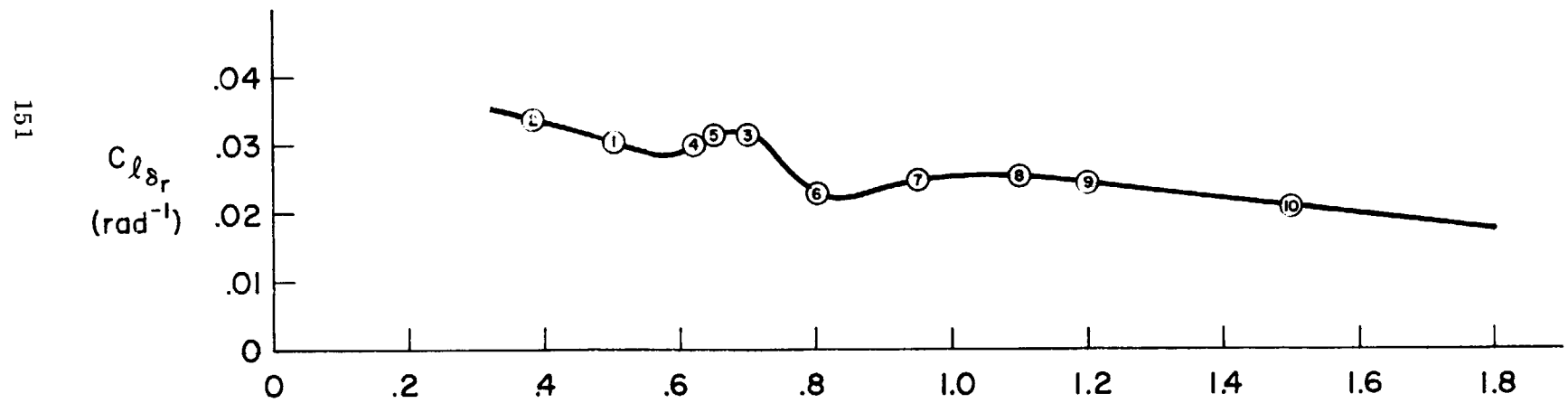
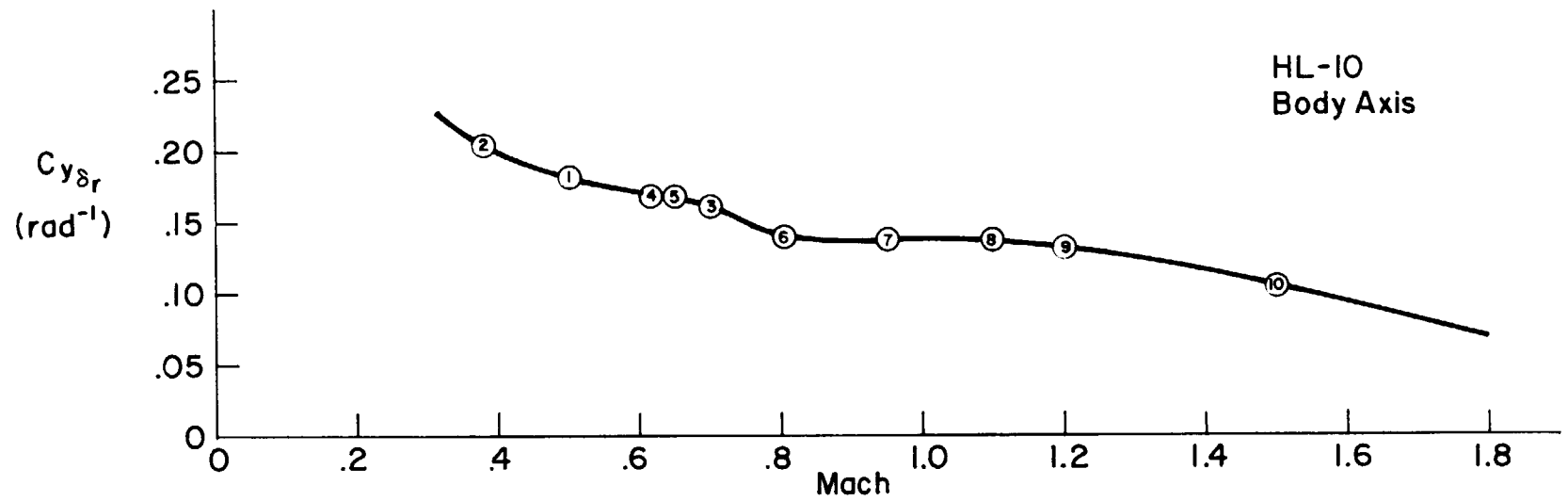


TABLE VI-1

HL-10 DIMENSIONAL, MASS AND FLIGHT CONDITION PARAMETERS

$$s = 160.0 \text{ sq ft}, \quad b = 13.60 \text{ ft}, \quad \bar{c} = 21.17 \text{ ft}$$
[illegible]

TABLE VI-2

HL-10 LONGITUDINAL DIMENSIONAL DERIVATIVES

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M	.500	.380	.700	.620	.350	.800	.950	1.10	1.20	1.50
XU *	-.0509	-.0401	-.0260	-.0227	-.0191	-.0325	-.0648	-.0597	-.0200	-.0272
ZU *	.0383	-.0122	.0417	.0175	-.0143	.0128	.0182	.00842	.00754	.000762
MU *	.00463	.00596	.00225	.00217	.00136	.00400	.00479	.00478	.00120	.00258
XW	.164	.140	.0637	.0777	.0727	.0148	.0851	.0763	.0242	.0293
ZW	-.916	-.481	-.742	-.453	-.291	-.432	-.417	-.334	-.111	-.137
MW	-.0305	-.0166	-.0141	-.0102	-.00548	-.0139	-.0148	-.0162	-.00663	-.00493
ZWD	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ZQ	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MWD	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MQ	-.662	-.321	-.472	-.305	-.205	-.139	-.155	-.199	-.0714	-.0945
XDE	38.1	25.6	33.7	27.7	22.1	29.6	33.5	21.6	4.01	5.84
ZDE	-212.	-74.2	-180.	-98.7	-65.0	-117.	-133.	-85.3	-28.5	-27.5
MDE	-28.0	-9.51	-16.8	-12.3	-7.97	-16.2	-16.1	-14.2	-4.23	-4.53
	+	+	+	+	+	+	+	+	+	+

TABLE VI-3

HL-10 ELEVATOR TRANSFER FUNCTION FACTORS

SAS Off

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50
DE NUMINATOR										
Z(DET)1	.469	.283	.403	.363	.333	.526	.618	.697	.609	.362
W(DET)1	.0760	.117	.0581	.0632	.0651	.0676	.0636	.0531	.0209	.0345
Z(DET)2	.186	.145	.184	.145	.125	.0794	.0751	.0610	.0316	.0418
W(DET)2	4.19	2.68	3.25	2.56	1.89	3.35	3.79	4.25	2.81	2.80
NUMERATORS										
N(U /DE)										
A(U)	38.1	25.6	33.7	27.7	22.1	29.6	33.5	21.6	4.01	5.84
1/T(U)1	72.3	48.6	66.1	73.7	73.0	.138	108.	172.	172.	235.
Z(U)1	.346	.536	.862	.699	.601	(.402)	.539	.467	.602	.467
W(U)1	.440	.298	.369	.265	.196	(.104.)	.199	.158	.112	.106
N(W /DE)										
A(W)	-212.	-74.2	-180.	-98.7	-65.0	-117.	-133.	-85.3	-28.5	-27.5
1/T(W)1	.0158	48.8	-.0120	.00853	73.2	104.	108.	.0320	.00596	.00571
1/T(W)2	.0500	(.267)	.0422	.0191	(.369)	(.629)	(.991)	.0366	.0164	.0115
1/T(W)3	72.5	(.0690)	66.4	74.6	(.0378)	(.0296)	(.0341)	172.	172.	235.
N(TH/DE)										
A(TH)	-28.0	-9.51	-16.8	-12.3	-7.97	-16.2	-16.1	-14.2	-4.23	-4.53
1/T(TH)1	.0440	.0423	.0204	.0178	.0218	.0246	.0599	.0583	.0193	.0282
1/T(TH)2	.686	.334	.594	.378	.239	.332	.289	.231	.0661	.103
N(HD /DE)										
A(HD)	182.	76.2	165.	92.0	63.2	109.	123.	72.0	27.8	27.2
1/T(HD)1	.0650	.0207	.0325	.0265	.0199	.0292	.0632	.0647	.0215	.0167
1/T(HD)2	-6.69	-3.83	-6.16	-5.22	-4.07	-5.93	-5.50	-6.18	-3.31	-4.97
1/T(HD)3	7.34	4.18	6.60	5.51	4.28	6.06	5.67	6.37	3.38	5.07
N(AZP/DE)										
A(AZP)	-29.6	-12.4	-71.1	-18.5	-13.3	-11.5	-28.7	7.01	-1.00	1.98
1/T(AZP)1	.0218	-.0105	.0107	.00983	.00613	.00918	.00782	.0117	.00350	-.0108
1/T(AZP)2	.0445	.0301	.0229	.0175	.0142	.0218	.0571	.0552	.0186	.0256
Z(AZP)1	(-18.5)	(-9.66)	(-9.84)	(.12.2)	(-9.18)	(.17.8)	(.11.0)	.0142	(-17.9)	.00687
W(AZP)1	(.19.0)	(.9.95)	(.10.1)	(-12.3)	(.5.26)	(-19.4)	(-12.2)	22.0	(.18.1)	18.4

TABLE VI-4

HL-10 ELEVATOR TRANSFER FUNCTION FACTORS

SAS On

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50
DENOMINATOR										
Z(DET)1	(2.42)	.267	(2.86)	.342	.315	.503	.629	.705	.610	.363
W(DET)1	(10.3)	.109	(5.06)	.0561	.0594	.0625	.0600	.0514	.0207	.0341
Z(DET)2	.485	.792	.391	.986	.883	.969	.872	.706	.231	.361
W(DET)2	.0636	2.90	.0457	2.88	2.07	3.63	4.01	4.39	2.83	2.83
NUMERATORS										
N(U /DE)										
A(U)	38.1	25.6	33.7	27.7	22.1	29.6	33.5	21.6	4.01	5.84
1/T(U)1	72.3	48.6	66.1	73.7	73.0	.138	108.	172.	172.	235.
Z(U)1	.346	.536	.862	.699	.601	(.402)	.539	.467	.602	.447
W(U)1	.440	.298	.365	.265	.195	(104.)	.190	.158	.112	.106
N(W /DE)										
A(W)	-212.	-74.2	-180.	-98.7	-65.0	-117.	-133.	-85.3	-28.5	-27.5
1/T(W)1	.0158	48.8	-.0120	.00853	73.2	104.	108.	.0320	.00596	.00571
1/T(W)2	.0500	(.267)	.0422	.0191	(.369)	(.629)	(.991)	.0366	.0164	.0115
1/T(W)3	72.5	(.0690)	66.4	74.6	(.0378)	(.0296)	(.0341)	172.	172.	235.
N(THE /DE)										
A(THE)	-28.0	-9.51	-16.8	-12.3	-7.97	-16.2	-16.1	-14.2	-4.23	-4.53
1/T(THE)1	.0440	.0423	.0204	.0178	.0218	.0246	.0599	.0583	.0193	.0202
1/T(THE)2	.686	.234	.594	.378	.239	.332	.289	.231	.0661	.103
N(HD /DE)										
A(HD)	182.	76.2	165.	92.0	63.2	109.	123.	72.0	27.8	27.2
1/T(HD)1	.0650	.0207	.0325	.0263	.0199	.0292	.0632	.0647	.0215	.0167
1/T(HD)2	-6.69	-3.83	-6.16	-5.22	-4.07	-5.93	-5.50	-6.18	-3.21	-4.97
1/T(HD)3	7.34	4.18	6.60	5.51	4.28	6.06	5.67	6.37	3.38	5.07
N(AZP/DE)										
A(AZF)	-29.6	-12.4	-71.1	-18.5	-13.3	-11.5	-28.7	7.01	-1.00	1.98
1/T(AZP)1	.0218	-.0105	.0107	.00983	.00613	.00918	.00782	.0117	.00250	-.0108
1/T(AZP)2	.0445	.0301	.0225	.0173	.0142	.0218	.0571	.0552	.0186	.0256
Z(AZF)1	(-18.5)	(-9.66)	(-9.84)	(12.2)	(-5.18)	(17.8)	(11.9)	.0142	(-17.9)	.00687
W(AZP)1	(19.0)	(9.95)	(10.1)	(-12.3)	(5.26)	(-19.4)	(-12.2)	22.0	(18.1)	18.4

TABLE VI-5

HL-10 LONGITUDINAL HANDLING QUALITIES PARAMETERS

SAS Off

(Body Axis System)

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TABLE VI-6

HL-10 LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50
YV	-.354	-.173	-.322	-.203	-.140	-.160	-.204	-.173	-.0564	-.0851
YB	-218.	-69.3	-232.	-125.	-88.2	-124.	-187.	-184.	-66.0	-124.
LB'	-102.	-43.1	-75.1	-49.5	-34.0	-71.4	-71.5	-69.7	-25.5	-34.5
NB'	13.9	5.51	13.1	7.79	5.53	12.4	14.0	11.0	3.94	4.91
LP'	-1.49	-.723	-1.13	-.686	-.473	-.710	-.627	-.524	-.175	-.201
NP'	.0390	.0189	.0240	.0179	.0115	.0119	.00882	.00877	.00344	.00774
LR'	1.16	.561	.820	.532	.357	.477	.397	.351	.123	.180
NK'	-.498	-.248	-.382	-.235	-.162	-.245	-.234	-.185	-.0616	-.0795
Y*DA	-.00523	-.0104	-.00679	-.00547	-.00428	-.00899	-.00900	-.00590	.000154	0.
L'DA	36.0	15.5	30.7	17.0	10.7	18.5	18.2	16.7	5.23	7.52
N'DA	3.39	1.96	3.73	2.11	1.19	2.43	2.76	1.83	.0603	-.392
Y*DR	.0865	.0473	.0553	.0372	.0251	.0280	.0233	.0202	.00677	.00777
L'DR	13.0	5.12	13.0	6.67	4.87	5.82	6.53	6.50	2.45	3.94
N'DR	-10.6	-3.81	-8.77	-5.12	-3.51	-4.65	-4.60	-4.87	-1.75	-2.18

TABLE VI-7

HL-10 AILERON TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50
DENOMINATOR										
1/T(DEL)1	.0868	.0796	.0509	.0548	.0440	.0434	.0367	.0375	.0249	.0201
1/T(DEL)2	.812	.383	.683	.375	.265	.417	.389	.298	.171	.124
Z(DEL)1	.132	.0777	.107	.0762	.0579	.0602	.0574	.0518	.0179	.0320
W(DEL)1	5.64	4.39	5.17	4.55	4.02	5.44	5.57	5.27	2.73	3.46
NUMERATORS										
N(B /DA)										
A(B)	-.00523	-.0104	-.00679	-.00547	-.00428	-.00899	-.00900	-.00590	.000154	1.95
1/T(B)1	-.578.	-307.	-291.	-457.	-539.	-243.	-193.	-394.	.0874	
Z(B)1	-.172	.546	-.415	.337	.587	.182	.0638	.318	(.153)	.973
W(B)1	.587	.358	.541	.313	.216	.315	.313	.201	(.4339.)	.0726
N(P /DA)										
A(P)	36.0	15.5	30.7	17.0	10.7	18.5	18.2	16.7	5.23	7.52
1/T(P)1	.0218	-.00737	.0120	.00983	.00395	.00802	.00743	.0111	.00339	-.00987
Z(P)1	.103	.0817	.0862	.0695	.0591	.0537	.0535	.0481	.0282	.0473
W(P)1	4.83	3.23	4.69	3.67	2.98	4.60	4.92	4.26	2.05	1.75
N(R /DA)										
A(R)	3.39	1.96	3.73	2.11	1.19	2.43	2.76	1.83	.0603	-.392
1/T(R)1	.305	.247	.234	.192	.158	.167	.141	.115	.197	.0955
1/T(R)2	(.148)	(.0725)	(.120)	(.0750)	(.0519)	(.0586)	(.0556)	(.0484)	(.0240)	-3.50
1/T(R)3	(.6.55)	(.5.31)	(.5.81)	(.5.47)	(.5.19)	(.6.39)	(.6.31)	(.6.47)	(.7.14)	3.54
N(PHI/DA)										
A(PHI)	34.6	15.7	29.7	16.6	10.6	18.1	17.6	16.0	5.22	7.33
Z(PHI)1	.102	.0807	.0853	.0702	.0598	.0541	.0538	.0491	.0291	.0419
W(PHI)1	4.74	3.26	4.64	3.62	2.95	4.54	4.87	4.13	2.03	1.86
N(AYP/DA)										
A(AYP)	69.5	30.3	62.3	34.1	20.0	34.8	35.1	29.0	7.89	7.98
1/T(AYP)1	-.293	.271	-.234	.286	.200	.198	.207	.178	.0838	.0629
1/T(AYP)2	.569	-.387	.506	-.320	-.498	-.216	-.228	-.332	-.617	.114
1/T(AYP)3	(.123)	(.126)	(.0933)	(.111)	(.167)	(.0679)	(.0687)	(.0963)	(.283)	4.71
1/T(AYP)4	(.4.45)	(.3.10)	(.4.33)	(.3.22)	(.2.24)	(.4.54)	(.4.62)	(.3.34)	(.1.19)	-4.73

TABLE VI-8

HL-10 RUDDER TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	+
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K	
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50	
DENOMINATOR											
1/T(DET)1	.0868	.0796	.0509	.0548	.0440	.0434	.0367	.0375	.0249	.0201	
1/T(DET)2	.812	.383	.683	.375	.265	.417	.389	.298	.171	.124	
Z(DET)1	.132	.0777	.107	.0762	.0579	.0602	.0574	.0518	.0179	.0320	
W(DET)1	5.64	4.39	5.17	4.55	4.02	5.44	5.57	5.27	2.73	3.46	
NUMERATORS											
N(B /DR)											
A(B)	.0865	.0473	.0553	.0372	.0251	.0280	.0233	.0202	.00677	.00777	
1/T(B)1	.00111	-.0316	.00183	-.00794	-.0105	-.00222	.00348	.00498	-.00203	-.0116	
1/T(B)2	1.17	.527	.881	.506	.343	.544	.480	.396	.175	.161	
1/T(B)3	148.	112.	200.	181.	195.	213.	261.	313.	306.	380.	
N(P /DR)											
A(P)	13.0	5.12	13.0	6.67	4.87	5.82	6.53	6.50	2.45	3.94	
1/T(P)1	.0219	-.00739	.0120	.00985	.00395	.00802	.00743	.0111	.00339	-.00989	
1/T(P)2	7.91	4.83	6.01	5.27	4.17	6.42	5.90	6.26	3.73	-3.72	
1/T(P)3	-8.66	-5.22	-6.19	-5.53	-4.30	-6.75	-6.00	-6.39	-3.78	3.73	
N(R /DR)											
A(R)	-10.6	-3.81	-8.77	-5.12	-3.51	-4.65	-4.60	-4.87	-1.75	-2.18	
1/T(R)1	.306	.247	.234	.192	.159	.167	.141	.115	.196	.0955	
Z(R)1	.183	.0816	.172	.0952	.0685	.0830	.0855	.0713	.00461	.0345	
W(R)1	3.68	3.40	3.20	3.24	2.91	3.70	3.55	3.67	1.67	2.31	
N(PHI/DR)											
A(PHI)	17.3	4.79	15.4	7.61	5.13	6.71	7.50	8.34	2.67	2.88	
1/T(PHI)1	6.86	5.04	-5.42	4.86	4.03	5.90	-5.40	-5.28	3.56	4.53	
1/T(PHI)2	-6.98	-5.52	5.48	-4.98	-4.12	-6.07	5.41	5.33	-3.58	-4.63	
N(AYP/DR)											
A(AYP)	-3.07	1.32	.978	-.991	-.182	-.445	.609	-1.02	-.0183	2.69	
1/T(AYP)1	-.0430	-.123	-.0195	-.0404	-.0393	-.0557	-.0218	-.0131	-.0143	-.0207	
1/T(AYP)2	.681	.300	.613	.330	.235	.260	.297	.248	.132	.129	
1/T(AYP)3	(.170)	17.2	43.8	(.125)	(.187)	(.139)	39.9	(.0789)	(.196)	11.8	
1/T(AYP)4	(33.9)	-20.6	-63.9	(31.5)	(51.9)	(43.9)	-48.1	(36.5)	(92.8)	-12.1	

TABLE VI-9

HL-10 AILERON TRANSFER FUNCTION FACTORS

SAS On

(BODY AXIS SYSTEM)

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	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K	
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50	
DENOMINATOR											
1/T(DET)1	.00751	.00201	.00626	.00491	.00328	.00533	.00555	.00620	.00269	-.000773	
1/T(DET)2	.323	.275	.269	.224	.187	.214	.197	.156	.232	.120	
1/T(DET)3	2.55	14.8	2.10	19.9	13.6	17.7	17.0	18.7	6.75	8.54	
Z(DET)1	(7.48)	.638	(7.01)	.709	.533	.673	.706	.608	.439	.445	
W(DET)1	(42.0)	3.75	(34.0)	3.63	3.22	4.37	4.40	4.22	1.91	2.53	
NUMERATORS											
V(B /DA)											
A(B)	-.00523	-.0104	-.00679	-.00547	-.00428	-.00899	-.00900	-.00590	.000154	1.93	
1/T(B)1	.00374	.00600	.00352	.00391	.00399	.00442	.00408	.00295	.00277	.00272	
1/T(B)2	.304	.245	.232	.190	.157	.164	.138	.113	.196	.0951	
1/T(B)3	69.8	25.1	76.0	37.5	23.0	37.7	44.7	35.9	8.10	6.75	
1/T(B)4	-830.	-352.	-453.	-531.	-576.	-292.	-248.	-435.	4349.		
V(P /DA)											
A(P)	36.0	15.5	30.7	17.0	10.7	18.5	18.2	16.7	5.23	7.52	
1/T(P)1	.0225	-.00735	.0122	.00989	.00395	.00803	.00744	.0111	.00339	-.00981	
1/T(P)2	48.0	18.0	41.8	23.9	16.3	21.4	.561	22.1	6.93	7.93	
Z(P)1	.886	.755	.880	.790	.742	.914	(.645)	.891	.693	.640	
W(P)1	.394	.438	.414	.430	.424	.570	(22.0)	.519	.447	.357	
V(R /DA)											
A(R)	3.39	1.96	3.73	2.11	1.19	2.43	2.76	1.83	.0603	-.392	
1/T(R)1	.305	.247	.234	.192	.158	.167	.141	.115	.197	.0955	
1/T(R)2	.330	.330	.330	.330	.330	.330	.330	.330	.330	.330	
1/T(R)3	(.148)	(.0725)	(.120)	(.0750)	(.0519)	(.0586)	(.0556)	(.0484)	(.0240)	-3.50	
1/T(R)4	(6.65)	(5.31)	(5.81)	(5.47)	(5.19)	(6.39)	(6.31)	(6.47)	(7.14)	3.54	

TABLE VI-10

HL-10 RUDDER TRANSFER FUNCTION FACTORS

SAS On

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M	.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50
DENOMINATOR										
1/T(DET)1	.00751	.00201	.00626	.00491	.00328	.00533	.00555	.00620	.00269	-.000773
1/T(DET)2	.323	.275	.269	.224	.187	.214	.197	.156	.232	.120
1/T(DET)3	2.55	14.8	2.10	19.9	13.6	17.7	17.0	18.7	6.75	8.54
Z(DET)1	(7.48)	.638	(7.01)	.709	.533	.673	.706	.608	.439	.445
W(DET)1	(42.0)	3.75	(34.0)	3.63	3.22	4.37	4.40	4.22	1.91	2.53
NUMERATORS										
V(B /DR)										
A(B)	.0865	.0473	.0553	.0372	.0251	.0280	.0233	.0202	.00677	.00777
1/T(B)1	.0188	-.0117	.0107	.00724	.00150	.00658	.00693	.0103	.00251	-.0102
1/T(B)2	.330	.330	.330	.330	.330	.330	.330	.330	.330	.330
1/T(B)3	7.68	2.96	6.52	3.37	2.00	3.80	3.72	3.24	1.06	1.14
1/T(B)4	149.	113.	200.	182.	196.	214.	262.	314.	306.	380.
V(P /DR)										
A(P)	13.0	5.12	13.0	6.67	4.87	5.82	6.53	6.50	2.45	3.94
1/T(P)1	.0219	-.00739	.0120	.00985	.00395	.00802	.00743	.0111	.00339	-.00089
1/T(P)2	.330	.330	.330	.330	.330	.330	.330	.330	.330	.330
1/T(P)3	7.91	4.83	6.01	5.27	4.17	6.42	5.90	6.26	3.73	-3.72
1/T(P)4	-8.66	-5.22	-6.19	-5.53	-4.30	-6.75	-6.00	-6.39	-3.78	3.73
N(R /DR)										
A(R)	-10.6	-3.81	-8.77	-5.12	-3.51	-4.65	-4.60	-4.87	-1.75	-2.18
1/T(R)1	.321	.263	.238	.198	.162	.173	.142	.115	.210	.0961
1/T(R)2	.330	.330	.330	.330	.330	.330	.330	.330	.330	.330
Z(R)1	(1.90)	.631	(1.47)	.713	.497	.677	.710	.592	.330	.330
W(R)1	(7.53)	3.30	(6.88)	3.20	2.88	3.64	3.54	3.68	1.61	2.30

TABLE VI-10 Continued

N(PHI/DR)										
A(PHI)	17.3	4.79	15.4	7.61	5.13	6.71	7.50	8.34	2.67	2.88
1/T(PHI)1	.330	.330	.330	.330	.330	.330	.330	.330	.330	.330
1/T(PHI)2	-6.02	4.92	-4.86	-4.73	-4.06	-5.79	-5.11	-4.86	-3.54	4.30
1/T(PHI)3	7.87	-5.65	6.06	5.11	4.09	6.18	5.70	5.76	3.60	-4.90
V(AYP/DR)										
A(AYP)	-3.07	1.32	.978	-.991	-.182	-.445	.609	-1.02	-.0183	2.69
1/T(AYP)1	.0104	-.0400	.00743	.00105	-.00468	-.00313	.00350	.00814	-.822E-4	-.0115
1/T(AYP)2	.330	.330	.330	.330	.330	.330	.330	.330	.330	.330
1/T(AYP)3	6.26	1.20	-193.	2.08	1.23	1.61	2.41	2.11	.672	.937
1/T(AYP)4	11.1	9.30	(.954)	(.935)	24.6	20.4	17.3	(.630)	40.7	10.9
1/T(AYP)5	61.0	-34.1	(7.71)	(29.6)	103.	84.9	-99.3	(35.2)	208.	-12.9
		+	+	+	+	+	+	+	+	+

TABLE VI-11

HL-10 LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

SAS Off

(BODY AXIS SYSTEM)

		+	+	+	+	+	+	+	+	+	+
F/C #		1	2	3	4	5	6	7	8	9	10
H		03 K	16 K	22 K	30 K	40 K	38 K	45 K	51 K	75 K	72 K
M		.500	.380	.700	.620	.650	.800	.950	1.10	1.20	1.50
DR PERIOD (SEC)		1.12	1.44	1.22	1.38	1.56	1.16	1.13	1.19	2.30	1.82
1/C(1/2)		1.20	.706	.974	.693	.526	.547	.521	.470	.162	.290
SPIRAL (2) (SEC)		--	--	--	--	--	--	--	--	--	--
P(1)		25.8	11.1	31.2	21.5	9.65	--	--	--	7.43	2.44
P(2)		25.8	11.0	31.1	21.5	9.65	--	--	--	7.27	.860
P(3)		26.3	13.7	--	--	13.0	--	--	--	10.2	4.95
P(2)/P(1)		.000	.988	.997	.998	.000	--	--	--	.979	.353
P(OSC)/P(AV)		.00471	.0619	--	--	.0800	--	--	--	.0967	.622
W(PHI)/W(D)		.842	.743	.898	.795	.733	.836	.873	.784	.744	.537
DEL-B-MAX		.120	.290	.115	.201	.257	.123	.0928	.146	.179	.309
PHI TO BETA, PHASE		1.21	1.54	1.53	.799	.873	1.29	.497	.153	-358.	.202
PHI TO BETA		3.38	2.21	2.94	2.45	2.12	2.49	2.40	2.66	3.48	2.68
PHI TO VE		.368	.406	.332	.373	.389	.353	.339	.374	.793	.455
		+	+	+	+	+	+	+	+	+	+

HL-10 DATA SOURCES

1. Ladson, Charles L., and Acquilla S. Hill, Aerodynamics of a Model of the HL-10 Flight Test Vehicle at Mach 0.35 to 1.80, NASA TN D-6018, Feb. 1971
2. Pyle, Jon S., Lift and Drag Characteristics of the HL-10 Lifting Body during Subsonic Gliding Flight, NASA TN D-6263, Mar. 1971
3. Ware, George M., Full Scale Wind Tunnel Investigation of the Aerodynamic Characteristics of the HL-10 Manned Lifting Entry Vehicle, NASA TMX-1160, Oct. 1965.

SECTION VII
LOCKHEED JETSTAR

JETSTAR BACKGROUND

The Jetstar is a four engine utility transport. Controls consist of conventional ailerons, elevators, and rudder. Ailerons and elevators are mechanically actuated with hydraulic boost. The rudder is mechanically activated but assisted by a servo tab.

The primary source of aerodynamic data was NASA CR-544. Power approach aerodynamics were estimated using CR-544 and flight test data from FTC-TDR-62-24C-140. The control system description was based solely on flight test data from the latter reference.

JETSTAR

Nominal Configuration

Slipper Tanks Installed

Heavy Gross Weight

W = 38204 lb

c.g. at 0.25 $\bar{c}$, W.L. 94.2

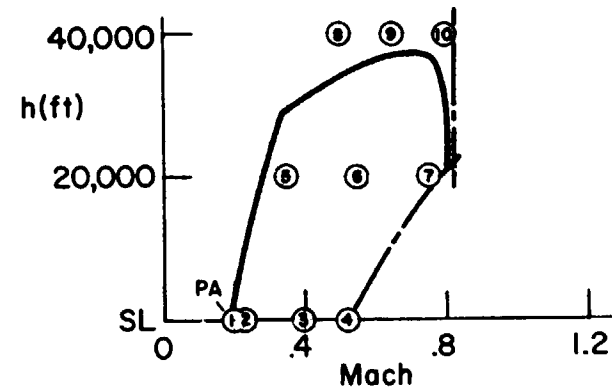
$I_x = 118773 \text{ slug-ft}^2$

$I_y = 135869 \text{ slug-ft}^2$

$I_z = 243504 \text{ slug-ft}^2$

$I_{xz} = 5061 \text{ slug-ft}^2$

Flight Envelope



———— Level Flight Envelope

----- Speed Restrictions

⑩ Transfer Function Case

Power Approach Configuration

Slipper Tanks Installed

Light Gross Weight

Gear Down

40% Flaps

1.4 V_S

W = 23904 lb

c.g. at 0.25 $\bar{c}$, W.L. 94.2

$I_x = 42273 \text{ slug-ft}^2$

$I_y = 126099 \text{ slug-ft}^2$

$I_z = 160104 \text{ slug-ft}^2$

$I_{xz} = 5470 \text{ slug-ft}^2$

Figure VII-1. Jetstar Flight Conditions

JETSTAR

$$S = 542.5 \text{ ft}^2$$

$$b = 53.75 \text{ ft}$$

$$\bar{c} = 10.93 \text{ ft}$$

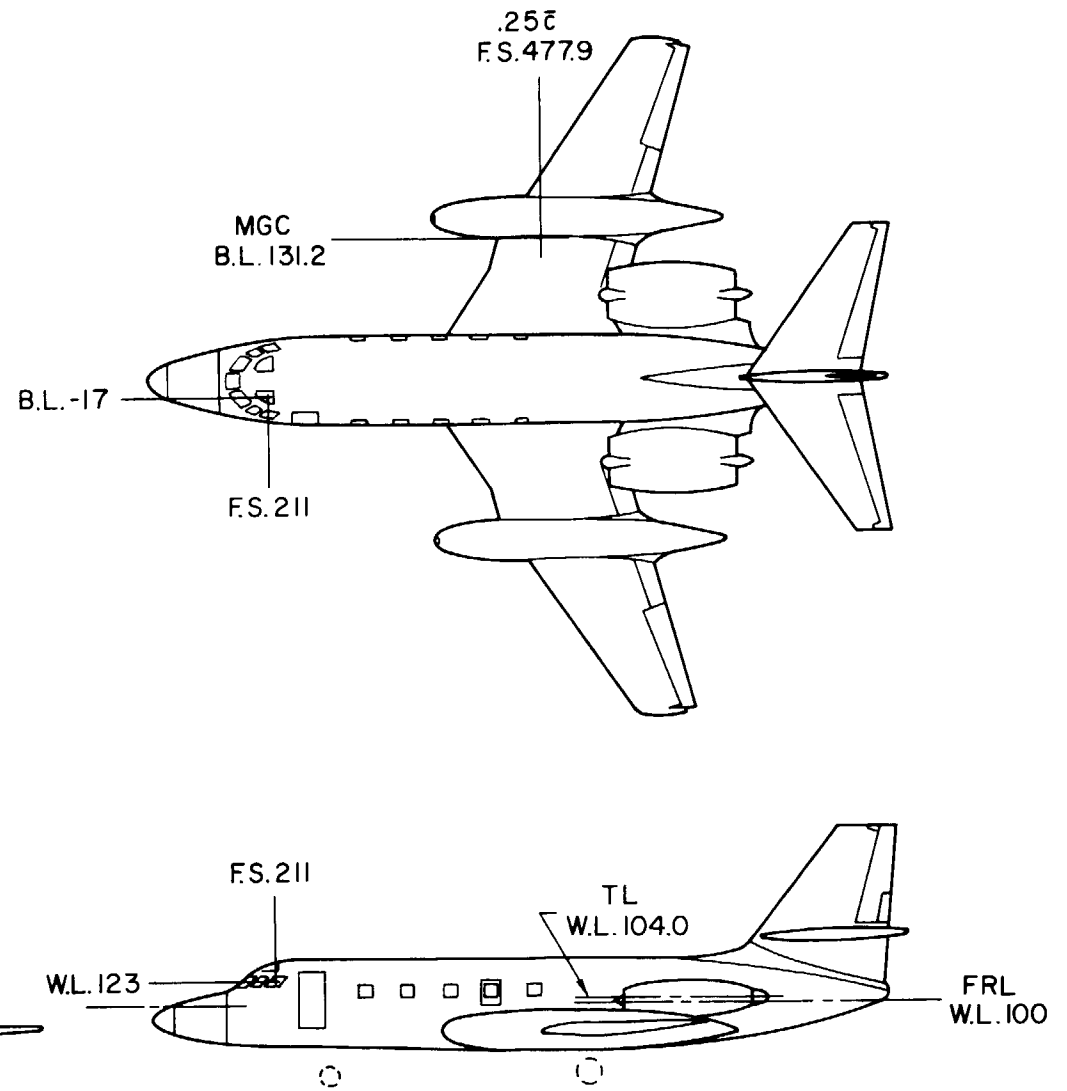
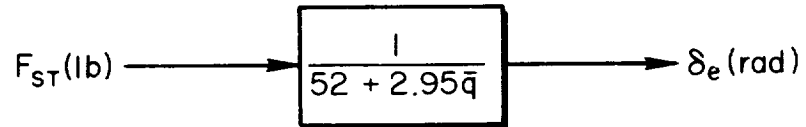


Figure VII-2. Jetstar General Arrangement

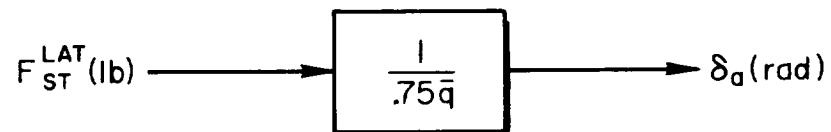
JETSTAR

PITCH AXIS



Note: Angle of attack effects on elevator hinge moment are neglected

ROLL AXIS



YAW AXIS

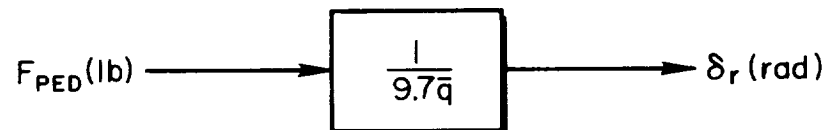


Figure VII-3. Jetstar Control System

TABLE VII-1

JETSTAR

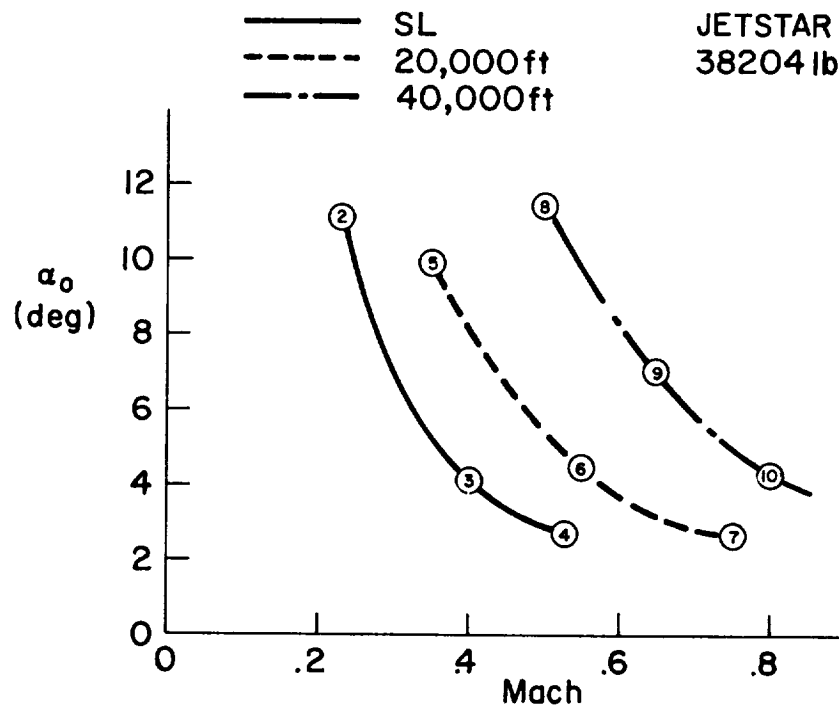
Power Approach Non-Dimensional Stability Derivatives

$$h = \text{sea level}$$

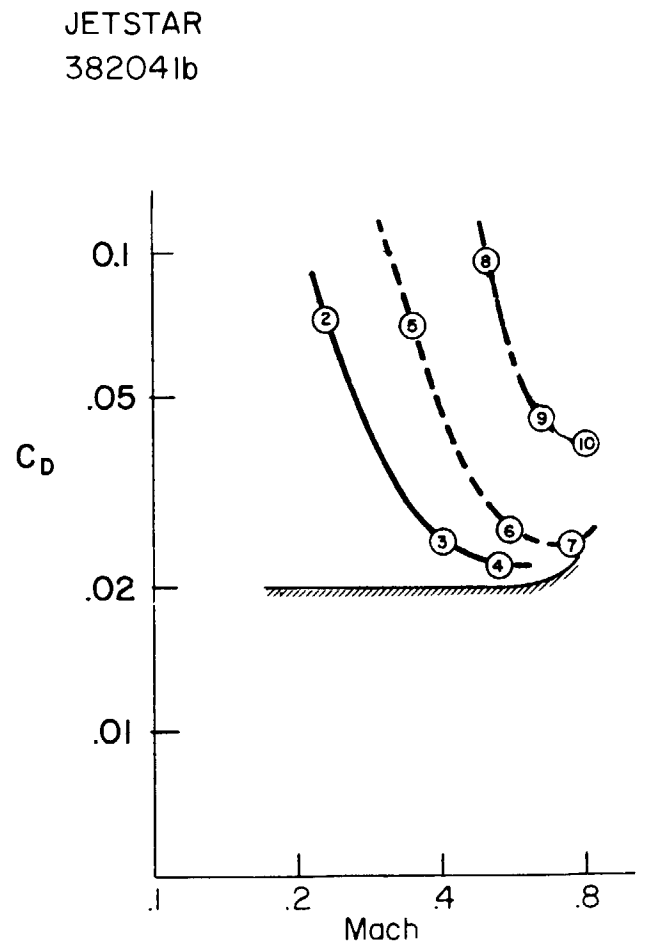
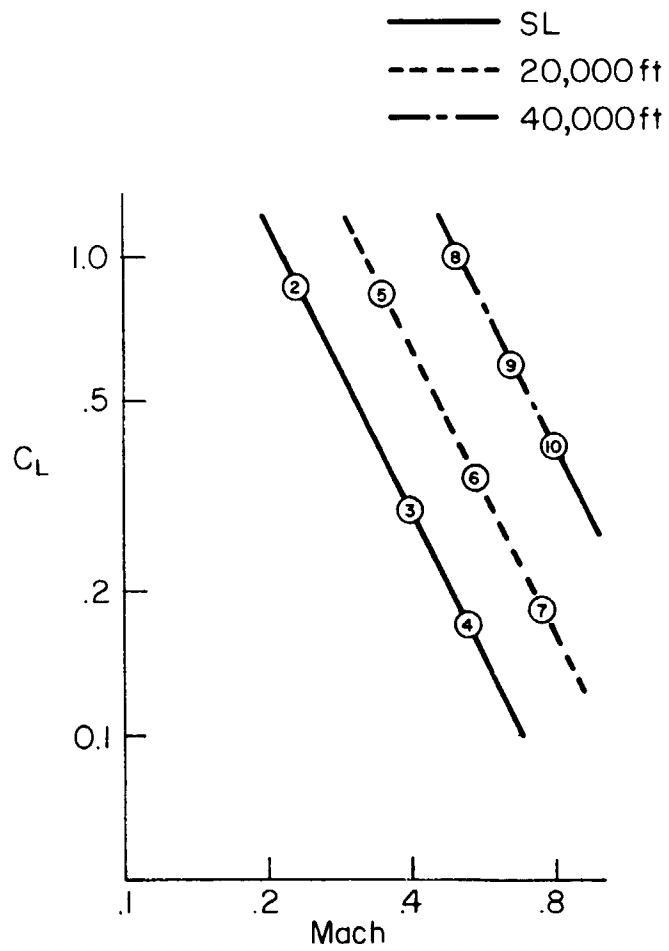
$$V_{T_0} = 224 \text{ ft/sec} = 132.5 \text{ kt}$$

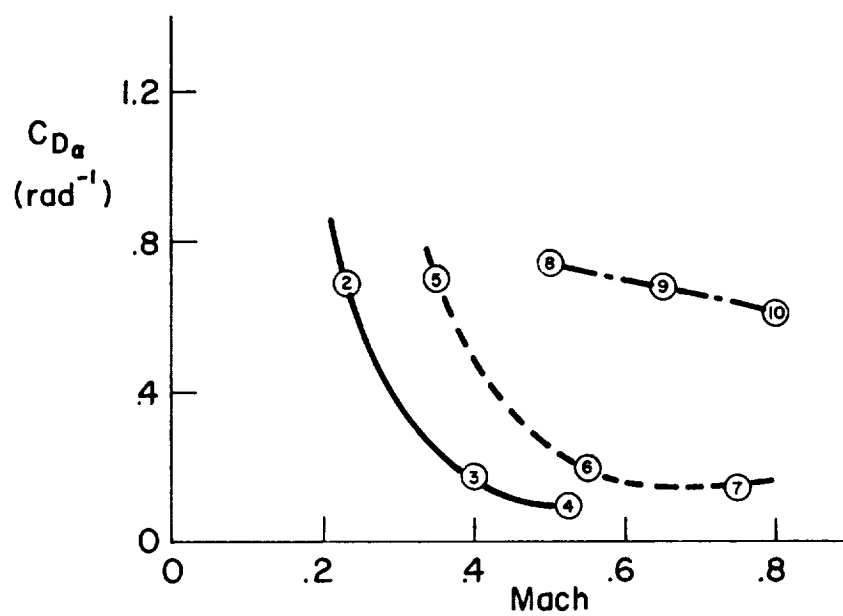
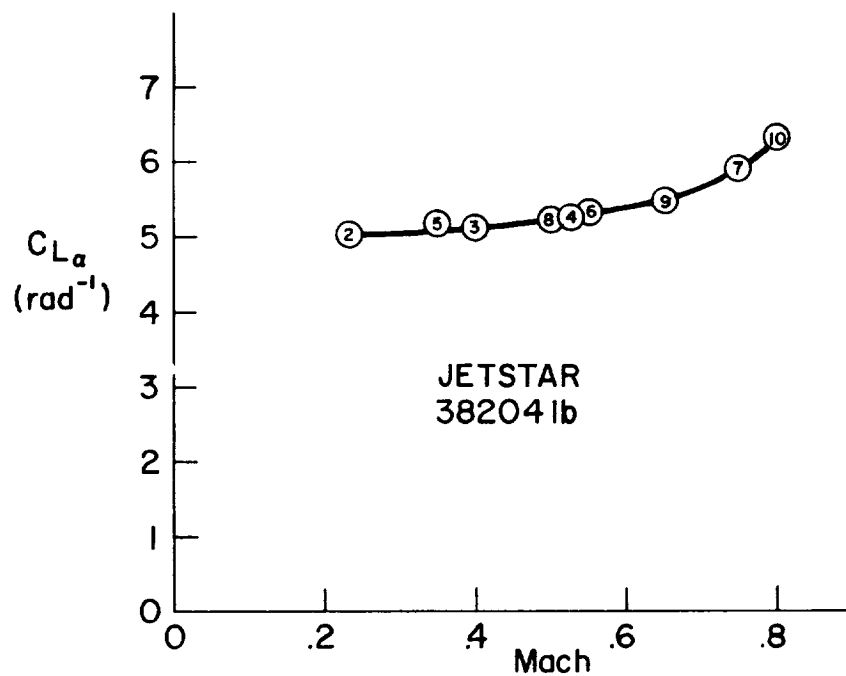
$$\alpha_0 = 6.5^\circ$$

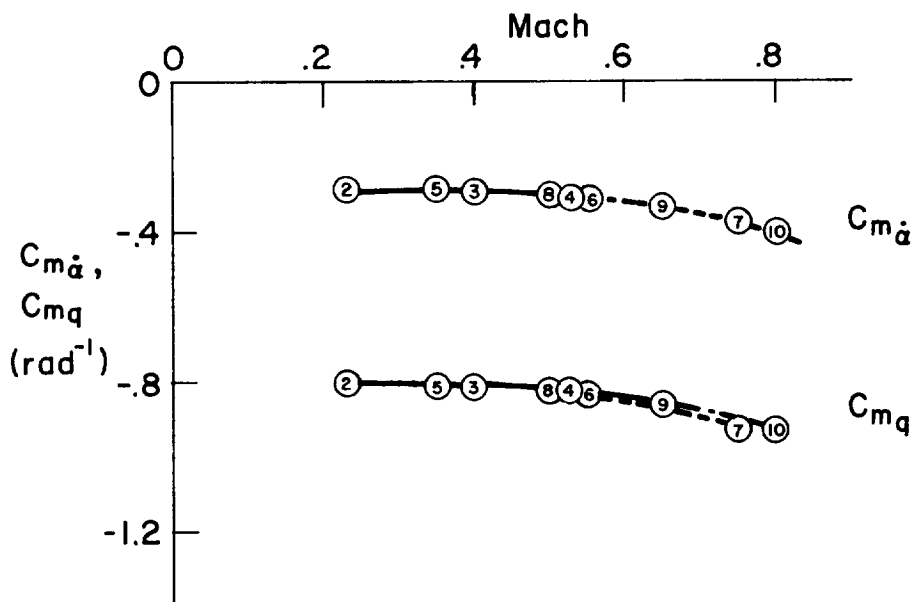
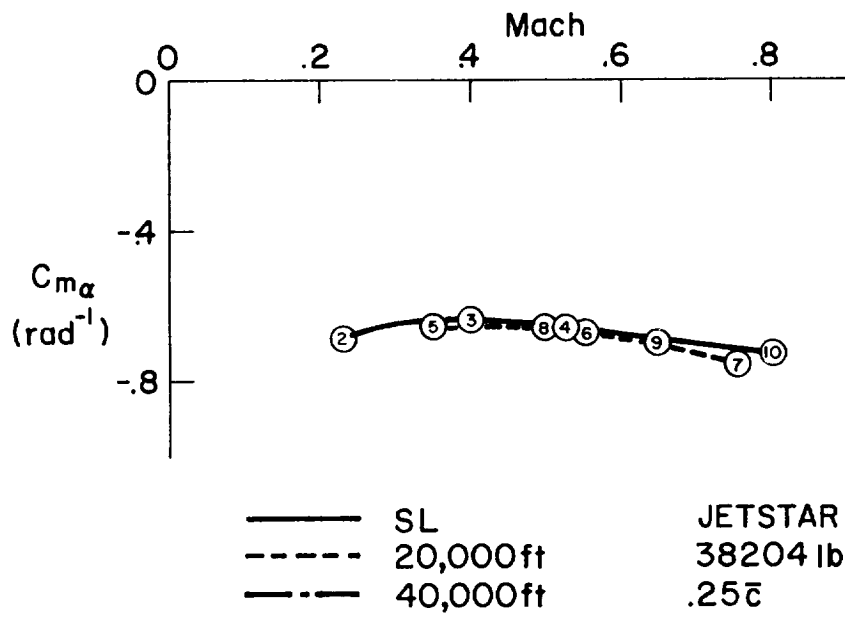
Longitudinal	Lateral-Directional (Body Axis)
$C_L = .737$	$C_{Y\beta} = -.72/\text{rad}$
$C_D = .095$	$C_{n\beta} = .137/\text{rad}$
$C_{L\alpha} = 5.0/\text{rad}$	$C_{\ell\beta} = -.103/\text{rad}$
$C_{D\alpha} = .75/\text{rad}$	$C_{\ell p} = -.37/\text{rad}$
$C_{m\alpha} = -.80/\text{rad}$	$C_{n p} = -.14/\text{rad}$
$C_{m\dot{\alpha}} = -3.0/\text{rad}$	$C_{\ell r} = .11/\text{rad}$
$C_{mq} = -8.0/\text{rad}$	$C_{n r} = -.16/\text{rad}$
$C_{L\delta_e} = .4/\text{rad}$	$C_{n\delta_a} = -.0075/\text{rad}$
$C_{m\delta_e} = -.81/\text{rad}$	$C_{\ell\delta_a} = .054/\text{rad}$
	$C_{Y\delta_r} = .175/\text{rad}$
	$C_{n\delta_r} = -.063/\text{rad}$
	$C_{\ell\delta_r} = .029/\text{rad}$

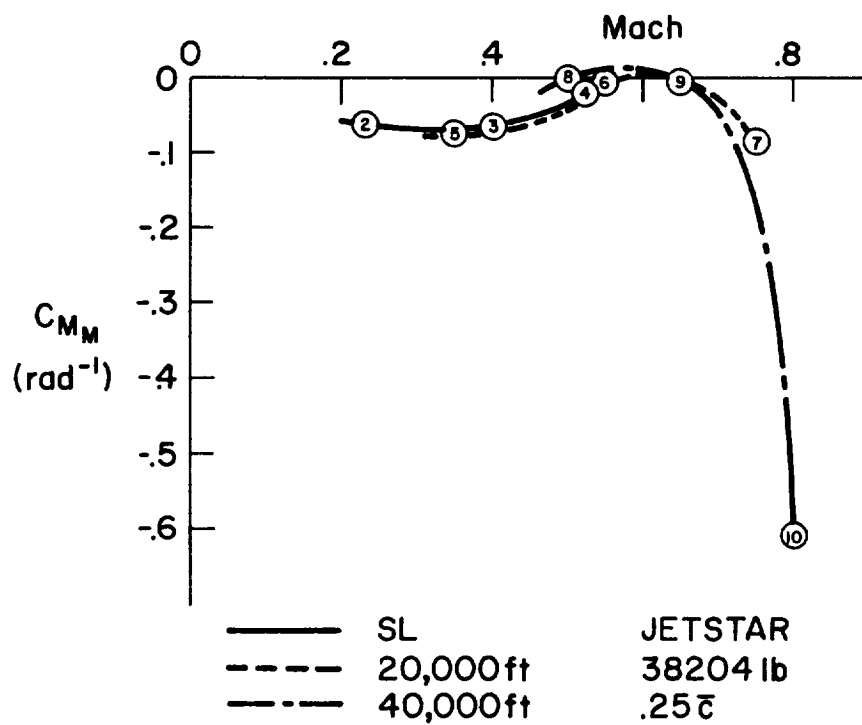
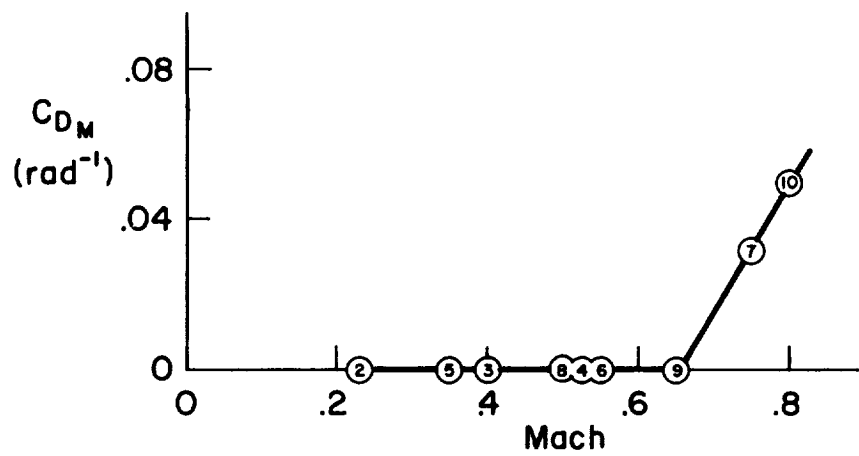


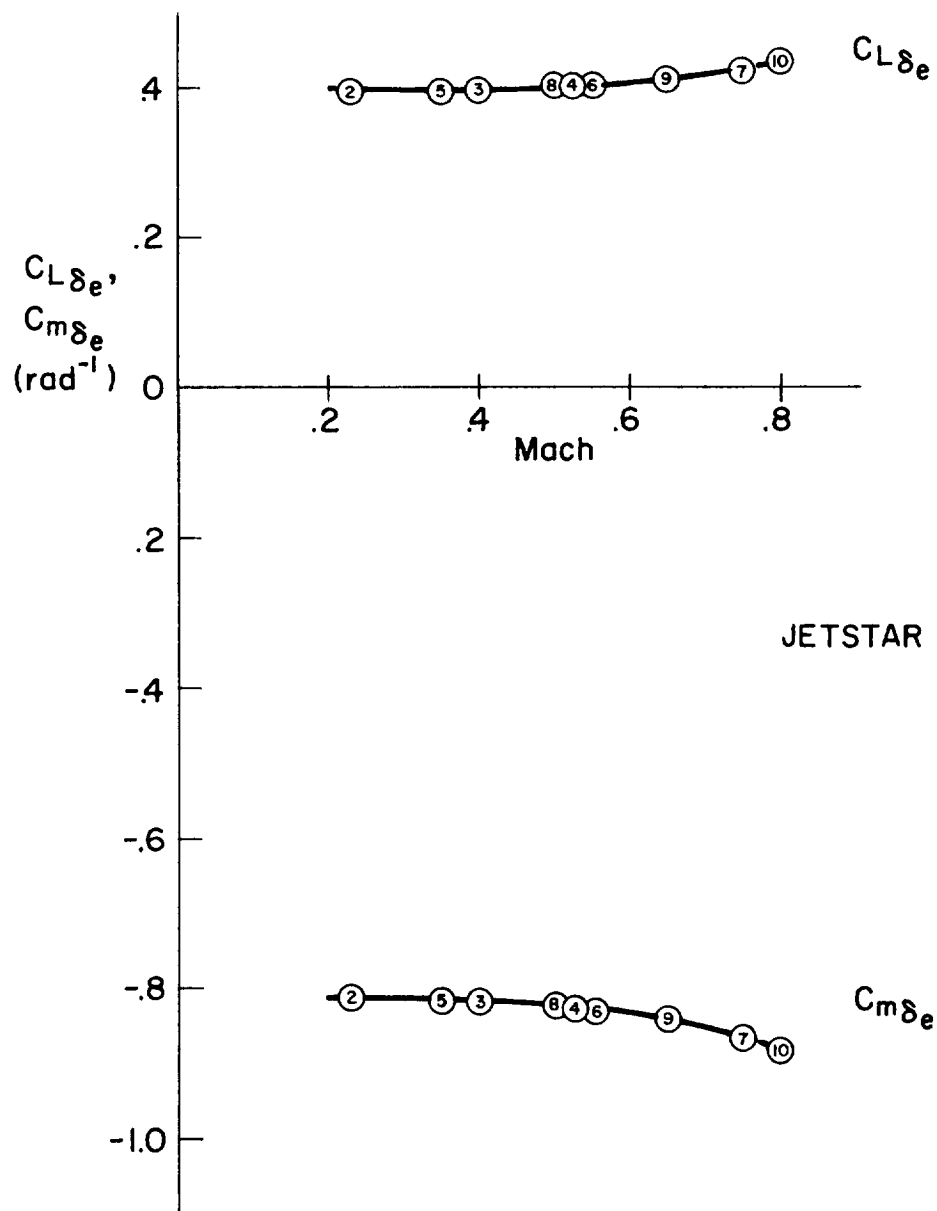
173

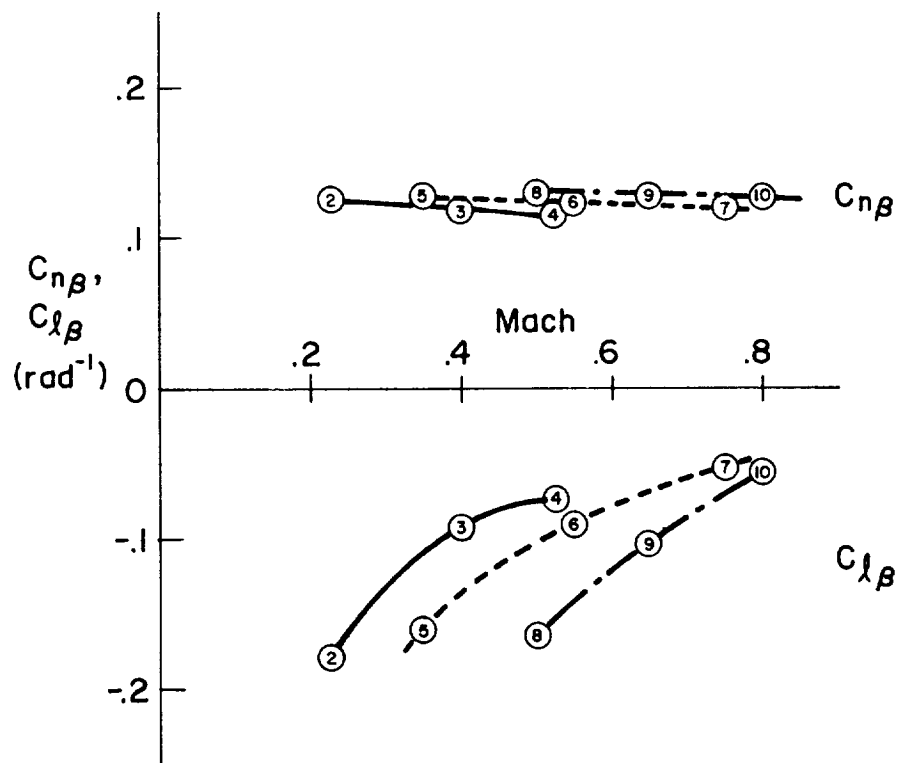
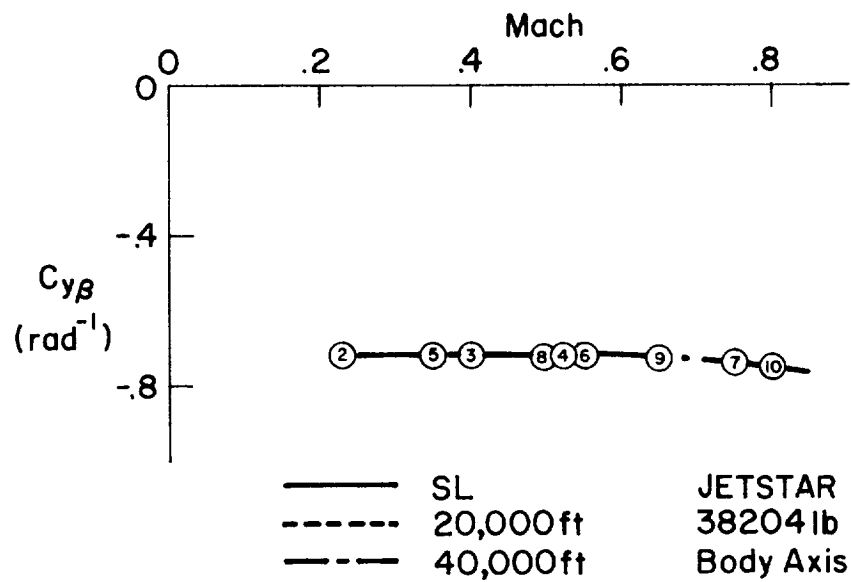


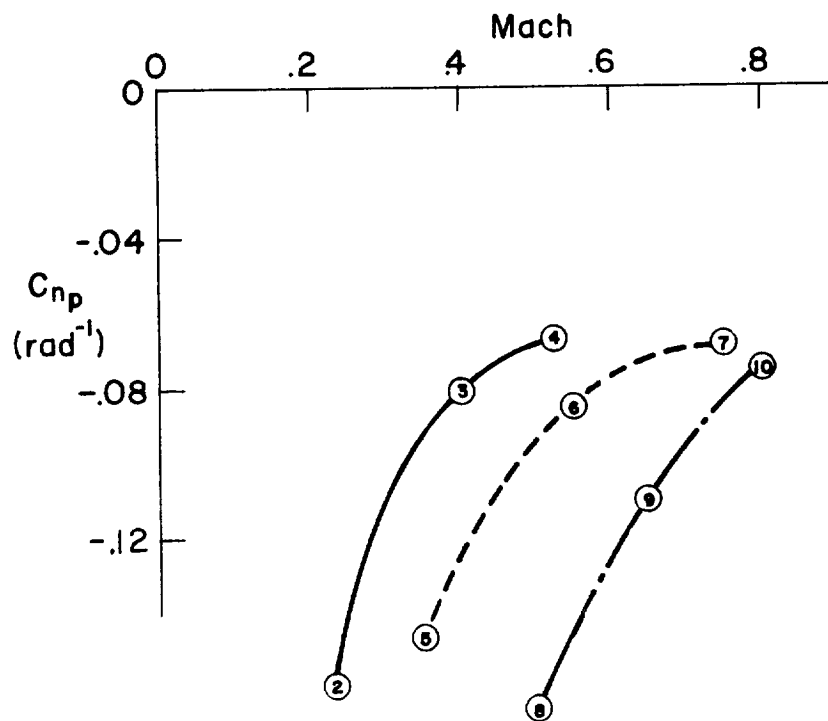
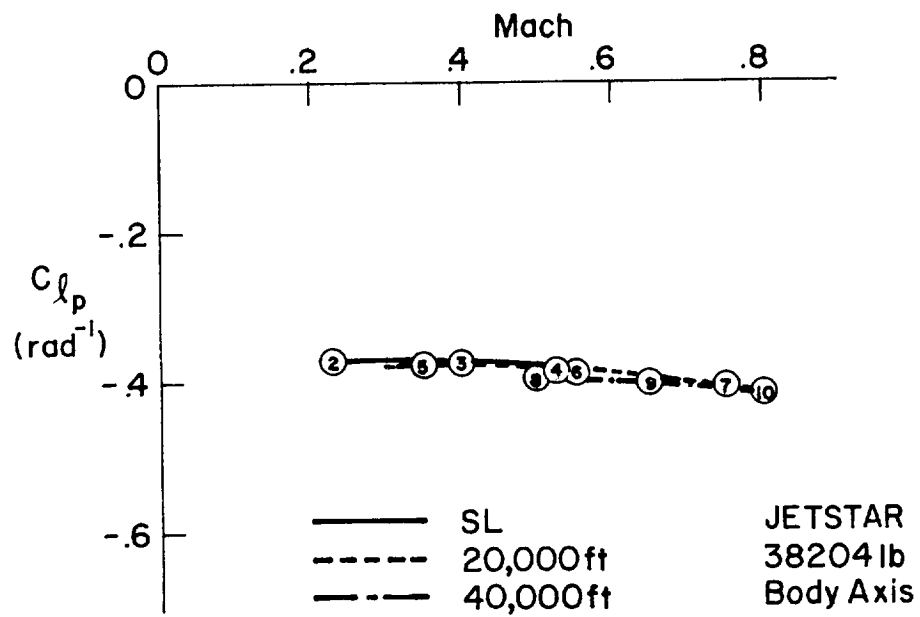


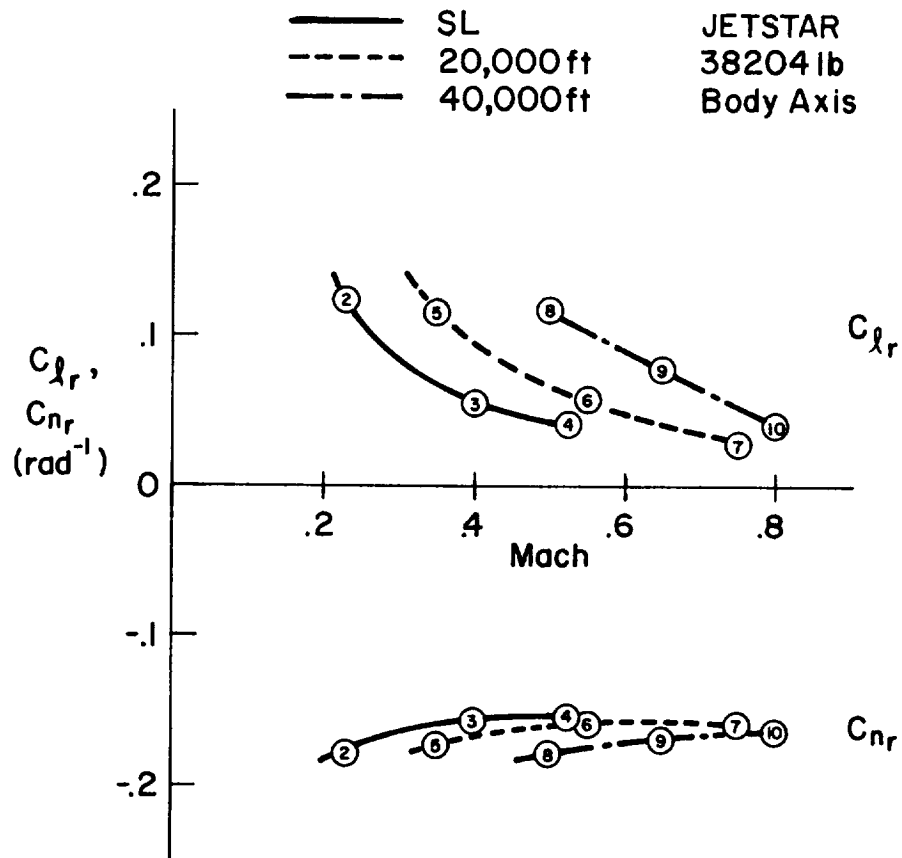


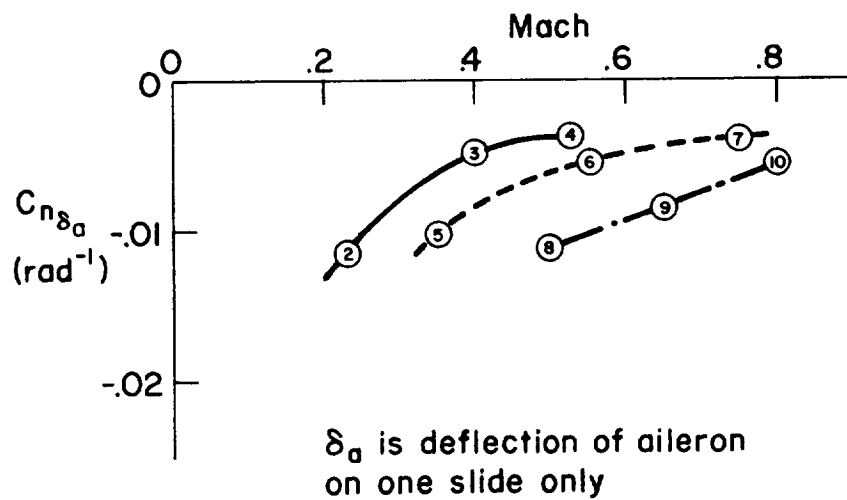
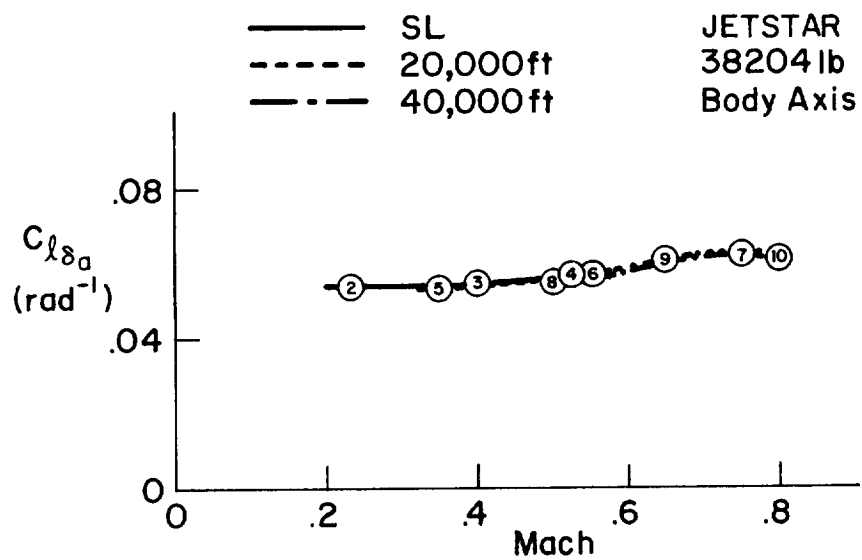


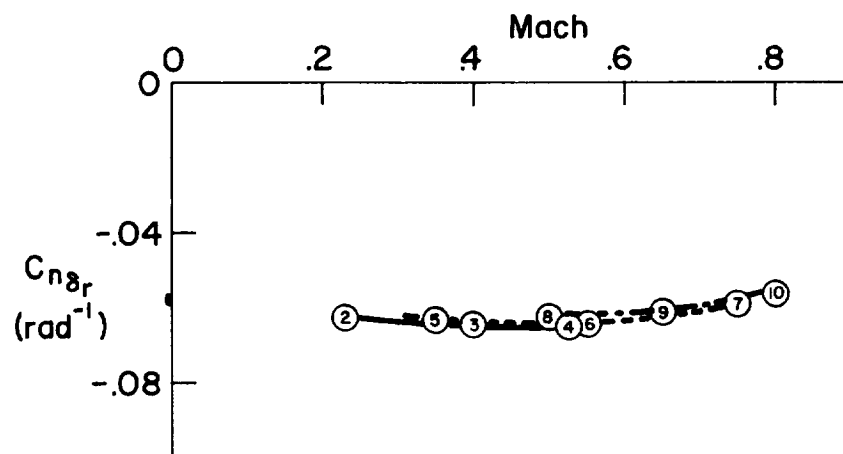
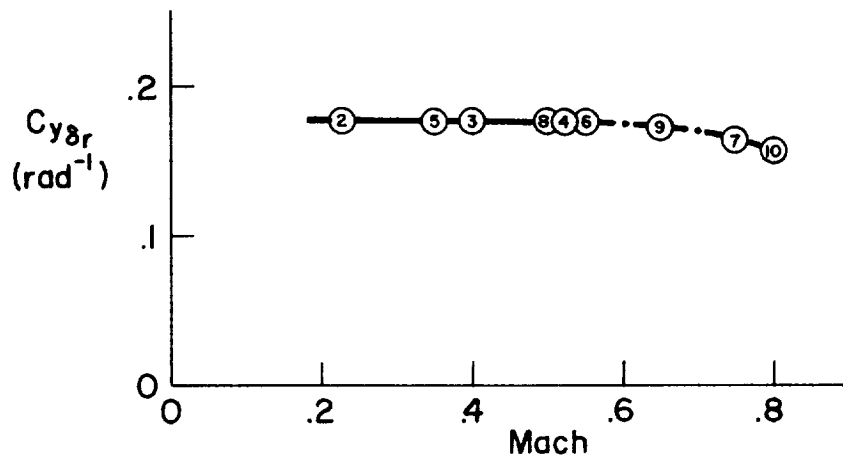












— SL
 - - - 20,000ft
 - · - 40,000ft

JETSTAR
 38204lb
 Body Axis

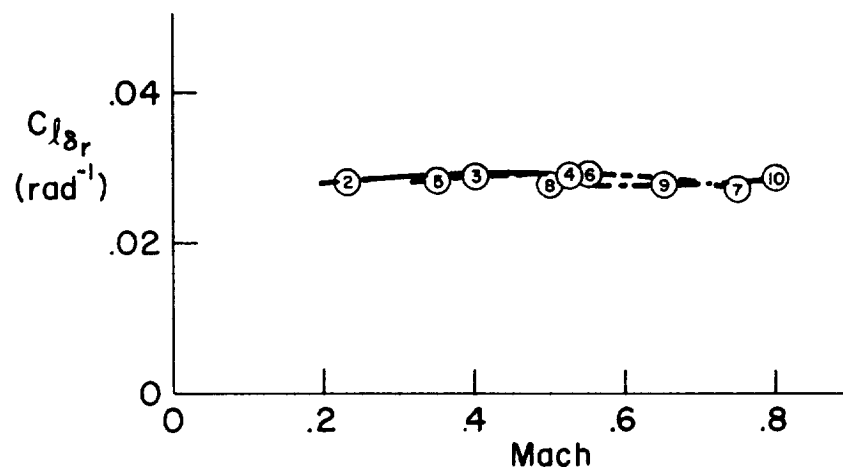


TABLE VII-2

JETSTAR DIMENSIONAL, MASS, AND FLIGHT CONDITION PARAMETERS

F/C #	1	2	3	4	5	6	7	8	9	10
H(FT)	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M(-)	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
VTO(FPS)	224.	257.	447.	586.	363.	570.	778.	484.	629.	774.
VTO(KTAS)	133.	152.	265.	347.	215.	338.	461.	287.	373.	459.
VTO(KCAS)	132.	152.	265.	347.	158.	252.	348.	146.	193.	243.
W(LBS)	23905.	38205.	38205.	38205.	38205.	38205.	38205.	38205.	38205.	38205.
C.G.(MGC)	.250	.250	.250	.250	.250	.250	.250	.250	.250	.250
IX (SLUG-FT SQ)	42275.	118779.	118779.	118779.	118779.	118779.	118779.	118779.	118779.	118779.
IY (SLUG-FT SQ)	126106.	135876.	135876.	135876.	135876.	135876.	135876.	135876.	135876.	135876.
IZ (SLUG-FT SQ)	160113.	243518.	243518.	243513.	243518.	243518.	243518.	243518.	243518.	243518.
IXZ(SLUG-FT SQ)	5470.	5061.	5061.	5061.	5061.	5061.	5061.	5061.	5061.	5061.
EPSILON(DEG)	-2.65	-2.32	-2.32	-2.32	-2.32	-2.32	-2.32	-2.32	-2.32	-2.32
Q(PSF)	59.4	78.4	237.	408.	83.5	206.	383.	69.0	117.	177.
QC(PSF)	60.0	79.4	247.	437.	86.0	222.	440.	73.4	129.	207.
ALPHA(DEG)	6.50	11.2	4.00	2.70	9.90	4.50	2.60	11.4	7.00	4.20
GAMMA(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LXP(FT)	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2	22.2
LZP(FT)	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40	-2.40
ITH(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
XI(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LTH(FT)	-.820	-.820	-.820	-.820	-.820	-.820	-.820	-.820	-.820	-.820

TABLE VII-3

JETSTAR LONGITUDINAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
XU *	-.0166	-.00456	-.0102	-.0136	-.00324	-.00697	-.0157	-.00353	-.00168	-.211E-5
ZU *	-.175	-.103	-.0593	-.0305	-.0804	-.0436	-.0212	-.0614	-.0408	-.0348
MU *	.00131	.00175	.000549	.000727	.00102	.000815	-.000470	.000902	.000747	-.00425
XW	.108	.164	.118	.100	.111	.0918	.0689	.0858	.0498	.0266
ZW	-1.01	-.723	-1.24	-1.65	-.565	-.881	-1.33	-.354	-.475	-.655
MW	-.00991	-.00902	-.0146	-.0201	-.00665	-.0107	-.0154	-.00401	-.00561	-.00760
ZWD	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ZQ	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MWD	-.000910	-.000834	-.000848	-.000906	-.000447	-.000482	-.000574	-.000207	-.000237	-.000280
MQ	-.546	-.582	-1.03	-1.33	-.439	-.724	-1.09	-.279	-.380	-.506
XDE	1.97	2.78	3.02	3.51	2.62	2.96	3.34	2.49	2.66	2.54
ZDE	-17.2	-14.0	-43.2	-74.3	-15.0	-37.5	-73.5	-12.4	-21.7	-34.6
MDE	-2.26	-2.80	-8.38	-14.6	-2.95	-7.47	-14.5	-2.47	-4.27	-6.73
XD TH	.00135	.000842	.000842	.000842	.000842	.000842	.000842	.000842	.000842	.000842
ZD TH	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
PD TH	-.650E-5	-.604E-5	-.604E-5	-.604E-5	-.604E-5	-.604E-5	-.604E-5	-.604E-5	-.604E-5	-.604E-5
	+	+	+	+	+	+	+	+	+	+

TABLE VII-4

JETSTAR ELEVATOR TRANSFER FUNCTION FACTORS

Bare Airframe
(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
DE Nominator										
1/T(DET)1	(.0521)	(.0293)	(.0626)	(.102)	(.0386)	(.0498)	-.0195	(.0600)	(.0492)	.102
1/T(DET)2	(.188)	(.160)	(.0797)	(.0644)	(.115)	(.0751)	.0339	(.0937)	(.0709)	-.134
Z(DET)1	.528	.456	.475	.477	.355	.362	.382	.252	.259	.289
W(DET)1	1.66	1.66	2.75	3.75	1.64	2.60	3.77	1.44	1.93	2.45
NUMERATORS										
N(U /DE)										
A(U)	1.97	2.78	3.02	3.51	2.62	2.96	3.34	2.49	2.66	2.54
1/T(U)1	28.5	50.2	86.4	115.	70.1	113.	154.	94.7	123.	151.
Z(U)1	.590	.384	.258	.252	.410	.274	.335	.378	.558	.727
W(U)1	1.11	.670	1.10	1.35	.529	.773	1.07	.340	.434	.596
N(W /DE)										
A(W)	-17.2	-14.0	-43.2	-74.5	-15.0	-37.5	-73.5	-12.4	-21.7	-34.6
1/T(W)1	29.7	50.9	87.4	116.	70.7	114.	155.	95.1	124.	152.
Z(W)1	.0612	.00143	.0704	.146	.0105	.0581	.270	.0191	.00430	-.0104
W(W)1	.161	.118	.0662	.0427	.0867	.0515	.0273	.0659	.0476	.0237
N(TH/DE)										
A(TH)	-2.25	-2.79	-8.34	-14.5	-2.94	-7.45	-14.5	-2.47	-4.27	-6.77
1/T(TH)1	.0360	.0297	.0160	.0155	.0199	.0118	.0158	.0198	.00589	.00210
1/T(TH)2	.919	.653	1.17	1.57	.515	.824	1.25	.317	.443	.626
N(HD /DE)										
A(HD)	17.4	14.3	43.3	74.6	15.3	37.7	73.5	12.6	21.8	34.7
1/T(HD)1	-.00931	-.0168	.00715	.0118	-.0104	.00405	.0143	-.00751	-.00553	-.00182
1/T(HD)2	-4.77	-5.36	-9.36	-12.5	-5.73	-9.17	-13.1	-5.34	-7.10	-9.36
1/T(HD)3	5.57	6.19	10.8	14.4	6.36	10.2	14.7	5.74	7.64	10.1
N(AZP/DE)										
A(AZP)	32.7	47.9	142.	248.	50.3	128.	249.	42.4	75.1	116.
1/T(AZP)1	.0198	.0196	-.00611	-.00277	.0129	-.00549	-.00195	.0126	.00390	.00177
1/T(AZP)2	-.0297	-.0372	.0132	.0145	-.0237	.00951	.0162	-.0207	-.00952	-.00361
Z(AZP)1	.140	.106	.0999	.100	.0790	.0746	.0777	.0550	.0533	.0565
W(AZP)1	3.76	3.11	5.54	7.33	3.29	5.23	7.54	2.98	4.02	5.32

TABLE VII-5

JETSTAR THRUST TRANSFER FUNCTION FACTORS

Bare Airframe

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
DE Nominator										
1/T(DET)1	(.0521)	(.0293)	(.0626)	(.102)	(.0386)	(.0498)	-.0195	(.0600)	(.0492)	.102
1/T(DET)2	(.188)	(.160)	(.0797)	(.0644)	(.115)	(.0751)	.0339	(.0937)	(.0709)	-.134
Z(DET)1	.528	.456	.475	.477	.355	.362	.382	.252	.259	.289
W(DET)1	1.66	1.66	2.79	3.75	1.64	2.60	3.77	1.44	1.93	2.45
NUMERATORS										
N(U /DTH)										
A(U)	.00135	.000842	.000842	.000842	.000842	.000842	.000842	.000842	.000842	.000842
1/T(U)1	.0430	.0409	.0327	.0251	.0348	.0261	.0199	.0273	.0225	.0208
Z(U)1	.546	.547	.508	.499	.474	.417	.411	.475	.387	.347
W(U)1	1.68	1.67	2.80	3.75	1.66	2.61	3.78	1.46	1.98	2.55
N(W /DTH)										
A(W)	-.00165	-.00157	-.00272	-.00354	-.00220	-.00345	-.00469	-.00290	-.00379	-.00468
1/T(W)1	(-.440)	(-.752)	(-.354)	(-.905)	(-.648)	(-.922)	.00970	(-.776)	(-.958)	-.000608
1/T(W)2	(.160)	(.135)	(.0676)	(.0436)	(.0954)	(.0541)	.0747	(.0752)	(.0521)	.595
N(TH/DTH)										
A(TH)	-.636E-5	-.590E-5	-.599E-5	-.600E-5	-.597E-5	-.600E-5	-.601E-5	-.600E-5	-.601E-5	-.602E-5
1/T(TH)1	-.502	-.357	-.150	-.135	-.226	-.167	.0450	-.178	-.151	.438
1/T(TH)2	1.22	.822	1.32	1.70	.646	.937	1.37	.408	.522	.820
N(HD /DTH)										
A(HD)	.000152	.000164	.587E-4	.397E-4	.000145	.661E-4	.382E-4	.000166	.000103	.617E-4
1/T(HD)1	4.70	3.32	-.203	-.159	-.611	-.224	.0482	-.454	-.238	.610
1/T(HD)2	(-.615)	(-.672)	-5.24	-9.48	-1.16	-4.95	-10.7	-1.18	-2.90	-6.06
1/T(HD)3	(1.14)	(.951)	8.94	13.5	3.40	7.61	13.9	2.67	4.48	7.32
N(AZP/DTH)										
A(AZP)	.000141	.000131	.000133	.000133	.000133	.000133	.000133	.000133	.000133	.000134
1/T(AZP)1	-.0157	-.0235	-.00493	-.00257	-.0147	-.00437	-.00191	-.0126	-.00611	-.00307
1/T(AZP)2	-.809	-.483	-.167	-.141	-.292	-.183	.0457	-.233	-.175	.542
Z(AZP)1	-.0231	.0565	.0980	.115	.0370	.0725	.0307	.0163	.0408	.0521
W(AZP)1	2.81	2.70	4.93	6.55	2.91	4.71	6.82	2.66	3.62	4.79

TABLE VII-6

JETSTAR LONGITUDINAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
N	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
STICK FIXED										
D(G)/D(U) (DEG/KT)	.0278	.0502	-.0215	-.0354	.0310	-.0122	-.0429	.0225	.0166	.00545
NZA (G/RAD)	6.32	5.24	16.0	28.0	5.86	14.5	30.0	4.90	8.64	15.0
DE/G (DEG/G)	10.8	10.4	3.29	1.94	8.64	3.53	1.85	9.43	5.70	3.49
CAP (RAD/SEC/SEC/G)	.425	.506	.478	.492	.444	.459	.468	.406	.425	.413
PHUGOID(2) (SEC) (TUCK(2))	--	--	--	--	--	--	(35.4)	--	--	(5.17)
1/C(1/10)	1.70	1.40	1.47	1.48	1.04	1.06	1.13	.711	.731	.825
	+	+	+	+	+	+	+	+	+	+

TABLE VII-7

JETSTAR LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
YV	-.140	-.100	-.175	-.229	-.0756	-.119	-.167	-.0469	-.0618	-.0781
YB	-31.2	-25.8	-78.0	-134.	-27.5	-67.8	-130.	-22.7	-38.9	-60.5
LB'	-4.05	-3.42	-5.27	-7.28	-3.23	-4.43	-4.93	-2.75	-2.93	-2.27
NB'	1.34	1.10	3.30	5.47	1.21	2.99	5.63	1.02	1.75	2.66
LP'	-1.85	-.752	-1.30	-1.75	-.582	-.935	-1.34	-.380	-.492	-.635
NP'	-.245	-.173	-.164	-.187	-.121	-.119	-.137	-.0840	-.0758	-.0682
LR'	.517	.234	.181	.170	.169	.124	.0868	.105	.0936	.0551
NR'	-.190	-.172	-.261	-.333	-.125	-.178	-.252	-.0804	-.0994	-.120
Y*CA	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
L'CA	2.21	1.04	3.14	5.71	1.10	2.88	5.83	.929	1.71	2.64
N'CA	-.00557	-.0864	-.0767	-.0524	-.0770	-.0759	-.0624	-.0716	-.0831	-.0720
Y*CR	.0340	.0244	.0424	.0557	.0184	.0289	.0371	.0114	.0144	.0162
L'CR	1.11	.533	1.61	2.77	.568	1.40	2.43	.444	.766	1.21
N'CR	-.644	-.580	-1.81	-3.12	-.618	-1.55	-2.66	-.511	-.836	-1.16

TABLE VII-9

JETSTAR RUDDER TRANSFER FUNCTION FACTORS

Bare Airframe

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K	
M	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800	
DENOMINATOR											
1/T(DET)1	-.0112	.00318	.00535	.00467	.000351	.00242	.00186	-.000600	-.000248	-.000201	
1/T(DET)2	1.95	.658	1.45	1.89	.741	1.04	1.42	.499	.576	.680	
Z(DET)1	.0832	.0229	.0729	.0856	.0147	.0499	.0690	.00352	.0267	.0453	
W(DET)1	1.45	1.39	1.97	2.47	1.37	1.86	2.45	1.26	1.46	1.69	
NUMERATORS											
N(B /DR)											
A(B)	.0340	.0244	.0424	.0557	.0184	.0289	.0371	.0114	.0144	.0162	
1/T(B)1	-.0255	-.0312	-.00240	.000986	-.0201	-.00232	.00211	-.0164	-.00599	.00100	
1/T(B)2	2.13	.670	1.43	1.90	.668	1.02	1.45	.434	.548	.698	
1/T(B)3	22.5	27.7	45.4	58.5	38.5	57.4	74.6	51.7	64.2	77.1	
N(P /DR)											
A(P)	1.11	.533	1.61	2.77	.568	1.40	2.43	.444	.766	1.21	
1/T(P)1	-.0161	-.0246	-.00502	-.00259	-.0154	-.00444	-.00177	-.0134	-.00627	-.00303	
Z(P)1	(.974)	(1.55)	(-1.58)	(-1.54)	(1.47)	(-1.34)	.252	(1.43)	(-1.19)	.0848	
W(P)1	(-1.05)	(-1.67)	(1.67)	(1.77)	(-1.55)	(1.42)	.496	(-1.48)	(1.20)	.695	
N(R /DR)											
A(R)	-.644	-.580	-1.81	-3.12	-.618	-1.55	-2.66	-.511	-.836	-1.16	
1/T(R)1	2.25	.803	1.42	1.90	.604	1.01	-.0455	.363	.506	-.156	
1/T(R)2	(.147)	(.134)	(.181)	(.287)	(.116)	(.161)	.131	(.0982)	(.120)	.186	
1/T(R)3	(.333)	(.607)	(.344)	(.264)	(.553)	(.310)	1.47	(.579)	(.365)	.717	
N(PHI/DR)											
A(PHI)	1.03	.418	1.48	2.63	.460	1.28	2.31	.341	.663	1.12	
Z(PHI)1	(.933)	(1.65)	(1.68)	(-1.64)	(1.57)	(1.43)	.179	(1.58)	(1.26)	.0464	
W(PHI)1	(-1.20)	(-2.09)	(-1.72)	(1.76)	(-1.85)	(-1.47)	.502	(-1.81)	(-1.34)	.721	
N(AYP/DR)											
A(AYP)	-4.05	-5.33	-17.4	-30.0	-5.68	-14.6	-24.3	-4.77	-7.66	-10.3	
1/T(AYP)1	-.0407	-.0666	-.00942	-.00202	-.0408	-.00693	.00234	-.0317	-.0118	.00207	
1/T(AYP)2	3.99	.550	1.37	1.92	.453	.951	1.56	.269	.457	.765	
Z(AYP)1	.233	.243	.111	.0878	.173	.0953	.0472	.126	.0872	.0394	
W(AYP)1	1.20	1.57	2.20	2.83	1.59	2.07	2.68	1.50	1.67	1.87	
	+	+	+	+	+	+	+	+	+	+	+

TABLE VII-10

JETSTAR LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.200	.230	.400	.525	.350	.550	.750	.500	.650	.800
DR PERIOD (SEC)	4.34	4.51	3.20	2.55	4.60	3.39	2.57	4.97	4.29	3.72
1/C(1/2)	.757	.208	.662	.778	.134	.453	.627	.0319	.242	.411
SPIRAL (2) (SEC)	62.1	--	--	--	--	--	--	866.	2795.	3443.
P(1)	.865	.594	1.60	2.45	.756	1.89	3.87	.753	1.53	3.47
P(2)	.464	.0631	1.51	2.44	.352	1.88	3.73	.385	1.53	3.36
P(3)	.904	.709	1.88	2.78	1.03	2.40	3.84	1.17	2.31	3.53
P(2)/P(1)	.537	.106	.943	.996	.466	.999	.962	.510	.996	.970
P(USC)/P(AV)	.311	.823	.0705	.0342	.434	.0639	.0173	.428	.114	.0200
W(PHI)/W(D)	.804	.664	.912	.949	.740	.919	.970	.724	.874	.958
DEL-B-MAX	.381	.368	.251	.207	.356	.262	.203	.374	.339	.261
PHI TO BETA, PHASE	57.9	-327.	31.8	31.8	-333.	26.0	25.7	-340.	19.7	20.0
PHI TO BETA	1.22	1.48	1.13	.983	1.50	1.11	.708	1.53	1.21	.693
PHI TO VE	.314	.330	.145	.0961	.325	.153	.0714	.365	.222	.103
	+	+	+	+	+	+	+	+	+	+

JETSTAR DATA SOURCES

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Models C-140A and VC-140B Aircraft, T. O. 1C-140A-2

SECTION VIII
CONVAIR 880M

CONVAIR 880M BACKGROUND

The Convair 880M is a medium-size four engine jet transport. Longitudinal and directional control consists of servo tab deflected elevators and rudder. Lateral control consists of servo tab deflected ailerons plus hydraulic actuated spoilers.

Elevator, aileron, and rudder transfer functions are in terms of respective primary surface deflections with tab losses included. Although the control system diagram shows a lag in the spoiler actuator, none was used in computing transfer functions.

CV-880M

Nominal Configuration

W = 155000 lb
c.g. at .25 $\bar{c}$, W.L. -19.2
 $I_x = 1510000 \text{ slug-ft}^2$
 $I_y = 2510000 \text{ slug-ft}^2$
 $I_z = 4100000 \text{ slug-ft}^2$

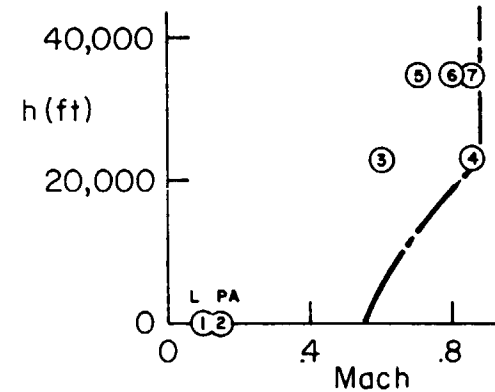
Power Approach Configuration

Flaps 30°
Gear Up
W = 126000 lb
c.g. at .195 $\bar{c}$, W.L. -19.2
 $I_x = 1150000 \text{ slug-ft}^2$
 $I_y = 2450000 \text{ slug-ft}^2$
 $I_z = 4070000 \text{ slug-ft}^2$

Landing Configuration

Same as Power Approach except:
Flaps 50°
Speed Brakes 8°
Gear Down

Flight Envelope



— — — — — Speed Restrictions

① Transfer Function Case

Figure VIII-1. Convair 880M Flight Conditions

CV-880M

$S = 2000 \text{ ft}^2$

$b = 120 \text{ ft}$

$\bar{c} = 18.94 \text{ ft}$

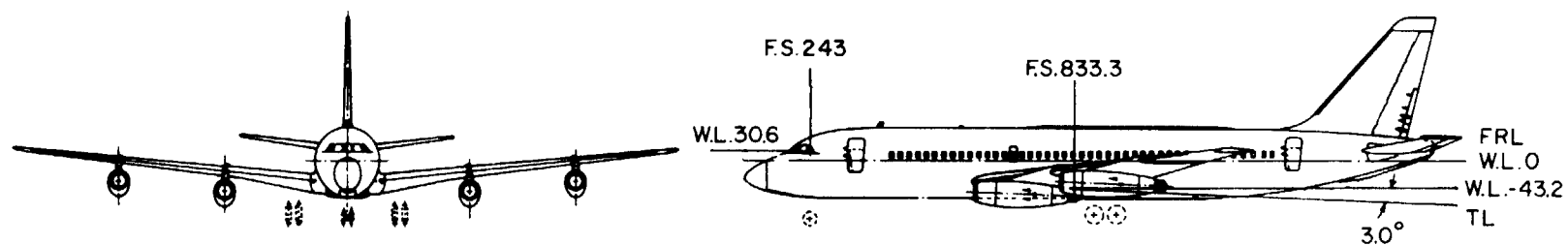
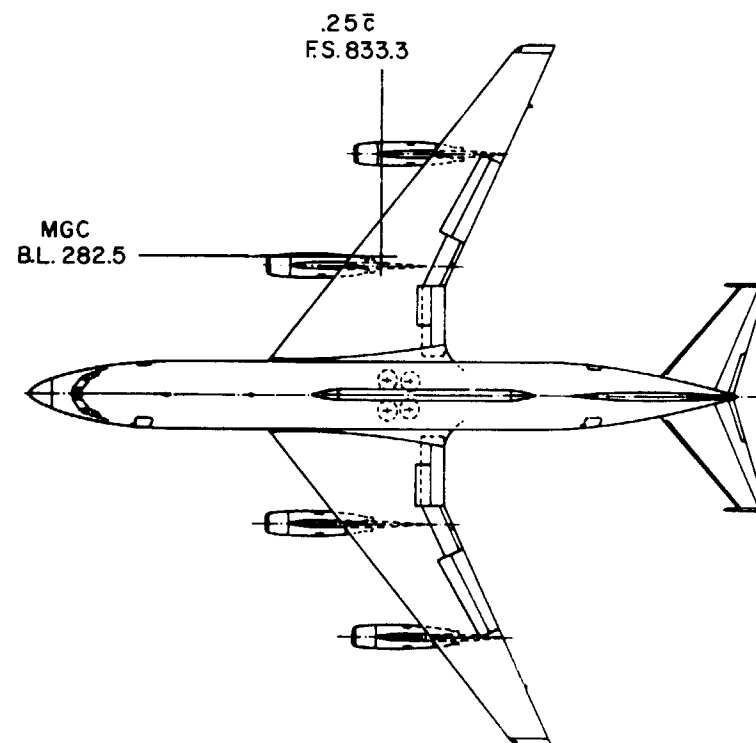
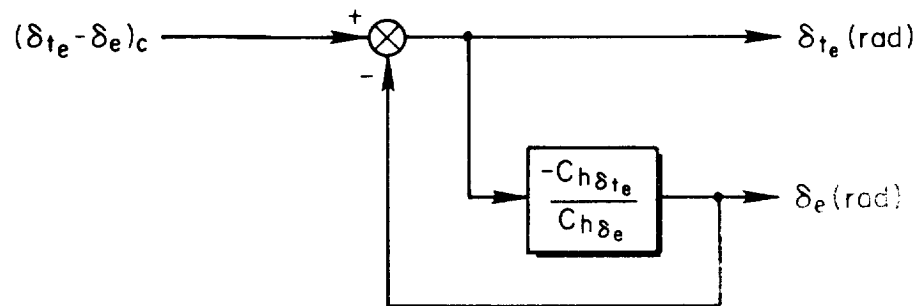


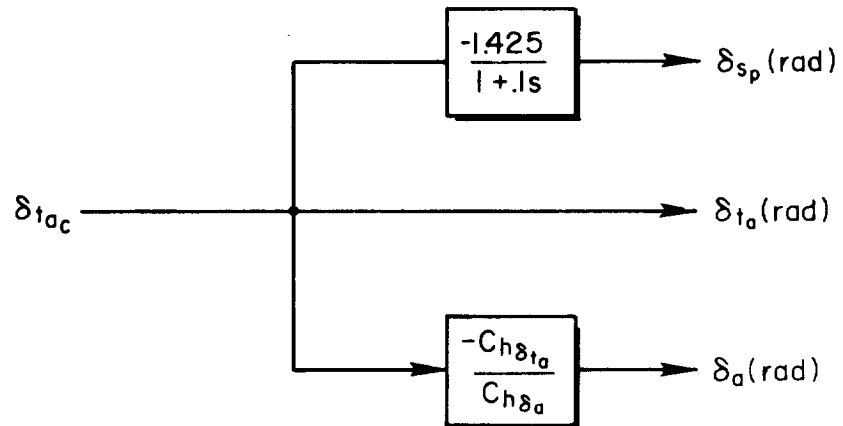
Figure VIII-2. Convair 880M General Arrangement

CV-880M

PITCH AXIS



ROLL AXIS



YAW AXIS

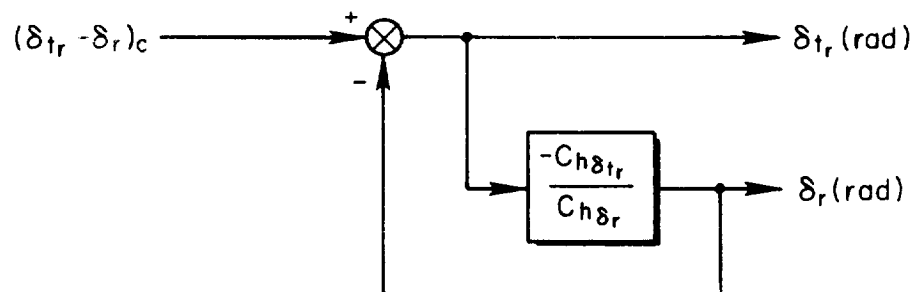


Figure VIII-3. CV-880M Control System

TABLE VIII-1

CV-880M**Longitudinal Non-Dimensional Stability Derivatives**

Flight Condition	1	2	3	4	5	6	7
Configuration	L	PA					
Speed	134 KTAS	165 KTAS	.6M	.86M	.7M	.8M	.86M
Altitude	SL	SL	23K	23K	35K	35K	35K
α_o (Deg)	5.2	4.3	5.3	2.8	8.3	4.7	4.0
C_L	1.03	0.68	0.36	0.175	0.454	0.347	0.301
C_D	0.154	0.080	0.022	0.019	0.025	0.024	0.023
$C_{L\alpha}$ (1/rad)	4.66	4.52	4.28	4.41	4.62	4.8	4.9
$C_{D\alpha}$ (1/rad)	0.43	0.27	0.14	0.07	0.18	0.15	0.13
$C_{m\alpha}$ (1/rad)	-0.381	-0.903	-0.522	-0.572	-0.568	-0.65	-0.74
$C_{L\dot{\alpha}}$ (1/rad)	2.7	2.7	2.44	2.5	2.75	2.75	2.9
C_{Lq} (1/rad)	7.92	7.72	6.76	6.37	7.51	7.5	7.62
$C_{m\dot{\alpha}}$ (1/rad)	-4.17	-4.13	-4.16	-4.66	-4.4	-4.5	-4.6
C_{mq} (1/rad)	-12.2	-12.1	-11.5	-11.8	-12.	-12.	-12.
$C_{L\delta_e}$ (1/rad)	0.22	0.213	0.193	0.141	0.203	0.190	0.180
$C_{m\delta_e}$ (1/rad)	-0.657	-0.637	-0.586	-0.438	-0.618	-0.57	-0.532
$C_{h\delta_e}$ (1/rad)	-0.326	-0.328	-0.336	-0.278	-0.342	-0.31	-0.285
$C_{L\delta_{te}}$ (1/rad)	0.055	0.0532	0.0482	0.0352	0.0508	0.047	0.0450
$C_{m\delta_{te}}$ (1/rad)	-0.164	-0.159	-0.146	-0.11	-0.155	-0.14	-0.134
$C_{h\delta_{te}}$ (1/rad)	-0.287	-0.285	-0.297	-0.343	-0.312	-0.335	-0.352

TABLE VIII-2

CV-880M**Lateral-Directional Non-Dimensional Derivatives
(Stability Axis System)**

Flight Condition	1	2	3	4	5	6	7
Configuration	L	PA					
Speed	134 KTAS	165 KTAS	.6M	.86M	.7M	.8M	.86M
Altitude	SL	SL	23K	23K	35K	35K	35K
C_{Y_β} (1/rad)	-1.015	-0.877	-0.788	-0.815	-0.807	-0.8125	-0.842
C_{l_β} (1/rad)	-0.239	-0.196	-0.163	-0.145	-0.181	-0.177	-0.179
C_{n_β} (1/rad)	0.145	0.139	0.128	0.122	0.129	0.129	0.133
C_{l_p} (1/rad)	-0.395	-0.381	-0.329	-0.243	-0.341	-0.312	-0.294
C_{n_p} (1/rad)	-0.087	-0.049	-0.0173	-0.0031	-0.023	-0.011	-0.0054
C_{l_r} (1/rad)	0.309	0.198	0.146	0.088	0.180	0.153	0.146
C_{n_r} (1/rad)	-0.218	-0.185	-0.163	-0.189	-0.166	-0.165	-0.165
$C_{Y_{\delta a}}$ (1/rad)	0	0	0.0019	0.0745	0.0044	0.00775	0.00975
$C_{l_{\delta a}}$ (1/rad)	-0.0487	-0.0384	-0.0466	-0.0452	-0.0479	-0.0497	-0.0479
$C_{n_{\delta a}}$ (1/rad)	0.01862	0.0172	0.00746	0.01061	0.007	0.00803	0.00975
$C_{h_{\delta a}}$ (1/rad)	-0.607	-0.481	-0.236	-0.258	-0.2233	-0.2005	-0.258
$C_{Y_{\delta ta}}$ (1/rad)	0	0	0	0	0	0	0
$C_{l_{\delta ta}}$ (1/rad)	-0.0072	-0.0056	-0.0068	-0.0068	-0.0071	-0.0075	-0.0071
$C_{n_{\delta ta}}$ (1/rad)	0	0	0	0	0	0	0
$C_{h_{\delta ta}}$ (1/rad)	-0.249	-0.227	-0.215	-0.2125	-0.226	-0.235	-0.213
$C_{Y_{\delta s}}$ (1/rad)	-0.078	-0.0315	-0.0189	-0.0175	-0.0189	-0.0189	-0.0175
$C_{l_{\delta s}}$ (1/rad)	0.0805	0.0405	0.029	0.0281	0.0324	0.0329	0.0339
$C_{n_{\delta s}}$ (1/rad)	0.0258	0.0129	0.01146	0.0109	0.00975	0.01004	0.00917
$C_{Y_{\delta r}}$ (1/rad)	0.223	0.2155	0.1904	0.1394	0.199	0.184	0.1685
$C_{l_{\delta r}}$ (1/rad)	0.0207	0.0226	0.0176	0.0183	0.0165	0.0187	0.0193
$C_{n_{\delta r}}$ (1/rad)	-0.0905	-0.0958	-0.0845	-0.0534	-0.0848	-0.0756	-0.0644
$C_{h_{\delta r}}$ (1/rad)	-0.2140	-0.2125	-0.1626	-0.1844	-0.1345	-0.1491	-0.1924
$C_{Y_{\delta tr}}$ (1/rad)	0.0493	0.0467	0.0374	0.0215	0.0404	0.0355	0.0316
$C_{l_{\delta tr}}$ (1/rad)	0.0021	0.0027	0.0016	0.0018	0.0014	0.0019	0.0020
$C_{n_{\delta tr}}$ (1/rad)	-0.020	-0.019	-0.015	-0.0077	-0.016	-0.0134	-0.011
$C_{h_{\delta tr}}$ (1/rad)	-0.255	-0.253	-0.267	-0.254	-0.27	-0.267	-0.265

TABLE VIII-3

CV-880M DIMENSIONAL, MASS, AND FLIGHT CONDITION PARAMETERS

	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	
H (FT)	SL	SL	23 K	23 K	35 K	35 K	35 K	
M (-)	.203	.249	.600	.860	.700	.800	.860	
VTO (FPS)	226.	278.	615.	881.	681.	779.	837.	
VTO (KTAS)	134.	165.	364.	522.	404.	461.	496.	
VTO (KCAS)	134.	165.	259.	381.	235.	272.	295.	
W (LBS)	126007.	126007.	155008.	155008.	155008.	155008.	155008.	
C.G. (MGC)	.195	.195	.250	.250	.250	.250	.250	
IX (SLUG-FT SQ)	.115E+7	.115E+7	.151E+7	.151E+7	.151E+7	.151E+7	.151E+7	
IY (SLUG-FT SQ)	.245E+7	.245E+7	.251E+7	.251E+7	.251E+7	.251E+7	.251E+7	
IZ (SLUG-FT SQ)	.359E+7	.359E+7	.410E+7	.410E+7	.410E+7	.410E+7	.410E+7	
IXZ (SLUG-FT SQ)	0.	0.	0.	0.	0.	0.	0.	
EPSILON (DEG)	0.	0.	0.	0.	0.	0.	0.	
Q (PSF)	60.8	92.2	216.	444.	171.	224.	259.	
QC (PSF)	61.4	93.6	236.	532.	193.	262.	310.	
ALPHA (DEG)	5.20	4.32	5.30	2.80	8.30	4.65	4.04	
GAMMA (DEG)	0.	0.	0.	0.	0.	0.	0.	
LXP (FT)	48.1	48.1	49.1	49.1	49.1	49.1	49.1	
LZP (FT)	-4.15	-4.15	-4.15	-4.15	-4.15	-4.15	-4.15	
ITH (DEG)	3.00	3.00	3.00	3.00	3.00	3.00	3.00	
XI (DEG)	3.00	3.00	3.00	3.00	3.00	3.00	3.00	
LTH (FT)	2.00	2.00	2.00	2.00	2.00	2.00	2.00	
	+	+	+	+	+	+	+	+

TABLE VIII-5
CV-880M ELEVATOR DIMENSIONAL DERIVATIVES
Bare Airframe
(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7
H	SL	SL	23 K	23 K	35 K	35 K	35 K
M	.203	.249	.600	.860	.700	.800	.860
DENOMINATOR							
Z(DET)1	.120	.0628	.0361	.0819	.0351	.0443	.0513
W(DET)1	.131	.137	.0659	.0452	.0528	.0538	.0504
Z(DET)2	.793	.599	.494	.492	.400	.399	.381
W(DET)2	.818	1.29	1.42	2.13	1.37	1.56	1.78
NUMERATORS							
N(U /DE)							
A(U)	.443	.531	1.14	1.00	1.51	1.09	1.08
1/T(U)1	18.1	23.1	67.2	98.5	74.9	84.3	88.2
Z(U)1	.345	.304	.192	.187	.236	.212	.209
W(U)1	1.08	1.11	.593	.822	.401	.535	.577
N(W /DE)							
A(W)	-4.87	-7.03	-12.3	-20.5	-10.4	-13.5	-15.4
1/T(W)1	19.7	24.7	67.7	99.2	75.3	84.9	88.7
Z(W)1	.0965	.0783	.0429	.105	.0554	.0533	.0641
W(W)1	.180	.143	.0508	.0329	.00961	.0397	.0366
N(TH/DE)							
A(TH)	-.439	-.642	-1.37	-2.33	-1.17	-1.48	-1.64
1/T(TH)1	.0841	.0505	.0121	.00977	.00815	.00932	.00876
1/T(TH)2	.597	.697	.596	.884	.477	.545	.595
N(HD /DE)							
A(HD)	4.89	7.05	12.3	20.5	10.5	13.5	15.4
1/T(HD)1	.0161	.0124	.00289	.00675	.00101	.00304	.00377
1/T(HD)2	3.34	3.94	6.11	9.02	5.78	6.56	7.00
1/T(HD)3	-3.89	-4.65	-6.72	-9.82	-6.30	-7.15	-7.63
N(AZP/DE)							
A(AZP)	16.3	23.9	54.9	93.5	46.9	59.3	65.3
1/T(AZP)1	-.0250	-.0154	-.00636	-.00200	-.00703	-.00428	-.00334
1/T(AZP)2	.0405	.0277	.00921	.00874	.00798	.00730	.00710
Z(AZP)1	.260	.250	.145	.143	.125	.124	.125
W(AZP)1	1.97	2.32	3.02	4.40	2.83	3.26	3.54

TABLE VIII-6

CV-880M THRUST DIMENSIONAL DERIVATIVES

Bare Airframe

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	
H	SL	SL	23 K	23 K	35 K	35 K	35 K	
M	.203	.249	.600	.860	.700	.800	.860	
DEACMINATOR								
Z(DET)1	.120	.0628	.0361	.0815	.0351	.0443	.0513	
W(DET)1	.131	.137	.0659	.0452	.0528	.0538	.0504	
Z(DET)2	.793	.599	.494	.492	.400	.399	.381	
W(DET)2	.818	1.29	1.42	2.13	1.37	1.56	1.78	
NUMERATORS								
N(L /DTH)								
A(L)	.000255	.000255	.000207	.000207	.000207	.000207	.000207	
1/T(U)1	-.104	-.0586	-.0453	-.0284	-.0438	-.0348	-.0293	
Z(U)1	.776	.590	.435	.461	.281	.335	.328	
W(L)1	.858	1.30	1.42	2.13	1.34	1.55	1.77	
N(W /DTH)								
A(W)	-.128E-4	-.129E-4	-.107E-4	-.107E-4	-.107E-4	-.108E-4	-.108E-4	
1/T(W)1	-8.64	-12.9	-43.6	-62.9	-.00906	-55.8	-60.1	
Z(W)1	-.400	-.0568	.590	.811	(.0937)	.750	.899	
W(W)1	.228	.158	.0456	.0295	(-49.5)	.0350	.0317	
N(TH/DTH)								
A(TH)	.842E-6	.839E-6	.807E-6	.805E-6	.806E-6	.803E-6	.803E-6	
1/T(TH)1	(.955)	(.838)	.130	.0388	.0850	.111	.113	
1/T(TH)2	(.398)	(.580)	.598	.943	.536	.559	.625	
N(HD /DTH)								
A(HD)	.359E-4	.320E-4	.298E-4	.209E-4	.405E-4	.275E-4	.253E-4	
1/T(HD)1	.137	.210	.0973	.0776	.0668	.0865	.0906	
Z(HD)1	.658	.501	.249	.210	.187	.197	.188	
W(HD)1	2.15	2.70	3.52	5.96	2.92	3.92	4.44	
N(AZP/DTH)								
A(AZP)	-.533E-4	-.532E-4	-.503E-4	-.503E-4	-.503E-4	-.502E-4	-.502E-4	
1/T(AZP)1	-.0155	-.00919	-.00469	-.00181	-.00686	-.00344	-.00276	
1/T(AZP)2	.163	.353	.113	.0845	.0864	.0991	.102	
Z(AZP)1	.549	.416	.201	.185	.145	.165	.162	
W(AZP)1	1.70	1.99	2.56	3.72	2.39	2.76	3.01	
	+	+	+	+	+	+	+	+

TABLE VIII-7

CV-880M LONGITUDINAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(BODY AXIS SYSTEM)

+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	
H	SL	SL	23 K	23 K	35 K	35 K	35 K	
M	.203	.249	.600	.860	.700	.800	.860	
STICK FIXED								
D(G)/D(U) (DEG/KT)	-.0488	-.0376	-.00873	-.0203	-.00306	-.00918	-.0114	
NZA (G/RAD)	4.69	6.47	11.6	24.4	10.1	13.3	15.6	
DE/G (DEG/G)	18.8	23.5	7.36	4.60	9.03	7.09	7.11	
CAP (RAD/SEC/SEC/G)	.144	.264	.176	.187	.184	.184	.204	
PHUGOID(2) (SEC) (TUCK(2))	--	--	--	--	--	--	--	
1/C(1/10)	3.55	2.04	1.55	1.55	1.19	1.19	1.13	
+	+	+	+	+	+	+	+	+

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7
H	SL	SL	23 K	23 K	35 K	35 K	35 K
M	.203	.249	.600	.860	.700	.800	.860
YV	-.139	-.148	-.115	-.170	-.0842	-.0969	-.108
YB	-31.5	-41.3	-70.7	-150.	-57.4	-75.5	-90.4
LB'	-3.19	-3.96	-5.98	-10.6	-5.38	-6.64	-7.72
NB'	.499	.763	1.42	2.98	1.02	1.50	1.82
LP'	-1.39	-1.62	-1.14	-1.15	-.863	-.884	-.893
NP'	-.113	-.0857	-.0416	-.0105	-.0453	-.0240	-.0165
LR'	.980	.756	.434	.409	.364	.384	.401
NR'	-.215	-.232	-.188	-.327	-.130	-.156	-.169
Y*DA	-.0371	-.0161	-.00458	-.00774	-.00303	-.00364	-.00512
L'DA	3.84	2.81	2.85	6.00	2.30	2.93	4.00
N'DA	.401	.202	.230	.321	.192	.142	.195
Y*DR	.0250	.0298	.0245	.0259	.0187	.0196	.0187
L'DR	.335	.507	.806	1.36	.563	.824	.892
N'DR	-.327	-.480	-.926	-1.22	-.747	-.870	-.829

TABLE VIII-9
CV-880M AILERON TRANSFER FUNCTION FACTORS
Bare Airframe
(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	
H	SL	SL	23 K	23 K	35 K	35 K	35 K	
M	.203	.249	.500	.860	.700	.800	.860	
DENOMINATOR								
1/T(DET)1	.00912	.0123	.00785	.0184	.00553	.00790	.00837	
1/T(DET)2	1.50	1.69	1.12	1.17	.792	.871	.875	
Z(DET)1	.119	.136	.112	.132	.105	.0903	.0931	
W(DET)1	1.02	1.11	1.41	1.88	1.33	1.43	1.54	
NUMERATORS								
N(B /DA)								
A(B)	-.0371	-.0161	-.00458	-.00774	-.00303	-.00364	-.00512	
1/T(B)1	.316	.315	-8.17	-1.26	.294	.253	-17.1	
Z(B)1	(-2.74)	(-3.87)	.981	(1.68)	(.423)	(.868)	.987	
W(B)1	(5.42)	(4.78)	.952	(4.71)	(-46.9)	(-26.6)	.578	
N(P /DA)								
A(P)	3.84	2.81	2.85	6.00	2.30	2.93	4.00	
1/T(P)1	-.0122	-.00835	-.00475	-.00176	-.00682	-.00333	-.00269	
Z(P)1	.266	.223	.127	.141	.107	.105	.104	
W(P)1	.938	1.05	1.39	1.90	1.21	1.35	1.49	
N(R /DA)								
A(R)	.401	.202	.230	.321	.192	.142	.195	
1/T(R)1	.951	1.05	.576	.786	.325	.504	.549	
Z(R)1	-.251	-.211	.0451	.0865	.0193	-.0167	.0190	
W(R)1	1.09	1.26	1.46	1.76	1.59	1.75	1.77	
N(PHI/DA)								
A(PHI)	3.87	2.82	2.87	6.01	2.33	2.94	4.01	
Z(PHI)1	.261	.219	.126	.141	.104	.104	.103	
W(PHI)1	.934	1.04	1.39	1.90	1.22	1.35	1.49	
N(AYP/DA)								
A(AYP)	26.8	16.9	20.2	33.8	16.9	16.3	21.9	
1/T(AYP)1	.360	.352	-.210	-.273	.180	.196	.223	
1/T(AYP)2	-.596	-.561	.292	.363	-.214	-.269	-.244	
Z(AYP)1	.159	.152	.118	.133	.108	.105	.104	
W(AYP)1	.987	1.09	1.40	1.90	1.26	1.38	1.51	
	+	+	+	+	+	+	+	+

TABLE VIII-10

CV-880M RUDDER TRANSFER FUNCTION FACTORS

Bare Airframe

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	
H	SL	SL	23 K	23 K	35 K	35 K	35 K	
M	.203	.249	.600	.860	.700	.800	.860	
DENOMINATOR								
1/T(DET)1	.00912	.0123	.00789	.0184	.00553	.00790	.00837	
1/T(DET)2	1.50	1.69	1.12	1.17	.792	.871	.875	
Z(DET)1	.119	.136	.112	.132	.105	.0903	.0931	
W(DET)1	1.02	1.11	1.41	1.88	1.33	1.43	1.54	
NUMERATORS								
N(B /DR)								
A(B)	.0250	.0298	.0245	.0255	.0187	.0196	.0187	
1/T(B)1	-.0753	-.0398	-.0164	-.00295	-.0205	-.0136	-.0115	
1/T(B)2	1.53	1.71	1.12	1.17	.819	.871	.880	
1/T(B)3	14.4	17.5	41.0	49.5	44.1	47.9	47.9	
N(P /DR)								
A(P)	.335	.507	.806	1.36	.563	.824	.892	
1/T(P)1	-.0123	-.00846	-.00481	-.00177	-.00685	-.00334	-.00270	
1/T(P)2	1.29	1.50	2.16	2.53	2.26	2.21	2.20	
1/T(P)3	-2.12	-2.06	-2.54	-2.60	-2.70	-2.51	-2.45	
N(R /DR)								
A(R)	-.327	-.480	-.926	-1.22	-.747	-.870	-.829	
1/T(R)1	1.53	1.71	.974	1.05	.444	.721	.751	
Z(R)1	.0813	.114	.275	.221	.367	.228	.209	
W(R)1	.498	.462	.504	.495	.698	.547	.543	
N(PHI/DR)								
A(PHI)	.305	.471	.720	1.30	.454	.753	.834	
1/T(PHI)1	1.28	1.49	2.22	2.56	2.42	2.26	2.24	
1/T(PHI)2	-2.36	-2.24	-2.79	-2.65	-3.20	-2.70	-2.59	
N(AYP/DR)								
A(AYP)	-8.68	-12.7	-27.1	-31.4	-21.6	-24.1	-21.4	
1/T(AYP)1	-.0930	-.0595	-.0270	-.0161	-.0303	-.0238	-.0226	
1/T(AYP)2	1.63	1.78	.859	.956	.516	.634	.654	
Z(AYP)1	.227	.204	.184	.103	.184	.146	.134	
W(AYP)1	1.03	1.12	1.57	2.24	1.63	1.70	1.85	
	+	+	+	+	+	+	+	+

TABLE VIII-11

CV-880M LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	
H	SL	SL	23 K	23 K	35 K	35 K	35 K	
N	.203	.249	.600	.860	.700	.800	.860	
DR PERIOD (SEC)	6.20	5.69	4.49	3.37	4.75	4.41	4.10	
1/C(1/2)	1.08	1.24	1.03	1.21	.956	.822	.848	
SPIRAL (2) (SEC)	--	--	--	--	--	--	--	
P(1)	2.52	1.56	2.36	4.85	2.37	2.95	4.11	
P(2)	1.57	1.19	--	4.85	2.12	2.72	3.92	
P(3)	2.12	1.38	--	4.85	2.21	2.79	3.95	
P(2)/P(1)	.624	.764	--	1.00	.893	.924	.953	
P(OSC)/P(AV)	.192	.105	--	.107E-4	.0395	.0263	.0143	
W(PHI)/W(D)	.915	.937	.984	1.01	.914	.947	.967	
DEL-B-MAX	.669	.272	.0578	.0237	.140	.105	.0839	
PHI TO BETA, PHASE	-302.	-304.	34.1	23.1	-333.	-333.	24.9	
PHI TO BETA	1.96	1.94	2.45	2.68	2.64	2.85	2.90	
PHI TO VE	.497	.400	.329	.251	.398	.376	.357	
	+	+	+	+	+	+	+	+

CV-880M DATA SOURCES

McNeill, Walter E., Calculated and Flight Measured Handling-Qualities
Factors of Three Subsonic Jet Transports, NASA TN D-4832, Nov. 1968.

Brooks, Peter W., The World's Airlines, London, Putnam, 1962.

SECTION IX

BOEING 747

BOEING 747 BACKGROUND

The Boeing 747 is a very large four-fanjet intercontinental transport designed to operate from existing international airports. To obtain the necessary low speed characteristics the wing has triple-slotted trailing flaps and Krueger type leading edge flaps. The Krueger flaps outboard of the inboard nacelle are variable cambered and slotted while the inboard Krueger flaps are standard unslotted. Longitudinal control is obtained through four elevator segments and a movable stabilizer. The lateral control employs five spoiler panels, an inboard aileron between the inboard and outboard flaps, and an outboard aileron which operates with flaps down only on each wing. The five spoiler panels on each wing also operate symmetrically as speedbrakes in conjunction with the most inboard sixth spoiler panel. Directional control is obtained from two rudder segments.

Information for this aircraft was obtained solely from a 747 simulator description (Boeing D6-30643).

Nominal Configuration

Load to Max Zero Fuel Weight

TOGW less 40% Fuel

W = 636,600 lb

c.g. at $0.25 \bar{c}$ $I_x = 18.2 \times 10^6 \text{ slug-ft}^2$ $I_y = 33.1 \times 10^6 \text{ slug-ft}^2$ $I_z = 49.7 \times 10^6 \text{ slug-ft}^2$ $I_{xz} = 0.97 \times 10^6 \text{ slug-ft}^2$

Body Axis

Power Approach Configuration

Max Landing Weight

20° Flaps

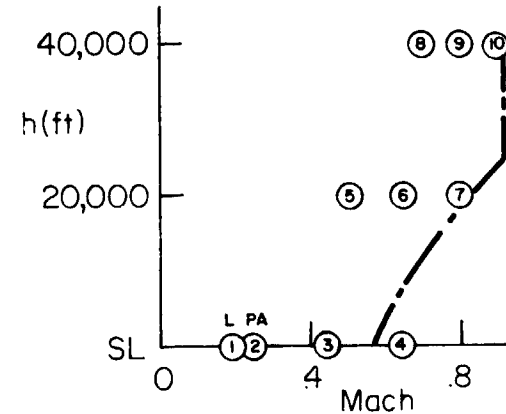
Gear Up

 $1.4 V_S$

W = 564,000 lb

c.g. at $0.25 \bar{c}$ $I_x = 13.7 \times 10^6 \text{ slug-ft}^2$ $I_y = 30.5 \times 10^6 \text{ slug-ft}^2$ $I_z = 43.1 \times 10^6 \text{ slug-ft}^2$ $I_{xz} = 0.825 \times 10^6 \text{ slug-ft}^2$

Body Axis

**Landing Configuration**

Same as Power Approach except:

30° Flaps

Gear Down

 $1.2 V_S$

Figure IX-1. B-747 Flight Conditions

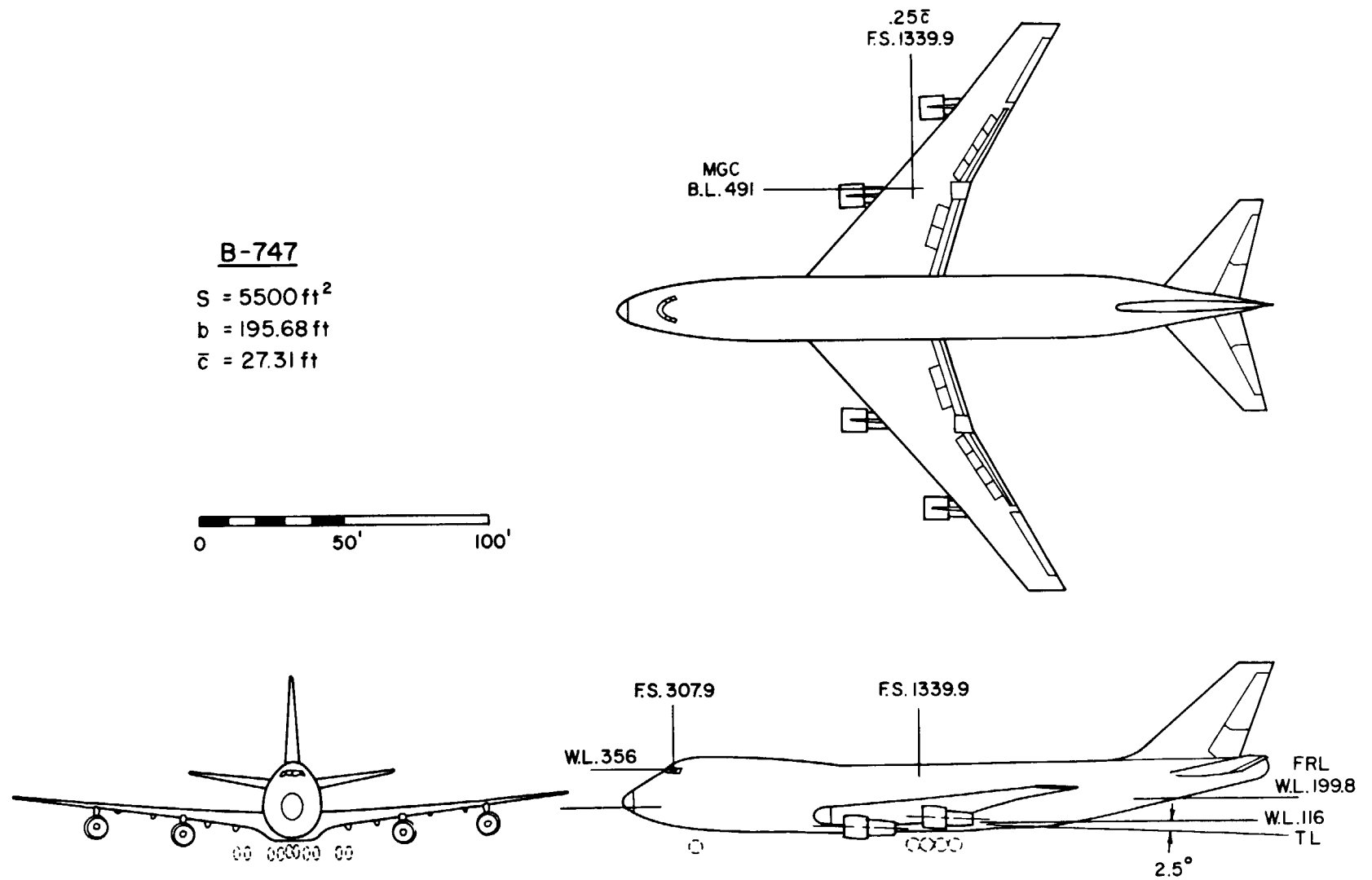
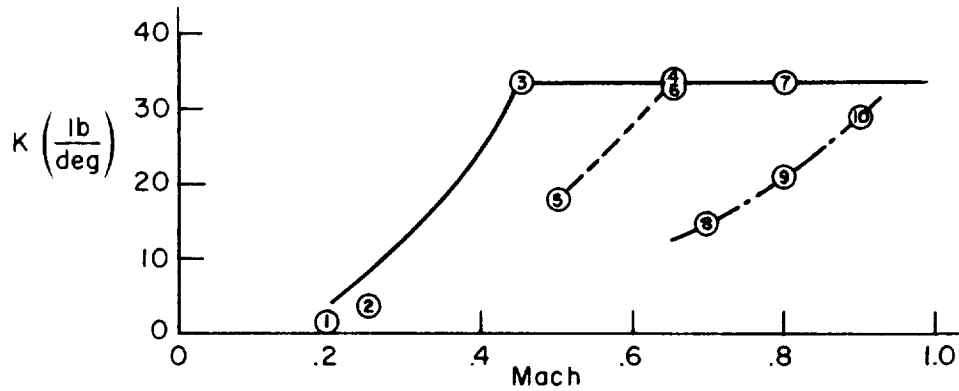
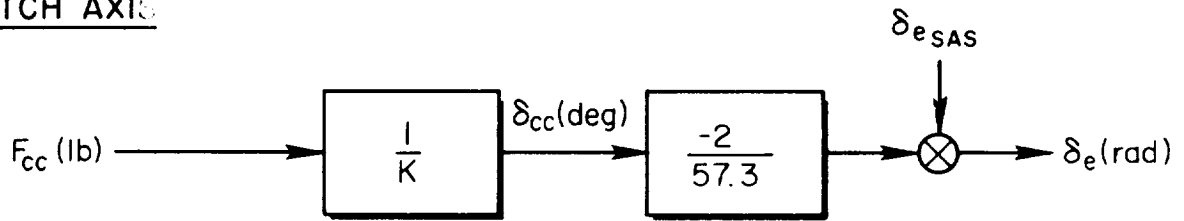


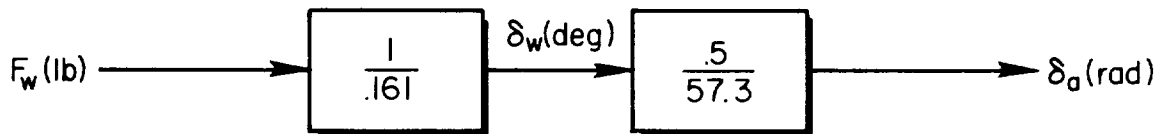
Figure IX-2. B-747 General Arrangement

B-747

PITCH AXIS



ROLL AXIS



YAW AXIS

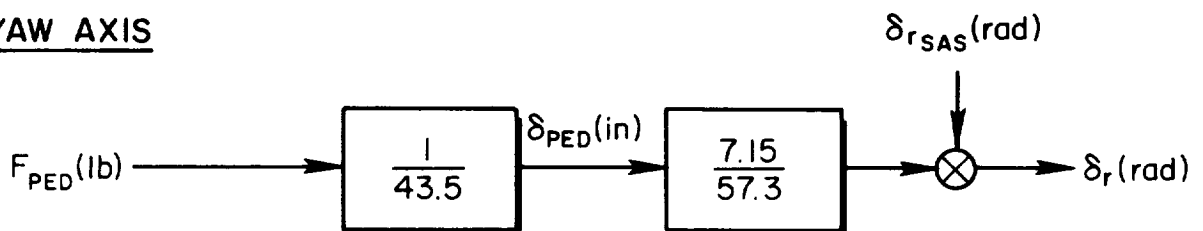
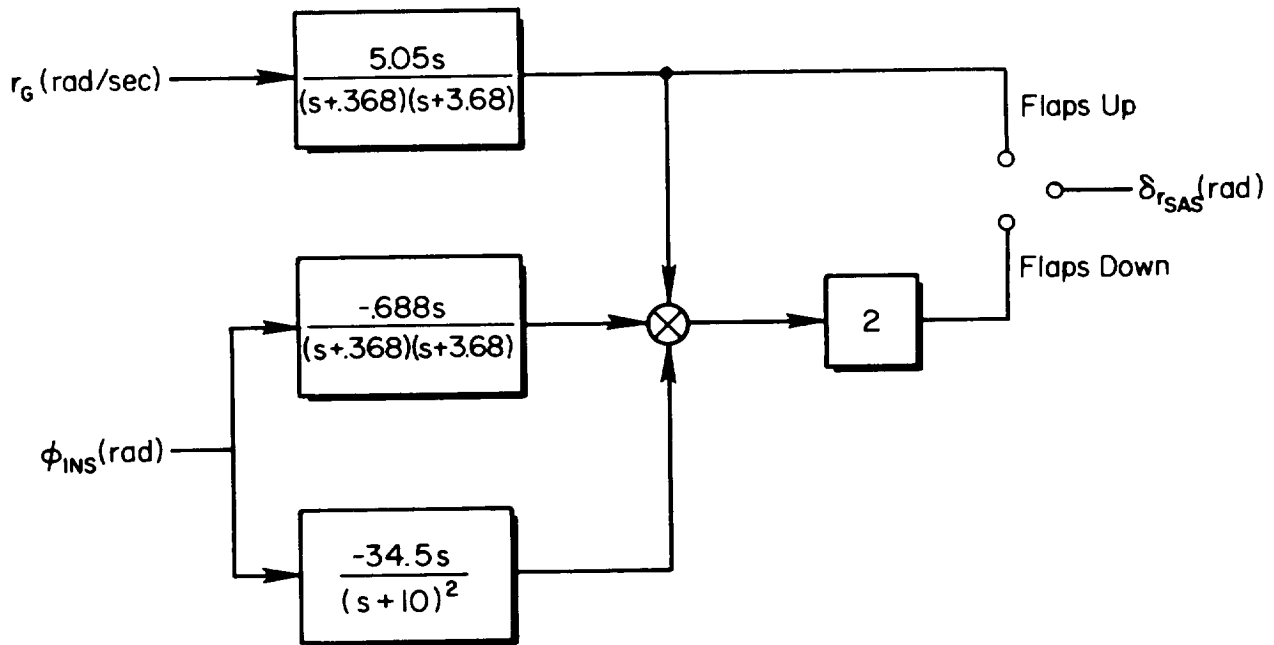


Figure IV-5. B-747 Control System

B-747

YAW SAS



$$r = r$$

$$\phi_{INS} = \int p \, dt$$

(Gyro and INS Aligned with FRL)

Figure IX-4. B-747 SAS

TABLE IX-1

B-747

Landing Configuration Non-Dimensional Derivatives

h = sea level

 $V_{T_0} = 131$ KTAS $\alpha_0 = 8.5^\circ$ $\delta_s = -6.3^\circ$

Longitudinal	Lateral-Directional
$C_L = 1.76$	$C_{Y\beta} = -1.08/\text{rad}$
$C_D = .263$	$C_{l\beta} = -.281/\text{rad}$
$C_{L\alpha} = 5.67/\text{rad}$	$C_{n\beta} = .184/\text{rad}$
$C_{D\alpha} = 1.13/\text{rad}$	$C_{l_p} = -.502/\text{rad}$
$C_{m\alpha} = -1.45/\text{rad}$	$C_{n_p} = -.222/\text{rad}$
$C_{L\dot{\alpha}} = -6.7/\text{rad}$	$C_{l_r} = .195/\text{rad}$
$C_{m\dot{\alpha}} = -3.3/\text{rad}$	$C_{n_r} = -.36/\text{rad}$
$C_{Lq} = 5.65/\text{rad}$	$C_{l\delta_a} = .0530/\text{rad}$
$C_{mq} = -21.4/\text{rad}$	$C_{n\delta_a} = .0083/\text{rad}$
$C_{LM} = -1.1$	$C_{Y\delta_r} = .179/\text{rad}$
$C_{mM} = .36$	$C_{l\delta_r} = 0$
$C_{L\delta_e} = .356/\text{rad}$	$C_{n\delta_r} = -.112/\text{rad}$
$C_{m\delta_e} = -1.40/\text{rad}$	

δ_a = total deflection of right inboard aileron plus left inboard aileron with the effect of outboard ailerons included

TABLE IX-2

B-747

**Power Approach Configuration
Non-Dimensional Derivatives**

h = sea level

$V_{T_O} = 165 \text{ KTAS}$

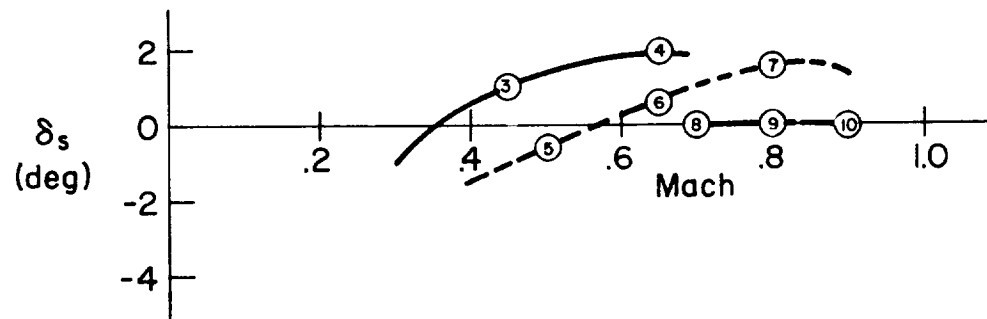
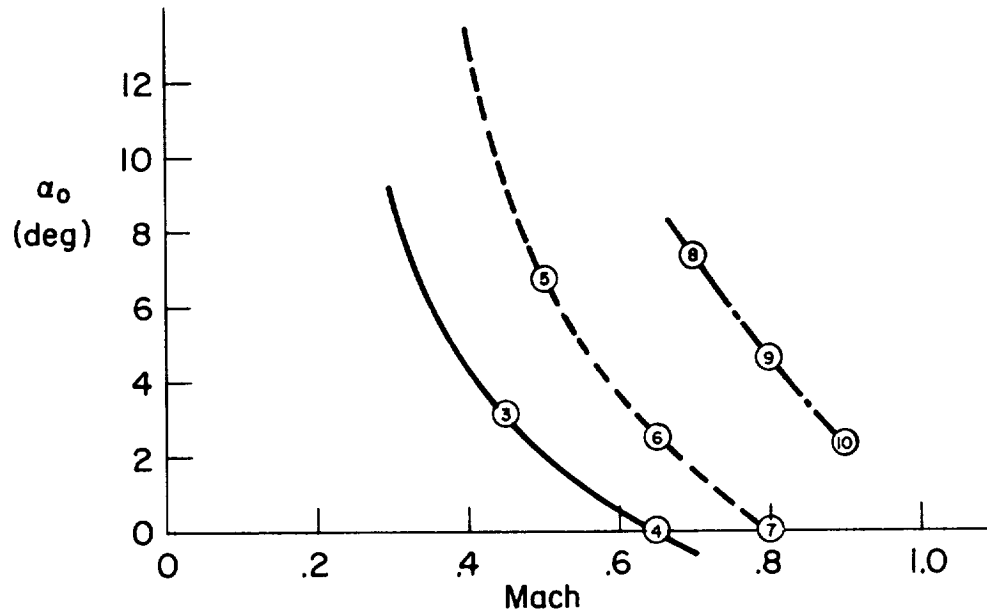
$\alpha_O = 5.7^\circ$

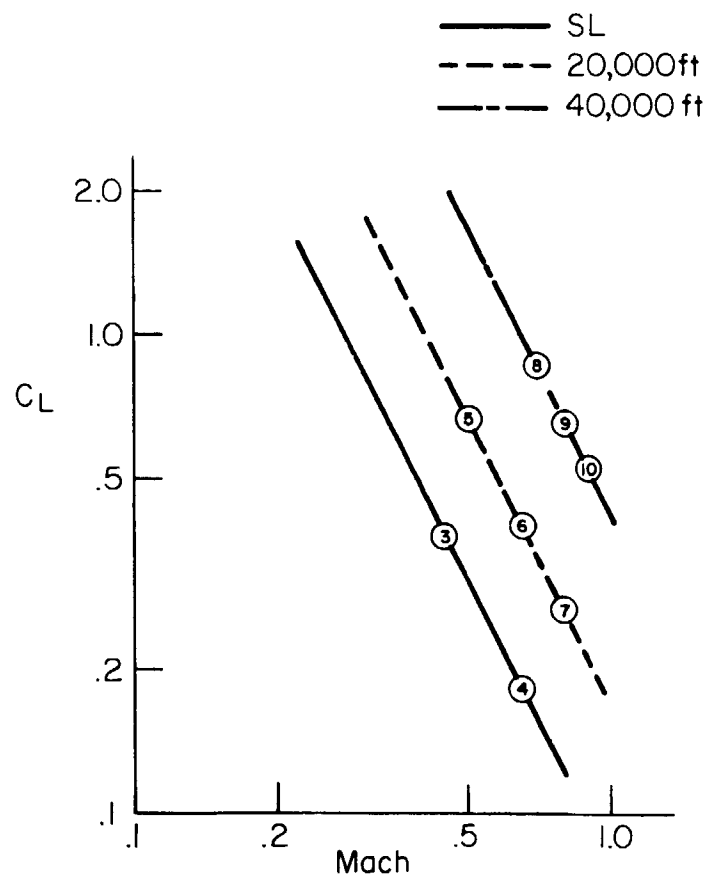
$\delta_S = -2.1^\circ$

Longitudinal	Lateral-Directional
$C_L = 1.11$	$C_{Y\beta} = -.96/\text{rad}$
$C_D = .102$	$C_{l\beta} = -.221/\text{rad}$
$C_{L\alpha} = 5.70/\text{rad}$	$C_{n\beta} = .150/\text{rad}$
$C_{D\alpha} = .66/\text{rad}$	$C_{l_p} = -.45/\text{rad}$
$C_{m\alpha} = -1.26/\text{rad}$	$C_{n_p} = -.121/\text{rad}$
$C_{L\dot{\alpha}} = -6.7/\text{rad}$	$C_{l_r} = .101/\text{rad}$
$C_{m\dot{\alpha}} = -3.2/\text{rad}$	$C_{n_r} = -.30/\text{rad}$
$C_{Lq} = 5.4/\text{rad}$	$C_{l\delta_a} = .0461/\text{rad}$
$C_{mq} = -20.8/\text{rad}$	$C_{n\delta_a} = .0064/\text{rad}$
$C_{L_M} = -.81$	$C_{Y\delta_r} = .175/\text{rad}$
$C_{m_M} = .27$	$C_{l\delta_r} = .007/\text{rad}$
$C_{L\delta_e} = .338/\text{rad}$	$C_{n\delta_r} = -.109/\text{rad}$
$C_{m\delta_e} = -1.34/\text{rad}$	

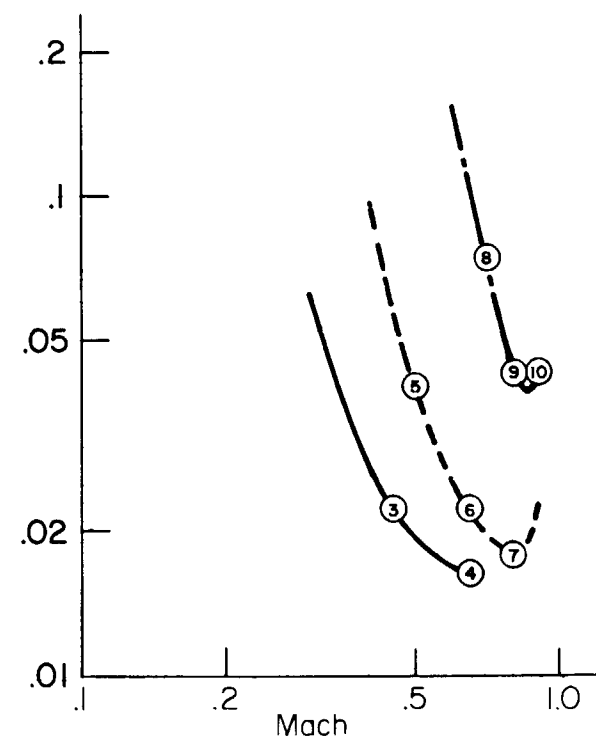
δ_a = total deflection of right inboard aileron plus left inboard aileron with the effect of outboard ailerons included

—	SL	B-747
- - -	20,000 ft	636600 lb
- · - · -	40,000 ft	.25 $\bar{c}$
		Flexible

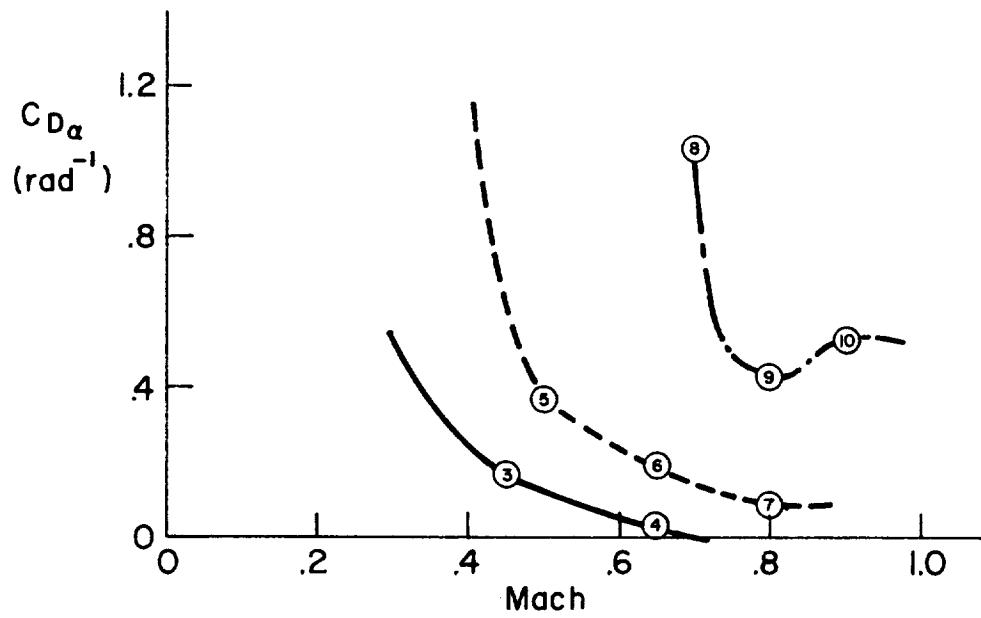
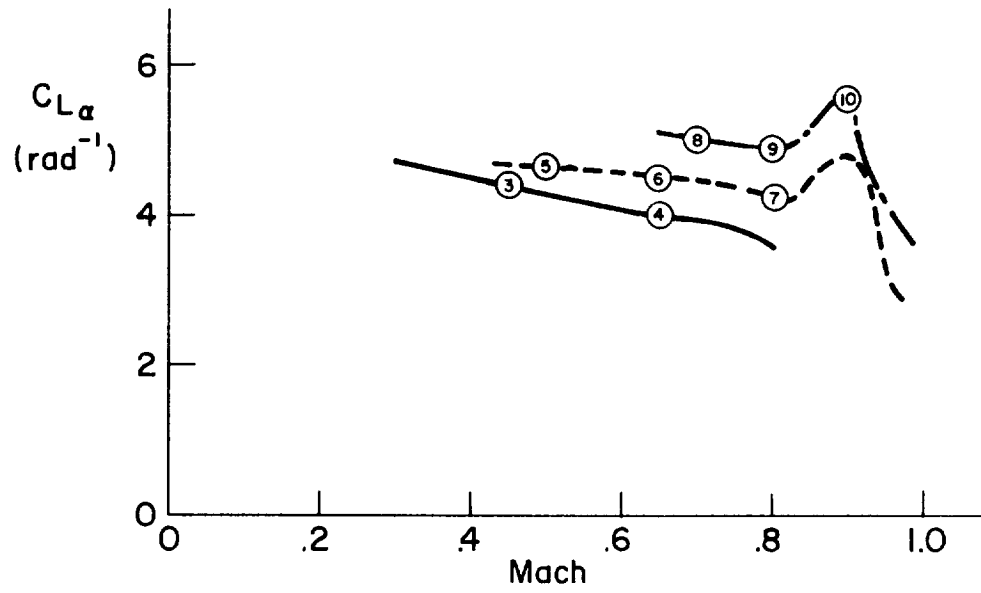


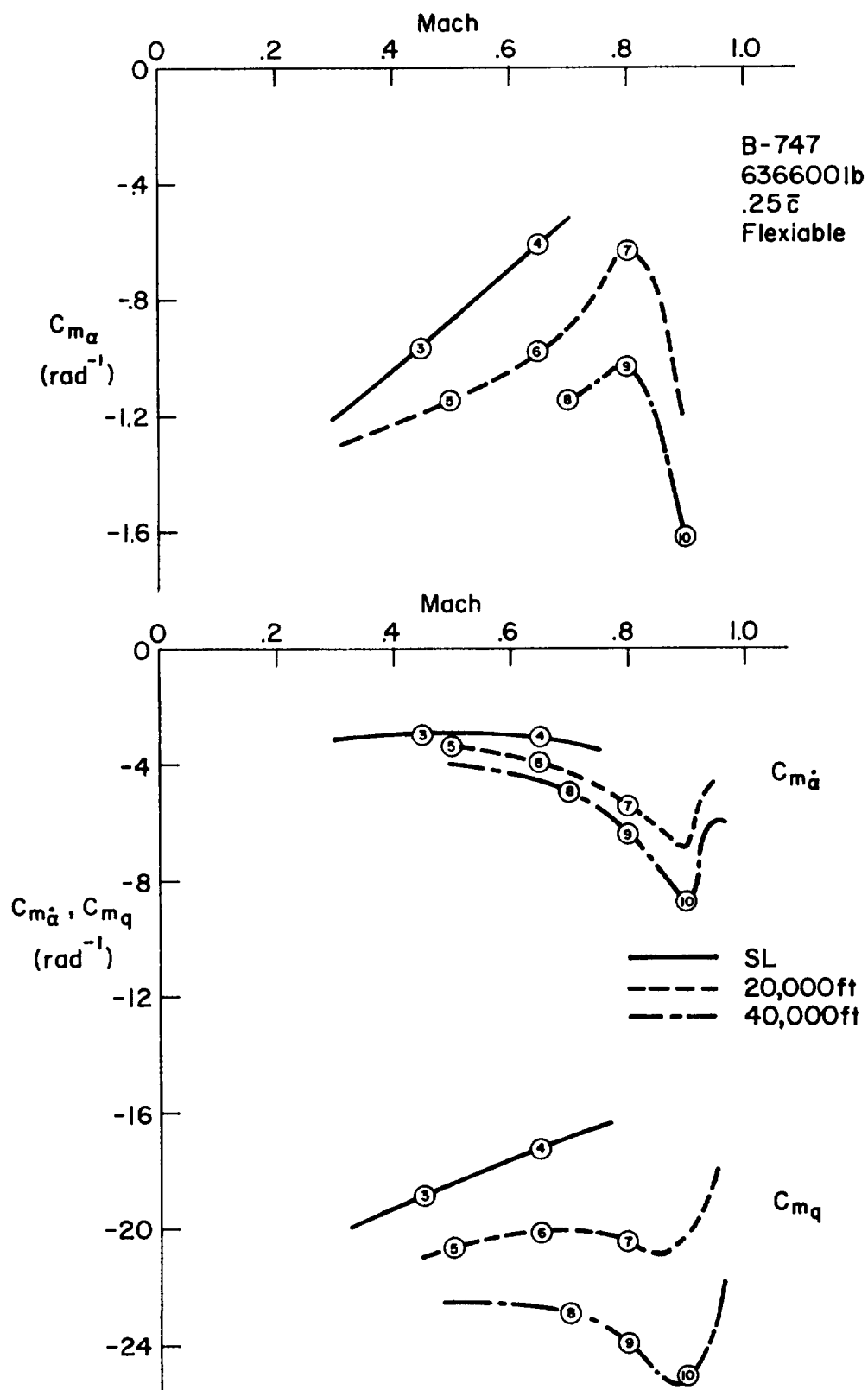


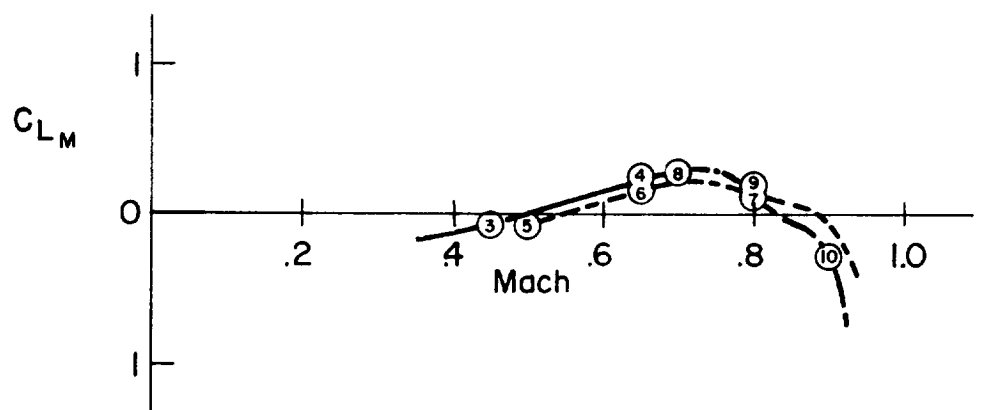
B-747
636600 lb



————	SL	B-747
-----	20,000 ft	636600 lb
- - - - -	40,000 ft	Flexible

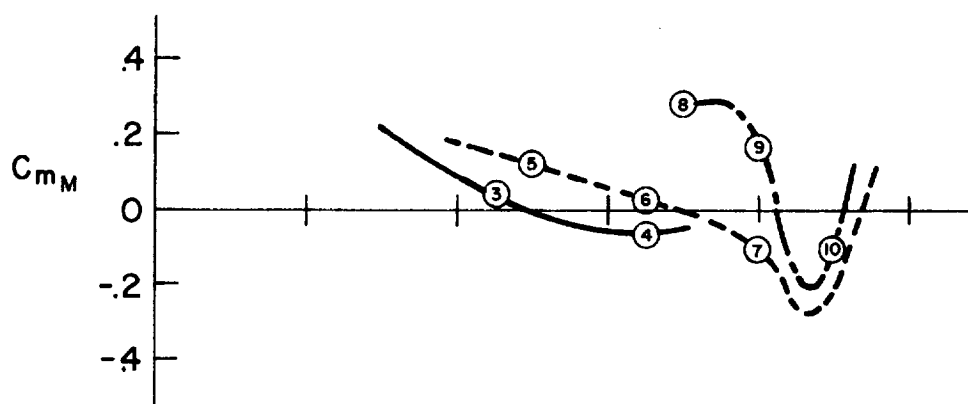
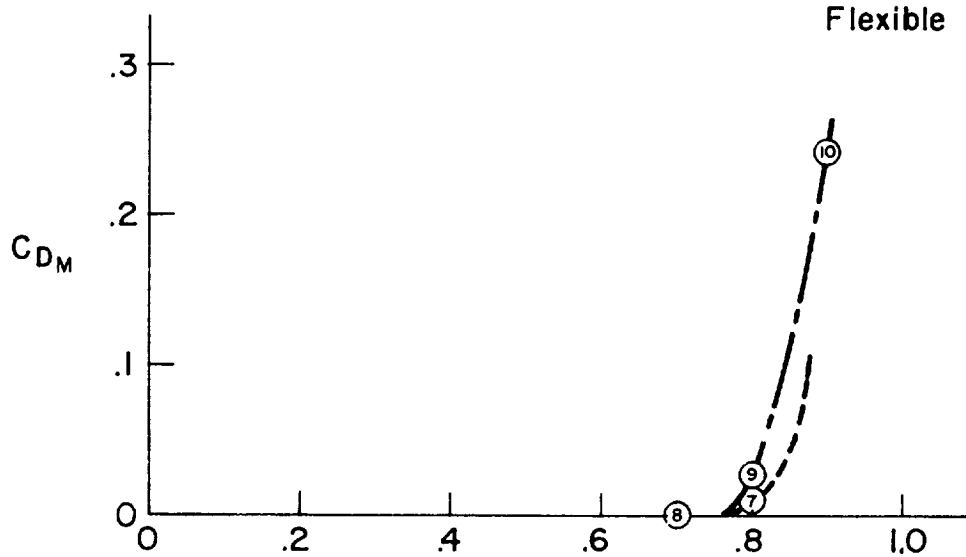






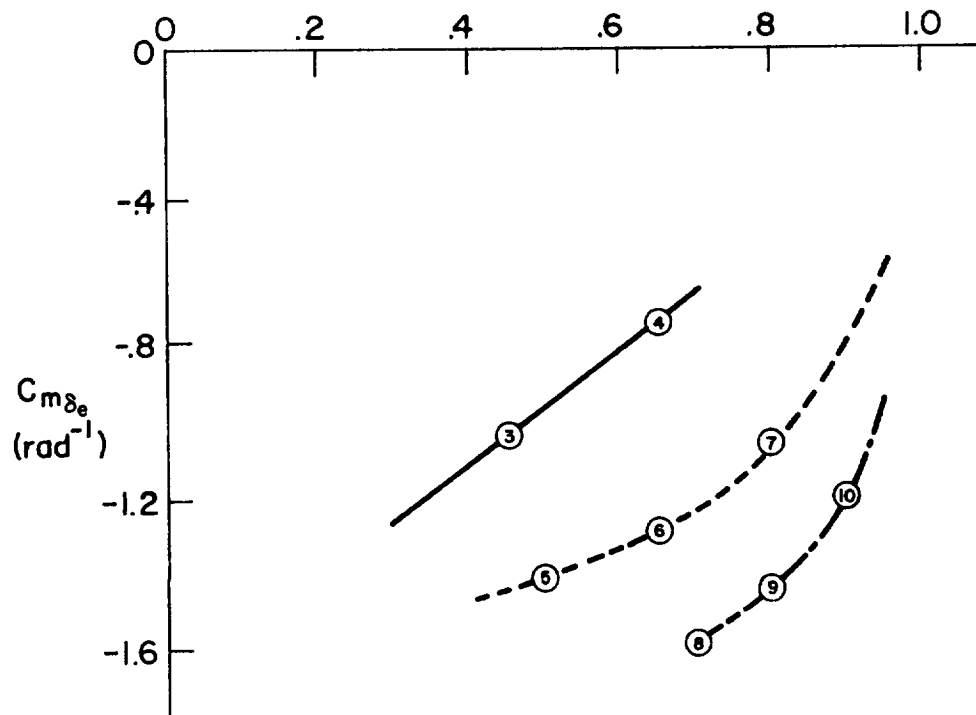
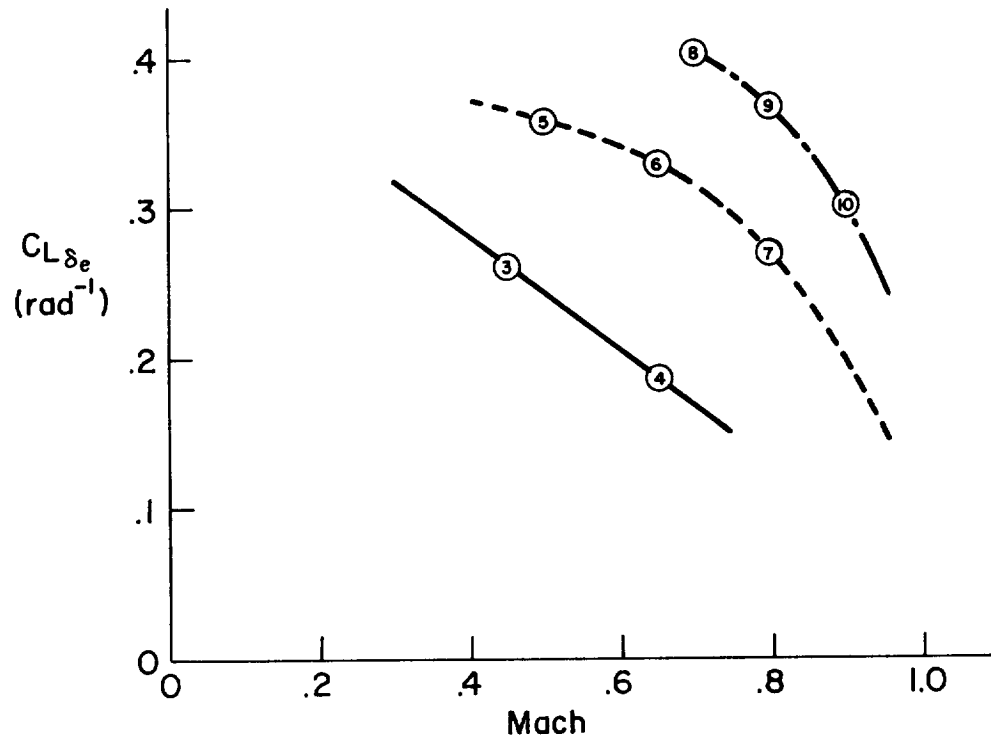
— SL
 - - - 20,000 ft
 - · - 40,000 ft

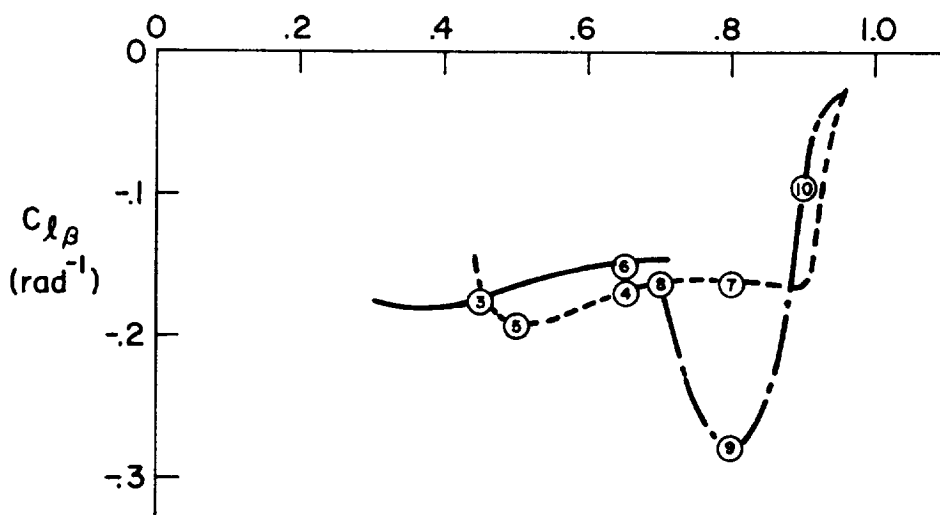
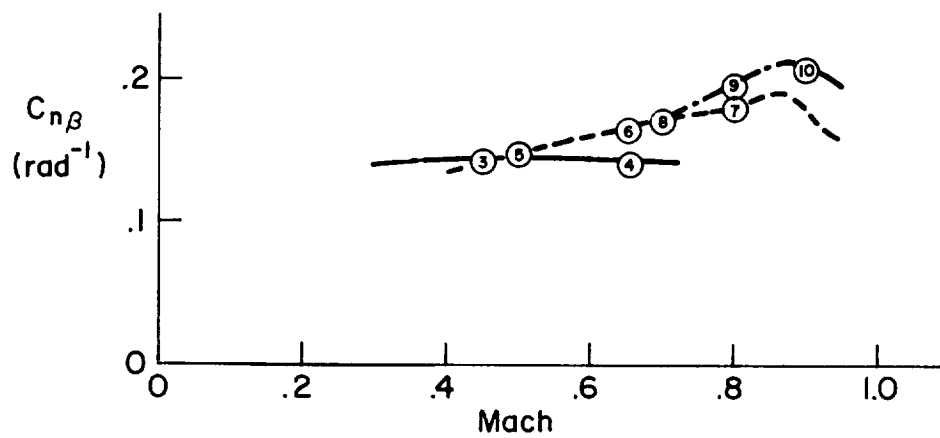
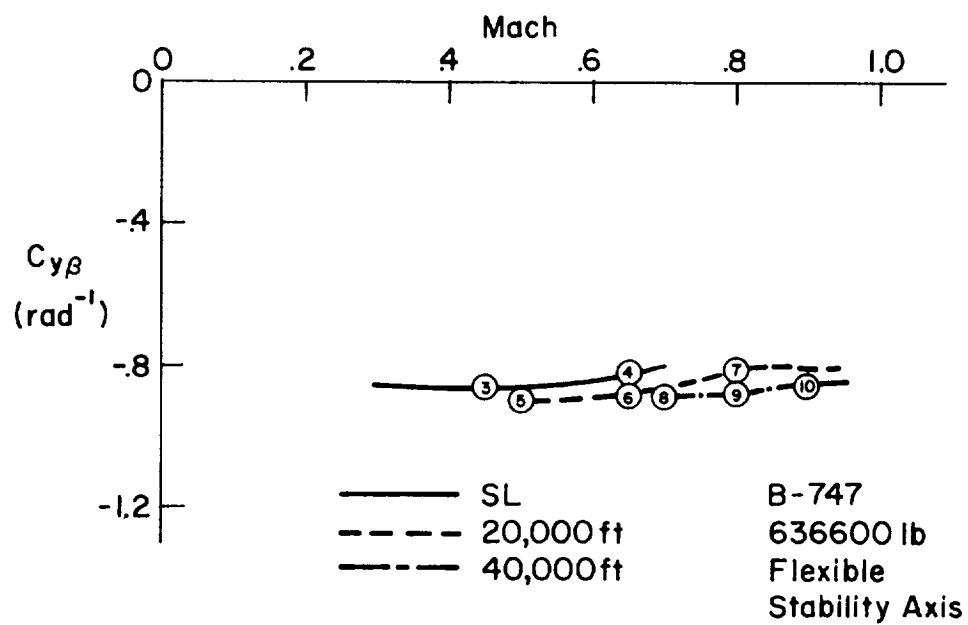
B-747
 636600 lb
 .25 $\bar{c}$
 Flexible

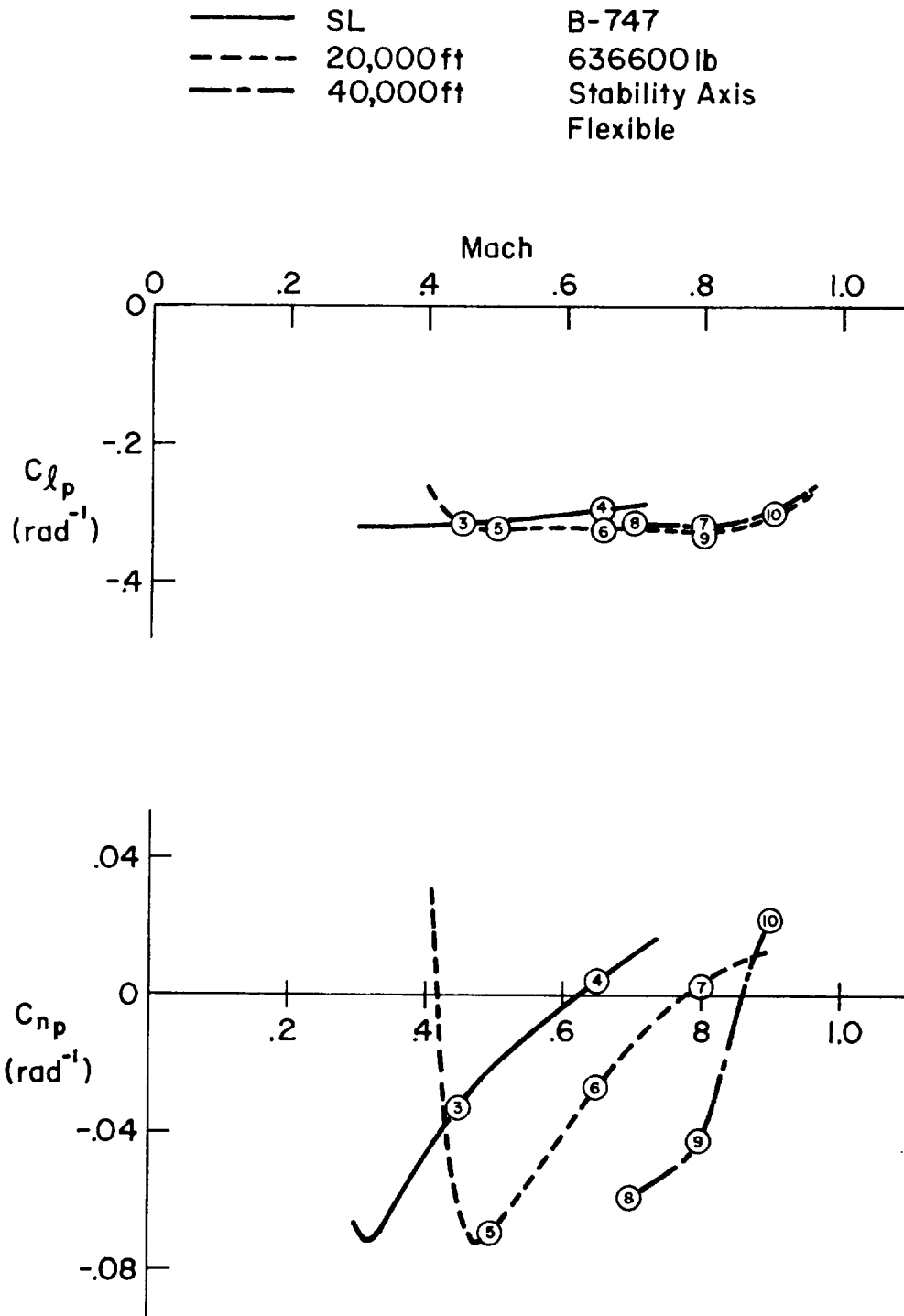


— SL
 - - - 20,000ft
 - · - 40,000ft

B-747

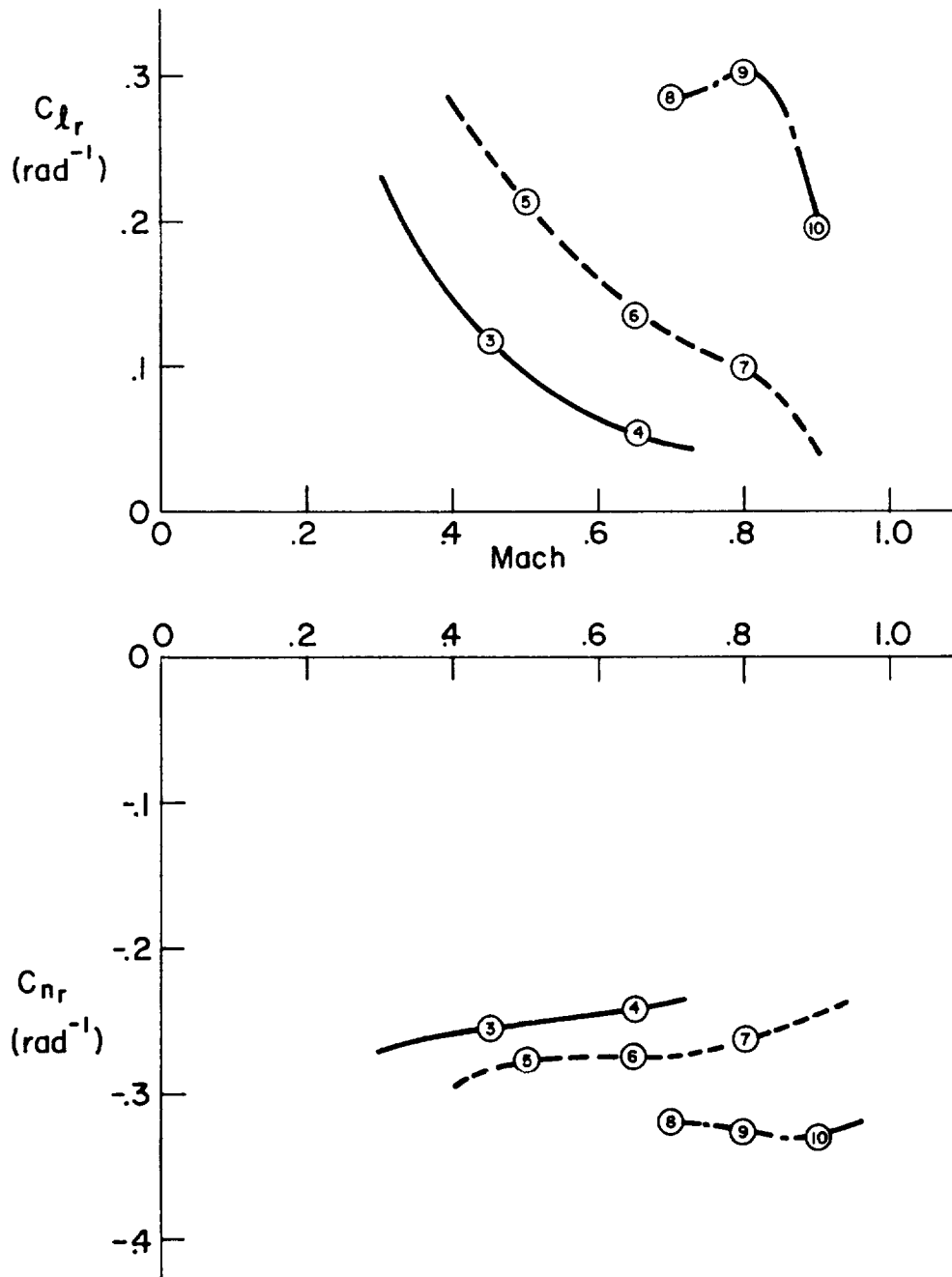




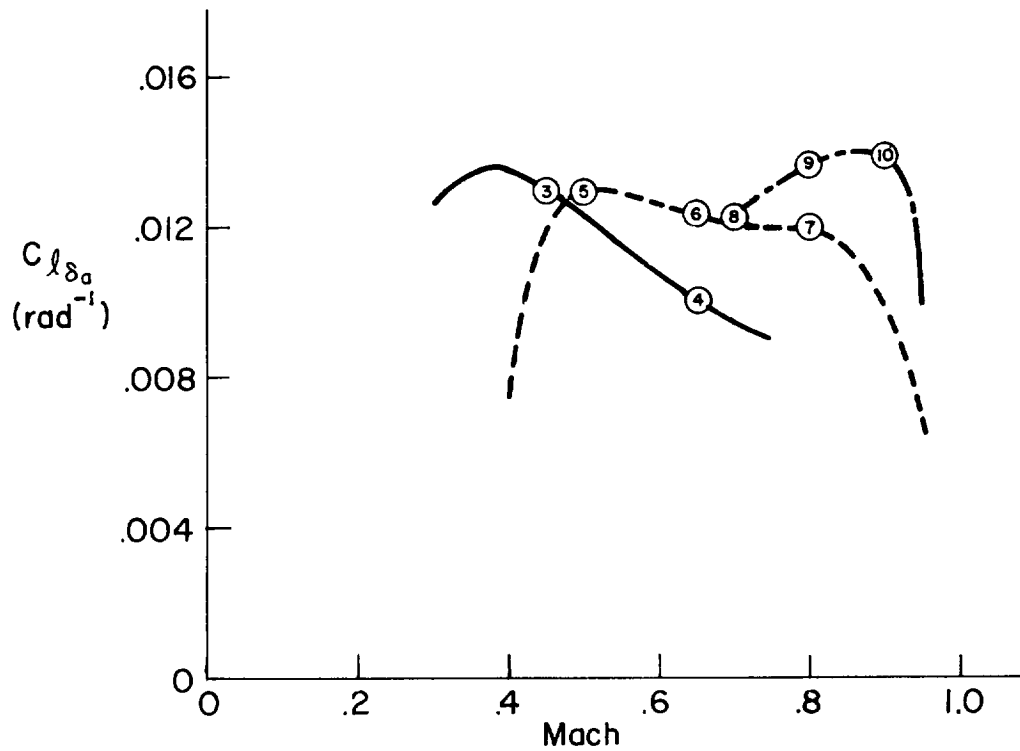


— SL
 - - - 20,000ft
 - - - 40,000ft

B-747
 636600lb
 Stability Axis
 Flexible

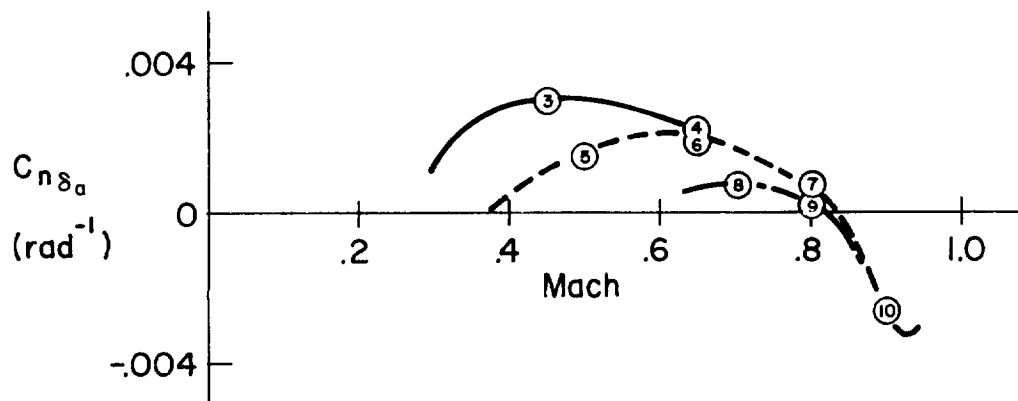


————	SL	B-747
- - - -	20,000 ft	636600 lb
- · - ·	40,000 ft	Flexible



Note:

- Because spoilers operate around a dead band their effect is neglected here
- δ_a is the total differential deflection of right and left inboard ailerons



— SL
 --- 20,000ft
 - - - 40,000ft

B-747
 636600lb
 Flexible

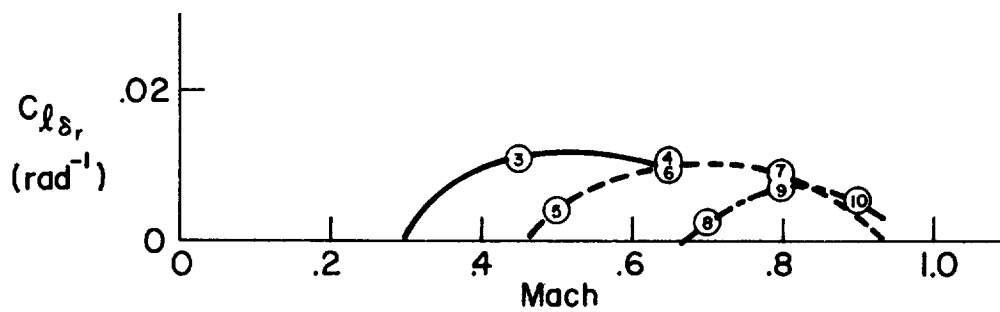
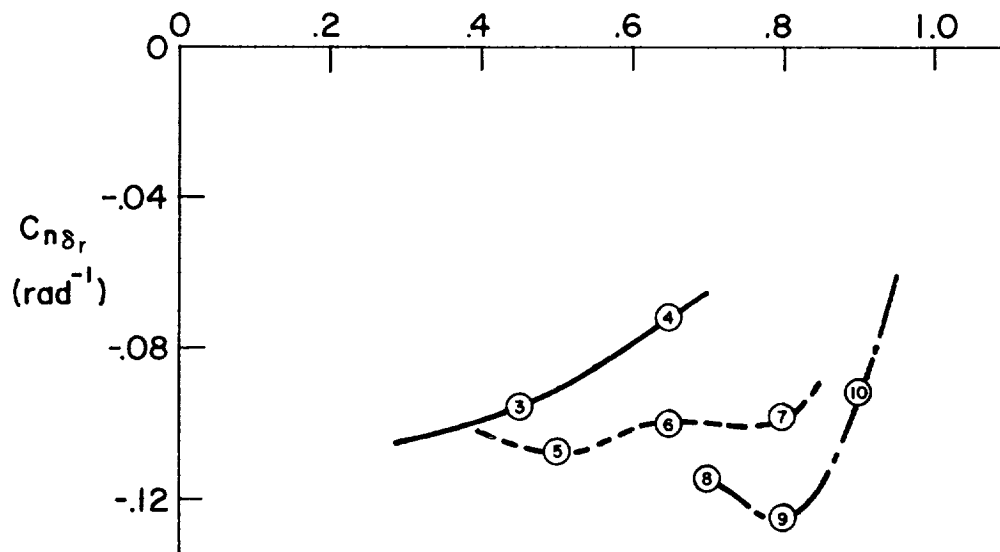
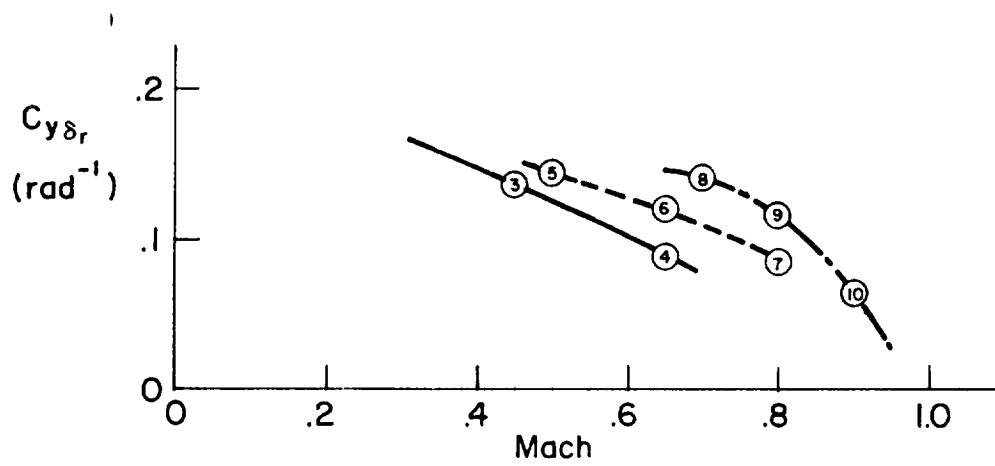


TABLE IX-3

B-747 DIMENSIONAL, MASS AND FLIGHT CONDITION PARAMETERS

 $s = 5500 \text{ sq ft}, \quad b = 195.68 \text{ ft}, \quad \bar{c} = 27.31 \text{ ft}$

F/C #	1	2	3	4	5	6	7	8	9	10
H(FT)	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M(-)	.138	.249	.450	.650	.500	.650	.800	.700	.800	.900
VTD(FPS)	221.	278.	502.	726.	518.	674.	830.	678.	774.	871.
VTD(KTAS)	131.	165.	298.	430.	307.	399.	492.	402.	459.	516.
VTD(KCAS)	131.	165.	298.	430.	228.	299.	373.	210.	243.	278.
W(LBS)	564032.	564032.	636636.	636636.	636636.	636636.	636636.	636636.	636636.	636636.
C.G.(MGC)	.250	.250	.250	.250	.250	.250	.250	.250	.250	.250
IX (SLUG-FT SQ)	.142E+8	.142E+8	.182E+8	.182E+8	.182E+8	.182E+8	.182E+8	.182E+8	.182E+8	.182E+8
IY (SLUG-FT SQ)	.323E+8	.323E+8	.331E+8	.331E+8	.331E+8	.331E+8	.331E+8	.331E+8	.331E+8	.331E+8
IZ (SLUG-FT SQ)	.454E+8	.454E+8	.497E+8	.497E+8	.497E+8	.497E+8	.497E+8	.497E+8	.497E+8	.497E+8
IXZ(SLUG-FT SQ)	870050.	870050.	970056.	970056.	970056.	970056.	970056.	970056.	970056.	970056.
EPSILON(DEG)	-1.60	-1.60	-1.76	-1.76	-1.76	-1.76	-1.76	-1.76	-1.76	-1.76
Q(PSF)	58.1	92.2	300.	626.	170.	288.	436.	135.	177.	224.
QC(PSF)	58.7	93.6	315.	695.	181.	320.	510.	153.	207.	273.
ALPHA(DEG)	8.50	5.70	3.10	0.	6.80	2.50	0.	7.30	4.60	2.40
GAMMA(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LXP(FT)	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0	86.0
LZP(FT)	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0	-10.0
ITH(DEG)	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50
XI(DEG)	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50
LTH(FT)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
	+	+	+	+	+	+	+	+	+	+

(BODY AXIS SYSTEM)

[illegible]

TABLE IX-5

B-747 ELEVATOR TRANSFER FUNCTION FACTORS

Bare Airframe
(BODY AXIS SYSTEM)

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F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.108	.249	.450	.650	.500	.650	.800	.700	.800	.900
DENOMINATOR										
Z(DET)1	.0417	.0228	.0319	.110	.0241	.0264	.323	.0636	.0489	.304
W(DET)1	.152	.127	.0753	.0363	.0823	.0653	.00984	.0781	.0673	.0311
Z(DET)2	.616	.629	.575	.637	.446	.473	.567	.357	.387	.351
W(DET)2	.771	.910	1.37	1.63	1.04	1.26	1.30	.879	.964	1.35
NUMERATORS										
N(U /DE)										
A(U)	.993	1.01	1.22	-1.15	2.05	1.17	-.873	1.95	1.45	.785
1/T(U)1	11.9	15.2	21.1	3.71	32.7	42.0	1.83	42.8	49.6	56.0
Z(U)1	.441	.390	.269	(-14.9)	.306	.335	(-25.5)	.705	.434	.783
W(U)1	.725	.857	.926		.469	.758		.323	.399	.578
N(W /DE)										
A(W)	-6.65	-10.1	-22.5	-33.3	-17.2	-26.8	-33.2	-15.2	-18.0	-18.7
1/T(W)1	12.9	16.4	32.3	46.6	33.2	43.0	52.7	43.2	50.1	56.9
Z(W)1	.0814	.0514	.0401	.0518	.0238	.0338	.0537	.00781	.0435	.260
W(W)1	.171	.133	.0728	.0728	.0666	.0635	.0593	.0593	.0531	.0387
N(TH /DE)										
A(TH)	-.377	-.572	-1.40	-2.07	-1.09	-1.68	-2.07	-.968	-1.16	-1.21
1/T(TH)1	.0801	.0396	.0136	.0124	.0158	.0107	.0105	.00419	.0113	.0217
1/T(TH)2	.440	.574	.711	.952	.400	.511	.606	.272	.295	.373
N(HD /DE)										
A(HD)	6.72	10.1	22.5	33.3	17.3	26.8	33.2	15.3	18.1	18.8
1/T(HD)1	-.0118	-.00415	.00240	.00646	-.00302	.000539	.00454	-.0151	-.00166	.0161
1/T(HD)2	-2.17	-2.75	-4.21	-5.79	-3.35	-4.29	-5.15	-3.19	-3.64	-4.36
1/T(HD)3	2.71	3.39	5.31	7.43	3.97	5.09	6.13	3.59	4.08	4.82
N(AZP/DE)										
A(AZP)	25.7	39.1	97.7	144.	76.3	118.	145.	68.1	81.7	85.7
1/T(AZP)1	.0339	.0189	-.00577	0.	.00927	-.00445	0.	.00154	.00532	-.00188
1/T(AZP)2	-.0468	-.0233	.00814	.00645	-.0124	.00497	.00454	-.0167	-.00705	.0179
Z(AZP)1	.213	.197	.140	.127	.109	.103	.0984	.0831	.0771	.0843
W(AZP)1	1.24	1.55	2.27	3.15	1.73	2.23	2.69	1.61	1.81	2.15

TABLE IX-6

B-747 THRUST TRANSFER FUNCTION FACTORS

Bare Airframe

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.198	.249	.450	.650	.500	.650	.800	.700	.800	.900
DENOMINATOR										
Z(DET)1	.0417	.0228	.0319	.110	.0241	.0264	.323	.0636	.0489	.304
W(DET)1	.152	.127	.0753	.0368	.0823	.0653	.00984	.0781	.0673	.0311
Z(DET)2	.616	.629	.575	.637	.446	.473	.567	.357	.387	.351
W(DET)2	.771	.910	1.37	1.63	1.04	1.26	1.30	.879	.964	1.35
NUMERATORS										
N(U /DTH)										
A(U)	.571E-4	.570E-4	.505E-4	.505E-4	.505E-4	.505E-4	.505E-4	.505E-4	.505E-4	.505E-4
1/T(U)1	-.173	-.141	-.0823	-.0700	-.0943	-.0715	-.0713	-.114	-.0803	-.0502
Z(U)1	.592	.605	.540	.646	.323	.433	.586	.159	.251	.301
W(U)1	.784	.928	1.37	1.67	1.00	1.25	1.32	.740	.908	1.29
N(W /DTH)										
A(W)	-.287E-5	-.277E-5	-.236E-5	-.227E-5	-.234E-5	-.227E-5	-.224E-5	-.227E-5	-.225E-5	-.223E-5
1/T(W)1	-19.4	-27.6	-62.6	-93.8	-65.1	-87.5	-109.	-88.1	-102.	-116.
Z(W)1	-.0347	.0126	.0785	-.355	.286	.0642	-.360	.360	.320	.0817
W(W)1	.189	.139	.0728	.0775	.0627	.0637	.0621	.0551	.0508	.0400
N(TH/DTH)										
A(TH)	.312E-6	.312E-6	.303E-6	.303E-6	.303E-6	.303E-6	.303E-6	.303E-6	.303E-6	.303E-6
1/T(TH)1	(.876)	.197	.0898	.0326	.115	.0808	.0125	.110	.0932	.0484
1/T(TH)2	(.340)	.504	.721	.955	.383	.500	.603	.233	.270	.379
N(HD /DTH)										
A(HD)	.113E-4	.842E-5	.508E-5	.227E-5	.830E-5	.448E-5	.224E-5	.866E-5	.629E-5	.434E-5
1/T(HD)1	.118	.102	.0686	.0243	.0739	.0601	.00572	.0652	.0626	.0372
Z(HD)1	.433	.330	.170	.139	.176	.158	.158	.161	.153	.127
W(HD)1	1.97	2.74	4.98	10.0	3.06	5.18	8.60	2.75	3.58	5.13
N(AZP/DTH)										
A(AZP)	-.297E-4	-.296E-4	-.284E-4	-.283E-4	-.284E-4	-.283E-4	-.283E-4	-.283E-4	-.283E-4	-.283E-4
1/T(AZP)1	-.0276	-.0137	-.00376	C.	-.00810	-.00231	0.	-.00689	-.00372	-.00169
1/T(AZP)2	.155	.122	.0751	.0243	.0884	.0641	.00572	.0761	.0692	.0401
Z(AZP)1	.362	.302	.203	.195	.164	.160	.161	.135	.128	.124
W(AZP)1	1.16	1.42	2.06	2.84	1.58	2.03	2.42	1.46	1.65	1.98

+ + + + + + + + + +

TABLE IX-7

B-747 LONGITUDINAL HANDLING QUALITIES PARAMETERS

Bare Airframe

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.198	.249	.450	.650	.500	.650	.800	.700	.800	.900
STICK FIXED										
D(G)/D(U) (DEG/KT)	.0349	.0123	-.00726	-.0154	.00900	-.00166	-.0137	.0452	.00494	-.0486
NZA (G/RAD)	3.27	5.00	10.8	20.8	6.50	10.7	15.4	5.76	7.22	10.1
DE/G (DEG/G)	25.9	15.7	6.80	3.43	8.45	4.95	2.98	7.75	6.25	8.45
CAP (RAD/SEC/SEC/G)	.170	.157	.166	.124	.160	.145	.108	.131	.127	.179
PHUGOID(2) (SEC) (TUC K(2))	--	--	--	--	--	--	--	--	--	--
1/C(1/10)	2.13	2.21	1.92	2.26	1.36	1.46	1.88	1.04	1.14	1.02
	+	+	+	+	+	+	+	+	+	+

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TABLE IX-8

B-747 LATERAL-DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.158	.249	.450	.650	.500	.650	.800	.700	.800	.900
YV	-.0890	-.0997	-.143	-.197	-.0822	-.104	-.120	-.0488	-.0558	-.0606
YB	-19.7	-27.8	-71.7	-143.	-42.6	-70.4	-99.4	-33.1	-43.2	-52.8
LB'	-1.33	-1.63	-3.19	-5.45	-2.05	-2.96	-4.12	-1.45	-3.05	-1.32
NB'	.168	.247	.810	1.82	.419	.923	1.62	.404	.598	.971
LP'	-.975	-1.10	-1.12	-1.47	-.652	-.804	-.974	-.404	-.465	-.459
NP'	-.166	-.125	-.0706	-.0214	-.0701	-.0531	-.0157	-.0366	-.0316	.00284
LR'	.327	.198	.379	.256	.376	.317	.292	.312	.388	.280
NR'	-.217	-.229	-.246	-.344	-.140	-.193	-.232	-.0963	-.115	-.141
Y*CA	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
L'CA	.227	.318	.229	.372	.128	.210	.310	.0964	.143	.186
N'CA	.0264	.0300	.0285	.0371	.0177	.0199	.0127	.00875	.00775	-.00611
Y*CR	.0148	.0182	.0226	.0213	.0131	.0142	.0124	.00777	.00729	.00464
L'CR	.0636	.110	.254	.318	.148	.211	.183	.115	.153	.100
N'CR	-.151	-.233	-.614	-.970	-.391	-.616	-.922	-.331	-.475	-.442

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.198	.249	.450	.650	.500	.650	.800	.700	.800	.900
DENOMINATOR										
1/T (DET) 1	.0427	.0465	.0194	.0203	.00903	.0108	.0103	-.00234	.00730	-.00777
1/T (DET) 2	1.11	1.23	1.23	1.56	.745	.913	1.06	.462	.562	.478
Z (DET) 1	.0878	.107	.126	.153	.0693	.0823	.0981	.0568	.0349	.0929
W (DET) 1	.735	.746	1.06	1.40	.863	1.07	1.31	.788	.947	1.02
NUMERATORS										
N(B /DA)										
A(B)	.00740	.00171	-.0161	-.0371	-.00243	-.0107	-.0127	.00358	.00373	.0139
1/T (B) 1	.154	.176	.448	-.168	.174	.230	.333	.0981	.0995	.103
1/T (B) 2	7.10	25.9	-.605	.981	-3.62	-.910	-.694	1.61	2.17	.628
N(P /DA)										
A(P)	.227	.318	.229	.372	.128	.210	.310	.0964	.143	.186
1/T (P) 1	-.0199	-.0108	-.00335	0.	-.00722	-.00205	0.	-.00601	-.00331	-.00154
Z (P) 1	.308	.274	.197	.181	.166	.149	.135	.122	.111	.100
W (P) 1	.591	.653	1.12	1.56	.846	1.11	1.35	.734	.877	.967
N(R /DA)										
A(R)	.0264	.0300	.0285	.0371	.0177	.0199	.0127	.00875	.00775	-.00611
1/T (R) 1	.499	.593	.849	1.68	.442	.718	1.46	.330	.435	-1.22
Z (R) 1	-.482	-.395	-.0874	-.142	-.128	-.201	-.347	-.153	-.217	.893
W (R) 1	.895	.507	.855	.791	.842	.920	1.08	.919	1.16	.925
N(PHI/DA)										
A(PHI)	.231	.321	.230	.372	.130	.211	.310	.0975	.143	.186
Z (PHI) 1	.284	.264	.196	.181	.162	.148	.135	.117	.109	.0991
W (PHI) 1	.586	.650	1.12	1.56	.844	1.11	1.35	.735	.878	.968
N(AYP/DA)										
A(AYP)	4.54	5.76	4.74	6.91	2.80	3.82	4.19	1.72	2.10	1.34
1/T (AYP) 1	.257	.279	-.143	-.0950	-.151	-.137	-.126	-.134	-.135	-.166
1/T (AYP) 2	-.331	-.313	.406	.656	.216	.264	.334	.141	.149	.173
Z (AYP) 1	.0459	.121	.145	.146	.109	.0983	.0955	.0731	.0529	.0405
W (AYP) 1	.643	.705	1.07	1.42	.847	1.07	1.29	.762	.907	1.11

TABLE IX-11

B-747 AILERON TRANSFER FUNCTION FACTORS

SAS On

(Body Axis System)

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F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.198	.249	.450	.650	.500	.650	.800	.700	.800	.900
DENOMINATOR										
1/T(DET)1	.0770	.198	.0148	.0160	.00688	.00843	.00801	-.00185	.00514	-.00687
1/T(DET)2	.471	.858	.804	1.63	.495	.696	1.28	.386	.467	2.83
1/T(DET)3	1.20	1.53	(.524)	(.799)	1.09	1.67	(.685)	.808	1.10	(.973)
1/T(DET)4	3.10	2.50	(.900)	(.791)	2.91	1.90	(.748)	3.07	2.68	(.540)
1/T(DET)5	9.26	9.03	(.978)	(.692)	(.267)	(.477)	(.729)	(.234)	(.255)	(.472)
1/T(DET)6	10.7	10.9	(1.94)	(2.28)	(.794)	(.908)	(2.10)	(.716)	(.838)	(.961)
Z(DET)1	.472	.849								
w(DET)1	.576	.287								
NUMERATORS										
N(B /DA)										
A(B)	.00740	.00171	-.0161	-.0371	-.00243	-.0107	-.0127	.00358	.00373	.0130
1/T(B)1	11.1	13.0	.0905	.101	.0706	.0633	.0535	.0470	.0413	.0413
1/T(B)2	13.7	39.4	.841	-.381	.465	.708	-1.17	.371	.464	3.27
Z(B)1	.790	.150	(-1.21)	(1.78)	(4.90)	(-1.57)	(1.58)	.783	.754	.906
w(B)1	.289	.444	(3.97)	(3.25)	(-5.32)	(4.03)	(3.17)	3.47	3.90	.806
Z(B)2	.901	.849								
w(B)2	3.69	2.88								
N(P /DA)										
A(P)	.227	.318	.229	.372	.128	.210	.310	.0864	.143	.186
1/T(P)1	-.0198	-.0107	-.00335	0.	-.00721	-.00204	0.	-.00600	-.00331	-.00154
1/T(P)2	.863	1.48	.613	.466	.619	.611	.621	.612	.706	.525
1/T(P)3	3.04	2.43	1.80	1.28	2.86	2.07	.907	3.04	2.67	2.84
1/T(P)4	9.99	9.99	(.837)	(.610)	(.577)	(.740)	(.690)	(.456)	(.581)	(.475)
1/T(P)5	10.0	10.0	(1.24)	(2.35)	(.741)	(1.15)	(2.00)	(.626)	(.743)	(.921)
Z(P)1	.594	.616								
w(P)1	.426	.402								
N(R /DA)										
A(R)	.0264	.0300	.0285	.0371	.0177	.0199	.0127	.00875	.00775	-.00611
1/T(R)1	3.96	12.6	.368	.368	.368	.368	.368	.330	.368	.368
1/T(R)2	6.22	(.573)	.849	1.68	.442	.718	1.46	.262	.435	-1.22
1/T(R)3	12.7	(.251)	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68
Z(R)1	.773	.634	-.0874	-.142	-.128	-.201	-.347	-.153	-.217	.893
w(R)1	.410	1.49	.855	.791	.842	.920	1.08	.919	1.16	.925
Z(R)2	.0977	.952								
w(R)2	1.02	4.22								

TABLE IX-11 (Concluded)

N(PHI /DA)										
A(PHI)	.231	.321	.230	.372	.130	.211	.310	.0975	.143	.186
1/T(PHI)1	.838	1.46	.616	.466	.615	.612	.621	.606	.702	.526
1/T(PHI)2	3.06	2.45	1.83	1.28	2.87	2.09	.907	3.05	2.68	2.84
1/T(PHI)3	9.59	9.59	(.834)	(.610)	(.563)	(.736)	(.690)	(.445)	(.574)	(.474)
1/T(PHI)4	10.4	10.4	(1.22)	(2.35)	(.739)	(1.14)	(2.09)	(.629)	(.744)	(.922)
Z(PHI)1	.585	.620								
w(PHI)1	.428	.402								
N(AYP/DA)										
A(AYP)	4.54	5.76	4.74	6.91	2.80	3.82	4.19	1.72	2.10	1.34
1/T(AYP)1	-.249	-.175	.147	.145	.118	.103	.0823	.0835	.0700	.0698
1/T(AYP)2	3.35	2.87	-.212	-.151	-.217	-.217	-.226	-.202	-.227	-.273
1/T(AYP)3	7.90	7.16	.821	(.464)	.495	.694	(.524)	.440	.518	.699
1/T(AYP)4	11.7	12.2	2.94	(1.40)	3.30	2.94	(1.26)	3.32	3.08	2.51
Z(AYP)1	.794	.483	.425	.949	.348	.391	.932	.336	.382	.474
w(AYP)1	.360	.379	1.09	1.96	.870	1.11	1.78	.775	.942	1.20
Z(AYP)2	.502	.857								
w(AYP)2	.723	1.02								
	+	+	+	+	+	+	+	+	+	+

TABLE IX-12

B-747 RUDDER TRANSFER FUNCTION FACTORS

SAS On

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.198	.249	.450	.650	.500	.650	.800	.700	.800	.900
DENOMINATOR										
1/T(DET)1	.0770	.198	.0148	.0160	.00688	.00843	.00801	-.00185	.00514	-.00687
1/T(DET)2	.471	.858	.804	1.63	.495	.696	1.28	.386	.467	2.83
1/T(DET)3	1.20	1.53	(.524)	(.795)	1.09	1.67	(.685)	.808	1.10	(.973)
1/T(DET)4	3.10	2.50	(.900)	(.791)	2.91	1.90	(.748)	3.07	2.68	(.540)
1/T(DET)5	9.26	9.03	(.978)	(.692)	(.267)	(.477)	(.729)	(.234)	(.255)	(.432)
1/T(DET)6	10.7	10.9	(1.94)	(2.28)	(.794)	(.908)	(2.10)	(.716)	(.838)	(.961)
Z(DET)1	.472	.849								
W(DET)1	.576	.287								
NUMERATORS										
N(B /DR)										
A(B)	.0148	.0182	.0226	.0213	.0131	.0142	.0124	.00777	.00720	.00464
1/T(B)1	-.0503	-.0192	-.0182	-.00420	-.0359	-.0162	-.00957	-.0366	-.0320	-.0208
1/T(B)2	.368	.368	.368	.368	.368	.368	.368	.368	.368	.368
1/T(B)3	1.05	1.17	1.16	1.50	.665	.830	.995	.411	.478	.471
1/T(B)4	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68
1/T(B)5	9.94	9.97	28.0	45.8	31.0	44.0	74.4	44.3	66.8	96.3
1/T(B)6	10.1	10.0								
1/T(B)7	11.0	12.6								
N(P /DR)										
A(P)	.0636	.110	.254	.318	.148	.211	.183	.115	.153	.100
1/T(P)1	-.0209	-.0113	-.00340	0.	-.00728	-.00206	0.	-.00601	-.00332	-.00153
1/T(P)2	.368	.368	.368	.368	.368	.368	.368	.368	.368	.368
1/T(P)3	1.42	1.64	2.28	3.58	1.83	2.41	3.68	1.57	2.45	1.74
1/T(P)4	-2.18	-1.99	-3.09	3.68	-2.77	-3.24	3.75	-2.41	-3.63	-2.83
1/T(P)5	3.68	3.68	3.68	-4.18	3.68	3.68	-5.15	3.68	3.68	3.68
Z(P)1	1.00	(9.98)								
W(P)1	10.0	(10.0)								
N(R /DR)										
A(R)	-.151	-.233	-.614	-.970	-.391	-.616	-.922	-.331	-.475	-.442
1/T(R)1	.368	.368	.368	.368	.368	.368	.368	.368	.368	.368
1/T(R)2	1.05	1.17	1.16	1.50	.621	.865	1.11	.393	.497	.524
1/T(R)3	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68
1/T(R)4	9.99	9.98	(.130)	(.0796)	(.144)	(.0522)	(-.0468)	(.0789)	(.0245)	(-.0283)
1/T(R)5	10.0	10.0	(.297)	(.370)	(.434)	(.382)	(.364)	(.397)	(.488)	(.278)
Z(R)1	.0790	.0895								
W(R)1	.416	.284								

TABLE IX-12 (Continued)

N(PHI/DR)										
A(PHI)	.0410	.0867	.221	.318	.101	.185	.183	.0727	.115	.0815
1/T(PHI)1	.368	.368	.368	.368	.368	.368	.368	.368	.368	.368
1/T(PHI)2	1.48	1.69	2.35	3.58	2.01	2.50	3.68	1.79	2.69	1.83
1/T(PHI)3	-3.31	-2.48	-3.47	3.68	3.68	-3.58	3.75	-3.40	3.68	-3.30
1/T(PHI)4	3.68	3.68	3.68	-4.18	-3.74	3.68	-5.15	3.68	-4.44	3.68
1/T(PHI)5	0.00	(1.00)								
1/T(PHI)6	10.0	(10.0)								
N(AYP/DR)										
A(AYP)	-9.12	-13.9	-38.9	-64.7	-25.3	-41.3	-67.1	-22.1	-33.7	-33.0
1/T(AYP)1	-.0646	-.0386	-.0268	-.0103	-.0373	-.0210	-.0126	-.0330	-.0234	-.0200
1/T(AYP)2	.368	.368	.368	.368	.368	.368	.368	.312	.334	.368
1/T(AYP)3	.958	1.07	.973	1.32	.491	.669	.848	.368	.368	.427
1/T(AYP)4	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68	3.68
Z(AYP)1	.247	.208	.191	.137	.193	.147	.118	.136	.135	.0797
W(AYP)1	.668	.740	1.11	1.45	.984	1.10	1.22	.860	1.05	.873
Z(AYP)2	1.00	1.00								
W(AYP)2	10.0	10.0								
	+	+	+	+	+	+	+	+	+	+

TABLE IX-13

B-747 LATERAL-DIRECTIONAL HANDLING QUALITIES PARAMETERS

SAS Off

(Body Axis System)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	40 K
M	.198	.249	.450	.650	.500	.650	.800	.700	.800	.900
DR PERIOD (SEC)	8.59	8.47	5.98	4.53	7.30	5.87	4.83	7.99	6.64	6.19
1/C(1/2)	.799	.978	1.16	1.41	.630	.749	.894	.516	.317	.846
SPIRAL (2) (SEC)	--	--	--	--	--	--	--	296.	--	89.2
P(1)	.178	.235	.211	.304	.162	.241	.302	.156	.188	.363
P(2)	.0285	.0867	.171	.253	.134	.215	.287	.153	.175	.359
P(3)	.111	.148	.182	.268	.155	.233	.299	.187	.221	.381
P(2)/P(1)	.160	.369	.811	.832	.832	.891	.949	.979	.935	.990
P(OSC)/P(AV)	.671	.377	.0691	.0618	.0819	.0494	.0238	.0569	.0755	.0174
W(PHI)/W(D)	.797	.871	1.05	1.11	.978	1.03	1.03	.933	.927	.950
DEL-B-MAX	.161	.136	.00830	.0178	.0219	.00936	.00425	.0301	.0234	.0316
PHI TO BETA, PHASE	-304.	-306.	43.0	37.0	-322.	35.4	32.5	-331.	-332.	-333.
PHI TO BETA	1.54	1.69	2.07	2.07	2.26	2.12	2.03	2.09	3.07	1.18
PHI TO VE	.399	.349	.236	.163	.343	.247	.192	.355	.456	.156
	+	+	+	+	+	+	+	+	+	+

B-747 DATA SOURCES

Hanke, C. Rodney and Donald R. Nordwall, The Simulation of a Large Jet Transport Aircraft, Boeing Rept. No. D6-30643, Vols. I and II, Sept. 1970.

SECTION X

C-5A

C-5A BACKGROUND

The C-5A is a very large military logistics transport powered by four turbofan engines. Longitudinal control consists of elevators in four sections with an all-movable stabilizer for trim, roll control employs ailerons and spoilers, and yaw control a conventional rudder. All control surfaces are irreversible.

A bobweight is used in the longitudinal feel system. The effective bobweight position is assumed to be at the pilot.

The C-5A employs stability augmentation about all axes. A description of the SAS is not included here.

C-5A

Nominal Configuration

220,000 lb Cargo

TOGW less 40% Fuel

W = 654,362 lb

c.g. at 0.30 $\bar{c}$, W.L. 265

$I_x = 27.8 \times 10^6$ slug-ft²

$I_y = 31.8 \times 10^6$ slug-ft²

$I_z = 56.2 \times 10^6$ slug-ft²

$I_{xz} = 2.46 \times 10^6$ slug-ft²

} Body Axis

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Power Approach Configuration

220,000 lb Cargo

TOGW less 80% Fuel

30° Flaps

Gear Down

1.4 V_s

W = 580,723 lb

c.g. at 0.30 $\bar{c}$, W.L. 265

$I_x = 19.1 \times 10^6$ slug-ft²

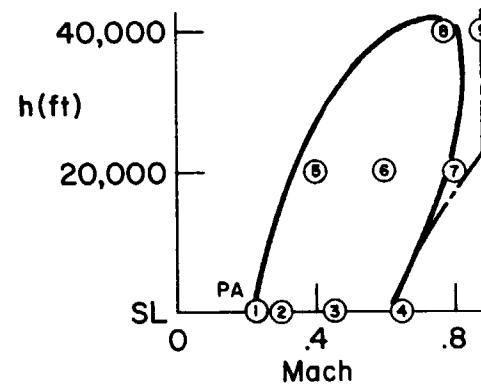
$I_y = 31.3 \times 10^6$ slug-ft²

$I_z = 47.0 \times 10^6$ slug-ft²

$I_{xz} = 2.5 \times 10^6$ slug-ft²

} Body Axis

Flight Envelope



- Level Flight Envelope (Nominal Configuration)
- - - - - Speed Restrictions
- ⑨ Transfer Function Case n

Figure X-1. C-5A Flight Conditions

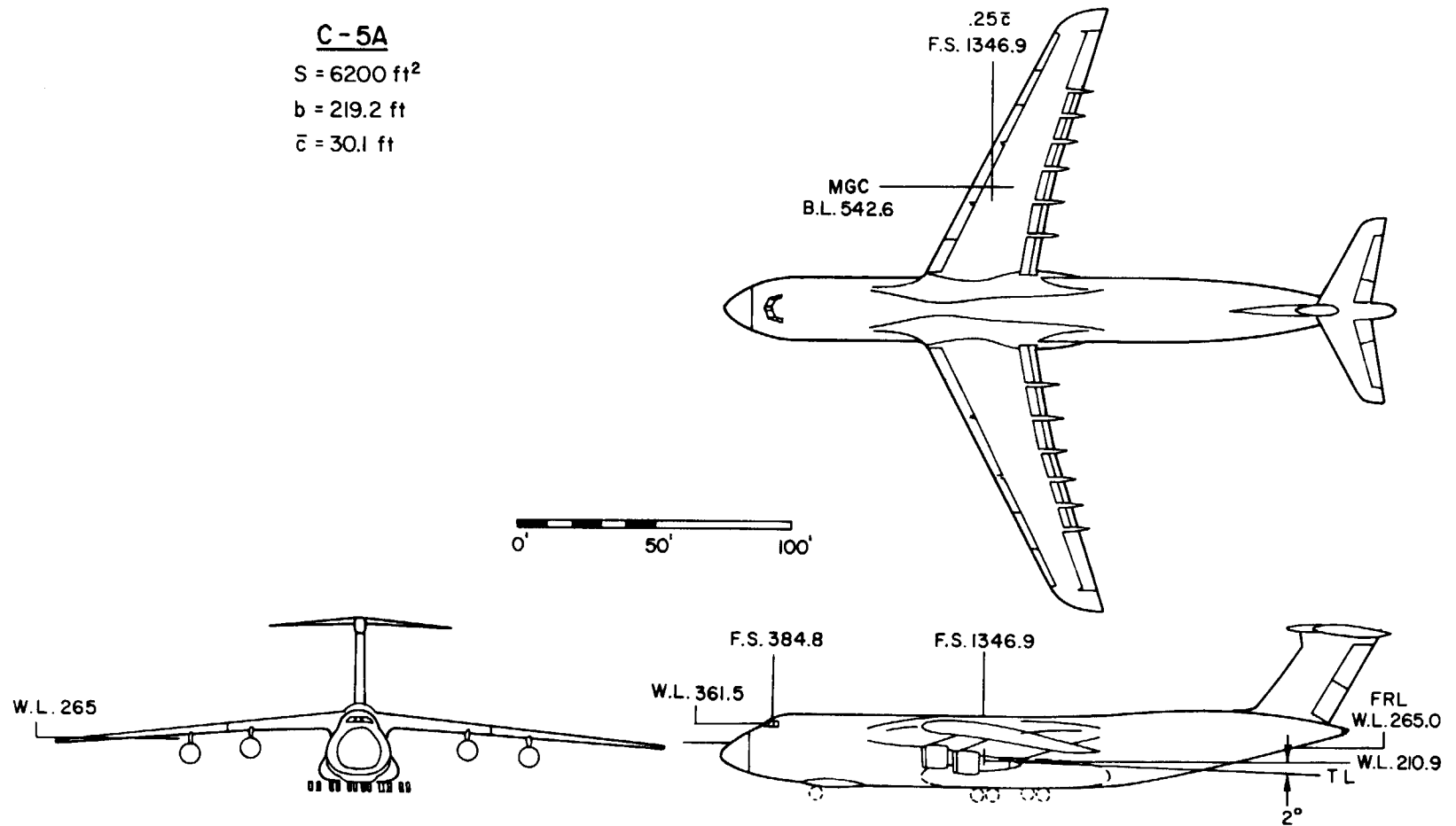
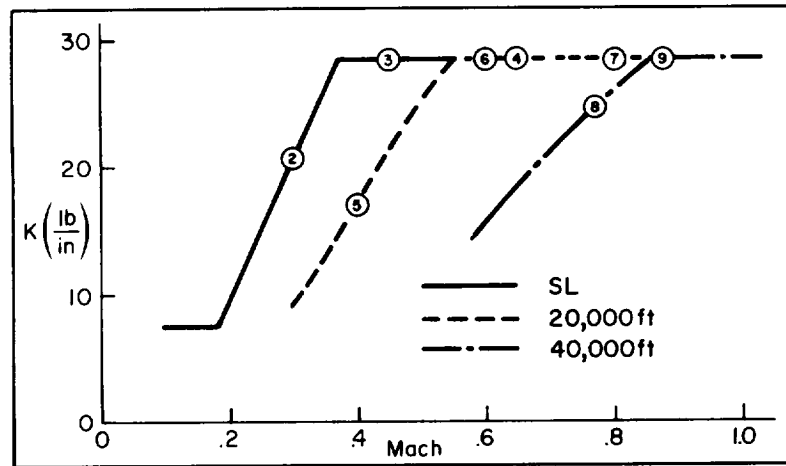
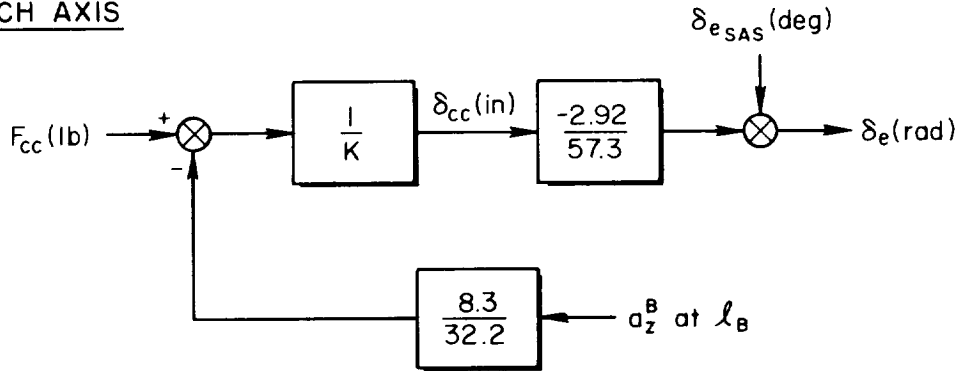


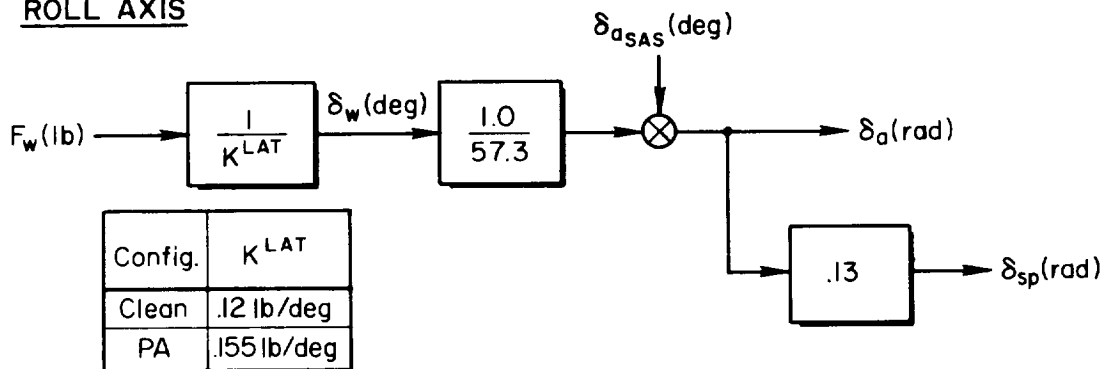
Figure X-2. C-5A General Arrangement

C-5A

PITCH AXIS



ROLL AXIS



YAW AXIS

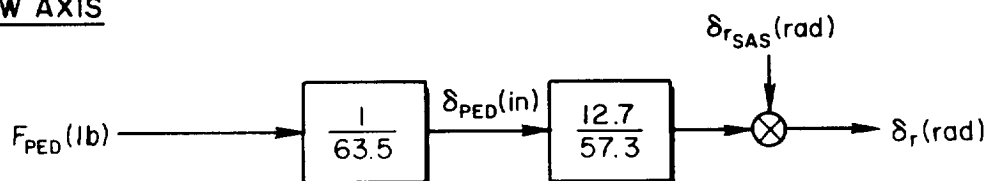


Figure X-3. C-5A Control System

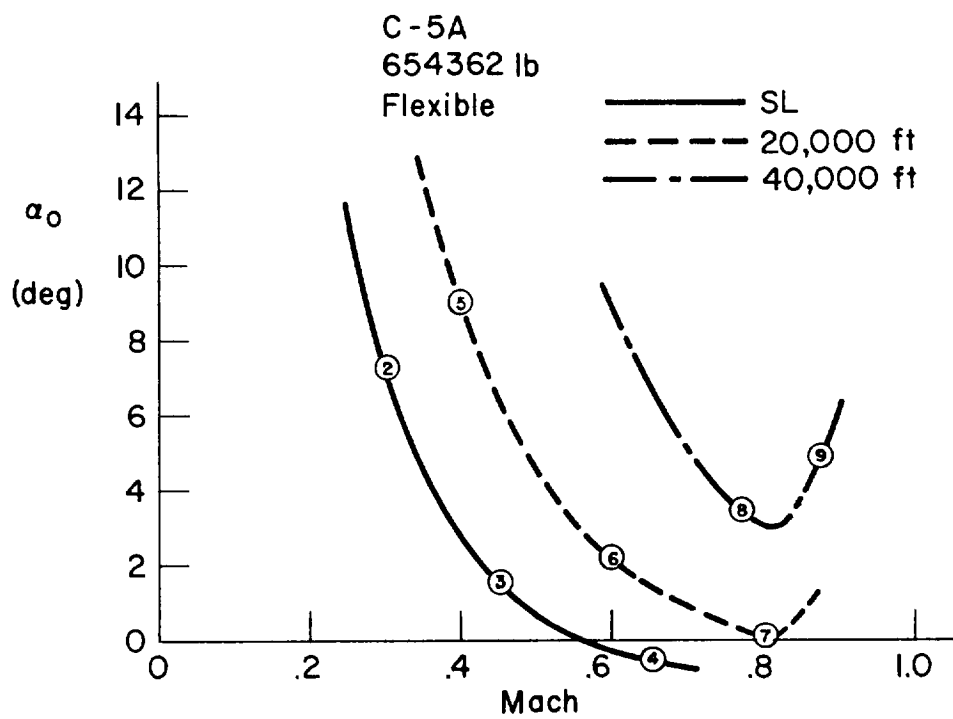
TABLE X-1

C-5A**Power Approach Non-Dimensional Derivatives**

h = sea level

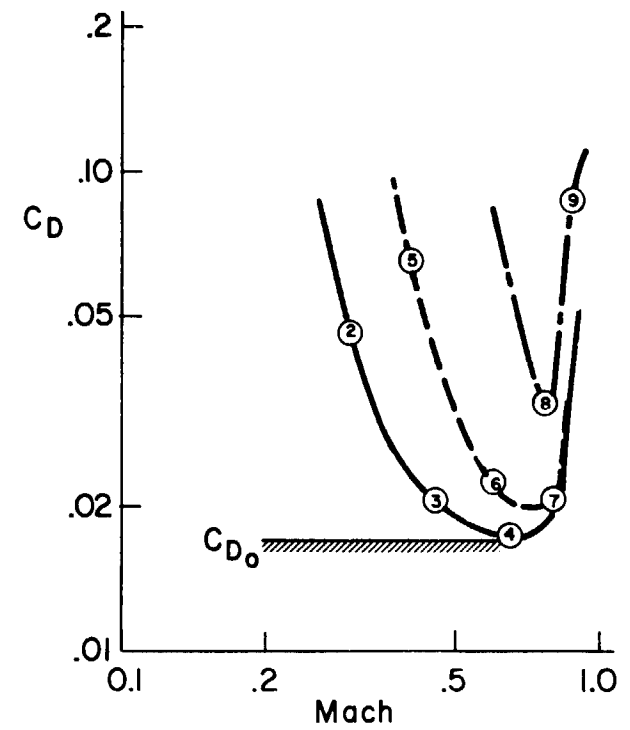
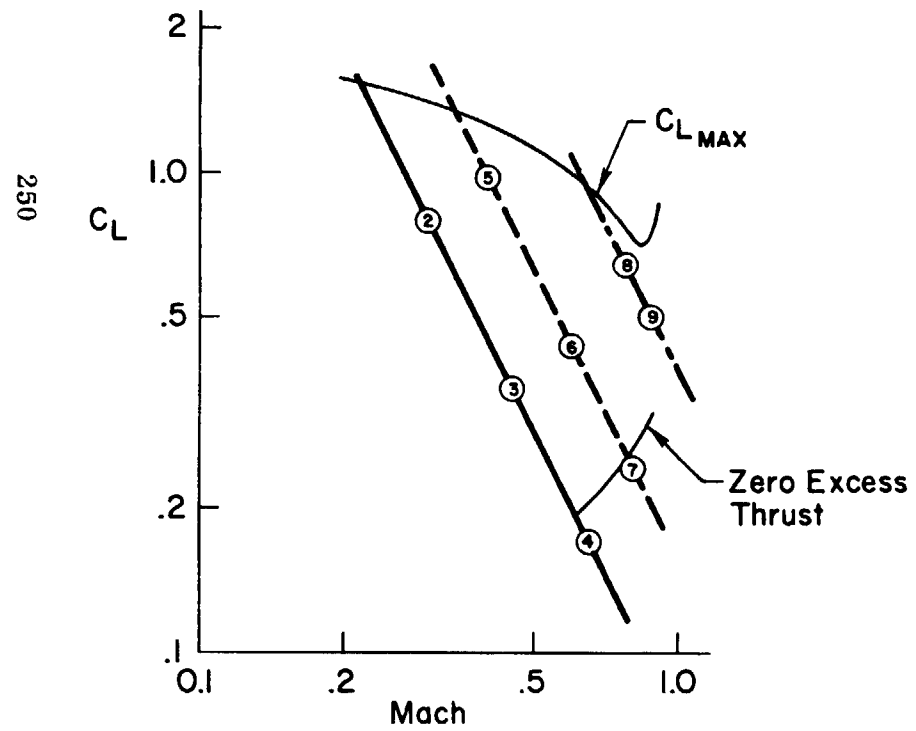
 $V_{T_0} = 247 \text{ ft/sec} = 146 \text{ kt}$ $\alpha_0 = 2.7^\circ$

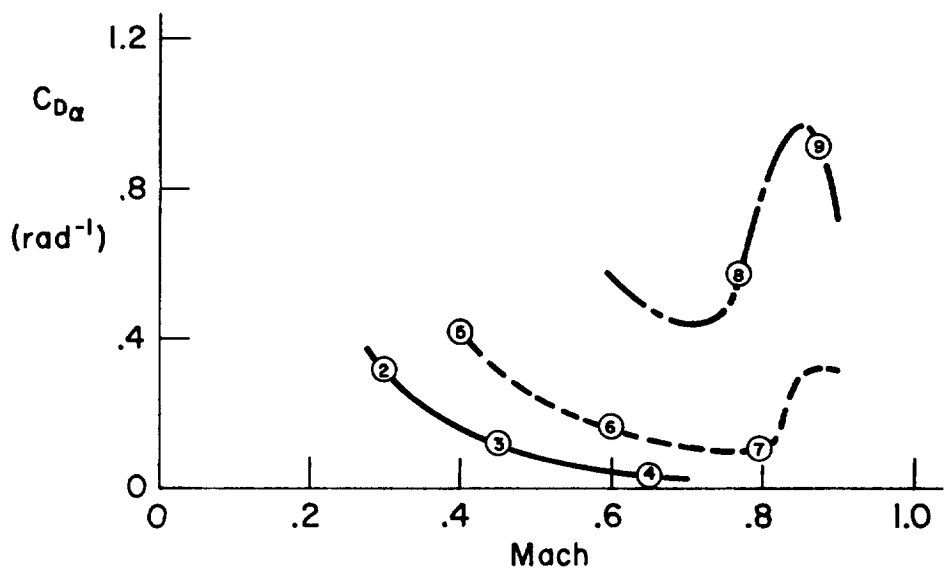
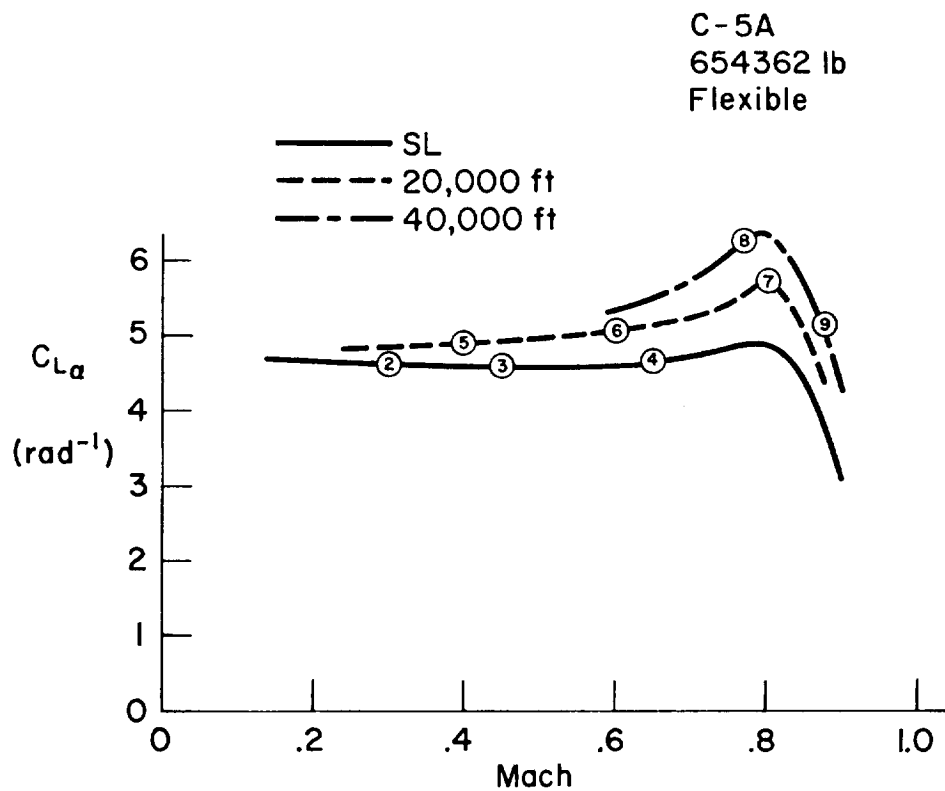
Longitudinal	Lateral-Directional (Stability Axis)	
$C_L = 1.29$	$C_{y\beta} = -.77/\text{rad}$	
$C_D = .145$	$C_{n\beta} = .075/\text{rad}$	
$C_{L\alpha} = 6.08/\text{rad}$	$C_{l\beta} = -.123/\text{rad}$	
$C_{D\alpha} = .622/\text{rad}$	$C_{l_p} = -.458/\text{rad}$	
$C_{m\alpha} = -.827/\text{rad}$	$C_{n_p} = -.098/\text{rad}$	
$C_{m\dot{\alpha}} = -8.3/\text{rad}$	$C_{l_r} = .290/\text{rad}$	
$C_{m_q} = -23.2/\text{rad}$	$C_{n_r} = -.293/\text{rad}$	
$C_{L\delta_e} = .385/\text{rad}$	$C_{y\delta_a} = -.0044/\text{rad}$	} Spoiler Effects Included
$C_{m\delta_e} = -1.6/\text{rad}$	$C_{n\delta_a} = .0091/\text{rad}$	
	$C_{l\delta_a} = .089/\text{rad}$	
	$C_{y\delta_r} = .211/\text{rad}$	
	$C_{n\delta_r} = -.106/\text{rad}$	
	$C_{l\delta_r} = .0209/\text{rad}$	

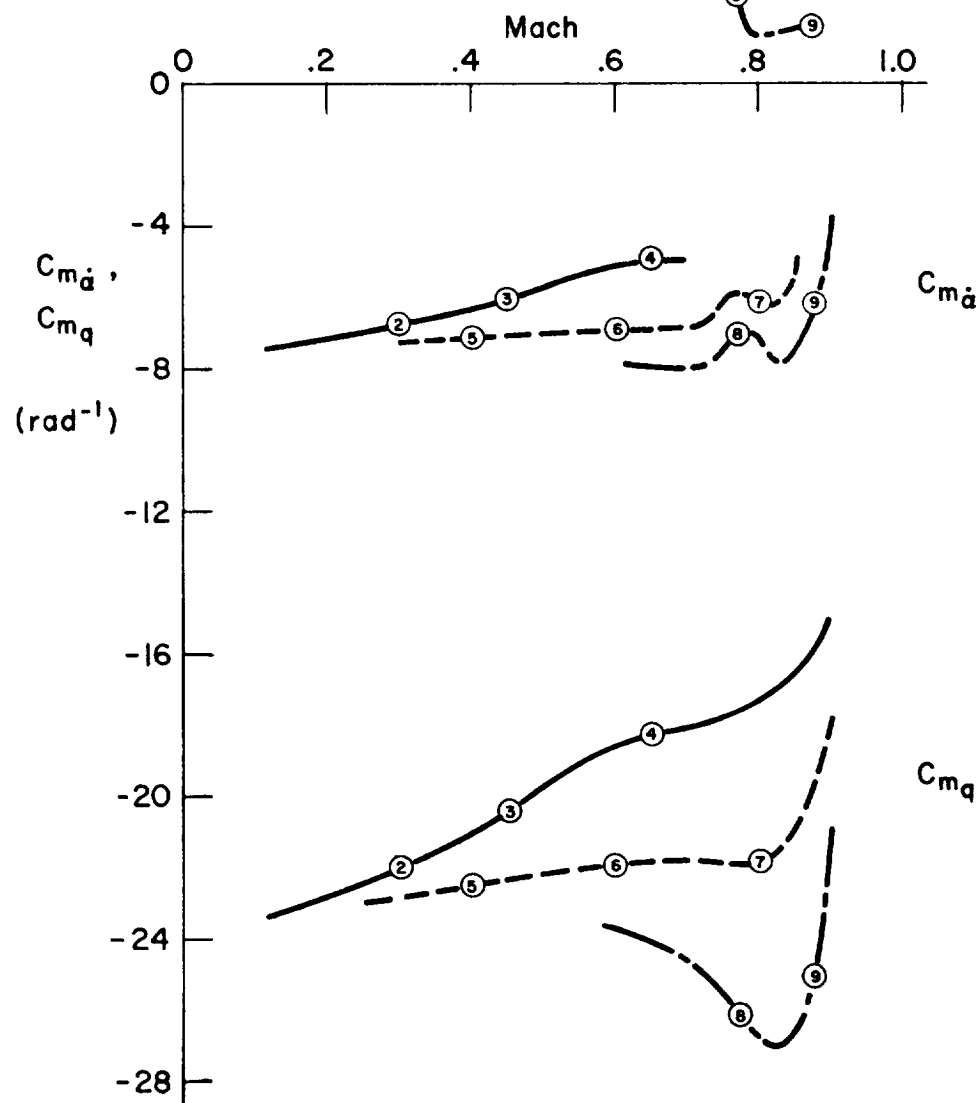
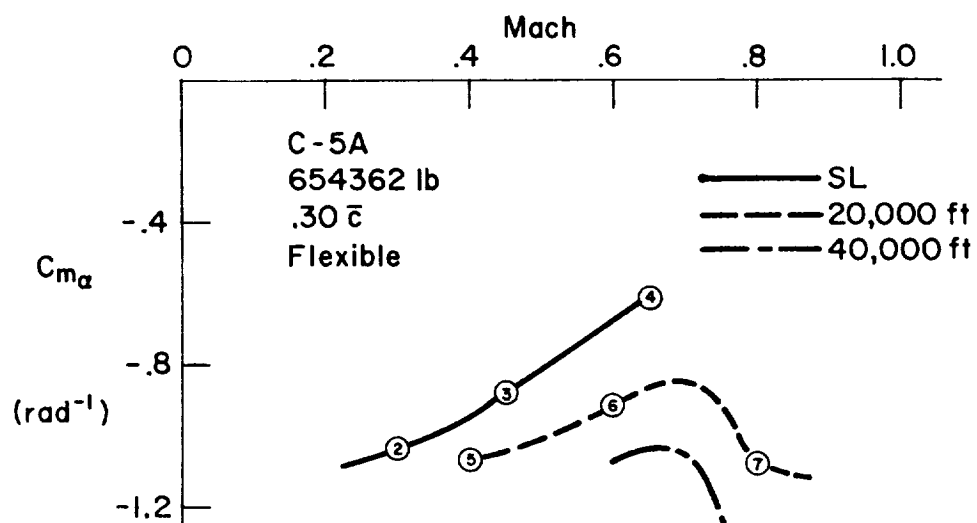


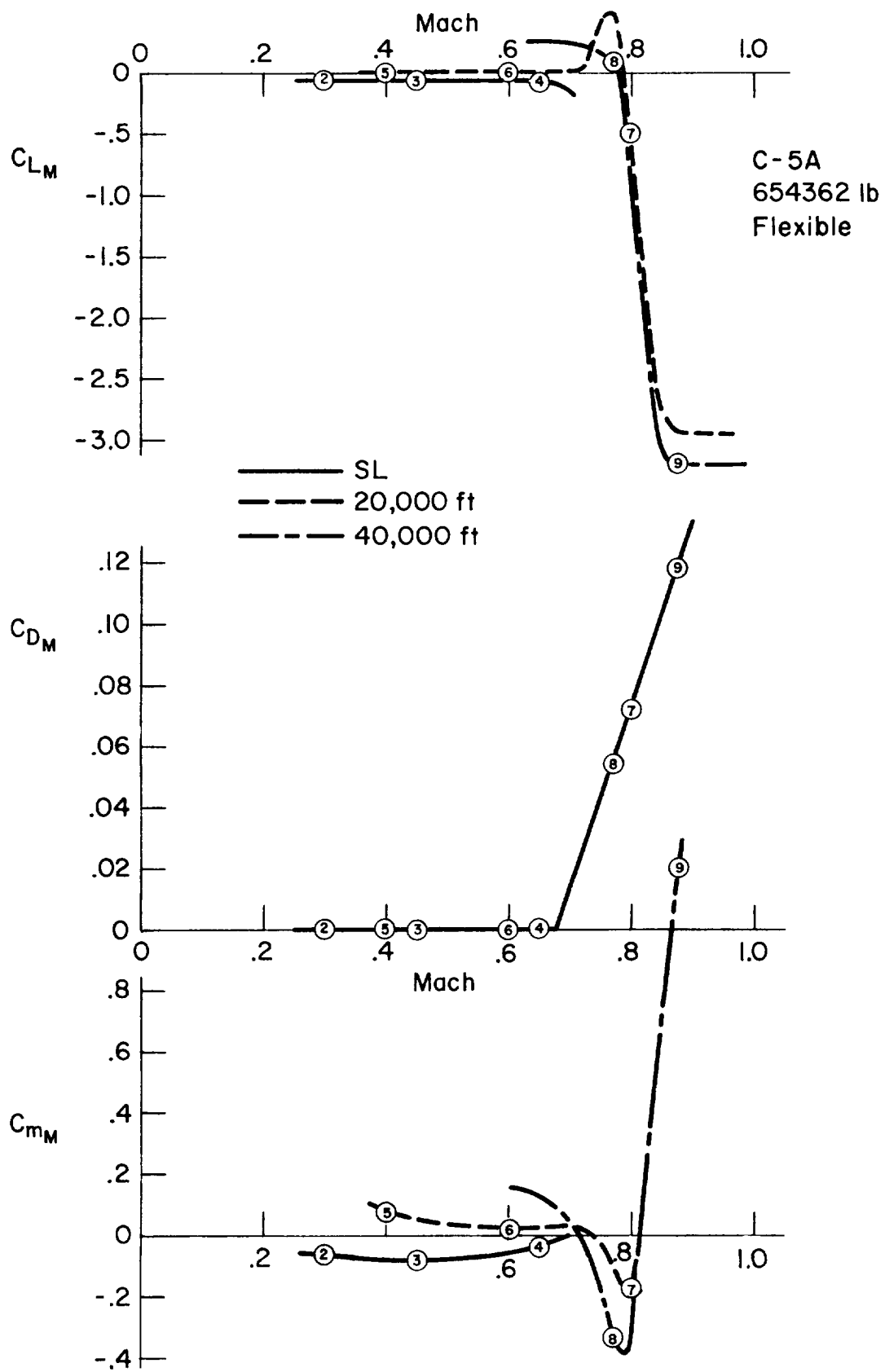
C-5A
654362 lb

— SL
- - - 20,000 ft
- · - 40,000 ft



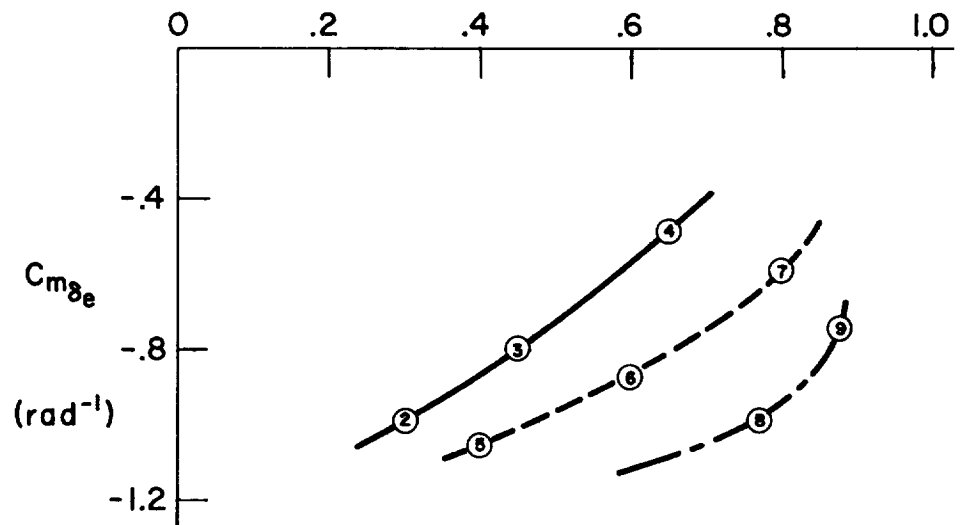
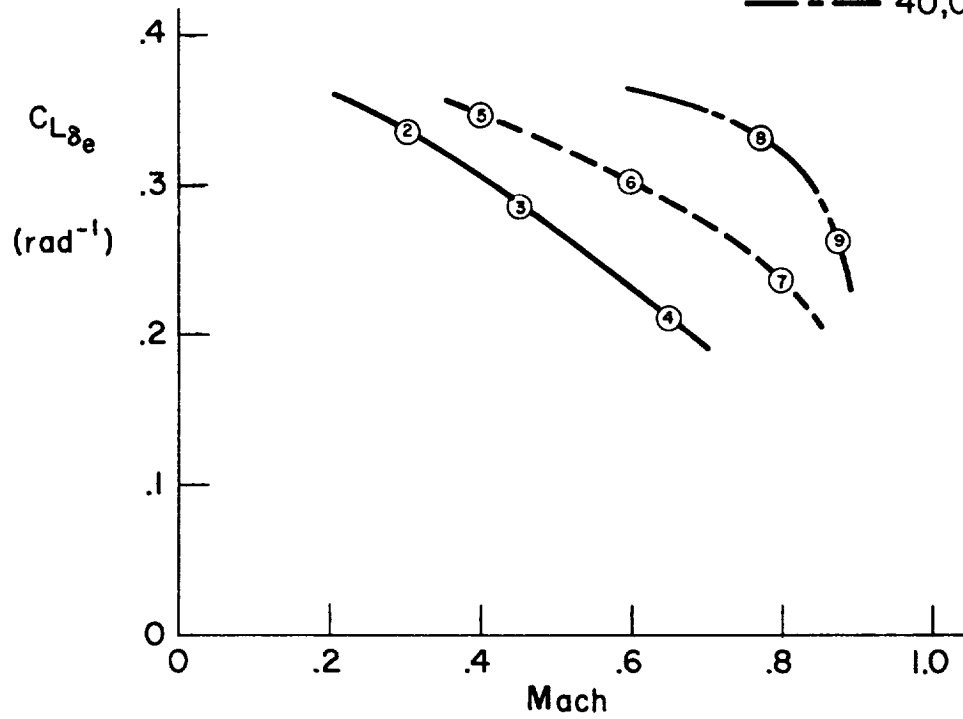


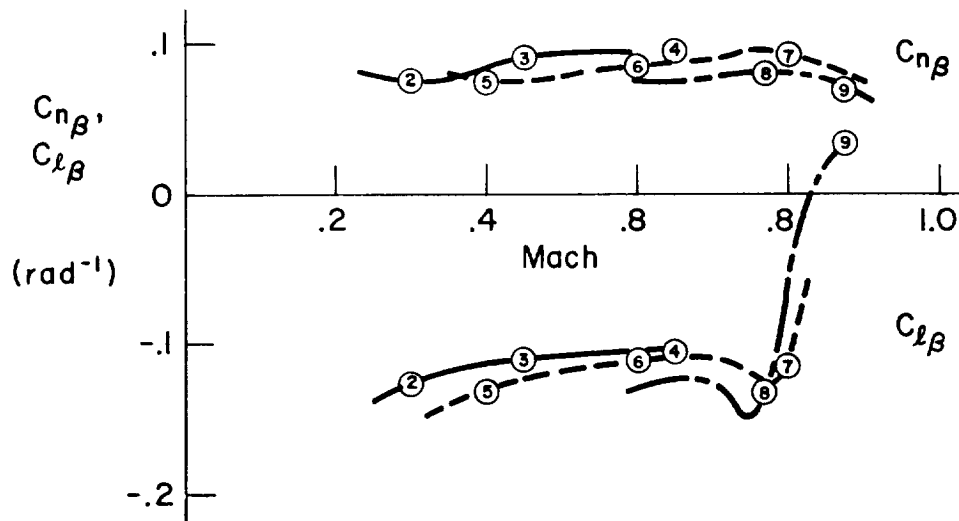
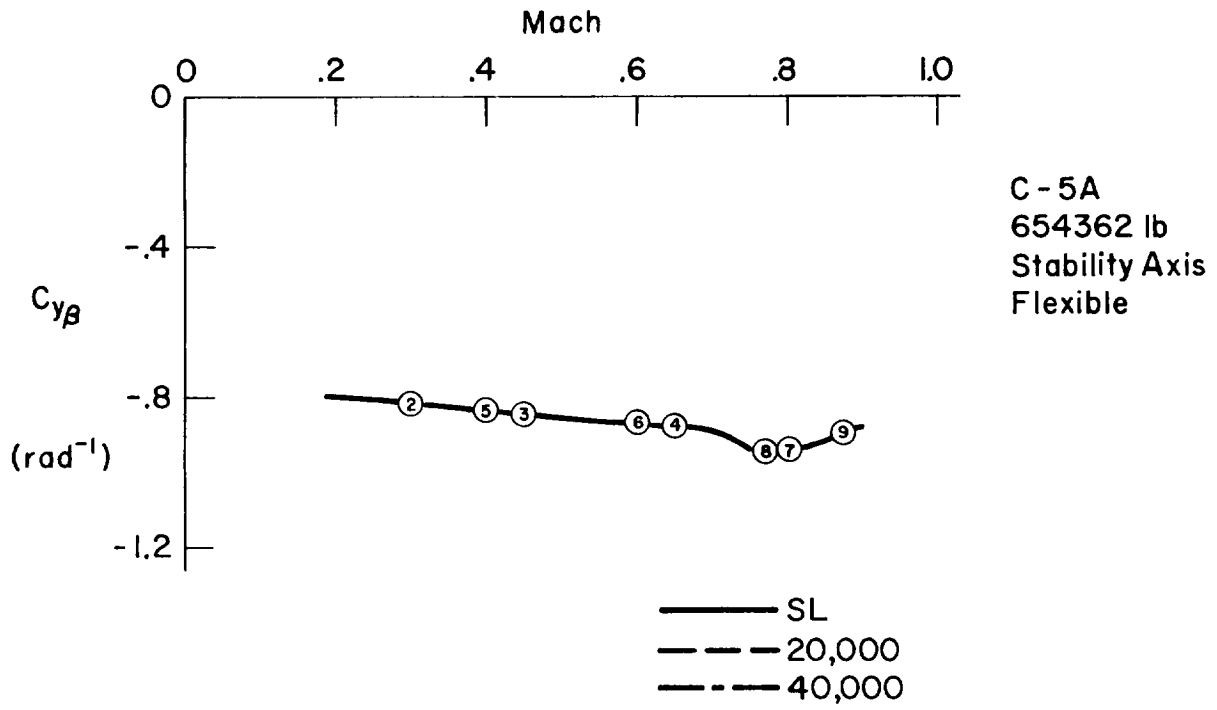


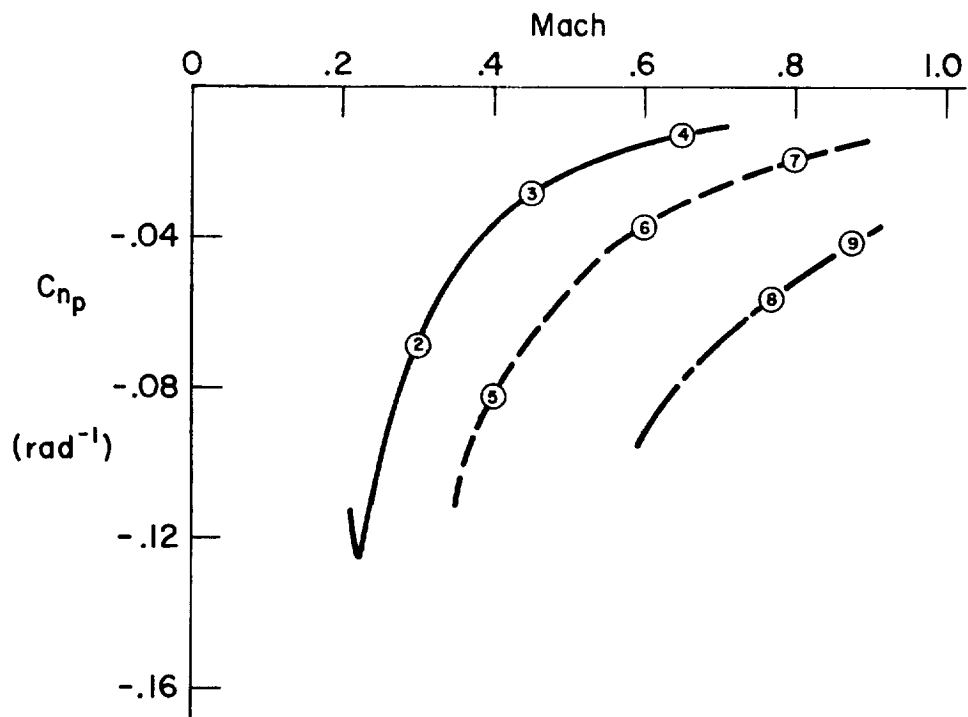
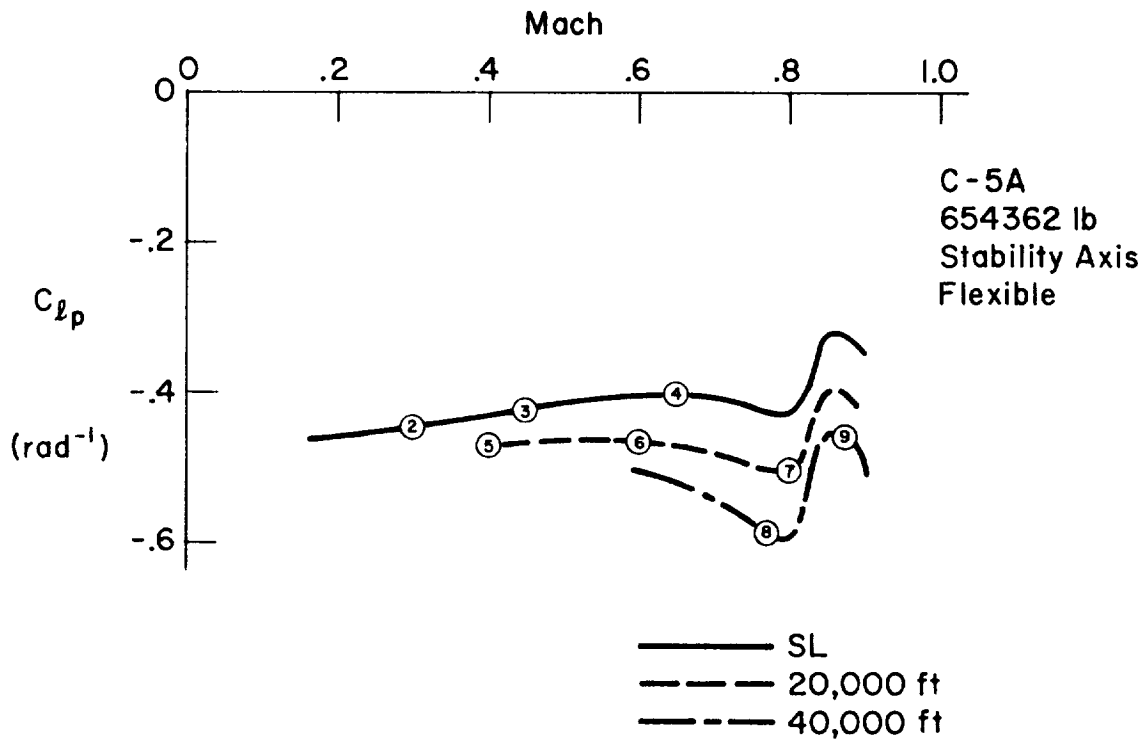


C-5A
654362 lb
Flexible

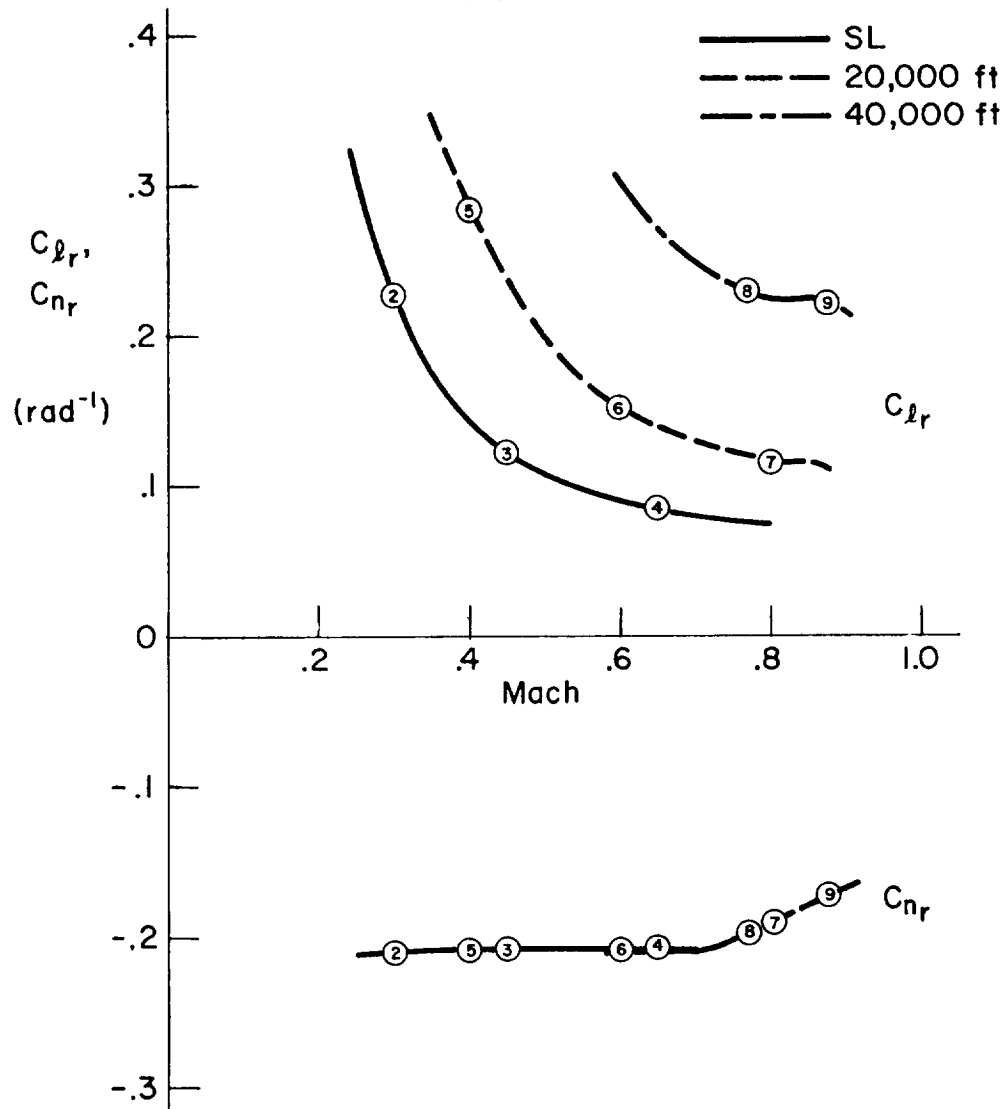
— SL
- - - 20,000 ft
- - - 40,000 ft





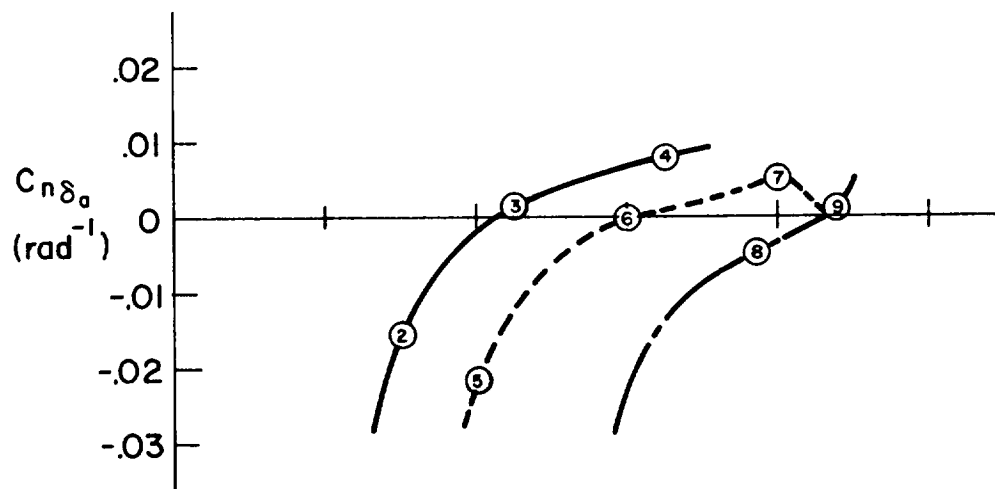
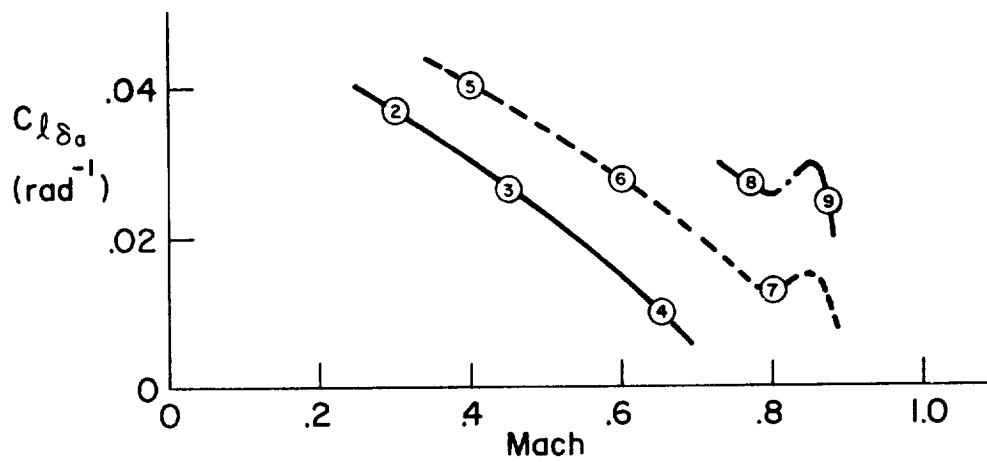


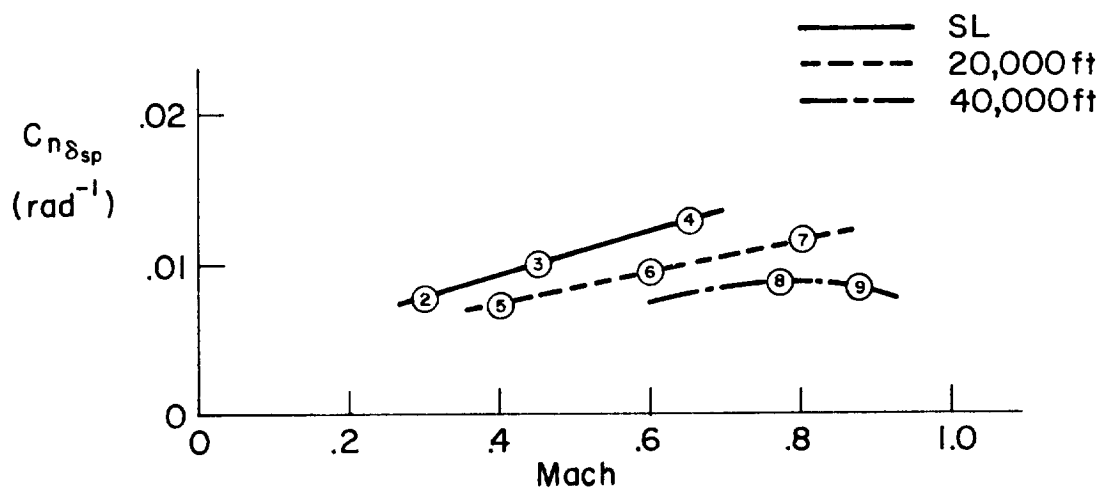
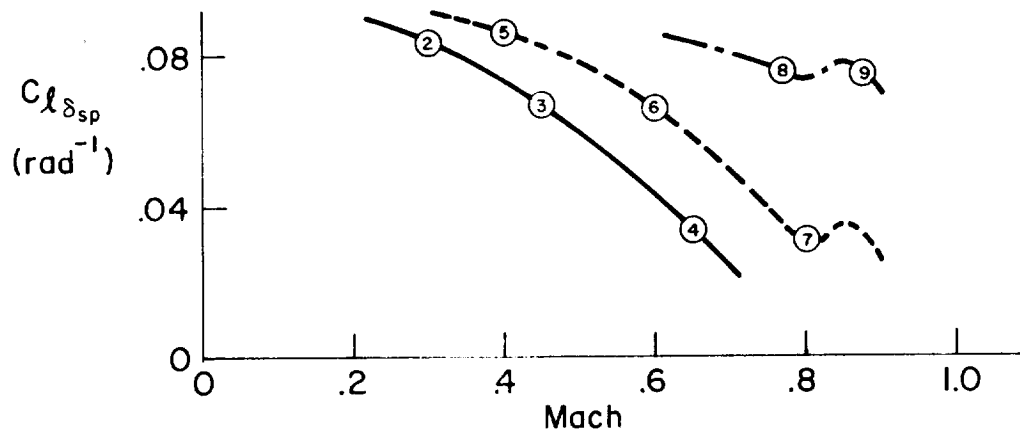
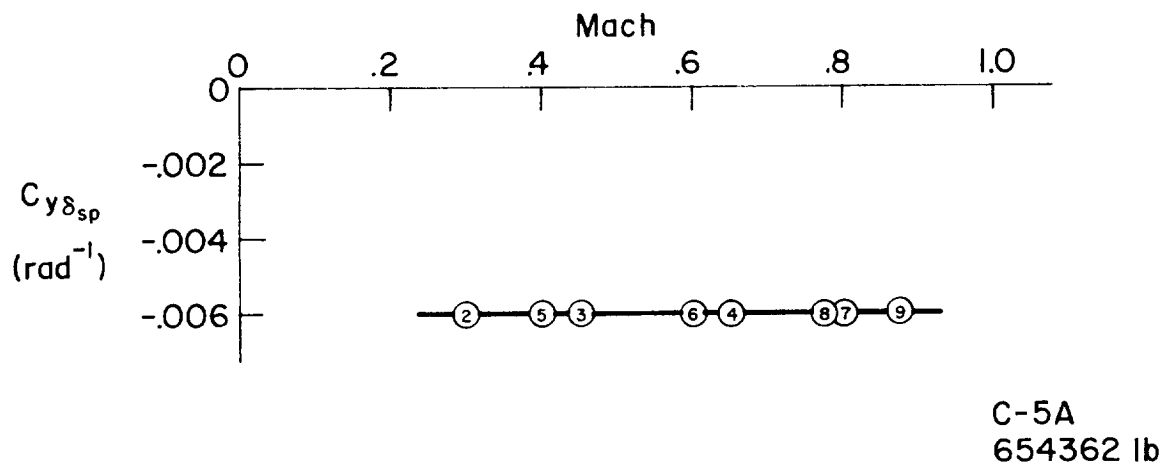
C-5A
 654362 lb
 .30 c
 Stability Axis
 Flexible

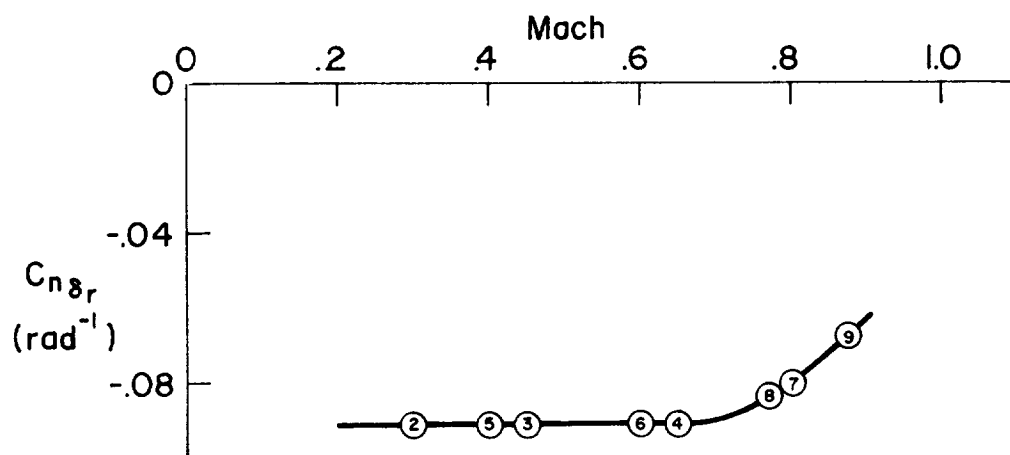
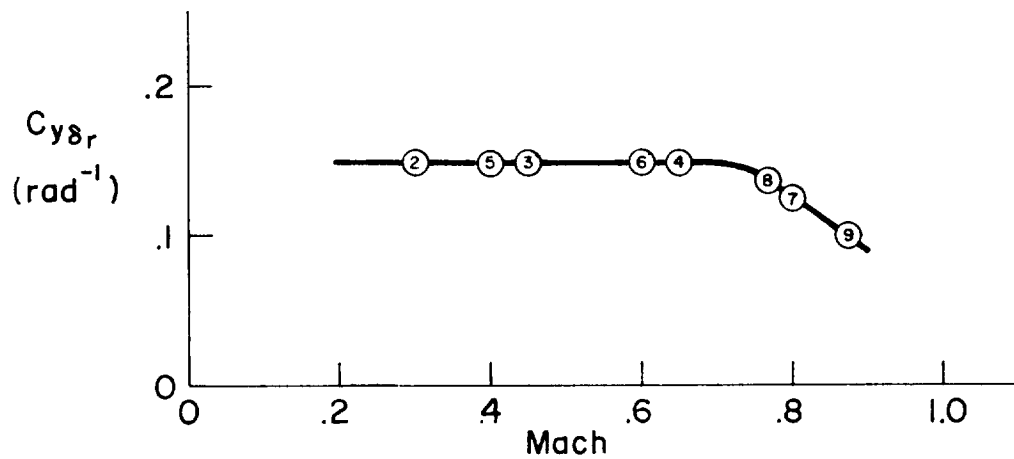


— SL
 --- 20,000 ft
 - - - 40,000 ft

C-5A
 654326 lb







— SL
 --- 20,000 ft
 - - - 40,000 ft

C-5A
 654362 lb
 Stability Axis
 Rigid

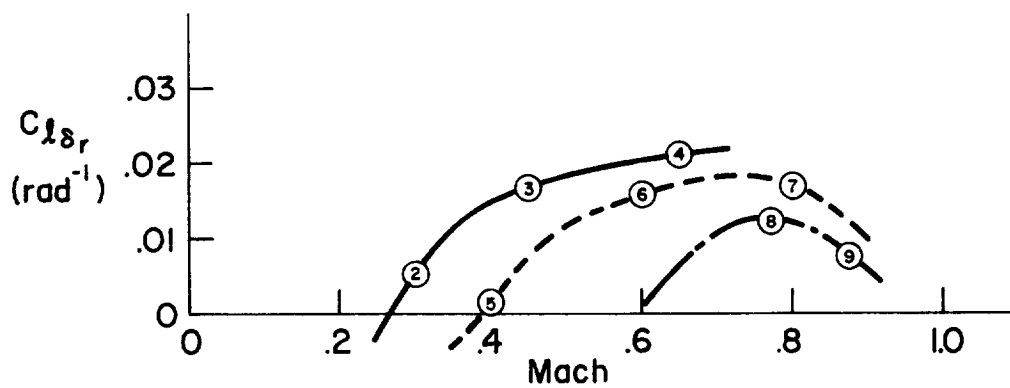


TABLE X-2

C-5A DIMENSIONAL, MASS AND FLIGHT CONDITION PARAMETERS

 $s = 6200 \text{ sq ft}, \quad b = 219.20 \text{ ft}, \quad \bar{c} = 30.10 \text{ ft}$

F/C #	1	2	3	4	5	6	7	8	9
H(FT)	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K
M(-)	.221	.300	.450	.650	.400	.600	.800	.770	.875
VTO(FPS)	246.	335.	502.	726.	415.	622.	830.	745.	847.
VTO(KTAS)	146.	198.	298.	430.	246.	369.	492.	442.	502.
VTO(KCAS)	146.	198.	298.	430.	181.	275.	373.	233.	269.
W(LBS)	580756.	654399.	654399.	654399.	654399.	654399.	654399.	654399.	654399.
C.G.(MGC)	.300	.300	.300	.300	.300	.300	.300	.300	.300
IX (SLUG-FT SQ)	.191E+8	.278E+8	.278E+8	.278E+8	.278E+8	.278E+8	.278E+8	.278E+8	.278E+8
IY (SLUG-FT SQ)	.313E+8	.318E+8	.318E+8	.318E+8	.318E+8	.318E+8	.318E+8	.318E+8	.318E+8
IZ (SLUG-FT SQ)	.470E+8	.562E+8	.562E+8	.562E+8	.562E+8	.562E+8	.562E+8	.562E+8	.562E+8
IXZ(SLUG-FT SQ)	.250E+7	.246E+7	.246E+7	.246E+7	.246E+7	.246E+7	.246E+7	.246E+7	.246E+7
EPSILON(DEG)	-5.08	-4.91	-4.91	-4.91	-4.91	-4.91	-4.91	-4.91	-4.91
Q(PSF)	72.2	133.	300.	626.	109.	245.	436.	164.	211.
QC(PSF)	73.0	136.	315.	695.	113.	268.	510.	189.	255.
ALPHA(DEG)	2.70	7.30	1.60	-5.00	9.00	2.20	.100	3.50	4.90
GAMMA(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.
LXP(FT)	81.7	81.7	81.7	81.7	81.7	81.7	81.7	81.7	81.7
LZP(FT)	-8.10	-8.10	-8.10	-8.10	-8.10	-8.10	-8.10	-8.10	-8.10
ITH(DEG)	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
XI(DEG)	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
LTH(FT)	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50	4.50

TABLE X-3

C-5A LONGITUDINAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

[illegible]

TABLE X-4

C-5A ELEVATOR TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K
M	.221	.300	.450	.650	.400	.600	.800	.770	.875
DE NOMINATOR									
Z(DET)1	.100	.0351	.0612	.232	.0283	.0271	(-.0487)	.0506	.453
W(DET)1	.119	.104	.0594	.0213	.0969	.0638	(.0648)	.0110	.0716
Z(DET)2	.843	.706	.712	.752	.577	.608	.570	.435	.373
W(DET)2	.865	1.12	1.57	1.99	.947	1.34	1.93	1.23	1.40
NUMERATORS									
N(U /DE)									
A(U)	.450	1.73	.728	-.350	1.79	.851	.0545	1.00	1.46
1/T(U)1	15.6	19.1	26.1	1.92	24.4	34.2	2.29	42.4	.287
1/T(U)2	(.482)	(.202)	(.196)	-2.58	(.239)	(.442)	(.736)	(.593)	.530
1/T(U)3	(1.40)	(.639)	(1.36)	37.9	(.439)	(.892)	(18.3)	(.532)	45.6
N(W /DE)									
A(W)	-9.53	-13.5	-26.1	-.40.1	-11.3	-22.4	-31.2	-16.4	-17.0
1/T(W)1	18.4	19.8	28.2	33.3	24.8	35.5	41.1	43.1	-.0651
1/T(W)2	(.0730)	(.0308)	(.0428)	(.0751)	(.0304)	(.0323)	(.854)	(.0493)	.0806
1/T(W)3	(.170)	(.107)	(.0795)	(.0609)	(.0852)	(.0642)	(.00922)	(.0490)	46.1
N(TH/DE)									
A(TH)	-.680	-.765	-1.39	-1.74	-.667	-1.24	-1.50	-.938	-.015
1/T(TH)1	.0610	.0342	.0149	.0115	.0302	.00913	.0161	.00817	.0216
1/T(TH)2	.582	.505	.777	1.17	.353	.578	.862	.394	.350
N(HD /DE)									
A(HD)	9.54	13.6	26.1	40.1	11.5	22.4	31.2	16.5	17.1
1/T(HD)1	.00211	-.00248	.00448	.00852	-.00376	-.000206	.0159	-.000655	.0422
1/T(HD)2	-2.88	-2.63	-3.94	-5.26	-2.66	-4.00	-5.24	-3.78	-3.73
1/T(HD)3	3.73	3.70	5.34	7.02	3.37	5.01	6.55	4.43	4.30
N(AZP/DE)									
A(AZP)	46.0	49.0	87.7	102.	43.2	79.2	91.2	60.2	57.7
1/T(AZP)1	-.0179	.0189	-.00337	.000558	.0169	.00414	-.682E-4	.00430	-.00226
1/T(AZP)2	.0197	-.0215	.00784	.00795	-.0211	-.00435	.0153	-.00501	.0440
Z(AZP)1	.193	.124	.121	.124	.0990	.0980	.104	.0783	.0556
W(AZP)1	1.50	1.65	2.50	3.81	1.52	2.38	3.43	2.14	2.20

TABLE X-5

C-5A THRUST TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K
M	.221	.300	.450	.650	.400	.600	.800	.770	.875
DENOMINATOR									
Z(DET)1	.100	.0351	.0612	.232	.0283	.0271	(-.0487)	.0506	.453
W(DET)1	.119	.104	.0594	.0213	.0969	.0638	(.0648)	.0110	.0716
Z(DET)2	.843	.706	.712	.752	.577	.608	.579	.435	.373
W(DET)2	.865	1.12	1.57	1.99	.947	1.34	1.93	1.23	1.40
NUMERATORS									
N(U /DTH)									
A(U)	.554E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4
1/T(U)1	-.0712	-.0525	-.0336	-.0290	-.0552	-.0351	-.0241	-.0308	-.0253
Z(U)1	.836	.666	.703	.758	.508	.596	.583	.399	.339
W(U)1	.896	1.13	1.58	2.00	.939	1.33	1.93	1.21	1.31
N(W /DTH)									
A(W)	-.193E-5	-.172E-5	-.172E-5	-.172E-5	-.172E-5	-.172E-5	-.172E-5	-.172E-5	-.172E-5
1/T(W)1	-11.0	-24.0	-37.5	-55.6	-31.4	-48.5	-.00482	-59.3	-.0142
1/T(W)2	(-.726)	(-.0564)	(-.589)	(-.934)	(.414)	(.0188)	-.138	(-.847)	.602
1/T(W)3	(.223)	(.108)	(.0862)	(.0667)	(.0776)	(.0651)	-66.9	(.0552)	-74.7
N(THE/DTH)									
A(THE)	.148E-6	.147E-6	.143E-6	.142E-6	.145E-6	.143E-6	.142E-6	.142E-6	.142E-6
1/T(THE)1	(.930)	(.887)	.143	.0282	(.847)	.157	-.116	.0164	.141
1/T(THE)2	(.398)	(.397)	.728	1.19	(.314)	.529	.945	.358	.868
N(HD /DTH)									
A(HD)	.454E-5	.795E-5	.309E-5	.129E-5	.938E-5	.360E-5	.180E-5	.471E-5	.591E-5
1/T(HD)1	.137	.172	.0967	.0227	.159	.109	-.114	.00332	.286
Z(HD)1	.715	.451	.345	.235	.355	.276	.112	.229	-.117
W(HD)1	2.71	2.24	4.82	10.3	1.87	4.19	7.98	3.37	3.05
N(AZP/DTH)									
A(AZP)	-.140E-4	-.137E-4	-.134E-4	-.133E-4	-.136E-4	-.134E-4	-.133E-4	-.133E-4	-.134E-4
1/T(AZP)1	-.00740	-.0130	-.00191	.000443	-.0131	-.00209	-.674E-4	-.00405	-.00308
1/T(AZP)2	.147	.217	.103	.0220	.206	.117	-.114	.00828	.333
Z(AZP)1	.501	.310	.269	.250	.245	.214	.179	.164	.00630
W(AZP)1	1.53	1.56	2.26	3.21	1.41	2.12	2.93	1.91	1.87

+ + + + + + + + + +

TABLE X-6

C-5A STICK FORCE TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Closed

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K
M	.221	.300	.450	.650	.400	.600	.800	.770	.875
DENOMINATOR									
Z(DET)1	.110	.0376	.0646	.247	.0304	.0285	(-.0454)	.0462	.459
W(DET)1	.110	.101	.0566	.0197	.0928	.0503	(.0613)	.0104	.0693
Z(DET)2	.783	.682	.677	.693	.553	.575	.543	.415	.361
W(DET)2	.936	1.16	1.65	2.17	.991	1.42	2.06	1.30	1.45
NUMERATORS									
N(U /FST)									
A(U)	-.00201	-.00430	-.00132	.000637	-.00542	-.00156	-.989E-4	-.00211	-.00263
1/T(U)1	15.6	19.1	26.1	1.92	24.4	34.2	2.29	42.4	.287
1/T(U)2	(.482)	(.202)	(.196)	-2.58	(.239)	(.442)	(.736)	(.593)	.539
1/T(U)3	(1.40)	(.639)	(1.36)	37.9	(.439)	(.892)	(18.2)	(.532)	45.6
N(W /FST)									
A(W)	.0427	.0335	.0472	.0730	.0342	.0405	.0567	.0345	.0307
1/T(W)1	18.4	19.8	28.2	33.3	24.8	35.5	41.1	43.1	-.0651
1/T(W)2	(.0730)	(.0308)	(.0428)	(.0751)	(.0304)	(.0323)	(.864)	(.0493)	.0806
1/T(W)3	(.170)	(.107)	(.0795)	(.0605)	(.0852)	(.0642)	(.00922)	(.0490)	46.1
N(THF/FST)									
A(THF)	.00305	.00190	.00252	.00317	.00202	.00225	.00272	.00197	.00165
1/T(THF)1	.0610	.0342	.0149	.0115	.0302	.00913	.0161	.00817	.0316
1/T(THF)2	.582	.505	.777	1.17	.353	.578	.862	.394	.350
N(HD /FST)									
A(HD)	-.0428	-.0338	-.0472	-.0730	-.0347	-.0405	-.0567	-.0345	-.0308
1/T(HD)1	.00211	-.00248	.00448	.00852	-.00376	-.000206	.0159	-.000655	.0422
1/T(HD)2	-2.88	-2.68	-3.94	-5.26	-2.66	-4.00	-5.24	-3.78	-3.73
1/T(HD)3	3.73	3.70	5.34	7.02	3.37	5.01	6.55	4.43	4.39
N(AZP/FST)									
A(AZP)	-.206	-.122	-.159	-.186	-.131	-.143	-.166	-.126	-.104
1/T(AZP)1	-.0179	.0189	-.00337	.000558	.0169	.00414	-.682E-4	.00430	-.00226
1/T(AZP)2	.0197	-.0215	.00784	.00796	-.0211	-.00436	.0159	-.00501	.0440
Z(AZP)1	.198	.124	.121	.124	.0990	.0980	.104	.0783	.0556
W(AZP)1	1.50	1.65	2.50	3.81	1.52	2.38	3.42	2.14	2.20
	+	+	+	+	+	+	+	+	+

TABLE X-7
C-5A THRUST TRANSFER FUNCTION FACTORS
 SAS Off — Bobweight Loop Closed
 (BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K
M	.221	.300	.45C	.65C	.400	.600	.800	.770	.875
DE Nominator									
Z(DET)1	.110	.0376	.0646	.247	.0304	.0285	(-.0454)	.0462	.459
W(DET)1	.110	.101	.0566	.0197	.0928	.0603	(.0613)	.0104	.0693
Z(DET)2	.783	.682	.677	.693	.553	.575	.543	.415	.361
W(DET)2	.936	1.16	1.65	2.17	.991	1.42	2.05	1.30	1.45
NUMERATORS									
N(U /DTH)									
A(U)	.554E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4	.491E-4
1/T(U)1	-.0636	-.0501	-.0307	-.0248	-.0518	-.0317	-.0213	-.0282	-.0238
Z(U)1	.781	.644	.669	.699	.486	.563	.547	.380	.327
W(U)1	.959	1.17	1.67	2.18	.983	1.41	2.06	1.28	1.36
N(W /DTH)									
A(W)	-.195E-5	-.173E-5	-.174E-5	-.175E-5	-.173E-5	-.173E-5	-.174E-5	-.173E-5	-.173E-5
1/T(W)1	-.211	-23.6	-36.7	-.0351	-30.9	-47.6	-.00478	-58.3	-.0138
1/T(W)2	-.252	(-.168)	(-.744)	-.117	(.292)	(-.115)	-.142	(-.944)	.628
1/T(W)3	-10.5	(.111)	(.0875)	-53.9	(.0798)	(.0659)	-65.4	(.0559)	-73.9
N(TH/DTH)									
A(TH)	.146E-6	.146E-6	.142E-6	.141E-6	.144E-6	.142E-6	.141E-6	.142E-6	.142E-6
1/T(TH)1	(.853)	(.867)	.145	.0288	(.826)	.163	-.117	.0168	.137
1/T(TH)2	(.402)	(.400)	.717	1.20	(.316)	.519	.964	.353	.906
N(HD /DTH)									
A(HD)	.456E-5	.796E-5	.311E-5	.132E-5	.540E-5	.362E-5	.183E-5	.473E-5	.592E-5
1/T(HD)1	.135	.169	.0962	.0228	.155	.109	-.113	.00322	.284
Z(HD)1	.713	.449	.345	.235	.353	.275	.112	.228	-.116
W(HD)1	2.73	2.26	4.84	10.2	1.90	4.21	7.98	3.39	3.07
N(AZP/DTH)									
A(AZP)	-.139E-4	-.137E-4	-.134E-4	-.132E-4	-.135E-4	-.133E-4	-.133E-4	-.133E-4	-.133E-4
1/T(AZP)1	-.00740	-.0130	-.00191	.000443	-.0131	-.00209	-.674E-4	-.00405	-.00308
1/T(AZP)2	.146	.216	.103	.0220	.206	.117	-.114	.00828	.334
Z(AZP)1	.486	.307	.267	.249	.242	.212	.180	.163	.0118
W(AZP)1	1.54	1.57	2.28	3.25	1.42	2.14	2.96	1.93	1.88

+ + + + + + + + + +

(BODY AXIS SYSTEM)

[illegible]

TABLE X-10

C-5A AILERON TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K
M	.221	.300	.450	.650	.400	.600	.800	.770	.875
DENOMINATOR									
1/T(DET)1	.0283	.0162	.0161	.0139	.00800	.0103	.00788	.00520	-.0264
1/T(DET)2	1.13	1.04	1.44	1.96	.766	1.07	1.51	.793	.582
Z(DET)1	.226	.184	.209	.227	.103	.138	.144	.0564	.197
w(DET)1	.530	.608	.875	1.25	.549	.771	1.03	.618	.605
NUMERATORS									
N(B /DA)									
A(B)	-.000443	-.947E-4	-.000142	-.000205	-.625E-4	-.937E-4	-.000125	-.522E-4	-.593E-4
1/T(B)1	.203	.0473	.292	.0455	.0184	.145	-.0595	.0389	.0932
1/T(B)2	-2.75	1.78	-1.09	1.48	1.11	-1.95	.917	2.72	-1.64
1/T(B)3	72.7	-564.	253.	844.	-1046.	191.	676.	-232.	183.
N(P /DA)									
A(P)	.461	.321	.516	.446	.284	.434	.370	.298	.357
1/T(P)1	-.00541	-.0105	-.00167	.000367	-.0106	-.00190	-.655E-4	-.00257	-.00318
Z(P)1	.422	.382	.284	.256	.349	.222	.194	.165	.163
w(P)1	.456	.368	.877	1.62	.238	.749	1.22	.515	.596
N(R /DA)									
A(R)	.0522	-.0126	.0500	.165	-.0212	.0343	.0850	.00618	.0414
1/T(R)1	.505	-.224	.796	1.72	-.133	.574	1.20	.327	.333
Z(R)1	-.560	(.258)	-.295	.0953	(.164)	-.413	-.0352	(-.612)	-.215
w(R)1	.645	(4.88)	.771	.410	(2.35)	.782	.448	(-2.69)	.584
N(PHI/DA)									
A(PHI)	.464	.320	.518	.444	.281	.435	.370	.298	.360
Z(PHI)1	.415	.340	.284	.254	.276	.221	.194	.159	.160
w(PHI)1	.452	.364	.875	1.62	.235	.748	1.22	.515	.595
N(AYP/DA)									
A(AYP)	7.89	1.54	8.20	17.3	.542	6.26	9.84	2.88	6.22
1/T(AYP)1	.273	.0515	-.334	-.0418	.0191	.178	-.0507	.0453	.110
1/T(AYP)2	-.451	-3.35	.339	1.31	-7.49	-.396	.774	-.882	-.289
Z(AYP)1	.186	.284	.209	.114	.353	.208	.0941	.305	.220
w(AYP)1	.595	.753	.971	1.45	.693	.866	1.22	.728	.698
	+	+	+	+	+	+	+	+	+

TABLE X-11

C-5A RUDDER TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	
H	SL	SL	SL	SL	20 K	20 K	20 K	40 K	40 K	
M	.221	.300	.450	.650	.400	.600	.800	.770	.875	
DENOMINATOR										
1/T(DEF)1	.0283	.0162	.0161	.0139	.00800	.0103	.00788	.00520	-.0264	
1/T(DEF)2	1.13	1.04	1.44	1.96	.766	1.07	1.51	.793	.582	
Z(DEF)1	.226	.184	.209	.227	.103	.138	.144	.0564	.197	
W(DEF)1	.530	.608	.875	1.25	.549	.771	1.03	.618	.605	
NUMERATORS										
N(B /DR)										
A(B)	.0212	.0181	.0271	.0392	.0119	.0179	.0200	.00910	.00760	
1/T(B)1	-.0559	-.0424	-.0120	-.00394	-.0423	-.0123	-.00571	-.0140	-.0162	
1/T(B)2	1.25	1.03	1.42	1.93	.718	1.03	1.47	.733	.644	
1/T(B)3	10.4	16.3	24.1	34.6	20.1	29.7	41.7	36.4	45.6	
N(P /DR)										
A(P)	.105	.0852	.229	.500	.0625	.187	.292	.112	.107	
1/T(P)1	-.00568	-.0117	-.00173	.000377	-.0119	-.00194	-.665E-4	-.00260	-.00336	
Z(P)1	(.719)	(1.19)	(1.70)	(2.42)	(1.17)	(1.55)	(2.16)	(1.44)	-.262	
W(P)1	(-1.78)	(-2.39)	(-2.38)	(-2.94)	(-2.32)	(-2.16)	(-2.78)	(-2.03)	1.18	
N(R /DR)										
A(R)	-.213	-.282	-.639	-1.34	-.231	-.522	-.830	-.324	-.338	
1/T(R)1	1.20	1.02	1.43	1.95	.694	1.04	1.49	.733	-.141	
1/T(R)2	(.0541)	(.201)	(.211)	(.293)	(.190)	(.151)	(.192)	(.117)	.181	
1/T(R)3	(.251)	(.276)	(.251)	(.243)	(.282)	(.242)	(.232)	(.242)	.676	
N(PHI/DR)										
A(PHI)	.0949	.0490	.212	.511	.0259	.167	.290	.0924	.0783	
Z(PHI)1	(.704)	(1.21)	(1.70)	(2.42)	(1.29)	(1.58)	(2.16)	(1.49)	-.404	
W(PHI)1	(-2.01)	(-4.16)	(-2.57)	(-2.88)	(-5.22)	(-2.39)	(-2.79)	(-2.39)	1.38	
N(AYP/DR)										
A(AYP)	-11.3	-16.3	-36.7	-77.2	-13.4	-30.0	-48.9	-18.8	-20.3	
1/T(AYP)1	-.0688	-.0481	-.0180	-.00808	-.0442	-.0163	-.00822	-.0162	-.0156	
1/T(AYP)2	1.32	.599	1.39	1.89	.645	.981	1.42	.663	.721	
Z(AYP)1	.0988	.170	.0992	.0816	.180	.0991	.0887	.111	.0362	
W(AYP)1	.577	.770	1.09	1.58	.745	1.04	1.38	.924	.871	
	+	+	+	+	+	+	+	+	+	

C-5A DATA SOURCES

C-5 Flight Control Report (Aerospace Vehicle) Stability and Control,
Lockheed-Georgia Rept. No. LG1US42-1-1, 8 Feb. 1966

SECTION XI

XB-70A

XB-70A BACKGROUND

The XB-70A was originally designed as a weapons systems with long range supersonic cruise capabilities. The two aircraft built became research aircraft to explore SST-related problems.

The two XB-70A's were identical except that the first airplane (XB-70A-1) had zero geometric dihedral while the second had 5 deg geometric dihedral. The first airplane is considered here.

Pitch control employs interconnected elevon and canard surfaces except in takeoff and landing where the canard is locked and a fixed canard flap is used. Roll control is obtained through differential action of the elevons. Yaw control is provided by rotation of the vertical stabilizers about a 45 deg hinge line.

The airplane is equipped with stability augmentation in all axes.

Data shown here is a composite of many sources. The object was to use flight test data where possible.

Nominal Configuration

Tips Folded According to Flight Condition

50% Internal Fuel

 $W = 784524 \text{ lb}$ c.g. at $0.218 \bar{c}$, W.L. -7.2

$$\left. \begin{aligned} I_x &= 1.8 \times 10^6 \text{ slug-ft}^2 \\ I_y &= 19.9 \times 10^6 \text{ slug-ft}^2 \\ I_z &= 22.1 \times 10^6 \text{ slug-ft}^2 \\ I_{xz} &= -0.88 \times 10^6 \text{ slug-ft}^2 \end{aligned} \right\} \text{Body Axis}$$

Power Approach Configuration

Tips Extended

19% Internal Fuel

Canard Flaps Down

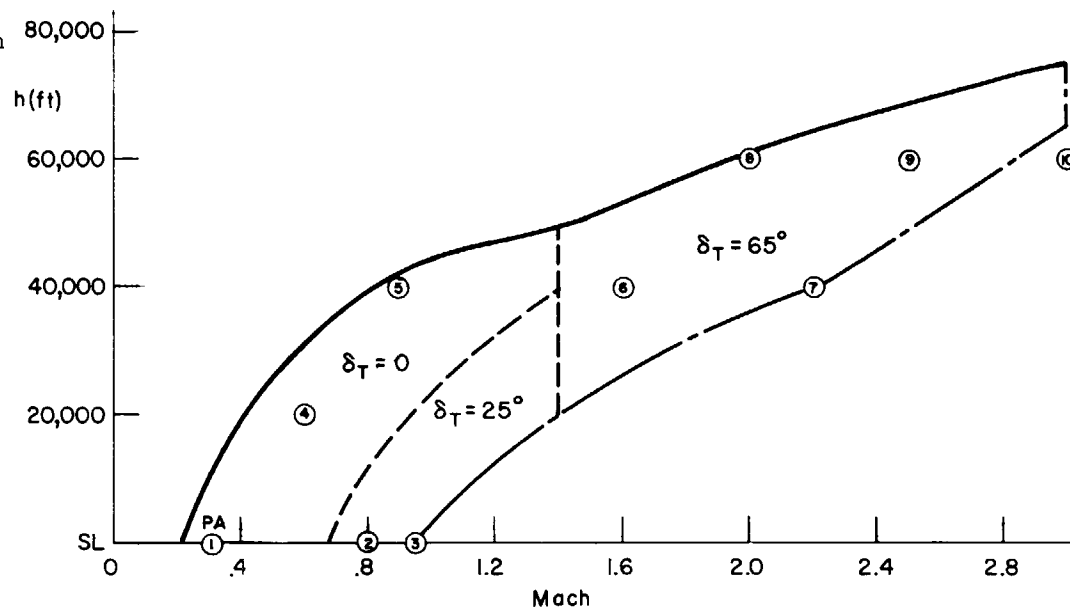
Gear Down

 $W = 300000 \text{ lb}$ c.g. at $0.235 \bar{c}$

$$\left. \begin{aligned} I_x &= 1.45 \times 10^6 \text{ slug-ft}^2 \\ I_y &= 16 \times 10^6 \text{ slug-ft}^2 \\ I_z &= 17.2 \times 10^6 \text{ slug-ft}^2 \\ I_{xz} &= -0.6 \times 10^6 \text{ slug-ft}^2 \end{aligned} \right\} \text{Body Axis}$$

XB-70A

Flight Envelope



— Level Flight Envelope
 - - - Speed Restrictions
 (n) Transfer Function Case

Figure XI-1. XB-70A Flight Conditions

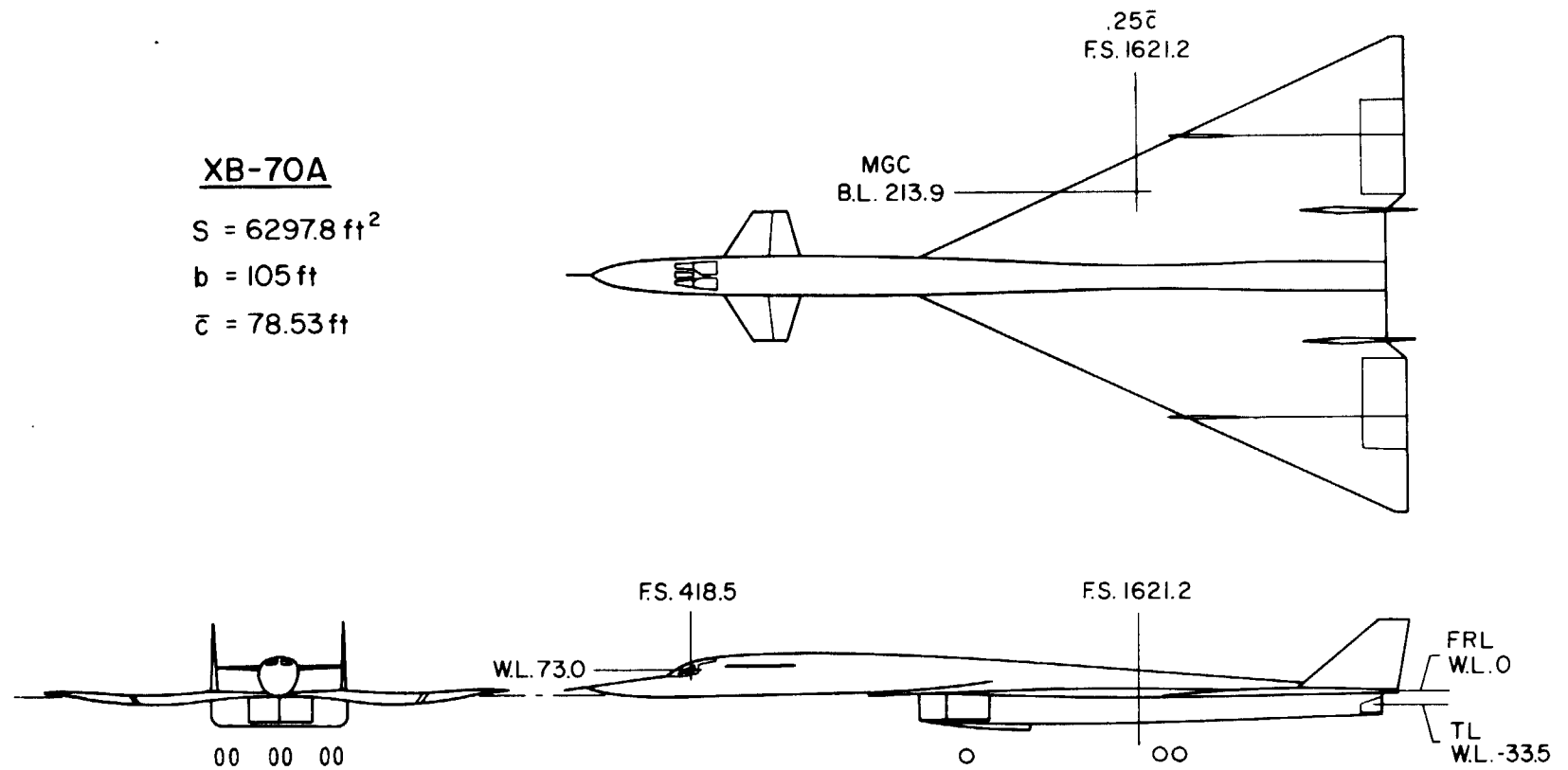
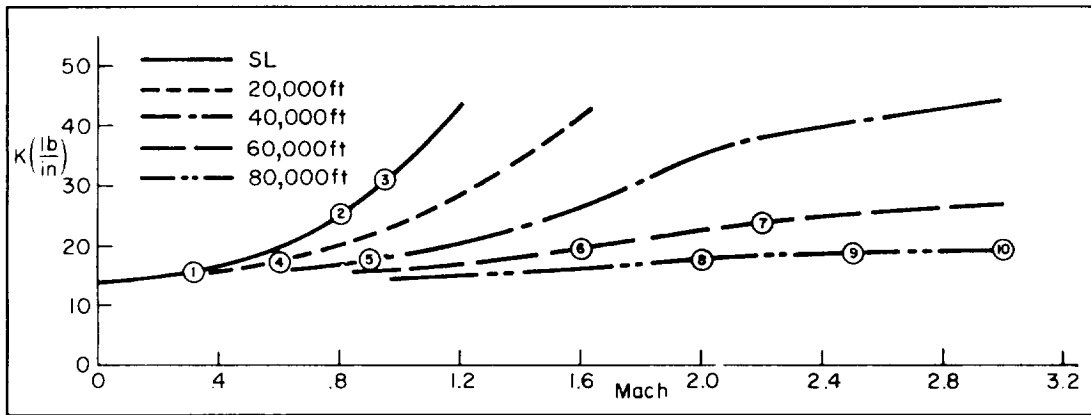
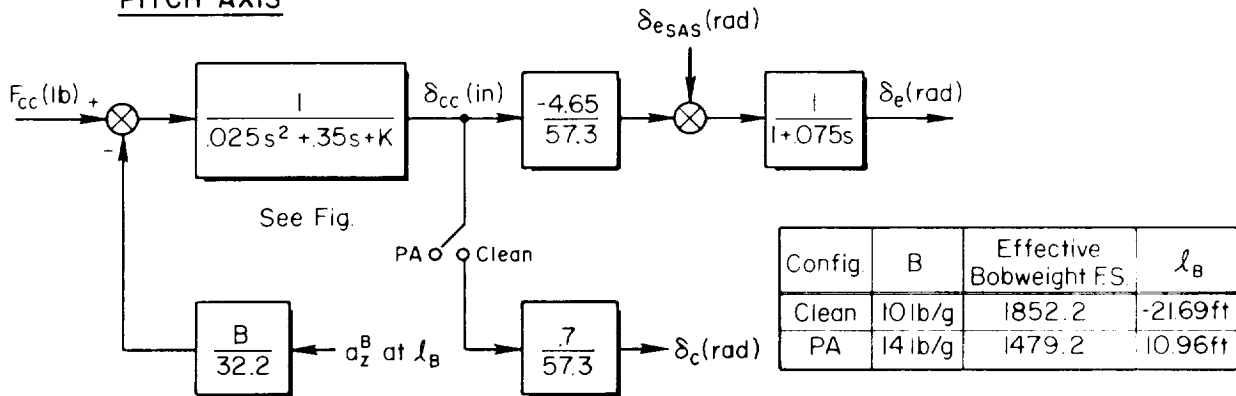


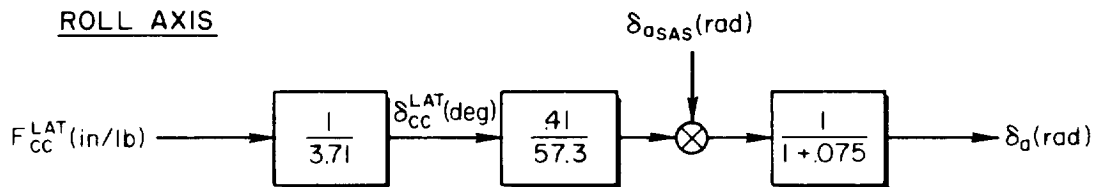
Figure XI-2. XB-70A General Arrangement

XB-70A

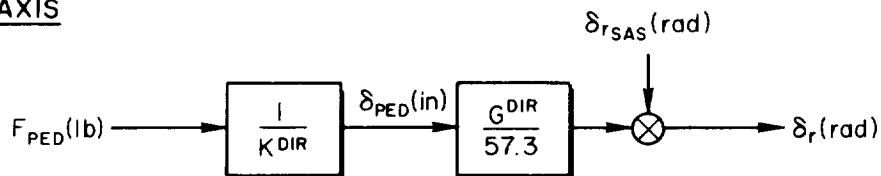
PITCH AXIS



ROLL AXIS



YAW AXIS

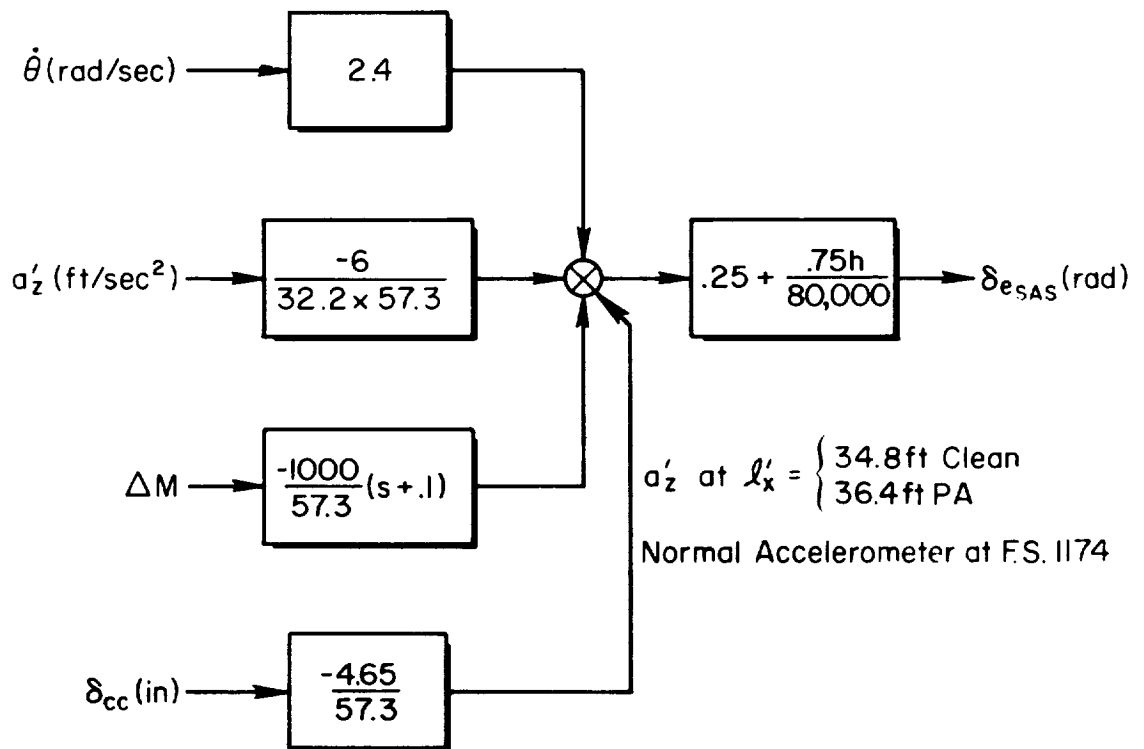


Config.	K^{DIR}	G^{DIR}
Gear UP	28 lb/in	.96 deg/in
Gear DN	31 lb/in	4.0 deg/in

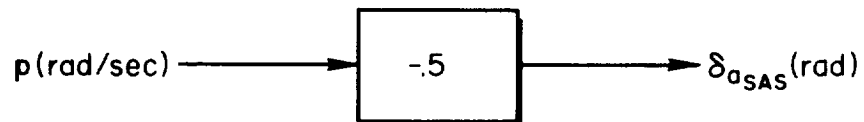
Figure XI-3. XB-70A Control System

XB-70A

PITCH SAS



ROLL SAS



YAW SAS

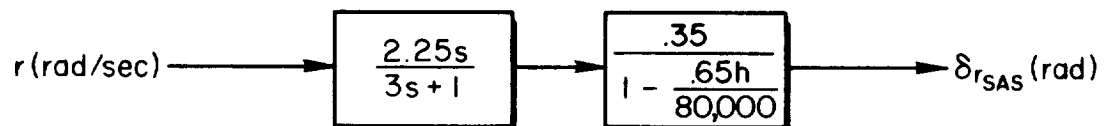


Figure XI-4. XB-70A SAS

TABLE XI-1

XB-70A

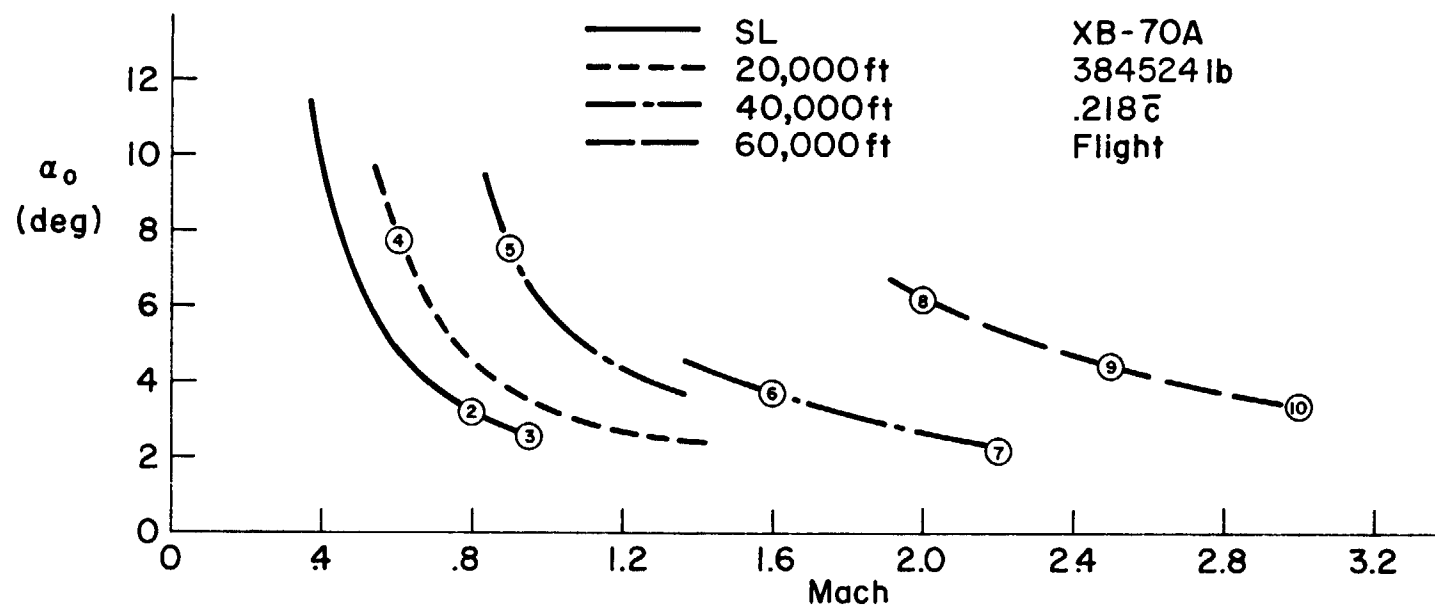
Power Approach Nondimensional Stability Derivatives

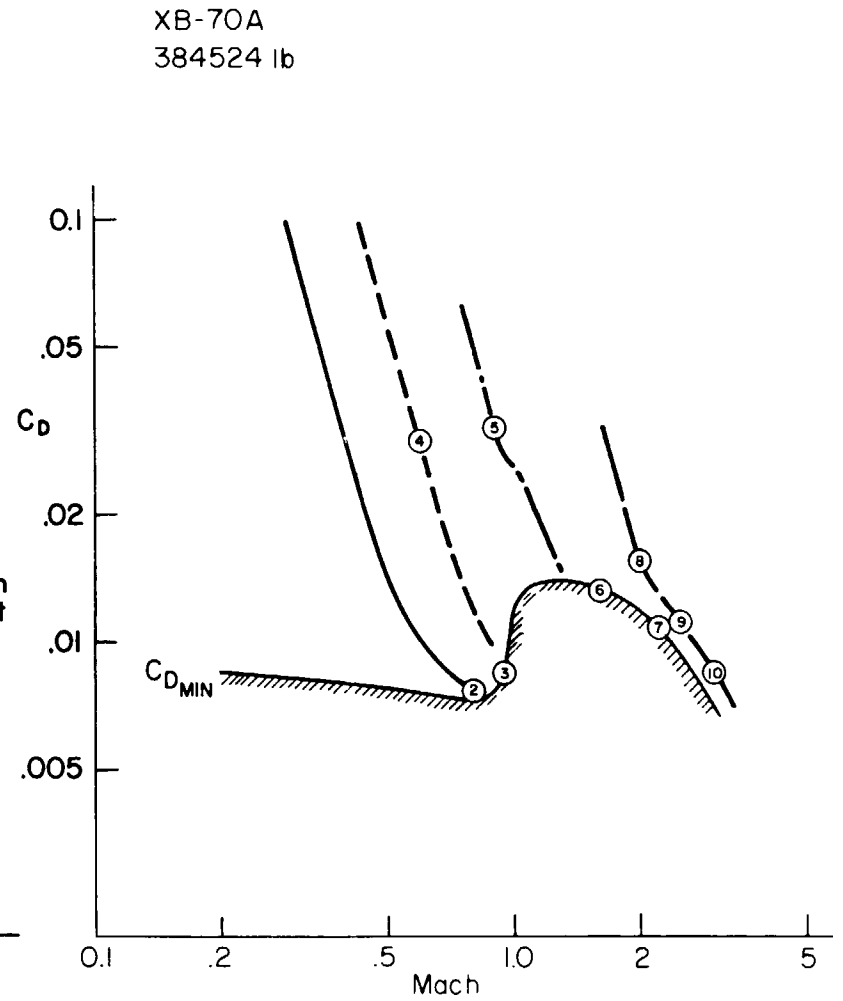
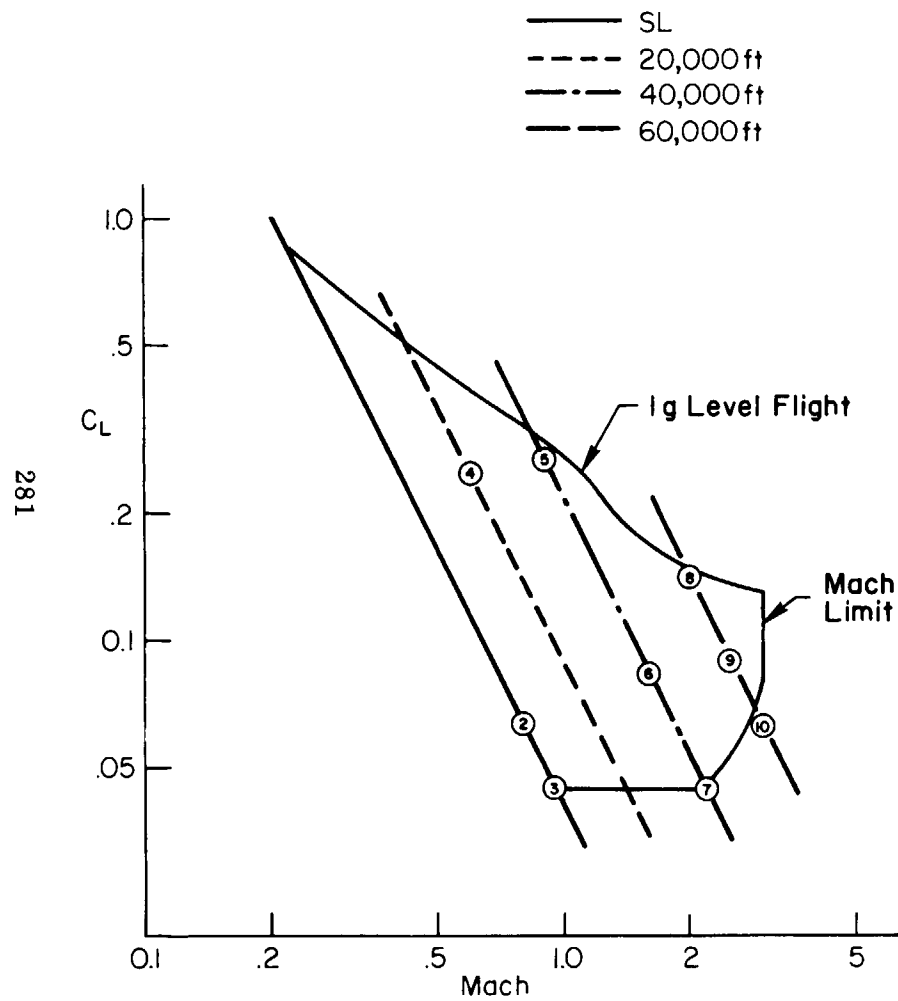
$$h = \text{sea level}$$

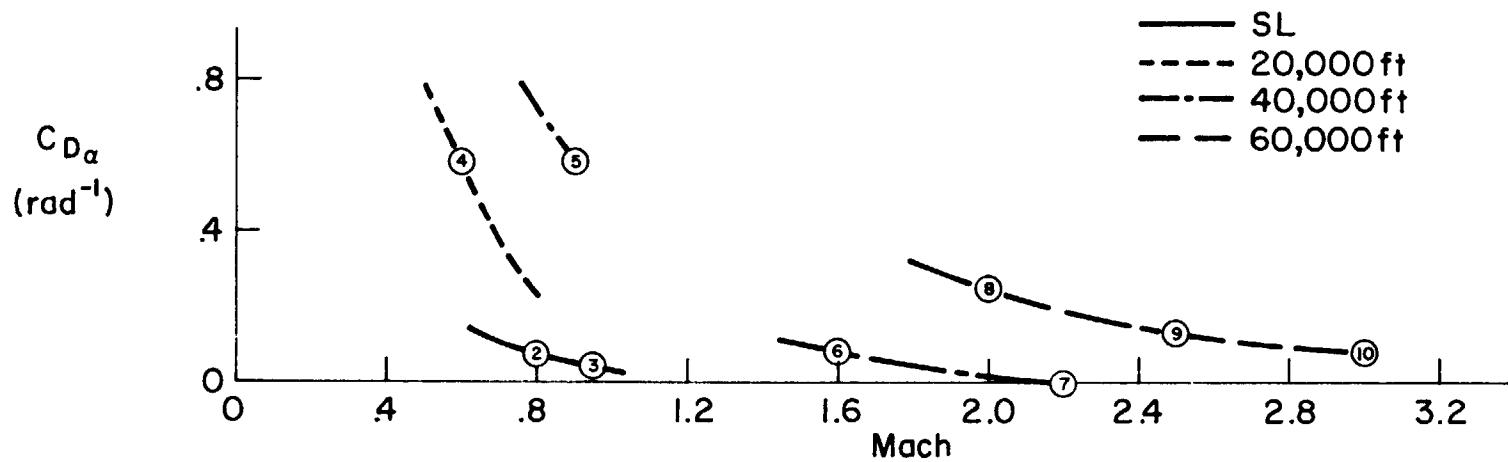
$$V_{T_0} = 347 \text{ ft/sec} = 205 \text{ kt}$$

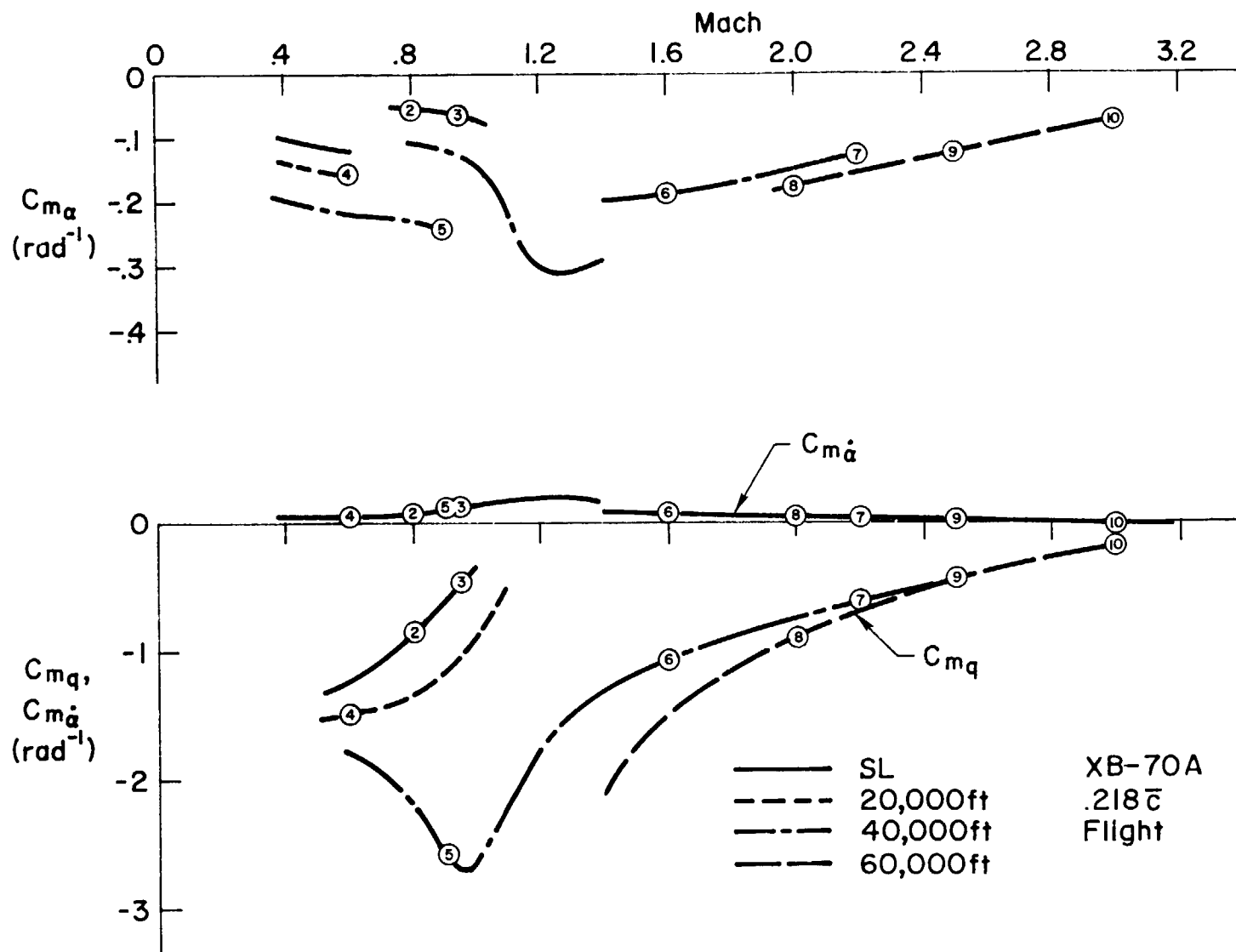
$$\alpha_0 = 7.5 \text{ deg}$$

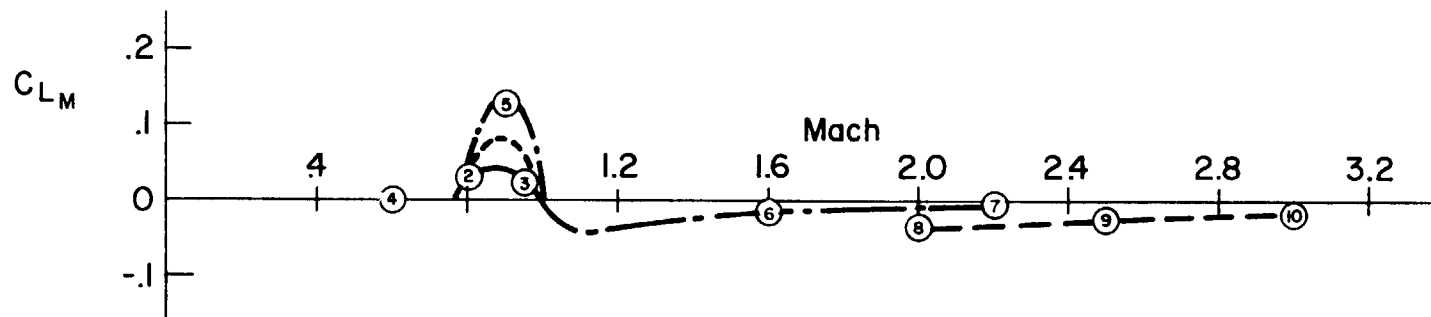
Longitudinal	Lateral-Directional (Body Axis)
$C_L = .333$	$C_{Y_\beta} = -.183/\text{rad}$
$C_D = .055$	$C_{N_\beta} = .132/\text{rad}$
$C_{L_\alpha} = 2.6/\text{rad}$	$C_{\ell_\beta} = -.072/\text{rad}$
$C_{D_\alpha} = .56/\text{rad}$	$C_{\ell_p} = -.18/\text{rad}$
$C_{m_\alpha} = -.23/\text{rad}$	$C_{n_p} = -.26/\text{rad}$
$C_{m_{\dot{\alpha}}} = +.05/\text{rad}$	$C_{\ell_r} = -.03/\text{rad}$
$C_{m_q} = -1.5/\text{rad}$	$C_{n_r} = -.25/\text{rad}$
$C_{I_{\delta_e}} = .46/\text{rad}$	$C_{Y_{\delta_a}} = -.063/\text{rad}$
$C_{m_{\delta_e}} = -.19/\text{rad}$	$C_{\ell_{\delta_a}} = .042/\text{rad}$
	$C_{n_{\delta_a}} = -.0052/\text{rad}$
	$C_{Y_{\delta_r}} = .12/\text{rad}$
	$C_{\ell_{\delta_r}} = -.0018/\text{rad}$
	$C_{n_{\delta_r}} = -.103/\text{rad}$





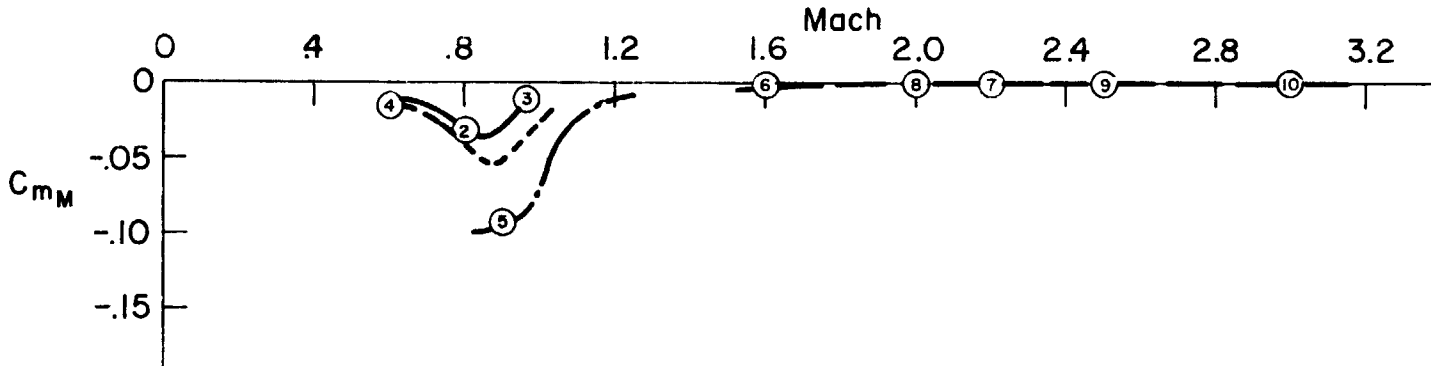
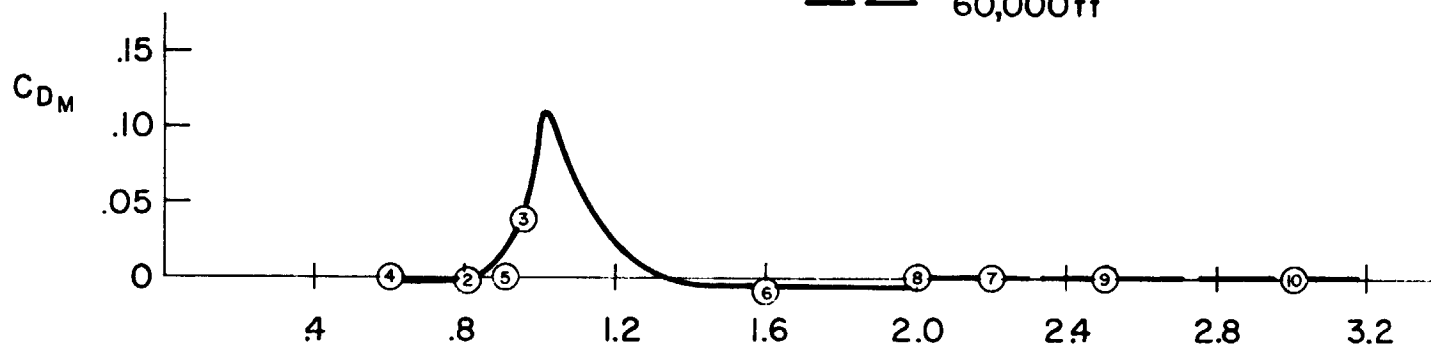


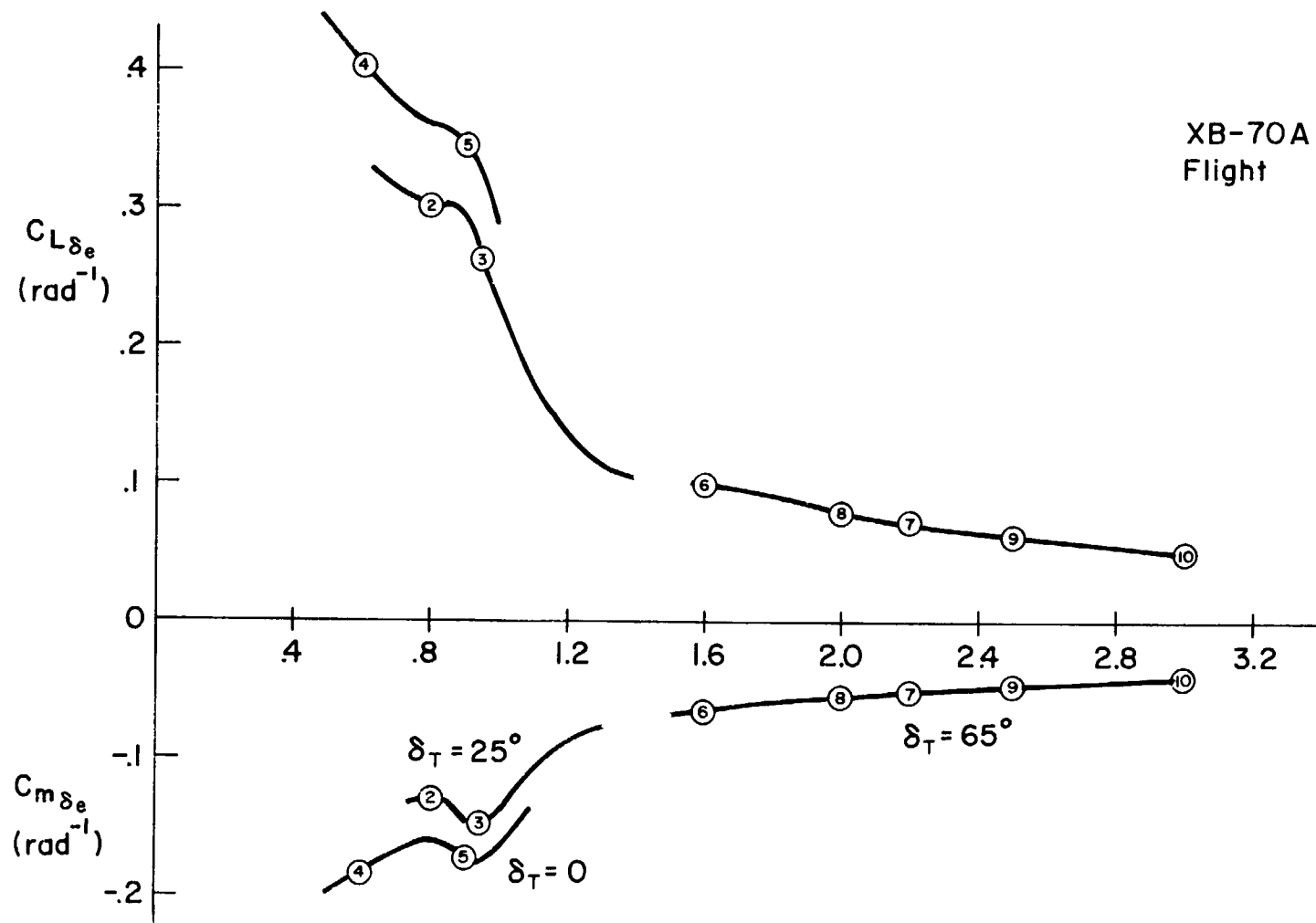


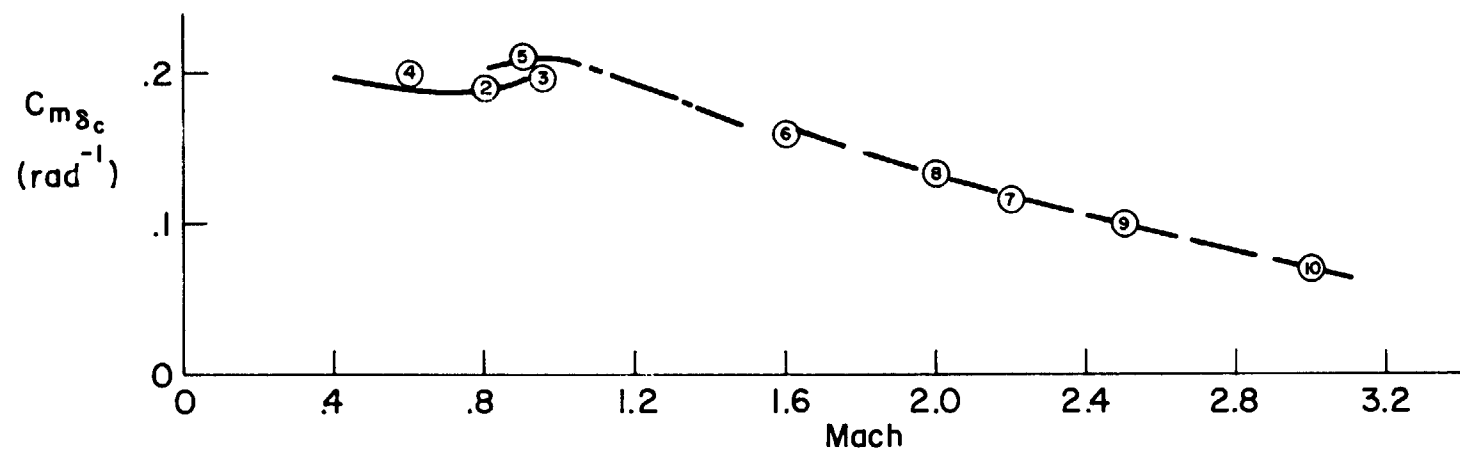
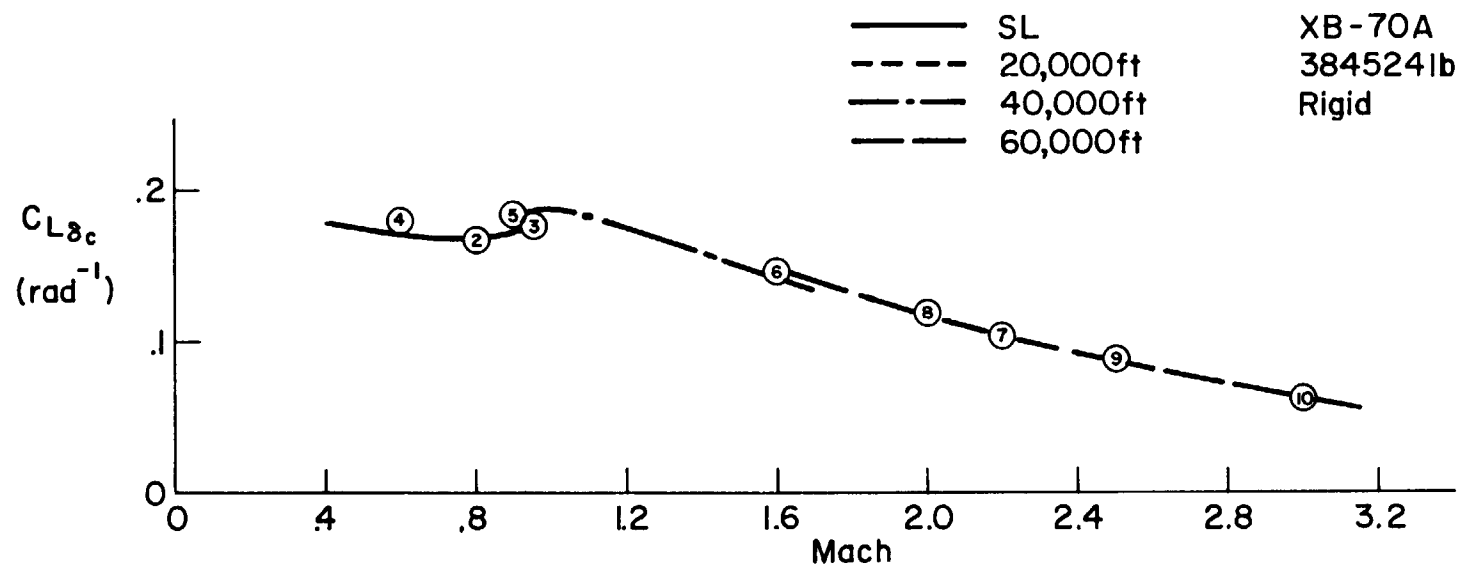


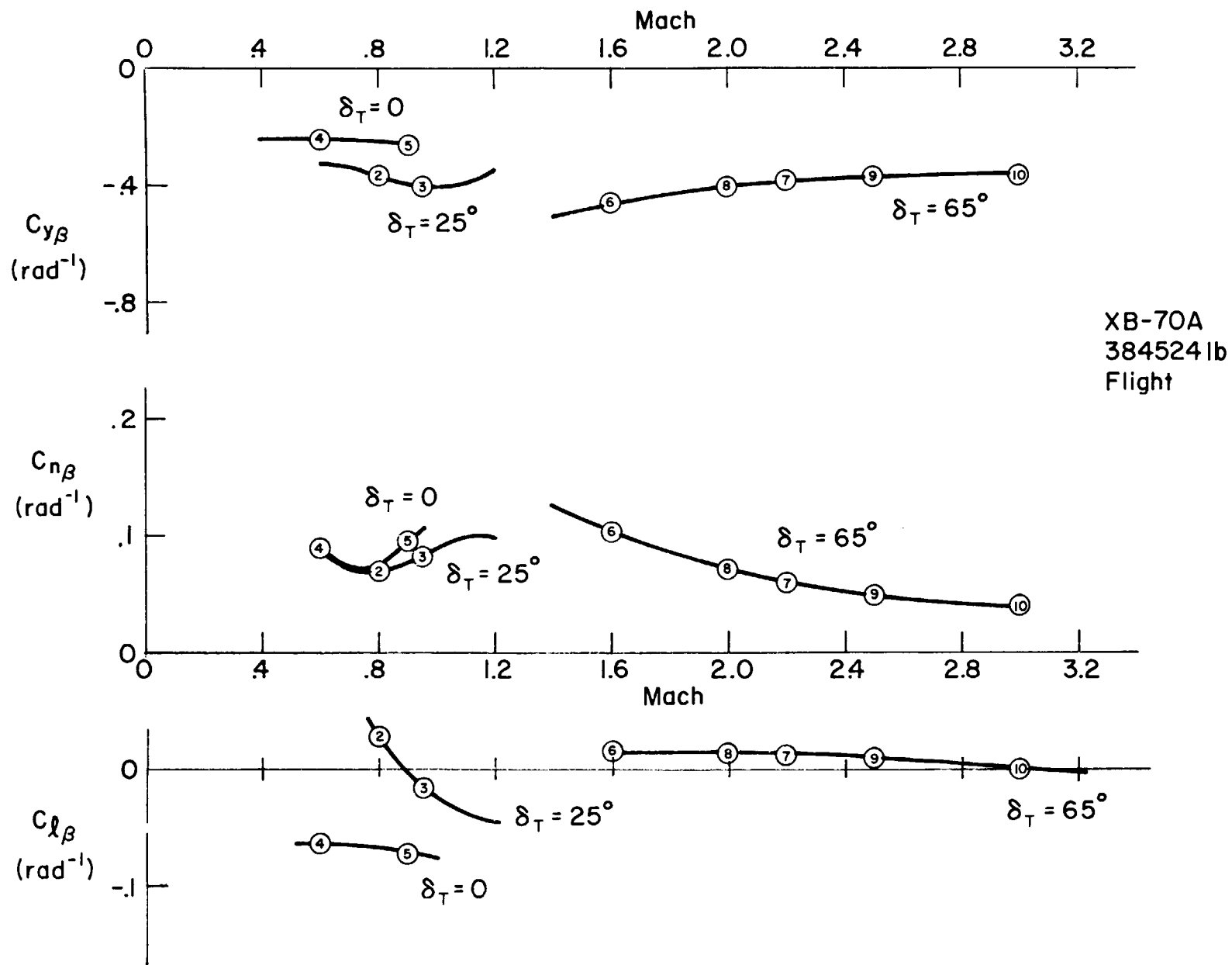
— SL
 --- 20,000 ft
 -.- 40,000 ft
 --- 60,000 ft

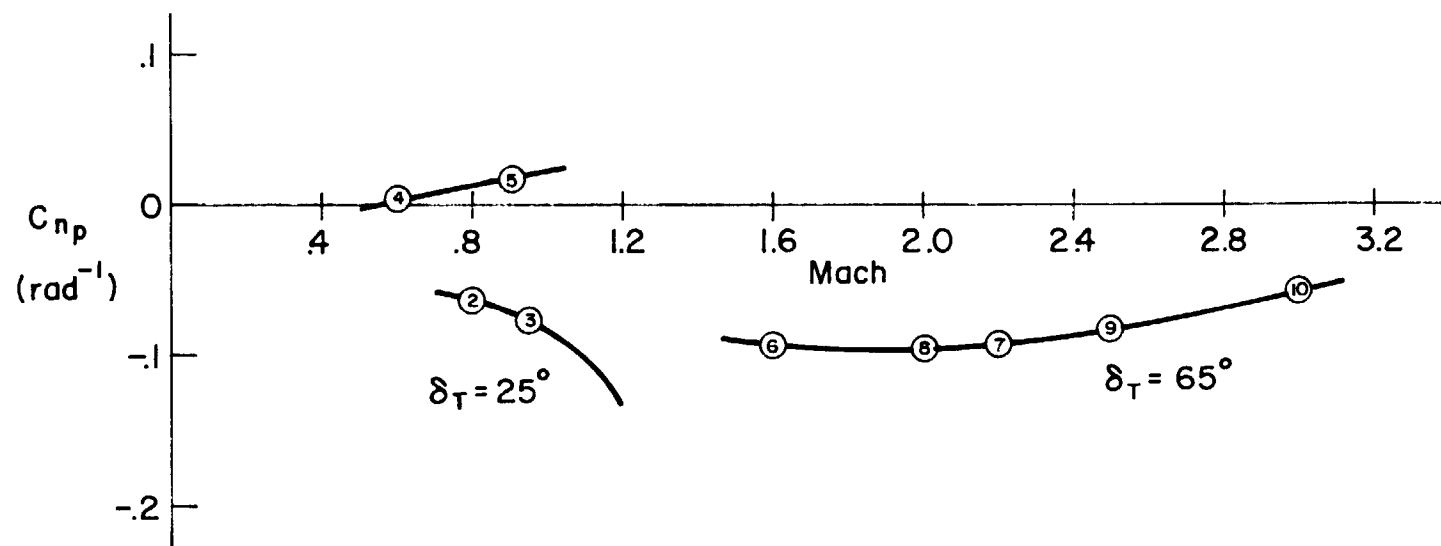
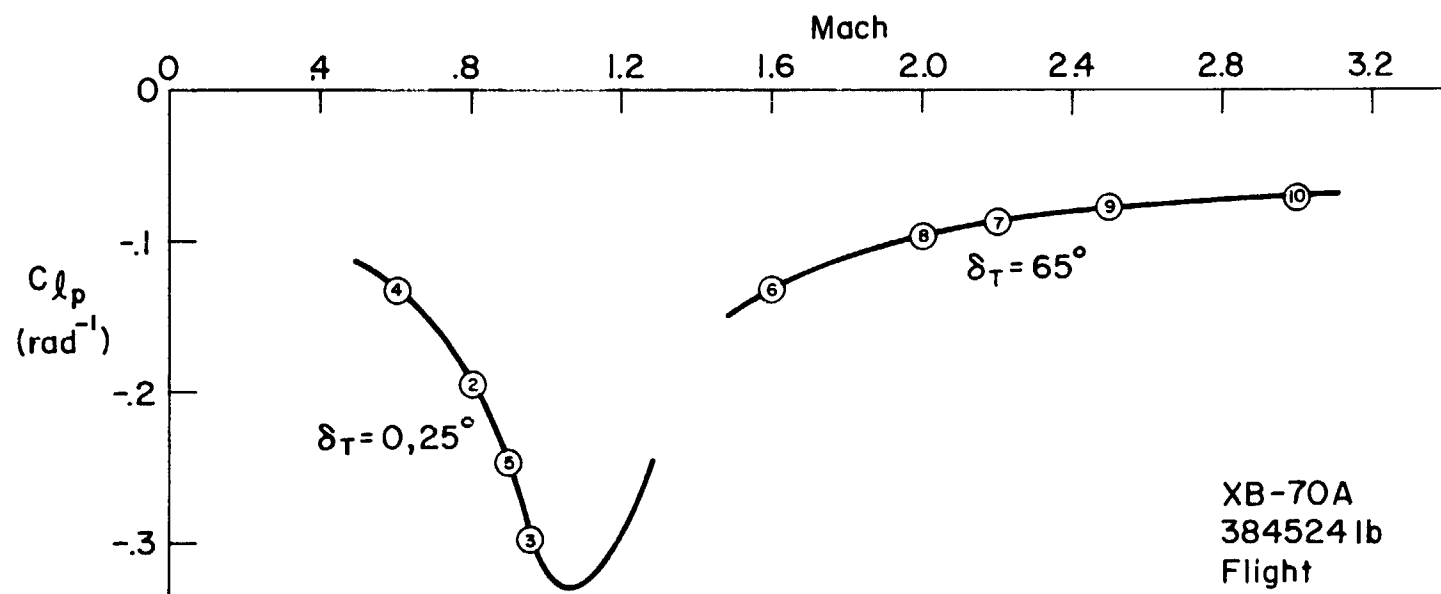
XB-70A
 384524 lb
 Rigid

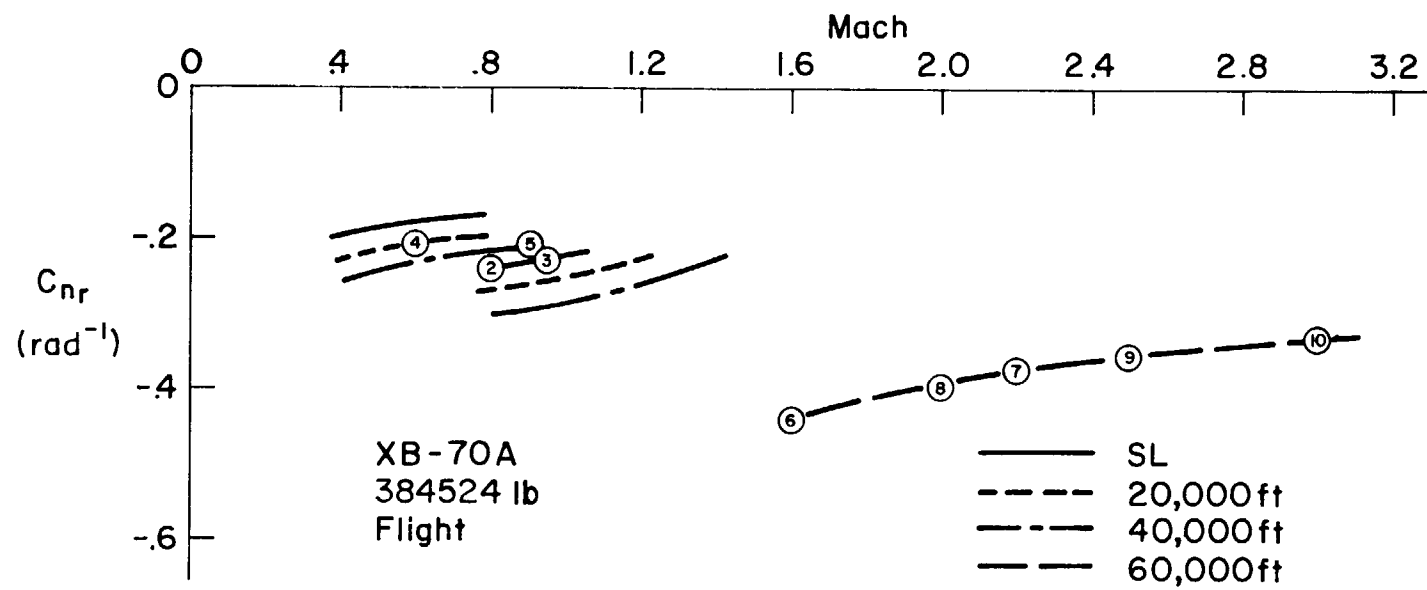
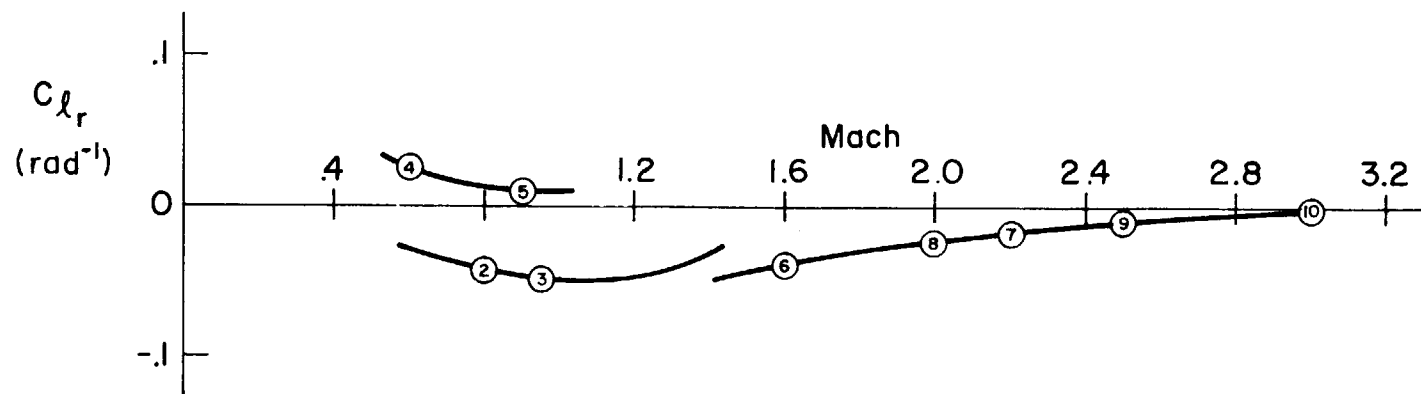


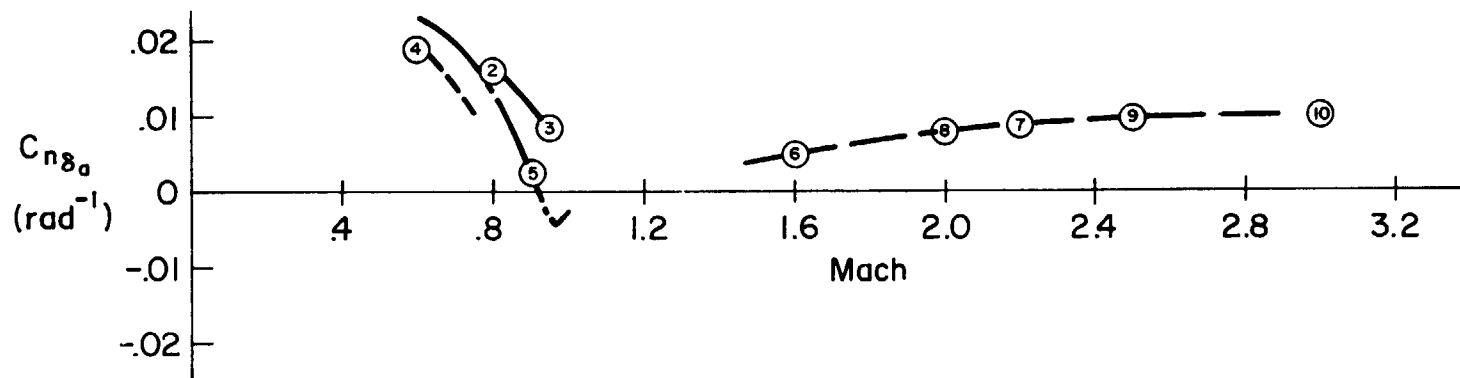
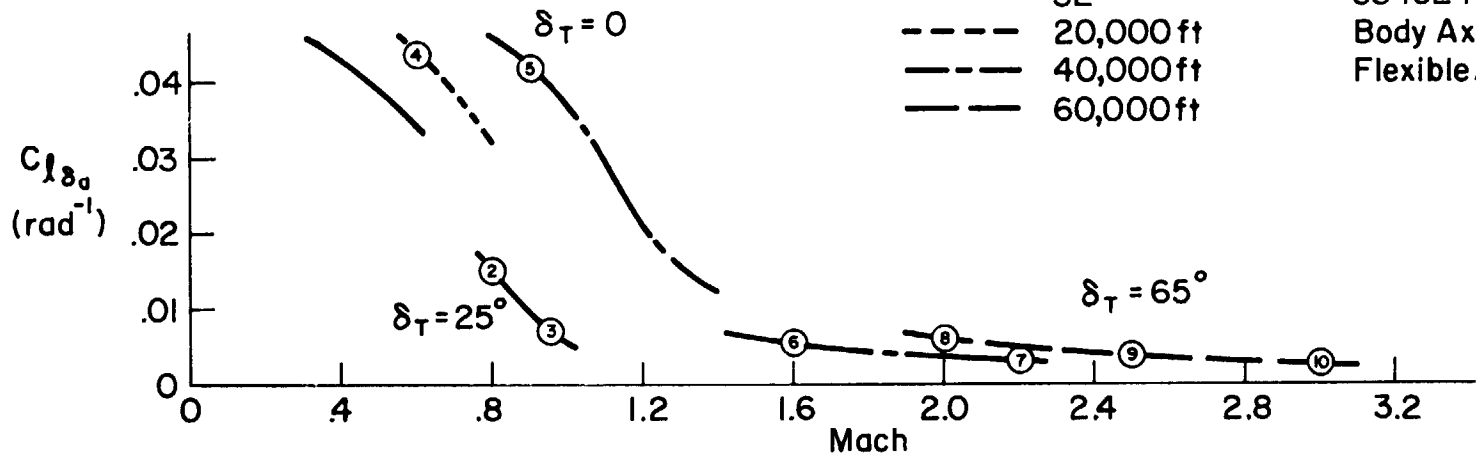
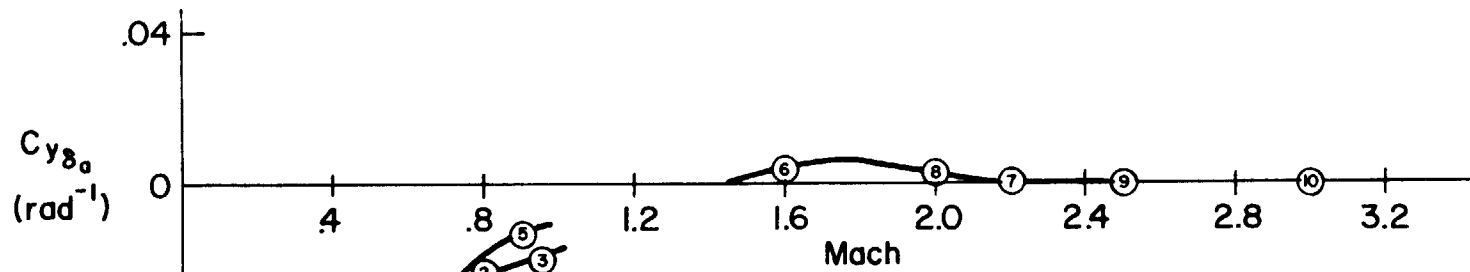












XB-70A
384524 lb
Body Axis
Flexible / Flight

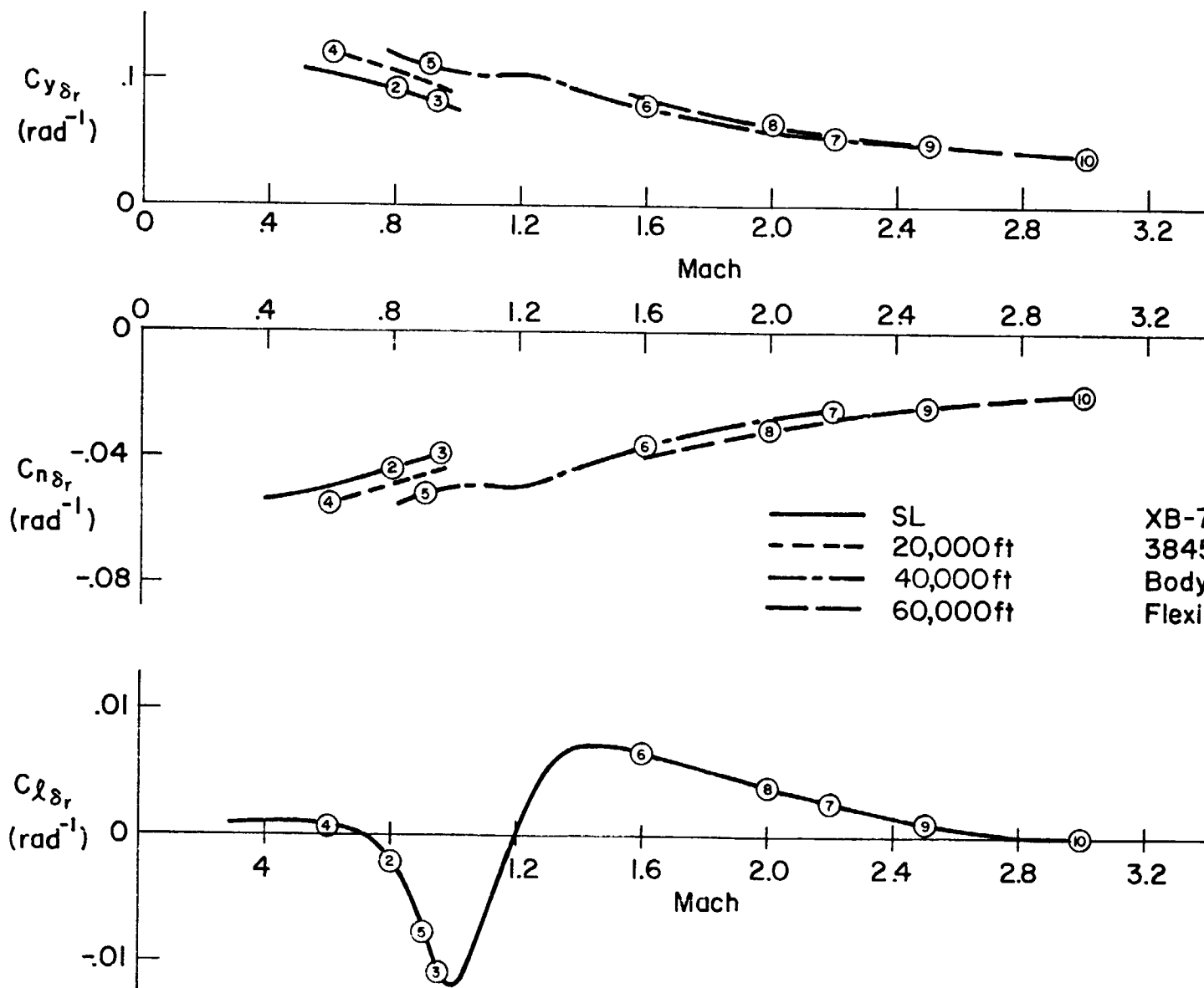


TABLE XI-2
XB-70A DIMENSIONAL, MASS, AND FLIGHT CONDITION PARAMETERS
s = 6297.8 sq ft, b = 105.0 ft, $\bar{c}$ = 78.53 ft

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H(FT)	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M(-)	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00
VTO(FPS)	346.	893.	1060.	622.	871.	1548.	2129.	1936.	2420.	2904.
VTO(KTAS)	205.	529.	628.	369.	516.	918.	1261.	1147.	1433.	1720.
VTO(KCAS)	205.	529.	628.	275.	278.	528.	710.	432.	535.	630.
W(LBS)	300017.	384546.	384546.	384546.	384546.	384546.	384546.	384546.	384546.	384546.
C.G.(MGC)	.235	.218	.218	.218	.218	.218	.218	.218	.218	.218
IX (SLUG-FT SQ)	.145E+7	.180E+7	.180E+7	.180E+7	.180E+7	.180E+7	.180E+7	.180E+7	.180E+7	.180E+7
IY (SLUG-FT SQ)	.160E+8	.100E+8	.100E+8	.100E+8	.100E+8	.100E+8	.100E+8	.100E+8	.100E+8	.100E+8
IZ (SLUG-FT SQ)	.172E+8	.221E+8	.221E+8	.221E+8	.221E+8	.221E+8	.221E+8	.221E+8	.221E+8	.221E+8
IXZ(SLUG-FT SQ)	-600035.	-880050.	-880050.	-880050.	-880050.	-880050.	-880050.	-880050.	-880050.	-880050.
EPSILON(DEG)	2.18	2.48	2.48	2.48	2.48	2.48	2.48	2.48	2.48	2.48
Q(PSF)	142.	948.	1336.	245.	224.	707.	1335.	424.	663.	954.
QC(PSF)	146.	1109.	1666.	268.	273.	1105.	2253.	703.	1139.	1675.
ALPHA(DEG)	7.50	3.20	2.60	7.70	7.50	3.70	2.30	6.20	4.40	3.40
GAMMA(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LXP(FT)	99.0	97.7	97.7	97.7	97.7	97.7	97.7	97.7	97.7	97.7
LZP(FT)	-6.70	-6.70	-6.70	-6.70	-6.70	-6.70	-6.70	-6.70	-6.70	-6.70
ITH(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
XI(DEG)	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
LTH(FT)	2.20	2.20	2.20	2.20	2.20	2.20	2.20	2.20	2.20	2.20
	+	+	+	+	+	+	+	+	+	+

TABLE XI-3

XB-70A LONGITUDINAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00
XU *	-.0105	-.00514	-.0352	.000472	.00212	-.00221	-.00780	-.00166	-.00267	-.00285
ZU *	-.0893	-.0188	-.00588	-.0271	-.0399	-.00543	-.00141	-.00494	-.00149	.00135
MU *	.000343	-.00113	-.000452	.000199	-.000644	.796E-4	.000143	.000152	.486E-4	-.376E-4
XW	.0327	.0629	.0698	.00819	.00700	.0282	.0349	.00776	.00969	.00983
ZW	-.757	-1.19	-1.50	-.580	-.380	-.424	-.515	-.192	-.204	-.218
MW	-.00290	-.00285	-.00376	-.00302	-.00316	-.00429	-.00403	-.00189	-.00163	-.00114
ZWD	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ZQ	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MWD	.715E-4	.000161	.000277	.610E-4	.567E-4	.285E-4	.114E-4	.655E-5	0.	0.
MQ	-.749	-1.75	-1.10	-1.13	-1.30	-.930	-.731	-.383	-.213	-.128
XDE	5.77	7.67	7.61	6.55	4.87	1.87	1.53	1.50	1.26	1.22
ZDE	-43.8	-137.	-168.	-48.4	-37.0	-29.0	-38.0	-13.8	-16.4	-20.6
MDE	-.836	-7.46	-11.9	-2.61	-2.24	-3.11	-4.62	-1.61	-2.06	-2.45
XDES	5.77	8.37	8.44	7.01	5.30	2.40	1.98	1.93	1.61	1.49
ZDES	-43.8	-150.	-186.	-51.9	-40.3	-37.2	-49.2	-17.8	-20.9	-25.1
MDES	-.836	-6.10	-9.92	-2.24	-1.90	-2.27	-3.50	-1.20	-1.57	-1.98
XDTH	.000107	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4
ZDTH	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
MDTH	.138E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6

TABLE XI-4

XB-70A ELEVATOR TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	60 K	80 K	100 K	120 K	140 K
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	2.00
DENOMINATOR										
1/T(DEL)1	(.0652)	-.0942	-.0470	(.0136)	-.0278	(.0494)	(.217)	(.0743)	(.118)	-.00101
1/T(DEL)2	(.103)	.0960	.0806	(.0499)	-.0430	(.0161)	(.0157)	(.0121)	(.0096)	.0107
Z(DEL)1	.591	.654	.487	.526	.460	.247	.204	.145	.106	.0950
W(DEL)1	1.25	2.14	2.37	1.59	1.78	2.65	2.99	1.93	1.59	1.82
NUMERATORS										
U(U /DE)										
A(U)	5.77	7.67	7.61	6.55	4.87	1.87	1.53	1.50	1.26	1.22
1/T(U)1	6.60	49.6	75.7	.386	.280	165.	259.	226.	205.	346.
Z(U)1	.922	.404	.314	(.508)	(.322)	.437	.0234	.832	.689	.605
W(U)1	.652	.848	.982	(34.0)	(53.4)	.352	.426	.164	.192	.107
W(U /DE)										
A(W)	-43.8	-137.	-168.	-48.4	-37.0	-29.0	-38.0	-13.8	-16.4	-20.6
1/T(W)1	7.30	-.00705	-.00175	34.4	53.7	167.	260.	226.	205.	-.00255
1/T(W)2	(.0531)	.0113	.0358	(-.0460)	(-.0251)	(.0548)	(.620)	(.0196)	(.187)	.00560
1/T(W)3	(.0241)	50.2	76.4	(.0394)	(.0326)	(.0112)	(.00584)	(.0100)	(.00472)	346.
W(THE/DE)										
A(THE)	-830	-7.48	-11.9	-2.61	-2.25	-3.11	-4.52	-1.42	-2.06	-2.65
1/T(THE)1	.0104	.00612	.0354	-.000936	-.000709	.00258	.00793	.00174	.00273	.00270
1/T(THE)2	.601	1.14	1.44	.523	.328	.384	.482	.175	.191	.208
W(HD /DE)										
A(HD)	44.2	137.	168.	48.9	37.3	29.0	38.0	13.9	16.4	20.6
1/T(HD)1	-.0166	.00425	.0341	-.0111	-.00902	.000906	.00724	-.002634	.00160	.00223
1/T(HD)2	-1.59	-6.47	-10.0	-3.59	-3.50	-7.55	-10.6	-5.08	-7.51	-8.42
1/T(HD)3	2.37	8.28	10.9	4.71	4.76	8.43	11.5	6.45	7.73	8.55
W(AZP/DE)										
A(AZP)	39.2	593.	000.	207.	182.	275.	414.	144.	185.	210.
1/T(AZP)1	.00537	-.00200	-.00138	-.002633	-.000407	-.00147	-.000513	.00147	-.00103	-.000640
1/T(AZP)2	-.0242	.00624	.0354	-.0104	-.00857	.00236	.00735	-.00214	.00264	.00287
Z(AZP)1	.115	.144	.181	.0991	.0401	.0642	.0682	.0410	.0420	.0416
W(AZP)1	2.08	3.57	4.27	2.01	1.86	2.59	3.39	1.94	2.27	2.60

TABLE XI-5

XB-70A THRUST TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Open

(BODY AXIS SYSTEM)

295

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
DENOMINATOR											
1/T(DET)1	(.0652)	-.0942	-.0470	(.0136)	.0278	(.0494)	(.217)	(.0743)	(.118)	-.00901	
1/T(DET)2	(.103)	.0760	.0806	(.0495)	-.0430	(.0161)	(.0157)	(.0181)	(.00886)	.0107	
Z(DET)1	.591	.654	.487	.526	.460	.747	.204	.145	.105	.0050	
W(DET)1	1.25	2.14	2.37	1.59	1.73	2.55	2.90	1.93	1.99	1.83	
NUMERATORS											
W(U) /DTH)											
A(U)	.000107	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	
1/T(U)1	-.0283	-.0231	-.0236	-.0262	-.0149	-.00546	-.00545	-.00540	-.00554	-.00540	
Z(U)1	.586	.623	.465	.482	.384	.200	.168	.00494	-.00168	-.00201	
W(U)1	1.24	2.14	2.37	1.54	1.75	2.54	2.99	1.89	1.86	1.80	
W(W) /DTH)											
A(W)	.372E-4	.000194	.000231	.000133	.000186	.000339	.000458	.000423	.000521	.000628	
1/T(W)1	.0689	-.00352	-.00267	(.902)	-.0103	.00248	-.000257	-.000435	-.00054	.000500	
1/T(W)2	.0901	-.439	-.138	(.0299)	-.265	.0279	.0615	.0584	.0203	-.0129	
W(W) /DTH)											
A(W)	.136E-6	.219E-6	.219E-6	.219E-6	.219E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6	
1/T(W)1	(.798)	-.411	-.132	.149	-.158	.0571	.0670	.0977	.0264	-.0140	
1/T(W)2	(.647)	1.18	1.49	.506	.290	.399	.510	.154	.135	.221	
W(W) /DTH)											
A(W)	.140E-4	.467E-5	.380E-5	.112E-4	.109E-4	.540E-5	.336E-5	.304E-5	.642E-5	.496E-5	
1/T(W)1	.344	-.364	-.125	.0388	-.0880	.0411	.0584	.0541	.0189	-.0133	
Z(W)1	.459	.233	.130	.290	.316	.116	.0655	.0777	.0456	.0200	
W(W)1	1.98	7.50	9.89	3.08	3.19	5.81	8.99	3.56	4.57	5.40	
W(AZP) /DTH)											
A(AZP)	-.135E-4	-.214E-4	-.214E-4	-.214E-4	-.214E-4	-.215E-4	-.215E-4	-.215E-4	-.215E-4	-.215E-4	
1/T(AZP)1	-.0124	-.00201	-.00138	-.00711	-.00471	-.00135	-.000508	-.00182	-.00103	-.000662	
1/T(AZP)2	.492	-.402	-.131	.125	-.119	.0534	.0653	.0779	.0257	-.0141	
Z(AZP)1	.369	.187	.189	.166	.110	.0821	.0772	.0500	.0460	.0425	
W(AZP)1	1.71	3.32	4.04	1.94	1.97	2.50	3.35	1.94	2.25	2.54	

TABLE XI-6

XB-70A STICK FORCE TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Closed)

(BODY AXIS SYSTEM)

296

F/C #	1	2	3	4	5	6	7	8	9	10
H M	SL .310	SL .800	SL .950	20 K .600	40 K .900	40 K 1.60	40 K 2.20	60 K 2.00	60 K 2.50	60 K 3.00
DEACMINATOR										
1/T(DET)1	14.5	-1.0579	-.0201	11.2	.0243	11.8	11.9	12.4	12.3	-.0068
1/T(DET)2	(.0474)	.0596	.0536	(-.0275)	-.0383	(.0516)	(.272)	(.0644)	(.126)	.0008
1/T(DET)3	(.0910)	10.0	10.5	(.0409)	11.6	(.0139)	(.0134)	(.0165)	(.0085)	12.0
Z(DET)1	.514	.229	.0150	.383	.373	.157	.0762	.107	.0551	.0241
W(DET)1	1.34	3.97	5.05	2.12	2.15	3.25	3.05	2.20	2.30	2.60
Z(DET)2	.255	.284	.267	.310	.296	.282	.259	.201	.274	.274
W(DET)2	25.3	32.2	35.6	26.2	26.6	28.0	31.1	26.6	27.6	29.0
NUMERATORS										
W(U) /FST)										
A(U)	-250.	-337.	-329.	-283.	-211.	-81.1	-66.0	-64.8	-54.5	-52.0
1/T(U) 1)	6.60	49.6	75.7	.386	.280	165.	259.	226.	205.	346.
Z(U) 1)	.422	.404	.314	(.508)	(.322)	.437	.0234	.832	.689	.605
W(U) 1)	.652	.848	.982	(.34.0)	(.53.4)	.352	.425	.164	.152	.197
W(W) /FST)										
A(W)	1897.	5937.	7250.	2095.	1602.	1254.	1543.	596.	709.	291.
1/T(W) 1)	7.30	-.00705	-.00175	34.4	53.7	167.	260.	226.	205.	-.00250
1/T(W) 2)	(.0531)	.0113	.0358	(-.0460)	(-.0251)	(.0548)	(.620)	(.0196)	(.187)	.00069
1/T(W) 3)	(.0941)	50.2	76.4	(.0394)	(.0326)	(.0112)	(.00584)	(.0100)	(.00472)	346.
W(TH) /FST)										
A(TH)	36.3	324.	517.	113.	97.2	125.	200.	69.9	89.4	106.
1/T(TH) 1)	.0104	.00619	.0354	-.000935	-.000709	.00258	.00733	.00174	.00277	.00276
1/T(TH) 2)	.601	1.14	1.44	.523	.328	.384	.482	.175	.131	.208
W(HD) /FST)										
A(HD)	-1913.	-5946.	-7257.	-2114.	-1616.	-1256.	-1645.	-500.	-710.	-897.
1/T(HD) 1)	-.0186	.00425	.0341	-.0111	-.00902	.000906	.00724	-.000624	.00140	.00223
1/T(HD) 2)	-1.50	-6.67	-10.0	-3.59	-3.50	-7.55	-10.8	-6.08	-7.51	-8.42
1/T(HD) 3)	2.37	8.28	10.9	4.71	4.76	8.43	11.5	6.45	7.72	8.55
W(AZP) /FST)										
A(AZP)	-1627.	-25684.	-43278.	-8945.	-7897.	-11900.	-17913.	-6233.	-8022.	-9484.
1/T(AZP) 1)	.00537	-.00200	-.00130	-.000633	-.000407	-.00147	-.000513	.00147	-.00106	-.000642
1/T(AZP) 2)	-.0242	.00624	.0354	-.0104	-.00857	.00234	.00785	-.00214	.00264	.00086
Z(AZP) 1)	.115	.144	.101	.0291	.0401	.0042	.0692	.0417	.0670	.0616
W(AZP) 1)	2.08	3.57	4.27	2.01	1.36	2.59	3.39	1.04	2.27	2.40

TABLE XI-7

XB-70A THRUST TRANSFER FUNCTION FACTORS

SAS Off — Bobweight Loop Closed

(BODY AXIS SYSTEM)

297

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M	.310	.200	.950	.600	.900	1.60	2.20	2.00	2.50	2.00
DENOMINATOR										
1/T(DEF)1	14.5	-.0579	-.0201	11.2	.0243	11.8	11.9	12.4	12.3	-.00698
1/T(DEF)2	(.0474)	.0596	.0536	(-.0275)	-.0383	(.0516)	(.272)	(.0646)	(.126)	.00884
1/T(DEF)3	(.0919)	10.0	10.5	(.0409)	11.6	(.0139)	(.0124)	(.0165)	(.00855)	12.2
Z(DEF)1	.514	.229	.0150	.393	.373	.157	.0762	.107	.0556	.0241
W(DEF)1	1.34	3.97	5.05	2.12	2.15	3.25	3.95	2.20	2.30	2.30
Z(DEF)2	.255	.284	.267	.310	.296	.282	.259	.281	.274	.274
W(DEF)2	25.3	32.2	35.6	26.2	26.6	28.0	31.1	26.6	27.6	28.0
NUMERATORS										
N(U /DTH)										
A(U)	.000107	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4	.837E-4
1/T(U)1	-.0251	-.0101	-.00750	-.0195	-.0125	-.00515	-.00373	-.00557	-.00442	-.00426
1/T(U)2	14.5	10.1	10.5	11.2	11.7	11.9	11.0	12.5	12.4	12.2
Z(U)1	.508	.212	.00227	.338	.301	.115	.0462	-.0223	-.0522	-.0758
W(U)1	1.33	3.96	5.04	2.07	2.13	3.23	3.94	2.15	2.34	2.36
Z(U)2	.255	.284	.267	.310	.296	.282	.259	.281	.274	.274
W(U)2	25.3	32.2	35.6	26.2	26.6	28.0	31.1	26.6	27.6	28.0
N(W /DTH)										
A(W)	.372E-4	.000194	.000231	.000123	.000186	.000339	.000438	.000423	.000531	.000628
1/T(W)1	18.9	-.00353	-.00264	12.0	-.00963	.00517	-.000236	-.000292	.143E-4	.000780
1/T(W)2	(-.220)	-.500	-.164	(.118)	-.318	.0165	.0588	.0538	.0186	-.0112
1/T(W)3	(.101)	11.0	11.1	(.0362)	12.4	12.7	12.7	13.0	13.0	12.0
Z(W)1	.221	.250	.226	.290	.279	.260	.235	.268	.259	.257
W(W)1	19.5	32.8	36.0	26.6	26.8	28.0	31.1	26.7	27.6	28.0

TABLE XI-7 (Concluded)

N(THE) / DTH)										
A(THE)	.136E-6	.219E-6	.219E-6	.219E-6	.219E-6	.220E-6	.220E-6	.220E-6	.220E-6	.220E-6
1/T(THE)1	12.1	-1.425	-1.134	.165	-1.175	.0594	.0673	.105	.0266	-0.0140
1/T(THE)2	(.705)	1.34	1.75	.490	.276	.406	.520	.146	.202	.228
1/T(THE)3	(.678)	11.3	11.2	12.3	12.6	12.8	12.8	13.1	13.0	13.0
Z(THE)1	.302	.248	.225	.287	.277	.260	.235	.268	.259	.257
W(THE)1	25.4	32.0	35.2	26.1	26.6	27.9	31.0	26.6	27.6	27.9
N(HD) / DTH)										
A(HD)	.140E-4	.467E-5	.380E-5	.112E-4	.109E-4	.540E-5	.336E-5	.904E-5	.642E-5	.496E-5
1/T(HD)1	.310	-1.324	-1.109	.0770	-1.0817	.0383	.0549	.0508	.0187	-0.0126
1/T(HD)2	15.7	9.86	10.5	10.8	11.3	11.7	11.8	12.4	12.2	12.1
Z(HD)1	.443	.0186	-1.108	.222	.266	.0514	-1.0219	.0519	.0102	-0.0168
W(HD)1	1.91	9.06	11.6	3.63	3.59	6.39	9.75	7.81	4.90	6.04
Z(HD)2	.232	.216	.302	.319	.302	.291	.274	.284	.279	.281
W(HD)2	25.4	32.6	36.2	26.2	26.6	28.1	31.2	26.6	27.6	28.0
N(AZF) / DTH)										
A(AZF)	-.135E-4	-.214E-4	-.214E-4	-.214E-4	-.214E-4	-.215E-4	-.215E-4	-.215E-4	-.215E-4	-.215E-4
1/T(AZF)1	-.0124	-.00201	-.00138	-.00711	-.00471	-.00135	-.000608	-.00182	-.00103	-.000667
1/T(AZF)2	.491	-.403	-.131	.124	-.119	.0534	.0663	.0779	.0257	-0.0141
1/T(AZF)3	13.3	11.1	11.1	12.1	12.5	12.7	12.7	13.0	13.5	12.9
Z(AZF)1	.369	.164	.165	.152	.0931	.0763	.0709	.0671	.0433	.0396
W(AZF)1	1.71	3.63	4.42	2.03	1.93	2.65	3.43	1.97	2.28	2.59
Z(AZF)2	.277	.255	.231	.291	.280	.262	.237	.269	.260	.259
W(AZF)2	25.3	32.0	35.3	26.1	26.6	27.9	31.0	26.6	27.6	27.9

TABLE XI-8

XB-70A ELEVATOR TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Open

(BODY AXIS SYSTEM)

299

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	+
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.809	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
DENOMINATOR											
1/T(DET)1	13.3	13.3	13.3	13.3	1.48	13.3	13.3	1.54	1.53	1.45	
1/T(DET)2	(.211)	(.575)	(.656)	(.390)	3.10	(.296)	(.317)	3.06	3.39	3.55	
1/T(DET)3	(.101)	(.0533)	(.0722)	(.0726)	13.3	(.0496)	(.0542)	13.3	13.3	13.3	
Z(DET)1	.687	.811	.735	.825	.404	.818	.786	.362	.358	.361	
W(DET)1	1.41	3.09	3.78	2.10	.0501	2.79	3.23	.0394	.0407	.0436	
Z(DET)2	.275	.222	.201	.272	.267	.253	.278	.265	.256	.252	
W(DET)2	25.5	31.2	34.5	25.5	25.9	27.3	30.3	26.1	27.0	27.3	
NUMERATORS											
N(U /DE)											
A(U)	5.26	3.29	1.99	3.84	2.51	.480	.136	.628	.421	.389	
1/T(U)1	6.60	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	
1/T(U)2	13.3	63.7	124.	37.4	62.5	359.	1320.	351.	541.	588.	
Z(U)1	.022	.401	.313	.497	.988	.431	.0204	.821	.679	.507	
W(U)1	.652	.875	1.02	.451	.307	.363	.442	.167	.186	.201	
Z(U)2	.277	.226	.205	.273	.269	.256	.230	.266	.257	.253	
W(U)2	25.3	31.9	35.1	26.1	26.5	27.7	30.8	26.5	27.4	27.8	
N(W /DE)											
A(W)	-29.9	-58.8	-43.8	-28.4	-19.1	-7.51	-3.45	-5.77	-5.48	-6.54	
1/T(W)1	7.30	.0245	-.0150	13.3	13.3	.0150	-.0167	.00865	-.0122	-.0125	
1/T(W)2	13.3	-.0258	.0432	37.9	62.8	-.0178	.0202	-.0102	.0122	.0132	
1/T(W)3	(.0531)	13.3	13.3	(-.111)	(-.104)	13.3	13.3	13.3	13.3	13.3	
1/T(W)4	(.0941)	64.5	125.	(.0748)	(.0268)	362.	1303.	352.	540.	580.	
Z(W)1	.277	.226	.205	.273	.269	.256	.230	.266	.257	.253	
W(W)1	25.3	31.9	35.1	26.1	26.5	27.7	30.8	26.5	27.4	27.8	

TABLE XI-8 (Concluded)

N(THE/DE)										
A(THE)	-4.764	-4.28	-5.32	-1.75	-1.40	-1.80	-2.17	-1.08	-1.25	-1.35
1/T(THE)1	.0104	.00623	.0354	.726E-4	.000599	.00253	.00729	.00253	.00310	.00296
1/T(THE)2	.401	1.17	1.50	.535	.336	.398	.503	.180	.197	.215
1/T(THE)3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
Z(THE)1	.277	.220	.200	.269	.265	.252	.227	.264	.255	.252
W(THE)1	25.3	31.8	35.0	26.0	26.4	27.7	30.8	26.5	27.4	27.8
N(HD/DE)										
A(HD)	40.3	58.9	43.8	28.7	19.2	7.53	3.47	5.80	5.50	6.55
1/T(HD)1	-.0186	.00473	.0343	-.00912	-.00630	.00182	.00727	.00122	.00280	.00304
1/T(HD)2	-1.50	8.62	12.2	-4.32	4.39	8.46	13.3	5.34	6.81	7.87
1/T(HD)3	2.37	-8.67	13.3	4.54	-4.70	13.3	14.5	-11.7	13.3	13.3
1/T(HD)4	13.3	13.3	-15.3	13.3	13.3	-16.5	-42.3	13.3	-15.3	-15.5
Z(HD)1	.277	.218	.196	.267	.262	.240	.201	.257	.246	.244
W(HD)1	25.3	32.2	35.6	26.2	26.7	28.7	32.3	27.0	28.0	28.4
N(AZP/DE)										
A(AZP)	35.7	359.	476.	142.	118.	168.	208.	59.3	115.	125.
1/T(AZP)1	.00537	-.00189	-.00136	.548E-4	.000427	-.00114	-.000551	-.00162	-.000863	-.000542
1/T(AZP)2	-.0242	.00661	.0356	-.00923	-.00680	.00295	.00784	.00279	.00364	.00357
1/T(AZP)3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
Z(AZP)1	.115	.194	.229	.151	.121	.153	.142	.147	.137	.122
W(AZP)1	2.08	3.55	4.23	2.02	1.87	2.58	3.35	1.95	2.27	2.60
Z(AZP)2	.277	.220	.200	.270	.265	.253	.228	.265	.255	.252
W(AZP)2	25.3	31.7	35.0	25.9	26.4	27.7	30.7	26.5	27.4	27.8
	+	+	+	+	+	+	+	+	+	+

TABLE XI-9

XB-70A THRUST TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Open

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	+
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
DENOMINATOR											
1/T(DET)1	13.3	13.3	13.3	13.3	1.48	13.3	13.3	1.54	1.58	1.45	
1/T(DET)2	(.211)	(.575)	(.656)	(.390)	3.10	(.226)	(.317)	2.06	3.38	3.05	
1/T(DET)3	(.101)	(.0533)	(.0722)	(.0726)	13.3	(.0496)	(.0542)	13.3	12.3	12.3	
Z(DET)1	.687	.811	.735	.925	.404	.818	.796	.242	.358	.341	
W(DET)1	1.41	3.09	3.78	2.10	.0501	2.79	3.23	.0394	.0407	.0436	
Z(DET)2	.275	.222	.201	.272	.267	.253	.228	.265	.256	.252	
W(DET)2	25.5	31.2	34.5	25.5	25.9	27.3	30.3	26.1	27.0	27.3	
NUMERATORS											
N(U /DTH)											
A(U)	.000105	.822E-4	.825E-4	.810E-4	.808E-4	.824E-4	.828E-4	.822E-4	.825E-4	.827E-4	
1/T(U)1	-.0222	-.00785	-.00523	-.0142	-.00956	-.00424	-.00265	-.00470	-.00342	.00281	
1/T(U)2	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	
Z(U)1	.664	.705	.629	.756	.752	.536	.525	.443	.474	.522	
W(U)1	1.33	3.07	3.77	1.92	1.94	2.71	3.24	1.96	2.12	2.22	
Z(U)2	.275	.222	.201	.273	.269	.255	.229	.267	.258	.255	
W(U)2	25.5	31.2	34.5	25.4	25.9	27.3	30.3	26.1	27.0	27.3	
N(W /DTH)											
A(W)	.160E-4	.261E-4	.254E-4	.197E-4	.219E-4	.191E-4	.207E-4	.135E-4	.145E-4	.158E-4	
1/T(W)1	.0153	-.00445	-.00283	-.00850	-.00569	-.00211	-.00113	-.00231	-.00146	-.00104	
1/T(W)2	.0254	.0539	.0871	.0824	.0519	.0928	.0975	.0927	.0927	.0927	
1/T(W)3	9.52	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	
1/T(W)4	13.3	45.0	66.2	33.6	49.9	112.	174.	160.	217.	268.	
Z(W)1	.278	.221	.200	.272	.266	.252	.227	.264	.255	.251	
W(W)1	25.1	31.9	35.2	26.1	26.5	27.9	31.0	26.6	27.5	27.9	

TABLE XI-9 (Concluded)

N(THE/DTH)										
A(THE)	.443E-6	.129E-5	.158E-5	.107E-5	.125E-5	.139E-5	.170E-5	.113E-5	.131E-5	.147E-5
1/T(THE)1	13.3	.0463	.0846	.108	.0761	.0956	.0977	.101	.0910	.0884
1/T(THE)2	(.967)	1.14	1.44	.509	.305	.360	.465	.162	.187	.209
1/T(THE)3	(.403)	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
Z(THE)1	.278	.220	.199	.269	.264	.251	.226	.263	.254	.251
W(THE)1	25.2	31.9	35.2	26.0	26.5	27.9	31.0	26.6	27.5	27.9
N(HD/DTH)										
A(HD)	-.212E-5	-.215E-4	-.216E-4	-.863E-5	-.112E-4	-.138E-4	-.174E-4	-.456E-5	-.809E-5	-.109E-4
1/T(HD)1	.181	.0455	.0829	.0922	.0631	.0905	.0957	.0887	.0875	.0877
1/T(HD)2	3.82	-7.65	10.3	5.01	4.52	7.02	9.49	5.45	6.52	7.50
1/T(HD)3	13.3	8.07	-10.9	-8.60	-7.53	-8.30	-10.3	13.3	-11.4	-10.8
1/T(HD)4	-19.3	13.3	13.3	13.3	13.3	13.3	13.3	-15.4	13.3	13.3
Z(HD)1	.343	.218	.198	.257	.255	.248	.225	.247	.249	.247
W(HD)1	23.4	32.1	35.4	26.9	27.2	28.2	31.2	27.4	28.0	28.2
N(AZP/DTH)										
A(AZP)	-.278E-4	-.595E-4	-.000129	-.848E-4	-.000101	-.000116	-.000145	-.970E-4	-.000113	-.000128
1/T(AZP)1	-.0123	-.00201	-.00138	-.00697	-.00485	-.00134	-.000507	-.00180	-.00102	-.000657
1/T(AZP)2	.221	.0471	.0847	.103	.0704	.0948	.0975	.0966	.0916	.0900
1/T(AZP)3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
Z(AZP)1	.251	.155	.192	.132	.0695	.0724	.0751	.0702	.0652	.0614
W(AZP)1	2.00	3.63	4.33	2.05	1.89	2.59	3.41	1.95	2.28	2.62
Z(AZP)2	.277	.220	.199	.265	.265	.251	.226	.264	.254	.251
W(AZP)2	25.3	31.8	35.2	26.0	26.5	27.9	30.9	26.6	27.5	27.9
	+	+	+	+	+	+	+	+	+	+

TABLE XI-10

XB-70A ELEVATOR TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Closed

(BODY AXIS SYSTEM)

303

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	2.00	
DENOMINATORS											
1/T(DET)1	14.4	10.9	11.6	11.5	1.84	12.2	12.4	1.85	2.05	1.04	
1/T(DET)2	(.187)	(.494)	(.637)	(.348)	3.22	(.277)	(.302)	3.06	3.27	3.84	
1/T(DET)3	(.0930)	(.0435)	(.0587)	(.0660)	11.8	(.0454)	(.0486)	12.4	12.4	12.3	
Z(DET)1	.616	.622	.540	.863	.374	.760	.702	.346	.360	.338	
w(DET)1	1.47	4.18	4.99	2.50	.0470	3.20	3.75	.0372	.0375	.0397	
Z(DET)2	.256	.258	.229	.201	.287	.270	.241	.277	.260	.245	
w(DET)2	25.6	31.2	34.4	25.4	25.8	27.2	30.2	26.1	27.0	27.2	
NUMERATORS											
N(U /DE)											
A(U)	-4.26	-3.97	-3.13	-6.66	-6.72	-2.87	-1.94	-3.41	-2.60	-2.10	
1/T(U)1	6.60	(.402)	(.314)	.397	.275	(.440)	52.6	41.1	38.0	35.3	
1/T(U)2	66.8	(.866)	(1.01)	.501	.329	(.353)	115.	110.	163.	215.	
Z(U)1	.922	.810	.756	.909	.899	.926	.0270	.828	.601	.604	
w(U)1	.652	52.9	64.7	36.1	39.7	62.6	.431	.163	.152	.158	
N(W /DE)											
A(W)	32.3	71.1	69.1	49.3	51.1	44.5	48.3	31.5	33.7	36.8	
1/T(W)1	7.30	.0233	-.0130	(-.0905)	(-.0733)	.0108	-.0121	.00267	-.00860	-.00870	
1/T(W)2	66.8	-.0240	.0423	(.0362)	(.0282)	-.0123	.0169	-.00346	.00332	.0110	
1/T(W)3	(.0531)	(.819)	(.753)	(.902)	(.900)	(.926)	52.2	41.1	38.0	35.3	
1/T(W)4	(.0941)	(52.2)	(64.9)	(36.2)	(39.7)	(62.6)	116.	110.	164.	215.	
N(TH/DE)											
A(TH)	.620	2.90	3.70	2.13	2.41	2.72	3.44	2.12	2.52	2.91	
1/T(TH)1	.0104	.00623	.0354	-.000258	.000242	.00254	.00753	.00221	.00286	.00280	
1/T(TH)2	.601	1.16	1.48	.525	.327	.380	.484	.173	.150	.200	
1/T(TH)3	66.7	77.1	75.4	48.8	38.4	42.0	40.5	35.3	34.7	33.4	

TABLE XI-10 (Concluded)

N(HD/DE)											
A(HD)	-32.6	-71.2	-69.1	-49.8	-51.5	-44.6	-48.3	-31.7	-33.8	-26.0	
1/T(HD)1	-.0186	.00469	.0343	-.00975	-.00704	.00153	.00725	.000478	.00235	.00272	
1/T(HD)2	-1.58	-7.64	12.3	-3.85	-3.82	7.98	-12.8	5.17	6.54	7.71	
1/T(HD)3	2.37	8.55	-12.4	4.48	4.35	-8.59	(-.979)	-6.62	-8.23	-9.18	
1/T(HD)4	66.8	49.8	41.7	38.7	30.2	22.1	(15.3)	23.1	22.3	22.6	
N(AZP/DE)											
A(AZP)	-29.0	-212.	-293.	-159.	-184.	-221.	-287.	-176.	-214.	-247.	
1/T(AZP)1	.00537	-.00130	-.00137	-.000189	.000163	-.00123	-.000578	-.00179	-.000027	-.000576	
1/T(AZP)2	-.0242	.00658	.0356	-.00959	-.00725	.00276	.00785	.00222	.00325	.00330	
1/T(AZP)3	66.7	86.1	83.4	51.8	40.7	45.9	44.3	37.5	36.6	35.0	
Z(AZP)1	.115	.185	.222	.134	.0866	.123	.123	.107	.102	.0914	
W(AZP)1	2.08	3.56	4.24	2.02	1.87	2.58	3.37	1.95	2.28	2.61	
	+	+	+	+	+	+	+	+	+	+	+

TABLE XI-11

XB-70A THRUST TRANSFER FUNCTION FACTORS

SAS On — Bobweight Loop Closed

(BODY AXIS SYSTEM)

305

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
DENOMINATOR											
1/T(DET)1	14.4	10.9	11.6	11.5	1.84	12.2	12.4	1.85	2.05	1.00	
1/T(DET)2	(.187)	(.494)	(.637)	(.348)	3.22	(.277)	(.302)	3.06	3.27	2.04	
1/T(DET)3	(.0930)	(.0435)	(.0507)	(.0660)	11.8	(.0454)	(.0485)	12.4	12.4	12.3	
Z(DET)1	.616	.622	.540	.863	.374	.760	.709	.344	.340	.320	
W(DET)1	1.47	4.18	4.99	2.50	.0470	3.20	3.75	.0372	.0378	.0297	
Z(DET)2	.256	.253	.229	.301	.287	.270	.241	.277	.268	.265	
W(DET)2	25.6	31.2	34.4	25.4	25.8	27.2	30.2	26.1	27.0	27.3	
NUMERATORS											
N(U /DTH)											
A(U)	.000105	.822E-4	.825E-4	.810E-4	.808E-4	.824E-4	.828E-4	.822E-4	.826E-4	.827E-4	
1/T(U)1	-.0206	-.00593	-.00394	-.0128	-.00889	-.00375	-.00225	-.00433	-.00306	-.00241	
1/T(U)2	14.4	11.1	11.8	11.6	12.1	12.4	12.5	12.7	12.7	12.6	
Z(U)1	.592	.516	.439	.674	.701	.476	.449	.412	.431	.461	
W(U)1	1.44	4.11	4.93	2.30	2.20	3.09	3.72	2.15	2.37	2.53	
Z(U)2	.255	.259	.230	.303	.290	.271	.243	.279	.270	.267	
W(U)2	25.6	31.2	34.5	25.4	25.8	27.3	30.3	26.1	27.0	27.3	
N(W /DTH)											
A(W)	.160E-4	.261E-4	.254E-4	.197E-4	.219E-4	.191E-4	.207E-4	.135E-4	.145E-4	.158E-4	
1/T(W)1	(-.0667)	-.00427	-.00277	-.00358	-.00573	-.00210	-.00112	-.00231	-.00146	-.00104	
1/T(W)2	(.0312)	.0567	.0881	.0773	.0490	.0924	.0976	.0933	.0923	.0925	
1/T(W)3	(.987)	14.0	14.1	13.5	13.5	13.7	13.7	13.5	13.5	13.5	
1/T(W)4	(11.1)	44.5	65.8	33.7	45.8	112.	174.	160.	217.	268.	
Z(W)1	.297	.219	.196	.271	.265	.247	.221	.261	.252	.249	
W(W)1	25.0	31.7	34.9	26.0	26.5	27.8	30.9	26.5	27.5	27.9	

TABLE XI-11 (Concluded)

N(THE/DTH)										
A(THE)	.442E-6	.122E-5	.158E-5	.107E-5	.125E-5	.139E-5	.170E-5	.113E-5	.131E-5	.147E-5
1/T(THE)1	13.0	.0467	.0845	.110	.0772	.0958	.0977	.102	.0909	.0880
1/T(THE)2	(.923)	1.11	1.39	.499	.298	.352	.453	.159	.185	.208
1/T(THE)3	(.408)	13.7	13.8	13.4	13.4	13.6	13.7	13.5	13.5	13.5
Z(THE)1	.285	.214	.192	.268	.262	.246	.221	.261	.252	.248
W(THE)1	25.2	31.9	35.2	26.0	26.5	27.9	31.0	26.6	27.5	27.9
N(HD/DTH)										
A(HD)	-.212E-5	-.215E-4	-.216E-4	-.863E-5	-.112E-4	-.138E-4	-.174E-4	-.456E-5	-.809E-5	-.109E-4
1/T(HD)1	.173	.0443	.0820	.0858	.0620	.0896	.0953	.0876	.0868	.0872
1/T(HD)2	-22.3	-7.69	-11.0	4.86	4.47	7.27	-10.4	5.42	6.68	7.40
1/T(HD)3	(.927)	8.30	(.996)	-8.35	-7.44	-8.38	10.8	13.6	-11.4	-10.9
1/T(HD)4	(5.66)	13.1	(11.8)	14.6	13.9	12.9	11.7	-15.4	13.3	13.2
Z(HD)1	.475	.218	.203	.233	.244	.253	.233	.241	.249	.249
W(HD)1	27.6	32.1	35.4	26.8	27.1	28.2	31.2	27.4	28.0	28.2
N(AZP/DTH)										
A(AZP)	-.278E-4	-.695E-4	-.000129	-.848E-4	-.000101	-.000116	-.000145	-.970E-4	-.000113	-.000128
1/T(AZP)1	-.0123	-.00201	-.00138	-.00697	-.00485	-.00134	-.000627	-.00180	-.00102	-.000657
1/T(AZP)2	.221	.0471	.0847	.103	.0704	.0948	.0976	.0966	.0916	.0900
1/T(AZP)3	13.3	13.9	14.0	13.5	13.5	13.8	13.8	13.5	13.5	13.5
Z(AZP)1	.260	.159	.197	.130	.0689	.0748	.0787	.0710	.0664	.0627
W(AZP)1	2.00	3.55	4.22	2.04	1.88	2.55	3.35	1.94	2.26	2.61
Z(AZP)2	.277	.211	.189	.267	.262	.243	.219	.260	.251	.249
W(AZP)2	25.3	31.9	35.2	26.0	26.5	27.9	30.9	26.6	27.5	27.9
	+	+	+	+	+	+	+	+	+	+

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F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M	.310	.800	.950	.600	.900	1.50	2.20	2.00	2.50	3.00
Bobweight Loop Open										
D(G)/D(U) (DEG/KT)	.0554	-.0128	-.103	-.0333	.0270	-.00273	-.0218	.00100	-.00481	-.00672
NZA (G/RAD)	5.78	30.4	46.9	9.70	8.59	18.3	31.8	10.5	14.3	18.8
DE/G (DEG/G)	16.6	1.13	.570	5.52	3.34	7.02	3.48	12.5	7.68	4.15
CAP (RAD/SEC/SEC/G)	.243	.147	.119	.252	.366	.381	.281	.353	.277	.178
PHUSOID(2) (SEC) (TRUCK(2))	--	(7.36)	(14.7)	--	(16.1)	--	--	--	--	(77.0)
1/C(1/10)	2.00	2.36	1.52	1.65	1.42	.696	.569	.400	.287	.260
Bobweight Loop Closed										
FST/KT (LB/KT)	-.338	.0807	.0256	-.0513	.0582	-.0191	-.0173	-.0494	-.0112	.00797
FST/G (LB/G)	71.2	16.2	13.8	30.2	45.3	39.4	28.0	57.7	41.4	27.4

TABLE XI-13

XB-70A LATERAL DIRECTIONAL DIMENSIONAL DERIVATIVES

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
YV	-.0508	-.213	-.266	-.0499	-.0352	-.113	-.129	-.0473	-.0548	-.0623	
YB	-17.6	-190.	-282.	-31.0	-30.6	-175.	-275.	-91.6	-133.	-181.	
LB'	-5.04	9.67	-9.19	-6.11	-6.18	2.90	4.81	1.94	1.99	-5.69	
NB'	.898	1.60	3.73	.889	.881	2.04	2.21	.811	.912	1.16	
LP'	-1.71	-4.02	-7.36	-1.05	-1.26	-1.16	-1.03	-.393	-.413	-.438	
NP'	-.156	.0533	.145	.0417	.0572	-.0219	-.0507	-.0170	-.0193	-.0115	
LR'	-.213	-.636	-1.01	.259	.0927	-.202	-.0625	-.0399	.0212	.0849	
NR'	-.200	-.375	-.415	-.140	-.0883	-.307	-.367	-.134	-.151	-.174	
Y*DA	-.0175	-.0129	-.0133	-.00914	-.00176	.000481	0.	.231E-4	0.	0.	
L'DA	2.78	5.24	3.54	4.01	3.54	1.51	1.67	.966	.993	1.07	
N'DA	-.125	-.0386	-.201	-.0936	-.188	-.166	-.107	-.0638	-.0395	-.0427	
Y*DR	.0333	.0515	.0531	.0249	.0149	.0183	.0182	.00750	.00721	.00693	
L'DR	.118	-.0881	-4.71	.260	-.455	2.10	1.75	.800	.481	.285	
N'DR	-.568	-1.24	-1.41	-.421	-.330	-.845	-1.07	-.425	-.485	-.582	
	+	+	+	+	+	+	+	+	+	+	+

TABLE XI-14

XB-70A AILERON TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
DENOMINATOR											
1/T(DET)1	.0287	-.0158	.00706	.0270	.0133	-.00576	-.0131	-.0178	-.0152	-.000645	
1/T(DET)2	1.77	4.15	7.27	.678	.745	1.19	.966	.396	.395	.436	
Z(DET)1	.0615	.184	.197	.217	.266	.145	.200	.126	.137	.108	
W(DET)1	1.27	1.28	1.94	1.23	1.16	1.38	1.43	.779	.875	1.10	
NUMERATORS											
N(B /DA)											
A(B)	-.0175	-.0129	-.0133	-.00914	-.00176	.000481	.174	.231E-4	.116	.106	
1/T(B)1	.0485	-21.9	.0451	-68.1	.0685	.0359	.0390	.0320	.0348	.0406	
1/T(B)2	1.94	(.559)	3.58	(.786)	.241	1.06	1.37	.394	.484	.464	
1/T(B)3	-27.9	(.505)	-23.0	(.213)	-368.	547.		7269.			
N(P /DA)											
A(P)	2.78	5.24	3.54	4.01	3.54	1.51	1.67	.966	.993	1.07	
1/T(P)1	-.0119	-.00193	-.00131	-.00691	-.00483	-.00132	-.000596	-.00179	-.00102	-.000653	
Z(P)1	.184	.216	.211	.118	.0851	.144	.156	.0960	.103	.109	
W(P)1	.829	1.32	1.83	.866	.743	1.55	1.60	.970	.998	1.07	
N(R /DA)											
A(R)	-.125	-.0386	-.201	-.0936	-.188	-.166	-.107	-.0638	-.0395	-.0427	
1/T(R)1	.407	-5.01	-.505	.430	.288	.283	.319	.148	.166	.180	
1/T(R)2	-.607	(.991)	.655	1.50	1.12	-.795	-.776	-1.01	-1.07	-1.06	
1/T(R)3	5.55	(1.28)	5.18	-2.54	-1.19	1.98	2.41	1.56	1.86	1.67	
N(PHI/DA)											
A(PHI)	2.76	5.24	3.53	4.00	3.52	1.50	1.67	.959	.990	1.07	
Z(PHI)1	.157	.215	.208	.115	.0804	.140	.155	.0923	.101	.108	
W(PHI)1	.834	1.32	1.83	.874	.752	1.55	1.60	.979	1.00	1.08	
N(AYP/DA)											
A(AYP)	.146	19.9	-10.0	12.0	3.84	-5.34	.791	.279	2.79	3.01	
1/T(AYP)1	.0691	.157	.0251	-.150	.127	.0423	.0419	.0381	.0390	.0441	
1/T(AYP)2	-497.	-1.93	19.0	.234	-.181	3.90	-19.1	-11.3	-1.84	-1.96	
Z(AYP)1	-.198	.245	.133	.0817	-.196	-.192	.124	.0259	.430	.586	
W(AYP)1	.511	1.44	1.87	1.56	1.99	1.41	2.00	1.27	1.14	1.18	

TABLE XI-15

XB-70A RUDDER TRANSFER FUNCTION FACTORS

SAS Off

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00
DENOMINATOR										
1/T(DET)1	.0287	-.0158	.00706	.0270	.0133	-.00576	-.0131	-.0178	-.0152	-.000645
1/T(DET)2	1.77	4.15	7.27	.678	.745	1.19	.966	.396	.395	.436
Z(DET)1	.0615	.184	.197	.217	.266	.145	.200	.126	.137	.108
W(DET)1	1.27	1.28	1.94	1.23	1.16	1.38	1.43	.779	.875	1.10
NUMERATORS										
N(B /DR)										
A(B)	.0333	.0515	.0531	.0249	.0149	.0183	.0182	.00750	.00721	.00693
1/T(B)1	.00130	.00337	-.00282	-.0153	-.0114	.0140	.00807	.00860	.00281	-.000649
1/T(B)2	1.73	4.07	10.5	.955	1.56	1.13	1.08	.395	.419	.436
1/T(B)3	17.5	24.4	19.8	18.4	17.8	53.9	62.9	68.0	72.3	86.6
N(P /DR)										
A(P)	.118	-.0881	-4.71	.260	-.455	2.10	1.75	.800	.481	.285
1/T(P)1	-.0121	-.00197	-.00135	-.00694	-.00485	-.00132	-.000599	-.00180	-.00102	.0485
1/T(P)2	4.77	6.67	(.0939)	2.62	(.0860)	(.146)	(.128)	(.0821)	(.0631)	(-.0555)
1/T(P)3	-4.91	-20.7	(2.57)	-3.43	(2.31)	(1.80)	(2.28)	(1.36)	(1.71)	(.773E-4)
N(R /DR)										
A(R)	-.568	-1.24	-1.41	-.421	-.330	-.845	-1.07	-.425	-.485	-.582
1/T(R)1	1.55	-.274	7.71	.570	.473	.243	.266	.142	.152	.105E-6
Z(R)1	.178	(.303)	.444	.317	.566	(-.455)	(-.367)	(-.497)	(-.367)	(.0485)
W(R)1	.538	(4.14)	.292	.708	.757	(1.49)	(1.30)	(.813)	(.688)	(.444)
N(PHI/DR)										
A(PHI)	.0433	-.158	-4.78	.203	-.498	2.05	1.71	.754	.443	.250
1/T(PHI)1	6.53	6.28	(.114)	2.83	(.107)	(.138)	(.123)	(.0735)	(.0540)	.0485
1/T(PHI)2	-9.93	-12.3	(2.56)	-4.16	(2.23)	(1.83)	(2.31)	(1.41)	(1.78)	-.0614
N(AYP/DR)										
A(AYP)	-43.9	-76.2	-113.	-23.9	-22.3	-40.2	-54.0	-21.7	-26.7	-34.9
1/T(AYP)1	-.0747	.0133	-.0119	-.0606	-.0352	.0259	.0148	.0170	.00732	-.000651
1/T(AYP)2	1.07	4.23	5.78	.383	.332	1.88	1.54	.755	.583	.439
Z(AYP)1	.715	-.125	.0668	.337	.406	-.194	-.103	-.169	-.0610	.00299
W(AYP)1	.516	1.38	1.59	1.07	1.10	1.26	1.74	.891	1.19	1.55
	+	+	+	+	+	+	+	+	+	+

TABLE XI-16

XB-70A AILERON TRANSFER FUNCTION FACTORS

SAS On

(BODY AXIS SYSTEM)

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	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00
DENOMINATOR										
1/T (DET) 1	.0153	-.0115	.00513	.00647	.00415	-.00392	-.00683	-.00774	-.00663	-.000649
1/T (DET) 2	.466	.397	.388	.351	.350	.345	.345	.382	.371	.353
1/T (DET) 3	3.00	6.73	8.86	2.75	2.65	2.00	1.90	.953	.943	.955
Z (DET) 1	.377	.781	.633	.304	.335	.215	.278	.147	.193	.217
W (DET) 1	1.05	1.12	1.88	.957	.865	1.38	1.42	.777	.855	1.06
NUMERATORS										
V(B /DA)										
A(B)	-.0175	-.0129	-.0133	-.00914	-.00176	.000481	.173	.231E-4	.115	.106
1/T(B)1	.0319	.0723	.0288	.147	.0511	.0337	.0354	.0290	.0310	.0351
1/T(B)2	.442	-21.7	.498	-68.0	.266	.333	.340	.257	.271	.268
1/T(B)3	2.25	(.533)	4.05	(.961)	.405	1.13	1.48	.563	.669	.667
1/T(B)4	-27.5	(1.09)	-21.3	(.320)	-367.	545.		7261.		
V(P /DA)										
A(P)	2.78	5.24	3.54	4.01	3.54	1.51	1.67	.966	.993	1.07
1/T(P)1	-.0118	-.00193	-.00131	-.00691	-.00483	-.00132	-.000596	-.00179	-.00102	-.000653
1/T(P)2	.504	.444	.357	.348	.354	.341	.344	.349	.352	.353
Z(P)1	.537	.773	.711	.176	.150	.206	.251	.166	.190	.209
W(P)1	.674	1.14	1.77	.846	.721	1.53	1.57	.947	.971	1.04
V(R /DA)										
A(R)	-.125	-.0386	-.201	-.0936	-.183	-.166	-.107	-.0638	-.0395	-.0427
1/T(R)1	.333	.333	.333	.333	.288	.283	.319	.148	.166	.180
1/T(R)2	.407	-5.01	-5.05	.430	.333	.333	.333	.333	.333	.333
1/T(R)3	-.607	(.991)	.655	1.50	1.12	-.795	-.776	-1.01	-1.07	-1.06
1/T(R)4	5.55	(1.28)	5.18	-2.54	-1.19	1.98	2.41	1.56	1.86	1.67

TABLE XI-16 Continued

N(PHI/DA)										
A(PHI)	2.76	5.24	3.53	4.00	3.52	1.50	1.67	.959	.990	1.07
1/T(PHI)1	.501	.445	.358	.348	.354	.341	.344	.349	.352	.353
Z(PHI)1	.507	.772	.708	.172	.145	.202	.249	.162	.188	.208
W(PHI)1	.680	1.14	1.77	.654	.730	1.54	1.58	.956	.975	1.04
N(AYP/DA)										
A(AYP)	.146	19.9	-10.0	12.0	3.84	-5.34	.791	.279	2.79	3.01
1/T(AYP)1	.0504	.0734	.0206	-.160	.113	.0411	.0393	.0358	.0357	.0392
1/T(AYP)2	.454	.724	.426	.211	-.206	.338	.341	.313	.309	.304
1/T(AYP)3	-457.	-2.08	16.8	.360	.344	3.73	-16.8	-9.47	-1.79	-1.89
Z(AYP)1	-.0762	.734	.445	.137	-.0790	-.174	.187	.108	.513	.660
W(AYP)1	.534	1.37	1.94	1.53	1.94	1.45	2.18	1.48	1.25	1.33
	+	+	+	+	+	+	+	+	+	+

TABLE XI-17

XB-70A RUDDER TRANSFER FUNCTION FACTORS

SAS On

(BODY AXIS SYSTEM)

	+	+	+	+	+	+	+	+	+	+	+
F/C #	1	2	3	4	5	6	7	8	9	10	
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K	
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00	
DENOMINATOR											
1/T(DET)1	.0153	-.0115	.00513	.00647	.00415	-.00392	-.00683	-.00774	-.00663	-.000649	
1/T(DET)2	.466	.397	.388	.351	.350	.345	.345	.382	.371	.353	
1/T(DET)3	3.00	6.73	8.86	2.75	2.65	2.00	1.90	.953	.943	.955	
Z(DET)1	.377	.781	.633	.304	.335	.215	.278	.147	.193	.217	
W(DET)1	1.05	1.12	1.88	.957	.865	1.38	1.42	.777	.855	1.06	
NUMERATORS											
N(B /DF)											
A(B)	.0333	.0515	.0531	.0249	.0149	.0183	.0182	.00750	.00721	.00693	
1/T(B)1	-.00454	.00126	-.00251	-.00985	-.00747	.00944	.00464	.00368	.000840	-.000654	
1/T(B)2	.333	.333	.333	.333	.333	.333	.333	.333	.333	.333	
1/T(B)3	3.07	6.73	(.967)	2.72	4.10	1.60	1.79	.750	.864	.947	
1/T(B)4	17.6	24.3	(16.3)	18.7	17.0	54.1	63.1	68.1	72.4	86.6	
N(P /DF)											
A(P)	.118	-.0881	-4.71	.260	-.455	2.10	1.75	.800	.481	.285	
1/T(P)1	-.0121	-.00197	-.00135	-.00694	-.00485	-.00132	-.000599	-.00180	-.00102	.0485	
1/T(P)2	.333	.333	.333	.333	.333	.333	.333	.333	.333	.333	
Z(P)1	(4.77)	(6.67)	.0939	(2.62)	.0860	.146	.128	.0821	.0631	-.0555	
W(P)1	(-4.91)	(-20.7)	2.57	(-3.43)	2.31	1.80	2.28	1.36	1.71	.773E-4	
N(R /DF)											
A(R)	-.568	-1.24	-1.41	-.421	-.330	-.845	-1.07	-.425	-.485	-.582	
1/T(R)1	.333	-.198	.333	.333	.333	.219	.227	.129	.131	.482E-7	
1/T(R)2	2.95	.258	9.82	2.74	2.92	.333	-.280	.333	-.269	.0485	
1/T(R)3	(.214)	.333	(.494)	(.389)	(.514)	-.379	.333	-.389	.333	.333	
1/T(R)4	(.390)	6.73	(.258)	(.322)	(.305)	1.99	2.00	1.14	1.09	.969	

TABLE XI-17 Continued

V(PHI/DF)											
A(PHI)	.0433	-.158	-4.78	.203	-.498	2.05	1.71	.754	.443	.250	
1/T(PHI)1	.333	.333	.333	.333	.333	.333	.333	.333	.333	.0485	
1/T(PHI)2	5.67	6.68	(.119)	2.61	(.144)	(.134)	(.119)	(.0643)	(.0428)	-.134	
1/T(PHI)3	-11.5	-11.5	(2.56)	-4.50	(2.23)	(1.83)	(2.31)	(1.41)	(1.78)	.333	
V(AYP/DF)											
A(AYP)	-43.9	-76.2	-113.	-23.9	-22.3	-40.2	-54.0	-21.7	-26.7	-34.9	
1/T(AYP)1	-.0475	.00777	-.00867	-.0371	-.0213	.0200	.00934	.00885	.00314	-.000655	
1/T(AYP)2	.333	.333	.333	.333	.333	.333	.333	.333	.333	.333	
1/T(AYP)3	2.86	6.73	7.80	2.76	2.46	2.43	2.34	1.22	1.11	.967	
Z(AYP)1	.362	-.0860	.0496	.367	.435	-.168	-.0869	-.144	-.0620	.00968	
W(AYP)1	.423	1.37	1.65	.534	.553	1.25	1.76	.933	1.22	1.54	
	+	+	+	+	+	+	+	+	+	+	+

TABLE XI-18

XB-70A LATERAL DIRECTIONAL HANDLING QUALITIES PARAMETERS

SAS Off

(BODY AXIS SYSTEM)

F/C #	1	2	3	4	5	6	7	8	9	10
H	SL	SL	SL	20 K	40 K	40 K	40 K	60 K	60 K	60 K
M	.310	.800	.950	.600	.900	1.60	2.20	2.00	2.50	3.00
DR PERIOD (SEC)	4.94	4.98	3.30	5.25	5.60	4.60	4.50	8.13	7.25	5.75
1/C(1/2)	.559	1.70	1.82	2.02	2.51	1.32	1.85	1.15	1.25	.989
SPIRAL (2) (SEC)	--	43.9	--	--	--	120.	52.8	38.9	45.6	1075.
P(1)	1.11	1.45	.461	2.20	1.67	1.70	2.28	4.06	3.32	2.34
P(2)	-.0279	1.38	.398	1.59	.869	1.55	2.22	3.89	--	2.33
P(3)	1.04	1.50	.435	2.82	2.04	1.70	2.40	4.72	--	2.35
P(2)/P(1)	-.0251	.954	.863	.724	.521	.909	.973	.958	--	.995
P(OSC)/P(AV)	1.05	.0329	.0593	.160	.315	.0473	.0135	.0605	--	.00265
W(PHI)/W(D)	.655	1.03	.943	.713	.646	1.13	1.12	1.26	1.15	.979
DEL-B-MAX	.607	.0689	.0658	.452	.447	.219	.175	.510	.303	.165
PHI TO BETA, PHASE	48.7	244.	55.2	22.8	386.	211.	197.	194.	190.	22.3
PHI TO BETA	1.86	1.90	.652	3.42	3.56	1.31	2.17	3.15	2.57	.405
PHI TO VE	.308	.122	.0352	.432	.471	.0973	.117	.302	.198	.0259

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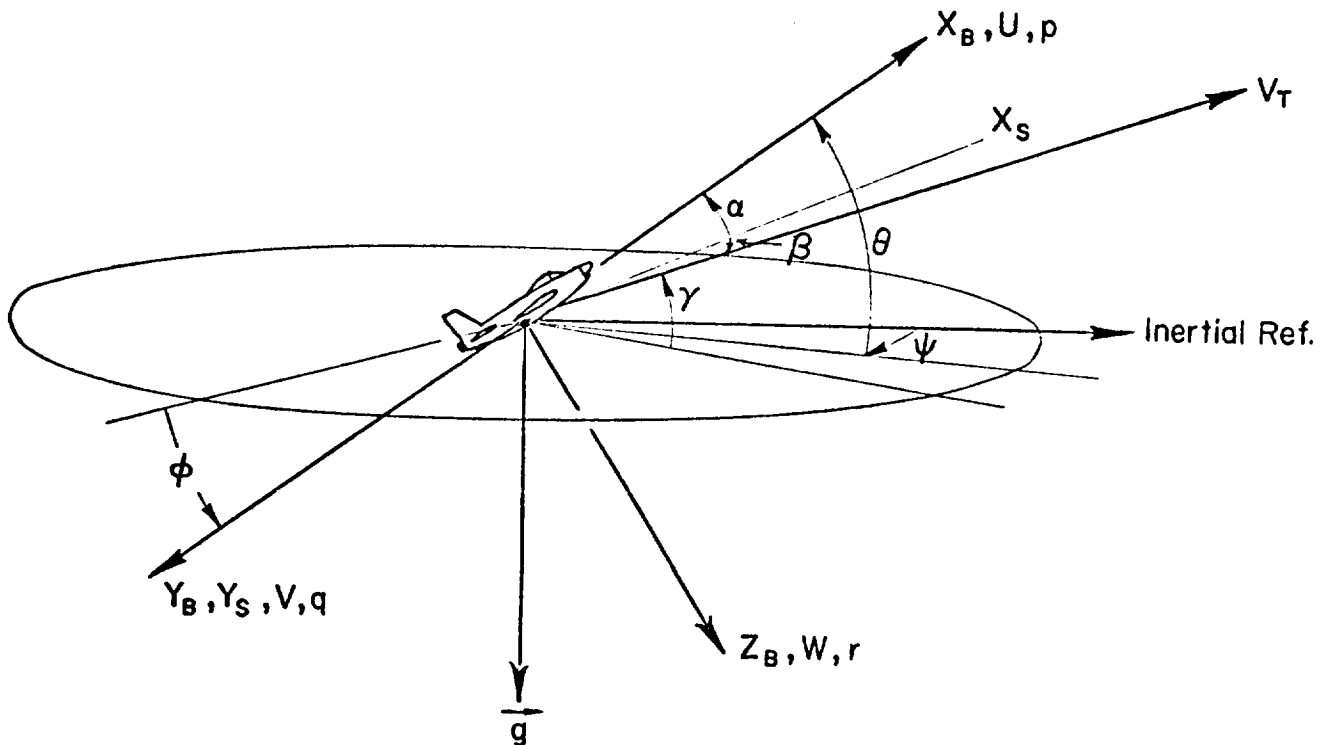
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APPENDIX A

AXIS SYSTEMS, SYMBOLS, COMPUTER MNEMONICS, AND DERIVATIVE DEFINITIONS

1. AXIS SYSTEMS



X_B, Y_B, Z_B - The Body-Axis System consists of right-handed, orthogonal axes whose origin is fixed at the nominal aircraft center of gravity. It's orientation remains fixed with respect to the aircraft, the X_B and Z_B axes being in the plane of symmetry. The exact alignment of X_B axis is arbitrary, herein it is taken along the body centerline reference.

X_S, Y_S, Z_S - The Stability-Axis System is that particular body-axis system for which the X_S -axis is coincident with the projection of the total steady-state velocity vector (V_{T_0}) on the aircraft's plane of symmetry. It's orientation remains fixed with respect to the aircraft.

2. SYMBOLS

a	Speed of sound in air	ft/sec
a_y	Lateral acceleration along the y-body axis at the center of gravity (positive out right wing)	ft/sec ²
a_y'	Lateral acceleration parallel to the y-body axis at a distance l_x and l_z from the c.g., $a_y' = a_y + l_x \dot{r} - l_z \dot{p}$	ft/sec ²
a_z'	Normal acceleration parallel to the z-body axis at a distance l_x from the c.g., $a_z' = a_z - l_x \dot{q}$	ft/sec ²
a_z^B	Normal acceleration parallel to the z-body axis at a distance l_B from the c.g.	
b	Reference wing span	ft
B	Bobweight gain	lb/g
B.L.	Buttock line	
$\bar{c}$	Reference chord	ft
C	Longitudinal feel system damping	lb/in./sec
c.g.	Center of gravity	
D	Aerodynamic force (drag) along the total velocity vector (positive aft)	lb
FRL	Fuselage reference line (parallel to x-body axis)	
F.S.	Fuselage station	
	Longitudinal control column force (+ aft)	lb
F_{ST}	Longitudinal stick force (+ aft)	lb
F_{ST}^{LAT}	Lateral stick force (+ right)	lb
F_{ped}	Rudder pedal force (+ right)	lb
g	Acceleration due to gravity	ft/sec ²
G	Pilot control to surface gearing	deg/in. or deg/deg

h	Altitude	ft
I	Longitudinal feel system inertia	lb/in./sec ²
I_x, I_y, I_z	Moments of inertia referred to body axis (unless otherwise specified)	slug-ft ²
I_{xz}	Product of inertia referred to body axis (unless otherwise specified)	slug-ft ²
$j\omega$	The imaginary portion of the complex variable $s = \sigma \pm j\omega$	rad/sec
ℓ_B	Effective distance of bobweight from c.g. (positive forward)	ft
ℓ_x	Distance along the x-body axis from the c.g. (positive forward)	ft
ℓ_{th}	Perpendicular distance from c.g. to thrust line (positive for nose-up pitching moment due to thrust)	ft
ℓ_z	Distance along the z-body axis from the c.g. (positive down)	ft
K	Longitudinal feel system spring constant	lb/in.
KTAS	Knots true airspeed	
KCAS	Knots calibrated airspeed	
K'	Feel system spring constant per unit dynamic pressure	(lb/in.)/psf
L	Rolling moment about the x-axis due to aerodynamic torques (positive right wing down)	ft-lb
L	Aerodynamic force (lift) perpendicular to the total velocity vector in the aircraft's plane of symmetry (positive up)	lb
m	Mass	slugs
M	Mach number	
M	Pitching moment about the y-axis due to aerodynamic torques (positive nose up)	ft-lb
MAC	Mean aerodynamic chord	ft
MGC	Mean geometric chord	ft

N	Aerodynamic normal force along the z-body axis, <u>but</u> positive up	lb
N	Yawing moment about z-axis due to aerodynamic torques (positive nose right)	ft-lb
p	Roll rate, angular velocity about x-axis (positive right wing down)	rad/sec
q	Pitch rate, angular velocity about y-axis (positive nose up)	rad/sec
$\bar{q}$	Dynamic pressure, $1/2 \rho V_{T0}^2$	lb/ft ²
r	Yaw rate, angular velocity about z-axis (positive nose right)	rad/sec
r_{RG}	Yaw rate gyro signal	rad/sec
s	Laplace operator, $\sigma + j\omega$	rad/sec
S	Reference wing area	ft ²
TED	Trailing edge down	
TEU	Trailing edge up	
TL	Thrust line	
u	Linear perturbed velocity along the x-axis (positive forward)	ft/sec
U_0	Linear steady-state velocity along the x-axis (positive forward)	ft/sec
v	Linear perturbed velocity along the y-axis (positive out right wing)	ft/sec
V_s	Stall speed	
V_{T0}	Total linear steady-state velocity (positive forward)	kt
w	Linear perturbed velocity along the x-axis (positive down)	
W.L.	Water line	in.
W	Weight	lb
W_0	Linear steady-state velocity along the z-axis (positive down)	ft/sec

X	Aerodynamic force along the x-axis (positive forward)	
Y	Aerodynamic force along y-axis (positive out right wing)	lb
Z	Aerodynamic force along z-axis (positive down)	lb
α	Perturbed angle of attack	rad
α_0	Steady-state (trim) angle of attack relative to the FRL	deg
β	Sideslip angle	rad
γ_0	Steady-state flight path angle	deg
δ_a	Aileron control surface deflection (includes spoiler effects, etc.) (positive for positive rolling moment)	rad
δ_e	Elevator surface deflection from trim (positive for nose-down pitching moment for aft surface)	rad
δ_{e0}	Trim elevator deflection	deg
δ_{cc}	Longitudinal control column deflection from trim (positive aft)	deg
δ_{ST}	Longitudinal stick deflection from trim (positive aft)	in.
δ_{ST}^{LAT}	Lateral stick deflection from trim (positive right)	in.
δ_{ped}	Rudder pedal deflection from trim (positive right pedal forward)	in.
δ_w	Lateral wheel deflection from trim (positive about x-axis)	deg
δ_s	Stabilizer surface deflection from trim (positive for TED)	rad
δ_{sp}	Spoiler surface deflection (positive up)	rad
δ_v	Vertical tail deflection from trim (positive for nose-left yawing moment)	rad
δ_r	Rudder deflection [positive for nose-left yawing moment (negative N)]	rad

Δ	Denominator of airframe transfer function	
ϵ	Angle between principle inertia axis and FRL (positive about y-axis)	deg
ζ_i	Damping ratio of linear second-order mode particularized by the subscript	
θ	Pitch angle, $\int q \, dt$ for straight and level flight, positive nose up	rad
i_{TH}	Inclination of thrust line with FRL [positive gives negative (-) z force]	deg
ρ	Mass density of air	slugs/ft ³
σ	The real portion of the complex variable $s = \sigma \pm j\omega$	rad/sec
ϕ	Roll angle, $(\cos \theta_0 \int p \, dt - \sin \theta_0 \int r \, dt)$ in straight and level flight (positive right wing down)	rad
ω_i	Undamped natural frequency of a second-order mode, particularized by subscript	rad/sec

Special Subscript

a	Aileron
cc	Control column
d	Dutch roll
e	Elevator
G	Gyro
INS	Inertial navigation system
p	Phugoid
r	Rudder
R	Roll subsidence
s	Spiral
SAS	Stability augmentation system
sp	Short period
ST	Stick

Special Superscript

DIR	Directional control system (e.g., rudder pedal)
LAT	Lateral control system

Symbols Unique to Specific Aircraft

ARI	Aileron-rudder interconnect (F-4)	
BLC	Boundary layer control (F-104, F-4)	
$K_{\text{FLEX}}^{\text{DIR}}$	Rudder flexure coefficient (F-4)	
P_{BF}	Bellows force parameter (F-4)	ft^2
q_{B}	Bellows pressure (F-4)	lb/ft^2
δ_{d}	Yaw damper surface deflection (F-104) (positive for nose-left yawing moment)	rad
δ_{ta}	Aileron tab deflection (CV-880M)	rad
δ_{ta_c}	Commanded aileron tab deflection (CV-880M)	rad
δ_{te}	Elevator tab deflection (CV-880M)	rad
$(\delta_{\text{te}} - \delta_{\text{e}})_c$	Commanded elevator-elevator servo tab combination (input linkage) (CV-880M)	rad
δ_{tr}	Rudder tab deflection (CV-880M)	rad
$(\delta_{\text{tr}} - \delta_{\text{r}})_c$	Commanded rudder-rudder servo tab combination (input linkage)(CV-880M)	rad

3. COMPUTER PRINTOUT MNEMONICS

a. DIMENSIONAL, MASS, AND FLIGHT CONDITION PARAMETERS

<u>COMPUTER PRINT OUT</u>	<u>STANDARD NOTATION, DEFINITION</u>
S	S, wing reference area
B	b, wing span
C	$\bar{c}$, mean geometric chord
F/C#	Flight Condition number
H(FT)	h, altitude, feet
SL	Sea Level
M(-)	M, Mach number
VTO(FPS)	V_{TO} , true airspeed, knots
VTO(KTAS)	V_{TO} , true airspeed knots
VTO(KTCS)	V_{TO} , calibrated airspeed, knots
W(LBS)	W, weight, pounds
C.G.(MGC)	c.g., center of gravity relative to mean geometric chord
IX	$\left. \begin{array}{l} I_x \\ I_y \\ I_z \\ I_{xz} \end{array} \right\} \text{Body axis (FRL) moments of inertia, slugs-ft}^2$
IY	
IZ	
IXZ	
EPSILON(DEG)	ϵ , inclination of principle axis with respect to FRL, degrees
Q(PSF)	q, dynamic pressure, psf
QC(PSF)	q_c , impact pressure, psf
ALPHA(DEG)	α_o , FRL angle of attack, degrees
GAMMA(DEG)	γ_o , flight path angle, degrees
LXP(FT)	ℓ_x , x distance to pilot, ft
LZP(FT)	ℓ_z , z distance to pilot, ft
ITH(DEG)	i_{th} , thrust incidence with respect to FRL, degrees
XI(DEG)	ξ_o , $i_{th} + \alpha_o$, degrees
LTH(FT)	ℓ_{th} , perpendicular distance to thrust line from c.g., ft

b. LONGITUDINAL PARAMETERS

<u>COMPUTER PRINT OUT</u>	<u>STANDARD NOTATION, DEFINITION</u>
XU*	X_u^* 1/sec
ZU*	Z_u^* 1/sec
MU*	M_u^* 1/sec-ft
XW	X_w 1/sec
ZW	Z_w 1/sec
MW	M_w 1/sec-ft
ZWD	$Z_{\dot{w}}$ 1/sec ²
ZQ	Z_q 1/sec
MWD	$M_{\dot{w}}$ 1/sec-ft
MQ	M_q 1/sec
†XDDD	X_δ ft/sec ² -rad
ZDDD	Z_δ ft/sec ² -rad
MDDD	M_δ 1/sec ²
DTH	δ_{th} Thrust
FST	Fst Stick force
U	u fps
W	w fps
THE	θ rad
HD	$\dot{h}$ fps
AZP	a_z' ft/sec ² at $X = l_x$

†DDD signifies a control surface, e.g., for elevator DDD = DE; for aileron DDD = DA

c. LATERAL-DIRECTIONAL PARAMETERS

<u>COMPUTER PRINT OUT</u>	<u>STANDARD NOTATION, DEFINITION</u>	
YV	Y_V	1/sec
YB	Y_β	ft/sec ²
LB'	L'_p	1/sec ²
NB'	N'_p	1/sec ²
LP'	L'_p	1/sec
NP'	N'_p	1/sec
LR'	L'_r	1/sec
NR'	N'_r	1/sec
[†] Y*DDD	Y_δ^*	1/sec
L'DDD	L'_δ	1/sec ²
N'DDD	N'_δ	1/sec ²
B	β	rad
P	p	rad/sec
R	r	rad/sec
PHI	ϕ	rad
AYP	a'_y	ft/sec ² at l_x, l_z

[†]DDD signifies a control surface, e.g., for elevator DDD = DE; for aileron DDD = DA.

d. TRANSFER FUNCTION PARAMETERS

The following shorthand notation is used to print the factored polynomials for all transfer functions*:

$$(s + 1/T_X)_i = 1/T_{X_i} \quad , \quad i = 1 \text{ to } k$$

$$(s^2 + 2\zeta_n s + \omega_n^2)_j = \zeta_j; \omega_{nj} \quad , \quad j = 1 \text{ to } \ell$$

where $k + 2\ell = n$, the order of the polynomial

<u>COMPUTER PRINT OUT</u>		<u>STANDARD NOTATION, DEFINITION</u>
DET		Roots of the denominator
N(X/Y)		Numerator N_y^x
A(X)		Gain of the transfer function x/y
[†] 1/T(X)I		$1/T_{X_i}$, rad/sec
[†] Z(X)J		ζ_j
[†] W(X)J		ω_{nj} , rad/sec
For example:	DENOMINATOR 1/T(DET)1 .0318 1/T(DET)2 2.20 Z(DET)1 .0609 W(DET)1 1.13	
	NUMERATORS N(B /DR) A(B) .0295 1/T(B)1 -.0494 1/T(B)2 2.05 1/T(B)3 42.3	
Translates to:	$\frac{\beta}{\delta_r} = \frac{.0295(s - .0494)(s + 2.05)(s + 42.3)}{(s + .0318)(s + 2.20)(s^2 + 2 \times .0609 \times 1.13s + 1.13^2 s^2)}$	

*The transfer function x/y is written as:

$$x/y = \frac{N_y^x}{\Delta} = \frac{A_x(s^m + s^{m-1} + \dots s^0)}{(s^n + s^{n-1} + \dots s^0)}$$

[†]Any roots enclosed in parentheses imply the opposite order of what is specified, e.g., $Z(DET)1 = (0.00132) \Rightarrow 1/T(DET)1 = 0.00132$

e. LONGITUDINAL HANDLING QUALITY PARAMETERS

<u>COMPUTER PRINT OUT</u>	<u>STANDARD NOTATION, DEFINITION</u>	<u>EQUATION</u>
D(G)/D(U) (DEG/KT)	$\partial\gamma/\partial u$, degrees/knot	$(1.689)(57.3) \frac{\left[N_{\delta}^g(s) + \frac{W_o}{V_{T_o}^2} N_{\delta}^u(s) - \frac{U_o}{V_{T_o}^2} N_{\delta}^w(s) \right]}{\frac{U_o}{V_{T_o}} N_{\delta}^u(s) + \frac{W_o}{V_{T_o}} N_{\delta}^w(s)}, \text{ for } s = 0$
NZA (G/RAD)	$N_{z\alpha}$, g/rad	$\frac{-U_o}{g} \frac{\hat{N}_{\delta}^{az}(s)}{\hat{N}_{\delta}^w(s)}, \text{ for } s = 0$
DE/G (DEG/G)	δ_e/g , degrees/g	$57.3 \left(\frac{1}{g} \frac{\hat{N}_{\delta}^{az}(s)}{\hat{\Delta}(s)} \right)^{-1}, \text{ for } s = 0$
CAP (RAD/SEC/SEC/G)	Control anticipation parameter, rad/sec ² /g	$-\left(\frac{s^2 \hat{N}_{\delta}^g(s)}{\hat{\Delta}(s)} \right) \bigg _{s=\infty} \bigg/ \left(\frac{1}{g} \frac{\hat{N}_{\delta}^{az}(s)}{\hat{\Delta}(s)} \right) \bigg _{s=0}$
PHUGOID(2) (TUCK(2))	The phugoid time to double amplitude, seconds	$\frac{\ln 2}{ \zeta_{ph} \omega_{n_{ph}} }, \text{ for } \zeta_{ph} < 0$
1/C(1/10)	Short period inverse cycles to 1/10 amplitude	$\frac{2\pi}{\ln 10} \sqrt{\frac{\zeta_{sp}^2}{1 - \zeta_{sp}^2}} \text{ for } 0 \leq \zeta_{sp} < 1$
FST/KT (LB/KT)	Stick force per knot, pounds/knot	$1.689 \left[\frac{u}{F_{st}}(s) \right]^{-1} \text{ for } s = 0$
FST/G (LB/G)	Stick force per g, pounds per g	$\left[\frac{1}{g} \frac{\hat{N}_{F_{st}}^{az}}{\hat{\Delta}} \right]^{-1} \text{ for } s = 0$
--	The parameter has no meaning or is not defined at this flight condition	

*The hat ($\hat{\cdot}$) notation implies constant speed ($u = \theta_o = 0$).

f. LATERAL-DIRECTIONAL HANDLING QUALITY PARAMETERS

COMPUTER PRINT OUT	STANDARD NOTATION, DEFINITION	EQUATION
DR PERIOD (SEC)	Dutch roll period, seconds	$2\pi/\omega_{nd} \sqrt{1 - \zeta_d^2}$
1/C(1/2)	Dutch roll inverse cycles to 1/2 amplitude	$\frac{2\pi}{\ln 2} \sqrt{\frac{\zeta_d^2}{1 - \zeta_d^2}}, \text{ for } \zeta_d \geq 0$
SPIRAL (2) (SEC)	Spiral time to double amplitude, seconds	$T_s \ln 2, \text{ for } 1/T_s \leq 0$
P(I)	Roll rate at peak I for a unit step input of δ_a	
P(OSC)/P(AV)	A measure of the oscillatory to the average roll rate	$\frac{p_1 + p_3 - 2p_2}{p_1 + p_3 + 2p_2}, \text{ for } \zeta_d \leq 0.2$
		$\frac{p_1 - p_2}{p_1 + p_2}, \text{ for } \zeta_d > 0.2$
W(PHI)/W(D)	Ratio of the roll frequency to the dutch roll frequency	ω_{nr}/ω_{nd}
DEL-B-MAX	$\Delta\beta_m$: Maximum sideslip excursion at the c.g., occurring within two seconds or one half-period of the dutch roll, whichever is greater for a step aileron-control command	
PHI TO BETA, PHASE	$\angle \varphi/\beta$ at $s = (\zeta; \omega_n)_d$, degrees	
PHI TO BETA	$ \varphi/\beta $ at $s = (\zeta; \omega_n)_d$, rad/rad	
PHI TO VE	$ \varphi/v_e $ at $s = (\zeta; \omega_n)_d$, deg/fps	
$*v_e = (\beta)(v_{EAS}), v_{EAS} = \sqrt{\frac{2q}{\rho_0}}$		

4. NONDIMENSIONAL DERIVATIVE DEFINITIONS

a) Longitudinal Body Axis

$$C_N = \frac{N}{\bar{q} S}, \text{ positive up}$$

$$C_X = -\frac{X}{\bar{q} S}, \text{ positive aft}$$

$$C_{N\alpha} = \partial C_N / \partial \alpha$$

$$C_{N\dot{\alpha}} = \frac{2V_{T_0}}{c} \partial C_N / \partial \dot{\alpha}$$

$$C_{NM} = \partial C_N / \partial M$$

$$C_{N\delta} = \partial C_N / \partial \delta$$

$$C_{X\alpha} = \partial C_X / \partial \alpha$$

$$C_{XM} = \partial C_X / \partial M$$

$$C_{X\delta} = \partial C_X / \partial \delta$$

$$C_M = \frac{M}{\bar{q} S c}$$

$$C_{M\alpha} = \partial C_M / \partial \alpha$$

$$C_{M\dot{\alpha}} = \frac{2V_{T_0}}{c} \partial C_M / \partial \dot{\alpha}$$

$$C_{MM} = \partial C_M / \partial M$$

$$C_{Mq} = \frac{2V_{T_0}}{c} \partial C_M / \partial q$$

b) Longitudinal Stability Axis

$$C_L = \frac{L}{\bar{q} S}, \text{ positive up}$$

$$C_D = \frac{D}{\bar{q} S}, \text{ positive aft}$$

$$C_{L\alpha} = \partial C_L / \partial \alpha$$

$$C_{L\dot{\alpha}} = \frac{2V_{T_0}}{c} \partial C_L / \partial \dot{\alpha}$$

$$C_{LM} = \partial C_L / \partial M$$

$$C_{L\delta} = \partial C_L / \partial \delta$$

$$C_{D\alpha} = \partial C_D / \partial \alpha$$

$$C_{DM} = \partial C_D / \partial M$$

$$C_{D\delta} = \partial C_D / \partial \delta$$

Pitching moment

derivatives are

identical to

those for body axis

c) Lateral Body and Stability Axis

Though physically and numerically different,* see Appendix B, the same symbols are used for body axis and stability axis lateral rolling and yawing moment derivatives. The sideforce derivatives (C_y , etc.) are physically and numerically the same in both axis systems. When the rolling or yawing moment derivatives are given in this report the axis system is specified. When using the following all quantities should be for the same axis system.

$$\begin{array}{lll}
 C_y = \frac{Y}{\bar{q}S} & C_l = \frac{L}{\bar{q}Sb} & C_n = \frac{N}{\bar{q}Sb} \\
 C_{y\beta} = \partial C_y / \partial \beta & C_{l\beta} = \partial C_l / \partial \beta & C_{n\beta} = \partial C_n / \partial \beta \\
 C_{y\delta} = \partial C_y / \partial \delta & C_{lp} = \frac{2V_{T_0}}{b} \partial C_l / \partial p & C_{np} = \frac{2V_{T_0}}{b} \partial C_n / \partial p \\
 & C_{lr} = \frac{2V_{T_0}}{b} \partial C_l / \partial r & C_{nr} = \frac{2V_{T_0}}{b} \partial C_n / \partial r \\
 & C_{l\delta} = \partial C_l / \partial \delta & C_{n\delta} = \partial C_n / \partial \delta
 \end{array}$$

*The exception is the zero trim angle of attack condition.

5. DIMENSIONAL STABILITY DERIVATIVE DEFINITIONS

The same symbols are used for body- and stability-axis dimensional derivatives. Care should be exercised so that a consistent set of quantities are used.

a) Longitudinal Body Axis

$$X_u^* = X_u + T_u \cos \xi_0 \quad 1/\text{sec}$$

$$X_u = \frac{\rho S U_0}{m} \left(-\frac{M}{2} C_{X_M} - C_X + \frac{W_0}{2 U_0} C_{X_\alpha} \right) \quad 1/\text{sec}$$

$$X_w = \frac{\rho S U_0}{2m} \left[-C_{X_\alpha} - 2 \frac{W_0}{U_0} \left(C_X + \frac{M}{2} C_{X_M} \right) \right] \quad 1/\text{sec}$$

$$X_{\delta_e} = -\frac{\rho S V_{T_0}^2}{2m} C_{X_{\delta_e}} \quad \frac{\text{ft}}{\text{sec}^2 \text{rad}}$$

$$Z_u^* = Z_u - T_u \sin \xi_0 \quad 1/\text{sec}$$

$$Z_u = \frac{\rho S U_0}{m} \left(-\frac{M}{2} C_{N_M} - C_N + \frac{W_0}{2 U_0} C_{N_\alpha} \right) \quad 1/\text{sec}$$

$$Z_w = \frac{\rho S U_0}{2m} \left[-C_{N_\alpha} - 2 \frac{W_0}{U_0} \left(C_N + \frac{M}{2} C_{N_M} \right) \right] \quad 1/\text{sec}$$

$$Z_{\dot{w}} = -\frac{\rho S c}{4m} \frac{U_0}{V_{T_0}} C_{N_{\dot{\alpha}}}$$

$$Z_{\delta_e} = -\frac{\rho S V_{T_0}^2}{2m} C_{N_{\delta_e}} \quad \frac{\text{ft}}{\text{sec}^2 \text{rad}}$$

$$M_u^* = M_u + \frac{\ell_{th}}{I_y} T_u \quad \frac{1}{\text{sec-ft}}$$

$$\begin{aligned}
M_u &= \frac{\rho S c U_o}{I_y} \left[\frac{M}{2} C_{mM} + C_m - \frac{W_o}{2U_o} C_{m\alpha} \right] & \frac{1}{\text{sec-ft}} \\
M_w &= \frac{\rho S c U_o}{2I_y} \left[C_{m\alpha} + \frac{2W_o}{U_o} \left(C_m + \frac{M}{2} C_{mM} \right) \right] & \frac{1}{\text{sec-ft}} \\
M_{\dot{w}} &= \frac{\rho S c^2}{4I_y} \frac{U_o}{V_{T_o}} C_{m\alpha} & \frac{1}{\text{sec-ft}} \\
M_{\alpha} &= U_o M_w & 1/\text{sec}^2 \\
M_{\dot{\alpha}} &= U_o M_{\dot{w}} & 1/\text{sec} \\
M_q &= \frac{\rho S c^2 V_{T_o}}{4I_y} C_{mq} & 1/\text{sec} \\
M_{\delta_e} &= \frac{\rho S c V_{T_o}^2}{2I_y} C_{m\delta_e} & 1/\text{sec}^2 \\
T_u &= \frac{1}{a_m} \partial T / \partial M & 1/\text{sec}
\end{aligned}$$

b) Lateral Body Axis

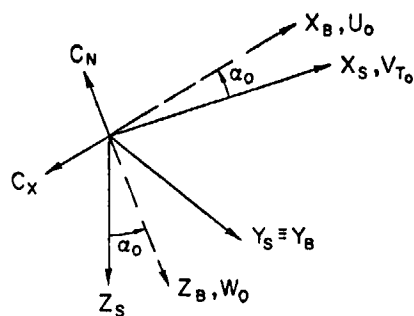
$$\begin{aligned}
Y_v &= (\rho S V_{T_o} / 2m) C_{Y\beta} & 1/\text{sec} \\
Y_{\beta} &= V_{T_o} Y_v & \text{ft/sec}^2 \\
Y_{\delta_a} &= (\rho S V_{T_o}^2 / 2m) C_{Y\delta_a} & \text{ft/sec}^2 \\
Y_{\delta_r} &= (\rho S V_{T_o}^2 / 2m) C_{Y\delta_r} & \text{ft/sec}^2 \\
Y_{\delta_r}^* &= (\rho S V_{T_o} / 2m) C_{Y\delta_r} & 1/\text{sec} \\
L_{\beta} &= (\rho S V_{T_o}^2 b / 2I_x) C_{L\beta} & 1/\text{sec}^2 \\
L_p &= (\rho S V_{T_o} b^2 / 4I_x) C_{Lp} & 1/\text{sec} \\
L_r &= (\rho S V_{T_o} b^2 / 4I_x) C_{Lr} & 1/\text{sec}
\end{aligned}$$

$$\begin{aligned}
L_{\delta_a} &= (\rho S V_{T_O}^2 b / 2 I_x) C_{l_{\delta_a}} & 1/\text{sec}^2 \\
L_{\delta_r} &= (\rho S V_{T_O}^2 b / 2 I_x) C_{l_{\delta_r}} & 1/\text{sec}^2 \\
Y_{\delta_a}^* &= (\rho S V_{T_O} / 2 m) C_{y_{\delta_a}} & 1/\text{sec} \\
N_{\beta} &= (\rho S V_{T_O}^2 b / 2 I_z) C_{n_{\beta}} & 1/\text{sec}^2 \\
N_p &= (\rho S V_{T_O} b^2 / 4 I_z) C_{n_p} & 1/\text{sec} \\
N_r &= (\rho S V_{T_O} b^2 / 4 I_z) C_{n_r} & 1/\text{sec} \\
N_{\delta_a} &= (\rho S V_{T_O}^2 b / 2 I_z) C_{n_{\delta_a}} & 1/\text{sec}^2 \\
N_{\delta_r} &= (\rho S V_{T_O}^2 b / 2 I_z) C_{n_{\delta_r}} & 1/\text{sec}^2 \\
L_{\beta}' &= (L_{\beta} + I_{xz} N_{\beta} / I_x) G & 1/\text{sec}^2 \\
L_p' &= (L_p + I_{xz} N_p / I_x) G & 1/\text{sec} \\
L_r' &= (L_r + I_{xz} N_r / I_x) G & 1/\text{sec} \\
L_{\delta_r}' &= (L_{\delta_r} + I_{xz} N_{\delta_r} / I_x) G & 1/\text{sec}^2 \\
L_{\delta_a}' &= (L_{\delta_a} + I_{xz} N_{\delta_a} / I_x) G & 1/\text{sec}^2 \\
N_{\beta}' &= (N_{\beta} + I_{xz} L_{\beta} / I_z) G & 1/\text{sec}^2 \\
N_p' &= (N_p + I_{xz} L_p / I_z) G & 1/\text{sec} \\
N_r' &= (N_r + I_{xz} L_r / I_z) G & 1/\text{sec} \\
N_{\delta_r}' &= (N_{\delta_r} + I_{xz} L_{\delta_r} / I_z) G & 1/\text{sec}^2 \\
N_{\delta_a}' &= (N_{\delta_a} + I_{xz} L_{\delta_a} / I_z) G & 1/\text{sec}^2 \\
G &= \frac{1}{1 - \frac{I_{xz}^2}{I_x I_z}}
\end{aligned}$$

APPENDIX B

TRANSFORMATION OF STABILITY AXIS DERIVATIVES TO BODY AXIS

a. NON-DIMENSIONAL STABILITY AXIS TO BODY AXIS



$$U_0 = V_{T_0} \cos \alpha_0$$

$$W_0 = V_{T_0} \sin \alpha_0$$

B-1

LONGITUDINAL

Body Axis

$$C_N = C_L \cos \alpha_0 + C_D \sin \alpha_0$$

$$C_X = C_D \cos \alpha_0 - C_L \sin \alpha_0$$

$$C_{N_\alpha} = C_{L_\alpha} \cos \alpha_0 - C_L \sin \alpha_0 + C_{D_\alpha} \sin \alpha_0 + C_D \cos \alpha_0$$

$$C_{N_{\dot{\alpha}}} = C_{L_{\dot{\alpha}}} \cos \alpha_0$$

$$C_{N_q} = C_{L_q} \cos \alpha_0$$

$$C_{N_M} = C_{L_M} \cos \alpha_0 + C_{D_M} \sin \alpha_0$$

$$C_{N_\delta} = C_{L_\delta} \cos \alpha_0 + C_{D_\delta} \sin \alpha_0$$

$$C_{X_\alpha} = C_{D_\alpha} \cos \alpha_0 - C_D \sin \alpha_0 - C_{L_\alpha} \sin \alpha_0 - C_L \cos \alpha_0$$

$$C_{X_{\dot{\alpha}}} = -C_{L_{\dot{\alpha}}} \sin \alpha_0$$

$$C_{X_q} = -C_{L_q} \sin \alpha_0$$

$$C_{X_M} = C_{D_M} \cos \alpha_0 - C_{L_M} \sin \alpha_0$$

$$C_{X_\delta} = C_{D_\delta} \cos \alpha_0 - C_{L_\delta} \sin \alpha_0$$

$$C_m, C_{m_\alpha}, C_{m_{\dot{\alpha}}}, C_{m_q}, C_{m_M}, C_{m_\delta} - \text{UNCHANGED}$$

LATERAL

Body Axis

$$(C_{l_\beta})_B = C_{l_\beta} \cos \alpha_0 - C_{n_\beta} \sin \alpha_0$$

$$(C_{l_p})_B = C_{l_p} \cos^2 \alpha_0 - (C_{l_r} + C_{n_p}) \sin \alpha_0 \cos \alpha_0 + C_{n_r} \sin^2 \alpha_0$$

$$(C_{l_r})_B = C_{l_r} \cos^2 \alpha_0 - (C_{n_r} - C_{l_p}) \sin \alpha_0 \cos \alpha_0 - C_{n_p} \sin^2 \alpha_0$$

$$(C_{l_\delta})_B = C_{l_\delta} \cos \alpha_0 - C_{n_\delta} \sin \alpha_0$$

$$(C_{n_\beta})_B = C_{n_\beta} \cos \alpha_0 + C_{l_\beta} \sin \alpha_0$$

$$(C_{n_p})_B = C_{n_p} \cos^2 \alpha_0 - (C_{n_r} - C_{l_p}) \sin \alpha_0 \cos \alpha_0 - C_{l_r} \sin^2 \alpha_0$$

$$(C_{n_r})_B = C_{n_r} \cos^2 \alpha_0 + (C_{l_r} + C_{n_p}) \sin \alpha_0 \cos \alpha_0 + C_{l_p} \sin^2 \alpha_0$$

$$(C_{n_\delta})_B = C_{n_\delta} \cos \alpha_0 + C_{l_\delta} \sin \alpha_0$$

$$C_{y_\beta}, C_{y_{\delta_r}}, C_{y_{\delta_a}} - \text{UNCHANGED}$$

b. TRANSFORMATION OF DIMENSIONAL DERIVATIVES
FROM STABILITY AXIS TO BODY AXIS

Longitudinal

$$\begin{aligned}
 (X_u)_b &= X_u \cos^2 \alpha_0 - (X_w + Z_u) \sin \alpha_0 \cos \alpha_0 + Z_w \sin^2 \alpha_0 \\
 (\dot{X}_u)_b &= \dot{Z}_w \sin^2 \alpha_0 \\
 (X_w)_b &= X_w \cos^2 \alpha_0 + (X_u - Z_w) \sin \alpha_0 \cos \alpha_0 - Z_u \sin^2 \alpha_0 \\
 (\dot{X}_w)_b &= \dot{X}_w \cos^2 \alpha_0 - \dot{Z}_w \sin \alpha_0 \cos \alpha_0 \\
 (X_{q;\delta})_b &= X_{q;\delta} \cos \alpha_0 - Z_{q;\delta} \sin \alpha_0 \\
 (Z_u)_b &= Z_u \cos^2 \alpha_0 - (Z_w - X_u) \sin \alpha_0 \cos \alpha_0 - X_w \sin^2 \alpha_0 \\
 (\dot{Z}_u)_b &= -\dot{Z}_w \sin \alpha_0 \cos \alpha_0 \\
 (Z_w)_b &= Z_w \cos^2 \alpha_0 + (Z_u + X_w) \sin \alpha_0 \cos \alpha_0 + X_u \sin^2 \alpha_0 \\
 (\dot{Z}_w)_b &= \dot{Z}_w \cos^2 \alpha_0 + \dot{X}_w \sin \alpha_0 \cos \alpha_0 \\
 (Z_{q;\delta})_b &= Z_{q;\delta} \cos \alpha_0 + X_{q;\delta} \sin \alpha_0 \\
 (M_u)_b &= M_w \cos \alpha_0 - M_u \sin \alpha_0 \\
 (\dot{M}_u)_b &= -\dot{M}_w \sin \alpha_0 \\
 (M_w)_b &= M_w \cos \alpha_0 + M_u \sin \alpha_0 \\
 (\dot{M}_w)_b &= \dot{M}_w \cos \alpha_0 \\
 (M_{q;\delta})_b &= M_{q;\delta} \\
 (I_y)_b &= I_y
 \end{aligned}$$

Lateral-Directional

$$(Y_V; \delta)_b = Y_V; \delta$$

$$(Y_{\dot{V}})_b = Y_{\dot{V}}$$

$$(Y_P)_b = Y_P \cos \alpha_0 - Y_R \sin \alpha_0$$

$$(Y_R)_b = Y_R \cos \alpha_0 + Y_P \sin \alpha_0$$

$$(L_V'; \delta)_b = L_V'; \delta \cos \alpha_0 - N_V'; \delta \sin \alpha_0$$

$$(L_{\dot{V}}')_b = L_{\dot{V}}' \cos \alpha_0 - N_{\dot{V}}' \sin \alpha_0$$

$$(L_P')_b = L_P' \cos^2 \alpha_0 - (L_R' + N_P') \sin \alpha_0 \cos \alpha_0 + N_R' \sin^2 \alpha_0$$

$$(L_R')_b = L_R' \cos^2 \alpha_0 - (N_R' - L_P') \sin \alpha_0 \cos \alpha_0 - N_P' \sin^2 \alpha_0$$

$$(N_V'; \delta)_b = N_V'; \delta \cos \alpha_0 + L_V'; \delta \sin \alpha_0$$

$$(N_{\dot{V}}')_b = N_{\dot{V}}' \cos \alpha_0 + L_{\dot{V}}' \sin \alpha_0$$

$$(N_P')_b = N_P' \cos^2 \alpha_0 - (N_R' - L_P') \sin \alpha_0 \cos \alpha_0 - L_R' \sin^2 \alpha_0$$

$$(N_R')_b = N_R' \cos^2 \alpha_0 + (L_R' + N_P') \sin \alpha_0 \cos \alpha_0 + L_P' \sin^2 \alpha_0$$

$$(I_X)_b = I_X \cos^2 \alpha_0 + 2I_{XZ} \sin \alpha_0 \cos \alpha_0 + I_Z \sin^2 \alpha_0$$

$$(I_Z)_b = I_Z \cos^2 \alpha_0 - 2I_{XZ} \sin \alpha_0 \cos \alpha_0 + I_X \sin^2 \alpha_0$$

$$(I_{XZ})_b = (I_Z - I_X) \sin \alpha_0 \cos \alpha_0 + I_{XZ}(\cos^2 \alpha_0 - \sin^2 \alpha_0)$$

APPENDIX C

EQUATIONS OF MOTION, TRANSFER FUNCTIONS, AND COUPLING NUMERATORS

1. Longitudinal

a. Equations

$$\begin{bmatrix} (1 - X_{\dot{u}})s - X_u^* & -X_{\dot{w}}s - X_w & (-X_q + W_0)s + g \cos \theta_0 \\ -Z_{\dot{u}}s - Z_u^* & (1 - Z_{\dot{w}})s - Z_w & (-Z_q - U_0)s + g \sin \theta_0 \\ -M_{\dot{u}}s - M_u^* & -(M_{\dot{w}}s + M_w) & s^2 - M_q s \end{bmatrix} \begin{bmatrix} u \\ w \\ \theta \end{bmatrix} = \begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \\ M_{\delta_e} \end{bmatrix} \begin{bmatrix} \delta_e \end{bmatrix}$$

$$q = s\theta$$

$$\dot{h} = -w \cos \theta_0 + u \sin \theta_0 + (U_0 \cos \theta_0 + W_0 \sin \theta_0)\theta$$

$$a_z = sw - U_0 q + (g \sin \theta_0)\theta$$

$$a_z' = a_z - l_x s^2 \theta$$

$$\dot{h}' = \dot{h} + l_x \cos \theta_0 \dot{\theta}$$

b. Transfer Functions

$$\frac{\theta}{\delta_e} = \frac{N_{\delta_e}^{\theta}}{\Delta}$$

$$1) \text{ Denominator, } \Delta = As^4 + Bs^3 + Cs^2 + Ds + E$$

$$A = (1 - Z_{\dot{w}})$$

$$B = -(M_q + X_u^*)(1 - Z_{\dot{w}}) - Z_w - M_{\dot{u}}$$

$$C = M_q Z_w - M_{\dot{u}} + X_u^*[(M_q)(1 - Z_{\dot{w}}) + Z_w + M_{\dot{u}}] \\ - X_w Z_u^* + W_0[M_{\dot{w}} Z_u^* + M_u^*(1 - Z_{\dot{w}})] + g M_{\dot{w}} \sin \theta_0$$

NOTE: Terms including $X_{\dot{u}}$, $Z_{\dot{u}}$, $M_{\dot{u}}$, $X_{\dot{w}}$ are neglected in polynomial expressions.

$$D = -X_u^*(M_q Z_w - M_\alpha) - M_u^* X_\alpha + M_q X_w Z_u^* + g[M_w Z_u^* + M_u^*(1 - Z_w)] \cos \theta_o + W_o(M_w Z_u^* - M_u^* Z_w) \\ + g(M_w - M_w X_u^*) \sin \theta_o$$

$$E = g(M_w Z_u^* - M_u^* Z_w) \cos \theta_o + g(M_u^* X_w - M_w X_u^*) \sin \theta_o$$

2) Numerators

$$N_\delta^\theta = A_\theta s^2 + B_\theta s + C_\theta$$

$$A_\theta = Z_\delta M_w + M_\delta(1 - Z_w)$$

$$B_\theta = X_\delta[M_w Z_u^* + M_u^*(1 - Z_w)] + Z_\delta(M_w - M_w X_u^*) - M_\delta[Z_w + X_u^*(1 - Z_w)]$$

$$C_\theta = X_\delta(M_w Z_u^* - M_u^* Z_w) + Z_\delta(M_u^* X_w - M_w X_u^*) + M_\delta(Z_w X_u^* - X_w Z_u^*)$$

$$N_\delta^u = A_u s^3 + B_u s^2 + C_u s + D_u$$

$$A_u = X_\delta(1 - Z_w)$$

$$B_u = -X_\delta[M_q(1 - Z_w) + Z_w + M_\alpha] + Z_\delta X_w - W_o[Z_\delta M_w + M_\delta(1 - Z_w)]$$

$$C_u = X_\delta(M_q Z_w - M_\alpha) - Z_\delta(g M_w \cos \theta_o + M_q X_w) + M_\delta[X_\alpha - (g \cos \theta_o)(1 - Z_w)] \\ + W_o(Z_w M_\delta - M_w Z_\delta) + g X_\delta M_w \sin \theta_o$$

$$D_u = g(Z_w M_\delta - M_w Z_\delta) \cos \theta_o + g(X_\delta M_w - M_\delta X_w) \sin \theta_o$$

$$N_\delta^w = A_w s^3 + B_w s^2 + C_w s + D_w$$

$$A_w = Z_\delta$$

$$B_w = -Z_\delta(M_q + X_u^*) + U_o M_\delta + X_\delta Z_u^*$$

$$C_w = X_u^*(Z_\delta M_q - U_o M_\delta) + W_o(Z_\delta M_u^* - M_\delta Z_u^*) - g M_\delta \sin \theta_o + X_\delta(M_u^* U_o - Z_u^* M_q)$$

$$D_w = g(Z_\delta M_u^* - M_\delta Z_u^*) \cos \theta_o + g M_\delta X_u^* \sin \theta_o - X_\delta M_u^* g \sin \theta_o$$

$$N_{\delta}^h = A_h^* s^3 + B_h^* s^2 + C_h^* s + D_h^*$$

$$A_h^* = -\cos \theta_O A_w + \sin \theta_O A_u$$

$$B_h^* = -\cos \theta_O B_w + \sin \theta_O B_u + (U_O \cos \theta_O + W_O \sin \theta_O) A_{\theta}$$

$$C_h^* = -\cos \theta_O C_w + \sin \theta_O C_u + (U_O \cos \theta_O + W_O \sin \theta_O) B_{\theta}$$

$$D_h^* = -\cos \theta_O D_w + \sin \theta_O D_u + (U_O \cos \theta_O + W_O \sin \theta_O) C_{\theta}$$

$$N_{\delta}^{a_z'} = A_{a_z'} s^4 + B_{a_z'} s^3 + C_{a_z'} s^2 + D_{a_z'} s + E_{a_z'}$$

$$A_{a_z'} = A_w - l_x A_{\theta}$$

$$B_{a_z'} = B_w - l_x B_{\theta} - U_O A_{\theta}$$

$$C_{a_z'} = C_w - l_x C_{\theta} - U_O B_{\theta} + g \sin \theta_O A_{\theta}$$

$$D_{a_z'} = D_w - U_O C_{\theta} + g \sin \theta_O B_{\theta}$$

$$E_{a_z'} = + g \sin \theta_O C_{\theta}$$

To obtain a_z , let $l_x = 0$.

2. Lateral

a. Equations

$$\begin{bmatrix} s - Y_v & -\frac{W_O s + g \cos \theta_O}{V_{T_O}} & \frac{U_O s - g \sin \theta_O}{V_{T_O} s} \\ -L_{\beta}^* & s(s - L_p^*) & -L_r^* \\ -N_{\beta}^* & -N_p^* s & s - N_r^* \end{bmatrix} \begin{bmatrix} \beta \\ \frac{p}{s} \\ r \end{bmatrix} = \begin{bmatrix} Y_{\delta_a}^* & Y_{\delta_r}^* \\ L_{\delta_a}^* & L_{\delta_r}^* \\ N_{\delta_a}^* & N_{\delta_r}^* \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}$$

$$v = V_{T_O} \beta$$

$$a_y = sv + U_O r - W_O p - g(\cos \theta_O) \phi$$

$$\phi = \frac{p}{s} + \frac{r}{s} \tan \theta_O$$

$$a_y' = a_y + l_{x_{lat}} sr - l_{z_{sp}}$$

$$\psi = \frac{1}{\cos \theta_O} \frac{r}{s}$$

b. Transfer Functions

$$\frac{\varphi}{\delta_a} = \frac{N_{\delta a}^{\varphi}}{\Delta_{lat}} \quad ; \quad \frac{r}{\delta_r} = \frac{N_{\delta r}^r}{\Delta_{lat}} \quad ; \quad \text{etc.}$$

$$1) \text{ Denominator, } \Delta_{lat} = as^4 + bs^3 + cs^2 + ds + e$$

$$a = 1$$

$$b = -(Y_v + L_p^i + N_r^i)$$

$$c = \frac{U_o}{V_{T_o}} N_{\beta}^i + L_p^i (Y_v + N_r^i) - N_p^i L_r^i + Y_v N_r^i - \frac{W_o L_{\beta}^i}{V_{T_o}}$$

$$d = \frac{U_o}{V_{T_o}} (N_p^i L_{\beta}^i - L_p^i N_{\beta}^i) + Y_v (N_p^i L_r^i - L_p^i N_r^i) - \frac{g}{V_{T_o}} (L_{\beta}^i \cos \theta_o + N_{\beta}^i \sin \theta_o) \\ + \frac{W_o}{V_{T_o}} (L_{\beta}^i N_r^i - N_{\beta}^i L_r^i)$$

$$e = \frac{g}{V_{T_o}} [(L_{\beta}^i N_r^i - N_{\beta}^i L_r^i) \cos \theta_o - (N_p^i L_{\beta}^i - L_p^i N_{\beta}^i) \sin \theta_o]$$

$$2) \delta (\delta_a \text{ or } \delta_r) \text{ Numerators}$$

$N_{\delta}^{\beta} = A_{\beta} s^3 + B_{\beta} s^2 + C_{\beta} s + D_{\beta}$
$A_{\beta} = Y_{\delta}^*$
$B_{\beta} = -Y_{\delta}^* [L_p^i + N_r^i] - N_{\delta}^i \frac{U_o}{V_{T_o}} + \frac{W_o}{V_{T_o}} L_{\delta}^i$
$C_{\beta} = Y_{\delta}^* (L_p^i N_r^i - N_p^i L_r^i) + L_{\delta}^i \frac{g}{V_{T_o}} \cos \theta_o + (N_{\delta}^i L_p^i - L_{\delta}^i N_p^i) \frac{U_o}{V_{T_o}} \\ + \frac{W_o}{V_{T_o}} (N_{\delta}^i L_r^i - L_{\delta}^i N_r^i) + N_{\delta}^i \frac{g}{V_{T_o}} \sin \theta_o$
$D_{\beta} = \frac{g}{V_{T_o}} (N_{\delta}^i L_r^i - L_{\delta}^i N_r^i) \cos \theta_o + \frac{g}{V_{T_o}} (N_p^i L_{\delta}^i - N_{\delta}^i L_p^i) \sin \theta_o$

$$N_{\delta}^p = A_p s^3 + B_p s^2 + C_p s + D_p$$

$$A_p = L_{\delta}'$$

$$B_p = Y_{\delta}^* L_{\beta}' - L_{\delta}'(N_r' + Y_v) + N_{\delta}' L_r'$$

$$C_p = Y_{\delta}^*(L_r' N_{\beta}' - L_{\beta}' N_r') + L_{\delta}' Y_v N_r' - N_{\delta}' Y_v L_r' + (L_{\delta}' N_{\beta}' - N_{\delta}' L_{\beta}') \frac{U_o}{V_{T_o}}$$

$$D_p = -\frac{g}{V_{T_o}} (L_{\delta}' N_{\beta}' - N_{\delta}' L_{\beta}') \sin \theta_o$$

$$N_{\delta}^r = A_r s^3 + B_r s^2 + C_r s + D_r$$

$$A_r = N_{\delta}'$$

$$B_r = Y_{\delta}^* N_{\beta}' + L_{\delta}' N_p' - N_{\delta}'(Y_v + L_p')$$

$$C_r = Y_{\delta}^*(L_{\beta}' N_p' - N_{\beta}' L_p') - L_{\delta}' Y_v N_p' + N_{\delta}' Y_v L_p' + \frac{W_o}{V_{T_o}} (L_{\delta}' N_{\beta}' - N_{\delta}' L_{\beta}')$$

$$D_r = \frac{g}{V_{T_o}} (L_{\delta}' N_{\beta}' - N_{\delta}' L_{\beta}') \cos \theta_o$$

$$N_{\delta}^{\phi} = A_{\phi} s^2 + B_{\phi} s + C$$

$$A_{\phi} = A_p + A_r \tan \theta_o$$

$$B_{\phi} = B_p + B_r \tan \theta_o$$

$$C_{\phi} = C_p + C_r \tan \theta_o$$

$$N_{\delta y}^{a'} = A_{a_y}' s^4 + B_{a_y}' s^3 + C_{a_y}' s^2 + D_{a_y}' s + E_{a_y}'$$

$$A_{a_y}' = V_{T_O} A_{\beta} + l_{x_{lat}} A_r - l_z A_p$$

$$B_{a_y}' = V_{T_O} B_{\beta} + U_O A_r - W_O A_p + l_{x_{lat}} B_r - l_z B_p$$

$$C_{a_y}' = V_{T_O} C_{\beta} + U_O B_r - W_O B_p - g \cos \theta_O A_{\phi} + l_{x_{lat}} C_r - l_z C_p$$

$$D_{a_y}' = V_{T_O} D_{\beta} + U_O C_r - W_O C_p - g \cos \theta_O B_{\phi} + l_{x_{lat}} D_r - l_z D_p$$

$$E_{a_y}' = U_O D_r - W_O D_p - g \cos \theta_O C_{\phi}$$

To obtain a_y , let $l_{x_{lat}} = l_z = 0$.